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Notes on al-Battānī's *Zīj*

from *Al-Battānī sive Albatenii Opus astronomicum* (Milan, 1899–1907)

translated from the Latin

Nallino's edition of al-Battānī's *Zīj* gives the Arabic text (Part III, 1899), a Latin translation of the canons with a commentary on them (Part I, 1903), and the tables in Latin with notes on them, a glossary and indexes (Part II, 1907). The canons themselves are translated into English from the Arabic in the English *Zīj* of this edition; this volume translates Nallino's own writing, which is in Latin and has not been translated before. It is made from the line-by-line transcription of his Latin in this edition (S03, S04 and S07). Every sentence is translated, in Nallino's order; nothing is summarised or left out. His Arabic, Greek and Hebrew quotations stand in their scripts, as he prints them, with his translations of them in English. Proper names are given in a modern transliteration. In the headings of the commentary, «p.» and «l.» are the pages and lines of Nallino's Latin translation (Part I, pp. 1–150); «t. III» is the Arabic text, whose pages the English *Zīj* gives in its margin; in the heads of the notes on the tables, «p.» is a page of the tables (Part II). Nallino's footnotes are numbered; the few notes of the translator are lettered. Nallino's own corrections in the «Addenda et emendanda» of Part II are followed and noted. The pages of Nallino's volumes are given in the margin.

The translation is in progress. This version holds the whole of the commentary of Part I (pp. 151–327) and the notes on the tables of Part II (pp. 189–255). It is AI-translated (Claude), 2026; no person has proofread it. Nallino's work is in the public domain; the translation is dedicated to the public domain (CC0 1.0).

Commentary

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On chapter 3

a) — pp. 9–10.

All the propositions on *chords* that the author gives are proved in the *Almagest* I, 9 (ed. Halma, vol. I, pp. 26–37). If we denote the radius by the letter R, they are expressed in our algebraic notation as follows:

(α) $\operatorname{chord}\frac{360^\circ}{6}$ or $\operatorname{chord} 60^\circ = R$

(β) $\operatorname{chord}\frac{360^\circ}{3}$ or $\operatorname{chord} 120^\circ = \sqrt{4R^2 - \operatorname{chord}^2\frac{360^\circ}{6}}$

(γ) $\operatorname{chord}(180^\circ - a) = \sqrt{4R^2 - \operatorname{chord}^2 a}$

(δ) $\operatorname{chord}\frac{360^\circ}{4}$ or $\operatorname{chord} 90^\circ = \sqrt{2R^2}$

(ε)[a] $\operatorname{chord}\frac{360^\circ}{10}$ or $\operatorname{chord} 36^\circ = \sqrt{R^2 + \left(\frac{2R}{4}\right)^2} - \frac{2R}{4}$

(ζ) $\operatorname{chord}\frac{360^\circ}{5}$ or $\operatorname{chord} 72^\circ = \sqrt{\operatorname{chord}^2\frac{360^\circ}{10} + R^2}$

(η) $\operatorname{chord}(a-b) = \frac{\operatorname{chord} a \operatorname{chord}(180^\circ - b) - \operatorname{chord} b \operatorname{chord}(180^\circ - a)}{2R}$

(ϑ) $\operatorname{chord}\frac{a}{2} = \sqrt{\frac{2R - \operatorname{chord}(180^\circ - a)}{2} \times 2R}$

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To find the chord of the sum of two arcs, Ptolemy (ed. Halma, vol. I, pp. 32–33), whom our author follows, has nothing like the rule (η), but goes by way of the chord of the supplement[1]; hence

(ι)[b] $\operatorname{chord}(180^\circ - [a+b]) = \frac{\operatorname{chord}(180^\circ - a)\operatorname{chord}(180^\circ - b) - \operatorname{chord} a \operatorname{chord} b}{2R}$

(κ) $\operatorname{chord}(a+b) = \sqrt{4R^2 - \operatorname{chord}^2(180^\circ - [a+b])}$

Almost all the historians of mathematics describe the way by which Ptolemy came to compute the chord of 1°; of the most recent, one may consult M. Cantor, *Vorlesungen über Geschichte der Mathematik*, Leipzig 1880 ff., vol. I, pp. 353–355 (or 390–392 of the second edition, of 1894); R. Wolf, *Handbuch der Astronomie, ihrer Geschichte und Litteratur*, Zürich 1890–93, vol. I, pp. 164–165, § 61*b*; A. von Braunmühl, *Vorlesungen über Geschichte der Trigonometrie*, Leipzig 1900 ff., vol. I, p. 21. The error, which F. Woepcke determined with Lagrange's interpolation formula (*Discussion de deux méthodes arabes pour déterminer une valeur rapprochée de sin 1°*, in the Journal de Mathématiques pures et appliquées published by J. Liouville, vol. XIX, 1854, p. 165), is 0.00000009, that is, less than 1/10,000,000; it therefore becomes noticeable only in the thirds,

[1] Delambre, *Astron. ancienne*, vol. II, p. 39, is mistaken (as Schiaparelli points out to me) in ascribing to Ptolemy the rule

$$\operatorname{crd}(p+q) = \frac{\operatorname{crd} p \operatorname{crd}(180^\circ - q) + \operatorname{crd} q \operatorname{crd}(180^\circ - p)}{2R}.$$

[*Translator:* The print has – between the two products; Nallino's Addenda et emendanda (Part II, p. XXI) correct it to +, which gives the chord of the sum.]

which Ptolemy leaves out of his table of chords. — A value of the sine of 1° nearer the truth was found by Abū l-Wafāʾ (940–998), whose error shows only in the fifths, that is, in the ninth decimal place (sin 30′ = $0^p\,31'\,24''\,55'''\,54^{IV}\,55^{V}$ instead of $54^{V}\,47^{VI}\,16^{VII}$); see Woepcke, *Recherches sur l'histoire des sciences mathématiques chez les Orientaux*, in the Journal Asiatique, 5th series, vol. XV, 1860, pp. 286–303; Cantor, I, 641–642 (= 703 of the second edition); von Braunmühl, I, 56–57. — According to what Mīram Čelebī (d. 931 H = 1524–25) has handed down to us in his commentary on Ulugh Beg (in Sédillot's French translation, pp. 69–77, note), other, early Arabs, whose names are not known, carried Ptolemy's method through and found a value of the sine of 1° that exceeds the truth by less than 1/100,000,000; for the error is 37/416 (or about 1/11) of Ptolemy's error, since they make sin 1° = $1^p\,2'\,49''\,43'''\,13^{IV}\,5^{V}\,30^{VI}$ instead of $11^{IV}\,14^{V}\,44^{VI}$. But had they not deliberately neglected certain small fractions in their computations, they could have found a far better number, whose error would have been 1/64 of Ptolemy's. See Woepcke, *Discussion de deux méthodes* etc., pp. 155–157, 159–167. — Lastly, Ghiyāth al-Dīn[1] Jamshīd al-Kāshī, in whom other Arabs had probably gone before him, computed sin 1° = $1^p\,2'\,49''\,43'''\,11^{IV}$ correctly and most elegantly, by solving the equation of the trisection of the angle approximately, that is, by expanding the equation of the third degree in a series; his method, handed down to us by Mīram Čelebī, loc. cit., pp. 77–83, was explained by Woepcke, *Discussion*, pp. 167–176, 301–303. See also H. Hankel, *Zur Geschichte der Mathematik im Alterthum und Mittelalter*, Leipzig 1874, pp. 290–293; Cantor, I, 671–672 (= 737 of the second edition); von Braunmühl, I, 73–75.

b) — pp. 10–11.

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The technical word *jayb*, by which the Arabs denote the sine, al-Battānī never mentions, although it was already in common use at the beginning of the ninth century, as appears from the unpublished Latin translation of the astronomical work of Muḥammad ibn Mūsā al-Khwārizmī, made by Adelard of Bath in the twelfth century. For there the Arabic name *elgeib* is kept (with the Arabic article *al* or *el* put before it), and «*elgeib elmustewi* or plane» [*al-jayb al-mustawī*], that is, the right sine, and «*elgeib elmacus* or diminished»[2], that is, the versed sine, are named. Likewise Ḥabash, a contemporary of al-Battānī, always uses the word *jayb*. But our author would not take up the word, which is of foreign origin and wholly alien to Greek usage. Not a few historians of mathematics, then, and among the best, are wrong when they ascribe the use of the word *jayb* to our author and think that Plato of Tivoli brought the word *sinus* to Europe with his translations; of the more recent I shall cite only Hankel, p. 280; Wolf, *Handb. der Astron.*, I, 166, § 62*a*; Cantor, I, 632 (= 693 of the second edition), II, 101 and 249. The old Kaestner, however, had already corrected the error, *Geschichte der Mathematik*, Göttingen 1796, vol. I, p. 522; see now M. Koppe, *Die Behandlung der Logarithmen und der Sinus*, Programm des Andreas-Realgymnasium, Berlin 1893, p. 32; von Braunmühl, I, 50. — The word *sinus* occurs only once in Plato of Tivoli's translation: «si autem per has cordas [*sic*] versas arcus scire desideras, si chorda quam habueris minus 60 fuerit, eam de 60 minue, et residui arcum scito, quem de 90 minues, et quod remanserit, erit arcus *sinus* versi» («but if you wish to know the arcs from these versed chords: if the chord you have is less than 60, take it from 60 and find the arc of the remainder; take that from 90, and what remains will be the arc of the versed *sinus*»; fol. 7v of the Nuremberg edition; the passage corresponds to p. 11, ll. 28–30 of my translation). But, as von Braunmühl, I, 50, note 3, rightly judges, the word must be taken for a gloss of Regiomontanus that has crept into the text by mistake.

[1] Some indeed call him, wrongly, ʿAtāb al-Dīn, reading عتاب ʿAtāb for غياث Ghiyāth. He died about the year 840 H, 1436–37, as H. Suter, *Die Mathematiker und Astronomen der Araber*, Leipzig 1900, pp. 173–4, no. 429, rightly established; to his arguments one may add that the little work on finding the sine of 1° (which still survives in the Cairo Library) is already praised by Qāḍīzāde al-Rūmī, who died between 840 and 850 H, as Mīram Čelebī relates.

[2] Adelard's error; *maʿkūs* = versed, inverted. The passage was cited by Chasles, *Recherches sur l'astronomie ancienne*, in the Comptes rendus de l'Académie des Sciences, vol. XXIII, session of 2 November 1846, p. 850.

Since chords are hardly ever used in trigonometric computations, it is no wonder that al-Battānī called the halves of the chords of the double arcs by the name of chords; a similar usage is found among the Indians as early as the fourth century, of whom one reads the following in the notes of Burgess and Whitney on the book entitled *Sūrya-siddhānta*: «In this passage, the sine is called *jyārdha*, 'half-chord'; hereafter, however, that term does not once occur, but *jyā* 'chord' (literally 'bowstring') is itself employed, as are also its synonyms *jīvā*, *māurvikā*, to denote the sine. The usage of Albategnius is the same» (*Translation of the Sūrya-Siddhānta, a text-book of Hindu astronomy, by* E. Burgess, note on ch. II, verse 28, in the Journal of the American Oriental Society, vol. VI, New Haven 1860, p. 201). — Nor should it be passed over that among the Arabs the name of the whole *chord* too comes from the usage of the Indians, as the passage just quoted shows; among the Greeks the chord was called ὑποτείνουσα, «subtending», ἡ ἐν τῷ κύκλῳ εὐθεῖα, «the straight line in the circle», εὐθεῖα ἐν κύκλῳ μὴ διὰ τοῦ κέντρου οὖσα, «a straight line in the circle not passing through the centre», εὐθεῖα ἐλάττων τοῦ διαμέτρου, «a straight line smaller than the diameter».

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The origin of the technical word *jayb* was explained satisfactorily by Munk, whom F. Woepcke, *Essai sur la propagation des chiffres indiens*, in the Journal Asiatique, 6th series, vol. I, 1863, p. 478, note, and almost all later writers have followed. From the Indian word *jīvā*, chord and sine, which I mentioned a little above, the Arabs, who have no letter *v*, made *jīb* جيب; for the Sanskrit *j* is pronounced like the Arabic *j*. But since *jīb* has no meaning in Arabic, while the letters جيب can also be read *jayb* without any change to the writing (for in ordinary writing the small signs that show the true pronunciation of the vowels are almost always left out), the Arabs preferred this form of the name, which in their language means not only «the round opening of a garment through which a man's neck and head pass», but also «the fold or pocket made in a garment for carrying small things»; occasionally, but rarely, it also means a man's bosom itself. We have already seen that Adelard of Bath kept the Arabic name *elgeib* in his translations; but his contemporary Gerard of Cremona (1114–1187), in translating the astronomy written by Jābir ibn Aflaḥ, used the word *sinus*, which came into our languages by this route, as Koppe, loc. cit., rightly observed[1]. J. Ruska, *Zur Geschichte des «Sinus»* (in the Zeitschrift für Mathematik und Physik, vol. XL, Leipzig 1895, historisch-literarische Abtheilung, p. 128), cites two Syriac writers of the thirteenth century[2] in whom the technical word *jayb* is always rendered by the Syriac *ʿūbā* ܥܘܒܐ, which means a man's bosom; but he is wrong when he writes: «Die Uebersetzung durch *sinus* ist demnach als richtig durch die syrische Belege gesichert. Die verständigere Uebersetzung aber, welche einen Mann voraussetzt, dem der mathematische Inhalt und dessen klarer Ausdruck wichtiger ist als pedantische Wörtlichkeit, lautet natürlich *chorda* . . . . . Beide Uebersetzungen finden sich vor: die zweite, wie Koppe nachgewiesen hat, bei Plato von Tivoli» («The translation by *sinus* is thus confirmed as correct by the Syriac evidence. The more sensible translation, however, which presupposes a man to whom the mathematical content and its clear expression matter more than pedantic literalness, is of course *chorda* . . . . . Both translations occur: the second, as Koppe has shown, in Plato of Tivoli»). For Plato of Tivoli writes *chorda* not from any striving after a more rational translation (how foreign that was to him and to the other translators of his age, their translations show plainly enough everywhere), but because al-Battānī uses this very word.

That the Arab mathematicians learned from the Indians the terms unknown to the Greeks — the chord, the sine, the right sine (Sanskrit *kramajyā*), the versed sine (Sanskrit *utkramajyā*), the sagitta (Sanskrit *śara* or *ishu*) — can hardly be doubted. To this one may add what the Arab writers relate of the Indian book entitled *al-Sindhind* (that is, the Sanskrit *Brahma-sphuṭa-siddhānta*), which was brought to the caliph at Baghdad in the year 156 H, 773. When all this is weighed, the

[1] But from the fact that it was Gerard of Cremona, not Plato of Tivoli, who spread the use of this word among us, I do not see how Koppe can call into doubt the origin of the Arabic *jayb* that Munk explained.

[2] They belong, then, to a late age not only of Syriac but also of Arabic letters, and not, as von Braunmühl, I, 50, note 1, writes, «in den ersten Zeiten des Mittelalters», to the first times of the Middle Ages.

history of the trigonometric functions can be told as follows. The Indians, moved above all by their use of orthogonal projections (von Braunmühl, I, 31–32), used right and versed sines instead of the chords of the Greeks; they computed them for the twenty-fourth parts of the quadrant, that is, for every 3° 45′, taking a radius of 3438 minutes. Once they had received the first notion of sines from the Indians, and had come to know Ptolemy's books at about the same time, the Arabs rightly understood that it was more convenient and more exact to construct new sines by halving Ptolemy's chords (which are computed for every half degree and a radius of sixty parts). This had already been done before al-Battānī.

c) — p. 11.

The *interpolation* for finding the sines of arcs that fall between two consecutive arcs of the table is the same as the one found in Ptolemy for computing chords: the change of the sines and of the chords is taken to be proportional to the change of the arcs within the interval of the table, half a degree. What Cantor, I, 355 (= 392 of the second edition), writes of Ptolemy, «ein Beweis für diese angenommene Proportionalität in engem Bereiche ist dagegen nicht vorhanden» («a proof of this proportionality, assumed within a narrow range, is however not given»), may rightly be said also of al-Battānī, of Ḥabash (fol. 18v) and of the other Arabs. More exact interpolations they used only in constructing tables. — Tables of sines for every minute of the quadrant, computed by Ibn Yūnus, survive in the Arabic codex 5752 of the Royal Library of Berlin; Ulugh Beg also computed others.

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Versed sines, whose use the Indians knew before the Arabs, the author defines:

$$\text{versed sine } a = R - \cos a.$$

He thought it superfluous to construct a table of them, which on the other hand we find in Ḥabash. At the foot of the page two formulas are given; one is

(α) $\quad \cos a = \sqrt{R^2 - \sin^2 a},$

which follows from the theorem of Pythagoras and, with $R = 1$, corresponds to our familiar rule

$$\cos a = \sqrt{1 - \sin^2 a}.$$

The other,

(β) $\quad \sin \dfrac{90^\circ - a}{2} = \sqrt{30^p\,(R - \sin a)},$

comes from the rule of al-Battānī and of Ptolemy mentioned in note *a* (p. 153),

$$\text{chord } \frac{a}{2} = \sqrt{\frac{2R - \text{chord}\,(180^\circ - a)}{2} \times 2R},$$

which, when sines are put for chords and $R = 60^p$ is taken, becomes

$$\sin \frac{a}{2} = \sqrt{30^p\,(R - \cos a)}.$$

Why al-Battānī took the complement of *a* in place of *a*, I do not know.

On chapter 4

a) — p. 12.

Of the Arabs who measured the *obliquity of the ecliptic* before the year 1000, Ibn Yūnus relates the following in the eleventh chapter of his work[1]: «Between Ptolemy and the authors of the *Tested Tables* (*al-mumtaḥan*) I know of no observations of the obliquity except one made a little after[2] the year 160 of the Hijra [which began on 18 October 776], whose author gives the obliquity as 23° 31′. As for the value found by the authors of the *Tested Tables*[3] in observations close to our own age: Yaḥyā ibn Abī Manṣūr, Sanad ibn ʿAlī, al-ʿAbbās ibn Saʿīd al-Jawharī and a group of learned and eminent men (وطايفة من اهل العلم والفضل) investigated it in [the quarter called] al-Shammāsiyya of the city of Baghdad, in the time of [the caliph] al-Maʾmūn, and found 23° 33′, as Muḥammad ibn Mūsā al-Khwārizmī in his astronomical work (في زيجه) and Muḥammad ibn Kathīr al-Farghānī in his book on the making of the astrolabe report on their authority; and the matter is well known[4]. Those who observed at Damascus after the death of Yaḥyā ibn Abī Manṣūr, with the instrument that al-Maʾmūn had ordered to be made when he set out for the land of the Byzantines (بالالة التي امر باتخاذها المامون حين توجـه الى بـلاد الروم) — they are Khālid ibn ʿAbd al-Malik al-Marwarrūdhī, Abū l-Ṭayyib Sanad ibn ʿAlī, ʿAlī ibn ʿĪsā[a] al-Asṭurlābī and others — report that they found the greatest declination of the ecliptic to be 23° 33′ 52″; and the measurement made with this instrument was in the year 201 of Yazdajird» [which began on 27 April 832][5]. I 158

«Abū l-Ḥasan Thābit ibn Qurra says: I found ancient operations from before Ptolemy which show (تدل) that the greatest declination is 23° 35′[6]. Muḥammad ibn Jābir ibn Sinān al-Battānī relates that he found 23° 35′ by his observations. The noble and eminent Abū l-Qāsim ʿAlī ibn al-Ḥusayn ibn Muḥammad ibn ʿĪsā, known as Ibn al-Aʿlam, a man of learning and of outstanding intelligence, says that he measured the greatest declination and, taking the utmost care in the measurement, found 23° 34′ 2″. Abū l-Ḥasan al-Ṣūfī ʿAbd al-Raḥmān ibn ʿUmar relates that he measured the greatest declination and, having carried out the measurement with the greatest accuracy, found 23° 33′ 45″[7]; that man held an eminent place in the science of geometrical proofs. Help comes from God.»

Ibn Yūnus, ch. IV (in Caussin, p. 120), says that the value 23° 35′ was chosen in the *Tested Tables*; and al-Farghānī reports the same in ch. V (p. 20 of Golius's edition, fol. 121r of the Paris codex 2504). But Ibn Rusta, who copies the whole of this chapter of his contemporary al-Farghānī, has

[1] P. 222 of the Leiden codex (of which the Paris codex no. 2495 is a most careful copy, keeping the same division of the pages). I compared the passage with the Paris codex 2496 (which gives selected chapters of Ibn Yūnus's work and is written more carelessly), fol. 25r.

[2] What I have rendered «a little after» is in Arabic نيف, that is, «more than two but fewer than ten» [years after].

[3] This is the astronomical work that the astronomers of the caliph al-Maʾmūn (reigned 813–833) compiled from their observations under the direction of Yaḥyā ibn Abī Manṣūr.

[4] Ibn Yūnus himself confirms this in the introduction to his work (in Caussin, p. 56), saying that in the year 214 H (which began on 10 March 829), 198 of Yazdajird (which began on 28 April 829), Yaḥyā ibn Abī Manṣūr, al-ʿAbbās ibn Saʿīd al-Jawharī, Sanad ibn ʿAlī and others found 23° 33′ in the city of Baghdad. — Al-Zarqālī, in the unpublished Latin translation of the *Canons* or *Toledan Tables*, and the Jew Abraham ibn Ezra (in the second half of the twelfth century) say that Yaḥyā ibn Abī Manṣūr found 23° 33′ 30″; see Delambre, p. 177 (whose codex of the Toledan Tables gave the name in a corrupt form, «Almeon fils d'Albumazar»), and M. Steinschneider, *Études sur Zarkali*, in the Bullettino di bibliografia e di storia delle scienze matematiche e fisiche, vol. XX, 1887–88, p. 87 (where a codex of the Tables is cited that has the name «Jahiben filius Albumasaris»).

[5] He relates the same in the introduction (Caussin, 56); the year there is 217 H (which began on 7 February 832), 201 of Yazdajird.

[6] But in his own works Thābit chose 23° 33′, as appears from Ḥabash, fol. 90v, and Delambre, p. 73.

[7] The Paris codex 2496 wrongly writes 23. 35. 45. 38.

[a] The print has ʿAlī ibn Mūsā; Nallino's Addenda et emendanda (Part II, p. XXI) correct it to ʿAlī ibn ʿĪsā.

23° 33′ for 23° 35′[1]; and 23° 33′ is also read in the Latin translation of al-Farghānī made by John of Seville and printed at Nuremberg in 1537 (fol. 5r). This seems the more probable when one compares what Ibn Yūnus himself relates of the measurement made by the authors of the *Tested Tables*, and what Rabiçag of Toledo writes in his *Libro de las armellas*[2], which is drawn wholly from Arabic sources: «Et sabet que en este nuestro tiempo segund el nostro rectificar. bien se paresció por los rectificamentos antigos et los nuevos. que lo que ha entre ellos [that is, the interval between the pole of the ecliptic and the pole of the equator] non es uno todauia. Et esto es porque Abrachis [= Hipparchus] falló la declinacion que era .XXIII. grados et .LI. menudos. et falláronla los rectificadores [that is, the authors of the Tested Tables] en el tiempo de Memun Abdalla. fiio de Haron Arraxit .XXIII. grados et XXXIII. menudos. Et fallóla Mohamat. fiio de Aben el Bateni [= Muḥammad ibn Sinān al-Battānī] .XXIII. grados et .XXXV. menudos» («And know that in this our time, by our rectifying, it has appeared clearly from the ancient and the new rectifications that what lies between them is not always one. And this is because Abrachis found the declination to be 23 degrees and 51 minutes; and the rectifiers found it in the time of Memun Abdalla, son of Haron Arraxit, to be 23 degrees and 33 minutes; and Mohamat, son of Aben el Bateni, found it to be 23 degrees and 35 minutes»). — Thābit ibn Qurra says the same in the book *On the motion of the eighth sphere, or of access and recess*, which survives in two different Latin translations in the Vatican Library; one (cod. Lat. 4275, f. 84v, col. I) says: «Et eius [that is, of the whole obliquity] summa est 24 graduum secundum quod translatum est ab Indis, et similiter reperit Ptholomeus 23 g. et 51 minut., et consideratores [that is, اصحاب الممتحن, the authors of the Tested Tables] invenerunt in tempore Maymonis 23 g. et 33 m.» («and its amount is 24 degrees according to what has been handed down from the Indians, and likewise Ptolemy found 23 degrees and 51 minutes, and the observers found in the time of al-Maʾmūn 23 degrees and 33 minutes»); the other (cod. Lat. 4083, f. 56v, col. II): «pervenit talis declinatio ad 24 gradus secundum quod ab Indis traditum fuit. Ptolomeus quoque invenit eam 23 graduum et 51 mt. Alii autem inspectores post Ptolomeum in partem Mamonis eam 23 graduum 33 minutorum invenerunt»[a] («this declination reached 24 degrees according to what was handed down by the Indians. Ptolemy too found it to be 23 degrees and 51 minutes. But other observers after Ptolemy, on the side of al-Maʾmūn, found it to be 23 degrees 33 minutes»). — Lastly, it should not be passed over that Thābit ibn Qurra, as we saw in note 1, chose 23° 33′; and so did Qusṭā ibn Lūqā, who died about 910 or 912, in his book on the construction and use of the celestial globe[3]. — They would hardly have chosen 23° 33′ had they not found this value in the *Tested Tables*.

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Of those who flourished after the authors of the *Tested Tables*, down to the last years of al-Battānī, we know that the family of the Banū Mūsā[4], al-Nayrīzī[5] and the family of the Banū Amājūr[6] found the obliquity to be 23° 35′ by observation. So did, among later astronomers, Abū l-Wafāʾ, in the city of Baghdad, in the year 987 (on the evidence of Delambre, p. 156), and Ibn Yūnus in the year 391 H (which began on 30 November 1000; ch. XI, in Caussin, p. 238; more exactly he had found 23° 34′ 52″). — Many others adopted this obliquity of 23° 35′ in their tables, such as Ḥabash (a

[1]Ibn Rusta, *Kitâb al-Aʿlâk an-Nafisa VII*, edited by M. J. de Goeje, Leiden 1889, p. 16 [= Bibliotheca Geographorum Arabicorum, vol. VII].

[2]Book I, ch. IV = *Libros del saber de astronomia del Rey D. Alfonso X de Castilla, copilados, anotados y comentados por D. Manuel Rico y Sinobas*, Madrid 1863–67, vol. II, p. 6. It belongs to the second half of the thirteenth century.

[3]Ch. XXXIV of the Spanish translation (*Libros del saber de astronomia*, vol. I, p. 187), ch. XXX of the unpublished Hebrew translation (in Baldi, *Vite di matematici arabi con note di* M. Steinschneider, Rome 1874, p. 70, note).

[4]Ibn Yūnus, ch. V, in Caussin, p. 152. — One of these Banū Mūsā died in the month of Rabīʿ I 259 H = January 873; on them see Suter, *Die Mathematiker und Astronomen der Araber und ihre Werke*, Leipzig 1900, pp. 20–21, no. 43 (= Abhandlungen zur Geschichte der mathematischen Wissenschaften, part X; supplement to the 45th year of the Zeitschrift für Mathematik und Physik).

[5]Ibn Yūnus, ch. IV, in Caussin, p. 120. — He died about the year 310 H (922–23); see Suter, p. 45, no. 88.

[6]Ibn Yūnus, ch. V, in Caussin, p. 152. — They observed in the years 272–321 H (885–933); see Suter, pp. 49–50, no. 99.

[a]The print has «quos» for «quoque» and «tempore» for «partem»; Nallino's Addenda et emendanda (Part I, p. LXXII) correct both.

contemporary of al-Battānī), al-Kharaqī in his *Muntahā al-idrāk*[1], the Hebrew-Spanish Abraham bar Ḥiyya, writing in 1135[2], and his contemporary ʿAbd al-Raḥmān al-Khāzinī[3]; likewise Naṣīr al-Dīn al-Ṭūsī in the books he wrote before the *Īlkhānī Tables*[4], the Greek Chrysococces in his *Persian Tables*[5], and lastly, at the beginning of the fourteenth century, al-Jaghmīnī (as appears from pp. 264 and 265). Barhebraeus, or Abū l-Faraj, in his Syriac Astronomy written in 1279, uses round numbers, and so in places he gives the obliquity as 24°[6]; but where he wants more exact numbers he takes it as 23° 35′, as can easily be inferred from his computations of the right and the oblique ascensions (pp. 165–167 of the Syriac text).

Abū ʿAlī al-Ḥasan (that is, Aboul Hhassan), ch. XXIII, pp. 175 and 176, says that the obliquity varies between the limits 23° 33′ and 23° 53′; but in ch. XXIV, pp. 180–187, and in his tables, he adopted 23° 35′.

It should be noted that an error of two minutes in fixing the obliquity is of little consequence in the computations derived from it; for in the right ascensions it can produce a discrepancy only in the seconds of arc[7]; in the oblique ascensions the greatest difference never reaches three minutes of arc (especially since the latitudes that the Arabs had to use were always smaller than 45°); and lastly, in computing the greatest half arc of the day of the sun in the lands under Arab rule, the difference can never exceed ten seconds of an hour. Such differences could be neglected in those times.

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So that one may judge with what care the Arabs observed the greatest declination of the ecliptic, I have made a small table of the chief observations, using in my computations Bessel's formula, which for t years before 1750 is

$$\varepsilon = 23^\circ\,28'\,18.0'' + 0.48368''\,t - 0.00000272295''\,t^2.$$

With an asterisk I mark the years of the observations that are certain; where the year is not known, I have put the epoch of the author's tables, or the year that can be assumed approximately. I have given the errors to the second; but everyone knows that these numbers cannot always be taken absolutely, since sometimes the year of the observation is not known for certain, or the Arab observer deliberately neglected the seconds[8].

[1] Laurentian codex (Orient.-Palat. 110), f. 40r (maqāla I, bāb XIV, faṣl I).

[2] *Sphaera mundi auctore Rabbi Abrahamo Hispano filio R. Haijae. Osw. Schreckenfuchsius vertit in linguam Latinam, Seb. Munsterus illustravit annotationibus.* Basel 1546, ch. X.

[3] Vatican Arabic codex 761, f. 10v and in many places.

[4] As in the *Tadhkira*, Laurentian codex (Palat.-Orient. 271), f. 38v (bāb II, faṣl IV).

[5] In Ismael Bullialdus, *Astronomia Philolaica*, Paris 1545, part II, p. 224. — Bullialdus thinks that he flourished in the first half of the thirteenth century (loc. cit., p. 213); but Krumbacher, *Geschichte der Byzantinischen Litteratur*, 2nd ed., Munich 1897, says that he lived about the year 1346.

[6] This value was known to the Greeks before Eratosthenes; the Indians always used it. — See also below, p. 161, note 3.

[7] [The greatest difference is in the octants, that is, at the right ascensions 45°, 135°, 225°, 315°, where an error of one minute in fixing the obliquity produces an error of 0.217′ (or very nearly 13″) in the right ascension. — Schiaparelli]

[8] The small tables that, after Lalande (I think), circulate among our writers (the most careful Houzeau and Wolf not excepted) contain errors whose origin I do not know; for they ascribe to Eratosthenes 23° 45′ 7″, to al-Battānī 23° 35′ 41″, to Ibn Yūnus 23° 34′ 26″, to Ulugh Beg 23° 31′ 48″, to Abū ʿAlī al-Ḥasan (Aboul Hhassan) 23° 33′ 45″!

Observers	*Place of the observation*	*Year*	*Obliquity observed*	*True (mean) obliquity*	*Error*
Eratosthenes	Alexandria	230 BC	23° 51′ 20″	23° 43′ 45.01″	+ 7′ 35″
Hipparchus[1]	Rhodes	130 BC	23 51 20	23 42 56.62	+ 8 23
The authors of the Tested Tables[2]	Baghdad	* 829	23 33	23 35 41.16	− 2 41
Other astronomers of the caliph al-Maʾmūn	Damascus	* 832	23 33 52	23 35 39.71	− 1 48
Banū Mūsā	Baghdad	860	23 35	23 35 26.32	− 0 26
al-Battānī	al-Raqqa	880	23 35	23 35 16.74	− 0 17
Ibn al-Aʿlam	Baghdad?	960	23 34 2	23 34 37.92	− 0 36
ʿAbd al-Raḥmān al-Ṣūfī	Baghdad?	965	23 33 45	23 34 35.50	− 0 50
Abū l-Wafāʾ	Baghdad	* 987	23 35	23 34 25.46	+ 0 35
Wayjan ibn Rustam al-Kūhī[3]	Baghdad	* 988	23 51 1	23 34 24.98	+ 16 36
Ibn Yūnus	Cairo	* 1001	23 34 52	23 34 18.74	+ 0 33
Alfonsine Tables[4]	Toledo	1250	23 32 29	23 32 19.16	+ 0 10
Naṣīr al-Dīn al-Ṭūsī[5]	Marāgha	1270	23 30	23 32 9.49	− 2 9
Ibn al-Shāṭir	Damascus	1363	23 31	23 31 24.78	− 0 24
Ulugh Beg	Samarqand	1437	23 30 17	23 30 49.13	− 0 32

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The author's words, p. 12, ll. 14–24, fully confirm what I had already said in note 9, p. 5: that al-Battānī could by no means have inferred a *continuous* decrease of the obliquity of the ecliptic from the uncertain observations of the Greeks. How could he have argued it, when from Eratosthenes to Theon, that is, over an interval of about 600 years, all the Greeks always say they found 23° 51′ 20″?

I 162

To the later Arabs, who could compare many good observations with one another, the hypothesis of a continuous decrease was not unknown, although the small discrepancies could have been put down to errors of the observers and of the instruments. Indeed, we read the following in the astronomical book entitled the Sanjarī (*al-zīj al-sanjarī*), which Abū Manṣūr ʿAbd al-Raḥmān al-Khāzinī wrote about the year 520 H (1126)[6]: «In [fixing] the obliquity of the ecliptic there are many discrepancies, about whose cause opinions differ. The most likely of all is that they arise only from the refraction of the rays (وغالب الظنّ ان سببه انعطاف الشعاع دون سايره), so that, when the latitude of two observers is one and the same, the same obliquity appears to both. The greatest declination is 24° 0′ with the Indians, 23° 51′ with Ptolemy and Hipparchus, 23° 35′ with the authors of the Tested [Tables] and al-Battānī, 23° 33′ with some of the moderns (بعض المحدثين). Because of these discrepancies, and because of the irregularity in the motion of the fixed stars, some thought fit (ظهر لبعضهم) to posit a motion in the space between the two spheres, that of the universe and that of the fixed stars, which would complete the whole circle every 1285 (١٢٨٥) solar years, and whose poles, like the poles of the fixed stars too, would not lie on the axis of the universe; so that the poles of the signs of the zodiac would approach the equators or recede from them, and the motion of the fixed stars would be quickened or slowed, in a way that it

[1] Ptolemy, in the year 140, keeps the same value; his error is therefore + 10′ 10″. The error of Theon, who wrote about the year 360, is + 12′ 1″.

[2] If 23° 33′ 30″ is admitted, as some say (see p. 157, note 4), the error becomes − 2′ 11″. [*Translator:* The print refers to «p. 6, note 3»; Nallino's Addenda et emendanda (Part II, p. XXI) correct the reference to p. 157, note 4.]

[3] According to the passage of Ibn al-Qifṭī that Casiri, *Bibliotheca Arabico-hispana Escurialensis*, Madrid 1760–70, vol. I, p. 441, and Sédillot, in the preface to the text of Oloug Beg, pp. LII–LIII, cite. The codex of Ibn al-Qifṭī kept at Rome in the Library of the Accademia dei Lincei gives the same reading.

[4] So in many places in the *Libro de las taulas alfonsies* (*Libros del saber de astronomia*, vol. IV) and in the *Libro del quadrante para rectificar* (ibid., vol. III, pp. 296–297).

[5] According to Sédillot's note on the text of Oloug Beg, p. 63; the passage he cites is found on f. 105r of the Florence codex of these same *Īlkhānī Tables*. The first declinations (ff. 120r–121r) and the second (ff. 121v–122v) are computed with this value.

[6] Vatican Arabic codex no. 761, in the third chapter (*bāb*) of the preface, f. 10v.

would take too long to set out now. But since the obliquity is 23° 35′ with most [authors], we shall use this value in our reckonings, as one rightly accepted (استعملناه في الاعتبار مُسلا)». — Naṣīr al-Dīn al-Ṭūsī, in his Memorandum of what is chiefly to be remembered in astronomy[1], relates the following: «The total declination found by the ancient and by the more recent observations is not one and the same; the value the ancients found is greater than the one the more recent have measured. What a more recent observer found is sometimes taken to be less than what an earlier one found, the greatest value found not reaching 24°[2], while the least does not go below 23° 1/2 + 1/(2 × 10) [= 23° 33′][3]. Most (الجمهور) set it at 23° 1/4 + 1/3 [= 23° 35′]. Because of these discrepancies some have thought that the ecliptic moves in latitude and approaches the equator; if this were true, another sphere would have to be posited that carries the zodiac with such a motion. If the ecliptic moved, it could complete the circle; or it could not complete it, but reach a certain limit from which it would afterwards turn back. Any such limit could lie after the two falls of the ecliptic onto the equator, or at the time of the second fall, or between the two falls; and this last [in three ways, namely] after half the circle has been completed, or exactly at the half circle, or before the semicircle is completed. Lastly, the ecliptic could also not reach the space between the two falls, if it is thought to turn back either at the time of its first fall onto the equator or before it. There are, then, eight hypotheses. By the first five, the two halves of the zodiac, the northern and the southern, would exchange places, together with their properties; and by three of these [five], each half of the zodiac would [at times] fall onto each half of the equator. By the first seven hypotheses, a half of the ecliptic would coincide with the half of the equator nearest to it; whenever this happened, days and nights would be equal in all places of the earth, and the different seasons of the year would fail. By the last [that is, the eighth] hypothesis[4], however, there would follow only an increase or a decrease in the altitudes [of the sun] and in the lengths of the days and the nights».

Quṭb al-Dīn Maḥmūd al-Shīrāzī, who died in the month of Ramaḍān 710 (February 1311), says in his *al-Tuḥfa al-shāhiyya* (ff. 5v–6r) that no definite rate of decrease is found among the different values of the obliquity measured at different times.

Those who have written the history of astronomy say that in the second half of the sixteenth century some first recognized the continuous decrease of the obliquity, such as Ignazio Danti in his *Trattato dell'Astrolabio*, Florence 1569, though many others of that age called it into doubt; that Tycho Brahe (1546–1601) inferred it from the changed latitudes of the stars; and that, lastly, Euler (1707–1783) proved it by the laws of celestial mechanics.

b) — p. 12, ll. 27–32.

[1] *Al-Tadhkira fī ʿilm al-hayʾa*, in the fourth chapter (*faṣl*) of the second section (*bāb*), ff. 38r–39r of the codex of the Laurentian Library, which also contains the commentary of Qāḍīzāde. — A French summary of the passage is given by Carra de Vaux, *Les sphères célestes selon Nasîr-eddîn Attûsî*, at the end of P. Tannery's *Recherches sur l'histoire de l'astronomie ancienne*, Paris 1893, p. 346.

[2] The Indian astronomers always have 24°. — Qāḍīzāde remarks on this passage: «Some think that the sages of the Indians found it [24°], and that this same value was very well known in the time of Euclid, which is why [Euclid] in his book, after computing the side of the hexagon inscribed in a circle, derived the side of the pentadecagon inscribed in a circle, since 24 is the fifteenth part of the circumference». — So too Quṭb al-Dīn Maḥmūd al-Shīrāzī, *al-Tuḥfa al-shāhiyya*, f. 5v: «Some think that all the Indians agree on this value [24°], and that 24° was the obliquity in the time of Euclid; which is why he derived the side of the pentadecagon in his book, since 24 is the fifteenth part of the circumference». — Strange enough that some saw the obliquity of the ecliptic foreshadowed in the fifteenth and sixteenth propositions of the fourth book of Euclid, on inscribing regular hexagons and pentadecagons in a circle! [*Translator:* Nallino's Addenda et emendanda (Part II, pp. XXI–XXII) trace this story to Heron of Alexandria's commentary on Euclid, as quoted by al-Nayrīzī, and cite Proclus and, after Eudemus, Theon of Smyrna, who give the distance between the two poles as the side of the fifteen-sided polygon, 24°; the Addenda are translated with Part II.]

[3] Qāḍīzāde: «So at the time this book was written; but afterwards the author himself, with the new observations that he set up [in the city of] Marāgha, found 23° 30′, which is the least value found so far». And in fact Naṣīr al-Dīn adopted the obliquity 23° 30′ in his *Īlkhānī* tables.

[4] By which the ecliptic is thought never to coincide with the equator.

The rule for finding the **declination** of the *sun*, or of any degree of the ecliptic, follows from Ptolemy's construction, *Almag.* I, 12 (ed. Halma, vol. I, pp. 56–58). Let ε be the obliquity of the ecliptic, δ the declination of the given degree, L the longitude of the sun or of the given degree; by the well-known theorem of Menelaus Ptolemy proves

$$\frac{\text{chord}\,180^\circ}{\text{chord}\,2\varepsilon} = \frac{\text{chord}\,180^\circ}{\text{chord}\,2\delta} \times \frac{\text{chord}\,2L}{\text{chord}\,180^\circ}.$$

Since chord 180° is the diameter (2R), if sines are used instead of chords, there results

$$\frac{2R}{2\sin\varepsilon} = \frac{2R}{2\sin\delta} \times \frac{2\sin L}{2R},$$

whence al-Battānī's formula[1]

$$\sin\delta = \frac{\sin L\ \sin\varepsilon}{R}.$$

c) — p. 13, ll. 15–18.

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Nothing like this is read in Ptolemy, but it is found in Theon's *Canon*, ed. Halma, vol. I, p. 54: «For when the sun is between 1° and 90° we shall say that it is descending towards the north [*sic*, βοῤῥᾶν κατιέναι]; when between 91° and 180°, that it is descending to the south; between 181° and 270°, that it is ascending towards the south [*sic*, νότον ἀνιέναι]; between 271° and 360°, that it is ascending to the north». Likewise in the tables of the declinations of the sun, under the columns of the degrees of the ecliptic, Theon, I, 144, writes:

From 0° to 90°:	Βοῤῥαῖος καταβαθμός, northern descent.
From 90° to 180°:	Νότιος καταβαθμός, southern descent.
From 180° to 270°:	Νότιος ἀναβαθμός, southern ascent.
From 270° to 360°:	Βοῤῥαῖος ἀναβαθμός, northern ascent.

As in al-Battānī, we read in the *Libro de las taulas alfonsies*, ch. XX (*Libros del saber de astronomia*, vol. IV, p. 136): «et si el logar fuer de un grado fata tres signos. el sol es suuiente en septentrion. et si fuer de tres signos fata .VI. es decendiente en septentrion. et si fuer de .VI. signos fata .IX. es decendiente en mediodía. et si fuer de .IX. signos fata .XII. es suuiente en mediodía» («and if the place is from one degree to three signs, the sun is ascending in the north; and if it is from three signs to six, it is descending in the north; and if it is from six signs to nine, it is descending in the south; and if it is from nine signs to twelve, it is ascending in the south»).[a]

[1] With Ptolemy and al-Battānī the radius is of sixty parts; in our trigonometric formulas, on the other hand, $R = 1$ is taken, so that in our usage the author's rule becomes $\sin\delta = \sin L \sin\varepsilon$.

[a] Nallino's Addenda et emendanda (Part I, p. LXXII) add to this note that Theon's strange definitions are found also in the Astrologia of Vettius Valens, and that Julianus of Laodicea agrees with al-Battānī's usage; the Addenda are translated with Part I's front matter.

On chapter 5

a) — pp. 13–14.

The computation of the *right ascension.* Let δ be the declination of the given degree of the ecliptic, α its right ascension, ε the obliquity of the ecliptic; Ptolemy, *Almag.* I, 13 (ed. Halma, vol. I, pp. 60–61), proves by the theorem of Menelaus

$$\frac{\text{chord}\, 2\alpha}{\text{chord}\, 180^\circ} = \frac{\text{chord}\, 2\delta}{\text{chord}\, (180^\circ - 2\delta)} \times \frac{\text{chord}\, (180^\circ - 2\varepsilon)}{\text{chord}\, 2\varepsilon}.$$

Using sines instead of chords, one derives

$$\frac{2\sin\alpha}{2R} = \frac{2\sin\delta}{2\cos\delta} \times \frac{2\cos\varepsilon}{2\sin\varepsilon},$$

whence al-Battānī's formula

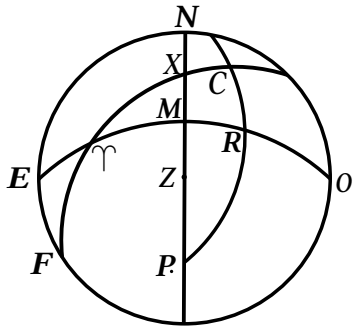
$$\sin\alpha = R\,\frac{\sin\delta\,\cos\varepsilon}{\cos\delta\,\sin\varepsilon}.$$

Ḥabash, who lived in the same age as al-Battānī, had constructed tables of tangents and cotangents (taking a radius of 60 parts), and so could teach in his work (fol. 88r), more briefly[a],

$$\sin\alpha = \frac{\text{tg}\,\delta\,\,\text{cotg}\,\varepsilon}{R}.$$

b) — p. 14, ll. 10–12.

[Now that the computation of the right ascensions of any point of the ecliptic is known, the author wants to show how the time of the culmination of those points is found, that is, the degree of the celestial equator that rises from the horizon of the given city at the time of the culmination.



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Let the celestial sphere be projected onto the plane of the horizon FENO of the given city; let P be the pole, PN the meridian, EMO the equinoctial circle, ♈ the beginning of Aries, F♈C the ecliptic, F the ascendant, C the beginning of Capricorn, PC the solstitial colure. Necessarily ♈R, that is, the arc of the celestial equator cut off between the point ♈ of the vernal equinox and the point R of the solstitial colure, is of 90 degrees; likewise the arc EM, cut off by the horizon and the meridian, must always be 90°. Clearly, then, at any time E♈ = MR. — Let X be the point of the ecliptic whose time of culmination we want to know; XC will be its longitude counted from

[a]The print has sin α = R tg δ cotg ε; Nallino's Addenda et emendanda (Part II, p. XXIV) correct it to tg δ cotg ε : R, which is right when the sine and the tangents are all taken to the radius R.

the beginning of Capricorn, MR its right ascension from the winter solstitial colure. Since we have seen that MR equals E♈ (which is the oblique ascension corresponding to the arc ♈F of the ecliptic), the number of degrees contained in the arc MR will equal the degree of the equator then rising from the horizon; and this in all places of the earth. The computation of the arc MR, since XC is given, reduces very easily to the computation of the arc M♈ when X♈ is given, which is the problem al-Battānī solved a little before. So the author's «one and the same number» means that at any time and in any place of the earth there holds the relation

right ascension of the culminating degree X of the ecliptic, counted from the colure of Capricorn = oblique ascension of the rising degree F of the ecliptic,

that is,

90° + right ascension of the culminating degree, counted from the beginning of Aries = oblique ascension of the rising degree.

Al-Battānī's tables of right ascensions, fol. 179v–181r, therefore count the ascensions from the colure of Capricorn, not, as is the common usage of the Greeks and our own, from the beginning of Aries. In this way, to give an example, one learns at once from the tables that the end of the tenth degree of Pisces (that is, 340° from the beginning of Aries) culminates at the same time as the point of the equinoctial circle that is 71° 34′ from the vernal equinox rises. — Schiaparelli.]

Ḥabash always counts the right ascensions from the beginning of Aries, as the other Arab astronomers published so far usually do; Abū ʿAlī al-Ḥasan, however (Aboul Hhassan, ch. XXXII, pp. 218–227 and in the tables), follows al-Battānī's usage. Ibn Yūnus (on the evidence of Delambre, p. 122) made a double table of right ascensions, counting them both from the beginning of Aries and from the solstitial colure of Capricorn; and so did Ulugh Beg (Oloug Beg, p. 67 of the text, 93 of the translation, in ch. VII of the second part), who afterwards uses the ascensions counted from the winter solstitial colure in the astrological computations of the direction of the significator (p. 161 of the text, 214 of the translation) and of the projection of the rays (p. 158 of the text, 210 of the translation), in which it is more convenient to take the distances from the solstitial colure than from the equinoctial one. Tables of right ascensions beginning from the winter solstitial colure, for every minute of the ecliptic, were also constructed by ʿAbd al-Qādir ibn Muḥammad al-Fayyūmī, who died in 1022 H (1613–14); they are kept in the Cairo Library; see Suter, *Die Mathematiker und Astronomen der Araber*, p. 193, no. 479.

The oblique ascensions, on the other hand, are always counted from the beginning of Aries.[a]

On chapter 6

a) — p. 17, ll. 10–11.

I 165

Plato wrongly has 3500. — Those ancients, then, reckoned the degree of the earth's circumference at 75 miles; the 65 (in Plato 85) that are read at the end of the chapter seem to be an error, as I said on p. 19, note 8. Such a degree is of Graeco-Syrian origin. The Roman mile (1479.5 metres) contains seven *philetaerian* stadia (213.048 m), which were in very wide use in Egypt and Syria

[a] Nallino's Addenda et emendanda add to this note further tables of right ascensions counted from the beginning of Capricorn (Part I, p. LXXII: al-Khāzinī, Kūshyār ibn Labbān, al-Zarqālī, Ibn al-Musharraf, al-Akhṣāṣī; Part II, p. XVII: Theon's tables, and later Arabic manuscripts that call them «the ascensions of the culmination» or «the ascensions of the Cupola»); the Addenda are translated with the front matter of the two parts.

in the decline of the Roman empire. The Graeco-Syrian writers believed that the stadia of which Ptolemy gives 180,000 to the circumference of the earth were philetaerian; they therefore held that the equator of the earth contains 27,000 miles, and the degree 75 miles. Some Arabs, however, confusing the Roman miles with the Arabic ones (1973 m), took over the numbers of the Syrian writers without change. Compare what I discussed in *Il valore metrico del grado di meridiano secondo i geografi arabi*, in Guido Cora's *Cosmos*, vol. XI, Turin 1892, pp. 50–53 (= pp. 9–11 of the offprint); add that Jacob of Edessa, who died in 708 or 710, writes in his Syriac *Hexaemeron* that some make the degree 90 miles, others 75 (A. Hjelt, *Études sur l'Hexaméron de Jacques d'Édesse*, Helsingfors 1892, p. 20).

Al-Battānī does not mention the measurement of a degree of the meridian that Arab astronomers carried out in the reign of al-Ma'mūn (813–833) between the cities of Palmyra (Tadmur) and al-Raqqa, and in the plain of Sinjār (in Mesopotamia). They had found that, between about 35° and 36° of latitude, the degree is 56 2/3 Arabic miles, that is, 111,814.67 m; their error was therefore only 877 metres. See what I wrote, loc. cit., pp. 53–63, 105–121[a] (= pp. 12–39 of the offprint).

b) — p. 17, l. 22.

The Indians held that the island of *Laṅkā* (Taprobane, today Ceylon), placed on the equator, occupied the middle place in the longitude of the earth; the Arabs, not rightly understanding the Indian tradition, held that the central meridian of the earth passed through an island known by no name and lying on the equator between India and Ethiopia. The place where the central meridian cuts the equator they called *qubbat al-arḍ*, «the dome of the earth», as the author says, or simply *al-qubba*, «the dome, the cupola»; whence the corrupt word *Catha* in the old Latin translation of the astrology of ʿAlī ibn Abī l-Rijāl[1].

It was also known by the name *qubbat Uzayn* (or *Azīn* ازين); for the Indians held that the first meridian of the island of Laṅkā passed through the city of *Ujjayinī*, very famous among them (Ptolemy's Ὀζήνη, today Ujjain اجين in the province of Mālwa in central India, 75° 51′ 45″ E of Greenwich, 23° 11′ 10″ N), which flourished greatly in trade in the first three centuries of the Christian era and was perhaps the chief seat of astronomical studies in India. The Arabs mostly confused *Uzayn* or *Azīn* (for, with the signs of the vowels left out as usual, ازين can be read either way) with the island of Laṅkā itself, and put it on the equator; later writers, leaving out a dot in the writing, pronounced it *Arīn* (ارين), whence *Arin*, *Arim*, *Arym* in our writers of the Middle Ages. Read the excellent discussion of Reinaud, *Introduction générale à la Géographie d'Aboulféda*, Paris 1848, pp. CCXXXVI–CCLIV; of the Arabs, al-Bīrūnī above all should be consulted (*India*, pp. 158–159 of the text, vol. I, pp. 306–308 of the English translation).

c) — p. 17, ll. 22–27.

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I noted already in the translation that lines 7–25 are read in the same words in al-Masʿūdī, *Prairies*, vol. I, pp. 179–181, who ascribes them to «the sages» (*al-ḥukamā'*); al-Battānī, on the other hand, to anonymous «ancients» (see l. 3). We shall see in note *d* that the first source of this sixth chapter is Graeco-Syrian. Our author reports what he has received, neither approving nor disapproving it, as in ch. L (pp. 120–123) on the distances and the sizes of the celestial spheres and bodies; it is strange, all the same, that he did not reject the two gross errors of these lines. [For the area of a spherical quadrilateral bounded by two parallels and two meridians is not found by multiplying

[1] *Albohazen Haly filii Abenragel, De iudiciis astrorum libri octo, doctorum aliquot virorum opera in Latinum sermonem conversi, postremo a barbarie vindicati et puritati linguae donati, per Antonium Stupam.* Basel 1571, book VIII, ch. 8, p. 374.

[a] The print has 195–121.

the angular extent in latitude by the angular extent in longitude, since the degrees of one parallel, in linear measure, necessarily differ from the degrees of another. But even if one could compute areas as the author says, the computation would be wrong: half the circumference ought to be multiplied not by a sixth of the whole 360°, but by a third of the 180° that the latitude spans from pole to pole, so that the area found would be not 1/12 but 1/6 of the earth[1]. — The area of the spherical zone between the equator and the parallel of 60° is to the whole sphere as the sine of 60° is to the diameter; so half that zone will be to the whole sphere as the sine of 60° (that is, ½√3, or 0.8660) to twice the diameter (that is, 4, the radius being taken as 1). Instead of 0.8660 : 4, or 0.2165 : 1, those ancients found 1 : 12, or 0.0833 : 1, which is nearly three times less than the truth. — Schiaparelli.]

d) — pp. 17–19.

The description of the seas and the lands. — In the translation I have already pointed out, in their places, the Arab authors who drew from the same unknown source from which al-Battānī took his information; and I said that our author's passage had already been edited and translated into French by Reinaud. It remains to inquire what this common source is.

The oldest of all the Arab writers who tell such things about the seas in their books still extant today are Ibn Rusta, who wrote between the years 285 and 300 H (898–913), and Qudāma ibn Jaʿfar, who died in the year 337 H (which began on 10 July 948); al-Kharaqī, on the other hand, as we shall see below, says that he borrowed his information from al-Jayhānī (who wrote a little after 301 H, 913–14) «and other sages». All of them, therefore, belong to the age of our author. But a passage of al-Masʿūdī, *Tanbīh*, p. 51, sheds more light for us; for of the Indian sea he tells the following: «According to those who have applied themselves to measuring and depicting the earth by its places in astronomical longitudes and latitudes, the length of this sea from east to west comprises 8000 miles; its breadth to the north 2700, and [also] 1900 is said. So Ptolemy reports, and others who lived before and after him; the last who has attained these things in Islam is Yaʿqūb ibn Isḥāq al-Kindī, in his epistle on the seas and their ebb and flow, and his pupil Aḥmad ibn al-Ṭayyib in his epistle on the uses of the seas, the mountains and the rivers». Al-Masʿūdī mentions the same men in the *Prairies*, vol. I, p. 276, where he speaks of the breadth of the sea of the Byzantines[2]. In this way we reach a time before the age of al-Battānī; for al-Kindī, who had ordered a translation of Ptolemy's Geography to be made for himself[3], died about the year 260 H (which began on 26 October 873); his pupil Aḥmad ibn al-Ṭayyib al-Sarakhsī in the year 286 (which began on 16 January 899).

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Consider now that the degree of 75 miles comes from the Graeco-Syrian writers, as I showed in note *a*; add what I shall prove in the second part of this work[4], namely that unmistakable Syriac traces are to be recognized in al-Battānī's tables of the longitudes and latitudes of cities: it will not be a rash conjecture that the early Arabs received that geographical information and those measures of the seas from the Syrians. And the conjecture becomes quite certain, since in fact all these measures are found already in the famous Syriac writer Jacob of Edessa[5].

[1] The error is read again on p. 19, l. 21.

[2] Elsewhere (*Prairies*, I, p. 256), speaking again of the sea of the Byzantines, he cites al-Battānī: «Many authors of astronomical tables, among them Muḥammad ibn Jābir al-Battānī and others, say in their astronomical works that its length is 5000 miles, and that its breadth varies from 800 to 700 and 600 miles».

[3] See my *al-Ḫuwârizmî e il suo rifacimento della Geografia di Tolomeo*, p. 6, note 10 (Memorie della R. Accademia dei Lincei, Classe di scienze morali, 5th series, vol. II, 1894).

[4] I have already set this out in my little work *Le tabelle geografiche d'al-Battânî tradotte ed annotate*, Turin 1898, p. 3 (= Guido Cora's *Cosmos*, 2nd series, vol. XII, p. 163).

[5] He was born about 633 and died in 708 or 710. The geographical part is read in Syriac and in Latin in A. Hjelt, *Études sur l'Hexaméron de Jacques d'Édesse, notamment sur ses notions géographiques contenues dans le 3me Traité*, Helsingfors 1892.

If we now ask whence the Graeco-Syrians collected all this, we can only agree with the words of al-Masʿūdī quoted above, that it takes its first origin from Ptolemy's *Geography*. Let the miles be turned into degrees, taking the degree as 75 miles and making no distinction between the degree of the meridian and the degrees of the parallels (an error which, as we saw in note *c*, those authors had already made): everything will be found to agree perfectly with Ptolemy's map. It is surprising that Reinaud[1] and Lelewel[2] not only did not recognize this but at times denied it. For the former, considering the reading of the codex that gives the Indian sea 1900 (instead of 900) miles of breadth beyond the equator, writes on p. CCLXXXIV: «Mille et neuf cents milles, à raison de soixante et quinze milles par degré, donnaient un peu plus de vingt-cinq degrés. Voilà un des motifs qui me font croire qu'Albateny n'a pas admis le système de Ptolémée, relativement à la prolongation du continent africain du côté de l'est» («Nineteen hundred miles, at seventy-five miles to the degree, gave a little more than twenty-five degrees. This is one of the reasons that make me believe that Albateny did not accept Ptolemy's system as regards the extension of the African continent towards the east»). So, in drawing the picture of the world according to al-Battānī (or rather according to the ancient Greeks and Syrians whom al-Battānī copied), he makes the African coast, immediately south of the equator, turn west from the farthest peninsula of the Somalis in an almost straight line and join the coast of the Atlantic Ocean. Lelewel, vol. I, p. 37, note 86, attacks this, rightly observing, among other considerations, that to determine the breadth of the Indian Ocean is the same as to put land to the south of it; but afterwards he adds: «Mais nourrir l'idée d'une mer interne Hind, n'est pas suivre le système de Ptolémée ou copier sa carte, et je désire disculper Albateni d'une semblable accusation» («But to entertain the idea of an inner sea of India is not to follow Ptolemy's system or to copy his map, and I wish to clear Albateni of such an accusation»). And on p. 37, note 83, after reporting some of al-Battānī's measures of the seas, he remarks quite wrongly: «Comptant 75 milles au degré, ces nombres paraissent trop forts même pour le système de Ptolémée. La longueur des deux mers intérieurs [that is, the sea of the Byzantines and the Indian sea], 5000 et 8000, ferait évaluer la longueur de l'habitable sur le grand cercle à 185 1/3 degrés [*sic*, but 13000 : 75 = 173 1/3], à quoi il faudrait ajouter quelques degrés de plus. Pour adapter ces nombres à la carte ptoléméenne, il faudrait compter au moins 80 milles au degré du grand cercle» («Counting 75 miles to the degree, these numbers seem too large even for Ptolemy's system. The length of the two inner seas, 5000 and 8000, would put the length of the inhabited land on the great circle at 185 1/3 degrees, to which some degrees more would have to be added. To fit these numbers to Ptolemy's map one would have to count at least 80 miles to the degree of the great circle»). An error of arithmetical division, then, was perhaps the chief reason why Lelewel denied the Ptolemaic origin of this chapter of al-Battānī; a lesser reason was that at that time the Arabic and Syriac writers who give the same measures were not known.

To show better all I have said of the common, and Ptolemaic, origin of the descriptions of the seas, and so as to be able to correct our author's numbers here and there, I shall now give a survey of the measures reported by al-Battānī (with whom Ibn Rusta agrees), by al-Jayhānī (according to al-Kharaqī) and by Jacob of Edessa, comparing the other Arab writers too[3]. I 168 I 169

[1] In the book I cited on p. 17, note 1.

[2] J. Lelewel, *Géographie du moyen âge*, Breslau 1852, vol. I: Cartes de géographes, pp. 36–38.

[3] The fragments read in Qudāma, pp. 230–231, correspond to p. 17, ll. 29–33 (as far as the number 200), p. 18, ll. 9–16 and 20–24 (leaving out the miles of the circuit of the islands), and p. 19, ll. 2–4, of my translation of al-Battānī.

	al-Battānī and Ibn Rusta	*al-Ǧayhānī in al-Kharaqī*	*Ǧacob of Edessa*
The inhabited land:			
length	13,500 [= 180°]	13,500	13,500
breadth	. . . . [= 60°]	. . . .	. . . . [= 60°]
The sea of the Indians[1]:			
length	8000 [= 106° 2/3]	8000	8000
whole breadth	2700 [= 36°]	2700	2700
of which south of the equator	1900 [= 25° 1/3]	1700	900 [= 12°]
The Barbarian Gulf[2]:			
length	500 [= 6° 2/3]	500	. . . .
breadth	100 [= 1° 1/3]	100	. . . .
The sea of al-Qulzum[3]:			
length	1400 [= 18° 2/3]	1400	1400
breadth at the end	200 [= 2° 2/3]	200	200
breadth at the beginning	700 [= 9° 1/3]	700	400 [= 5° 1/3]
The sea of al-Baṣra[4]:			
length	1400 [= 18° 2/3]	1400	1400
breadth at the beginning	500 [= 6° 2/3]	500	700 [= 9° 1/3] (both)
breadth on the direct course	150 [= 2°]	150	
The land between the sea of al-Qulzum and the Persian Gulf:			
length	1500 [= 20°]	1500	. . . .
The Green Gulf:			
length	1500 [= 20°]	1500	. . . .
The sea of the Byzantines and of Egypt[5]:			
length	5000 [= 66° 2/3]	5000	5000
breadth	600 [= 8°]	600	
	700 [= 9° 1/3]	700	400 [= 5° 1/3] (all three)
	800 [= 10° 2/3]	800	
The Adriatic Gulf:			
length	500 [= 6° 2/3]	500	. . . .
The Gulf of Narbonne:			
length	200 [= 2° 2/3]	200	. . . .
The Pontus[6]:			
length	1060 [= 14° 2/15]	1300	1300 [= 17° 1/3]
breadth	300 [= 4°]	300	300
The Maeotic Lake[7]:			
length	300 [= 4°]	300	a little smaller
breadth	100 [= 1° 1/3]	100	than the Pontus (both)
The strait of Constantinople:			
breadth	3	3	. . . .
The sea of Jurǧān[8]:			
length	800 [= 10° 2/3]	800	1800 [= 24°]
breadth	600 [= 8°]	600	600

Of the two islands of the sea of Jurjān (the Caspian), unknown to Ptolemy, Jacob of Edessa, p. LXV, has the following: «Quo in mari, quia identidem multitudine aquarum, quae perpetuo infunduntur, nimis augetur, fieri non potuit ut insulae exstiterint. Vix enim ab antiquissimis temporibus usque ad hoc tempus duae solae insulae parvae ibi conspectae sunt, quae ad habitandum parum sunt aptae» («In this sea, because it is again and again greatly increased by the mass of the waters that pour into it continually, it has not been possible for islands to arise. For from the most ancient times down to the present hardly two small islands have been seen there, and they are little fit for habitation»). See what al-Masʿūdī, *Prairies*, II, 26; *Tanbīh*, p. 60; Aboulféda, pp. 47 ff. (text), etc., relate of volcanic islands in this sea.

	al-Battānī and Ibn Rusta	*al-Ǧayhānī in al-Kharaqī*	*Ǧacob of Edessa*
In the **Indian sea** there are islands[9]	1370	1370	1378
of which around Taprobane	59[10a]	19	19
The circuit of **Taprobane**	3000 miles	3000	4000

The circuits of the islands of Cyrnos, Sardinia, Cyprus, Sicily and Crete are read as in al-Battānī in the Astronomy of Barhebraeus, p. 135 of the text, except only that the 350 miles of Cyprus are wrongly given to Cyrnos, and the 200 of Cyrnos to Cyprus.

I shall now give the whole unpublished passage of al-Kharaqī that describes the seas[11].[b]

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[1]Ibn Rusta, Qudāma and al-Masʿūdī, as I said on p. 17, note 4, confirm al-Battānī's numbers; the 1700 of al-Kharaqī is a copyist's error (سبع 7 for تسع 9) that is very frequent in Arabic manuscripts. The Arabic source (al-Kindī?), then, already had 1900; but the true and oldest reading is 900, as appears from Ptolemy's map and from Jacob of Edessa («in latitudinem circa 900 milliaria a loco aequinoctiali meridiem versus irrumpit usque ad terram a terminis eius australibus versus meridiem sitam, quae ab hominibus incognita appellatur et prorsus inhabitabilis est», «in breadth it breaks in about 900 miles from the equinoctial towards the south, as far as the land that lies to the south of its southern borders, which is called unknown to men and is wholly uninhabitable»; Hjelt, *L'Examéron*, p. LXVI). See the passage of al-Masʿūdī, *Tanbīh*, p. 51, quoted above (p. 166).

[2]It is Ptolemy's Βαρβαρικὸν πέλαγος, but with these Arabs it seems to denote the Gulf of Aden. Ptolemy calls Βαρβαρία the coasts that stretch from Cape Guardafui to the island of Zanzibar. — The same numbers in al-Masʿūdī, *Prairies*, I, 231, and in Barhebraeus, p. 135 of the text.

[3]The breadth of 700 miles is also in Ibn Rusta and in al-Masʿūdī, *Prairies*, I, 237; it is therefore the reading of the ancient Arabic source. But Jacob of Edessa and Barhebraeus (loc. cit.) have 400, more correctly. — On this sea see note *e*.

[4]Ibn Rusta has the same numbers; and al-Masʿūdī, *Prairies*, I, 238, writes: «the length of this gulf is 1400 miles; its breadth at the beginning 500, but at times the breadth of its two extremities becomes 150 miles». But «of its two extremities» (طرفيه) is plainly a scribal error for the original «of its direct course» (طريقه) that is read in al-Battānī. The same error occurs in another passage of al-Masʿūdī; see note *e*.

[5]The same numbers in Ibn Rusta; al-Masʿūdī, *Tanbīh*, pp. 56–57; Barhebraeus, p. 135 of the text. In the *Prairies*, I, 256, we read: «Many authors of astronomical tables, among them Muḥammad ibn Jābir al-Battānī and others, say in their books on the stars that its length is 5000 miles, and that its breadth varies from 800 to 700 and 600». Al-Masʿūdī himself, on the other hand, *Prairies*, I, 276 (see *Tanbīh*, p. 56), reports the same numbers as Jacob of Edessa, on the authority of al-Kindī and of Aḥmad ibn al-Ṭayyib al-Sarakhsī.

[6]The length 1060 is an error; Ibn Rusta, al-Masʿūdī (*Tanbīh*, p. 66) and Barhebraeus (p. 137 of the text) have 1300, like al-Kharaqī and Jacob. In al-Masʿūdī, *Prairies*, I, 260, the length 1100 is an evident scribal error for 1300. — See p. 18, note 12.

[7]The same numbers in al-Masʿūdī, *Tanbīh*, p. 67.

[8]The same numbers in al-Masʿūdī, *Prairies*, I, 263, and *Tanbīh*, p. 60.

[9]Ibn Rusta and Barhebraeus (p. 136 of the text) also have 1370; but Jacob of Edessa's 1378 is Ptolemy's number (*Geography*, VII, 4, 11).

[10]The true reading is 19, as I said on p. 18, note 5.

[11]It is ch. II (*bāb*) of the second treatise (*maqāla*); Paris codex ff. 74r–76r, Laurentian codex ff. 57v–58v. — The author died in the city of Merv in the year 533 H (which began on 7 September 1138).

[a]The print has no call for Nallino's note 5; it belongs to the 59 of al-Battānī.

[b]Nallino's Addenda et emendanda (Part II, pp. XXII–XXIII) add to pp. 167–175 the like descriptions of the seas by Niẓām al-Dīn al-Naysābūrī and Fakhr al-Dīn al-Rāzī; they are translated with Part II.

«Chapter II: on the description of the seas and their position in the inhabited lands. Al-Jayhānī and other sages [versed] in the knowledge of the position and the extent of the seas relate that there are five known seas: the first, *the sea of the Indians*, also called *the sea of China*; the second, *the sea of the West* (*baḥr al-maghrib*); the third, *the sea of Syria, of the Byzantines and of Egypt*; the fourth, *the Pontus* (*bunṭus*); the fifth, *the sea of Jurjān* (that is, *the Caspian*). — *The length of the sea of the Indians* stretches from west to east, from the farthest part of *Ethiopia* (*arḍ al-ḥabasha*) to the farthest land of the *Indians* and the *Chinese*, and comprises 8000 miles; its breadth is 2700 miles, and it passes the line of the equator by 1700 miles. A gulf goes out of it near *Ethiopia* and stretches towards *al-Barbar*; it is called *the Barbarian Gulf*; its length is 500 miles, the breadth of its extremity[1] 100. — Out of it [the sea of the Indians] another gulf too goes out towards *Ayla*, namely *the sea of al-Qulzum*, whose length comprises 1400 miles, its breadth at its beginning 700, and its end towards the nearest extremity, which is called *the Green Sea*, 200[2]; at its almost pointed extremity is [the city of] *al-Qulzum*, from which the sea took its name; to the east of it is the land of *al-Yaman* and *ʿAdan*, to the south *Ethiopia*. — A gulf also goes out of it [the Indian sea] towards the land of *Persia* (*fāris*), called *the Persian Gulf*, or *the sea of al-Baṣra and of Persia*, on whose eastern [shore] are *Tīz* and [the province of] *Mukrān*, on the western ….. [3] *ʿUmān*; its length is 1400 miles, its breadth at its beginning 500, on the direct course 150. Between these two gulfs, that is, of *Ayla* and of *Persia*, are the land of *al-Ḥijāz*, *al-Yaman* and the other regions of the *Arabs*, 1500 miles in extent. — Another gulf too goes out towards the farthest lands of the *Indians*, which near its end is called *the Red Gulf*[4], and whose length is 1500 miles. — In this *sea of the Indians* there are 1370 islands, inhabited or deserted, among which is a great island at the far end of the sea, facing the land of the *Indians* from the east, near the region of *China*, called *Taprobane* or *Sarandīb*; it has a circuit of 3000 miles; in it are high mountains and many rivers; from it[5] come rubies and sapphires. Around it are 19 inhabited islands, rich in cities and villages; among the islands of this sea are the islands of *al-Zābij*[6], the islands of *al-Dībaḥāt*[7] and *Qumayr*[8], the island of *Kalah*[9], from which tin (الرصاص القلعي) is brought, and the island of *Sarbuza*[10], from which camphor comes. I 171

[1]The manuscripts have طرفه, but without doubt طريقه, «of its direct course», should be read, as in al-Battānī.

[2]On this passage see note *e*, pp. 175–177.

[3]I do not know how to correct the corrupt words of the manuscripts.

[4]So the manuscripts, with which one might compare the Greek name of the Indian Ocean (Ἐρυθρὰ θάλασσα, the Red Sea); but why a small gulf should have taken the name of the whole sea is not understood. When al-Battānī (p. 18, l. 4 and note 2) and Ibn Rusta are compared, there can hardly be a doubt that الاحمر, «red», is an error of the manuscripts, or of the author himself, for الاخضر, «green». [*Translator:* Nallino's Addenda et emendanda (Part II, p. XXIII) note that Fakhr al-Dīn al-Rāzī rightly calls this gulf the Green Gulf, al-khalīj al-akhḍar.]

[5]So the manuscripts, ومنها يخرج, rightly; al-Battānī leaves out the و, so that the pronoun must refer to the mountains and rivers.

[6]It is the island we call *Java* (in Malay *Jāwa* جاو, among the Arabs of the thirteenth century *Jāwah*); see Van der Lith's discussion in *Le livre des merveilles de l'Inde par le capitaine Bozorg fils de Chahriyâr, texte arabe et traduction française par P. A. Van der Lith et M. Devic*, Leiden 1883, pp. 231–33.

[7]The manuscripts are without dots; others read *al-Dībajāt* (ج j for ح ḥ). The name arose from the Indian *dvīpa*, that is, island; it denotes the islands we call today the *Laccadives* and the *Maldives* (*Dībat al-maḥal* among the Arabs of the fourteenth century); see Van der Lith, 216–17.

[8]I keep the reading of manuscript L (قمير); P has قمر. The name is also found written so in al-Bīrūnī, *India*, p. 103 of the text (= I, 210 of the translation), where without doubt it denotes the same region that the other Arabs call قمار *Qumār* (that is, *Cambodia*, whose ancient inhabitants were the *Khmēr*). The passage of al-Bīrūnī, which Sachau did not quite rightly understand, was better interpreted by de Goeje in his translation of Ibn Khurradādhbih, p. 50, note 1.

[9]It is the port called today *Kēdā* (*Quedah*), in the Malay peninsula (Malāka ملاك), as Van der Lith has shown, pp. 255–264; see also Ibn Rusta, p. 88, and al-Masʿūdī, *Tanbīh*, p. 61.

[10]It was a village or region in the island of Sumatra, on the banks of the river *Mūsī*, near the place where the city of *Palembang* (فلمبغ) now stands; in the Chinese annals it is called *Sam-bo-tsai*. See Van der Lith, pp. 247–253 (and 102, 105, 111, 124, 158–160, 165, 176, 190); al-Masʿūdī, *Tanbīh*, p. 61; *Prairies*, I, 243 (where the name is corrupted to *Serīra* سريرة for سريزة); Yāqūt, I, 61, III, 68 and 354; Aboulféda, pp. 22, 368, 375 (= II, 26, III, 126 and 132 of the translation), where it is always *Serīra*; al-Dimashqī, pp. 22, 199, 204 (*Serīra*), etc.

«*The sea of the West*, which is also called *the Encircling sea* and by the Greeks *Oceanus*, is joined to the *sea of the Indians*, and nothing of it is known except the part which, lying in the west and the north, faces the land of the *Russians* (*al-rūs*) and of the *Slavs* (*al-ṣaqāliba*)[1]. Beginning from its last limit in the south, it faces the *land of the Blacks* (*arḍ al-sūdān*) and passes along the borders of *Farthest al-Sūs*[2], *Tangier* (*Ṭanja*), *Tāhart*[3], *Spain*, *al-Ǧalāliqa*[4] and the *Slavs*[5] as far as *Britain* (*al-Barṭāniya*); then from there it stretches, beyond impassable mountains and uninhabited lands, towards the *sea of the East*. This sea, *Oceanus*, is not furrowed by ships and is sailed only near its shores; in it are six islands facing the land of the *Ethiopians*, which are called the *Eternal* (*al-khālidāt*) or the *Fortunate* (*al-suʿadāʾ*). Another island, called *Gadira*, lies opposite *Spain*, near the *gulf* that goes out of this sea; the gulf is 7 miles wide where it goes out, lies between *Spain* and *Tangier*, and was [anciently] called the *Ford of Hercules* (*maʿbarat Hiraqlas*); today it is called *al-Zuqāq* [that is, the narrow lane]; this gulf flows into the *sea of the Byzantines and of Syria*. In the northern stretch of *Oceanus* are the 12 islands of *Britain*. Out of it too, in the north of the *Slavs*, arises a great gulf, which is also called the *sea of the Varangians* (*warank*), after the people that inhabits its shores[6]; it stretches to the land of the Muslim *Bulgars* (*bulghār*), and extends 300 miles in length from east to west and 100 in breadth.

«*The sea of the Byzantines, of Africa, of Egypt and of Syria*, proceeding eastwards from the gulf that goes out of the *sea of the West* as far as *Tyre* and *Sidon*, has a length of 5000 miles, a breadth in one place of 600, in another of 700, in another of 800. A gulf goes out of it which, going northwards near *Rome*, extends 500 miles in length and is called *Adrias*[7]. Another gulf goes out towards the land of *Narbonne*; it has a length of 200 miles. In this sea there are 162 inhabited islands, of which 5[8] are large, namely *Rhodes*, *Cyprus*, *Samos*, *Sicily* and *Crete*.

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«*The Pontus* is called the land-locked (that is, inner) *sea*, and [also] *the sea of Trebizond*, after a port on its shore; it stretches from *Lādhiqa* beyond *Constantinople*, into the land of the *Russians* and the *Slavs*; it has a length of 1300 miles and a breadth of 300. Into this sea flows the river called *Tanais*, which runs from the northern side with respect to the *Maeotic sea*, which is the *sea of the Varangians* (*warank*)[9]. From the *Pontus*, near the walls of *Constantinople*, a gulf goes out like a river[10] and flows down into the *sea of Egypt*; near *Constantinople*, which is on its western

[1]By this name the Arab writers denote the Slavs and many Germanic tribes; see G. Jacob, *Studien in arabischen Geographen*, Berlin 1891–92, part I, p. 17.

[2]*Farthest al-Sūs* is the southern province of the empire of Morocco, whose chief city today is *Tārūdānt*.

[3]It was the name of two great cities five Arabic miles (about ten km) apart, of which one had the surname «the old» (*al-qadīma*), the other «the new» (*ḥadītha*). The former is called *Tiharet* (*Tiyāret*) on our maps, the other *Tāqdemt*; both are in western Algeria. Read de Goeje's discussion, *Specimen e litteris orientalibus exhibens descriptionem al-Magribi sumtam e Libro Regionum al-Ǧaʿqubii*, Leiden 1860, pp. 103 ff., and later in *Description de l'Afrique et de l'Espagne par Edrisi*, Leiden 1866, p. 100; see also R. Basset, *Documents géographiques sur l'Afrique septentrionale traduits de l'arabe*, Paris 1898, p. 23. [*Translator:* Nallino's Addenda et emendanda (Part I, p. LXXIII) add J. Canal, *Tiaret, monographie ancienne et moderne*, in the Bulletin trimestriel de géographie et d'archéologie d'Oran, vol. XX, 1900, pp. 1–44.]

[4]They are the *Callaeci* of the Romans, the *Gallaeci* of the Middle Ages, whose name survives in the present province of *Galicia* (in northern Spain).

[5]See note 1.

[6]It is the Baltic Sea, whose eastern shores were inhabited in the ninth century by the *Varangians* (German *Wäringer* or *Waräger*), a Norman people. The name is uncommon in the Arab writers; Aboulféda, p. 35 (= II, 42 of the translation), testifies that he found it written only in al-Bīrūnī and in Naṣīr al-Dīn's *Tadhkira*. — What follows in al-Kharaqī, however, strangely confuses, as we shall also see below, the Baltic Sea with the Maeotic Lake (the Sea of Azov), near which the Muslim Bulgars lived, and to which the measures given in this passage without doubt belong. [*Translator:* Nallino's Addenda et emendanda (Part II, p. XXIII) take back «uncommon»: the name is found also in al-Naysābūrī, al-Qazwīnī and al-Dimashqī, and C. M. Frähn, *Ibn Foszlan's und anderer Araber Berichte über die Russen älterer Zeit* (St Petersburg 1823), pp. 177–204, gathered the Arabic and Persian witnesses of this sea.]

[7]With the Syrians *Adrias* (ܐܕܪܝܐܣ *adryūs*) is the Mediterranean Sea; see F. Nau, *Notice sur le livre des Trésors de Jacques de Bartela*, in the Journal Asiatique, 9th series, vol. VII, 1896, pp. 309–312 (= 26–29 of the offprint).

[8]Manuscript P has خمسون, 50; but the sense and the grammatical construction (for جزائر عظام is read afterwards) demand خمس, 5, as manuscript L, al-Battānī and Ibn Rusta have.

[9]He confuses the Baltic Sea with the Maeotic Lake, as I said on p. 171, note 6.

[10]The manuscripts have the corrupt بحرى فيه مايه نهر, which I restore with al-Battānī and Ibn Rusta to يجري كانه نهر.

shore, it has a breadth of 3 miles, and grows ever narrower until it flows into the *sea of Syria and of Egypt*. On the southern shore of this sea [*of Egypt and of the Byzantines*] are the lands of *al-Maghrib*[1] and of *Africa* as far as *Alexandria* and *Egypt*; opposite, on the northern side, lie the lands of *Spain*, of the *Byzantines* and of the *Franks* (*faranja*) as far as *Antioch*, which *Syria* and *Palestine* follow.

«*The sea of Jurjān*, or of the [*Caucasian*] *Gate*, extends 800 miles in length from west to east and 600 in breadth; it contains two islands, which were inhabited in times past. This sea is also called *the sea of Ābaskūn*, after one of its ports[2]; it then reaches to [the provinces of] *Ṭabaristān*, *al-Daylam* and *Shirwān*, [the city of] *Bāb al-Abwāb*[3], and the region of the *Alans* (*al-Lān*)[4]; then to the *Khazars* (*al-khazar*) and the mouth of the river *Itil*[5]; lastly to the regions of the *Persians*, until it reaches *Ābaskūn* again. It is called by the name of each region that it washes; in ancient times it was called *the sea of Jurjān*, today it is called *the sea of Ābaskūn* and *the sea of the Khazars*; it does not flow into any other sea.

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«These are the great seas; the rest are lakes and marshes, such as the lake of *Khwārizm* (that is, of *Chorasmia*, today the *Aral*), the lake of *Tiberias* in *Syria*, and *Īssik-kūl* near *Barsukhān*[6]. *Aristotle*[7] relates that *Oceanus* surrounds [the whole] earth like a crown; that out of it arises a gulf on the western side, at the place called the *Ford of Hercules*[8], which, entering the inhabited land, proceeds eastwards and is the *sea of Egypt, of the Byzantines, of Africa and of Syria*. On the eastern side, on the other hand, another gulf comes out of it [*Oceanus*], entering the inhabited land, which stretches westwards and is the *sea of China, of the Indians, of Persia, of al-Qulzum, of the Barbarians* (*al-Barbar*). This is what we know of the seas».

I give the Arabic text of the whole passage of al-Kharaqī after the Paris (P) and the Laurentian (L) manuscripts:

﴿الباب الثاني﴾ في ذكر البحور وكيفيّة وَضْعها من المساكن. قد حكى الجَيْهانيُّ[1] وغيره من العلماء بمواقع البحور وامتداداتها ان البحور المعروفة خمسة احدها بحر الهند وهو الذي يقال له ايضًا بحـر الصين والثاني بحر المغرب والثالث بحر الشام والروم ومصر والرابع بحر بنطس[2] والخامس بحر جرجان فاما بحر الهند فانه يمتـدّ طوله من المغرب الى المشرق

[1]The *Mauretania* of the ancients; today the empire of Morocco, Algeria and the province of Tunis.

[2]Abuskūn, Abiskūn or Aboskūn was a great sea port of the region and the city of Jurjān (whose ruins stand near Pisarak پسرك). There are two opinions on the site of the place: B. Dorn, *Caspia, Ueber die Einfälle der alten Russen in Tabaristan*, St Petersburg 1875, pp. 5, 6, 7, puts it at the mouth of the river Gurgān گرگان, not far from the village of *Gümüsh-tepeh* گومش تپه, where Mouravieff (in the first quarter of the nineteenth century) found many ruins; Melgunof, *Das südliche Ufer des kaspischen Meeres*, Leipzig 1868, pp. 62, 63, 112, 119–120, around the present mausoleum (imām-zādeh) called *Kharāb-i-shehr*, 10 km from the shore and 6 east of the town of Gez جز (in the district of Anezān of the province of Asterābād). But it is always near the place where the eastern shore of the sea turns westwards.

[3]Today *Derbent*.

[4]They are the people called today *Ossetes*.

[5]What the ancients called the *Rha*, we call the *Volga*.

[6]The names are corrupt in the manuscripts. The city is known to the older Arab geographers, such as Qudāma, p. 203 (where it is corrupted to نوشجان Nūshajān for برسخان), and al-Muqaddasī, pp. 48, 263, 275, 340; the lake, on the other hand, is unknown to them, except to Qudāma, who mentions it without a name, loc. cit. Yet the lake, which lies in eastern Turkestan, is already called *Īssik-kūl* (ايسكول, or corruptly السكول) by a Persian book of an unknown author, written in the year 372 H (which began on 26 June 982) and entitled *Ḥudūd al-ʿālam*, «the Limits of the world»; see V. Barthold, *Otčet pojezdkije v srednjuju Aziju s naučnoju cjelju, 1893–94* (Report on a journey in Central Asia undertaken for a scientific purpose in the years 1893–94), St Petersburg 1897, pp. 31–32 (= Mémoires de l'Académie Impériale des Sciences de St.-Pétersbourg, 8th series, Classe historico-philologique, vol. I, no. 4). The Turkish name means «warm lake»; it is written in various ways: *īssīq*, *issīq*, *īssīgh*, *issīgh*. Abū Saʿīd ʿAbd al-Ḥayy Gardīzī (گردیزي) describes the city of Barsukhān and the lake (*īssigh-kūl*) at length, about the year 1050, in his book *Zayn al-akhbār*; his words are given in Persian in Barthold, pp. 89–90, and in Russian translation, pp. 113–114. The name *īssīgh-kūl* is written by Muḥammad Ḥaydar in his Persian «Rashīdī History» (*Taʾrīkh-i rashīdī*), in the sixteenth century; see Barthold, p. 50.

[7]Quṭb al-Dīn Maḥmūd al-Shīrāzī also gives this whole passage of Aristotle, *al-Tuḥfa al-shāhiyya*, f. 158r.

[8]See above, p. 171, l. 12, and below, on the Arabic text, pp. 174 and 175.

من اقصى ارض الحبشة الى اقصى ارض الهند والصين يكون مقدار ذلك ثمانية آلاف ميل وعرضه الفان وسبعمائة ميــل ويجاوز خطّ الاستواء الف وسبعمائة ميل يخرج منه خليج عند ارض الحبشة ويمتدّ الى ناحيــة البربر يُسمَّي ذلك الخليج البربريّ يكون طوله مقدار خمسمائة ميل وعرض طرفه مائة ميل ويخرج منه ايضًا خليج آخَرُ نحو أَيْلةَ وهو بحر القلزم طوله الف واربعمائة ميل وعرضه في الاصل سبعمائة ميل ومنتهاه الى طرف[3] الأَذْنَى الذي يُسمَّى البحر الاخضر مقدار مائتَيْ ميل وعلى طرفه المستدقّ القلزم فلذلك سُمِّيَ بــه على شرقيّه ارض اليمــن وعدن وعلى جنوبيّه ارض الحبشة ويخرج منه خليج آخَرُ نحــو ارض فارس يُسمَّى الخليج الفارسيّ وهو بحــر البصرة وفارس الذي على شرقيّه تِيز ومُكران وعلى غربيّه *وبحاله ى حمه[4] عمان طوله الف واربعمائة ميل وعرضه في الاصل خمسمائة ميل وعرض طريقه مائة وخمسون ميــلًا وبين هذَيْن الخليجَيْن اعني خليج ايلة وخليج فارس ارض الحجاز واليمن وسائر بــلاد العرب فيما بين مسافة الف وخمسمائة ميل ويخرج منه خليج آخَرُ الى اقصى ارض الهند عند تمامــه يُسمَّى الخليج الاحمر وطوله الف وخمسمائة ميل وفي بحر الهند هذا من الجزائر العامرة وغــير العامرة الف وثلثمائة وسبعون جزيرة

1) P الجبهانى, L الجهيانى — 2) P نبطس — 3) So the manuscripts, for طرفه or الطرف — 4) So P; L وبحالهِ ى خنبه

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منها جزيرة ضَخمة في اقصى البحر مقابل ارض الهند في ناحية المشرق عند بلد الصين تُسمَّى طبروباني[1] وهي سَرَنديب يحيط بها ثلثة آلاف ميل فيها جِبال عظيمة وانهار كثيرة ومنها يخرج الياقوت الاحمر ولون السماء وحَوْلَ هذه الجزيرة تسع عشرة جزيرة عامرة فيها مدائن وقُرَى كثيرة ومن جزائر هــذا البحر جزائر الزابج وجزائر الدِّيبَجات وقُمَيْر[2] ومنها جزيرة كَلَه التي يُجْلَب منها الرَّصاص القَلْعِيّ وجزيرة سَرْبُزَة[3] التي يُجْلَب منها الكافور واما بحر المغرب فهو الذي يُسمَّى المحيط وتُسَمّيه اليونانيّون اوقيانوس ويتَّصِل به بحر الهند ولا يُعْرَف منه إلّا في ناحية المغرب والشمال عند مُحاذاة ارض الروس والصقالبة فيأخذ من اقصى المُنتَهَى في الجنوب محاذيًا لارض السودان مارًّا على حدود السوس الأَقْصَى وطنجة وتاهَرْت[4] ثمّ الاندلس والجلالقة والصقالبة الى البرطانية ثمّ يمتــدّ من هناك وراء الجبال[5] المسلوكة والاراضي غير المسكونة نحو بحر المشرق وهذا البحر اعني اوقيانوس لا يجري فيه السُّفُن وانّما يُسْلَك بالقُرْب من ساحله وفيه ستّ جزائر مقابل ارض الحبشة تُسمَّى جزائر الخالدات وتُسمَّى ايضًا جزائر السَّعَداء وجزيرة اخرى تُسمَّى غديرة[6] مقابل الاندلس عند خليج يخرج من هذا البحر وعرضه حيث يخرج سبعة اميال وهــو بين الاندلس وطنجة كان يُعْرَف بمَعْبَرَة هِرَقْلَس[7] والآن يُعْرَف بالزُّقاق وهذا الخليج ينصبّ في بحر الروم والشام وفيه اعني بحر اوقيانوس من ناحية الشمال جزائر برطانية وهي اثنتا عشرة جزيرة ويخرج منه ايضًا خليج عظيم في شمال الصقالبة يُسمَّى ايضًا بحر وَرَنْك[8] وهم امّــة على ساحله ويمتدّ هــذا الخليج الى ارض بلغــار المسلمين طوله من المشرق الى المغرب ثلثمائة ميــل وعرضه مائة ميل واما بحر الروم وافريقية ومصر والشام طوله من عنــد الخليج الذي يخرج من بحر المغرب الى ناحية المشرق ويمتدّ الى صور وصَيْداء يكون ذلك مقدار خمسة الاف ميل وعرضه في مكان ستّمائة ميل وفي مكان سبعمائة ميل وفي مكان ثمانمائة ميل ويخرج منه خليج الى ناحية الشمال قريب من الرومية يكون طوله خمسمائة ميل يُسمَّى اذريس[9] ويخرج منه خليج آخَرُ الى ارض تربُونة[10] يكون طوله مائتَيْ ميل وفي هذا البحر مائة واثنتان وستّون جزيرةً عامرةً منها خمس[11] جزائر عظام وهي جزيرة

1) P طروبانى, L طرفانى — 2) P قمر, L قمير — 3) P سريره, L سربره — 4) P وماهرت, L وباهرت — 5) So the manuscripts, but without doubt غير should be added — 6) P عديره, L عدره — 7) P بمغره مرقلس, L بمعره هرقلس. See also p. 175, note 11; Yāqūt, I, 61, l. 19, معبرة هيرَقْلَس; Qāḍīzāde, in his commentary on the first chapter of the third section of the *Tadhkira*, f. 120v, معره هرقل; Quṭb al-Dīn Maḥmūd al-Shīrāzī, *al-Tuḥfa al-shāhiyya*, f. 158r, معبره هرقلس — 8) P وربله, L وربك — 9) The manuscripts خمسون, but afterwards جزاير — 10) P ادرس, L بربريه — 11) P بربره

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رودص وقبرس وسامس وسقلية واقريطش وامّا بحر بنطس الارضيّ فإنّه يُسمَّى طرابزند[1] لأنّها فُرْضة عليه ويمتدّ من

لاذقة[2] الى خَلْفِ قسطنطينيّة في ارض الروس والصقالبة طوله مقدار الف وثلثمائة ميل وعرضه ثَلثمائة ميل ويدخل فيه النهر الذي يُسَمّي طانيس وهو يجري من ناحية الشمال من البحر الذي يُسمَّى مايطس وهو بحر وَرَنْك وينفجر من بحر بنطس عند سُوَر قسطنطينيّة خليج يجري فيه مائة[3] نهر وينصبّ في بحر مصر وعرضه عند قسطنطينيّة قدر ثلثة اميال وقسطنطينيّة هي على شاطئه الغربيّ ولا يزال يتضايق حتّى يقع في بحر الشام ومصر الذي على جنوبيّهِ بلادُ المغرب وافريقيّة الى الاسكندريّة ومصر وبحِذائه في الشمال ارض الاندلس والروم وفَرَنْجة[4] الى انطاكِيَة ويليها بلاد الشام وفِلَسْطِين وامّا بحر جُرْجان وهو بحر الباب[5] طوله من المغرب الى المشرق ثمانمائة ميل وعرضه سّتّمائة ميل وفيه جزيرتان كانتا عامرتَيْن فيما مضى من الزمان ويُعْرَف هذا البحر ببحر آبَسْكُون[6] لأنّها على فُرْضته ثمّ يمتدّ الى طبرستان والدَّيْلَم وشِرْوان وباب الابواب وناحية اللّان ثمّ الخزر على مَصَبّ نهر إتِل ثمّ ديار الفُرْس الى ان يعود الى آبسكون[7] ويُسمَّى باسم كلّ بُقْعة حاذاها كان في القديم يُعْرَف ببحر جرجان والآن ببحر آبَسْكُون[8] وببحر الخزر وليس يتَّصِل ببحر آخَرَ وهذه هي البحور العظام فامّا غيرها بُحَيْرات وبطائح كبُحَيْرة خوارزم وبحيرة طَبَرِيَّة بأرض الشام وإِسِكّول[10] بالقرب من بَرْسُخَان[9] وقد حكى ارسطاطاليس انّ بحر اوقيانوس محيط الارض بمَنْزِلة إكْليل لها وانّه ينفتح منه خليج في ناحية المغرب من الموضع المسمّى بمَعْبَرَة هِرَقْلَس[11] فيدخل الى المعمورة ماراًّ نحو المشرق وهو بحر مصر والروم وافريقيّة والشام وينفتح منه ايضًا في ناحية المشرق خليج فيدخل الى المعمورة ماراًّ نحو المغرب وهـو بحر الصين والهند وفارس والقلزم والبربر فهذا ما عرفناه من امر البحور

a

1) P طراوند, L طراوبد — 2) P لادقيه, L لاذقيه — 3) So the manuscripts; but see p. 172, note 2. — 4) P فرخه, L فرحه — 5) The manuscripts الراب — 6) P السلون, L انسكون — 7) P السكون — 8) P السلون — 9) The manuscripts والسلول[b] — 10) P برسحان, L برشخان — 11) P بمعره مرفلس, L بمعره هرقلس; see above, p. 174, note 7.

e) — p. 17, l. 33.

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On this *Green Sea* opinions have differed. Reinaud, loc. cit., p. CCLXXXV, note 2, writes: «En grec, mer πρασώδης, du mot πράσον, signifiant *poireau*» («In Greek, the πρασώδης sea, from the word πράσον, meaning *leek*»); clearly, then, he took it to be the *Indian Ocean*, a part of which, washing the coasts of Africa from the island of Menuthias (Zanzibar?) to the farthest east, is called ἡ Πρασώδης θάλασσα by Ptolemy (see p. 18, note 2). But «the direct course, called the Green Sea» cannot lie outside the sea of al-Qulzum itself; moreover, the author afterwards denotes the part of the sea of al-Qulzum that adjoins the Indian Ocean by the words «at the beginning», to which, following Ptolemy, he gives the enormous breadth of 700 miles[1].

The parallel passage in Ibn Rusta, p. 84, is: «and the breadth near its end in the west, that is, at its nearest extremity, which is called the *Last sea*, 200» (وعند منتهاه في المغرب اعني طرفه الادنى الذي يسمى بالبحر الاخير مائتا ميل). With a small change of letters, «its extremity» (طرفه) answers to «its direct course» (طريقه), which al-Battānī has; likewise, in Arabic writing, الاخير, «Last», differs little from الاخضر, «Green».

Qudāma, p. 230, writes: «and the breadth near its end in the west, and at the place adjoining (or: joined to) the Green Sea, 200» (وعند منتهاه في المغرب والموضع المتصل بالبحر الاخضر مائتا ميل). In the French translation of Qudāma's book, p. 174, note 2, de Goeje explains this: «Reinaud, Aboulféda,

[1]Or rather 400; see p. 168, note 3.

[a]Nallino's Addenda et emendanda (Part I, p. LXXIII) exchange the note numbers 9 and 10 in the thirteenth line of this page: 9 belongs to وإِسِكّول, 10 to بَرْسُخَان, as the readings of the two notes show.

[b]The print has واسلول; Nallino's Addenda et emendanda (Part I, p. LXXIII) correct it to والسلول.

p. 211 (comp. aussi Ibn Khaldoun *Prolég.* I, 96) dit que ce nom appartient à l'Océan Indien. Mais on l'emploie souvent aussi pour désigner l'Atlantique, et même la Méditerranée porte parfois ce nom. On pourrait croire que Kodâma a supposé avec Bâtiny [*sic*, for al-Battānī] que l'Afrique, au sud de l'Abyssinie, tournait à l'ouest et se dirigeait vers l'océan Atlantique (comp. Reinaud, Introd., p. CCLXXXVI, note 2)[1], mais la comparaison du texte avec celui d'Ibn Rosteh prouve que Kodâma n'a pas compris sa source et que c'est en réalité à l'Atlantique que le nom de mer Verte appartient ici» («Reinaud, Aboulféda, p. 211 (compare also Ibn Khaldūn, *Prolégomènes* I, 96), says that this name belongs to the Indian Ocean. But it is often used also for the Atlantic, and even the Mediterranean sometimes bears this name. One might think that Qudāma supposed with al-Battānī that Africa, south of Abyssinia, turned west and ran towards the Atlantic Ocean, but a comparison of the text with that of Ibn Rusta proves that Qudāma did not understand his source, and that it is in reality to the Atlantic that the name of Green Sea belongs here»). But how can the Atlantic Ocean be called joined or adjoining to the sea of al-Qulzum, which we today call the Red Sea? Anyone can see that de Goeje's opinion too must be wholly rejected.

Let us now consider similar passages in other writers. Al-Masʿūdī, *Prairies*, I, 237, has the following: «its length is 1400 miles; the breadth of its two extremities, which is the nearest place in its breadth, 200; the breadth in the middle, lastly, 700, which is its greatest breadth» (طوله الف واربعمائة ميل وعرض طرفيه مائتا ميل وهو اقرب المواضع في عرضه وعرضه في الوسط سبعمائة ميل وهو اكبر العرض فيه). The words «of its two extremities» (طرفيه) are a small scribal error[2] for طريقه, «of its direct course»; so al-Masʿūdī's words agree with al-Battānī. — Al-Kharaqī, fol. 74v of manuscript P = 57v of manuscript L: «its end, towards the nearest extremity, which is called the *Green Sea*, is 200 miles» (الادنى الذي يسمّى البحر الاخضر مقدار مائتي ميل (codd. طرف) ومنتهاه الى طرفه).

Quṭb al-Dīn Maḥmūd al-Shīrāzī, *al-Tuḥfa al-shāhiyya*, f. 155v: «Two hundred parasangs [= 600 miles] is the breadth of the beginning of the second gulf, which is called the Red Gulf, and whose extremity is called the Green Sea; its length towards the north is 460 parasangs [= 1380 miles]; where it becomes narrower, and that is its end, it comprises 60 parasangs [= 180 miles]» وهي عرض بالبحر الاخضر وطوله في الشمال اربعمائة وستّون فرسخًا [add. يُعْرَف] اصل الخليج الثاني ويُسمَّى الخليج الاحمر وطَرَفه وحيث يستدقّ وهو منتهاه ستّون فرسخًا. Qāḍīzāde, in his commentary on ch. I (*faṣl*) of the third section (*bāb*) of the book *al-Tadhkira*, f. 120v, says that the Red Gulf comprises 200 parasangs [= 600 miles] near its beginning (عند اصله); «afterwards it becomes narrower, until its end comes to 60 parasangs [= 180 miles]» ثمّ انّه يستدقّ الى أن يصير منتهاه ستّين فرسخًا.

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«The nearest extremity», «the nearest direct course», etc., of almost all the Arab writers can only be explained as «nearest to us»; and since both the Arab authors and the Syrians (from whom the Arabs received those notions about the seas) lived in Syria and in Mesopotamia, the part of the sea of al-Qulzum nearest to them must plainly be the northern one. Nor can «the end» be understood otherwise, since it is opposed to the beginning, that is, to the strait we call Bāb el-Mandeb. Jacob of Edessa confirms it most certainly; for he writes in the *Hexaemeron* (p. LXVI of Hjelt's translation): «in latitudinem extrema parte australi 400 fere milliaria, septentrionali vero ad ducenta aut fortasse minus patet» («in breadth it extends about 400 miles at its southern end, but at the northern up to two hundred or perhaps less»). This being so, one might think that the true reading is the one Qudāma gives, and that in this passage the name *Green Sea* belongs to the Mediterranean (as the Arabs sometimes, though rarely, say), which can rightly be called adjoining the northern end of the sea of al-Qulzum. There are, however, difficulties: it seems hardly probable that Qudāma alone kept the good reading and that all the others, even those older than Qudāma, corrupted the text; nor could one explain why the name *Green Sea* is never mentioned in the passages of these descriptions that deal with the Mediterranean.

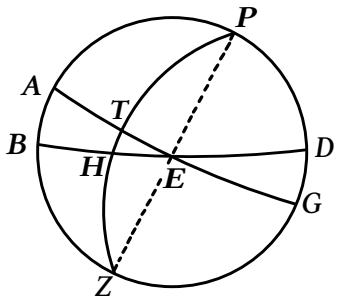
[1] But we have already seen (note *d*, p. 167) that this conjecture of Reinaud is wholly false.

[2] The same error in the description of the sea of al-Baṣra; see note *d*, p. 168, note 4.

The true interpretation can, I think, be drawn from a passage of the Syriac astronomy of Barhebraeus, edited very recently by Nau. The author, who wrote in 1279, used the oldest Syriac authors and the Arab writers alike in describing the seas, lakes and rivers; of the Red Sea (*yamā sūmāqā*) he has the following (p. 135 of the text): «The second [gulf that arises from the Indian Ocean] is the Red Sea, which is small and extends 400 miles in length; the breadth of its extremity, which is the *yamā dh-sūph*, comprises 200 miles». This *yamā dh-sūph* ܝܡܐ ܕܣܘܦ, which is read already in the oldest Syriac translations of the Old and the New Testament, is the Hebrew name *yam sūph* adapted to the nature of the Syriac language; it means «the sea of weeds» and denotes the stretch of the Red Sea in which Pharaoh, pursuing the Israelites, perished most miserably with his men. The first Syrian who compiled that description of the seas from Ptolemy's Geography would not leave out a name most famous in the sacred traditions; but the Arab who translated the Syriac, not knowing well what plant the word *sūph* answered to, rightly rendered it, for the sense, as «*Green Sea*». — So al-Battānī's reading is the best; Qudāma's المتصل, «joined, adjoining», is an error or a false correction for المسمى, «called».[a]

On chapter 7

a) — p. 20, ll. 22–30.



To compute the *solstitial amplitude*[1], given the greatest half arc of the day (P) and the obliquity of the ecliptic (ε).

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Let ABGD be the meridian circle, AEG the equinoctial circle (whose poles are P and Z), BED the horizon, H the solstitial point of the ecliptic; then TH = ε, AT = P, HE = the solstitial amplitude. With this construction Ptolemy, *Almag.* II, 2 (ed. Halma, vol. I, pp. 67–69), proves

$$\frac{\text{chord}\,2AT}{\text{chord}\,2AE} = \frac{\text{chord}\,2TZ}{\text{chord}\,2ZH} \times \frac{\text{chord}\,2HB}{\text{chord}\,2BE},$$

that is,

$$\frac{\text{chord}\,2P}{\text{chord}\,180^\circ} = \frac{\text{chord}\,180^\circ}{\text{chord}\,(180^\circ - 2\varepsilon)} \times \frac{\text{chord}\,(180^\circ - 2\ \text{sol. ampl.})}{\text{chord}\,180^\circ}.$$

[1] It is the arc of the horizon between the point of the east or the west and the place of the horizon from which the solstitial point of the ecliptic rises; if it is counted from the east point it is called the *rising amplitude* (*sa'at al-mashriq*), if from the west point, the *setting amplitude* (*sa'at al-maghrib*). The author can also call the amplitudes *azimuth*, since with the Arabs the beginning of the azimuths is taken from the east or the west point, not, as always with us, from the north or the south point.

[a] Nallino's Addenda et emendanda (Part II, pp. XXIII–XXIV) give a letter of de Goeje on this note, who finds Nallino's solution very ingenious but uncertain, and holds «طريقه» a corruption of «طرفه»; they are translated with Part II.

Using sines instead of chords, one finds

$$\frac{\sin P}{R} = \frac{R \cos(\text{sol. ampl.})}{\cos\varepsilon \times R},$$

whence al-Battānī's formula:

(α) $\quad \cos(\text{sol. ampl.}) = \dfrac{\cos\varepsilon \ \sin P}{R}.$

b) — p. 20, ll. 34–41.

To compute the *amplitude* of any point of the ecliptic that is not the solstitial point, given the half arc of the day of that point and the obliquity of the ecliptic.

In the formula of the preceding problem it is enough to call P the half arc of the day of the given point of the ecliptic, and to use δ, the declination of the same given point, instead of the whole obliquity. Al-Battānī's formula follows at once:

(α) $\quad \cos(\text{ampl.}) = \dfrac{\cos\delta \ \sin P}{R}.$

c) — p. 21, ll. 1–6.

Given the latitude φ of the city and the declination δ of the proposed point of the ecliptic, to compute the *amplitude*.

In the preceding figure BZ is plainly the altitude of the pole above the horizon of the given city, that is, the latitude φ. Ptolemy, *Almag.* II, 3 (ed. Halma, vol. I, pp. 71–72), proves geometrically

$$\frac{\text{chord}\, 2ZA}{\text{chord}\, 2AB} = \frac{\text{chord}\, 2ZT}{\text{chord}\, 2TH} \times \frac{\text{chord}\, 2HE}{\text{chord}\, 2EB},$$

that is[a],

$$\frac{\text{chord}\, 180^\circ}{\text{chord}\, (180^\circ - 2\varphi)} = \frac{\text{chord}\, 180^\circ}{\text{chord}\, 2\varepsilon} \times \frac{\text{chord 2 sol. ampl.}}{\text{chord}\, 180^\circ}.$$

With sines put for chords, the amplitude of any point of the ecliptic for that of the solstitial point, and so the declination δ of the given point for the whole obliquity ε, one finds

$$\frac{R}{\cos\varphi} = \frac{\sin(\text{ampl.})}{\sin\delta},$$

whence at once al-Battānī's formula:

(α) $\quad \sin(\text{ampl.}) = R\,\dfrac{\sin\delta}{\cos\varphi}.$

Ḥabash, a contemporary of al-Battānī, fol. 92r, has three other formulas besides this one. The first (since $\text{cosec}\ x = R^2/\sin x$, and Ḥabash, ff. 85r–87v, gives a table of the cosecants divided by *R*, that is, cosec/R = R/sin) is

$$\sin(\text{ampl.}) = \frac{\text{cosec}(90^\circ - \varphi)}{R} \ \sin\delta.$$

The second, since $\sin\delta = \sin\varepsilon \sin L/R$ (where L is the longitude of the point of the ecliptic), is

$$\sin(\text{ampl.}) = \frac{\sin\varepsilon \ \sin L}{\cos\varphi}.$$

The third was plainly found by orthographic projection:

$$\sin(\text{ampl.}) = \sqrt{(\text{tg}\,\varphi \ \sin\delta)^2 + \sin^2\delta}.$$

[a] In the numerator of the third fraction the print has «chord sol. ampl.» for the chord of the double arc, «chord 2 sol. ampl.», as in the formula before it; the 2 is set here.

On chapter 8

a) — p. 21.

To compute the *altitude* φ *of the pole*, given the difference between the equinoctial day (of 12 hours) and the longest day, that is, given the difference between the equinoctial half arc of the day (of 90 degrees) and the greatest half arc of the day.

First al-Battānī teaches how to find «the distance of the rising of the beginning of Cancer from the north point», which in the following figure is plainly represented by the arc HD, the same as the arc HE (that is, the solstitial amplitude) increased by ninety degrees; so the sine of that distance is the cosine of the solstitial amplitude, and al-Battānī's rule follows at once from the formula of note *a* on ch. VII:

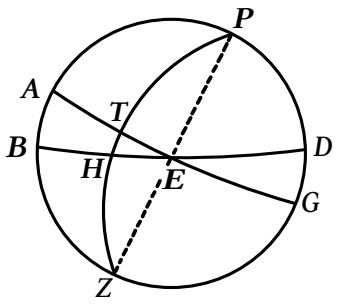
$$(\alpha) \qquad \sin(\text{dist. of the rising of } 0^\circ \text{ Cancer from the north point}) = \frac{\cos\varepsilon\,\sin P}{R}.$$

The arc EH is what the author now calls «the distance of the rising of the beginning of Cancer from the point of the rising of the equinox», that is, what he had called «the solstitial amplitude» in the preceding chapter. This premised, al-Battānī gives the formula

$$(\beta) \qquad \sin\varphi = \frac{\sin(P-90^\circ)\times\sin(\text{dist. of } 0^\circ \text{ Cancer from the north point})}{\sin(\text{dist. of Cancer from the equinoctial rising})}\times\frac{R}{\sin P},$$

which, since $\sin(P-90^\circ) = -\cos P$, may be written[1]

$$\sin\varphi = -\frac{\cos P}{\sin P}\times\frac{\cos(\text{sol. ampl.})}{\sin(\text{sol. ampl.})}\times R.$$



The formula comes from Ptolemy's construction, *Almag.* II, 3 (ed. Halma, vol. I, pp. 69–70). Let the figure be as in the notes on the preceding chapter; Ptolemy proves

$$\frac{\text{chord } 2ET}{\text{chord } 2AT} = \frac{\text{chord } 2EH}{\text{chord } 2HB}\times\frac{\text{chord } 2BZ}{\text{chord } 2ZA},$$

that is,

$$\frac{\text{chord}\,(180^\circ-2P)}{\text{chord } 2P} = \frac{\text{chord 2 sol. ampl.}}{\text{chord}\,(180^\circ-\text{2 sol. ampl.})}\times\frac{\text{chord } 2\varphi}{\text{chord } 180^\circ}.$$

With sines put for chords,

$$\frac{\cos P}{\sin P} = \frac{\sin(\text{sol. ampl.})}{\cos(\text{sol. ampl.})}\times\frac{\sin\varphi}{R},$$

[1] It is easily reduced to our $\operatorname{tg}\varphi = -\cos P\operatorname{cotg}\varepsilon$ if $\cos\varepsilon\sin P/R$ is put for cos sol. ampl., and $R\sin\varepsilon/\cos\varphi$ for sin sol. ampl., as the author taught in ch. VII.

whence

$$\sin\varphi = \frac{\cos P}{\sin P} \times \frac{\cos(\text{sol. ampl.})}{\sin(\text{sol. ampl.})} \times R,$$

where cos P is positive because Ptolemy considered the difference between the least half arc of the day and the equinoctial half arc of the day, that is, 90° – P instead of P – 90°.

On chapter 9

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a) — pp. 21–22.

Given the altitude φ of the pole, to find the *greatest* or the *least half arc of the day of the sun.*

With the preceding figure Ptolemy, *Almag.* II, 3 (ed. Halma, vol. I, pp. 70–71), proves

$$\frac{\text{chord}\, 2ZB}{\text{chord}\, 2BA} = \frac{\text{chord}\, 2ZH}{\text{chord}\, 2HT} \times \frac{\text{chord}\, 2TE}{\text{chord}\, 2EA},$$

that is,

$$\frac{\text{chord}\, 2\varphi}{\text{chord}\,(180^\circ - 2\varphi)} = \frac{\text{chord}\,(180^\circ - 2\varepsilon)}{\text{chord}\, 2\varepsilon} \times \frac{\text{chord}\,(180^\circ - 2P)}{\text{chord}\, 180^\circ}.$$

With sines put for chords,

$$\frac{\sin\varphi}{\cos\varphi} = \frac{\cos\varepsilon\ \sin(90^\circ - P)}{\sin\varepsilon \times R},$$

whence

$$\sin(90^\circ - P) = R\,\frac{\sin\varphi\ \sin\varepsilon}{\cos\varphi\ \cos\varepsilon}.$$

Ptolemy had considered the shortfall of the shortest day with respect to the equinoctial day, that is, 90° – *P*; al-Battānī, on the other hand, the excess of the longest day, that is, *P* – 90°, so that his formula is

(α) $\quad \sin(P - 90^\circ) = -\cos P = R\,\dfrac{\sin\varphi\ \sin\varepsilon}{\cos\varphi\ \cos\varepsilon}.$

If another arc is sought instead of the greatest half arc of the day, let the declination (δ) of the point of the ecliptic that the sun occupies on the given day be put in place of the obliquity ε; al-Battānī's formula becomes[1]

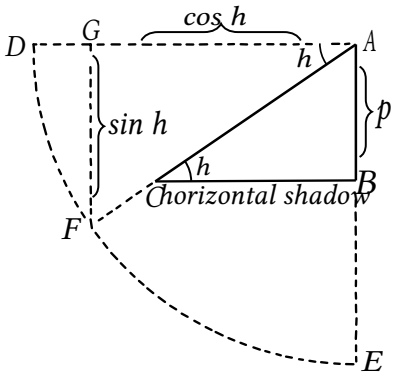
(β) $\quad -\cos P = R\,\dfrac{\sin\varphi\ \sin\delta}{\cos\varphi\ \cos\delta}.$

[1] We, using tangents and taking the radius $R = 1$, write more briefly $-\cos P = \text{tg}\,\varphi\ \text{tg}\,\delta$.

On chapter 10

a) — p. 22.

To compute the *altitude h of the sun*, or the *zenith distance* ζ (= 90° − *h*), from the shadow of the gnomon, divided into 12 parts.



I. From the *horizontal shadow*. Let the parts of the gnomon (that is, of the side AB) be called *p*; the author teaches, ll. 13–15, that

(α) horizontal shadow $= p\,\dfrac{\cos h}{\sin h}$.

The proposition can easily be derived in this way: with the radius AD of sixty parts (to which the author refers the trigonometric functions) let the quadrant of a circle EFD be described; from the similarity of the right-angled triangles ABC and AGF one gathers at once

$$CB : AB = GA : FG,$$

$$\text{horizontal shadow} : p = \cos h : \sin h.$$

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It could have been derived more easily from the theorem that in any plane triangle the sides are to one another as the sines of the angles opposite them. But this proposition seems to have been unknown to the Arabs of that age[1]; it is read later in Naṣīr al-Dīn al-Ṭūsī, who died in July 1274, as von Braunmühl, I, pp. 66–67, rightly says[a].

The author then teaches how to find «the hypotenuse[2] of the shadow triangle», that is, the side AC; plainly

(β) hypotenuse of the shadow triangle $= \sqrt{\text{horizontal shadow}^2 + p^2}$.

And since $\sin h = R\,AB/AC$, we shall also have

[1]The proposition «in any spherical triangle the sines of the angles are directly proportional to the sines of the opposite sides», on the other hand, as far as right-angled triangles are concerned, seems from a passage of the book on the quadrilateral written by Naṣīr al-Dīn al-Ṭūsī to have been known already before al-Battānī to the mathematician Thābit ibn Qurra, who died in the month of Ṣafar 288 H (February 901); see H. Suter, *Zur Geschichte der Trigonometrie*, in Bibliotheca Mathematica, edited by G. Enestrom, new series, vol. VII, Stockholm 1893, p. 7.

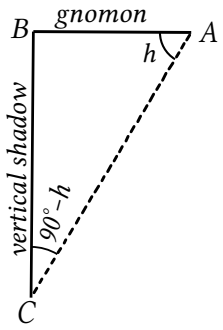
[2]Or «the diameter»; in Arabic *quṭr* means both things, for the hypotenuse is always the diameter of the circle circumscribed about a right-angled triangle.

[a]Nallino's Addenda et emendanda (Part I, p. LXXIII) add that al-Bīrūnī already mentions the theorem as a thing known to all (*Chronology*, p. 184 of the text, p. 166 of the translation).

(γ) $\quad \sin h = \dfrac{Rp}{\sqrt{\text{horizontal shadow}^2 + p^2}}.$

II. From the *vertical shadow*. The author, ll. 30–32, states it as

(δ) $\quad \text{vertical shadow} = p\,\dfrac{\sin h}{\cos h};$



then he gives two formulas:

(ε) $\quad \sin\zeta = \cos h = \dfrac{Rp}{\sqrt{\text{vertical shadow}^2 + p^2}},$

which is derived very easily, like (γ);

(ζ) $\quad \sin h = \dfrac{R \times \text{vertical shadow}}{\sqrt{\text{vertical shadow}^2 + p^2}},$

which only represents the well-known definition $\sin h = R\,BC/AC$.

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Had the author divided the gnomon into 60 parts (as he had already taken a radius of 60 parts, to which he refers the sine, the versed sine and the cosine), these shadows would plainly have become true trigonometric functions, and the formulas just found could have been written

(α′) $\quad \operatorname{cotg} h = R\,\dfrac{\cos h}{\sin h}$ $\qquad$ (β′) $\quad \operatorname{cosec} h = \sqrt{R^2 + \operatorname{cotg}^2 h}$

(γ′) $\quad \sin h = \dfrac{R^2}{\operatorname{cosec} h}$ $\qquad$ (δ′) $\quad \operatorname{tg} h = R\,\dfrac{\sin h}{\cos h}$

(ε′) $\quad \cos h = \dfrac{R^2}{\sec h}$ $\qquad$ (ζ′) $\quad \sin h = R\,\dfrac{\operatorname{tg} h}{\sec h};$

which, when $R = 1$ is taken, as is the present custom, correspond entirely to our trigonometric formulas. It is worth quoting here the words of Delambre, pp. 16–17: «Les Arabes avaient donc, au tems d'Albategni, des tables des quantités, ombre $= \sigma = \nu\,\frac{\cos h}{\sin h} = 12\cot h = 12\operatorname{tang} N$, et par conséquent des tables de tangentes, mais calculées pour le rayon 12, tandis que les sinus étaient calculés pour le rayon 60; mais en multipliant ces tangentes par 5, ou 10/2, on aurait eu les tangentes pour le rayon 60. ….. Voilà bien la première idée des tangentes; voilà des tables qui donnent les tangentes de tous les arcs, et par lesquelles tout arc peut être connu d'après sa tangente. Le malheur est ce choix du rayon de douze parties, qui étaient encore une imitation d'une pratique grecque[1]. Albategni ne sentit pas l'utilité de ces tangentes pour les calculs trigonometriques, et la preuve, c'est qu'il n'a pas changé le rayon» («So in al-Battānī's time the Arabs had tables of the quantities shadow = σ = ν cos h/sin h = 12 cot h = 12 tan N, and consequently tables of tangents, but computed for the radius 12, while the sines were computed for the

[1]More correctly von Braunmühl, I, p. 51: «Unzweifelhaft folgt er hierin indischen Vorbildern, denn, wie wir schon sahen (S. 39), teilten auch die Inder das Gnomon in 12 Finger, während Ptolemaeus die Einteilung in 60^p hat [*Almag.* II, 5, ed. Halma, vol. I, p. 75]» («Without doubt he follows Indian models in this, for, as we have already seen (p. 39), the Indians too divided the gnomon into 12 fingers, while Ptolemy has the division into 60 parts»).

radius 60; by multiplying these tangents by 5, or 10/2, one would have had the tangents for the radius 60. ….. Here indeed is the first idea of the tangents; here are tables that give the tangents of all arcs, and by which any arc can be known from its tangent. The misfortune is this choice of a radius of twelve parts, which was again an imitation of a Greek practice. Albategni did not feel the usefulness of these tangents for trigonometric computations, and the proof is that he did not change the radius»).

The historians of mathematics among us agree with one voice that Abū l-Wafāʾ (who died in the month of Rajab 388 = July 998) was the first to use the shadows as trigonometric functions and to make tables of cotangents for the radius 1 and for the radius 60^p. This universal opinion, however, does not seem right to me, since already in the astronomical work of Ḥabash, al-Battānī's contemporary, the shadows are referred to the radius 60^p, like the sines (fol. 97r–v); indeed, ff. 85r–87v give tables which, taking the radius as 60^p, contain for every degree of the quadrant the tangents (*ẓill maʿkūs*, «the inverted shadow»), the cotangents (*ẓill mustawī*, «the straight shadow»)[1], the tangents multiplied by $0^p\ 26'\ 11''\ 34'''$, and lastly the cosecants (*quṭr al-ẓill*, «the hypotenuse of the shadow») divided by the radius, that is, cosec/R = R/sin. And he uses these trigonometric functions, computed to the seconds, here and there in his formulas.

What al-Battānī calls *ẓill mabsūṭ*, «the extended or spread shadow», other Arabs call *ẓill thānī*, «the second shadow», or *ẓill mustawī*, «the straight shadow». The tangent, on the other hand, which al-Battānī calls *ẓill qāʾim*, «the standing shadow», or *ẓill muntaṣab*, «the erect shadow», others mostly call *ẓill awwal*, «the first shadow», or *ẓill maʿkūs*, «the inverted shadow».

With the Arabs a threefold division of the gnomon is found. The first is the one the Indians always use, which al-Battānī adopted as the most usual of all with the calculators; it comprises 12 parts, called digits (*aṣābiʿ*), and can be used only in dialling. Another, also set up for dialling but rarer than the first[2], counts 7, or 6 2/3, or 6 1/2 parts, called feet (*aqdām*). The third, lastly, which serves dialling and trigonometry at once, gives a division into sixtieths (*ajzāʾ*). — Oloug Beg, p. 61 of the text, 85 of the translation, says that the calculators always refer the first shadow (the tangent) to a gnomon of 60 parts, and the second shadow (the cotangent) indifferently to a gnomon of 60, 12 or 7 parts.

The gnomon shows the shadow of the upper limb of the solar disc, not of the centre of the sun, so that the altitude found with the gnomon exceeds the true altitude by about 16 minutes. Neither the Greeks nor al-Battānī seem to have noticed this; only later Ibn Yūnus, ch. XI, as Delambre, p. 102, relates, «dans un discours sur l'ombre, il démontre géometriquement que l'ombre du gnomon est celle du bord supérieur; il ne dit rien de la pénombre» («in a discourse on the shadow, proves geometrically that the shadow of the gnomon is that of the upper limb; he says nothing of the penumbra»).

On chapter 11

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a) — p. 23.

To find the *azimuth of the altitude*, that is, from the given altitude *h* of the sun to find *the azimuth of this circle of altitude* (that is, *the azimuth of the sun*) on the horizon of a place on the earth of known latitude φ.

[1] M. Curtze (in *Bibliotheca Mathematica herausgegeben von G. Enestrom*, new series, vol. X, 1896, p. 71) is mistaken when he says of the words umbra recta and umbra versa: «vor Abû 'l-Wafâ, der 998 starb, waren sie überhaupt als *termini technici* nicht bekannt» («before Abū l-Wafāʾ, who died in 998, they were not known at all as technical terms»).

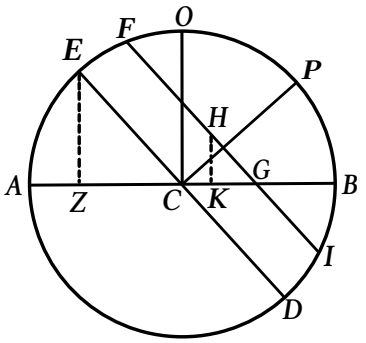
[2] And later; for, as far as the published Arab writers go, it is not found before the thirteenth century; see Aboul Hhassan, ch. XV, p. 152 (where 6 2/3); Barhebraeus, p. 178 of the text (where 7); al-Jaghmīnī, p. 270 (where 7 and 6 1/2); Oloug Beg, p. 61 of the text, 85 of the translation (where 7).

Leaving out the proofs, the author teaches the formulas:

(α) sine of the difference of the horizon $= \dfrac{\sin h \, \sin \varphi}{\cos \varphi}$

(β) $\sin(\text{azimuth}) = R \dfrac{\sin(\text{rising ampl.}) \pm \sin(\text{diff. of the horizon})}{\cos h}$.

Both were plainly derived by the orthographic projection of the sphere. The name *sine of the difference* (or *of the variation*) *of the horizon*, together with al-Battānī's formula (α), is read also in Ibn Yūnus, ch. XIX (in Delambre, p. 114); Abū l-Wafā'[1], Aboul Hhassan, ch. LXII, p. 308, and Oloug Beg, p. 78 of the text, 106–107 of the translation, on the other hand, call it the argument of the azimuth (*ḥiṣṣat al-samt*). Geometrically, as Abū l-Wafā' and Mīram Čelebī (in his commentary on Ulugh Beg) say, it is the line that joins the foot of the perpendicular drawn from the star to the plane of the horizon, that is, the foot of the sine of the altitude, with the line of intersection of the horizon and the parallel of the star.



Let the horizon ACB, the celestial equator ECD and the parallel circle FI, which the sun describes in its daily motion, be projected orthogonally onto the plane of the meridian circle AOBD. Suppose the sun to have reached H; HK will then be the projection of the altitude *h*. Let EZ be drawn perpendicular to AB, and so parallel to the line HK; since P is the pole of the equator, we shall have

$$KG = \text{sine of the difference of the horizon}, \qquad BP = EO = \varphi,$$

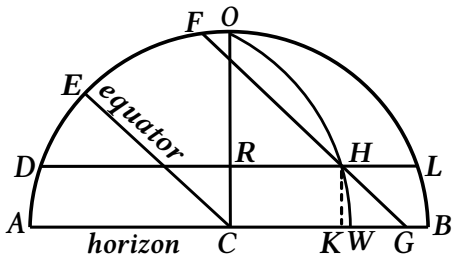
$$ZC = \sin \varphi, \qquad HK = \sin h, \qquad EZ = \cos \varphi.$$

From the similarity of the right-angled triangles EZC and HKG it follows at once that

$$EZ : HK = ZC : KG, \quad \text{that is,} \quad \cos \varphi : \sin h = \sin \varphi : \sin(\text{diff. horiz.}),$$

whence al-Battānī's rule

(α) $\sin(\text{diff. horiz.}) = \dfrac{\sin h \, \sin \varphi}{\cos \varphi}$.



It remains for us now to prove (β), by the same way that al-Battānī probably followed. Let the figure drawn here be the orthographic, or orthogonal, projection of the celestial sphere as seen from the east point; let AOB be the meridian, ACB the horizon, OC the vertical circle perpendicular to the meridian, EC the celestial equator, FG the parallel described by the sun, which we put at H. Through H let the circle of altitude OHW be drawn, and the circle DRL parallel to the horizon. Lastly, let the line HK be drawn perpendicular to the plane of the horizon.

The arcs CW and RH, cut off by the same vertical circles OC and OW, will contain the same number of degrees; so their sines, which the straight lines CW, RH represent in the figure, will be proportional to the radii CB, RL with which those arcs are described. Since LB = HW is the altitude h of the sun, and its sine is HK = RC, while the arc OH = OL is the complement of the altitude and has the sine RL, by what was said above we shall plainly have

$$CB : RL = CW : RH,$$

that is,

$$R : \cos h = \sin(\text{azimuth}) : RH,$$

whence

$$\sin(\text{azimuth}) = R\,\frac{RH}{\cos h}.$$

By the constructions made, RH = CK = CG – KG. But CG is plainly the sine of the rising amplitude, that is, the sine of the arc of the horizon between the east point C and the point from which the sun rises on the given day; KG is what the author calls the sine of the difference of the horizon; hence al-Battānī's formula

$$(\beta) \qquad \sin(\text{azimuth}) = R\,\frac{\sin(\text{rising ampl.}) - \sin(\text{diff. horiz.})}{\cos h},$$

which holds when the azimuth, the amplitude and the declination are counted positive towards the north. If the difference, which is the numerator of the fraction, is negative[1], the azimuth will be towards the south.

The same formula (β) is found in Ibn Yūnus, ch. XX (Delambre, p. 116), in Aboul Hhassan, ch. LXII, p. 308, and in Oloug Beg, p. 78 of the text, 106–107 of the translation. Ḥabash, f. 92v, puts for the sine of the difference of the horizon its value by rule (α), and so writes

$$\sin(\text{azimuth}) = \frac{R}{\cos h}\left(\frac{\sin\varphi}{\cos\varphi}\,\sin h - \sin(\text{rising ampl.})\right).$$

Let us put into formula (β) the value of the sine of the difference of the horizon found before (that is, $\sin h \sin\varphi/\cos\varphi$), and the value of the sine of the rising amplitude found in ch. VII (that is, $R\sin\delta/\cos\varphi$); there results

$$\sin(\text{azimuth}) = R\,\frac{R\sin\delta - \sin h\,\sin\varphi}{\cos h\,\cos\varphi}.$$

But since al-Battānī, like almost all the Arabs, counts the azimuth from the line of the rising and setting, while we proceed from the north point towards the east, the sine of the azimuth of the Arabs becomes with us $\sin(90^\circ - \text{azimuth}) = \cos(\text{azimuth})$. Lastly, taking the radius as one part, as is our custom, we find at once the present-day formula

$$\cos(\text{azimuth}) = \frac{\sin\delta - \sin h\,\sin\varphi}{\cos h\,\cos\varphi},$$

[1] Carra de Vaux, *L'Almageste d'Aboû 'l-Wéfa al-Bûzdjâni*, in the Journal Asiatique, 8th series, vol. 19, 1892, p. 429.

[1] When the sine of the difference of the horizon is greater than the sine of the rising amplitude. — When the sun is on the southern side of the equator, not the difference but the sum of the sines must be used.

which is an immediate consequence of the fundamental rule of spherical trigonometry (called by the Germans the *Cosinussatz*)

$$\cos a = \cos b \cos c + \sin b \sin c \cos A,$$

if one puts $a = 90^\circ - \delta$, $b = 90^\circ - h$, $c = 90^\circ - \varphi$, $A =$ azimuth.

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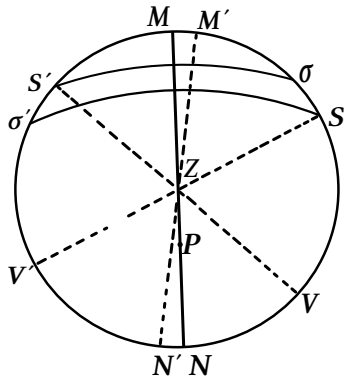
What Delambre writes on p. 17, mixing his algebraic explanations with al-Battānī's rules, was not rightly interpreted by Hankel, *Zur Gesch. der Mathem.*, p. 281, by Cantor, I, 633 (= 694 of the second edition), and by Suter, *Zur Geschichte der Trigonometrie* (in Bibliotheca Mathematica, ed. G. Enestrom, new series, vol. VII, Stockholm 1893, pp. 6 and 8), who say that al-Battānī knew the proposition $\cos a = \cos b \cos c + \sin b \sin c \cos A$. We can indeed easily derive it from (β); but the author neither made such a reduction nor wished to make it; rather, the very form of rule (β) shows that al-Battānī by no means regarded it as a general proposition for solving spherical triangles. Von Braunmühl, I, p. 53, judged better, but not absolutely enough.

Regiomontanus, in a long note on this chapter, which the printed editions wrongly give in the middle of chapter XII, rightly guessed the way that al-Battānī (or the Arabs before him) probably followed to find his two formulas[1]. For he writes: «Albategnius in demonstrationibus suis multis ad lineas rectas refugit, et ex similitudine triangulorum multa conclusit. Modus iste intellectui facilis videretur, nisi sectiones superficierum obscuritatem afferrent» («In many of his proofs Albategnius takes refuge in straight lines, and he concluded much from the similarity of triangles. This method would seem easy to the understanding, were it not that the sections of the surfaces bring obscurity»). He adds a proof that differs little from mine, but is perhaps more obscure.

On chapter 12

a) — p. 24, ll. 26–31.

[«*The sun moves slowly in declination*». For near the solstices the declination of the sun changes little in a day, so that the sun can then be said, to a close approximation, to run along a true parallel; which, on the other hand, would be false near the equinoxes, for between its rising and its setting the sun can vary by more than 11 minutes in declination.



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[1]Because of the form of the rule, and because of what al-Battānī proves geometrically elsewhere, the conjecture was not too difficult for Regiomontanus, who had pondered al-Battānī's work so assiduously. So I do not think one should agree with von Braunmühl, I, 52, note 1, when he writes: «Es scheint demnach, das Regiomontanus die Beweise des Albategnius, die in der Druckausgabe nich angeführt sind, kannte» («It seems, then, that Regiomontanus knew the proofs of Albategnius, which are not given in the printed edition»).

In what follows, the Arabic text ought to be translated literally: «It is also manifest that the time of midday cannot be determined truly, because of the hasty passage of the sun through (or in) the meridian circle .....». Plato: «Manifestum etenim est quod medii diei hora propter festinum Solis motum in medii diei circulo veraciter non terminatur, quae tamen vel ei propior apprehenditur» («For it is manifest that the hour of midday, because of the hasty motion of the sun in the circle of midday, is not truly determined, though it, or a time nearer to it, is grasped»). — But the interpretation we have adopted seems more probable, and rests on these considerations. When the sun rises at the point S′ of the horizon, its setting, if its declination were constant, would have to take place at the point σ of the same celestial parallel. But because of the daily motion in declination, the sun passes gradually from that parallel to another, σ′S, and sets at S. At the time of its rising it casts the shadow along the direction ZV, at the time of its setting along ZV′, which makes with the meridian line MN an angle different from the angle contained by the meridian itself and the shadow ZV. So by bisecting the angle V′ZV one finds not the meridian line MN but the line M′N′; whence it is clear that at the moment of time equally distant from the rising and from the setting («at the time of midday») the sun is not on the meridian circle, but in the vertical plane that bisects the angle V′ZV. In the regions of the Arabs, from the equator to the parallel of 45°, the small angle MZM′ can rise to 8′ or 9′; whence we understand why al-Battānī says that this drawing of the meridian line should be undertaken about the times of the solstices. — Schiaparelli.]

b) — p. 25, ll. 16–22, and p. 25, l. 37 – p. 26.

Given the rising amplitude and the meridian altitude, to compute *the altitude of the sun that has no azimuth*, that is, *the altitude on the prime vertical circle* (which passes through the east and the west points, from which the azimuths of the Arabs are counted).

So far the author had stated his formulas without proofs; now, on the other hand, he shows geometrically how the rule is found. Why he did so is not certainly known. It seems probable to me that he left out the earlier proofs as being already known from other authors, whereas this solution of the problem was new, and for this reason al-Battānī wished to prove it. For we have seen that the propositions of spherical astronomy contained in the earlier chapters either can easily be derived from Ptolemy's constructions (with sines put for chords) or are read also in Ḥabash[1], our author's contemporary; the problem of this chapter, on the other hand, I find solved in this way only in later writers. Likewise in chapter XXVI we read the proof of another trigonometric formula which, there can be no doubt, was first found by al-Battānī.

Al-Battānī's proof rests on the analemma, or the orthographic projection of the sphere, whose use is frequent with the Indians and more frequent with the Arabs[2]. — The last words of the proof (p. 26, ll. 27–29), which Plato leaves out, were spoiled by some unskilled reader, who disfigured the text in the same way in chapter XXVI (p. 38, ll. 27–28). [Al-Battānī meant: so if the line GF is diminished in the ratio of CM to CF, the line EM remains; likewise, if GF is diminished in the ratio of EC to CG, the line EM remains. That is, since $EC : CG = EM : GF = CM : CF$, plainly $EM = \frac{GF \times CM}{CF} = \frac{GF \times EC}{CG}$. — Schiaparelli.]

Since EC = the sine of the rising amplitude of the sun on the given day, GF = the sine of the meridian altitude of the sun, and CG = EC + EG = the sine of the rising amplitude + the cosine of the meridian altitude, al-Battānī's formula follows at once:

$$(\alpha) \qquad \sin(\text{alt. without azimuth}) = \frac{\sin(\text{rising ampl.}) \times \sin(\text{merid. alt.})}{\sin(\text{rising ampl.}) + \cos(\text{merid. alt.})},$$

which is read later, together with others, in Ibn Yūnus, ch. XVIII (Delambre, p. 114).

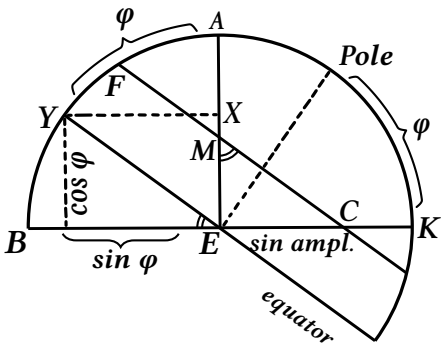
[1] Who, however, always leaves out the proofs.

[2] As von Braunmühl rightly understood, I, p. 38 (on the Indians) and in many places.

It is useless, and does not answer to the author's thought, to develop this formula as Delambre does, p. 19, putting for the sine of the rising amplitude its value $R \sin\delta / \cos\varphi$ found in chapter VII, for the sine of the meridian altitude the value $\cos(\varphi - \delta)$ that follows from chapter XV, and lastly $\sin(\varphi - \delta)$ for the cosine of the meridian altitude. One then obtains[1]

$$\sin(\text{alt. without azimuth}) = R\,\frac{\sin\delta}{\cos\varphi}\left(\frac{\cos(\varphi-\delta)}{\sin(\varphi-\delta) + R\,\frac{\sin\delta}{\cos\varphi}}\right) = \frac{R\sin\delta\,\cos(\varphi-\delta)}{\cos\varphi\,\sin(\varphi-\delta) + R\sin\delta}.$$

It should rather be noted that in place of al-Battānī's formula three others are found in Ḥabash, f. 93r:



I. $$\sin(\text{alt. without azimuth}) = \frac{\sin(\text{rising ampl.}) \times \cos\varphi}{\sin\varphi},$$

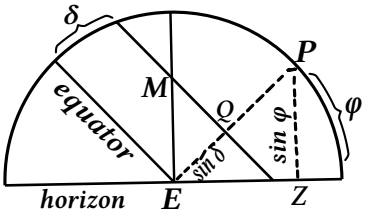
probably also found by orthographic projection; for from the similarity of the triangles MEC, YXE the proportion is easily gathered

$$EM : EC = EX : XY,$$

$$\sin(\text{alt. without azim.}) : \sin(\text{rising ampl.}) = \cos\varphi : \sin\varphi.$$

II. $$\sin(\text{alt. without azimuth}) = \frac{\sin(\text{rising ampl.})}{\frac{\sin\varphi}{\cos\varphi}},$$

which is only another way of writing formula I.



III. $$\sin(\text{alt. without azimuth}) = R\,\frac{\sin\delta}{\sin\varphi},$$

which can be derived algebraically from I, when its value $R \sin\delta / \cos\varphi$ is put for the sine of the rising amplitude; but it also follows directly in the orthographic projection, from the similarity of the triangles MEQ, EPZ:

$$EM : EQ = EP : PZ,$$

[1] Delambre puts $\sin(\varphi - \delta)$ in the numerator and $\cos(\varphi - \delta)$ in the denominator of the fraction; but it is a mere slip of the pen.

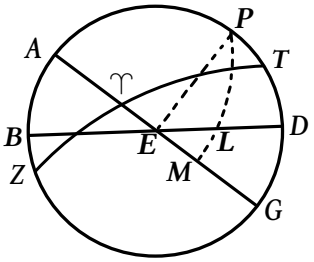
$$\sin(\text{alt. without azim.}) : \sin\delta = R : \sin\varphi.$$

Formula III is also in Aboul Hhassan, ch. LXI, p. 306.

On chapter 13

a) — p. 27, ll. 17–24.

From the right ascension of any degree of the ecliptic and the greatest length of the day in a given city, to find the *oblique ascension* of that degree of the ecliptic in that city.



The *oblique ascension*, which the Arabs call *the ascensions of the clime* or *in the clime*, is the arc of the equinoctial circle between the beginning of Aries and the point of the equinoctial circle that is on the horizon at the same time as the given star or degree of the ecliptic. Let P be the pole of the equator, AEG the equinoctial circle, ♈ the beginning of Aries, Z♈T the ecliptic, BED the horizon of the given place on the earth, L the given star, which rises from the horizon, PM the circle of declination passing through the star L; the whole arc ♈M will be the right ascension of the star, ♈E its oblique ascension, the difference EM the ascensional difference.

The problem plainly reduces to finding the ascensional difference EM, which al-Battānī calls *the argument of the difference of the day in the quadrant*[1]. The author gives two formulas. I 188

I. The author considers the oblique ascension of a given degree of the ecliptic, not of any star. Let *m* be the ascensional difference; P the greatest half arc of the day, or the greatest length of half the day, computed in degrees of the equator, in the given city, so that P – 90° is half the excess of the longest day over the equinoctial day (= 2 × 90° = 12^h); α the right ascension of the given degree of the ecliptic. The author's formula is

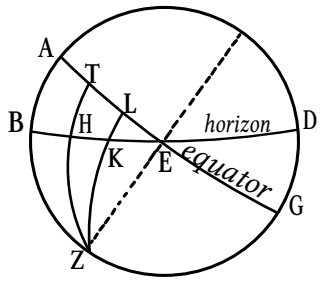
$$(\alpha) \qquad \sin m = \frac{\sin\alpha\ \sin(P - 90^\circ)}{R},$$

which we can write

$$\sin m = -\frac{\sin\alpha\ \cos P}{R}.$$

It is contained implicitly in Ptolemy's proof, *Almag.* II, 7 (ed. Halma, vol. I, pp. 97–98).

[1]The name plainly comes from the author's considering the ascension of a given degree of the ecliptic; so that, as appears from ch. XIX, the half arc of the day of that degree, that is, half the length of the day when the sun has reached that degree, is 90° ± the ascensional difference.



Let ABGD be the meridian circle; AEG the equator, whose southern pole is Z; BED the horizon; ZHT and LKZ two arcs of circles of declination; H the point common to the horizon and to the parallel that passes through the winter solstitial point; K the point common to the horizon and to the parallel that passes through the given place of the ecliptic. TEG is plainly the greatest half arc of the day in the place of the earth to which the celestial horizon BED corresponds. Ptolemy found

$$\frac{\text{chord}\,2TH}{\text{chord}\,2ZH} = \frac{\text{chord}\,2TE}{\text{chord}\,2EL} \times \frac{\text{chord}\,2KL}{\text{chord}\,2KZ},$$

that is,

$$\frac{\text{chord}\,2\varepsilon}{\text{chord}\,(180^\circ - 2\varepsilon)} = \frac{\text{chord}\,(2P - 180^\circ)}{\text{chord}\,2\text{ asc. diff.}} \times \frac{\text{chord}\,2\delta}{\text{chord}\,(180^\circ - 2\delta)},$$

and, with sines put for chords,

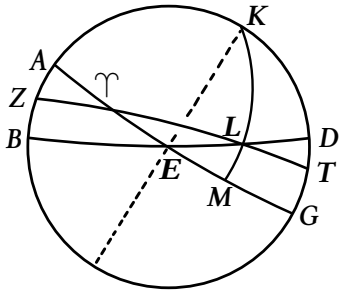
$$\frac{\sin\varepsilon}{\cos\varepsilon} = \frac{\sin(P - 90^\circ)}{\sin(\text{asc. diff.})} \times \frac{\sin\delta}{\cos\delta},$$

whence

$$\sin(\text{asc. diff.}) = -\cos P\,\frac{\sin\delta\,\cos\varepsilon}{\cos\delta\,\sin\varepsilon}.$$

But al-Battānī had taught in chapter V that $\sin\alpha = R\,\frac{\sin\delta\,\cos\varepsilon}{\cos\delta\,\sin\varepsilon}$, so that

$$\sin m = -\frac{\cos P\,\sin\alpha}{R}.$$



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II. The other formula is derived directly from Ptolemy's proof, *Almag.* II, 7 (ed. Halma, vol. I, pp. 92–94). Let ABGD be the meridian circle; BED the horizon; AEG the celestial equator; Z♈T the ecliptic, whose point ♈ is the beginning of Aries; K the northern pole of the equator; KM an arc of a circle of declination. ♈E will be the oblique ascension of the given point L of the ecliptic, ♈M its right ascension, EM the ascensional difference. Ptolemy proves

$$\frac{\text{chord}\,2KD}{\text{chord}\,2DG} = \frac{\text{chord}\,2KL}{\text{chord}\,2LM} \times \frac{\text{chord}\,2ME}{\text{chord}\,2EG},$$

that is,

$$\frac{\text{chord}\, 2\varphi}{\text{chord}\,(180^\circ - 2\varphi)} = \frac{\text{chord}\,(180^\circ - 2\delta)}{\text{chord}\, 2\delta} \times \frac{\text{chord}\, 2m}{\text{chord}\, 180^\circ}.$$

With sines put for chords, one obtains

$$\frac{\sin\varphi}{\cos\varphi} = \frac{\cos\delta}{\sin\delta} \times \frac{\sin m}{R},$$

whence al-Battānī's formula

(β) $$\sin m = R\,\frac{\sin\varphi\ \sin\delta}{\cos\varphi\ \cos\delta},$$

which Ḥabash, f. 104r, using trigonometric tangents, writes more briefly[a]

$$\sin m = \frac{\text{tg}\,\varphi\ \text{tg}\,\delta}{R},$$

where φ is the geographical latitude of the given city.

On chapter 14

a) — pp. 29–30.

To find the *geographical latitude* φ of a given place on the earth from the meridian altitude of the sun.

The rules are very well known; when the sun (⊙) is north of the celestial equator,

(α) $\varphi = 90^\circ - (\text{merid. alt. of } \odot - \text{declin. of } \odot)$;

when the sun is in the southern signs of the zodiac,

(β) $\varphi = 90^\circ - (\text{merid. alt. of } \odot + \text{declin. of } \odot)$.

We write them briefly

$$\varphi = \text{complement of the merid. alt.} \pm \text{declin.}$$

On chapter 15

a) — p. 30.

To find the *meridian altitude* of the sun from the geographical latitude of the given city and the declination of the sun.

Al-Battānī gives the very well known formulas in two ways; when the sun is in the northern signs of the zodiac,

(α) $\text{merid. alt.} = 90^\circ - (\text{latit.} - \text{declin.})$

(α′) $\text{merid. alt.} = \text{complement of the latit.} + \text{declin.}$;

when the sun is in the southern signs,

(β) $\text{merid. alt.} = 90^\circ - (\text{latit.} + \text{declin.})$

(β′) $\text{merid. alt.} = \text{complement of the latit.} - \text{declin.}$

[a] The print has sin m = R tg φ tg δ; Nallino's Addenda et emendanda (Part II, p. XXIV) correct it to tg φ tg δ : R.

On chapter 16

a) — p. 30.

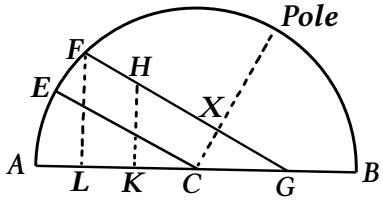
To find the *hours elapsed since the rising of the sun* at a given moment of time, from the altitude of the sun.

The problem reduces to finding the hour angle t (that is, the angle contained by the meridian and the circle of declination passing through the sun); it is added to the half arc of the day if the given time is after noon, or subtracted if the time is before noon. The sum or the remainder will be the arc of the revolution of the sphere, whose degrees are easily turned into equal or seasonal hours.

I 190

Let P be the half arc of the day, h the altitude of the sun at the given time; al-Battānī's formula is

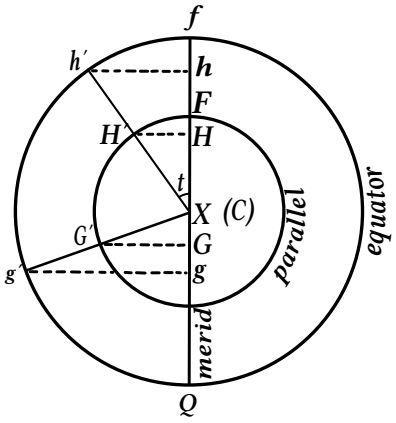
$$(\alpha) \qquad \operatorname{vers} t = \operatorname{vers} P - \frac{\sin h \operatorname{vers} P}{\sin(\text{merid. alt.})}.$$



This formula was plainly derived by the orthographic projection of the sphere. Onto the plane of the meridian circle AFPB let there be projected orthographically the horizon ACB, the celestial equator EC, and the parallel FG described by the sun, which is supposed to have reached H; the perpendicular FL will be the sine of the meridian altitude, HK the sine of the altitude at the given time. In the triangle FLG, since HK is parallel to the side FL, we have the proportion

$$FG : HG = FL : HK,$$

$$FG : HG = \sin(\text{merid. alt.}) : \sin h.$$



Now, to know what FG and HG are, let us imagine the orthographic projection of the figure onto the plane of the equinoctial circle, leaving out the horizon circle; FXQ will be the meridian, the

angle fXh' = t, the angle fXg' = P, hX = cos t, Xg = cos P. From the similarity of the triangles $h'h$X and H′HX one gathers

$$FH : fh = H'X : h'X,$$

that is,

$$FH : fh = FX : R;$$

but $fh = fX - hX = R - \cos t = \operatorname{vers} t$, therefore

$$FH = \operatorname{vers} t \, \frac{FX}{R}.$$

Similarly the proportion is derived

$$FG : fg = G'X : g'X,$$

or

$$FG : fg = FX : R;$$

but, since in the figure $P > 90°$,

$$fg = fX + Xg = R + \cos P = \operatorname{vers} P,$$

therefore $FG = \operatorname{vers} P \, \frac{FX}{R}$. Lastly

$$HG = FG - FH = \frac{FX}{R} \, (\operatorname{vers} P - \operatorname{vers} t).$$

The first proportion found therefore becomes

$$\frac{FX}{R} \operatorname{vers} P : \frac{FX}{R} \, (\operatorname{vers} P - \operatorname{vers} t) = \sin(\text{merid. alt.}) : \sin h,$$

whence $\operatorname{vers} P - \operatorname{vers} t = \frac{\sin h \, \operatorname{vers} P}{\sin(\text{merid. alt.})}$, and at last al-Battānī's formula:

$$(\alpha) \qquad \operatorname{vers} t = \operatorname{vers} P - \frac{\sin h \, \operatorname{vers} P}{\sin(\text{merid. alt.})}.$$

The same is read in Ḥabash, f. 99v (who calls vers P *jayb al-nahār*, «the sine of the day»), in Aboul Hhassan, ch. XLI, p. 261, and in Oloug Beg, p. 95 of the text, 122 of the translation. I 191

Algebraically it is not difficult to develop this formula further. By the rule of chapter IX

$$\operatorname{vers} P = R - \cos P = R + R \, \frac{\sin \varphi \, \sin \delta}{\cos \varphi \, \cos \delta};$$

from chapter XV one gathers

$$\sin(\text{merid. alt.}) = \sin(\text{compl. of } \varphi \pm \delta) = \frac{\cos \varphi \, \cos \delta \pm \sin \varphi \, \sin \delta}{R},$$

$$\sin(\text{merid. alt.}) = \cos(\varphi \mp \delta);$$

so that formula (α) can be written

$$\operatorname{vers} t = \left(R + R \, \frac{\sin \varphi \, \sin \delta}{\cos \varphi \, \cos \delta}\right) - \frac{\sin h \left(R + R \, \frac{\sin \varphi \, \sin \delta}{\cos \varphi \, \cos \delta}\right)}{\cos(\varphi \mp \delta)}$$

$$= \frac{R \, (\cos \varphi \, \cos \delta + \sin \varphi \, \sin \delta)}{\cos \varphi \, \cos \delta} - \frac{R \sin h \, (\cos \varphi \, \cos \delta + \sin \varphi \, \sin \delta)}{\cos \varphi \, \cos \delta \, \cos(\varphi \mp \delta)}.$$

But (as appears from chapter III) $\cos\varphi\cos\delta + \sin\varphi\sin\delta = R\cos(\varphi - \delta)$, therefore

$$\operatorname{vers} t = R^2\,\frac{\cos(\varphi-\delta)}{\cos\varphi\,\cos\delta} - R^2\,\frac{\sin h\,\cos(\varphi-\delta)}{\cos\varphi\,\cos\delta\,\cos(\varphi\mp\delta)},$$

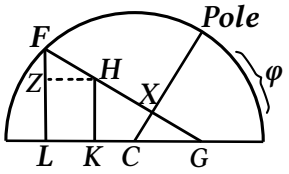
and at last

(I) $$\operatorname{vers} t = R^2\,\frac{\cos(\varphi-\delta)-\sin h}{\cos\varphi\,\cos\delta},$$

a formula which Delambre, p. 20, wrongly ascribes to al-Battānī himself (leaving out R^2, since we refer the trigonometric lines to $R = 1$). And since we derive the hour angle t directly from the well-known relation $\sin h = \sin\varphi\sin\delta + \cos\varphi\cos\delta\cos t$, that is,

$$\cos t = \frac{\sin h - \sin\varphi\,\sin\delta}{\cos\varphi\,\cos\delta},$$

where two multiplications have to be made instead of the single one (R^2 left out) found in (I), Delambre himself, p. 164, afterwards cites formula (I) as if it were the first and oldest example of *prosthaphaeresis*, the procedure by which our mathematicians, before logarithms were discovered, tried to put additions and subtractions in place of multiplications and divisions. Hankel, p. 281, and Cantor, I, p. 633 (= 694 of the second edition), repeat the same; but quite wrongly, for al-Battānī never converted formula (α), obtained by orthographic projection, into (I), as von Braunmühl, I, 53, note 2, very well perceived.



Formula (I), written

$$\operatorname{vers} t = R^2\,\frac{\sin(\text{merid. alt.})-\sin h}{\cos\varphi\,\cos\delta},$$

is in Oloug Beg, p. 96 of the text, 124 of the translation; who, however, without doubt derived it not algebraically from (α) but directly from the orthographic projection. In the first figure of this note let the perpendicular HZ be drawn; then ZL = HK = sin *h*, and FZ = sin merid. alt. – sin *h*. The angle XCG is the altitude of the pole above the horizon, that is, φ; therefore the angle FHZ = the angle XGC = 90° – φ. Plainly also CX = sin δ, FX = cos δ. Since the sides of any triangle are to one another as the sines of the opposite angles[1],

$$FZ : FH = \sin(\text{angle } H) : \sin(\text{angle } Z) = \cos\varphi : \sin 90^\circ = \cos\varphi : R;$$

therefore $FZ = FH\,\frac{\cos\varphi}{R}$. We saw above that $FH = \operatorname{vers} t\,\frac{FX}{R} = \operatorname{vers} t\,\frac{\cos\delta}{R}$; so

$$FZ = \frac{\operatorname{vers} t\,\cos\delta}{R}\times\frac{\cos\varphi}{R} = \frac{1}{R^2}\,\operatorname{vers} t\,\cos\delta\,\cos\varphi.$$

But $FZ = \sin(\text{merid. alt.}) - \sin h$; therefore

$$\sin(\text{merid. alt.}) - \sin h = \frac{1}{R^2}\,\operatorname{vers} t\,\cos\delta\,\cos\varphi,$$

whence

$$\operatorname{vers} t = R^2\,\frac{\sin(\text{merid. alt.})-\sin h}{\cos\delta\,\cos\varphi},$$

as was to be proved.

[1]Ulugh Beg was not unaware of the proposition; see p. 181, note 1.

On chapter 17

I 192

a) — p. 31.

To find the *altitude of the sun* from the hours of the day elapsed since its rising.

The problem is the converse of the preceding one; al-Battānī, then, inverting the formula of the preceding chapter, teaches

$$(\alpha) \qquad \sin h = \frac{(\operatorname{vers} P - \operatorname{vers} t)\ \sin(\text{merid. alt.})}{\operatorname{vers} P},$$

as we read also in Ḥabash, f. 100r, and in Aboul Hhassan, ch. XLIV, p. 266.

On chapter 18

a) — p. 31, ll. 30–32.

[«*But the stars on either side*» etc. — The whole sentence is very obscure. The declination is greater or smaller than the latitude in various ways in the different regions of the sky, and it is not understood how it could be useful to take the sum or the difference of two quantities which, as the author himself says just below, are counted on two different circles. Either it is a gloss of a very unskilled reader, or a fragment of a fuller passage that it is now impossible to restore. Perhaps the arc that we shall call β′ in what follows was defined in this place; but it is a mere conjecture. — Schiaparelli.]

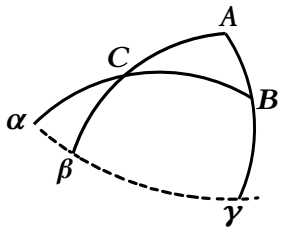
b) — pp. 31–32.

To compute the *declination* δ of a star whose longitude L and latitude β are known.

Let β′ be the latitude of the point of the celestial equator that has the same longitude as the star[1]; al-Battānī's rule is

$$(\alpha) \qquad \sin\delta = \frac{\sin(\beta \pm \beta')\ \cos\varepsilon}{\cos\beta'},$$

which Aboul Hhassan, ch. XXV, p. 189, also gives later, and, together with other formulas, Ibn Yūnus, ch. XXXIX (in Delambre, p. 141), and Oloug Beg, p. 64 of the text, 90 of the translation.



I 193

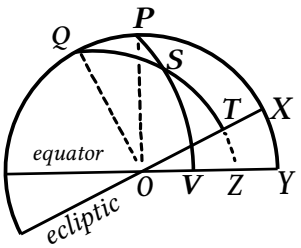
[This proposition was probably derived with the help of *the theorem of Menelaus*[2], which Ptolemy always uses to solve spherical triangles, and which can be stated in this way. Let ABC be a

[1]The quantity β′ al-Battānī calls «declination», that is, the second declination, as I have already said in the translation, p. 32, note 1.

[2]Menelaus, *Sphaerica* III, 2; it is also called *the rule of the six quantities.* The Arabs called its figure *al-shakl al-qaṭṭāʿ*, which the Latins of the Middle Ages translated *figura sector*, or, keeping the Arabic name, *figura alkata*; the Germans of today call it *Transversalenfigur*, and the proposition *Transversalensatz*.

spherical triangle whose sides are arcs of great circles; let a great circle αβγ cut the three sides of the triangle, produced as far as needed; among the six segments that can be conceived between each vertex of the triangle and the cutting arc αβγ there holds a relation which, putting sines for Menelaus' chords of the double arcs and using the present-day algebraic notation, is expressed in this way:

$$\sin A\beta \cdot \sin C\alpha \cdot \sin B\gamma = \sin C\beta \cdot \sin A\gamma \cdot \sin B\alpha.$$



Now let S be the given star, P the pole of the equator, Q the pole of the ecliptic; let the produced sides of the triangle QPS be cut by the equator OY; then

$$\sin QZ \cdot \sin SV \cdot \sin PY = \sin SZ \cdot \sin QY \cdot \sin PV.$$

But

$$QZ = QT + TZ = 90^\circ + \beta', \qquad QY = QX + XY = 90^\circ + \varepsilon,$$
$$SV = \delta, \qquad SZ = ST + TZ = \beta + \beta',$$
$$PY = 90^\circ, \qquad PV = 90^\circ.$$

So the equation found before becomes

$$\cos\beta' \ \sin\delta \times R = \sin(\beta + \beta') \ \cos\varepsilon \times R,$$

whence al-Battānī's rule. — Schiaparelli.]

But al-Battānī never teaches how β′ is computed; which is truly strange, unless a lacuna is supposed in the archetype of the Escorial codex and of Plato's translation; see note *a* on this very chapter (p. 192). In Ibn Yūnus, ch. XXXIX (Delambre, p. 141), in Aboul Hhassan, ch. XXIV, p. 181, and in Oloug Beg, pp. 63–64 of the text, 88 of the translation, the quantity β′ is found in two ways[a]:

$$\sin\beta' = \frac{\sin\delta'}{\cos d'}, \qquad \cos\beta' = \frac{\cos\varepsilon}{\cos d'},$$

where δ′ is the declination of the point *T* of the ecliptic (which is the longitude of the star), and *d′* the declination of the point of the ecliptic whose longitude equals the arc of the ecliptic between the point *T* and the nearest solstitial point. — But β′ can also be derived from the tables. Let the longitude *L* = *OT* be regarded as a right ascension; let it be increased by 90 degrees, since the author counts the right ascensions from the beginning of Capricorn, not of Aries, and let the sum be sought in the column of the degrees of the ascensions in the table of right ascensions (f. 179v ff.); the number corresponding to it in the column of the degrees of the ecliptic will be, after ninety degrees are taken off, the arc *OZ* of the equator, which we shall now regard as if it were an arc of the ecliptic. In the table of the declination of the sun (f. 177v f.) let us take the declination corresponding to the fictitious arc of the ecliptic *OZ*; the fictitious declination found will plainly be the arc *TZ*, that is, β′.

[a] The print has sin d′ in the first denominator. With sin δ′ = sin ε sin L and sin d′ = sin ε cos L, the second declination β′ (tg β′ = tg ε sin L) satisfies sin β′ = sin δ′ / cos d′, which is set here; the second formula is as printed.

In the spherical triangle *OTZ*, right-angled at *T*, plainly (since the angle *XOY* = *TOZ* = ε)

$$\sin(\text{angle } Z) = \frac{\cos\varepsilon}{\cos\beta'};$$

al-Battānī's formula can therefore be written

$$\sin\delta = \sin(\beta + \beta')\ \sin Z.$$

Delambre, pp. 289 and 290, seems to me to have had this in mind when he says that Regiomontanus, in his *Tabulae Directionum* (of 1475), started from al-Battānī's idea in constructing a table that gives, for all celestial longitudes, the *radix ascensionum* (that is, the arc *OZ*), the *arcus* (that is, β′, the second declination), sin *Z* and cotg *Z*. In this way the ecliptic coordinates (longitude and latitude) could easily be turned into equatorial ones (declination and right ascension), using the formulas[a]

$$\sin\delta = \sin(\beta + \beta')\ \sin Z,$$

$$\sin(OZ - \text{right asc.}) = \operatorname{tg}\delta\ \operatorname{cotg} Z.$$

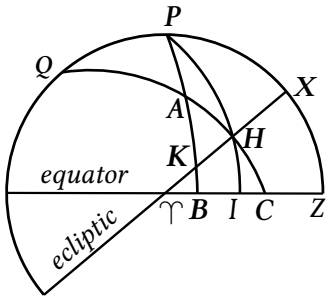
Nor should it be forgotten that the Arabs often use the second declination, and indeed constructed tables of this declination for every degree of the ecliptic.

c) — p. 32, ll. 13–17.

To derive *the degree of the ecliptic that culminates with a star* from the known latitude β, longitude *L* and declination δ of the star.

I 194

The author computes what he calls «the difference (or variation) of the passage of the star through the middle of the sky» (*ikhtilāf majāz al-kawkab fī wasaṭ al-samā'*); this quantity is added to the right ascension of the degree of the star's longitude, or subtracted from it, as the case requires; the sum or the remainder will be the right ascension of the degree that culminates with the star. The longitude of that degree is drawn from the tables of right ascensions.



[If the reading of the codex and of Plato were good, the author's rule would have to be expressed thus:

$$\sin(\text{diff. of passage}) = \frac{\sin\beta\ \sin\varepsilon\ \cos L}{R\ \cos\delta}.$$

Let us now establish what this difference of passage, or difference of culmination, is. Let P be the pole of the equator, Q the pole of the ecliptic; through the given star A let the circle of declination PAB and the circle of latitude QAC be drawn; through the point H of the ecliptic, which is the degree of the star's longitude, let the circle of declination PHI be described. Plainly the arc BI

[a]The print has cotg C for cotg Z: in the triangle SVZ, right-angled at V, sin VZ = tg δ cotg Z, with VZ = OZ – right ascension.

of the equator is the interval between the passage of the star A through the meridian and the passage of the point H of the ecliptic, which is the degree of the star; or the difference between the right ascension (ϒI) of the degree of the star and the right ascension (ϒB) of the star.

Since PB = PI = 90° by the constructions made, the arc BI (that is, the difference of passage sought) will be the measure of the angle APH. If we call D the declination of the point H, that is, the arc HI, we have

$$AH = \beta, \quad PH = 90^\circ - D, \quad PA = 90^\circ - \delta, \quad QA = 90^\circ - \beta, \quad QP = XZ = \varepsilon.$$

And since *QH* = *QX* = 90°, the arc *HX* (that is, 90° – *L*) will be the measure of the angle *PQA*. By the proportionality of the sines of the angles to the sines of the opposite sides, in the triangle AHP

$$\sin(\text{diff. of passage}) = \sin\beta\,\frac{\sin PAQ}{\cos D};$$

likewise in the triangle AQP

$$\sin PAQ = \sin\varepsilon\,\frac{\cos L}{\cos\delta};$$

therefore

$$(\alpha) \qquad \sin(\text{diff. of passage}) = \frac{\sin\beta\,\sin\varepsilon\,\cos L}{\cos D\,\cos\delta}.$$

It is necessary, then, to put cos D for the *R* that the Escorial codex and Plato have; otherwise the rule is false. As was noted on p. 48 of the Arabic text, the words «the chord of what remains of the declination of the degree in which the star is, to complete ninety» (that is, cos *D*), which perhaps filled a line, seem to have been left out by mistake by a copyist in the archetype; the passage so corrupted («divide the product by, and take the arc of what results») an unskilled reader, who spoiled other things too, thought to restore by adding «half the diameter» after «by».

I 195

Delambre, pp. 21–22, noticed no error in al-Battānī's rule, but interpreted it falsely. In our notation (for Delambre writes *B*, *D*, ω, λ, λ′ for our *H*, *B*, ε, β, β′), these are his words: «Cherchez cos *AC* et cos ϒ*H*, vous aurez cos *AB* cos *CB* = cos *AC*, ou cos $CB = \frac{\cos AC}{\cos AB}$, puis AR = ϒ*B* = ϒ*C* – *CB*; nous ferions tang *CB* = tang *AC* cos *C*, ou

$$\frac{\sin CB}{\cos CB} = \left(\frac{\sin AC}{\cos AC}\right)\cos C = \frac{\sin AC}{\cos AB\,\cos CB}\,\sin\varepsilon\,\cos\curlyvee H,$$

et

$$\sin CB = \frac{\sin AC\,\sin\varepsilon\,\cos\curlyvee H}{\cos AB} = \frac{\sin(\beta+\beta')\,\sin\varepsilon\,\cos L}{\cos\delta}.$$

….. On voit que les Arabes connaissaient les formules sin arc perp. = sin hypoténuse . sin angle à la base; cos hypoténuse = cos base . cos côté perpendiculaire»[a] («Find cos AC and cos ϒH; you will have cos AB cos CB = cos AC, or cos CB = cos AC / cos AB, then the right ascension = ϒB = ϒC – CB; we would put tan CB = tan AC cos C, or [the formulas above]. One sees that the Arabs knew the formulas: sine of the perpendicular arc = sine of the hypotenuse × sine of the angle at the base; cosine of the hypotenuse = cosine of the base × cosine of the perpendicular side»). But in all this Delambre arbitrarily puts the sine of the *adjusted latitude* (sin [β ± β′]) of the preceding problem for the sine of the *latitude of the star* (sin β); and he thinks that what al-Battānī calls *the difference of the passage of the star through the middle of the sky* is the arc BC. Wrongly indeed: the name could not then fit the thing, nor could such an interpretation be reconciled with the rule that follows: «add (or subtract) its difference of passage through the middle of the sky to the times *of the right ascensions that correspond to the degree in which the star is*», where plainly the point I is meant, not the point C, which in no way shows the right ascension of the degree of the

[a] The print has sin ω in the second fraction of the first formula; Nallino's Addenda et emendanda (Part I, p. LXXIII) correct it to sin ε.

star's longitude. Moreover, Delambre's interpretation takes as already known the arc ♈C, which al-Battānī, on the other hand, does not teach how to compute. In my interpretation the quantity D (or the arc IH) is drawn at once from the table of the declinations of the sun (f. 177v–178r) or from the rules of chapter IV. — Schiaparelli.]

On chapter 19

a) — p. 32.

To compute the *half arc of the day* P of a star of known declination δ.

The arc $P = 90° +$ the ascensional difference if the declination is northern; $P = 90° -$ the ascensional difference if the declination is southern.

The ascensional difference the author now calls *the difference* (or *variation*) *of the quadrant of the circle of the star*; for the quadrant (90°) must be increased or diminished by this quantity to give the half arc of the day, that is, the arc of the parallel that the star runs through on the given day from its rising to its passage through the meridian. In chapter XIII, where he had considered not any star but only the points of the ecliptic, he had called the ascensional difference *the argument of the difference of the day in the quadrant.* He repeats formula (β) of chapter XIII, which serves for all stars:

$$(\alpha) \qquad \sin(\text{asc. diff.}) = R\,\frac{\sin\varphi\,\sin\delta}{\cos\varphi\,\cos\delta}.$$

For the rest, he had already taught the computation of the half arc of the day of the sun in chapter IX, whose formula (β) is general and applies to any star.

On chapter 20

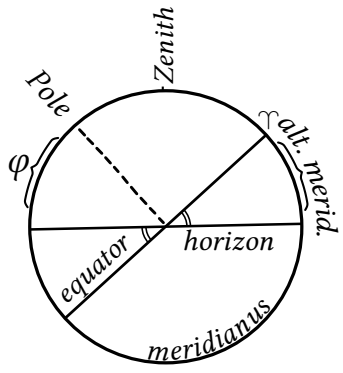
a) — p. 33.

The rules are very easy; they are set out more fully in Ptolemy, *Almag.*, II, 7 (ed. Halma, vol. I, pp. 90–108). In present-day astronomy, which does not use oblique ascensions, they are of no use.

On chapter 21

a) — pp. 33–34.

To compute *the hours of the night that have elapsed*, by any star.



First let the altitude of the star on the meridian circle be found, by adding the declination δ of the star to the [meridian] altitude of the beginning of Aries, or subtracting it, as the case requires. This rule of the author is the same as the rule set out in ch. XV for computing the meridian altitude of the sun:

(α) merid. alt. = complement of the latitude ± declin.;

for, as appears at once from the figure, the meridian altitude of the beginning of Aries (and of any other point of the equinoctial circle) is always the complement of the latitude φ of the given city, that is, the complement of the altitude of the pole of the equator above the horizon of the city[1].

This done, al-Battānī repeats the rule of chapter XVI:

$$(\beta) \qquad \operatorname{vers} t = \operatorname{vers} P - \frac{\sin h \operatorname{vers} P}{\sin(\text{merid. alt.})}.$$

By adding the arc *t* to the half arc of the day of the star, or subtracting it, according as the star is west or east of the meridian circle, the arc of the revolution of the sphere since the rising of the star will be found. Let this arc be increased (or, if the star rises from the horizon before sunset, diminished) by the difference between the oblique ascension of the star and the oblique descension (or the ascension of the nadir) of the sun; there will result the arc of the celestial sphere revolved since sunset, whose degrees are very easily turned into seasonal or equal hours.

If the computation is made by the moon, whose parallax is perceptible, not its true but its apparent coordinates must be used.

On chapter 22

a) — p. 35.

To find the *altitude h* of a star from the hours of the night that have elapsed.

The problem is the converse of the preceding one, whose formula is to be inverted thus:

$$(\alpha) \qquad \sin h = \frac{\sin(\text{merid. alt.}) \, (\operatorname{vers} P - \operatorname{vers} t)}{\operatorname{vers} P}.$$

On chapter 24

a) — pp. 35–36.

To compute the *declination* δ of a star and *the degree of the ecliptic that passes through the meridian circle at the same time as it*, when the degree of the ecliptic that rises at the same time as the star and the rising or setting amplitude of the star are known.

The first rule is

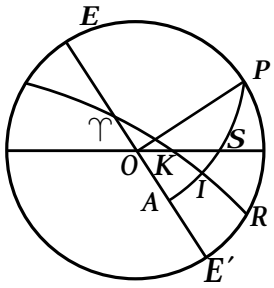
$$(\alpha) \qquad \sin \delta = \frac{\sin(\text{alt. of the beginning of Aries}) \times \sin(\text{ampl.})}{R}.$$

But the [meridian] altitude of the beginning of Aries, as the author has said in several places (see note 1 of p. 196), is the complement of the geographical latitude of the given city; so the formula

[1] See ch. VII (p. 20, ll. 1–2), XIV (p. 29, last line), XV (p. 30, l. 15).

can be written $\sin\delta = \cos\varphi \sin(\text{ampl.})/R$, which follows at once from the other formula of chapter VII, $\sin(\text{ampl.}) = R\sin\delta/\cos\varphi$. Once the declination is known, let the half arc of the day P of the star be computed by the rules of chapter XIX:

$$P = 90^\circ \pm \text{asc. diff.}, \qquad \sin(\text{asc. diff.}) = R\,\frac{\sin\varphi\,\sin\delta}{\cos\varphi\,\cos\delta}.$$



This done, al-Battānī teaches how to find the right ascension of the degree that culminates with the star, which is

obl. asc. of the degree rising with the star + half arc of the day of the star,

or obl. asc. of the degree setting with the star − half arc of the day of the star;

where the right ascension of the degree culminating with the star (which is also the right ascension of the star) is understood to be taken from the **beginning of Capricorn**, as will be clear from the figure drawn here to one who examines the matter more closely. For EE′ is the equator, S the place from which the star rises, ♈ the vernal equinox at that time, ♈R the ecliptic, K the degree of the ecliptic rising at the same time as the star, I the degree culminating with the star; so ♈A will be the right ascension, counted from the beginning of Aries, of the degree I culminating with the star, or the right ascension of the star itself; EA the half arc of the day of the star on the equator; ♈O the oblique ascension of the degree K that rises with the star, or the oblique ascension of the star S itself. Plainly ♈A = ♈O + OA; and since

$$OA = EA - 90^\circ,$$

we shall have ♈A = ♈O + EA − 90°, that is, 90° + ♈A = ♈O + EA. The tables of right ascensions will then give the degree of the ecliptic that corresponds to the right ascension so found.

On chapter 25

a) — pp. 36–37.

To compute the *longitude of a star*, given the declination of the star and the degree of the ecliptic that passes through the meridian circle at the same time as the star.

It is the converse of the problem of chapter XVIII, whose rules it would have been enough to invert. The author, on the other hand, seeks a new solution, and a most unhappy one, as will appear below. Let δ be the declination of the given star, δ′ the declination of the degree culminating with it, ε the obliquity of the ecliptic; the rules are these:

I. When δ and δ′ are on the same side:

$$\delta - \delta' \text{ (or } \delta' - \delta) = \text{adjusted distance.}$$

$$2R - \cos\varepsilon = \text{longest chord.}$$

$$2R - \cos\delta' = \text{exceeding chord.}$$

$$R\,\frac{\sin\varepsilon}{\cos\delta'} = \text{sine of the adjusted inclination.}$$

$$\frac{\sin(\text{adj. incl.}) \times \sin(\text{adj. dist.})}{\cos(\text{adj. dist.})} \times \frac{\text{exceed. chord}}{\text{longest chord}} \times$$

$$\times \frac{\cos\varepsilon}{\cos(\text{adj. incl.})} \times \frac{\sin(\text{solst. dist. of the culm. degree})}{R} =$$

$$= \text{sine of the variation of the star.}$$

It should be noted that the solstitial distance of the degree culminating with the star is counted in right ascensions.

If the degree culminating with the star belongs to the descending signs of the zodiac (from the beginning of Cancer to the end of Sagittarius), the arc of the variation of the star will be subtracted from the right ascension of the degree culminating with the star, or added to it, according as δ is northern or southern. If the culminating degree is in the ascending signs (from the beginning of Capricorn to the end of Gemini), addition will be used when δ is northern, subtraction when it is southern. Let the sum or the difference found be sought in the right ascensions; the degree of the ecliptic corresponding to it will be the longitude of the star.

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II. When δ and δ′ are on opposite sides:

$$\delta + \delta' = \text{adjusted distance.}$$

$$2R - \cos\varepsilon = \text{longest chord.}$$

$$2R - \cos\delta' = \text{exceeding chord.}$$

$$\frac{\sin\varepsilon\,\sin\delta}{\cos(\text{adj. dist.})} \times \frac{\cos\delta'}{R} = \text{sine of the adjusted inclination.}$$

$$\sin(\text{adj. incl.}) \times \frac{\sin\delta}{\cos\delta} \times \frac{\text{exceed. chord}}{\text{longest chord}} = \text{dividend.}$$

$$\cos(\text{adj. incl.}) \times \frac{\text{dividend}}{\cos\varepsilon} \times \frac{\sin(\text{solst. dist. of the culm. degree})}{R} =$$

$$= \text{sine of the variation of the star,}$$

or, putting in the value of the *dividend*,

$$\frac{\cos(\text{adj. incl.}) \times \sin(\text{adj. incl.})}{\cos\varepsilon} \times \frac{\sin\delta}{\cos\delta} \times \frac{\text{exceed. chord}}{\text{longest chord}} \times$$

$$\times \frac{\sin(\text{solst. dist. of the culm. degree})}{R} =$$

$$= \text{sine of the variation of the star.}$$

You use the variation of the star as was said above.

So far the Escorial codex, with which Plato's translation agrees, except for a few words that are plainly missing in Plato through the carelessness of the copyists. But the rules are wholly false; indeed, they can be called true monsters. Delambre, pp. 22–31, gave eight pages to drawing some sense or other out of the author's rule; but such great labour proved entirely useless. Nor does Regiomontanus seem to have been more successful before him; for, as Delambre relates of him elsewhere (p. 353), in a letter to Bianchini he «parle assez brièvement de la méthode d'Albategni

(chap. XXV), et c'est pour la trouver défectueuse. Il ajoute pourtant qu'elle ne serait admissible que pour le colure des solstices où la déclinaison et la latitude sont dans un même plan. Alors en effet $\frac{2-\cos \mathrm{AR}}{2-\cos L'} = \frac{2}{2} = 1$; cependant, à la manière dont il s'exprime, on serait tenté de croire que son manuscrit d'Albategni ne ressemblait guère aux éditions qu'on en a données»[1a] («speaks rather briefly of Albategni's method (ch. XXV), and that in order to find it defective. He adds, however, that it would be admissible only for the colure of the solstices, where the declination and the latitude are in one plane. Then indeed (2 – cos AR)/(2 – cos L′) = 2/2 = 1; yet from the way he expresses himself, one would be tempted to think that his manuscript of Albategni hardly resembled the editions that have been made of it»).

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[Those formulas are not only ridiculously difficult, but manifestly false. Delambre has already shown this in some measure; it will appear more clearly from the following considerations. 1°. The formulas in the two cases set out are different, whereas they ought to have been the same, with δ + δ′ put in the second for δ – δ′ (or δ′ – δ). Not to have understood this shows in the author of the formulas an unskilfulness that can in no way be imputed to al-Battānī, considering what he himself does on similar occasions. — 2°. The formulas appear absurd from the very fact that they cannot fit all the values of the quantities indicated in them. In the second rule the sine and the cosine of the *adjusted inclination* occur; but the sine of this inclination is defined by the formula $\frac{\sin \varepsilon \, \sin \delta}{\cos(\text{adj. dist.})} \times \frac{\cos \delta'}{R}$, which can lead to an absurdity whenever the adjusted distance (that is, δ + δ′) becomes equal to 90°; and no one will deny that this can happen. Then cos (δ + δ′) = nothing, and the sine of the adjusted inclination will be sought in vain in the tables, being infinitely great. Inept, moreover, are the names *longest chord* and *exceeding chord*, given to two quantities that always exceed *R*, and to which no possible geometrical representation in the sphere answers; so that it must be strongly doubted that they could ever be used in solving such problems. The factor determined by their ratio to each other is called «assez étrange», «rather strange», by Delambre, p. 27, and it does indeed look unusual.

It is also very strange that al-Battānī in this chapter did not invert the formulas already found in order to solve the converse problem of chapter XVIII, as he had done elsewhere, and did not see that to draw the latitude and the longitude from the right ascension and the declination is the same as to find the altitude and the azimuth from the declination and the hour angle, since in this last problem it is enough to put the distance between the zenith and the pole = ε, that is, the latitude = 90° – ε.

Lastly, the division of the computation into many small parts designated by special names seems foreign to al-Battānī's usage; he does this very rarely in the other chapters, and only when moved by evident reasons of convenience, as in ch. XXXIX, pp. 78–79[b].

When all this is weighed, I think that what teaches the computation of the *variation of the star*, or of the longitude, does not come from al-Battānī and is a most foolish interpolation. It is an easy conjecture that in the copy from which the Escorial codex and Plato's codex were afterwards derived, a leaf containing al-Battānī's true rules had perished, and that its owner, endowed more with good will than with astronomical knowledge, tried to fill the gap according to his own (alas,

[1] I have not been able to read this letter; but I think it hardly credible that Regiomontanus had before his eyes a text other than the one printed afterwards. The Nuremberg edition seems to have been made from the manuscript to which Regiomontanus himself had added notes here and there in his own hand; and in this chapter it does not differ from the Arabic text except by small gaps, which are manifest mistakes of the copyists. The differences that Delambre noted are probably different interpretations of Regiomontanus, arising from the almost incredible obscurity and ambiguity of Plato's words, or conjectures of Regiomontanus.

[a] Nallino's Addenda et emendanda (Part I, p. LXXIII) correct this passage and the note: the letter is not Regiomontanus' to Bianchini, as Delambre says, but Bianchini's to Regiomontanus, edited by M. Curtze, *Urkunden zur Geschichte der Mathematik im Mittelalter und der Renaissance*, Leipzig 1902, vol. I, pp. 240–41; the Addenda are translated with Part I's front matter.

[b] The words «as in ch. XXXIX, pp. 78–79» are added from Nallino's Addenda et emendanda (Part I, p. LXXIII).

how small!) powers. — Schiaparelli.]

b) — p. 37, ll. 21–25.

To compute the *latitude* β of a star when its declination and longitude are known.

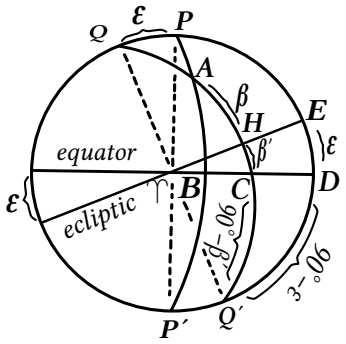
The solution of the problem depends on knowing an arc, which the author defines thus:

(α) $\quad \sin(\text{arc}) = \dfrac{\sin\delta\ \cos\beta'}{\cos\varepsilon},$

where β′ is what al-Battānī calls «the declination of the degree in which you found the star», that is, without doubt, the *second* declination of the degree of the star's longitude, which is the latitude of the degree of the equator that has the same longitude as the star. Then

(β) $\quad \beta = \beta' - \text{arc}, \qquad \text{or} \qquad \beta = \text{arc} - \beta',$

according as β′ is greater or smaller than the arc.



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The way by which the author arrived at his formulas is probably this. Let QPEDP′ be the meridian circle, ϒE the ecliptic, ϒD the equator, PAB the circle of declination passing through the star A, QACQ′ the circle of latitude of the same star A. So AB = δ, AH = β, HC = β′, and AC is the arc sought. From the triangle ABC, right-angled at B, one derives

$$\sin AC = R\,\frac{\sin AB}{\sin(\text{angle } C)} = R\,\frac{\sin\delta}{\sin(\text{angle } C)}.$$

The triangle CDQ′, right-angled at D, gives

$$\sin(\text{angle } C) = R\,\frac{\sin DQ'}{\sin CQ'} = R\,\frac{\cos\varepsilon}{\cos\beta'},$$

whence

(α) $\quad \sin AC = \dfrac{R\sin\delta}{R\,\frac{\cos\varepsilon}{\cos\beta'}} = \dfrac{\sin\delta\ \cos\beta'}{\cos\varepsilon}.$

But neither here nor before, in chapter XVIII (see p. 193), does the author teach how the necessary quantity β′ is computed. It can be drawn from the tables: let the arc of the ecliptic 90° + ϒH be imagined to be an arc of the equator, that is, the right ascension[1] of the point C; the tables will give the corresponding arc (90° + ϒC) of the fictitious ecliptic. When the 90° are thrown away and the tables of the declination of the sun (f. 177v–178r) are entered with the arc ϒC, the fictitious declination β′ will be found.

[1] Those ninety degrees must be added to the arc ϒH because the author counts the right ascensions from the colure of the winter solstice (the beginning of Capricorn), not from the beginning ϒ of Aries. See p. 164.

On chapter 26

a) — p. 38.

To compute *the distance between two stars*, on the great circle that passes through both.

The author puts a geometrical proof before his rules, which are set out on pp. 39–40, as he had already done in chapter XII; and to the proof he also adds an example in numbers, as he does later in chapter XXX, pp. 59–61. Already in note *b* on chapter XII (p. 186) I remarked that it is a probable conjecture that al-Battānī was the first to establish these formulas, and for that reason thought it necessary to prove their truth geometrically.

Regiomontanus strongly disapproves of the author's construction; «verum esset», he says, «si arcus GE esset circuli magni, et ipse primo bene inventus. Sed dico tibi ille processus intricatus est, et modicae reputationis. Utitur enim lineis curvis tanquam rectis, quod tametsi Ptolemaeum fecisse constet, ipse tamen arcus breves loco linearum rectarum accepit, noster vero indifferenter quantoscumque» («it would be true if the arc GE were of a great circle, and had itself first been rightly found. But I tell you that this procedure is intricate and of little worth. For he uses curved lines as if they were straight; and though it is certain that Ptolemy did this, Ptolemy took short arcs in place of straight lines, while our author takes arcs of any size indifferently»). He then proves that the quadrilateral of the four chords AB, BG, GE, EA is symmetric, and can therefore be inscribed in a circle; lastly he proves that chord $GE = R\,\frac{\text{chord of the arc } AB}{\sin(\text{arc } FG)}$, where GE is an arc of a smaller circle.

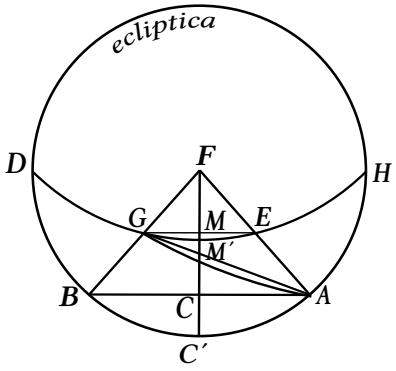
Delambre, pp. 33–34, judges no less harshly that the author's solution is «décidément mauvaise, sans aucune ambiguité» («decidedly bad, without any ambiguity»); he solves the numerical example with logarithmic tables and finds cos AG = cos 64° 20′ 27″; «Albategni la trouve de 54° 19′, trop faible de 10° 1′ 27″» («Albategni finds it to be 54° 19′, too small by 10° 1′ 27″»); and he concludes: «Albategni, en terminant ce chapitre, dit qu'il sert, comme le précédent, pour les nativités..... Pour la gloire d'Albategni, il faut croire ces deux derniers chapitres interpolés par quelque astrologue ignorant et charlatan» («Albategni, in ending this chapter, says that it serves, like the preceding one, for nativities..... For Albategni's glory we must believe these two last chapters to have been interpolated by some ignorant astrologer and charlatan»).

But had Delambre considered the numbers of al-Battānī's computation attentively, he would have found that 54° 19′, which Plato has, is a slip of the pen or of the typesetter for 64° 19′; so, neglecting the seconds, which the author leaves out, the error is only one minute, and it plainly comes from the difference in interpolating the tables of sines. Likewise, if the rules of pp. 39–40 are written in the present trigonometric manner, one finds the fundamental rules by which we solve spherical triangles. It is impossible, then, that the geometrical constructions from which a true computation and sure rules are derived should be false.

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From the numerical example it is very clear that by the name «line *x*» the author denotes not the arc *x* or the chord of the arc *x*, but the straight line that lies under the arc *x*, that is, the sine of that arc. The same expression occurs several times in chapter XXVIII. Nor should it be forgotten that the figure of p. 38, which Plato's translation shows with curved lines, is drawn with straight lines in the Escorial codex, as I have had it printed. Hence it becomes certain that al-Battānī had in mind the orthographic projection of the sphere, which he had already used elsewhere (as I showed in the notes on chapters XI, XII and XVI), and derived the rules from it. The ambiguities would easily be avoided by drawing al-Battānī's figure a little differently, so that the point M of the chord appears at once distinct from M of the arc, and C of the chord from C of the arc; Schiaparelli proposed to me the diagram printed here. Let the sphere be projected orthogonally onto the plane of the ecliptic; AHDB will be the ecliptic, whose pole is F; FG, FE, FB, FA will at one and the same time be the complements of the latitude and the sines of these complements.

Through the two points G, E, of the same latitude, let a *great circle*[1] DH be described, whose arc GE will be the distance between G and E; let the straight lines GE, BA be drawn, bisected by the line FC; lastly let the arc AG of a *great circle* and its chord be marked. Let the midpoints of the chords be distinguished by the letters M and C, the midpoints of the arcs by the letters M′ and C′.



This done, the proportion GM : BC = FG : FB[2] is manifest in the plane triangle FBC; that is, if L and L′ are called the longitudes of the stars and β the latitude,

$$\sin\frac{GE}{2} : \sin\frac{L-L'}{2} = \sin(90^\circ - \beta) : \sin 90^\circ.$$

But $\sin 90^\circ = R$, and the arc GE/2 is half the distance sought; therefore

$$\sin\frac{GE}{2} = \sin\frac{L-L'}{2} \times \frac{\cos\beta}{R},$$

which is al-Battānī's rule, stated on p. 40, ll. 1–2 (and 39, ll. 26–28).

There are also some corruptions or interpolations in the text, which I have pointed out in their places in my translation; they cannot be ascribed to al-Battānī himself, since they are most foolish, wholly superfluous and of no consequence in the author's numerical computation or in the rules of pp. 39–40. The hand of an unskilled reader appears, whose traces we find plainly elsewhere too.

According to the end of the chapter, these rules are most useful for computing the profections in nativities; but that they serve non-astrological questions too appears from al-Battānī's own words where he teaches how to compute the distance of the moon from the sun in a certain and absolute way (ch. XLI, p. 87, ll. 11–13).

b) — p. 39, ll. 1–12.

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Once the distance of the point G from E on a great circle is known by the formula

$$\sin\tfrac{1}{2}\,\text{distance} = \sin\frac{L-L'}{2} \times \frac{\cos\beta}{R},$$

[1]Not a smaller circle, that is, a circle parallel to the great circle BA, as Regiomontanus and Delambre interpreted it.

[2]The codex and Plato add «= FM : FC», which, though true, appears entirely useless, for the knowledge of these lines is neither required nor taught. It must therefore be regarded as an interpolation, which arose because the writer of the archetype from which the Escorial codex and Plato's codex derive wrongly put, in lines 23, 24, 30 and 31, FM ف م for FG ف ط, and FC ف ج for FB ف ب. The corruption pointed out on p. 38, note 6, also comes from this.

the author seeks the distance of the point G from A, also on a great circle. So he observes that in the *rectilinear* quadrilateral GEBA (which is symmetric, and so can be inscribed in a circle) Ptolemy's theorem, stated at the beginning of the chapter (p. 37, ll. 38–40), holds; so that one has

$$\overline{GA}^2 = \overline{GB}^2 + GE \times BA,$$

where the sides GB, BA are chords of known arcs, and GE has just been determined[a]. So the diagonal GA becomes known, which is the chord of the arc GA of a *great* circle, that is, of the distance sought. The solution can rightly be called elegant.

c) — pp. 39–40.

He now states the general rules for computing the *distances between two stars* (in arcs of great circles).

In spherical quadrilaterals, of the two sides whose chords are parallel he calls the one nearer the ecliptic (or the ecliptic itself) *the first side*, the one nearer the pole *the second side*. The other two sides, which are equal and whose chords are not parallel, he calls *the third* and *the fourth side*.

He distinguishes four cases:

I. — When one of the stars is on the ecliptic and the other has the latitude β. — The solution is the one I have already set out in note *b*, after al-Battānī himself; he puts $L - L'$ (that is, the difference of the longitudes) = the first side, $2 \arc\big[\sin\big(\frac{L-L'}{2}\big) \times \frac{\cos\beta}{R}\big]$ = the second side (that is, the arc GE, on which see note *a*), and lastly the chord of β = the third or the fourth side. So he states the rule:

(α) crd distance = $\sqrt{\text{crd first side} \times \text{crd second side} + \text{square of the third side}}$.

From this formula of al-Battānī it is not difficult to derive our cos dist. = cos β cos (L – L′). Since chord $x = 2 \sin \frac{x}{2}$, if we put sines for chords and square both members of equation (α), there results[b]

$$4 \sin^2 \frac{\text{distance}}{2} = 2 \sin\Big(\frac{L-L'}{2}\Big) 2 \sin\Big(\frac{L-L'}{2}\Big) \frac{\cos\beta}{R} + 4 \sin^2 \frac{\beta}{2} =$$
$$= 4 \sin^2\Big(\frac{L-L'}{2}\Big) \frac{\cos\beta}{R} + 4 \sin^2 \frac{\beta}{2}.$$

But $\sin^2 \frac{x}{2} = \frac{R^2 - R\cos x}{2}$; so, dividing everything by 4,

$$\frac{R^2 - R\cos(\text{dist.})}{2} = \frac{R^2 - R\cos(L-L')}{2} \times \frac{\cos\beta}{R} + \frac{R^2 - R\cos\beta}{2},$$

whence, multiplying by 2 and dividing by *R*,

$$R - \cos(\text{dist.}) = \cos\beta - \frac{\cos\beta\ \cos(L-L')}{R} + R - \cos\beta,$$

and lastly $\cos(\text{distance}) = \frac{\cos\beta\ \cos(L-L')}{R}$.

II. — When the stars have the same latitude. — The chord of the distance is then the second side itself; that is, as he taught before and as is derived from the geometrical construction of p. 38 (on which see note *a*):

(β) chord of the distance = $2 \sin\Big(\frac{L-L'}{2}\Big) \frac{\cos\beta}{R}$,

or $\sin \frac{\text{dist.}}{2} = \sin\big(\frac{L-L'}{2}\big) \frac{\cos\beta}{R}$.

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[a] The print has 2 GB². Ptolemy's theorem in the symmetric quadrilateral (GA × EB = GE × BA + GB × EA, with GA = EB and GB = EA) gives GB², as rule (α) below has it; GB² is set here.

[b] The print has cos β / 2R three times in these formulas; Nallino's Addenda et emendanda (Part I, p. LXXIII) correct it to cos β / R.

III. — When the latitudes β and β′ of the stars are different, but on the same side of the ecliptic. Al-Battānī then sets

$$\text{first side} = L - L', \qquad \text{second side} = \text{double arc}\Big[\sin\Big(\frac{L-L'}{2}\Big)\frac{\cos(\beta-\beta')}{R}\Big],$$

$$\text{third side} = \text{fourth side} = \beta - \beta',$$

$$\text{first side of the longitude} = \text{double arc}\Big[\sin\Big(\frac{L-L'}{2}\Big)\frac{\cos\beta}{R}\Big],$$

$$\text{second side of the longitude} = \text{double arc}\Big[\sin\Big(\frac{L-L'}{2}\Big)\frac{\cos\beta'}{R}\Big].$$

Al-Battānī's rule is:

$$(\gamma) \qquad \text{chord of the distance} = \sqrt{\text{crd I side of long.} \times \text{crd II side of long.} + \text{crd}^2 \text{ III side}},$$

which is plainly proved geometrically in the same way as rule (α) (see note *b*). The fundamental formula of our spherical trigonometry is contained in it implicitly. For by squaring both members of the equation and bringing in sines one finds

$$4\sin^2\Big(\frac{\text{distance}}{2}\Big) = 2\sin\Big(\frac{L-L'}{2}\Big)\frac{\cos\beta}{R} \times 2\sin\Big(\frac{L-L'}{2}\Big)\frac{\cos\beta'}{R} + 4\sin^2\Big(\frac{\beta-\beta'}{2}\Big).$$

Dividing everything by 4, and considering that $\sin^2\frac{x}{2} = \frac{R^2 - R\cos x}{2}$,

$$\frac{R^2 - R\cos(\text{distance})}{2} = \sin^2\Big(\frac{L-L'}{2}\Big)\frac{\cos\beta\,\cos\beta'}{R^2} + \frac{R^2 - R\cos(\beta-\beta')}{2} =$$

$$= \frac{R^2 - R\cos(L-L')}{2} \times \frac{\cos\beta\,\cos\beta'}{R^2} + \frac{R^2 - R\cos(\beta-\beta')}{2}.$$

Let everything be multiplied by 2 and divided by *R*; there will result

$$R - \cos(\text{distance}) = \frac{\cos\beta\,\cos\beta'}{R} - \frac{\cos\beta\,\cos\beta'\,\cos(L-L')}{R^2} + R - \cos(\beta-\beta').$$

But $\cos(\beta-\beta') = \frac{\sin\beta\,\sin\beta' + \cos\beta\,\cos\beta'}{R}$; so, taking away the equal terms of opposite sign,

$$\cos(\text{distance}) = \frac{\sin\beta\,\sin\beta'}{R} + \frac{\cos\beta\,\cos\beta'\,\cos(L-L')}{R^2},$$

which we, taking $R = 1$, now write

$$\cos(\text{dist.}) = \sin\beta\,\sin\beta' + \cos\beta\,\cos\beta'\,\cos(L-L').$$

If for β and β′ their complements Δ and Δ′ are taken, that is, the distances from the pole of the ecliptic, the formula becomes

$$\cos(\text{dist.}) = \cos\Delta\,\cos\Delta' + \sin\Delta\,\sin\Delta'\,\cos(L-L'),$$

which is derived at once from the consideration of the spherical triangle FDE, in which the angle DFE = L – L′, and which is always used today in this problem.

IV. — When the latitudes β and β′ are different and on two different sides. It differs from case III only in that β + β′ must be taken for the β – β′ of the third side. Then $\cos(\beta+\beta') = \frac{\cos\beta\,\cos\beta' - \sin\beta\,\sin\beta'}{R}$, so that the formula becomes

$$\cos(\text{dist.}) = -\frac{\sin\beta\,\sin\beta'}{R} + \frac{\cos\beta\,\cos\beta'\,\cos(L-L')}{R^2}.$$

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Al-Battānī's solution is most elegant, and all the more worth mentioning because, after S. Günther, *Studien zur Geschichte der mathematischen und physikalischen Geographie*, Halle 1877–79, part V, pp. 306–308, Georgius Amirucius is said to have been the first to solve this problem without dividing the figure into two right-angled spherical triangles[1]. Amirucius considers the geographical distance of two places on the earth; he uses chords, but follows a way different from al-Battānī's. Our author's method, however, so much older, seems to me also easier and more elegant than that of Amirucius. So this chapter XXVI, which Regiomontanus and Delambre attacked so bitterly, appears to be one of the chief parts of the work, once some mistakes of the copyists are cleared away and certain perhaps more obscure expressions are rightly understood.

On chapter 27

a) — p. 40, ll. 28–29.

Only one known author besides al-Battānī mentions such a *sidereal year of the Egyptians and the Babylonians*; for Ptolemy himself, whom al-Battānī cites afterwards, does not know it, as I shall say in note *b*.

The reading of the Arabic text, 365 1/4 + 1/120 (that is, $365^d\ 6^h\ 12^m$), differs from the printed translation of Plato, «ex 365 diebus et quarta ultraque parte ex 131 diei partibus» («of 365 days and a quarter, and beyond that one part of 131 parts of a day»)[a], which can only mean 365 1/4 + 1/131 (that is, $365^d\ 6^h\ 10^m\ 59.6^s$). So too a glossator (perhaps Regiomontanus) understood it, whose words in the margin of the Nuremberg edition are these: «ex 365 diebus, 15 minut. et 27 secund. et 30 ter. fere» («of 365 days, 15 minutes, 27 seconds and about 30 thirds»), that is, $365^d\ 6^h\ 11^m$[b]; and the Bologna edition took this marginal gloss into the text in place of Plato's true words. Nor, lastly, should it be passed over in silence that the Vatican codex of Plato has «ex 365 dierum [*sic*] et quarta unaque parte ex 130 diei partibus» («of 365 days and a quarter and one part of 130 parts of a day»), that is, $365^d\ 6^h\ 11^m\ 4.6^s$.

It is not easy to judge which is the genuine reading. In chapter LII (p. 127) the astrologers (Greek?) who supposed a trepidation of the fixed stars are credited with a (tropical?) year «exceeding 365 1/4 days by a fifth of an hour, or about a fifth», that is, 365 days $6^h\ 12^m$, or about this length. It is not impossible that those astrologers are the same as the ancient Egyptians and Babylonians of this chapter; but even if this were certain, we should have no criterion for settling the true reading. For «by a fifth of an hour» agrees exactly with the 12 minutes of the Escorial codex, while «or about a fifth» can agree not badly with Plato's $10^m\ 59.6^s$ (or rather $11^m\ 4.6^s$, as in the Vatican codex).

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The reading of the Arabic codex, however, seems more probable to me. Not only does the comparison I shall make below yield, among the Indians and the Persians, a year of 365 days, 6 hours, 12 minutes or a few seconds *more* (not less); but in a late Arab writer I find a passage that agrees excellently with the Escorial codex. Saḥnūn ibn ʿUthmān al-Maydawī al-Wānsharīshī, who wrote a commentary on the astronomical poem that ʿAbd al-Raḥmān al-Akhḍarī finished in the month

[1] R. Wolf, *Handbuch der Astronomie, ihrer Geschichte und Litteratur*, Zürich 1890–93, vol. II, p. 208, § 429, note *a*; Cantor, vol. II, pp. 416–417; von Braunmühl, vol. I, p. 134. — Amirucius, a Greek by nation, born at Trebizond, embraced the Islamic religion after 1461 and held a public office at Constantinople among the Turks; he died about 1490.

[a] Nallino's Addenda et emendanda (Part I, p. LXXIII) show that Plato wrote 1/130, not 1/131 as the editions have: the Vatican, Paris and Venice manuscripts and the book called *Almagesti parvum* agree; Plato seems to have found in his Arabic copy قل, 130, for قك, 120.

[b] The print has 360^d for 365^d.

of Dhū l-Qaʿda 939 H (June 1533), says the following: «What [al-Akhḍarī] relates of the non-Arab year[1] [that it comprises 365 1/4 days] is what is commonly set out. It is also said [by others that this year comprises] a part of 120 parts of a day; such is the opinion of the people of Egypt and Babylon (*qawl ahl miṣr wa bābil*), who used for this the conjunction of the sun with some fixed or wandering star until the time when the sun returned to it. This opinion is manifestly false in the highest degree. ‹For the truth is that the time of the year [is counted] from the conjunction of the sun with some fixed point of the sphere until its return to the same point, namely one of the equinoctial or solstitial points.› So relates al-Jādirī»[2]. If the whole passage of al-Battānī, p. 40, ll. 28–36, is compared with this, it will appear either that al-Battānī and the late al-Jādirī drew from the same source, or that al-Battānī himself is al-Jādirī's first source.

But let us now seek the origin of such a sidereal year. Relying on al-Battānī's authority and following the reading of the Bologna edition, Fréret (*Mémoires de l'Académie des Inscriptions*, old series, vol. XVI, pp. 214–215) ascribes the sidereal year of $365^d\ 6^h\ 11^m$ (which differs little from the value accepted today, $365.256374^d = 365^d\ 6^h\ 9^m\ 10.75^s$) to the Egyptians and the Chaldaeans; Montucla (*Histoire des Mathématiques*, 2nd ed., Paris 1799–1802, vol. I, p. 59), Bailly (*Histoire de l'Astronomie ancienne*, Paris 1775, pp. 149, 165–166, 392, 403), Lepsius (*Die Chronologie der Aegypter*, Berlin 1849, vol. I, p. 197) and others, the most recent of them P. Tannery (*Recherches sur l'histoire de l'astronomie ancienne*, Paris 1893, p. 306), to the Chaldaeans only. This sidereal year served well to prove that the Chaldaeans knew the annual precession of the equinoctial points long before Hipparchus[3].

Letronne, in his *Nouvelles recherches sur le calendrier égyptien*, memoir II, § 4, no. 2, pp. 103 ff. (Mémoires de l'Académie des Inscriptions, vol. XXIV, 2nd part, 1864), though he keeps the 11 minutes of the Bologna edition, thinks that al-Battānī's Egyptians and Babylonians are the very astrologers of chapter LII. Later, on p. 110, he writes: «Maintenant il est clair que la rétrogradation de 1 degré en 80 ans[4] donne un arc annuel de $40''$, parcouru en $16^m\ 19^s$ de temps. Si nous ajoutons 16^m à l'année tropique qu'Hipparque et Ptolémée ont conclu de leurs observations = 365 jours 5 heures 55^m, nous avons la fameuse année sidérale d'Albatégni, de 365 jours 6 heures 11^m. Il est à remarquer que c'est la même opération qui avait donné à Regiomontanus et à Purbach leur année sidérale si exacte de 365 jours, 6 heures $9^m\ 12^s$. Ils l'ont obtenue, disent-ils, dans l'hypothèse de la trépidation de l'astronome arabe Thebit….. Les anciens Égyptiens, comme les Chaldéens, sont tout à fait désintéressés dans la question. Nous n'avons ici qu'une rêverie astrologique postérieure à l'un des deux astronomes grecs, sinon à tous les deux» («Now it is clear that the retrogradation of 1 degree in 80 years gives a yearly arc of $40''$, run through in $16^m\ 19^s$ of time. If we add 16^m to the tropical year that Hipparchus and Ptolemy concluded from their observations, 365 days 5 hours 55^m, we have the famous sidereal year of Albatégni, of 365 days 6 hours 11^m. It should be noted that it is the same operation that had given Regiomontanus and Purbach their very exact sidereal year of 365 days 6 hours $9^m\ 12^s$. They obtained it, they say, on the hypothesis of the trepidation of the Arab astronomer Thebit….. The ancient Egyptians, like

[1] *Sana ʿajamiyya*, that is, the barbarian year, in the same sense in which the Greeks called all foreigners barbarians. From what follows in the text of the poem it is clear that the Roman year is meant.

[2] Saḥnūn al-Wānsharīshī, *Mufīd al-muḥtāj ʿalā l-manẓūma al-musammāh bi-l-Sirāj li-l-Akhḍarī*, Cairo 1314 H (1896–97), p. 33. — ʿAbd al-Raḥmān al-Jādirī (the name is also written الكادري) wrote an astronomical poem in the year 794 H (1391–92); Suter, *Die Mathematiker und Astronomen der Araber*, p. 172. — The words enclosed in ‹ › are also quoted, as taken from al-Jādirī, by Muḥammad ibn Saʿīd al-Marghītī al-Sūsī (who died at the end of the year 1089 H, a little before the beginning of 1679), *al-Mumtaʿ fī sharḥ al-maqnaʿ*, Fez 1313 H (1895–96), p. 3 of the third part. [*Translator:* The print has al-Mumtiʿ and al-muqniʿ; Nallino's Addenda et emendanda (Part I, p. LXXIII) correct them to al-Mumtaʿ and al-maqnaʿ.]

[3] For the Chaldaeans had a civil year of $365^d\ 6^h$ (365 1/4), which some ancients say was tropical. If, then, they had also known a sidereal year 11 or 12 minutes longer than that, the knowledge of precession could rightly have been ascribed to them. — But that precession was unknown before Hipparchus has been sufficiently proved by G. Schiaparelli, *Le sfere omocentriche di Eudosso, di Callippo e di Aristotele*, Milan 1875, pp. 20–23 (= Pubblicazioni del R. Osservatorio di Brera, no. IX), and by Th.-H. Martin in the dissertation that I shall cite below.

[4] It is the annual precession according to the astrologers of whom al-Battānī speaks in chapter LII.

the Chaldaeans, have nothing at all to do with the question. We have here only an astrological fancy later than one of the two Greek astronomers, if not than both»).

Th.-H. Martin thinks no differently, *Mémoire sur cette question: La précession des équinoxes a-t-elle été connue des Égyptiens ou de quelque autre peuple avant Hipparque?* (Mémoires présentés par divers savants à l'Académie des Inscriptions, 1st series: sujets divers, vol. VIII, 1st part, 1869); for in ch. II, § 3, pp. 369–372, he holds all that Letronne wrote to be true, and on p. 372 he ascribes that sidereal year to Alexandrian astrologers who lived between Hipparchus and Ptolemy and professed the trepidation of the fixed stars.

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Letronne's whole reasoning, however, depends on an error of computation that Martin did not notice[1]. For 1° (or 3600″) divided by 80 years gives a yearly arc of 45″, not 40″. The daily motion of the sun in Ptolemy is 0° 59′ 8″, or 3548″; divided by 45″, this shows that the arc of 45″ is run through in the 78.844th part of a day. Let the minutes of time that a day contains, that is, 1440^m, be divided by 78.84; there will result 18.264^m, or about $18^m\ 16^s$, not $16^m\ 19^s$ as Letronne has it. Let this space of time be added to Ptolemy's tropical year; one will find $365^d\ 5^h\ 55^m + 18^m\ 16^s = 365^d\ 6^h\ 13^m\ 16^s$, which differs from the sidereal year mentioned in al-Battānī (whichever reading is accepted as good, the 12^m of the codex or Plato's about 11^m). So I think that all that Letronne and Martin write about the origin of such a year must be rejected. It remains to inquire whether the length of the solar year that al-Battānī and al-Wānsharīshī (after al-Jādirī) ascribe to the Egyptians and the Babylonians (al-Battānī also to the astrologers who followed the theory of trepidation) is mentioned elsewhere too.

The Indian astronomers know a sidereal year that differs from the one just mentioned only by some added seconds of time. For they have cycles, which they call *yuga*, comprising some thousands of years, to which they assign a certain number of days; by dividing the number of days by the years of one *yuga*[a], the length of the year is easily derived. I take the table of the Indian years from J. Burgess, *Notes on Hindu Astronomy and the history of our knowledge of it* (in the Journal of the R. Asiatic Society of Great Britain, new series, vol. XXV, 1893, p. 722); I give the age of the books after G. Thibaut, *Astronomie, Astrologie und Mathematik* [*der Inder*], Strassburg 1899 (= Grundriss der indo-arischen Philologie, begründet von G. Bühler, fortgesetzt von F. Kielhorn, vol. III, part 9):

Name of the book	*Age after Christ*	*Length of the year*
Sūrya-Siddhānta[2]	before the 5th or 6th century	$365^d\ 6^h\ 12^m\ 36.56^s$
Parāśara-Siddhānta	before the 5th or 6th century	365 6 12 31.50
Pauliśa-Siddhānta	before the 5th or 6th century	365 6 12 36
Romaka-Siddhānta	before the 5th or 6th century	365 5 55 12
Prior Ārya-Siddhānta	499	365 6 12 30
Brahma-Siddhānta	628	365 6 12 9
Posterior Ārya-Siddhānta	650–1150	365 6 12 36.84
Siddhānta-Śiromaṇi	1150	365 6 12 9

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Except, then, for the *Romaka-Siddhānta*, which gives Ptolemy's tropical year, all the Indian (sidereal) years, if the seconds are neglected, have the same length that the Escorial codex mentions in this chapter, and the Escorial codex and Plato in chapter LII. Nor should it be forgotten that

[1] Martin, however, knew al-Battānī only from what Letronne reports; both, as we shall see in the notes on chapter LII, ascribe to al-Battānī certain things that he never said.

[2] In the older recension of the book the year seems to have been exactly $365^d\ 6^h\ 12^m\ 36^s$, as appears from *The Pañchasiddhāntikā, the astronomical work of Varāha Mihira, edited with an original commentary in sanskrit and an english translation and introduction by G. Thibaut and Sudhākara Dvivedī*, Benares 1889, p. 5 of the translation, ch. I, stanza 14.

[a] The Latin says that the years of one yuga are divided by the number of days.

Varāhamihira, who died in 587, gave in his *Pañcasiddhāntikā* the computation of the mean and the true longitude of the sun according to the old recension of the book *Pauliśa-Siddhānta*, from which a year of exactly $365^d\ 6^h\ 12^m$ is derived[1].

The Arab al-Bīrūnī, *India*, pp. 177, 178, 185, 222 of the text (= I, 350, 353, 368 and II, 39 of the translation), mentions the Indian year of $365^d\ 6^h\ 12^m\ 9^s$ on the authority of the book entitled *Brahma-Siddhānta*. It is known, moreover, that this *Brahma-Siddhānta*, written by Brahmagupta, was translated into Arabic, or rather reworked, by order of the caliph al-Manṣūr (754–775), and became known among the Arabs under the name *al-Sindhind* (a corruption of the Indian name); of it al-Masʿūdī, *Tanbīh*, p. 221, relates «that the solar year, according to the Indian *al-Sindhind*, consists of 365 days, a quarter of a day and a fifth of an hour, with 1/400 of an hour added», that is, $365^d\ 6^h\ 12^m\ 9^s$. The same appears from an Arab writer (in al-Bīrūnī, *Chron.*, p. 51 of the text, 60–61 of the translation) who added 729^s to the Roman year.

The year of $365^d\ 6^h\ 12^m$ seems to have been known to the ancient Persians too. Almost all the Arab writers who discuss the old chronology of the Persians relate that the Persians had a year of only 365 days, but intercalated one month of thirty days every 120th year, so that those 120 years were equal to 120 years of 365 1/4 days, since 120 : 4 = 30; and this until the Arab rule was set up in Persia (about 650); afterwards, however, they never used the intercalation. The scholars of our age have interpreted this Arab tradition in different ways; compare L. Ideler, *Handbuch der mathematischen und technischen Chronologie*, Berlin 1825–26, vol. II, pp. 540–550; Th. Benfey and M. A. Stern, *Ueber die Monatsnamen einiger alter Völker, insbesondere der Perser, Cappadocier, Juden und Syrer*, Berlin 1836, pp. 140–156; von Gutschmid, *Ueber das iranische Jahr* (in Berichte über die Verhandlungen der k. sächsischen Gesellschaft der Wissenschaften zu Leipzig, phil.-hist. Cl., XIV, 1862, pp. 1–9); lastly, as Schiaparelli points out to me, E. W. West, *Pahlavi texts translated*, vol. V, Oxford 1897 (= Sacred Books of the East, vol. XLVII), pp. XLIV–XLVI of the introduction, who, relying on solid arguments, puts the origin of this Persian calendar in the time of Darius Hystaspis (about 505 BC).

But another Persian year too, which the scholars just cited seem not to know, is mentioned by al-Bīrūnī, *Chron.*, p. 11 of the text, 12–13 of the translation: «The Persians reckoned the year as 365 days, without fractions, until, after 120 years had elapsed, a whole month [of 30 days] had been collected from the quarters of a day, and a whole day from the fifths of an hour that, with them, follow the quarter of a day. So they added a whole month to the year after every 116 years were completed. This usage of the Persians was followed by the ancient inhabitants of [the lands of] Khwārizm[2] and of Sogdiana, and by all who professed the Persian religion»[3]. It is, then, the year of 365 1/4 days + 1/5 hour (= $365^d\ 6^h\ 12^m$), which al-Battānī gives in the same numerical form in ch. LII.

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The same al-Bīrūnī, *Chron.*, p. 52 of the text, 61–62 of the translation, relates the following: «The

[1] *Pañchasiddhāntikā*, p. 14 of the translation, ch. III, stanzas 1–3. Another passage of this book (p. 5, ch. I, stanzas 11–13), where the year of the same *Pauliśa-Siddhānta* is discussed, is so obscure that the English translator has not dared to give any translation.

[2] It is the region south of the Aral lake, whose chief city is now called Khiva. Al-Bīrūnī was born in it, and he readily mentions the institutions of his homeland.

[3] He adds afterwards that this did not happen under the Pēshdādian dynasty, the oldest of all the Persian dynasties. This whole passage is read in the same words, but without al-Bīrūnī's name, in the description of Egypt written by al-Maqrīzī (d. 27 Ramaḍān 845 = 9 February 1442): al-Maqrīzī, *Kitāb al-mawāʿiẓ wa-l-iʿtibār fī dhikr al-khiṭaṭ wa-l-āthār*, Būlāq 1270 H, vol. I, p. 261. — Elsewhere al-Bīrūnī (*Chron.*, pp. 43–44 of the text, 53–55 of the translation, and in many places) mentions the simple intercalation of a month after 120 years, which depends on the year of $365^d\ 6^h$.

like[1] was set out by Ḥamza ibn al-Ḥasan al-Iṣfahānī[2] in his epistle on [the day of] *al-nayrūz*[3], where he defends the Persians as to their use of taking a solar year of 365 days + 6 hours + 1/5 + 1/400 of an hour [= $365^d\ 6^h\ 12^m\ 9^s$], while the Romans (*al-Rūm*), on the other hand, leave out the fraction that follows the six hours. He relies on the argument that the astronomer Muḥammad ibn Mūsā ibn Shākir[4] explained it and diligently investigated it in a book of his on the solar year, giving geometrical proofs and showing the errors of certain ancients. — But we have examined the observations of Muḥammad ibn Mūsā and of his brother Aḥmad, [and have found] that they show only that these fractions are less than six hours. The book that [Ḥamza] cites is the one ascribed to Thābit ibn Qurra; for Thābit was a protégé of this family, and collected what is read in this book from it and from those who polished its sciences [its books] for it[5]. The purpose of [this little book] is to prove the inequality and the difference of the solar years, since the apogee [of the sun] is movable [because of the precession of the equinoxes]; so that, to find the mean course of the sun in them, he would have needed equal revolutions and motions proportional to their times; but the only revolutions that appeared equal to him were those in the eccentric, from a given point of the eccentric to the return to the same point. This revolution, as Ḥamza says, contains fractions [of a day] exceeding six hours; but it is not called the solar year, since the year of the sun, as we have already determined, is the one in which the state, or the conditions, of nature for producing life and death [in all things] become again as they were before».

From what we have said so far it is clear that before the rule of the Arabs the Indians and the Persians knew, and indeed used, the sidereal year of $365^d\ 6^h\ 12^m\ 9^s$, or of only $365^d\ 6^h\ 12^m$, which is the value that al-Battānī ascribes in this chapter (when the reading of the codex and of al-Wānsharīshī, or rather of al-Jādirī, is accepted) to the Egyptians and the Babylonians, and in chapter LII to certain astrologers who flourished before Ptolemy.

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If a conjecture may be put forward in such obscure questions, I suspect that al-Battānī's information is true. Those Babylonians (from whom the astrologers of chapter LII probably do not differ) would then be Chaldaeans, contemporaries of Hipparchus or a little later, who could have derived such a sidereal year from observations carefully made over so many centuries. The most recent investigations show well enough that the astronomical and astrological science of the Chaldaeans flourished in the Hellenistic period too and did not neglect the old observations; so that, besides the Greek astronomy handed down to us by Ptolemy, another existed at the same time in Mesopotamia, fragments of which are preserved on clay tablets with cuneiform inscriptions[6]. A Babylonian, indeed, as Schiaparelli points out to me, was that Seleucus who, on the evidence of Pseudo-Plutarch (*De placitis philosophorum*, II, 1), dared after Aristarchus to teach the heliocentric system and the infinity of the world[7]. Much that is read in the Indian astronomers and does not agree with the doctrines of the Almagest, above all observational data, seems to

[1] He had spoken before of the Indian year.

[2] He flourished in the city of Baghdad in the first half of the fourth century H (the tenth century); he belonged to the party of the men called *shuʿūbiyya*, who tried to show in their historical and philological writings that the Persian race far surpassed the Arab stock in antiquity, nobility, language, manners and institutions. See I. Goldziher, *Muhammedanische Studien*, Halle a. S. 1889–90, vol. I, pp. 209–212.

[3] The Arabic form of the Persian name *nawrōz*, «new day», by which the Persians called the first day of the year, at the vernal equinox.

[4] He died in the month of Rabīʿ I 259 H (January 873).

[5] The text of al-Bīrūnī seems to me slightly corrupt, and not well interpreted by Sachau in his translation; I read: اذ كان صنيعة هولاء القوم ومِن بينهم وممن كان يُهذِّب لهم علومَهم جَمَلَ ما فى هذا الكتاب, with وممن for ومن and جَمَلَ for وجَمَلَ. — «The family» is the family of the sons of Mūsā ibn Shākir, of whom Muḥammad and Aḥmad are mentioned above; most skilled in the mathematical sciences and authors of many books, they spent the greatest diligence and liberality on furthering the disciplines. They greatly favoured Thābit ibn Qurra, who died in the month of Ṣafar 288 (February 901). See Suter, *Die Mathematiker*, no. 43 (pp. 20–21) and 66 (pp. 34–38).

[6] See what F. X. Kugler was able to derive from tablets not later than Hipparchus, *Die babylonische Mondrechnung. Zwei Systeme der Chaldäer über den Lauf des Mondes und der Sonne.* Freiburg i. Br. 1900.

[7] See G. Schiaparelli, *I precursori di Copernico nell'antichità*, Milan 1873, pp. 35–36 and 51 (= Pubblicazioni del R. Osservatorio di Brera, no. III). Seleucus lived before Hipparchus.

come from Chaldaeo-Hellenistic science. So it would be no wonder if the Chaldaeans had taught that sidereal year at the beginning of the Christian era to the Indians and the Persians, who seem to have given little attention to exact celestial observations[1]. And in fact Kugler, p. 91, shows that a sidereal year of $365.25953^d = 365^d\ 6^h\ 13^m\ 43.4^s$ was known to the Babylonians in the fourth and third centuries BC (what he says of al-Battānī on p. 92 should now be corrected).

b) — p. 40, l. 29 – p. 41, l. 2.

«*But Ptolemy says that they*» etc. — Ptolemy nowhere mentions that year of the Egyptians and the Babylonians; but in *Almag.* III, 2 (ed. Halma, vol. I, pp. 151–152) one reads the considerations, which al-Battānī reports, on the solar year that cannot be derived from the returns of the sun to the same fixed, non-wandering star.

The words «*so that he did not doubt the motion of the sun*» etc. seem to me to mean that Hipparchus nevertheless did not find such inequalities in the motion of the sun as to think it necessary, in order to represent them, to give the sun a third circle besides the ecliptic and the eccentric (or the epicycle). Al-Battānī had in mind, I think, this passage of the *Almagest*, where the precession of the fixed stars observed by Hipparchus is discussed: «What chiefly drives him [that is, Hipparchus] into that doubt is that from the returns of the sun to the solstitial and the equinoctial points the time of the year comes out less than 365 1/4 days, but from the returns to the fixed stars greater. Hence he conjectures that the sphere of the fixed stars too advances very slowly towards the following signs [that is, eastwards], as the planets also move in the direction contrary to the first motion [of the heavens], perpendicularly to the circle that passes through both poles, of the equator and of the ecliptic» (*Almag.* III, 2, ed. Halma, vol. I, p. 151).

c) — p. 42, ll. 6–8.

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The observation of the autumnal equinox. — Since al-Battānī follows the usage of the Byzantine indictions, putting the beginning of the Roman year on the first day of September, 19 Aylūl 1194 after Dhū l-Qarnayn will be 19 September of the year 882, not of 883. With this agrees 8 Pachōn, if it is referred to the vague years (without intercalation) of the era of Nabonassar[2]. From the epoch of the era of Nabonassar (26 February 748 BC) to the beginning of the year 882, 748 + 881 = 1629 Julian or Roman years elapsed, that is, 1629 vague years + 1629/4 days = 1630 vague years + 42 days. The first day of the month Thoth has therefore gone back 42 days (besides a whole vague year) with respect to 26 February; so 1 Thoth 1631 of Nabonassar = 15 January 882, and 8 Pachōn (which is the 248th day of the Egyptian year) = 19 September 882[a].

It remains to inquire how the words «4 3/4 hours before the rising of the sun on the 19th day of Aylūl» are to be understood, which, by the astronomical way of reckoning the day (that is, from noon), would mean 19 Sept., $13^h\ 15^m$. Had al-Battānī meant the astronomical day here, he

[1] I know of only two Persian observations between Ptolemy and the Hijra, which Ibn Yūnus reports, ch. VIII (in Caussin, p. 234), one about the year 470, the other about 630, for determining the longitude of the solar apogee. Whether they were true observations I hold uncertain.

[2] Not to the fixed years (of 365 days, with the Julian intercalation) of the Egypto-Coptic era mentioned in chapter XXXII and elsewhere, or of the era of Diocletian or of the Martyrs; both begin on 29 August, so that 8 Pachōn becomes 3 May. That al-Battānī meant the era of Nabonassar is clear from the very fact that in the Arabic text he calls the month *bākhūn*, after the Greek παχών, not after the Arabic *bashans*. Al-Battānī and the other Arabs give the Coptic months in the Greek form (which arose from the Memphitic Coptic) when they use the era of Nabonassar, in the Arabic form (from the Theban Coptic) when they use the eras of Augustus or of Diocletian. Likewise a little before, where he reports Ptolemy's observations of the equinoxes, *atūr* (ἀθύρ) and *fārmūth* (φαρμουθί) for the *hatūr* and *barmūdha* that are read in chapter XXXII (p. 67).

[a] By the usual reckoning the era of Nabonassar begins on 26 February 747 BC, and the vague year that begins on 15 January 882 is its 1630th; the dates 15 January and 19 September 882 are right.

would have referred the hour to noon, not to sunrise; it is probable, then, that the author had in mind the Muslim civil day, which begins at sunset. The observation was therefore in the morning hours of the civil 19th day, from whose noon only the astronomical 19th day begins; and since at the time of the equinox days and nights are equal, the time given will be 18 Sept. 882, $13^h\ 15^m$ mean time of al-Raqqa, in the astronomical usage. — This interpretation becomes quite certain from the interval of 743 vague years, 178 days and about $17^h\ 36^m$ that al-Battānī states to have elapsed between his observation and Ptolemy's, the difference of longitude between Alexandria and al-Raqqa being put at 2/3 of an hour. For the observations were:

Ptolemy: 25 Sept. 139, 19^h mean time of Alexandria $= 19^h\ 40^m$ al-Raqqa

al-Battānī: 18 Sept. 882, $13^h\ 15^m$ al-Raqqa;

so the interval is: (882 – 139) Julian years – (25 – 18) days – ($19^h\ 40^m$ – $13^h\ 15^m$) hours = 743 Julian years – 7 days – $6^h\ 25^m$ = 742 Julian years + 358 days – $6^h\ 25^m$.

From the year 139 to 882 there were 186 leap years; so the interval is:

742 vague years + (358 + 186) days – $6^h\ 25^m$ = 743 vague years + 179 days – $6^h\ 25^m$ = 743 vague years + 178 days + $17^h\ 35^m$,

as al-Battānī has it. Al-Battānī's observation was therefore 882 Sept. 18, $13^h\ 15^m$ mean time of al-Raqqa.

From the tables of R. Schram (*Hilfstafeln für Chronologie*, in the Denkschriften d. k. k. Akademie der Wissenschaften zu Wien, mathem. Classe, 1882, vol. XLV) one learns that in the year 882 the autumnal equinox fell on 18 September, $11^h\ 49^m$ mean time of Greenwich. According to the astronomical observations made by Chesney in 1835, the longitude of al-Raqqa is 39° 3′ 57″ E of Greenwich, $2^h\ 36^m$ in time; so the hour of al-Battānī's observation corresponds to $13^h\ 15^m$ – $2^h\ 36^m$ = $10^h\ 39^m$ mean time of Greenwich. Al-Battānī therefore observed the equinox $1^h\ 10^m$ before the true moment of time derived from Schram's tables.

[If we consider the astronomical instruments of that age, al-Battānī's observation must be judged excellent. Without doubt al-Battānī used the long alidade or the brick (the triquetrum and the mural quadrant), which are described in the last chapter of the book, or perhaps a very tall and most accurately made gnomon, to measure the meridian altitude of the sun on the days before and after the equinox; by making a proportion between the two measurements he could compute the moment of time between those two noons at which the meridian altitude of the sun would have been equal to the altitude of the celestial equator (or, as al-Battānī often says, to the altitude of the beginning of Aries) in the city of al-Raqqa (= 90° – the latitude of the city). Astronomers today, too, work in a similar way. — Schiaparelli.]

d) — p. 42, ll. 9–21.

The Arab astronomers mostly determined the *length of the year* by comparing their observations with Ptolemy's. Ibn Yūnus, ch. V (in Caussin, p. 158), however, says that it is better to divide the difference of the equinoxes found by the interval since the observations of Hipparchus.

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Ibn Yūnus, in the introduction and in ch. V (in Caussin, pp. 56 and 150 ff.), reports the mean motion of the sun in 365 days according to the Arab astronomers older than al-Battānī or of the same age as follows:

Aḥmad al-Nahāwandī, the calculator, in the city of Jundīsābūr in Susiana (today Shāhābād), by observations made at the beginning of the ninth century and by dividing the interval of time since Hipparchus (ch. V, Caussin 156–158)	359° 45′ 40″ 40‴.
The astronomers of the caliph al-Maʾmūn (Yaḥyā ibn Abī Manṣūr, al-ʿAbbās ibn Saʿīd al-Jawharī, Sanad ibn ʿAlī and others), at the observatory of al-Shammāsiyya[1] in the city of Baghdad, in the year 214 H (10 March 829 – 26 Febr. 830), 198 of Yazdajird (began 28 Apr. 829), by dividing the interval since Ptolemy[2]	359° 45′ 45″ 14‴ 24^{IV},
The astronomers of the same caliph (Sanad ibn ʿAlī, Khālid ibn ʿAbd al-Malik al-Marwarrūdhī, ʿAlī ibn ʿĪsā and others), in the city of Damascus, in the year 217 H (6 Febr. 832 – 25 Jan. 833), 201 of Yazdajird (began 27 Apr. 832)	359° 45′ 46″ 33‴ 50^{IV} 43^{V}.
The sons of Mūsā ibn Shākir, in the first half of the ninth century (ch. V, Caussin p. 150)	359° 45′ 39″ 58‴ 2^{IV}.
Aḥmad, son of Mūsā ibn Shākir, in the tables of the year 220 of Yazdajird (began 23 Apr. 851)	359° 45′ 40″.
Muḥammad ibn Aḥmad al-Samarqandī, in the year 234 of Yazdajird (began 19 Apr. 865), in the city of Samarqand (Maracanda)	359° 45′ 39″ 58‴ (var. 48‴).
The Banū Amājūr, who flourished in the same age as al-Battānī (ch. V, Caussin, p. 152)	359° 45′ 39″ 45‴.

Ḥabash, in his book of astronomy (fol. 157r), relates that the exceeding part of the revolution (*faḍl al-dawr*), that is, the amount (expressed in degrees) by which the whole circuit of the sun exceeds 365 days, was found by the astronomers of the caliph al-Maʾmūn:

at the observatory of al-Shammāsiyya	86° 43′ 39″ 36‴ 47^{IV}
in the city of Damascus	86° 41′ 25″ 14‴ 9^{IV}
by Yaḥyā ibn Abī Manṣūr	86° 35′ 13″ 46‴

In his tables Ḥabash adopted the first value, writing it in round figures 86° 43′ 39″ 37‴, or, with the degrees turned into time, 5^h 46^m 54^s 36^t. If the exceeding parts of the revolution are turned into hours and added to 365 days, the length of the tropical year comes out, according to

the observation of al-Shammāsiyya	365^d	5^h	46^m	54.640870^s
the observation of Damascus	365	5	46	45.682388
Yaḥyā ibn Abī Manṣūr	365	5	46	20.917777
the tables of Ḥabash	365	5	46	54.6

The mean motion in 365 days, according to the tables of Ḥabash (fol. 29v), is 359° 45′ 45″ 15‴ 27^{IV} 5^{V}; in a day 0° 59′ 8″ 20‴ 35^{IV} 25^{V} 0^{VI}. Astronomers today make the length of the tropical year of the sun 365^d 5^h 48^m 47.5^s (= 365.2422166 days); the mean motion in 365 days 359° 45′ 40.523″, in a day 0° 59′ 8.3302″.

On chapter 28

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a) — p. 43, ll. 11–20.

[1] The name of a gate and of a quarter of the city near the left bank of the Tigris, at the far northern end of the city; M. Streck, *Die alte Landschaft Babylonien nach den Arabischen Geographen*, Leiden 1900 ff., pp. 133–35; G. Le Strange, *Baghdad during the Abbasid Caliphate*, Oxford 1900, pp. 169–176.

[2] In one passage of the introduction (p. 56 Caussin) the seconds are 44″ (مد), but elsewhere (pp. 74 and 232) 45″ (مه).

The intervals of time between the equinoxes and the solstices. From many observations of his own he derived:

from the vernal equinox to the summer solstice:	a little less than $93^d\ 14^h$[1]
from the autumnal equinox to the vernal equinox:	about $178^d\ 14\ 1/2^h$
from the vernal equinox to the autumnal equinox:	about $186^d\ 14\ 1/2 + 1/4^h$
their sum:	about $365^d\ 5\ 1/4^h$.

In ch. XXVII, p. 42, he had established that the tropical year comprises $365^d\ 5^h\ 46^m\ 24^s$; so the difference is about $31^m\ 24^s$.

[It is strange that al-Battānī found so small a difference in the sum of two intervals, each of which is determined by observing two equinoctial times; for in these observations an uncertainty of several hours could very easily arise. Much more difficult indeed is the observation of the solstice, in which an error of half a day would not have been thought too great in al-Battānī's time. In his computation, then, he was helped by the happiest fortune, or rather he was able to use many very good observations of his own and of others. Near the solstices the declination of the sun changes by only a few seconds of arc in a day; and since the moment of time at which the declination becomes greatest has to be determined by observations, it is easily understood how hard this task was for the ancient astronomers, even though they used the shadows of long gnomons. Twelve hours before and after the solstice the declination of the sun is only 3 1/2 seconds less than at the time of the solstice; so that the Greeks not seldom erred by a whole day or more in determining these solstitial times. — For the rest, it should be noted that the error of 31 minutes that appeared in al-Battānī's computation is of very little consequence in this problem; for in that space of time the sun runs a little more than 1′ on the ecliptic, which was then the limit of the quantities that could be neglected. — Schiaparelli.]

To show the care of the observations, I shall now give the true intervals computed for the years 882[2] and 883 for mean Greenwich time, with the help of the tables of R. Schram, *Hilfstafeln für Chronologie* (Denkschriften der kais. Akademie der Wissenschaften zu Wien, mathem. Cl., 1882, vol. XLV):

	In the year 882	*In the year 883*
vernal equinox	74.9046^d = 15 March, $21^h\ 42.6^m$	75.1470^d = 16 March, $3^h\ 31.7^m$
summer solstice	168.4195 = 17 June, 10 4.1	168.6580 = 17 June, 15 46.5
autumnal equinox	261.4923 = 18 Sept., 11 48.9	261.7360 = 18 Sept., 17 39.8
winter solstice	350.6147 = 16 Dec., 14 45.2	350.8579 = 16 Dec., 20 35.4

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	In the year 882	*In the year 883*
From the vernal equinox to the summer solstice	$93^d\ 12^h\ 21.5^m$	$93^d\ 12^h\ 15.8^m$
From the vernal equinox to the autumnal equinox	186 14 6.3	186 14 8.0
From the autumnal equinox to the vernal equinox	178 15 42.5	178 15 40.6

Other equinoctial and solstitial observations of the Arabs, made at Damascus and Baghdad between the years 830 and 851, are handed down to us by Ibn Yūnus, ch. IV, in Caussin pp. 146–148, and by Delambre, pp. 85–86 (where 851 should be read for 831 on p. 86, l. 1).

b) — p. 44, l. 15.

[1] In the computation of p. 44, l. 5, he plainly chose $93^d\ 13^h\ 53.5^m$; for he states that in this interval of time the sun ran 92° 14′ 10″ instead of 92° 14′ 26″, which by al-Battānī's tables correspond to $93^d\ 14^h$. [*Translator:* The print has $93^d\ 14^m$ for $93^d\ 14^h$.]

[2] The year in which, as he himself relates in ch. XXVII, p. 42, he observed the autumnal equinox.

Anomaly, ἀνωμαλία (*ikhtilāf al-ḥaraka*, «variation, inequality of motion»), properly denoted the difference between the true longitude and the mean longitude of the sun or of a planet. Geometrically, on the hypothesis of the eccentric, it was the angle contained by the two lines drawn from the centre of the sun to the centre of the ecliptic, or of the world, and to the centre of the eccentric; on the hypothesis of the epicycle, the angle made by the two lines drawn from the centre of the ecliptic to the centre of the epicycle and to the centre of the sun. Very often, however, al-Battānī (as presently in what follows) and the other Arabs call this angle or this difference *taʿdīl*, «equation»[1], whence the present astronomical name equation of the centre. By the name anomaly[2], on the other hand, they denote, as the Greeks often did and the moderns always do, what would more rightly be called the *argument of the anomaly* or the *anomalistic motion*, that is, the arc of the eccentric or of the epicycle that the sun or the planet has run from the apogee to the given moment of time[3].

The greatest anomaly, or the greatest equation of the centre of the sun, al-Battānī found to be 1° 59′ 10″[4]; the eccentricity 2^p 4 3/4′, the radius of the eccentric being taken as 60 parts[5]. According to Leverrier, who put the greatest equation of the centre for the beginning of the year 1850 at 1° 55′ 18.31″ and the eccentricity of the solar (or rather the terrestrial) orbit at 0.016771063, the equation in the year 880 must have been 1° 58′ 2.85″, the eccentricity 0.017182700. — Other values of the equation adopted by authors older than al-Battānī or contemporary with him are these: I 214

Hipparchus and Ptolemy (*Almag.* III, 4, ed. Halma vol. I, pp. 188–190)	2° 23′
The Indian book entitled *Sūrya-siddhānta*, transl. Burgess and Whitney, p. 220, note (before the 5th or 6th century)	2° 10′ 31″
The astronomers of the caliph al-Maʾmūn, in the city of Baghdad, in the year 829 (Ibn Yūnus, introd., in Caussin p. 56)[6]	1° 59′
The same, in the year 832, at Damascus (Ibn Yūnus, loc. cit.)	1° 59′ 51″
The three brothers Banū Mūsā ibn Shākir (Ibn Yūnus, ch. V, in Caussin pp. 150 and 152)	2° 0′ 50″
Aḥmad, one of the three, for the year 220 of Yazdajird, which began on 23 April 851 (Ibn Yūnus, loc. cit.)	2° 0′ 8″
Ḥabash, Berlin codex ff. 30r–32r	1° 59′ 0″
The family of the Banū Amājūr (Ibn Yūnus, ch. V, in Caussin p. 152)	2° 0′ 50″

In the second half of the eleventh century al-Zarqālī, or Arzachel, was the first, it seems, who dared to profess a secular variation of the eccentricity and of the equation of the centre; Purbach (1423–1461) followed him in Europe. To the astronomers of the 16th and 17th centuries the decrease was not certain; Kepler, Longomontanus, Bullialdus and Riccioli denied it; Magini, Tycho Brahe, Snellius and Lansbergius affirmed it. Compare I. B. Riccioli, *Almagestum novum, astronomiam veterem novamque complectens*, Bologna 1651, vol. I, pp. 158–159.

[1] A word that Rudloff and Hochheim, al-Jaghmīnī pp. 238, 240, 248 and in many places, translate «Rectification».

[2] But when the moon and the other planets are concerned, al-Battānī almost always uses the word الحاصة *al-ḥāṣṣa*, «the [motion] falling to [the planet]», as does al-Bīrūnī, *India* (see Sachau's note in the translation, II, 363); others the word الخاصة *al-khāṣṣa*, «the proper or peculiar [motion]».

[3] In present-day astronomy the angle that the radius vector makes with the major axis of the elliptical orbit is called the *true anomaly*.

[4] The number 1° 58′ that is commonly given (Delambre p. 36, R. Wolf, *Handbuch*, I, 447, § 206) is an error of the printed translation of Plato; see p. 44, note 3.

[5] If the radius is taken as one part, one finds $\frac{2^p\ 4'\ 45''}{60^p} = 0.034652766\ldots$; Ptolemy had found $\frac{2^p\ 29'\ 30''}{60^p} = 0.0415277\ldots$, but, to make the computation more convenient, he chose the fraction 1/24 = 0.04166…. It should be noted that the eccentricity on the hypothesis of the eccentric is nearly the same as *twice* the eccentricity of Kepler's ellipse; so, if the numbers of the ancients are compared with the modern ones, they must be halved, and al-Battānī must be given the eccentricity 0.017326, Ptolemy 0.0207638… (or 0.02083….). For the ancients the eccentricity was the sine of the greatest equation of the centre.

[6] This value was adopted in the *Tested Tables*. Elsewhere Ibn Yūnus (ch. VI, in Caussin p. 232, and Delambre p. 93) wrongly puts this observation in the following year, that is, 199 of Yazdajird (which began on 28 Apr. 830), 215 H (which began on 27 Febr. 830).

c) — p. 44, ll. 29–37.

I. — *The observation of the apogee of the sun* in the year 1194 of Dhū l-Qarnayn (which began on 1 Sept. 882).

He relates that he found the longitude of the apogee to be 82° 17′. In the computation he puts the arc 7° 43′ as corresponding to the sine 8^p 4′; so 90° – 7° 43′ = 82° 17′. But in al-Battānī's tables, ff. 176v–177r, the arc corresponding to the sine 8^p 4′ is about 7° 44′, so that the longitude of the apogee would be 82° 16′. Since the author, as we shall see elsewhere, gives all the apogees the motion of the precession of the equinoxes, or of the fixed stars, that is, 54″ 33‴ a year, those two apogees will be 82° 15′ 16″ and 82° 14′ 16″ for the year 1191, to which al-Battānī's tables are mostly referred.

I cannot settle the doubt. The number 82° 15′ is confirmed by the passage of ch. XXXIII, p. 72: «on the first day of the month Ādhār of the year 1191 of Dhū l-Qarnayn [= 1 March 880] the apogee of the sun was at about 22° 15′ of Gemini, that is, 82° 15′ from the beginning of Aries»; and in the following lines the same number is repeated twice, and, as I shall show in note *d* on ch. XXXI, it is also drawn from Ibn al-Samḥ and al-Jaghmīnī. But at the end of ch. XLV, p. 114, he says that in the same year 1191 the apogee of Venus was «at 82° 14′, the apogee of the sun being as that of Venus»; and he gives this longitude, 82° 14′, to the apogee of Venus in the tables too, f. 217v. Likewise Ibn Yūnus (ch. V, Caussin p. 155) testifies that al-Battānī found the apogee of the sun at 22° 14′ of Gemini (82° 14′ from the beginning of Aries); and the same number, as regards Venus, is drawn from al-Bīrūnī, as I shall show in note *d* on ch. XXXI. — But in this computation a difference of 1 minute was of very little consequence in those times[1].

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[Al-Battānī's observation must be judged excellent, since the problem was a very difficult one at that time. By the formula $\omega = 280° 21′ 21.5″ - 61.6995″ \cdot t + 0.0001823″ \cdot t^2$, with which Leverrier determines the longitude ω of the *perigee* for *t* Julian years before 1850[2], the longitude of the *apogee* of the sun in the year 884 must have been 83° 50′ 51″; al-Battānī found 82° 17′, that is, a number 1 1/2° smaller than the true one. But since the problem depends on very uncertain elements, it would be unjust to demand more subtlety in these matters from the ancient astronomers. — Schiaparelli.]

II. — *The method of computing the longitude of the apogee of the sun.*

As he himself admits on p. 43, l. 11, al-Battānī follows the method of Ptolemy, which depends on knowing the equinoxes and the summer solstice, and also the length of the tropical year, from which the mean daily motion of the sun is derived. Let the mean motion of the sun in the spring of a given year be *v*, in the following summer *a*; the eccentricity of the orbit of the sun, or the greatest anomaly, is $FE = \sqrt{FS^2 + ES^2}$, where from al-Battānī's proof, with the radius of the eccentric = 60^p = *R*, it is clear that

$$FS = \sin\Big[v - \Big(90° + \frac{(v+a) - 180°}{2}\Big)\Big] = \sin\frac{v-a}{2},$$

$$ES = \sin\frac{(v+a) - 180°}{2} = -\cos\frac{v+a}{2}.$$

[1]Nor of any consequence in the computation of eclipses, in which the true longitude of the sun must be known. True longitude = mean longitude ± equation of the centre; the equation of the centre is found by entering the tables with the mean anomaly, which is the difference between the place of the apogee and the mean longitude. A change of 1 minute in the apogee (and so in the mean anomaly) brings into al-Battānī's equation of the centre a change that is at most 2 seconds (when the mean anomaly is about 1° or 360°), and that becomes 0″ when the mean anomaly comprises 110° or 250°. [*Translator:* With al-Battānī's eccentricity the change vanishes where the equation is greatest, at a mean anomaly of about 92° or 268°; at 110° and 250° it is about 0.65″.]

[2]But in his formula the hundredths of the seconds are not absolutely certain.

We can therefore write:

$$\text{eccentricity} = FE = \sqrt{\sin^2 \frac{v-a}{2} + \cos^2 \frac{v+a}{2}}.$$

The longitude of the apogee T is the arc AH, whose complement is the arc HB, or the angle FES. But plainly

$$\sin(90^\circ - \text{long. of the apogee}) = \sin FES = R \frac{ES}{FE},$$

where the quantities ES, FE have just been computed.

This method of Ptolemy is also set out at length by al-Bīrūnī, *Chron.*, pp. 183–85 of the text, 164–67 of the translation; afterwards he adds: «But when the *moderns*[1] had recognized that it is very difficult and almost impossible to determine the times of the solstices exactly, they preferred in their observations the points that mark the middles of the four seasons of the year, that is, the middles of the fixed signs [that is, 15° of Taurus, 15° of Leo, 15° of Scorpio, 15° of Aquarius]. The method, however, that my master Abū Naṣr Manṣūr ibn ʿIrāq[2], a client of the caliph, found for solving this problem rests on knowing any three points of the ecliptic and on knowing the length of the solar year exactly. In my book entitled *The request for testimony on the discrepancies in observations* I have shown that this method is as much better than the method of the modern astronomers as theirs is better than the method of the ancients»[3]. And in fact in this way points of the ecliptic can be chosen whose determination is easier than that of the solstices; it seems to be the same solution that Delambre proposed, *Hist. de l'astron. ancienne*, vol. II, pp. 121–133. — Of the difficulty of measuring the time of the solstice I have already spoken on p. 212[a]. I 216

III. — *The motion of the apogee of the sun.*

Delambre, p. 36, rightly denies that al-Battānī knew the *proper* motion of the solar apogee, as Bailly and many others wrongly say; in fact, at the end of chapters XXXIII and XLIV, he gives the apogee of the sun only the motion of the other apogees and of the fixed stars, that is, the motion of precession.

Delambre, however, is wrong to wonder that al-Battānī did not compare his solar apogee with that of Hipparchus and Ptolemy (who have 65° 30′), and so derive a peculiar annual motion of the apogee of 25 seconds (by comparison with Ptolemy's apogee) or of 21 (with Hipparchus's). Al-Battānī understood very well that the Greek longitude of the solar apogee was of no use; for Ptolemy[4], though he gives the motion of precession to the fixed stars and to the apogees of the planets, thinks the apogee of the sun immovable, affirming that in the year 463 after the death of Alexander (139) he found it at the same longitude, 65° 30′, at which Hipparchus had found it about 265 years before! Theon himself, lastly (*Tables manuelles*, ed. Halma, vol. I, p. 54), in the

[1] *Al-muḥdathūn*, that is, the astronomers who flourished in the time of al-Maʾmūn (813–833) and after.

[2] On him see Suter, *Die Mathematiker*, pp. 81–2, no. 186. — I think it very probable that the first part of this new solution is contained in the little work entitled «*The computation of the eccentricity, from the royal Almagest of Abū Naṣr ibn ʿIrāq*», which is kept in London, in the library called the «Library of the India Office».

[3] This whole passage of al-Bīrūnī on the apogee is also found edited, translated and explained in Sachau and Holetschek, *Eine Berechnung der Entfernung des Sonnen-Apogäums von dem Frühlingspunkte bei Albîrûnî* (Sitzungsberichte der philos.-histor. Classe der k. k. Akad. der Wissenschaften zu Wien, 1876, vol. LXXXII, pp. 243–266).

[4] *Almag.* III, 4 (ed. Halma vol. I, p. 184); likewise in the *Hypotheses*, ed. Halma, Paris 1820, pp. 44–45. — Proclus, *Hypotyposes*, ed. Halma, pp. 89–90 and 120, says that the astronomers found the apsides of the sun immovable, the apogee at 5 1/2° of Gemini (65° 30′) and the perigee at 5 1/2° of Sagittarius (245° 30′). But Proclus (5th century) also denies the precession of the equinoxes.

[a] Nallino's Addenda et emendanda (Part I, p. LXXIV) add that, on Snellius's authority, Montucla says that al-Zarqālī, about 1070–80, used observations of the sun at both equinoxes and at any third point, or at any three points, to compute the longitude of the solar apogee: the method of which al-Bīrūnī speaks; the Addenda are translated with Part I's front matter.

fourth century, always puts the apogee of the sun at 65° 30′[a].

How, then, could al-Battānī derive from the statements of the Greeks the small proper motion of the solar apogee, which is 11.46″ a year and in the interval of 745 years from Ptolemy to al-Battānī reaches only 2° 22′ 18″? It is already of the greatest consequence that the Arabs declared false the opinion of the Greeks on the immobility of the solar apogee, and recognized that it shares in the motion of the precession of the equinoxes, by which the longitudes of the fixed stars increase. Al-Masʿūdī writes, *al-Tanbīh*, p. 222: «The astrologers and astronomers of our time do not disagree with the sages of the Indians, the Chaldaeans, the Egyptians, the Greeks, the Romans and others in putting the apogee of the sun in the sign of Gemini; the disagreement among them concerns only its immobility and its motion, as we have said. Thābit ibn Qurra al-Ṣābiʾ al-Ḥarrānī[1] wrote an epistle to defend Hipparchus's opinion that the apogee of the sun moves, against what Ptolemy had said»[2]. — Caussin, p. 234, relates the following after Ibn Yūnus: «Dans ce même chap. VIII, Ebn Iounis rapporte deux observations faites par les Perses, postérieurement à Ptolémée, qui ont servi à reconnoître le mouvement de l'apogée du soleil que Ptolémée croyoit immobile. Par la première de ces observations, qui remonte à l'an 470 environ, ère vulgaire, l'apogée du Soleil fut trouvé dans 17° 55′ des gémeaux; et par la seconde, 160 ans environ après, 630 ère vulgaire, dans 20° des gémeaux» («In this same chapter VIII, Ibn Yūnus reports two observations made by the Persians after Ptolemy, which served to recognize the motion of the apogee of the sun, which Ptolemy believed immovable. By the first of these observations, which goes back to about the year 470 of the common era, the apogee of the sun was found at 17° 55′ of Gemini; and by the second, about 160 years later, 630 of the common era, at 20° of Gemini»). — Ibn Ezra[3], who died in the second half of the twelfth century, says in his preface to the Hebrew translation of the *Foundations of the tables of al-Khwārizmī*, written by a certain Ibn al-Muthannā, that Ptolemy wrongly believed the apogee of the sun to be immovable, whose motion, equal to the motion of the fixed stars, «the learned men of Ishmael», that is, the Arabs, were the first to recognize[4].

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Al-Masʿūdī, loc. cit., relates «that the Indians and others thought that the apogee of the sun moves 1° in every 100 years, so that it stays 3000 years in each sign of the zodiac and goes round the ecliptic in 36000 years»[5]. But al-Masʿūdī is gravely mistaken when he says that the Indians too ascribe to the apogee of the sun this motion of 1 degree in 100 years, or 36″ a year (which is the motion of precession according to Ptolemy). For the Indians do hold that the apogees of the sun and the planets move, but very slowly. In the book *Sūrya-siddhānta*, Burgess's translation, ch. I, st. 41–44, pp. 170–172, the apogee of the sun is said to complete its revolution in 11,162,790.7 years, that is, to run 1 minute in 516.8 years; Brahmagupta, on the other hand, and the Arabs who translated his *Brahmasphuṭa-siddhānta* in the eighth century, say that the revolution is completed in 9,000,000 years, so that 1′ is run in 416.66… years[6]. When they imagined these things is not clear[7]; but without doubt they are dreams for which astronomical observa-

[1]He died in the month of Ṣafar 288 H (February 901); Suter, *Die Mathematiker und Astronomen der Araber*, no. 66, pp. 34–38; M. Steinschneider, *Thabit ben Korra, bibliographische Notiz* (in the Zeitschrift für Mathematik und Physik, vol. XVIII, 1873, pp. 331–338).

[2]See also the passage of al-Bīrūnī that I quoted on p. 208, ll. 11 ff.

[3]The passage is given in Hebrew and in German in Steinschneider, *Zur Geschichte der Uebersetzungen aus dem Indischen ins Arabische* (in the Zeitschr. der deutsch. morgenländischen Gesellschaft, vol. XXIV, 1870, p. 356); but there the Hebrew *gabhūt* is wrongly translated «Höhe», height, instead of «Apogäum», apogee.

[4]See also the *Astronomy* of Barhebraeus, pp. 25–26 of the text.

[5]Al-Bīrūnī, *Chron.*, p. 9 of the text, 11 of the translation: «These returns [of the sun to some point of the ecliptic] are equal according to Ptolemy, since he found no motion in the apogee of the sun. But according to the authors of [the Indian book] *al-Sindhind* [= *siddhānta*] and other moderns they are unequal, for their observations show that the apogee has a motion». What motion the Indians gave the apogee we shall see a little below.

[6]Al-Bīrūnī, *India*, ch. L, p. 209 (transl. vol. II, p. 16).

[7]Thibaut says in his preface to the book *The Pañchasiddhântikâ, the astronomical work of Varâha Mihira, edited*

[a]Nallino's Addenda et emendanda (Part I, p. LXXIV) add here that Theon of Smyrna, in the first half of the second century, gives an anomalistic year of 365 1/2 days, so that the solar apogee would run round the whole zodiac in 1461 years; the Addenda are translated with Part I's front matter.

tions gave no foundation, as Burgess (and Whitney), loc. cit., and Thibaut, *Astronomie, Astrologie und Mathematik* [*der Inder*], Strassburg 1899, p. 33 (= Grundriss der indo-arischen Philologie, vol. III, part 9), rightly recognized[1]. — The Arabs, then, were the *first* to establish that *the motion of precession belongs to the apogee of the sun too.*

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The *proper* motion of the apogee, on the other hand, being very small, can be derived only from observations carried on over a very long space of time and with the help of the best instruments. So it is no wonder that the Arabs mostly did not know it; for the discrepancies, and very small ones, that appeared to them in the motion of the longitude of the apogee with respect to the motion of precession they necessarily put down to the fault of the observations, knowing that it is very difficult to measure the place of the solar apogee. Al-Zarqālī[2], however, on the evidence of Aboul Hhassan, ch. XII, p. 132, by comparing the observations of Ptolemy with al-Battānī's and his own, established that the apogee of the sun completes 1° in every 299 Roman years, that is, $12\frac{12}{299}''$ a year[a], which agrees wonderfully with the value 11.46″ accepted today; but he held that this motion takes place forward and backward alternately on the circumference of a small circle, because, as appears from his *Toledan Tables*, the apogee seemed to have advanced 16° 47′ in the 740 years from Ptolemy to al-Battānī, but to have gone back 4° 27′ in the 193 years from al-Battānī to al-Zarqālī. — The Jew Levi ben Gerson, or Leo de Balneolis (1288–1370), in his astronomy written in Hebrew, derived the motion of the apogee of the sun by diligently comparing his own observations with older ones as 1° in 43 2/3 years, that is, 1′ 19″ 39‴ 53.6^{IV} a year[3]. And since he had accepted al-Battānī's annual precession of 54″ 33‴, he put the *proper* motion of the apogee of the sun at 25″ 3‴. — Of the European astronomers, Kepler and Longomontanus were the first to distinguish the motion of the apogee of the sun from the motion of the fixed stars, that is, of precession; see I. B. Riccioli, *Almagestum novum astronomiam veterem novamque complectens*, Bologna 1651, vol. I, pp. 157–58.

IV. — *The longitude of the apogee of the sun* according to the astronomers older than al-Battānī or contemporary with him:

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with an english translation and introduction by G. Thibaut and M. Sudhâkara Dvivedî, Benares 1889, p. XVIII: «The Pañchasiddhântikâ says nothing about the revolutions of the apogees of the sun and planets, and it hence is possible that the old Sûrya Siddhânta was not yet acquainted with the theory held, on entirely insufficient grounds, by the modern treatise, and modern Hindû astronomers in general, that the apogees of the sun and the planets perform a certain number of revolutions within a mahâyuga or kalpa. On the other hand it might be supposed that Varâha Mihira, although acquainted with that doctrine, yet confined himself to stating the places which the apogees occupied at his time, since so much is sufficient for the purposes of a karaṇa-writer». — From what al-Bīrūnī relates in the *India*, ch. L, pp. 211–212 (transl. II, 18–19), he likewise seems to have found no motion of the apogees of the sun and the planets (the moon excepted) in the Indian book *Pulisa-siddhānta*, written before the fifth century, nor in the Arabic work that Abū l-Ḥasan al-Ahwāzī drew from Indian sources.

[1] [They made these revolutions extremely slow so that they could imagine all the celestial longitudes = 0° at a most remote epoch, which they took as the beginning of all times. — Schiaparelli.]

[2] The Arzachel of the Latin writers of the Middle Ages; he lived in the eleventh century (Suter, *Die Mathematiker und Astronomen der Araber*, no. 255, pp. 109–111). It is strange that so notable a passage of Aboul Hhassan (who took the proper motion of the apogee from al-Zarqālī) escaped those who have written the history of astronomy. [*Translator:* The print has «the tenth century»; Nallino's Addenda et emendanda (Part II, p. XXIV) correct it to the eleventh.]

[3] On the evidence of Augustinus Ricius, *De motu octavae sphaerae*, Casale 1513; see also Delambre 381. — 43 2/3 years are 43 years $243^d\ 8^h$; more precisely in the astronomy written (or translated?) in Latin by the same Leo de Balneolis, tract. I, ch. 58 (fol. 46v, col. *a* of the Vatican Latin codex 3098): «Nos vero invenimus eam [that is, the aux or apogee of the sun] moveri per 1 g. 43 annis, 232 diebus, 6 horis et circa 46 minutis» («but we have found that it moves 1 degree in 43 years, 232 days, 6 hours and about 46 minutes»).

[a] The print has 11″ 12/299; Nallino's Addenda et emendanda (Part II, p. XXIV) correct it to 12″ 12/299, as 3600″ : 299 requires.

The Babylonians in the third century BC[1]	76° 35′
All the Greeks from Hipparchus (about 125 BC) to Theon (4th century) and Proclus (5th century)	65° 30′
The first recension of the book *Sūrya-siddhānta*, before the 5th century, according to the *Pañcasiddhāntikā*, Thibaut's translation, ch. IX, st. 7, p. 57	80°
The later *Sūrya-siddhānta*, transl. Burgess and Whitney, notes, pp. 171 and 487	77° 15′
The book *Romaka-siddhānta*, before the 5th century, in the *Pañcasiddhāntikā*, ch. VIII, st. 2, p. 48 of the translation	75°
Āryabhaṭa, writing in 499, according to Thibaut's preface to the *Pañcasiddhāntikā*, p. XVIII	78°
The Arabic book *al-Sindhind*[2], on the evidence of Ibn Hibintā, ff. 1v and 2r	77° 15′
The tables called *al-Shāh*[3], on the evidence of Ibn Hibintā, f. 2r, and of al-Masʿūdī, *al-Tanbīh*, p. 222	80°
The astronomers of al-Maʾmūn, in the city of Baghdad, in the year 198 of Yazdajird, 214 H (829), on the evidence of Ibn Yūnus, introd., in Caussin p. 56[4]	82° 39′
The same, at Damascus, in the year 201 of Yazdajird, 217 H (832), on the evidence of Ibn Yūnus, loc. cit.	82° 1′ 37″
The three brothers Banū Mūsā ibn Shākir computed that the apogee was in the first year of Yazdajird (which began on 16 June 632), as Ibn Yūnus relates, ch. V, Caussin 150	80° 44′ 19″
Aḥmad, one of the three, in the year 220 of Yazdajird (which began on 23 Apr. 851), on the evidence of Ibn Yūnus, loc. cit., and ch. VIII (in Delambre, p. 99)	84° 33′
The family of the Banū Amājūr (Ibn Yūnus ch. V, Caussin p. 152)	83° 35′
Mufliḥ ibn Yūsuf, a client of that family (Ibn Yūnus, loc. cit.)	84° 5′

To these may be added the two *Persian* observations (?) which, taken from the work of Ibn Yūnus, I reported on p. 209, note 2; the later, of about the year 630, gave 80°, like the tables *al-Shāh* (translated into Arabic from Pehlevi, or middle Persian) and the first *Sūrya-siddhānta*.

d) — p. 46, l. 7.

I have kept the length of the line EF as it is in the Escorial codex and in Plato's translation, namely $61^p\ 48'\ 35''$. More exactly it was[a]

$$EF = \sqrt{EM^2 + FM^2} = \sqrt{(61^p\,48'\,2'')^2 + (1^p\,2'\,22\tfrac{1}{2}'')^2} =$$

[1] According to F. X. Kugler, *Die babylonische Mondrechnung*, Freiburg i. B. 1900, p. 90. The numbers have been reduced to our movable ecliptic; on the fixed ecliptic of the Babylonians they were 79° 49′.

[2] It was the translation, made in the eighth century, of the work that the Indian Brahmagupta wrote in 628 and called *Brahma-sphuṭa-siddhānta*. The longitude of the apogee is as in the later Sūrya-siddhānta. — Al-Masʿūdī, *al-Tanbīh* pp. 221–222, says that those who followed the method of the Sindhind put the apogee of the sun at 77° 55′ 14″ in the first year of the Hijra.

[3] Ibn Rusta also mentions them, p. 162, ll. 3–17, from which it is clear that, at least in the Arabic recension, they referred the celestial motions to the first year of Yazdajird (16 June 632). Without doubt they are what other Arabs call the *tables of al-Shahriyār*, for *shāh* and *shahriyār* mean «king» in Persian; at the end of the eighth century, or the beginning of the ninth, Abū l-Ḥasan ʿAlī ibn Ziyād al-Tamīmī had translated them from Persian; see *Kitāb al-Fihrist*, p. 241, l. 5, and 244, l. 29; *Hamzae Ispahanensis Annalium libri X*, edited by J. M. E. Gottwaldt, St Petersburg–Leipzig 1844–1848, vol. I, pp. 200 and 201; Haji Khalfa, *Lexicon bibliographicum et encyclopaedicum*, edited and translated by G. Flügel, Leipzig–London 1835–58, vol. III, p. 566, no. 6952. In al-Bīrūnī, *Chron.*, p. 6 of the text and of the translation, the name is *Tables of the Shahriyārān al-Shāh*. In the time when Abū Maʿshar (d. 886) wrote, they were very widespread among the Arabs, as appears from Ibn Rusta, loc. cit.; the astrologer Māshāʾallāh (d. about 815) relied on them, on the evidence of Ibn Hibintā, f. 224r. [Perhaps the Arabic translation of the Pehlevi book «Tables of Shatro-ayār», which Mānūshchihar mentions in his second epistle (about 880), in E. W. West, *Pahlavi texts translated*, vol. IV, Oxford 1892, p. XLVII = The sacred books of the East, vol. XXXVII. — Schiaparelli.]

[4] So for the year of the Hijra (which began on 15 July 622), having computed a precession of 3° 9′ 1″ in 214 Arab years, they put the longitude of the apogee at 79° 30′. This is the value adopted in their *Tested Tables* and by Ḥabash, f. 28r. — Al-Masʿūdī, *al-Tanbīh* p. 222, wrongly writes that this observation was in the year 217 H.

[a] The print has $(3472.5'')^2$ for $(3742.5'')^2$, whose square follows.

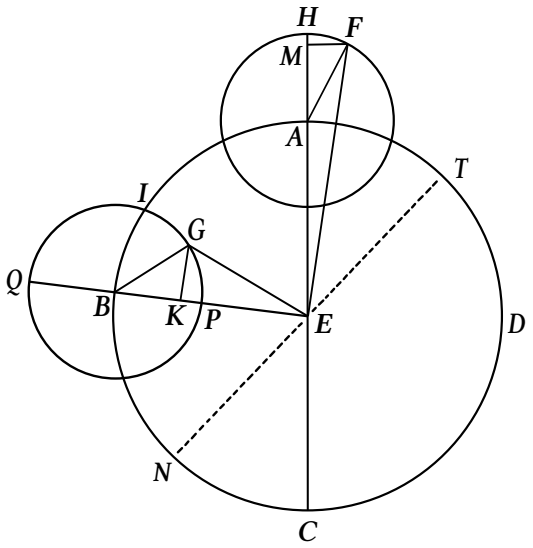
$$= \sqrt{(222\,482'')^2 + (3742.5'')^2} = \sqrt{49\,498\,240\,324^{\mathrm{IV}} + 14\,006\,306.25^{\mathrm{IV}}} =$$
$$= \sqrt{49\,512\,246\,630.25^{\mathrm{IV}}} = 222\,513.47\ldots'' = 61^p\,48'\,33.47''.$$

e) — p. 46, ll. 25–26.

The length of the line EG is given as in the codex and in Plato, namely $58^p\ 12'\ 34''$; but on p. 47, l. 32, the codex more rightly has $32''$ for the seconds. For[a]

$$EG = \sqrt{KG^2 + EK^2} = \sqrt{(1^p\,2'\,22\tfrac{1}{2}'')^2 + (58^p\,11'\,58'')^2} =$$
$$= \sqrt{(3742.5'')^2 + (209\,518'')^2} = \sqrt{14\,006\,306.25^{\mathrm{IV}} + 43\,897\,792\,324^{\mathrm{IV}}} =$$
$$= \sqrt{43\,911\,798\,630.25^{\mathrm{IV}}} = 209\,551.4\ldots'' = 58^p\,12'\,31.4''.$$

f) — p. 48, ll. 2–16.



To compute the *equation of the centre*.

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The geometrical constructions of pp. 46–47 are taken from *Almag.* III, 5 (ed. Halma vol. I, pp. 190–199); al-Battānī's formula, on the other hand, is not in Ptolemy:

$$\sin(\text{eq. of the centre}) = \frac{R\,\frac{e\sin m}{R}}{\sqrt{\left(\frac{e\sin m}{R}\right)^2 + \left(\frac{e\cos m}{R} + 60^p\right)^2}},$$

where *e* is the eccentricity or, on the hypothesis of the epicycle, the radius of the epicycle[1]; *m* the arc of the anomaly, that is, the arc of the eccentric or of the epicycle that the star has run from the apogee to the given moment of time. The rule is very easily derived from the constructions of Ptolemy and al-Battānī. On the hypothesis of the epicycle, where the equation of the centre = the angle HEF,

$$\sin HEF : MF = \sin FME : EF, \quad \text{that is,} \quad \sin(\text{eq. of the centre}) : MF = \sin 90^\circ : EF,$$

[1] In parts of the radius of the eccentric or of the orbit (= 60^p).

[a] The print has 43 911 798 625.25 for the sum, 43 911 798 630.25.

and since sin 90° = *R*,

$$\sin(\text{eq. of the centre}) = R\,\frac{MF}{EF}.$$

1) $$EF = \sqrt{\overline{MF}^2 + (AE + AM)^2} = \sqrt{\overline{MF}^2 + (60^p + AM)^2}.$$

$$AM : AF = \sin AFM : \sin AMF = \cos HAF : R,$$

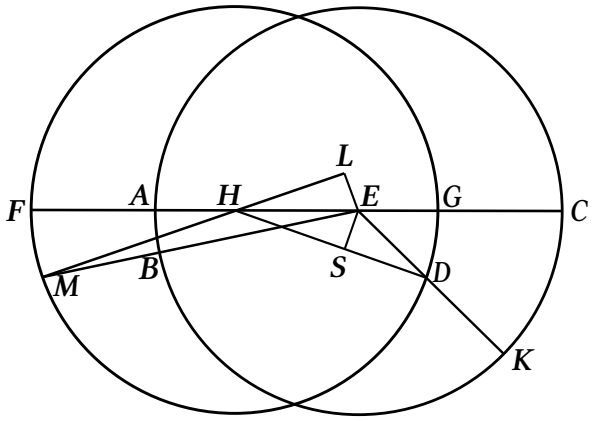
whence $AM = \frac{AF \times \cos HAF}{R} = \frac{e \cos m}{R}$.

$$MF : AF = \sin HAF : \sin AMF,$$

whence $MF = \frac{AF \times \sin HAF}{R} = \frac{e \sin m}{R}$. Let these values of the lines AM, MF be put into 1); there will result

$$EF = \sqrt{\left(\frac{e \sin m}{R}\right)^2 + \left(\frac{e \cos m}{R} + 60^p\right)^2},$$

which is the denominator of al-Battānī's formula.



Similarly on the hypothesis of the eccentric:

$$\sin(\text{eq. of the centre}) = \sin LME = R\,\frac{LE}{ME},$$

$$LE = HE\,\frac{\sin LHE}{R} = HE\,\frac{\sin FHM}{R} = \frac{e \sin m}{R},$$

$$LH = HE\,\frac{\cos LHE}{R} = HE\,\frac{\cos FHM}{R} = \frac{e \cos m}{R},$$

$$ME = \sqrt{\overline{LE}^2 + (HL + 60^p)^2} \quad \text{etc.}$$

Had he known the use of tangents, he would have found more simply, on the hypothesis of the epicycle,

$$\tan(\text{eq.}) = \tan HEF = R\,\frac{MF}{60^p + AM} = R\,\frac{e \sin m}{60^p R + e \cos m},$$

and on the hypothesis of the eccentric

$$\tan(\text{eq.}) = \tan LME = R\,\frac{LE}{60^p + HL} = R\,\frac{e \sin m}{60^p R + e \cos m}.$$

In all these formulas the eccentricity or the radius of the epicycle, that is, *e*, is in *sixtieth* parts of the radius of the eccentric or of the orbit, which is supposed to comprise 60^p. But if the radius of the eccentric or of the orbit is taken as 1^p, and *e* is expressed in its *decimal* parts, as is our custom (that is, if e/60 is put for the *e* of the Greeks and of al-Battānī), plainly 1 must be written everywhere for 60^p.

g) — p. 48, ll. 17–20.

The radii of the epicycles of the moon and the five planets. The correction of the numbers is easy, for these radii are the sines of the greatest equation of the centre (at the mean distance from the earth, if the five planets are concerned). Let the greatest equations (the mean ones for the planets) and their sines be taken from al-Battānī's tables; there will result: I 221

Planet	*Greatest equation (mean for the planets)*	*Its sine (radius of the epicycle)*	*Errors of the codex*	*Errors of Plato*	*Ptolemy*	*al-Farghānī*
Moon	5° 1′ 0″	5^p 15′	6^p	—	5^p 15′	6 1/3^p
Saturn	6° 13′	6^p 29′ 50″	—	7″	6^p 30′	6 1/2^p
Jupiter	11° 3′	11^p 30′	—	add. 5″	11^p 30′	11 1/2^p
Mars	41° 9′	39^p 27′ 22″	19^p 25′	55′	39^p 30′	39 1/2^p
Venus	45° 59′	43^p 9′	—	44^p 9′ 5″	43^p 10′	43 1/6^p
Mercury	22° 2′	22^p 30′ 30″	29^p	—	22^p 30′	22 1/2^p

From the reading of the codex and of Plato it seems that al-Battānī wrongly wrote the radius of the epicycle of Mars as 39^p 25′ 22″ for 39^p 27′ 22″; or perhaps ٥, 25, is a copyists' error for ٧, 27. — In the other planets he kept Ptolemy's numbers[1].

The astronomers of the caliph al-Maʾmūn in the *Tested Tables* (on the evidence of Ibn Yūnus, ch. VI, Caussin pp. 234–236) and Ḥabash, ff. 35r ff., have the mean equations of the planets as al-Battānī does, and the greatest equation of the moon as 5°, leaving out the minute. Ibn Yūnus, loc. cit., changed only a little the mean equation of Venus (46° 25′) and of Mercury (22° 24′).

On chapter 29

a) — p. 48, ll. 33–37.

Like Ptolemy, all the Arab astronomers known to me count the day from noon. Sédillot, however (from chapter I or III of Ibn Yūnus, which are unpublished?), p. 228 of the translation of Oloug Beg, says that Yaḥyā ibn Abī Manṣūr, the chief author of the *Tested Tables*, began the day at midnight, like Hipparchus; on the evidence of al-Bīrūnī, *Chron.* p. 6 of the text and of the translation, the author of the tables of the *Shahriyārān al-Shāh*, of which I have already spoken on p. 218, note 4, did the same.

b) — p. 49, ll. 8–13.

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[*The equation of time.* — In al-Battānī's theory of this equation some things are worth noting. In those times (and still today, nearly) the greatest retardation of the sun with respect to its mean motion came about 20° of Aquarius, the greatest advance a little after the beginning of Scorpio. This is what al-Battānī means by the words «*the place of the diminution is at about 2/3 of Aquarius*» etc. Astronomers today have defined mean time so that it holds the middle place between the oscillations of the solar motion, and so that the correction to be applied to true time to find mean time (that is, the equation of time) is sometimes additive and sometimes subtractive. Ptolemy did otherwise; like Theon after him in the *Handy Tables* (see Delambre, *Histoire de l'astronomie*

[1] In al-Farghānī the radius of the lunar epicycle is 6 1/3^p, both in Golius's edition (ch. XVI, p. 65) and in the Latin translation printed at Nuremberg (1537, fol. 16v). I have followed this edition in setting down the radius of the epicycle of Mars; Golius's edition gives 39 1/6^p.

ancienne, Paris 1817, vol. II, pp. 630–632), he determined the course of mean time so that it agreed with true time when true time had advanced more than in all the other parts of the year, that is, at the beginning of Scorpio. Then the two times agree; in all the other parts of the year true time will lag behind mean time, and the correction to be applied to it to find the true time, that is, the *equation of time*, will always be additive. Its greatest amount therefore fell at 20° of Aquarius, as is gathered from the table of this equation that Delambre gives after Ptolemy and Theon, *Astr. anc.*, II, 632.

Al-Battānī, on the other hand, for what reason I do not well know, supposes that mean and true time agree when true time is at its greatest retardation, that is, at 20° of Aquarius; for this is what the words mean, «*by this day-and-night* [20° of Aquarius] *compute the other days-and-nights of the year in this book*». Plainly, then, in all the other parts of the year true time will *precede* mean time; the correction, or equation, will always be subtractive, and will reach its greatest amount near the beginning of Scorpio, as appears from the table, vol. II, p. 64. Both ways lead to the same result; but Ptolemy's mean time *precedes* al-Battānī's mean time by the whole greatest equation, which Ptolemy makes $33^m\ 30^s$, al-Battānī $31^m\ 12^s$. For Ptolemy's mean time agrees with true time when true time is at its greatest advance, al-Battānī's when true time is at its greatest retardation. So al-Battānī, to make his astronomical tables comparable with Ptolemy's when both are entered with the same argument of time, had to increase all his epochs of the celestial motions by as much as those motions amount to in the interval of $31^m\ 12^s$. Such a correction is negligible in all the motions of the planets; for the sun it is a little more than 1′; for the moon, however, it reaches 18′. In this way at least it can be understood to some extent why he says below (ll. 28–29) *that he added 18′ to the mean longitude of the moon in his tables*. When al-Battānī's mean time is $0^h\ 0^m\ 0^s$, Ptolemy's reaches $0^h\ 32^m\ 12^s$. It also follows that the tables of Ptolemy and of al-Battānī, as regards the equation of time, cannot be compared directly with each other; for in the one the equation is nothing when in the other it is greatest. Had al-Battānī built his table on the elements that Ptolemy used, the sum of the two equations would always be $33^m\ 30^s$; but this sum is always slightly different, since al-Battānī's equation rests on somewhat different elements (another obliquity of the ecliptic, another eccentricity of the solar orbit, another length of the tropical year). — Schiaparelli.]

It is worth reporting here what Delambre, p. 98, relates from ch. III of Ibn Yūnus: «Il [Ibn Yūnus] compte cinq méthodes différentes pour calculer l'équation du tems. La première est celle de Ptolémée. Théon nous l'a expliquée de façon à ne rien laisser à désirer. La seconde est celle des auteurs des Tables qui commencent par le point de l'écliptique où l'équation est nulle. Suivant la troisième, on commence par le point où l'équation est soustractive (et sans doute la plus grande); dans la quatrième, on commence la Table au point où l'équation est additive; enfin, dans la cinquième, on commence au point équinoxial du printemps. Toutes ces méthodes au fond n'en sont qu'une» («He [Ibn Yūnus] counts five different methods of computing the equation of time. The first is Ptolemy's; Theon has explained it to us so as to leave nothing to be desired. The second is that of the authors of the tables that begin with the point of the ecliptic where the equation is nothing. By the third, one begins with the point where the equation is subtractive (and doubtless greatest); in the fourth, the table is begun at the point where the equation is additive; lastly, in the fifth, one begins at the vernal equinoctial point. All these methods are at bottom only one»)[a].

Ḥabash, f. 16v, follows Ptolemy's usage.

c) — p. 49, ll. 15 ff.

[a]Nallino's Addenda et emendanda (Part II, pp. XXIV–XXV) show from Muḥammad ibn Abī Bakr al-Fārisī that the fifth method is that of Yaḥyā ibn Abī Manṣūr and the authors of the *Tested Tables*, who began from the beginning of Aries; they are translated with Part II.

[«*The given mean place of the sun*» here means the initial mean place; «*the other mean place*», the mean place that the sun has reached at that time. Plainly, as in the *Almagest*, the problem is to correct a given interval of true time so as to know the interval of mean time corresponding to it. — Schiaparelli.]

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The rules are set out not very clearly both by Ptolemy, *Almag.* III, 8 (ed. Halma vol. I, pp. 207–210), and by al-Battānī; far better in the commentary of Theon, which Delambre reports in his notes on that passage of the Almagest (ed. Halma, vol. I, p. 28 of the notes). They can be described briefly in this way. Let a given interval of true time between t and t' be converted into an interval of mean time. For each of the times t and t' let the mean and the true longitudes be derived from the tables of the sun; the difference between the two mean longitudes will be the arc of the ecliptic run in mean motion, the difference between the two true longitudes the arc described in true motion. Let the intervals of right ascension that correspond to those two arcs of the ecliptic be taken; there will come out the arcs of the equator that measure the mean time and the true time between t and t'. The difference of these two arcs of the equator, turned into parts of an equal hour, will be the correction of true time by which mean time is found, that is, the equation of time; it will be added to the unequal, or true, time when the equatorial arc of the latter is shorter than the mean one, or subtracted from the unequal time when its arc is longer than the mean one. The given interval will come out expressed in mean, or equinoctial, days-and-nights.

«Suivant les modernes», says Delambre, loc. cit., «l'*équation* du temps est l'excès de la longitude moyenne sur *l'ascension droite vraie*, et cet excès converti en temps à raison de 15° par heure, est l'*équation* du temps, qui sert à convertir le temps vrai en temps moyen. Ce procédé est au fond le même que celui de Ptolémée, à la réserve que le précepte de Ptolémée donne la différence de deux *équations* du temps, au lieu que la règle moderne ne donne qu'une seule *équation.* Avec nos tables d'*équation du temps* nous corrigerions séparément les deux temps donnés, et nous prendrions la différence des deux temps corrigés, qui seroit l'intervalle exprimé en temps moyens» («According to the moderns, the equation of time is the excess of the mean longitude over the true right ascension, and this excess, turned into time at 15° to the hour, is the equation of time, which serves to convert true time into mean time. This procedure is at bottom the same as Ptolemy's, except that Ptolemy's rule gives the difference of two equations of time, whereas the modern rule gives only one equation. With our tables of the equation of time we would correct the two given times separately and take the difference of the two corrected times, which would be the interval expressed in mean time»).

Tannery, *Recherches sur l'astronomie ancienne*, Paris 1893, p. 178, says that the Ptolemaic method prevailed until Tycho Brahe. See Riccioli, *Alm. novum*, I, 177–78.

d) — p. 49, ll. 25–34.

[The wording of this passage is very obscure. First he means that *in the method he used the mean time will always be less than the time that the place of the sun indicates*, that is, than true time; this is clear from what I said in note *b*, pp. 221–22. — He then adds that *this will be so for a very long time*, until *the motion of the solar apogee* arising from precession *has* perceptibly *changed the course of the part of that equation that comes from the anomaly of the solar motion.* — Then he speaks of the 18 minutes by which he increased the longitude of the moon at the epoch of the tables, whose cause I explained in note *b*, p. 222. — Lastly, confirming what precedes, he indicates that the equation of his table must *always be subtracted* from true time to give mean time, and *added* to mean time to find true time.

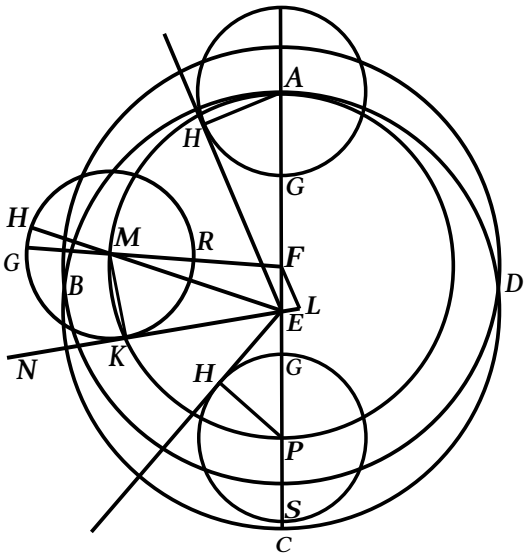
The words «*at the end of a long space of time, in which the variation of the apogee of the sun, which we have found on the ecliptic, will grow; so that what comes from the inequality of the motion of the sun will change*» point to two discoveries of great importance. The one, which I have already

mentioned, is that *the equation of time changes slowly over the years with the secular motion of the apogee*; the other, discussed in note *c* on ch. XXVIII (pp. 216–219), is that *Ptolemy's opinion of the unchanging distance of the solar apogee from the equinox is false.* — Schiaparelli.]

On chapter 30

a) — p. 50 – p. 51, l. 27.

Everything so far is taken from *Almag.* V, 2 (ed. Halma vol. I, pp. 288–292). The figure and the proof of p. 51, however, which are corrupt in the codex and in Plato, I have easily healed in the translation with very small changes, under Schiaparelli's guidance. The figure of the codex and of Plato is this:



[While what is read on pp. 52–54 agrees well enough with the doctrine of the *Almagest*, the figure and the proof of p. 51 (as the codex and Plato have them) are false and contrary to what is set out afterwards and to the figure of p. 53. Plainly the writer of the archetype from which the Escorial codex and Plato's codex arose, having drawn the figure carelessly, foolishly tried to fit the proof to it. — What the circle may be that has its centre at F and passes through B and D cannot be understood; it is wholly useless. But a point of the greatest consequence in Ptolemy's theory is missing, which is marked in the corrected figure with the letter F′; it is as far from the earth E as the centre F of the eccentric is on the opposite side, and it shows the true direction of the apogee H of the epicycle, from which the computation of the arc GH is derived. Instead of F′ the centre of the eccentric itself is considered, and by the two lines FM, EM an arc GH is defined which is falsely supposed to be the arc GH of the figure of p. 53. There is a similar error in the proof. When small corrections are made on p. 51, ll. 13–18, al-Battānī's text is well restored. — Schiaparelli.]

b) — p. 51, l. 36 – p. 52, l. 10.

All the numbers are taken from the *Almagest*, as I have noted in their place. For the 5° 1′ of the simple inequality the authors of the *Tested Tables* (see Ibn Yūnus ch. VI, Caussin p. 232) and Ḥabash, ff. 35r ff., adopted only 5°.

[The simple inequality at its greatest amount (5° 1′) is the *least* value of the greatest equation on the epicycle; the greatest is, according to Ptolemy, 7° 40′. The mean comes out 6° 20 1/2′; such would be the greatest equation of the centre of the moon if the second inequality, which we call the *evection*, did not exist. Today the mean value is put at 6° 18′, the second inequality at ± 1° 16′; so according to Ptolemy 5° 1′, 6° 20 1/2′, 7° 40′; according to the moderns 5° 2′, 6° 18′, 7° 34′. — Schiaparelli.]

The third inequality, which we call the *variation*, was unknown before Tycho Brahe; some wrongly said that Abū l-Wafāʾ (d. 388 H, which began on 2 Jan. 998) had already found it, as Carra de Vaux has sufficiently shown, *L'Almageste d'Aboû 'l-Wéfa al-Bûzdjâni*, pp. 434–471 (in the Journal Asiatique, 8th series, vol. 19, 1892). But in the tables of Ptolemy (and of the Arabs), as Tannery first noticed, *Rech. sur l'astr. ancienne*, p. 214, besides the term 1° 19′ 30″ sin (2 [☾ − ☉] − *m*), that is, besides the evection[1], there is also a term 18′ sin 2 (☾ − ☉), which is nearly half the variation. See also Carra de Vaux, p. 470.

c) — p. 53, ll. 18–20.

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[The words printed in italics seem to me interpolated. Al-Battānī had made the computation before, $42'\,38'' = 60'\,\frac{1^\circ\,53'}{2^\circ\,40'}$; now, as if it were something new and wonderful, we are taught that $2^\circ\,40' = 1^\circ\,53'\,\frac{60'}{42'\,38''}$. Having made this computation (which gives 2° 39′ for 2° 40′), our interpolator, not knowing what this number is, looks for it in the sixth column of the table and finds it under the argument 110°, which has nothing to do with this problem; entangled in his own error, he changes 110° into 120°, to get back to Ptolemy's number, which measures the angle AEM, that is, the mean anomaly of the eccentric. But in the table something entirely different corresponds to the number 120°. Plainly the author of these lines understood nothing of this computation, and he cannot possibly be al-Battānī. — Schiaparelli.]

d) — p. 54, ll. 19–30.

[*The lunar tables.* — When al-Battānī's tables are compared with Ptolemy's, what the author relates appears true. To the motion of the moon in longitude he added the correction that he had already applied before (ch. XXVII, p. 42) to the motion of the sun, so that the mean elongation (☾ − ☉) remains as in Ptolemy. The anomalistic motion too is unchanged.

Ptolemy's motion of the node, or of the latitude[2], he says he found diminished by 27′ in the interval of about 750 years between him and Ptolemy. Comparing the tables of the two, I find al-Battānī's motion in latitude smaller than Ptolemy's indeed, but not by exactly that amount; it is quite certainly gathered that al-Battānī diminished the motion in latitude by exactly three minutes in a century, which in 750 years makes 22 1/2′ instead of 27′. — The number 27′ of the codex and of Plato is, however, certain, for it is confirmed by what is related in ll. 28–30. If a correction of 27′ appeared in 750 years, in the roughly 1200 years that had elapsed since the era of Dhū l-Qarnayn the correction will be $27' \times \frac{1200}{750} = 43.2'$. These are the 3/4 of a degree by which, as al-Battānī says, the motion in latitude had to be diminished from the epoch of the era to «*this our time*», before the epochs of that motion, noted in the last column of vol. II, p. 72, were computed.

It may seem strange to some that al-Battānī found a correction of the motion in latitude (or of the motion of the node) of only 27 minutes (22 1/2′ in the tables) in 750 years, when in this space of time the correction of the motion in longitude had come to no less than 4° 31′. The matter

[1]By the sign ☾ I denote the mean longitude of the moon, by ☉ the mean longitude of the sun, by *m* the mean anomaly of the moon.

[2]The motion in latitude in al-Battānī is the sum of the motion in longitude and the motion of the node.

is explained thus. The ancient astronomers referred the motion of the moon in longitude to the motion of the sun, using the epochs of observed eclipses; so whatever error affected the motion of the sun affected the motion of the moon in the same interval of time. But the motion of the sun (or the length of the year) depended on observations of the equinoxes, each of which could give an error *of several hours*, indeed *of days*. — The motion of the nodes of the moon, on the other hand, was found from the position of the moon with respect to the nodes in the same eclipses; and this position could easily be determined within *a few minutes of arc* from the magnitude of the eclipses observed.

I add a comparison of the lunar tables of Ptolemy, al-Battānī and Hansen[1]:

	Inclination of the lunar orbit	*Mean motion in longitude in 100 Roman years*	*Mean motion in anomaly in 100 Roman years*	*Mean motion in latitude in 100 Roman years*
Ptolemy	5° 0′	307° 21′ 37″	198° 29′ 56″	82° 2′ 43″
al-Battānī	5 0	307 57 46	198 29 56	81 59 43
Hansen	5 8 2/3	307 53 40	198 50 38	82 2 40

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Al-Battānī, then, determined the motion in longitude much better than Ptolemy; he kept the inclination of the orbit and the motion in anomaly unchanged; he corrected the motion in latitude less well. But in this motion Hansen's number is not so certain that the comparison can be made safely. — Schiaparelli.]

The astronomers of the caliph al-Ma'mūn had already found, in the *Tested Tables* (see Ibn Yūnus ch. VI, Caussin p. 232), a motion in longitude better than Ptolemy's, but they had put the motion of the node too small. The lunar motions in a Persian year of 365 days (without intercalation) are these:

	Mean motion in longitude	*Mean motion in anomaly*	*Mean motion of the node*
Ptolemy[2]	129° 22′ 46″ 13‴ 50^{IV} 32^{V} 30^{VI}	88° 43′ 7″ 28‴ 41^{IV} 13^{V} 55^{VI}	19° 20′ 0″ 58‴ 53^{IV} 52^{V} 25^{VI}
Tested Tables	129 23 5 51	88 43 7 28 41	19 19 33 40
Ḥabash	129 23 6	88 43 7	19 19 39
al-Battānī	129 23 8	88 43 7	19 20 37[a]

e) — p. 54, l. 33.

«*And it ends beyond the sphere of Mercury*». — In this very chapter, p. 61, he computes that the shadow of the earth, when the sun is at its greatest geocentric distance, extends 254 2/3 earth radii beyond the centre of the earth. In chapter L, pp. 120–121, he finds

the nearest distance of Mercury	64 1/6 earth radii
the farthest distance of Mercury and the nearest distance of Venus	166 earth radii
the farthest distance of Venus	1070 earth radii

[1] [Whose numbers are taken from the *Monthly Notices of the Royal Astronomical Society*, vol. XV, p. 7. — Schiaparelli.]

[2] These numbers, computed to the sixths, are in *Almag.* IV, 3 (ed. Halma vol. I, pp. 222–25, and the tables, pp. 226–237). The motion of the node is not given there, but it is drawn at once by taking the mean motion in longitude from the mean motion in latitude.

[a] Al-Battānī's motion of the node in 365 days is 19° 19′ 38″ in Schiaparelli's computation of the fourth eclipse (p. 231), which takes it from his tables, and the motions in 100 Roman years in the table above likewise make it smaller than Ptolemy's; the 19° 20′ 37″ printed here is 59″ greater.

Hence it follows that in al-Battānī's system of the heavens the shadow of the earth passes beyond the sphere of Mercury and enters the sphere of Venus itself.

f) — p. 55, l. 15.

[Mercury and Venus are never seen in opposition to the sun; hence it is understood at once why neither planet can fall into the shadow of the earth. — Schiaparelli.]

g) — p. 56, ll. 9–21.

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The first eclipse (of the sun), observed on 8 Āb (August) 1202 of Dhū l-Qarnayn, 891, at $1^h\ 7\ 1/2^m$, in the city of al-Raqqa[1]. It bears the number 4995 in Oppolzer, *Canon der Finsternisse* (Denkschriften der k. Akad. der Wissenschaften zu Wien, Mathem. Cl., vol. LII, 1887). Halley, in the dissertation cited on p. 56, note 2, tried to clear the elements of all these eclipses of the countless errors of the editions of Plato, having collected the roots of the mean motions, corrected as far as was possible, from chapters XXVII, XXVIII and XXX. He restored much successfully; but he indicated only the emendations, leaving out the computations by which he had found them.

[To correct the numbers of the codex and of Plato properly, the author's computations must be repeated from his own tables and hypotheses. First, al-Battānī says that the time of the true conjunction preceded the apparent conjunction, that is, the middle of the eclipse, by about an eighth of an hour (see p. 56, ll. 16–17); it follows that the time of the true conjunction was $1^h\ 0^m$ after noon. Further, there is no doubt that al-Battānī's time, given in seasonal hours in the observation, is *true* time, not *mean*. With the sun at 19° of Leo, al-Battānī's table (vol. II, p. 63) gives the equation of days-and-nights $3°\ 16' = 13^m\ 4^s$, to be subtracted; hence the mean time of the true, or geocentric, conjunction is $0^h\ 47^m$ after noon, leaving out a few seconds of time, which are wholly negligible in this matter. The computation, then, taken from the volume of tables, pp. 72–83, is as follows:

Mean motions	*Mean long.* ☉	*Mean long.* ☾	*Mean anom.* ☾	*Asc. node* ☾
Dhū l-Qarnayn 1191	342° 48′ 50″	165° 40′ 17″	108° 20′ 0″	176° 23′ 0″
+ 11 years	359 21 47	9 35 37	282 2 11	212 42 13
Tammūz (July)	150 48 17	215 59 20	198 56 33	8 6 6
+ 8 days	7 53 7	105 24 40	104 31 12	0 25 25
+ 30 minutes	0 1 14	0 16 28	0 16 20	0 0 4
+ 15 minutes	0 0 37	0 8 14	0 8 10	0 0 2
+ 2 minutes	0 0 5	0 1 6	0 1 5	0 0 0
	140° 53′ 57″	137° 5′ 42″	334° 15′ 31″	37° 36′ 50″
				137° 5′ 42″
mean motion in lat. ☾				174° 42′ 32″

[1]From astronomical observations made in 1836 Chesney established the longitude of the city as 39° 3′ 57″ E of Greenwich and the latitude as 35° 55′ 35″ N; the place of observation was the north-west corner of the ruins of the palace once built by the caliph Hārūn al-Rashīd. [*Translator:* The print has 1835; Nallino's Addenda et emendanda (Part I, p. LXXIV) correct it to 1836.]

a) **Sun**[1]:				
mean long.	140° 53′ 57″	mean long. ☉	140° 53′ 57″	
apogee 1202	82 25 0	equation ☉	− 1 39 37	
mean anomaly ☉	58° 28′ 57″	true long. ☉	139° 14′ 20″	
b) **Moon**:				
mean long.	137° 5′ 42″	mean long. ☾	137° 5′ 42″	
mean long. ☉	140 53 57	simple equation	+ 2 5 48	
elongation ☾ − ☉	3 48 15	true long. ☾	139° 11′ 30″	
double elong.: 2 (☾ − ☉)	7° 36′ 30″			
mean anomaly ☾	334° 15′ 31″	mean motion in lat. ☾	174° 42′ 32″	
equation of the anomaly	− 1 5 24	simple equation	+ 2 5 48	
true anomaly ☾	333° 10′ 7″	true motion in lat.	176° 48′ 20″	
		dist. of ☾ from the node	3° 11′ 40″	at the true conj.
		subtract for 1/8 hour	4 8	
		dist. of ☾ from the node	3° 7′ 32″	at the app. conj.
		true lat. ☾	+ 0° 16′ 18″	at the app. conj.

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I have gathered the elements of the eclipse derived in this way in column I of the small table set out below; but we could not use them for our purpose. For al-Battānī did not take the longitude of the moon from his tables; rather, he supposed, as was certain from the observation, that the true longitudes of the sun and the moon were equal. So for the moon he took 139° 14′ instead of the 139° 11′ 30″ that the tables gave. Column II of the following table shows the numbers that al-Battānī must have found from this hypothesis of his; it will now be proper to compare them with the numbers of the codex and of Plato, set out in columns III and IV. Halley's emendations, mentioned above, have also been added in column V.

	Col. I	*Col. II*	*Esc. codex*	*Plato*	*Halley's corr.*
Mean longitude of the sun	140° 53′ 57″	140° 53′ 57″	140° 54′	140° 54′	…
True longitude of the sun	139 14 20	139 14 20	139 14	139 14	…
Mean longitude of the moon	137 5 42	137 8 32	*134* 7	137 *50*	137° 6′
True longitude of the moon	139 11 30	139 14 20	139 14	139 14	…
True anomaly of the moon	333 10 7	333 12 57	*332 57*	*332 57*	333 7
Mean motion in lat. of the moon	174 42 32	174 45 22	174 43	174 43	…
True motion in lat. of the moon	176 48 20	176 51 10	176 51	176 *11*	176 51
True latitude of ☾ at the middle of the eclipse	0 16 18	0 16 12	0 16	0 16	…
True motion in lat. of ☾ at the middle of the eclipse	176 52 25	176 55 15	176 55	*177 11*	176 55

Leaving aside small differences of one or two minutes, the comparison of column II with the numbers of the Escorial codex proves beyond all doubt that two corrections must be made in the codex, namely قلز 137° for قلد 134° in the mean longitude of the moon, and سلج يج 333° 13′ for سلب نز 332° 57′ in the true anomaly of the moon. — The corrupt numbers of the codex and of Plato are printed in italic, or sloping, type. — Schiaparelli.]

h) — p. 56, l. 22 – p. 57, l. 9.

The second eclipse (of the sun), observed on 22 Kānūn II 1212 of Dhū l-Qarnayn, that is, 22 January 901, at about 20 1/3^{h}[a] in the city of Antioch[2]. — In Oppolzer's canon it bears the number 5016.

[1] With the longitude of the apogee computed as 82° 15′ in the year 1191 of Dhū l-Qarnayn; see note *c* on ch. XXVIII (p. 214).

[2] According to M. Hartmann's map, *Karte des Liwa Haleb*, 1 : 220 000 (in the Zeitschrift der Gesellschaft für Erd-

[a] The print has 20^{h} 1/8. Al-Battānī puts the middle of the eclipse at Antioch about 3 2/3 equal hours before noon of 23 Kānūn II (p. 56), that is, at 20^{h} 20^{m} after noon of the 22nd, the time taken below.

[The noon of which al-Battānī speaks is certainly *true* noon, that is, the culmination of the sun[1]. At Antioch, then, the *true* time of the observation was $20^h\ 20^m$; in the city of al-Raqqa it was $20^h\ 30^m$, or a little more, as al-Battānī has it. Hence the *mean* time at al-Raqqa is found to be $20^h\ 29^m$, since the equation of days-and-nights for the eighth degree of Aquarius is found to be $15' = 1^m$ (see the tables in vol. II, p. 61). He says, moreover, that the true conjunction took place 50^m later; so the mean time of that conjunction was $21^h\ 19^m$ for the meridian of al-Raqqa, or a little more. For this reason the computations will have to be made for $21^h\ 20^m$.

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Mean motions	*Mean long.* ☉	*Mean long.* ☾	*Mean anom.* ☾	*Asc. node* ☾
Dhū l-Qarnayn 1211	343° 0′ 0″	299° 15′ 50″[a]	148° 1′ 59″	203° 11′ 23″
Kānūn I	301 36 34	71 58 41	37 53 5	16 12 10
+ 22 days	21 41 3	289 52 51	287 25 47	1 9 55
+ 20 hours	0 49 17	10 58 49	10 53 15	0 2 39
+ 1 hour	0 2 28	0 32 56	0 32 40	0 0 8
+ 1/3 hour	0 0 49	0 10 59	0 10 53	0 0 3
	307° 10′ 11″	312° 50′ 6″	124° 57′ 39″	220° 36′ 18″[b]
				312 50 6
mean motion in lat. ☾				173° 26′ 24″

a) **Sun**:

mean long.	307° 10′ 11″	mean long. ☉	307° 10′ 11″	
apogee	82 34 0	equation ☉	+ 1 25 40	
mean anom. ☉	224° 36′ 11″	true long. ☉	308° 35′ 51″	

b) **Moon**:

mean long. ☾	312° 50′ 6″	mean long. ☾	312° 50′ 6″	
mean long. ☉	307 10 11	simple equation	− 4 14 0	
elongation ☾ − ☉	5 39 55	true long. ☾	308° 36′ 6″	
double elongation	11° 19′ 50″			
mean anomaly ☾	124° 57′ 39″	mean motion in lat. ☾	173° 26′ 24″	
equation of the anomaly	+ 1 41 0	simple equation	− 4 14 0	
	126° 38′ 39″	true motion in lat. ☾	169° 12′ 24″	
		dist. of ☾ from the node	10° 47′ 36″	at the true conj.
		add for 50^m	27 33	
		dist. of ☾ from the node	11° 15′ 9″	at the app. conj.
		true lat. ☾	0° 58′ 25″	at the app. conj.

The elements of the eclipse found by this computation are entered in column I of the following table. The true longitudes of the sun and the moon differ from each other by only a few seconds; it is therefore plain that the time of the true conjunction derived from the observation agrees very well with al-Battānī's tables. Al-Battānī, however, put the true longitude of both bodies at 308° 35′, that is, about one minute less; accepting this hypothesis, I have tried to restore his computations in column II. Lastly, in the three following columns I have added the numbers of the codex, of Plato and of Halley for comparison.

kunde zu Berlin, vol. XXIV, 1894), the longitude of the city is 36° 8 2/3′ E of Greenwich, the latitude 36° 11′ 52″ N.

[1] [The ancient astronomers knew only one *noon*, namely the one that divides the natural day in the *middle*, and that was therefore called μεσημβρία, or *midday*. — Schiaparelli.]

[a] The print has 299° 15′ 30″; the computation of the fourth eclipse below has 299° 15′ 50″ for the same year, and the sum 312° 50′ 6″ is made with it.

[b] The print has 220° 36′ 10″; the motions of the node add up to 220° 36′ 18″, with which the motion in latitude, 173° 26′ 24″, is made.

	Col. I	*Col. II*	*Esc. codex*	*Plato*	*Halley*
Mean longitude of the sun	307° 10′ 11″	307° 9′ 20″	307° 9′	307° 9′	…
Mean longitude of the moon	312 50 6	312 49 0	312 49	312 49	…
True longitude of the sun	308 35 51	308 35 0	308 35	308 35	…
True longitude of the moon	308 36 6	308 35 0	308 35	308 35	…
True anomaly of the moon	126 38 39	126 37 33	126 *32*	*156 55*	126° 35′
Mean motion in lat. of the moon	173 26 24	173 25 18	173 25	173 *55*	173 25
True motion in lat. of the moon	169 12 24	169 11 18	*199* 11	169 *41*	169 11
The same at the middle of the eclipse	168 44 51	168 43 45	*198* 45	168 45	…
True latitude of the moon	0 58 25	0 58 29	0 59	0 59	…

The errors of the codex, then, are: in the anomaly of the moon لب 32′ for لز 37′; in the true motion in latitude of the moon قضط 199° for قصط 169°; in the true motion in latitude of the moon at the middle of the eclipse قضح 198° for قصح 168°. — Schiaparelli.]

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i) — p. 57, ll. 10–20.

The third eclipse (of the moon), observed on 23 Tammūz 1194 of Dhū l-Qarnayn, that is, 23 July 883, very little more than 8 equinoctial hours after noon, in the city of al-Raqqa. It bears the number 3232 in Oppolzer's canon.

[With the sun at 4° 2′ of Leo, the equation of days-and-nights is found to be 3° 4′, or $12^m\ 16^s$, to be subtracted; hence the true time[a] of the middle of the eclipse, which is also that of the true opposition, is very little more than $7^h\ 48^m$. Entering al-Battānī's tables with this time, I obtained numbers everywhere smaller than al-Battānī's. But weighing the matter more attentively, I saw that the word *very little* can be interpreted so that $7^h\ 55^m$ may rather be taken instead of $7^h\ 48^m$.

Mean motions	*Mean long.* ☉	*Mean long.* ☾	*Mean anom.* ☾	*Asc. node* ☾
Dhū l-Qarnayn 1191	342° 48′ 50″	165° 40′ 17″	108° 20′ 0″	176° 23′ 0″
+ 3 years	359 17 19	28 9 24	266 9 22	57 58 53
Ḥazīrān (June)	120 14 58	167 31 14	153 55 40	6 27 37
+ 23 days	22 40 11	303 3 26	300 29 41	1 13 5
+ 7 hours	0 17 15	3 50 35	3 48 38	0 0 56
+ 55 minutes	0 2 16	0 30 11	0 30 0	0 0 7
	125° 20′ 49″	308° 45′ 7″	113° 13′ 21″	242° 3′ 38″
mean long. ☾				308° 45′ 7″
mean motion in lat. ☾				190° 48′ 45″

a) *Sun*:			
mean long.	125° 20′ 49″	mean long. ☉	125° 20′ 49″
apogee	82 17 42	equation	− 1 19 17
mean anom.	43° 3′ 7″	true long. ☉	124° 1′ 32″
b) *Moon*:			
mean long. ☾	308° 45′ 7″	mean long. ☾	308° 45′ 7″
mean long. ☉	125 20 49	simple equation	− 4 43 43
elongation ☾ − ☉	183 24 18		304 1 24
double elongation	6° 48′ 36″		180
		true long. of the *shadow*	124° 1′ 24″
mean anom. ☾	113° 13′ 21″	mean motion in lat. ☾	190° 48′ 45″
equation of the anom.	+ 1 0 17	simple equation	− 4 43 43
true anom. ☾	114° 13′ 38″	true motion in lat. ☾	186° 5′ 2″
		latitude ☾	− 0° 31′ 52″

[a] So in the print. Subtracting the equation from the observed true time gives the mean time, as for the other three eclipses; the table on p. 232 gives $7^h\ 48^m$ as the mean time of this eclipse.

From this is derived the computation of the first column of the following table; there follow the numbers of the codex, of Plato and of Halley:

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	Col. I	*Esc. codex*	*Plato*	*Halley*
Mean longitude of the sun	125° 20′ 49″	125° 21′	125° *51′*	125° 21′
True longitude of the sun	124 1 32	124 2	124° *5*	124 1
Mean longitude of the moon	308 45 7	308 45	308 45	…
True longitude of the *shadow*	124 1 24	124 2	124 *5*	…
Mean anomaly of the moon	113 13 21	113 *0*	*93 0*	113 8
True anomaly of the moon	114 13 38	114 *9*	*94 10*	114 10
Mean motion in lat. of the moon	190 48 45	190 49	190 49	…
True motion in lat. of the moon	186 5 2	186 5	186 5	…
Southern latitude of the moon	0 31 52	0 32	0 32	…

In the mean anomaly of the moon I think the minutes have dropped out, for in place of ه 0′ one must read يج 13′. In the true anomaly one must read قيد يج 114° 13′, or قيد يد 114° 14′, instead of قيد ط 114° 9′. — Schiaparelli.]

k) — p. 57, ll. 21–32.

I. — *The fourth eclipse* (of the moon), observed at Antioch on 2 Āb 1212 of Dhū l-Qarnayn, that is, 2 August 901, 15 1/3^h after noon (in true time), or 15 1/3^h + 1/4^h if the time is reduced to the meridian of the city of al-Raqqa. It is number 3261 in Oppolzer's canon.

[With the sun at 14° 36′ of Leo, the equation of days-and-nights comes to 3° 10′, to be subtracted, or 13^m. Hence follows the mean time in the city of al-Raqqa, namely 15^h 22^m; the computations may be made for this moment of time.

Mean motions	*Mean long.* ☉	*Mean long.* ☾	*Mean anom.* ☾	*Asc. node* ☾
Dhū l-Qarnayn 1211	343° 0′ 0″	299° 15′ 50″	148° 1′ 59″	203° 11′ 23″
+ 1 year	359 45 46	129 23 8	88 43 7	19 19 38
Tammūz (July)	150 48 17	215 59 20	198 56 33	8 6 6
+ 2 days	1 58 17	26 21 10	26 7 48	0 6 21
+ 15 hours	0 36 58	8 14 7	8 9 56	0 1 59
+ 22 minutes	0 0 54	0 12 5	0 11 59	0 0 3
	136° 10′ 12″	319° 25′ 40″	110° 11′ 22″	230° 45′ 30″
				319 25 40
mean motion in lat. ☾				190° 11′ 10″

a) Sun:			
mean long.	136° 10′ 12″	mean long.	136° 10′ 12″
apogee	82 33 12	equation	− 1 33 59
mean anom.	53° 37′ 0″	true long.	134° 36′ 13″
b) Moon:			
mean long. ☾	319° 25′ 40″	mean long. ☾	319° 25′ 40″
mean long. ☉	136 10 12	simple equation	− 4 48 57
elongation	183 15 28	true long. ☾	314° 36′ 43″
double elongation	6° 30′ 56″		
mean anom. ☾	110° 11′ 22″	mean motion in lat. ☾	190° 11′ 10″
equation of the anomaly	+ 0 57 39	simple equation	− 4 48 57
true anom. ☾	111° 9′ 1″	true motion in lat.	185° 22′ 13″
		lat. ☾	− 0° 28′ 9″

This computation agrees well enough with al-Battānī's numbers, as appears from the following table:

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	Col. I	*Esc. codex*	*Plato*	*Halley*
Mean longitude of the sun	136° 10′ 12″	*134*° 10′	136° 10′	…
True longitude of the sun	134 36 13	134 36	134 36	…
Mean longitude of the moon	319 25 40	319 24	319 *54*	319° 24′
True longitude of the moon	314 36 43	314 36	314 36	…
Mean anomaly of the moon	110 11 22	110 *4*	110 7	…
True anomaly of the moon	111 9 1	111 *5*	*91 5*	111 5
Mean motion in lat. of the moon	190 11 10	190 10	*109* 10	190 10
True motion in lat. of the moon	185 22 13	185 21	185 *51*	185 21
True southern lat. of the moon	0 28 9	0 28	0 28	…

In the Escorial codex, then, the degrees of the mean longitude of the sun (قلو 136° for قلد 134°), the minutes of the mean anomaly (يا 11′ for د 4′) and the minutes of the true anomaly (ط 9′ for ه 5′, an easy error in Maghribi script) must be corrected. — Schiaparelli.]

II. — *General remarks on these four eclipses.*

[On these four eclipses, observed by al-Battānī and brought into computation, it is fitting to add a few remarks; and first on their excellent agreement with al-Battānī's tables. The following table shows, in the first place, for each eclipse the mean time of the true conjunction from the observation, for the meridian of the city of al-Raqqa; in the second and third places, the true longitudes of the sun and the moon according to the tables for the same moment of the true conjunction (or opposition), which would certainly have to be equal (or to differ from each other by 180°) if no error had crept into the observation and the tables. In the fourth place it shows the difference ☾ − ☉, which gives the error of the elongation of the moon arising from the tables; lastly, the difference between the time of the true conjunction (or opposition) taken from the observation (which I denote by the letter O) and the time of the true conjunction (or opposition) derived from the tables (which I mark with the letter T).

Eclipses	*Mean time from observation*	*True longitude from the tables*		*Difference in elongation* ☾ − ☉	*Difference of time at the true syzygy* O − T
		of the sun	*of the moon or shadow*		
I. 891, 8 Aug.[a]	$0^h\ 47^m$	139° 14′ 20″	139° 11′ 30″	− 2′ 50″	$-\ 5.7^m$
II. 901, 22 Jan.	21 20	308 35 51	308 36 6	+ 0 15	+ 0.5
III. 883, 23 Jul.	7 48	124 1 14	123 57 34	− 3 40	− 7.3
IV. 901, 2 Aug.	15 22	134 36 13	134 36 43	+ 0 30	+ 1.0[b]

The errors ☾ − ☉ and O − T must certainly be regarded as very small; indeed, they seem much smaller than should be expected in observations made without a telescope, at a time when the means of determining time accurately were also lacking. For this reason the suspicion arises that al-Battānī corrected his own tables to fit these eclipses.

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The matter stands otherwise if we compare the four eclipses with the more recent tables of the moon. The theory of the moon, although cultivated with the greatest labours by the astronomers

[a] The print has 7 Aug.; the eclipse is that of 8 August 891 (p. 226), $0^h\ 47^m$ after noon in mean time.

[b] The print has + 0.1^m. The other rows turn the difference in elongation into time at 2 minutes of time to 1′ (2′ 50″ into 5.7^m, 15″ into 0.5^m, 3′ 40″ into 7.3^m); at that rate 30″ give 1.0^m.

of our age, has not yet reached such certainty that the ancient observations of the Babylonians, Greeks and Arabs may safely be judged by it. On the contrary, a few years ago a man above all praise, Theodor Oppolzer, strove hard, in his tables of the syzygies and above all in the great *Canon of Eclipses*, to make those observations in some way help the theory. F. K. Ginzel afterwards took the same road: using the ancient observations of total eclipses of the sun, he very diligently sought to work out new corrections of Oppolzer's Canon[1], which R. Schram finally expressed in small tables admirably suited to the use of astronomers[2]. Thus, through the joint labours of three astronomers, tables have come into being, resting partly on a theoretical and partly on an empirical foundation, which must be considered to represent the course of the moon for ancient times more accurately than any others.

To make the comparison of our eclipses with these tables properly, one must first remember that al-Battānī's mean time is not the same as the mean time we use. It was shown earlier, in note *b* on ch. XXIX (pp. 221–22), that for al-Battānī the equation of days-and-nights is always subtractive and for Ptolemy additive, so that about 32^m must be added to al-Battānī's mean time to bring it to Ptolemy's usage. But the way in which our astronomers usually compute that equation is so arranged that it holds very nearly the middle between al-Battānī's method and Ptolemy's. Therefore, so as not to be entangled in overly intricate and unnecessary computations, it will be enough for our purpose to add to al-Battānī's mean time half the interval just mentioned, namely 16^m, to bring the author's numbers to our usage, which the authors of Oppolzer's Canon also followed.

The other reduction is that of longitude, from the meridians of the cities of Antioch and al-Raqqa, where the observations were made, to the meridian of Greenwich, which Oppolzer and his associates used. For Antioch (where the second and fourth eclipses were observed) I have taken the longitude east of Greenwich as $36^\circ\ 9' = 2^h\ 24^m\ 36^s$, on Hartmann's authority; for al-Raqqa (where the first and third eclipses were observed) $39^\circ\ 4' = 2^h\ 36^m\ 16^s$, according to Chesney's observations. With these things weighed, I have gathered the elements of the comparison in the following table.

Eclipses	*True time of the geocentric syzygy from al-Battānī*	*Equation of days + 16^m*	*Difference in longitude from Greenwich*	*Mean time of the geocentric syzygy at the meridian of Greenwich*		*Difference between al-Battānī and Ginzel*
				al-Battānī	*Oppolzer–Ginzel*	
I. 891, 7/8 Aug.	$1^h\ 0^m\ 0^s$	$+\ 2^m\ 56^s$	$2^h\ 36^m\ 16^s$	$22^h\ 26^m\ 40^s$	$22^h\ 13^m\ 12^s$	$+\ 13^m\ 28^s$
II. 901, 22 Jan.	21 10 0	+ 15 0	2 24 36	19 0 24	18 46 42	+ 13 42
III. 883, 23 Jul.	8 0 0	+ 3 44	2 36 16	5 27 28	5 9 24	+ 18 4
IV. 901, 2 Aug.	15 20 0	+ 3 20	2 24 36	12 58 44	12 31 18	+ 17 26[a]

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It is gathered at once that in all the eclipses the observation puts the syzygy about a quarter of an hour later than the tables do. Some part of this difference must, I have no doubt, be ascribed to the observations themselves. For the beginning of the obscuration cannot be perceived by the naked eye until some minutes of time have elapsed after the geometrical contact; and this must have happened above all to the ancient astronomers, who could not predict the beginning

[1] *Astronomische Untersuchungen über Finsternisse*, in the Sitzungsberichte der mathem.-naturwissenschaftliche Classe der k. Akad. der WW. zu Wien, vol. LXXXV, section II, 1882, pp. 663–747; vol. LXXXVIII, sect. II, 1883, pp. 629–755; vol. LXXXIX, sect. II, 1884, pp. 491–559.

[2] *Reductionstafeln für den Oppolzer'sche Finsterniss-Canon, zum Uebergang auf die Ginzel'schen empirischen Correctionen*, in the Denkschriften der k. Akad. der WW. zu Wien, mathem. Classe, vol. LVI, 1889.

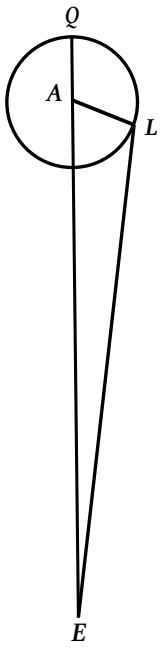
[a] So in the print: $12^h\ 58^m\ 44^s$ less $12^h\ 31^m\ 18^s$ is $27^m\ 26^s$, while the next paragraph speaks of about a quarter of an hour in all four eclipses, so that one of the three numbers of this row is misprinted.

of an eclipse accurately enough from tables. This uncertainty had no place in noting the end of the obscuration; so in the estimate of the middle of the eclipse there must necessarily have been some retardation. But whether the whole error of a full quarter of an hour must be ascribed to this cause I dare not affirm; for al-Battānī himself, in a rather rough manner, seems to have noted his times no more accurately than in intervals of ten minutes.

Lastly, it will be useful to compare the *magnitudes* of the lunar eclipses observed by al-Battānī with Oppolzer's computations. The magnitude of eclipse III, which al-Battānī estimated at 10 digits, was 11.2 digits according to Oppolzer. The magnitude of eclipse IV, which al-Battānī saw to be 12 digits or a little less, is found from Oppolzer's computations to be somewhat greater, namely 12.6 digits. — Al-Battānī did not report the magnitudes of the two solar eclipses, I and II. — Schiaparelli.]

l) — p. 58, ll. 1–4.

[«*The sun was towards the apogee*». — For in eclipse III the anomaly of the sun was 42°, in IV 52°.



«*The moon occupied the same place, about 1 1/2^p less than the mean distance*». — That is, about 1 1/2^p of the 60 parts into which the mean distance (AE in the figure of p. 51) of the moon in conjunction with the sun and in opposition to it is taken to be divided. In fact, in eclipse III the true anomaly of the moon on the epicycle was 114° 16′, in IV 111° 4′; and these are the amounts of the angle QAL in the figure drawn here, where E is the earth, A the centre of the epicycle, AE = 60^p, AL = 5.247^p. The computation of the triangle AEL gives EL = 58.04^p for eclipse III and EL = 58.32^p for IV. So what al-Battānī calls «about 1 1/2^p» was exactly 1.96^p and 1.68^p; plainly he made the computation approximately.

«*Between the first and the second latitude there were 0° 3′ 50″*». — According to al-Battānī's numbers the first latitude (eclipse III) «was about 0° 32′ towards the south», the second (eclipse IV) «about 0° 28′» (he omitted to indicate the side, but from the motion in latitude, 185° 21′, it is plain that it too was southern). The difference, then, is 4′ for 3′ 50″. But a more exact computation that I made gave 0° 31′ 52″ and 0° 28′ 3″[a] for those two latitudes; difference 3′ 49″, agreeing well with

[a]The computation of the fourth eclipse on p. 231 and the table on p. 232 have 0° 28′ 9″, with which the difference would be 3′ 43″.

al-Battānī's 3′ 50″. Plainly al-Battānī had made the computations of the eclipses more precisely, and afterwards gave round numbers in the book. — Schiaparelli.]

m) — p. 58, ll. 4–5.

«*The difference of the two eclipses is contained 8 3/4 times in the lunar diameter*». — This does not agree with the author's observations. In eclipse III the part obscured was 5/6 of the diameter; in IV almost the whole moon. The difference is 1 − 5/6 = 1/6; so he should have written 6 for 8 3/4. [I think that al-Battānī, so as not to be led to an absurd lunar diameter barely reaching 23 minutes (as would necessarily arise from the factor 6), wanted somehow to bring to a numerical ratio the «a little more than 5/6» of the third eclipse and the «very little less than its diameter» of the fourth; but since it was difficult to express these by numbers, he chose the factor 8 3/4, estimating the lunar diameter at 33 1/2′. Al-Battānī only wanted to show (not very clearly, it seems) that the diameter of the moon he proposed could be reconciled well enough with these two eclipses. So what al-Battānī says must not be considered a true determination of the lunar diameter, but only a proof that the observations do not conflict with the diameter he had adopted. — Schiaparelli.]

n) — p. 58, ll. 9–11.

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[«*The hourly motion of the moon at the times of the syzygies …*». — These hourly motions, the greatest 36′ 10″ and the least 30′ 12″ (Plato 15″, for in him 2 very often becomes 5), do not agree with the table of vol. II, p. 88, from which 36′ 4″ and 30′ 18″ are derived. I do not know the cause of the difference; perhaps al-Battānī's computation was made only approximately. — In the computations that follow he supposes that the greatest diameter is to the least as the greatest hourly motion to the least. This would be true if the hourly motion in *anomaly* were equal to the motion in *longitude*, as happens with the sun; but even in the diameter of the moon the error seems negligible. — Schiaparelli.]

o) — p. 58, ll. 16–18.

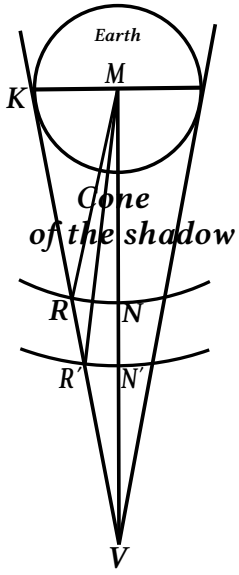
He puts the *semidiameter of the shadow* at about 0° 38 1/2′ when the moon is at its apogee, and at about 0° 46′ when the moon is at its perigee; the difference, then, is about 0° 7 1/2′.

Ibn Yūnus (introduction, Caussin pp. 70–72) makes these observations: «Abū l-ʿAbbās al-Faḍl ibn Ḥātim al-Nayrīzī[1] and Muḥammad ibn Jābir al-Battānī also err when they put the variation of the semidiameter of the shadow between the greatest and the least distance [of the moon] at 7′ 12″. Anyone even less versed in the studies of geometry than those two men will recognize that this opinion is false, if he considers Ptolemy's demonstration [*Almag.* V, 15] of the distance of the sun from the centre of the earth. It is impossible for the variation of the semidiameter of the shadow between the apogee distance of the moon and the perigee distance to be as small as they make it, or even 8′ or 9′; it must come out much greater if in determining it you follow the way of geometrical demonstration. I once made the computation, and found 0° 10′ 17″, putting the sun and the moon[2] at the greatest distance from the earth. The variation that affects the semidiameter of the shadow on account of the nearness of the sun to the earth cannot exceed 1 minute»[3].

[1] Anaritius in the Latin writers of the Middle Ages. He died about 922; Suter, *Die Mathematiker und Astronomen*, p. 45, no. 88.

[2] [Better: the sun at its apogee, the moon at the apogee of the eccentric. — Schiaparelli.]

[3] Al-Battānī about 0′ 50″.



[What Ibn Yūnus says is demonstrated in this way. Let RN be the radius of the shadow at the least distance, measured in minutes and seconds by the angle RMN; R′N′ the radius of the shadow at the greatest distance, measured by the angle R′MN′. KRM is the greatest parallax of the moon, amounting according to al-Battānī to 63′ 51″[a] at the syzygies; KR′M is the least parallax, according to al-Battānī 53′ 34″. Let φ be the angle KVM at the vertex of the cone of the shadow. In the triangle MRV we have

$$KRM = 63'\,51'' = \varphi + RMV,$$

$$KR'M = 53'\,34'' = \varphi + R'MV,$$

$$\text{difference} \quad \overline{10'\,17''} = RMV - R'MV = RMR',$$

and RMR′ is, as an angle, the difference between the greatest and the least shadow. Plainly the angle 10′ 17″ is always the same, although the distance of the sun and the angle of the cone of the shadow vary. — Schiaparelli.]

Al-Battānī plainly found his numbers in this way. He supposed (see p. 58, ll. 7–8) that the ratio of the diameter of the shadow to the diameter of the moon is the one that Ptolemy has, *Almag.* V, 14 (ed. Halma vol. I, p. 343), namely 2 3/5 to 1; with this ratio the diameters of the shadow 91.858′ and 76.7′ are derived from al-Battānī's greatest and least diameters of the moon (35 1/3′ and 29 1/2′), and so the semidiameters 45′ 54″ and 38′ 42″[b]. The difference is 7′ 12″, which Ibn Yūnus mentions; al-Battānī writes about 7 1/2′, since he puts the semidiameters in round numbers as 46′ and 38 1/2′.

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p) — p. 58, ll. 27–31.

From his observations of eclipses al-Battānī determined the *angular diameters of the moon* and *of the sun* better than Ptolemy. The diameter of the *sun* was always 31′ 20″ in the Greek author, who did not know that it varies. Ours recognized the variation of the diameter and found for the different distances from the earth:

[a] The print has 64′ 51″ here and in the first line below; 63′ 51″ less 53′ 34″ gives the difference 10′ 17″ that follows, which Ibn Yūnus found, and 53′ 34″ is the angle whose sine is 1/64;10, the parallax at al-Battānī's greatest distance of the moon, 64;10 earth radii.

[b] The halves of 91.858′ and 76.7′ are 45′ 56″ and 38′ 21″, whose difference is 7′ 35″, al-Battānī's «about 7 1/2′»; 38′ 42″ is not half of 76.7′, and the computation does not give the 7′ 12″ of Ibn Yūnus.

31′ 20″ at the greatest distance, 32′ 30″ at the mean, 33′ 40″ at the least.

The astronomers of our age[1] have established:

31′ 31.98″ at the greatest distance, 32′ 3.64″ at the mean, 32′ 36.48″ at the least.

The diameter of the *moon* according to

	at the greatest distance	*at the mean*	*at the least*
Ptolemy	31′ 20″	33′ 20″	35′ 20″
al-Battānī	29′ 30″	32′ 25″	35′ 20″
the moderns[2]	29′ 26″	31′ 8″	33′ 34″

This is a matter of no small moment, for from his measurements al-Battānī (as Delambre, p. 39, already remarked) infers in ll. 29–30 *that annular eclipses of the sun sometimes occur*; these, on the contrary, could not have happened according to Ptolemy's measurements, which make the diameter of the sun equal to the diameter of the moon at apogee; see Delambre, *Histoire de l'astronomie ancienne*, Paris 1817, vol. II, pp. 234–235, and P. Tannery, *Recherches sur l'astronomie ancienne*, Paris 1893, pp. 224 and 233. Among the Greeks, however, in the first century BC, the Peripatetic philosopher Sosigenes had already recognized the variation of the solar diameter and, on the testimony of Proclus Diadochus[3], had said that in perigee eclipses the sun «is not seen wholly covered, but exceeds the lunar disc with the rim of its circumference and gives light unhindered», ὁρᾶσθαι μὴ ὅλον ἐπιπροσθούμενον, ἀλλὰ τοῖς ἄκροις τῆς ἑαυτοῦ περιφερείας ὑπερβάλλειν τὸν κύκλον τῆς σελήνης καὶ φωτίζειν οὐκ ἐμποδιζόμενον. Proclus wants to prove this opinion false by the Ptolemaic measurements, saying that if it were true it would follow that the apparent diameter of the moon at the apogee points differs from the diameter of the sun; and this is contrary to Ptolemy's doctrine.

The Arabs after al-Battānī did not call annular eclipses of the sun into doubt. In 1279, in an astronomy written in Syriac, Barhebraeus relates, p. 88 of the text: «They found that the diameter of the sun varies from the apogee to the perigee from 31′ to 34′[4], the diameter of the moon from 29′ to 36′. When the two apparent diameters are equal, the whole sun is eclipsed without any shadow being found to exceed; when the diameter of the sun is greater than that of the moon, not only is there no exceeding shadow, but on the sun a luminous crown is seen around the moon that hides it».

Even at the beginning of the seventeenth century some Europeans doubted that a total eclipse of the sun could occur; see G. Schiaparelli, *Le sfere omocentriche di Eudosso, di Callippo e di Aristotele*, Milan 1875, p. 62, note 31 (= Pubblicazioni del R. Osservatorio di Brera, no. IX).

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[Perhaps already through the first eclipse, of 8 August 891, al-Battānī could have been confirmed in his opinion that *annular* eclipses sometimes occur; for according to Oppolzer's *Canon* it was annular along an oblique line crossing Asia Minor, Cilicia and Mesopotamia. News of the marvellous spectacle that had appeared in places not very far away could have reached him as he observed at al-Raqqa. — Two other annular eclipses occurred not far from al-Raqqa, on 26 April 906 and 5 April 916, as appears from Oppolzer's *Canon*, chart no. 101. — Schiaparelli.]

[1] I follow those who publish, at the Paris observatory, the very famous ephemerides *La Connaissance des Temps* and the *Annuaire du Bureau des Longitudes*.

[2] According to the Babylonians of the fourth and third centuries BC, 29′ 26.9″, 31′ 51.5″, 34′ 16.2″, which are better even than the Ptolemaic and Albatenian quantities. See Kugler, *Die babylonische Mondrechnung*, Freiburg i. Br. 1900, p. 126.

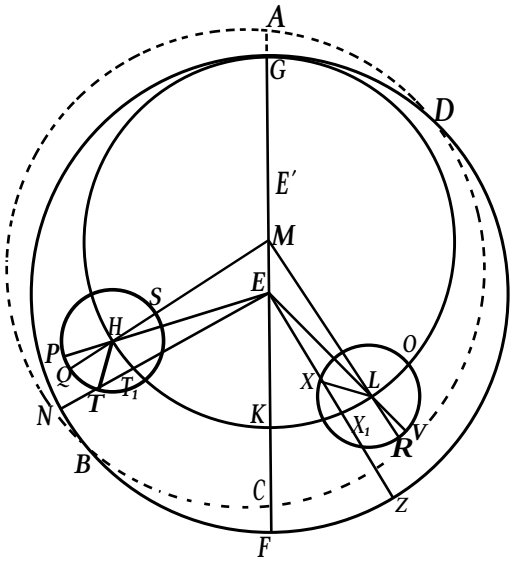
[3] *Hypothèses et époques des planètes de C. Ptolémée, et Hypotyposes de Proclus Diadochus publiées et traduites par Halma*, Paris 1820, p. 111. — See also Delambre, *Hist. de l'astr. anc.*, vol. I, p. 314.

[4] Barhebraeus almost always uses round numbers; if a given quantity consists of degrees, he leaves out the minutes; if of minutes, the seconds.

On chapter 31

a) — p. 64, l. 11 ff.

[*The spheres or circles of the planets.* — The figure is very badly drawn, and the proof has been spoiled by an unskilled hand. The circle ABCD, concentric with the earth and lying in the plane of the ecliptic, cannot lie in the same plane as the circle GBFD, which is indeed concentric with the earth but inclined at a small angle to the ecliptic. So if GBFD is drawn with the rest of the figure in the plane of the sheet, the circle ABCD, which falls outside the sheet, cannot be represented by a circle with centre E. — The figure also puts the nodes B, D of these circles 90 degrees from the line GK that determines the apsides of the eccentric; this happens in no planet, nor could it have seemed true to al-Battānī. — The lines ETN, EXZ ought not to be tangent to the epicycle, for the planet must be considered to occupy any place on the epicycle. Nor do they lie in the *ecliptic*, but in the plane of the *inclined circle* GBFD and of the eccentric GHKL; and even this is true only when the epicycle is *wrongly* supposed to lie in the plane of the eccentric.



I have tried to restore the figure as al-Battānī probably intended it; the figure printed on p. 64, following the codex and Plato, must be blamed on the unskilled writer of the archetype, who, as we have already seen elsewhere, afterwards tried to fit the text to a badly drawn figure. He represented the point A as if it lay in the plane GBFD; he also wrongly imagined that M, the centre of the eccentric, is the point about which the uniform motion of the centre of the epicycle takes place, whereas this motion takes place about the point E′, which is therefore the centre of the circle that the followers of Ptolemy called the *equant*[1]. The point E′ is such that ME′ = ME; see Delambre, *Hist. de l'astr. anc.*, vol. II, p. 316.

I think it not improbable that in al-Battānī's true figure there was a circle, the *equant*, drawn about the centre E′ (on which the centre of the epicycle runs through equal angles in equal times); that afterwards some reader turned it into a circle concentric with the earth and the ecliptic, and that everything thereby came out confused. It would then be clear why he placed the centre of such a circle on the line GF, and the nodes on the straight line perpendicular to GF, namely at B and D.

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[1] Among the Arabs, as in al-Farghānī ch. XIV, al-Jaghmīnī pp. 249–250, and others, *al-falak al-muʿaddil li-l-masīr*, «the sphere (or circle) that equalizes the motion». [*Translator:* The print has al-mumaththil; Nallino's Addenda et emendanda (Part I, p. LXXIV) correct it to al-muʿaddil.]

The whole proof is false in that, against Ptolemy's hypothesis, it puts M as the centre of the uniform motion of the epicycle H on the eccentric; E′ should have been considered instead of M. Other small corrections have already been indicated in their places, pp. 64–65, by which al-Battānī's true proof is perhaps restored. — Schiaparelli.]

b) — p. 65, ll. 17–23.

[«*The cause of the retrogradations of the planets is ...*». — On a given day let Δl be the motion of the planet in longitude on the ecliptic, Δm the mean daily motion, $\Delta\varepsilon$ the change of the equation; al-Battānī says that, since $\Delta l = \Delta m + \Delta\varepsilon$, if $\Delta\varepsilon$ is negative and greater than Δm, then Δl too is negative and the planet retrograde. This is the sense of lines 17–19. — What is read afterwards, «if the true place of the planet» as far as l. 24, is a most obscure repetition of the same thing; but I do not know how the passage is to be healed. Let l be the true longitude today and $l + \Delta l$ tomorrow, m the mean motion today and $m + \Delta m$ tomorrow, ε today's equation and $\varepsilon + \Delta\varepsilon$ tomorrow's; undoubtedly

$$l = m + \varepsilon, \qquad \Delta l = \Delta m + \Delta\varepsilon, \qquad l + \Delta l = m + \Delta m + \varepsilon + \Delta\varepsilon.$$

Which of these equalities is indicated in the text I do not understand. — Schiaparelli.]

c) — p. 65, ll. 23–34.

Delambre rightly writes, p. 40: «Les rétrogradations arrivent lorsque le mouvement sur l'épicycle est égal ou contraire au mouvement du centre de l'épicycle; ce qui a lieu, dit-il [i.e. al-Battānī], quand la planète est dans le rayon visuel tangent à l'épicycle, ce qui n'est pas exact, car alors le mouvement dans l'épicycle est nul, et il reste le mouvement du centre sans aucune compensation[1]; c'est une inadvertance» («The retrogradations occur when the motion on the epicycle is equal or contrary to the motion of the centre of the epicycle; which happens, he [al-Battānī] says, when the planet is on the visual ray tangent to the epicycle; this is not exact, for then the motion on the epicycle is nothing, and the motion of the centre remains without any compensation; it is an oversight»). — The same Delambre, p. 99, speaking of ch. IX of Ibn Yūnus, has this: «Il énonce, comme Albatégni, que la rétrogradation commence lorsque la planète se meut suivant la tangente à l'orbite. Si ce n'est pas une erreur positive, c'est au moins une expression bien impropre. Ptolémée s'était exprimé avec beaucoup plus de justesse» («He states, like Albategni, that the retrogradation begins when the planet moves along the tangent to the orbit. If this is not a positive error, it is at least a very improper expression. Ptolemy had expressed himself much more accurately»).

It is plain from this that al-Battānī and Ibn Yūnus meant to indicate the places of the retrogradations of the planets only approximately. — But another error too is found in our author, when he supposes that the points of contact of the tangents (X and T) are the same places in which the epicycle cuts the eccentric; from the very figure of the codex and of Plato (p. 64) this appears impossible.

Undoubtedly much in this chapter that serves to explain the theory of the planets does not come from al-Battānī. The author wrote this chapter above all to declare how he had corrected Ptolemy's planetary tables (pp. 65–66); he considered the cosmographical elements already known to his readers (as in the other parts of the book), and so perhaps had mentioned the spheres or circles of the planets only summarily, with the help of a figure. This brief exposition of the theory, I think, the usual unlearned reader tried to make fuller according to his small powers, and in this way he disfigured al-Battānī's text.

d) — p. 66, ll. 1–8, 12–15.

I. — *The motions in longitude of the superior planets*, he says (ll. 1–2), he found greater than those that Ptolemy had established. In fact the mean motions in an interval of 20 Julian years[1], or 20 Egyptian years (without intercalation) + 5 days, are gathered from the tables as follows:

	Saturn	*Jupiter*	*Mars*
al-Battānī	244° 42′ 44″ 20‴	247° 17′ 34″ 26‴	228° 22′ 10″ 49‴
Ptolemy	244 38 1 38	247 12 33 49	228 15 22 17
difference	0° 4′ 42″ 42‴	0° 5′ 0″ 37‴	0° 6′ 48″ 32‴

So in the roughly 740 Julian years between the epoch of Ptolemy's tables (137) and the epoch of al-Battānī's tables (880) the differences came to ♄ 2° 54′ 19″ 54‴, ♃ 3° 5′ 22″ 49‴, ♂ 4° 11′ 55″ 45‴.

Tested Tables and Ḥabash[2]	244° 43′ 14″	247° 17′ 46″ 20‴	228° 23′ 2″ 20‴
modern astronomers	244 42 5	247 15 31	228 20 14

It is plain that al-Battānī found better numbers than Ptolemy's, and almost beyond doubt that he also used the observations of the astronomers of the caliph al-Maʾmūn.

II. — *The apogees of the planets* he says he found in nearly the same places in which they should have been according to Ptolemy's tables (with the precession of the equinoxes in the interval of time added), except, however, that the longitude of the apogee of Jupiter was about 8° less than the Ptolemaic longitude.

The longitudes of the apogees for the year 1191 of Dhū l-Qarnayn (beginning 1 Sept. 879) are given at the end of chapter XLV (p. 114) and in the headings of the tables of the equations of the planets; by Ptolemy, *Almag.* IX, 3 (ed. Halma vol. II, pp. 128 ff.), for the year 137[a]. But before we compare them with each other, we must inquire whether the numbers of the codex are rightly written.

Al-Bīrūnī, in an unpublished little book on the elements of astronomy, f. 28r and v, writes: «The places of the apogees in this our time, namely in the year 420 of the hijra [beginning 19 Jan. 1029], according to what Muḥammad ibn Jābir al-Battānī found, must be: apogee of ☉ 24° 32′ Gemini, ♄ 6° 48′ Sagittarius, ♃ 16° 43′ Virgo, ♂ 8° 30′ Leo, ♀ 24° 39′ Gemini, ☿ 23° 43′ Libra». In the interval of 149 solar years the precession according to al-Battānī and al-Bīrūnī was 2° 15′; this difference, then, must be found in all the planets. A table of comparison gives:

	♄	♃	♂	♀	☿
al-Bīrūnī	246° 48′	166° 43′	128° 30′	84° 39′	203° 43′
al-Battānī	244 28	164 28	126 18	82 14	201 28
difference	2° 20′	2° 15′	2° 12′	2° 25′	2° 15′

[1]That is, at that place the planet shares in the motion of the epicycle on the eccentric.

[1]I choose this interval because only for it do al-Battānī's tables, vol. II, p. 103, give the thirds. In Ptolemy's numbers I leave out the fourths, fifths and sixths.

[2]Fols. 41r–49v. — The elements of the *Tested Tables*, that is, the mean motions in an Egyptian or Persian year of 365 days, are in Ibn Yūnus, ch. VI (Caussin pp. 234 ff.).

[a]The longitudes that are compared with al-Battānī's on p. 241 (224° 10′, 152° 9′, 106° 40′, 46° 10′, 181° 10′) are Ptolemy's apogees for the epoch of his tables, the beginning of the era of Nabonassar (747 BC). For his own time the *Almagest* (books IX–XI) puts them 8° 50′ further on (8° 51′ for Jupiter): ♏ 23°, ♍ 11°, ♋ 25° 30′, ♉ 25°, ♒ 10°, that is, 233°, 161°, 115° 30′, 55° and 190°.

If in al-Bīrūnī the minutes of Saturn are corrected to مج 43′ from مح 48′, and those of Venus to كط 29′ from لط 39′[a], the difference 2° 15′ appears everywhere, except only in *Mars*, where the emendation is difficult. Al-Battānī's numbers, then, seem right, except in Mars. I 240

In the *Libro de las láminas de las VII planetas*, whose first part (comprising 16 chapters) is a Spanish version of an Arabic work composed by Abulcacim Abnaçahm, that is, Abū l-Qāsim Aṣbagh Ibn al-Samḥ[1], part I, ch. XIII[2], taking [al-Battānī's] precession of one degree in 68 lunar years, gives the apogees of the planets for the year 416 H (beginning 3 March 1025), which plainly come from al-Battānī's. The numbers are these: ♄ 246° 40′, ♃ 166° 40′, ♂ 129° 10′, ☉ 84° 27′, ♀ = ☉ («tal cuemo ell auxe del sol», «just like the apogee of the sun»), ☿ 203° 40′. The interval between al-Battānī and Ibn al-Samḥ is 145 Julian years; the precession, then, is 2° 12′. Let this quantity be taken from those longitudes; there result the apogees for the epoch of al-Battānī's tables:

♄ 244° 28′, ♃ 164° 28′, ♂ 126° 58′, ☉ and ♀ 82° 15′, ☿ 201° 28′,

which agree entirely with al-Battānī's numbers, except for the apogee of Mars, where the minutes of the Escorial codex, يح 18′, seem to have to be changed to نח 58′[3].

In the second part of the same book *de las láminas de las VII planetas*, which is a version of a little work of the Arab astronomer Abuizac el Zarquiel, that is, Abū Isḥāq al-Zarqālī, ch. II (*Libros del saber de astronomia*, vol. III, pp. 272–273), the author says that he reduced the apogees of el Bateni (that is, of al-Battānī) to the year 473 (or 468?) of the hijra; but the Roman numerals teem with so many errors that they can in no way be restored.

Al-Jaghmīnī, p. 253, who takes al-Battānī's precession, also gives us help. His apogees are these:

☉ 87° 10′ 33″, ♄ 249° 23′ 33″, ♃ 169° 23′ 33″, ♂ 131° 53′ 46″, ♀ 87° 10′ 33″, ☿ 206° 23′ 33″[4].

All differ from al-Battānī's by exactly the quantity 4° 55′ 33″, except *Mars*, where the difference is 5° 35′ 46″. But if al-Battānī's minutes يح 18′ are corrected to نח 58′, as Ibn al-Samḥ, al-Kharaqī and Ptolemy too require, the difference again becomes 4° 55′ 46″, only the seconds disagreeing. — The precessional difference 4° 55′ 33″ answers to an interval of 325 Julian years and 30 days; al-Battānī's apogees are for the year 1191 of Dhū l-Qarnayn, beginning, by al-Battānī's usage, on 1 *September* 879; so al-Jaghmīnī reduced them to the year 1516, beginning, by the common usage, on 1 *October* 1204[5].

On the apogee of *Venus*, which al-Battānī puts equal to the apogee of the sun, and whose minutes are uncertain between 14′ and 15′, see note *c* on ch. XXVIII (p. 214).

With the minutes of *Mars* corrected to 58′, the comparison of al-Battānī with Ptolemy is this: I 241

[1] He died in the city of Granada on 18 Rajab 426 H (29 May 1035); see Suter, p. 85, no. 194. Some wrongly say that the year of his death was 476.

[2] *Libros del saber de astronomia del Rey D. Alfonso X de Castilla, copilados, anotados y comentados por D. Manuel Rico y Sinobas*, Madrid 1863–67, vol. III, p. 262.

[3] That the same correction is necessary is plain from what I shall say below and from al-Kharaqī, book I, ch. XV, faṣl V (Laurentian codex, f. 45v), who gives the apogees for the beginning of the year 1444 of Dhū l-Qarnayn, that is, for 1 Oct. 1132, plainly reducing al-Battānī's longitudes to his own time.

[4] Rudloff and Hochheim wrongly have لح 38″ for لج 33″ in this place.

[5] Rudloff and Hochheim wrongly translated that those apogees are «for the first year of Alexander [Dhū l-Qarnayn]»! — The codex kept in the National Library of Naples, written in a beautiful hand but full of errors in the numbers, has لاول سنة عشر لذي القرنين, «for the beginning of the year 1307 of Dhū l-Qarnayn». But عشر is a plain slip of the pen for غثيو 1516.

[a] The print has نط (59) for لط (39).

	Saturn	*Jupiter*	*Mars*	*Venus*	*Mercury*
al-Battānī	244° 28′	164° 28′	126° 58′	82° 14′	201° 28′
Ptolemy	224 10	152 9	106 40	46 10	181 10
difference	20° 18′	12° 19′	20° 18′	36° 4′	20° 18′

So a discrepancy of 8 degrees is in fact found in Jupiter, as al-Battānī had indicated; the great difference in Venus comes from al-Battānī's having established, against Ptolemy's opinion, that its apogee agrees in longitude with the apogee of the sun.

The difference 20° 18′ (due to precession), however, answers to an interval of 1340 Julian years, not to the 742 or 743 that lie between the epoch of Ptolemy and the epoch of al-Battānī. In the catalogue of the fixed stars (and see also ch. LI, p. 124) he had increased the Ptolemaic longitudes by 11° 10′; why he used an increase of 20° 18′ in the apogees of the planets I do not know. Plainly he relied on his own observations; but it is strange that he did not speak of these 20° 18′ − 11° 10′ = 9° 8′ added to the precession[a].

It should also be noted that the authors of the *Tested Tables*, who determined the apogees in the year 199 of Yazdegerd (beginning 28 Apr. 830), 215 H (beginning 27 Feb. 830), found longitudes exceeding Ptolemy's by much more than should have arisen from precession alone[b]. They are[1]:

	Saturn	*Jupiter*	*Mars*	*Venus*	*Mercury*
Tested Tables	242° 50′	172° 32′	124° 33′	82° 39′	201° 0′
Ptolemy	224 10	152 9	106° 40	46 10	181 10
difference	18° 40′	20° 23′	17° 53′	36° 29′	19° 50′

That the longitude of the apogee of the sun is the same as that of Venus had already been established by the *Tested Tables*; so too almost all the later Arabs[2].

[The comparison of the longitudes of the apogees of the superior planets found by Ptolemy and al-Battānī with the longitudes computed from the elements collected by Leverrier[3] is this:

	AD 140	*Ptolemy*[c]	*error*	*AD 880*	*al-Battānī*	*error*
Saturn	236° 41′	224° 10′	− 12° 31′	251° 9′	244° 28′	− 6° 41′
Jupiter	164 25	152 9	− 12 16	176 19	164 28	− 11 51
Mars	121 50	106 40	− 15 10	135 27	126 58	− 8 29

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It is difficult to pass judgement on these differences, for the eccentric of the Ptolemaic hypothesis does not answer exactly to the Keplerian ellipse. The circular and uniform motion of the planet

[1]The numbers of the *Tested Tables* are in Ibn Yūnus ch. VI (Caussin pp. 234–236, Delambre p. 93), but not without errors. With the help of the work of Ḥabash (who keeps those apogees), f. 28r (compared with 17v), I have corrected in Saturn 8^s 2° 50′ ن ب ح for Ibn Yūnus' 8^s 4° 30′ ل د ح, and in Mars 4^s 4° 33′ ج د ك for 3^s 3° 33′ ج ج ك.

[2]Likewise, among us, the *Alfonsine Tables* and Peurbach; see J. B. Riccioli, *Almagestum Novum astronomiam veterem novamque complectens*, Bologna 1651, vol. I, p. 563.

[3][His elements for 1 January 1850 are these: ♄ long. of the apogee 270° 7′, annual variation + 70.403″; ♃ apogee 191° 55′, var. + 57.910″; ♂ apogee 153° 18′, var. + 66.242″. — Schiaparelli.]

[a]Against Ptolemy's apogees for his own time (see the translator's note on p. 239) al-Battānī's are greater by 11° 28′ for Saturn, Mars and Mercury, by 3° 28′ for Jupiter, exactly 8° less, and by 27° 14′ for Venus. 11° 28′ is close to the precession of the 743 years at al-Battānī's 1° in 66 years (11° 15′) and only 18′ more than the 11° 10′ of his star catalogue; the 20° 18′, the 1340 years and the 9° 8′ come from comparing al-Battānī's apogees with Ptolemy's for the epoch of Nabonassar.

[b]Against Ptolemy's apogees for his own time the longitudes of the *Tested Tables* are greater by 9° 50′, 11° 32′, 9° 3′, 27° 39′ and 11° 0′, close, except for Venus, to the 10° 30′ of precession in the 693 years from 137 to 830 at 1° in 66 years.

[c]Ptolemy's column again holds his apogees for the epoch of Nabonassar; with those for his own time, 233°, 161° and 115° 30′, his errors are − 3° 41′, − 3° 25′ and − 6° 20′.

on the epicycle represents the elliptical and unequal motion of the earth about the sun only approximately, not precisely. So the ancients and the Arabs, to express the observations rightly, transferred the effect of the eccentricity and of the inequality of the motion of the earth to the eccentric; from which it follows that the eccentricities and apogees of Ptolemy and of the Arabs are nothing but compositions of the eccentricity and apogee of the planet with the eccentricity and apogee of the earth's orbit. If anyone wished to judge how much perfection those observations attained, he would have to undertake a long analysis for each planet. It is certain that the columns I have headed *errors* follow partly in fact from the imperfect theory, and partly from the fact that the data of the ancients cannot be compared directly with those of today.

As for *Venus*, whose apogee in al-Battānī and almost all the Arabs does not differ from the apogee of the sun, it must be noted that the eccentricity of the elliptical orbit that Venus describes about the sun is very small, so that this orbit can be considered an exactly circular epicycle; the deferent then becomes the orbit of the sun itself, about which, as we know today, the planet revolves. Although al-Battānī and the authors of the *Tested Tables* did not know this last fact, they nevertheless understood that the motion of Venus can be represented well when its *eccentric deferent* is given *the same eccentricity and the same apogee* as the orbit of the sun has. So there is not only the same longitude of the apogee, but also the same eccentricity and equation of the centre, as appears from al-Battānī's tables. Ptolemy, on the other hand, used the same eccentricity and equation of the centre (*Almag.* III, 5 and XI, 11, ed. Halma vol. I, p. 201, and II, 307), but put the apogee of Venus about 20° from the apogee of the sun[1a]. — Who judged more rightly cannot be decided until it has first been well established, with hard labour, how the eccentric of Venus represents the composition of the elliptical orbits of the earth and of Venus. — Schiaparelli.]

III. — *The anomalies of Venus and Mercury*, which he says (ll. 12–15) he found greater than the Ptolemaic ones. — Between the epoch of the tables of the Almagest (137) and the epoch of al-Battānī (879–880) there were about 740 Julian years. In this interval of time the motions in anomaly were, according to

	♀	☿
al-Battānī	317° 32′ 14″	175° 5′ 43″ 20‴
Ptolemy	313 3 47 16‴	174 9 3 0
difference	4° 28′ 26″ 44‴	0° 56′ 40″ 20‴

In Mercury, therefore, I have put 1° for the 2 1/2° of the codex and the 11 1/2° of Plato.

The authors of the *Tested Tables* (in Ibn Yūnus, ch. VI, Caussin 236) had already found the motion in anomaly of Venus a little greater than Ptolemy's; they had established 225° 2′ 0″ 2‴ in 365 days; Ptolemy 225° 1′ 32″ 28‴ 34^{IV}; Ḥabash (f. 53r) 225° 1′ 51″ 30‴ 53^{IV}.

[1]Moreover, he always makes the apogee of the sun immovable (see note *c* on ch. XXVIII, pp. 216–18), but gives the apogees of the planets the motion of precession.

[a]For his own time Ptolemy puts the apogee of Venus at ♉ 25° and that of the sun at ♊ 5° 30′, 10° 30′ apart; about 20° (19° 20′) is their distance with the apogee of Venus for the epoch of Nabonassar, 46° 10′, used on p. 241.

On chapter 32

a) — p. 67, ll. 16–22.

The *era of Alexander Dhū l-Qarnayn*[1] is the *Seleucid era*, whose epoch is commonly Monday 1 October 312 BC; but in al-Battānī, as I noted sufficiently on p. 67, note 1, it is *Saturday 1 September 312*. — See also Barhebraeus, *Syriac Astronomy*, p. 191 of the text: «The Greeks, in the time of paganism, began the year from the first Teshrī [October]; after they had accepted Christianity, from the month Ēlūl (= September)».

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The *era of Alexander of Macedon*, which al-Battānī mostly calls the *era from the death of Alexander*, is the one on which the *Handy Tables* (κανόνες πρόχειροι) of Ptolemy and Theon were built, which count the mean celestial motions «from the first year of Philip, son of Alexander[a], the founder of the city [of Alexandria]»[2]. It is the *era of Philip Aridaeus* (or Arrhidaeus), the successor, as they say, of Alexander the Great, whose brother by the same mother he was[b]; and so the Arabs mostly call it the *era of Philip*. — Years of this era are sometimes mentioned in the *Almagest*, where the observations of Hipparchus are treated; they are *vague* years there (that is, of 365 days, without the Julian intercalation), counted «from the death of Alexander», ἀπὸ τῆς Ἀλεξάνδρου τελευτῆς, and their epoch is Sunday 1 Thoth 425 of Nabonassar, that is, 12 November 324 BC. The same epoch, and with *vague* years too (whether Persian or Egyptian), is kept by al-Farghānī, Ḥabash, al-Masʿūdī, al-Bīrūnī, Ibn Yūnus, Kūshyār al-Jīlī and the rest; and so they put the interval between the Seleucid era and the era of Philip at 4341 days, that is, 11 vague years, 10 months and 26 days[3]. — But in Theon's *Handy Tables*, following an Alexandrian usage that arose in the first century AD, the years are *Julian*, and the *epoch of the era is 29 August 324* BC; al-Battānī (see p. 69, ll. 20–22) follows Theon, and so he puts the interval between the era of Dhū l-Qarnayn (counted from 1 September, not from 1 October) and the era of Alexander of Macedon at 12 Julian years less 3 days.

In the Persian work *Āyīn-i Akbarī*, which Abū l-Faẓl ʿAllāmī composed in 1594 by order of Akbar, emperor of India, this is read about the Roman era (*taʾrīkh-i rūmī*), that is, the era of Dhū l-Qarnayn: «Al-Battānī says that this era [is] from Philip, the father of Alexander Dhū l-Qarnayn, but names it after the name of his son, because of his fame. In his tables he sets the roots of the mean motions at this era»[4].

[1] That is, the Two-Horned. The name is in the *Qurʾān* XVIII, 82 ff.; it arose from the images in which Alexander the Great, as son of Jupiter Ammon, was portrayed with two horns. [*Translator:* XVIII, 82 is Flügel's numbering of the verse, 83 in the Cairo numbering now usual. Nallino's Addenda et emendanda (Part I, p. LXXIV) add that he had given this explanation after several scholars, chiefly because Clement of Alexandria complains that Alexander had himself portrayed in the likeness of a god, with horns; the true origin of the name is shown by Fr. Kampers, *Alexander der Grosse und die Idee des Weltimperiums in Prophetie und Sage*, Freiburg i. Br. 1901, pp. 71–85: in the legends of Alexander current in the East, above all among the Syrians, in the first centuries of the Christian era, he appears as the restorer of a new age and the saviour of the world, like the Messiah of the Hebrews, and is therefore given the two horns that Jewish writings call the emblem of the Messiah. All this flows from very ancient Babylonian beliefs (E. Schrader, *Die Keilinschriften und das Alte Testament*, 3rd ed. by H. Zimmern and H. Winckler, Berlin 1902, pp. 380–82), for which Naram-Sin, king of Babylonia, was already portrayed with two horns 25 centuries before Alexander. The Addenda are translated with Part I's front matter.]

[2] Ἀπὸ τοῦ πρώτου ἔτους Φιλίππου τοῦ μετ' Ἀλέξανδρον τὸν κτίστην; *Commentaires de Théon d'Alexandrie sur les tables manuelles astronomiques de Ptolémée*, publiés et traduits par Halma, Paris 1822, vol. I, p. 2 (Ptol.) and 27 (Theon).

[3] In al-Masʿūdī, *Tanbīh* 198, wrongly 12 years, 10 months, 20 days.

[4] *The Áin i Akbari of Abul Fazl i Mubárak i 'Allámí*, edited by H. Blochmann, Calcutta 1867–77, vol. I, p. 275 (= translated by H. Blochmann and H. S. Jarrett, Calcutta 1873–94, vol. II, p. 25; but in this place the translation is not good).

[a] The Greek of note 1, Φιλίππου τοῦ μετ' Ἀλέξανδρον τὸν κτίστην, says «of Philip, who came after Alexander the founder»; Philip was not Alexander's son.

[b] Philip Arrhidaeus was the son of Philip II by Philinna of Larissa, Alexander's half-brother by the same father; «uterinus», by the same mother, does not fit.

I have already spoken of the *names of the Coptic months* among the Arabs on p. 209, note 3; now I add a survey of them[a]:

Memphitic	*Greek*	*Arabic*[1]	*Theban*	*Arabic*[2]
ⲑⲱⲟⲩⲧ	θώθ	*tūt*	ⲑⲟⲟⲩⲧ	*tūt*
ⲡⲁⲟⲡⲓ	φαωφί	*bawūfī*	ⲡⲁⲟⲡⲉ	*bābah*
ⲁⲑⲱⲣ	ἀθύρ	*atūr*	ϩⲁⲑⲱⲣ	*hatūr*
ⲭⲟⲓⲁⲕ	χοϊάκ	*shuwāq*	ⲭⲟⲓⲁϩⲕ	*kiyahk*
ⲧⲱⲃⲓ	τυβί	*ṭūbī*	ⲧⲱⲃⲉ	*ṭūbah*
ⲙⲉⲭⲓⲣ	μεχίρ	*mākīr*	ⲙ̄ϣⲓⲣ	*amshīr*
ⲫⲁⲙⲉⲛⲱⲑ	φαμενώθ	*fāmīnūth*	ⲡⲁⲣⲙⲏϩⲁⲧ	*barmahāt*
ⲫⲁⲣⲙⲟⲩⲑⲓ	φαρμουθί	*fārmūthī*	ⲡⲁⲣⲙⲟⲩⲧⲉ	*barmūdhah*
ⲡⲁϣⲱⲛⲥ	παχώμ	*bākhūn*	ⲡⲁϣⲱⲛⲥ	*bashans*
ⲡⲁⲱⲛⲓ	παϋνί	*bāwūnī*	ⲡⲁⲱⲛⲉ	*bawūnah*
ⲉⲡⲏⲡⲓ	ἐπιφί	*efīfī*	ⲉⲡⲏⲡ	*abīb*
ⲙⲉⲥⲱⲣⲏ	μεσορῆ	*māsūrī*	ⲙⲉⲥⲱⲣⲏ	*misrī*

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b) — p. 68, l. 23 ff.

The era of the Copts. — Among the other Arabic writers known to me three eras are designated by this name: 1) the era of Nabonassar, 2) the era of Philip Aridaeus, 3) the era of Diocletian or of the Martyrs[3]. But al-Battānī means none of these. From what he relates later in this chapter and in ch. XLII it is plain that it is an era with *Julian* years, whose epoch is Friday 29 August 25 BC, for it is 287 Julian years less 3 days distant from the era of Dhū l-Qarnayn.

It is, then, a form of the *era of Augustus* or *of Actium.* Ptolemy mentions this era incidentally, *Almag.* III, 6 (ed. Halma vol. I, p. 204), from which passage Ideler, *Handbuch der mathematischen und technischen Chronologie*, Berlin 1825–26, vol. I, p. 116, deduced that its epoch was 1 Thoth 719 of Nabonassar, that is, Sunday 31 August 30 BC, and that its years were vague Egyptian years[4].

The difference that al-Battānī has in the epoch of the era must, I think, be explained in this way. Theon relates in the *Handy Tables* (ed. Halma, vol. I, pp. 30–32; see Ideler I, 158 ff.) that the first day of the month Thoth of the vague Egyptian year coincides again with the first day of Thoth of the fixed Alexandrian (or Julian) year after a period of 1460 fixed years, or 1461 vague years; and that this happened «*in the fifth year of Augustus*» (so on p. 30), or «*five years after the beginning of the reign of Augustus*» (so on p. 32). For this reason, Theon adds, one starts from this fifth year when days of the vague year have to be turned into days of the fixed Alexandrian year.

Theon's passage gives two pieces of information that disagree with each other. For if that period was completed in the fifth year of Augustus, the epoch of Theon's era becomes *Friday 30 August*

[1]When the vague years of Nabonassar are concerned.

[2]When they are not referred to the era of Nabonassar; they are still in use in Egypt.

[3]The era of Nabonassar and the era of Philip are in al-Farghānī, ed. Golius, ch. I, p. 6; that of Diocletian in Ḥabash f. 4v, Ibn Yūnus (Delambre 96) and al-Maqrīzī (*al-Mawāʿiẓ wa-l-iʿtibār fī dhikr al-khiṭaṭ wa-l-āthār*, Būlāq 1270 H, vol. I, pp. 261–262); the era of Nabonassar in al-Masʿūdī, *Prairies* vol. III, p. 400; those of Nabonassar, Philip and Diocletian in al-Masʿūdī, *Tanbīh* p. 198. — Abū l-Faḍl ʿAllāmī says only that it is ancient (vol. I, p. 274 = transl. II, 23).

[4]Petavius and the book *L'art de vérifier les dates* have 29 Aug. 30 BC, because they wrongly think that the years of the era were Julian years from the beginning. — To the first year of Augustus Ptolemy refers the mean places of the sun, the moon and the planets in the *Canobic Inscription*, which follows the Hypotheses (*Hypothèses de C. Ptolémée, publiées et traduites par Halma*, Paris 1820, p. 59).

[a]Nallino's Addenda et emendanda (Part I, p. LXXIV) add that the origin of the names of several Coptic months is very well discussed by A. Wiedemann, «Zu den ägyptischen Monatsnamen», in the Orientalistische Litteratur-Zeitung, vol. VI, 1903, cols. 5–7.

26 BC[a]; and so Mommsen, Lepsius and Th. H. Martin[1] think it was, relying on many historical arguments. But if the period was completed 5 years after the beginning of the reign of Augustus, the epoch of the era becomes *Friday 29 Aug. 25* BC; and in this way, but not well, Halma, Biot and Letronne among the moderns, and al-Battānī among the ancients, understood it. The years of this era are Julian.

So the Copts who used this era (which should be called *Alexandrian* rather than *of Augustus*), beginning the year on 29 August, preceded by three days, as al-Battānī says, the Greek-Egyptians, whose first day of the year was 1 September.

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Al-Battānī adds that in every fourth year the difference is not of *three* but only of *two* days: «they [i.e. the Greek-Egyptians] in every fourth year run ahead of them [i.e. the Copts] by one day». I find the explanation of this in a chronological fragment of the emperor Heraclius that Halma edited (*Commentaires de Théon sur les tables manuelles*, vol. III, p. 106). For it relates that «the Alexandrians put the intercalary day at the end of the year that precedes the Roman leap year; then, therefore, they begin their year not on the third but only on the second day before September». Al-Battānī plainly means this usage, which is set out more fully by al-Bīrūnī, *Chron.*, pp. 141 and 196 of the text (137 and 176 of the translation), and is still in force among the Copts today; so, after the Coptic intercalary year, up to the end of February of the Gregorian leap year, the correspondence of the months changes by one day[2]:

1 tūt	= 11 Sept.;	in the intercalary years:	12 Sept.
1 bābeh	= 11 Oct.	» » »	12 Oct.
1 hatūr	= 10 Nov.	» » »	11 Nov.
1 kiyahk	= 10 Dec.	» » »	11 Dec.
1 ṭūbah	= 9 Jan.	» » »	10 Jan.
1 amshīr	= 8 Feb.	» » »	9 Feb.
1 barmahāt	= 10 Mar.		

Martin, loc. cit., pp. 282–283, explains the origin of the Alexandrian usage in this way: it is probable that the Alexandrian tetraëteris, or four-year cycle, imitated a Sothiac four-year cycle that perhaps had its fourth year as the leap year; and that at the time when the fixed Alexandrian year was established, the Sothiac leap year was the third year of the Julian four-year cycle.

The era of Augustus of whatever kind, the Ptolemaic as well as the Theonian, was in force in Egypt itself for no long time. So it is rarely mentioned by Oriental writers, and not always rightly, it seems. From what Ḥabash, ff. 3r and 5r, and al-Bīrūnī, *Chron.*, pp. 29, 137, 141 and 196 of the text (33, 133, 137, 176 of the translation), say about the era of Augustus (*taʾrīkh Oghusṭos*), it is plain that they interpreted Theon's passage in the same way as al-Battānī, and put the epoch of the era on 29 Aug. 25 BC. But Kūshyār al-Jīlī[3], Ibn Yūnus[4], al-Khāzinī (*al-Zīj al-Sanjarī*, book I, section I, ch. II, f. 22v) and al-Kharaqī (treatise III, ch. VI, Paris codex f. 118v) reckon the era of Augustus in such a way that its epoch is deduced to be, beyond doubt, Thursday 13 November 30 BC; so it does not even agree with Ptolemy's Sunday 31 Aug. 30 BC. The cause of this difference I do not

[1]Who treated the question at length in his *Sur la date historique d'un renouvellement de la période sothiaque*, pp. 239–244 and 293–294 (in Mémoires présentés par divers savants à l'Académie des Inscriptions, 1st series, vol. VIII, part 1, 1869). R. Schram follows this opinion implicitly, *Hilfstafeln für Chronologie* (in Denkschriften der kais. Akad. der Wissenschaften zu Wien, mathem. Cl., vol. XLV, 1892).

[2]The following numbers hold from 1 March 1900 to 28 February 2100, in which interval of time the Gregorian calendar runs 13 days ahead of the Julian. — The Copts today use the era of Diocletian or of the Martyrs, whose epoch is Friday 29 Aug. 284.

[3]In Ideler, *Handb. d. mathem. Chron.*, II, 628, 632–633. — He wrote about 400 H (1009–1010); see Suter, *Die Mathem.*, pp. 83–84, no. 192.

[4]In ch. I, in Delambre p. 96, where, in line 5 from the foot of the page, 241287 is a typesetter's error for 241281.

[a]30 August 26 BC was a Thursday; 29 August 25 BC, 365 days later, was a Friday.

know. Abū l-Faḍl ʿAllāmī (*Āyīn-i Akbarī*, vol. I, p. 275 = II, 25 of the translation) says only that the era of Augustus begins at the beginning of the reign (that is, in 30 BC), and that its years are Julian, but its months Coptic.

c) — p. 70, ll. 18–23.

To compute on what day of the Roman year of the era of Dhū l-Qarnayn (or of the Seleucids) the first day of a given Persian year (according to the era of Yazdegerd) *falls.* — From 366, says al-Battānī, let there be taken 77 days + as many days as there are Roman years in a quarter of the years of the era of Dhū l-Qarnayn elapsed since the epoch of Yazdegerd; let what remains from the subtraction be counted, beginning from the first day of Aylūl (September); the first day of the Persian year will come out.

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The rule will serve until the remainder is found to be 0, that is, until the equation $366 = 77 + \frac{x}{4}$ arises, from which is deduced $\frac{x}{4} = 366 - 77 = 289$, and $x = 289 \times 4 = 1156$ Julian years. So the rule can be used up to the first day of the Julian year 1156 of Yazdegerd = 1788, which was amply sufficient for al-Battānī's readers.

Once 1156 Roman years have elapsed after Yazdegerd, that is, when the remainder $366 - \left(77 + \frac{x}{4}\right)$ is negative, one must not go forward from 1 Aylūl (September) with this remainder, but back.

On chapter 33

a) — p. 71, ll. 14–15.[a]

[«*But do not count with them the year in which you are, unless the noon of the last day of Subāṭ* [February] *has already passed*».

To avoid the inconvenience that arises from an intercalary day placed in the middle of the year, al-Battānī set up the epochs of the tables for a fictitious year, whose beginning is not 1 Aylūl (September) but the preceding 1 Ādhār (March); and so in the tables of the monthly motions (vol. II, pp. 74, 87, 104, 107) he always put Ādhār as the first month, not Aylūl. In this way the intercalary day is the last of his fictitious year; the numbers that answer to the single months in the tables remain unchanged in leap years and in common years alike, except for Subāṭ, which, however, never needs to be used, for when Subāṭ is complete the year too is complete.

So the epochs of the mean motions for the single years on p. 72 of vol. II of the tables are not for the beginning of Aylūl of that year, but for the noon of the first day of the preceding Ādhār (March). This will be plainer if one considers that for the year 1191 of Dhū l-Qarnayn the longitude of the sun is given as 342° 48′ 50″, which can never happen on the first day of September in the Julian reckoning, unless times most remote from al-Battānī and from us are to be thought of.

It is also to be noted that on p. 72 of vol. II the mean longitude of the sun in the year 1211 of Dhū l-Qarnayn is a whole number of degrees, without minutes and seconds; so I think that al-Battānī, in constructing his table, set out from the *year 1211* of Dhū l-Qarnayn as from a fundamental epoch, and derived the other epochs from it with the help of the mean motions. — For the anomaly of the moon and the motion of the node, on the other hand, he seems to have taken the *year 1191* as the fundamental epoch, for under this year the numbers are round. — Schiaparelli.]

[a] Nallino's Addenda et emendanda (Part I, p. LXXIV) add to this note: see also Samps Puchar, cited in the notes on the tables, vol. II, pp. 72–77.

On chapter 35

a) — pp. 73–74.

The computation of the twelve astrological houses.

What Delambre has on pp. 45–48 does not serve to explain this chapter, for the division of the houses that Delambre chose is not the one that al-Battānī means. I shall first set out briefly what these houses are, following above all V. Nabod, *Enarratio elementorum astrologiae, in qua praeter Alcabicii, qui Arabum doctrinam in compendio prodidit, expositionem atque cum Ptolemaei principiis collationem, de verae artis praeceptorum origine et usu satis disseritur*, Cologne 1560, pp. 116–122[1].

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Astrologers have various methods by which the whole heaven is divided in various ways into twelve parts, called *houses*[2]; all, however, agree in this, that six great circles, cutting one another in two opposite places and having a common diameter that runs from south to north, are thought to divide the whole heaven into twelve parts. But the ends of the diameter, at which those six circles cut one another, different astrologers place differently. Alcabitius and those who wrote on the division of the celestial houses before Campanus (thirteenth century) held, as Nabod relates, that these two points are the poles of the equator. Let there be two great circles, the one the meridian, the other the circle that passes through the point of the ecliptic lying on the horizon and cuts the equator at right angles. The *equinoctial circle* then appears divided into four parts, of which each two opposite ones are equal. Let all these four parts be cut into three equal parts, and through the points of section and the poles of the equator let four circles be drawn (thus four *circles of declination*), which with the two former will make the twelve houses[3]. In this way the ecliptic is divided into twelve parts, of which only the opposite ones are equal; only once a day will the ecliptic appear divided into four equal parts, namely at the moment of time at which the beginnings of Aries and Libra fall on the horizon.

Others put the ends of the common diameter at the points of intersection of the horizon and the meridian; the other four great circles are therefore those that are called *circles of position*. But on the method of drawing these circles they disagree, as Nabod relates[4]:

a) Campanus and Gazulus[5] take the *vertical* circle that passes through the points of rising and setting and is divided by the meridian and the horizon into four equal quadrants; they divide each quadrant of the vertical into three equal sections and draw the *circles of position* through the points of section. So the twelve houses are twelve equal lunes.

b) Regiomontanus, in problem XIV of the *Tabulae directionum*, supposes the *equinoctial circle* divided into equal quadrants by the meridian and the oblique horizon; the quadrants are then cut into three equal parts of 30 degrees, and the *circles of position* are drawn through the points of section.

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Others, lastly, such as Johannes Schöner[6], think that the *ecliptic* is divided into four quadrants by the horizon and by the great circle passing through both poles of the ecliptic and through the

[1] It is a long commentary, but without mathematical proofs, on the very well-known little book of Alcabitius (ʿAbd al-ʿAzīz al-Qabīṣī), who flourished in the tenth century; Suter, pp. 60–61, no. 132.

[2] Or *domicilia*; in Greek τόποι, «places»; in Firmicus *loca*; among the Arabs *bayt* (pl. *buyūt*), «house», very rarely *makān* (pl. *amkina*), «places».

[3] Delambre, p. 501, seems wrongly to think that Alcabitius used circles of position.

[4] Regiomontanus sets out the first three methods at length (Iohannes de Monteregio, *Tabulae directionum profectionumque*, Venice 1485, problems XIV–XVI).

[5] Iohannes Gazulus of Ragusa lived at the beginning of the fifteenth century.

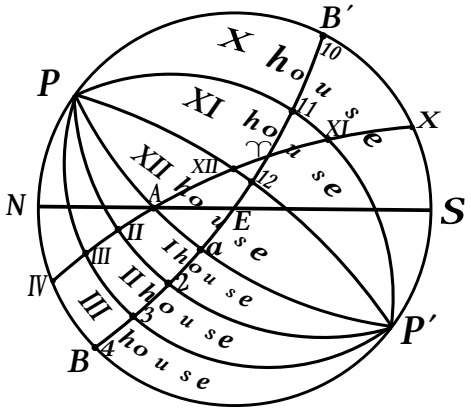
[6] Born 1477, died 1547. A dissertation on the celestial houses is found also at the end of his work entitled *Tabulae resolutae astronomicae* I. Schoneri, Wittenberg 1587.

zenith; that the quadrants are cut into three equal parts; that through the points of section and the poles of the ecliptic four other *circles of latitude* are drawn; and that in this way the ecliptic and the celestial sphere are divided into twelve equal houses. «That all astrologers have adapted their judgements to these», says Nabod, «has been handed down by many. And that this is true cannot be denied, if it is understood of Firmicus, Albohali[1], Albumasar[2], Albubater[3], Bonatus[4] and the like». But that Ptolemy divided the *equator* into 12 equal parts (of 30 degrees, that is, of two equinoctial hours each), Nabod tries to deduce from various passages of the Tetrabiblos, against the opinion of other authors.

These twelve points of the *ecliptic*, however they are determined, are called *cusps* or *beginnings of the houses*. Of them four are of the greatest moment; they have the names κέντρα, γωνίαι, *cardines geniturae*, *anguli* (the cardines of the nativity, the angles), and in Arabic *pegs* (*watad*, pl. *awtād*): they are 1) the *ascendant*, *the horoscoping part* (ἀνατολή, ἀνατέλλων, μοῖρα ὡροσκοποῦσα, ὡροσκόπος, *ṭāliʿ*), namely the point of the ecliptic on the eastern horizon at a given moment of time; 2) the *lower heaven* (ὑπόγειον, ἀντιμεσουράνημα, ἀντιμεσουρανῶν, ὑποχθόνιον κέντρον, *watad al-arḍ*, «the peg of the earth», *al-rābiʿ*, «the fourth»), namely the point of the ecliptic on the lower meridian; 3) the *setting* (δύσις, δύνων, *al-ghārib*, «the setting one»), or the point of the ecliptic on the western horizon; 4) the *middle heaven* (μεσουράνημα, μεσουρανῶν, *watad wasaṭ al-samāʾ*, «the peg of the middle of heaven», *al-ʿāshir*, «the tenth»), which is the place of intersection of the ecliptic and the upper meridian. In every division the order of the pegs and of the houses is always the same, namely in the succession of the signs.

On the virtues of each house the books of astrology may be consulted.

As appears from the very rules of the computation, al-Battānī's division of the houses is beyond doubt the one that Alcabitius and the Jew Ibn ʿEzrā (Avenezra, in Nabod, loc. cit.) teach; I find the same in Ḥabash ff. 102r and 108v, in Abū l-Wafāʾ[5], in Ulugh Beg pp. 145–146 of the text (195–196 of the translation), and here and there in the *Libros del saber de astronomia* that King Alfonso X had compiled from Arabic sources. Ptolemy seems to have had it in mind in the *Tetrabiblos*; see Nabod, loc. cit., and what I shall cite below from the *Libros del saber*.



Let PSP′N be the meridian, NES the horizon (where N is the north point, S the south, E the east), BB′ the celestial equator, A♈X the ecliptic, ♈ the beginning of Aries, P and P′ the poles of the equator, A the point of the ecliptic that is then on the eastern horizon, namely the ascendant. —

[1]That is, Abū ʿAlī Yaḥyā al-Khayyāṭ; he flourished at the end of the eighth and the beginning of the ninth century; Suter pp. 9–10, no. 17.

[2]Abū Maʿshar, d. 28 Ramaḍān 272 H (8 March 886); Suter pp. 28–30, no. 53.

[3]Al-Ḥasan ibn al-Khaṣīb, ninth century; Suter pp. 32–33, no. 62.

[4]Guido Bonatti lived about 1230–1300.

[5]Carra de Vaux, *L'Almageste d'Abû 'l-Wéfa al-Bûzdjâni* (Journal Asiatique, 8th series, vol. 19, 1892, p. 428).

Before all else al-Battānī bids us take the temporal hours of the point A[1] («*the hour-times of the degree of the ascendant in that clime*») and double them; which is the same as dividing the arc *a*B′ of the equator (the semidiurnal arc of the point A), the arc *a*B (the seminocturnal arc) and their opposites in the western hemisphere into three equal parts each. When circles of declination are drawn through the points of section, two unequal series of lunes arise; in one of them each lune comprises two diurnal temporal hours of the ascendant, in the other two nocturnal hours. The equator appears cut at *a*, 2, 3, 4 … 10, 11, 12; the ecliptic at A (the ascendant), II, III, IV (the peg of the earth) … X (the middle heaven), XI, XII, which are the beginnings or cusps of the houses. Once the arcs of the equator are known, the tables of right ascensions will at once give the corresponding arcs of the ecliptic.

Knowing the semidiurnal and seminocturnal arcs of the ascendant is also knowing the right ascensions of the pegs, that is, of the cusps of the I, IV, VII and X houses. The others will be found by al-Battānī's rules[2]: I 249

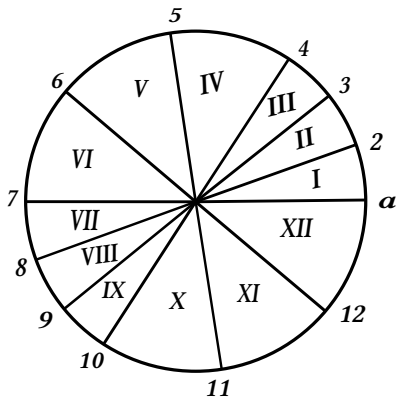
$$\text{RA of the beginning of XI} = \text{RA of the midheaven} + \frac{\text{semid. arc of the ascendant}}{3}$$

$$\text{RA of the beginning of XII} = \text{RA of the beginning of XI} + \frac{\text{semid. arc of the ascendant}}{3}$$

$$\text{RA of the beginning of II} = \text{RA of the ascendant} + \left(60^\circ - \frac{\text{semid. arc of the ascendant}}{3}\right)$$

$$\text{RA of the beginning of III} = \text{RA of the beginning of II} + \left(60^\circ - \frac{\text{semid. arc of the ascendant}}{3}\right)$$

The beginnings of V, VI, VIII and IX are the nadirs of XI, XII, II and III.



So in this partition of the houses not only do two unequal series of lunes arise, but an absurdity is also found, which Delambre, p. 502, speaking of Alcabitius, notes in these words[3]: «Les six maisons dernières sont toujours diamétralement opposées aux six premières; il en résulte cependant une espèce d'absurdité. Le quart de l'équateur, entre le méridien et l'horizon occidental, se trouve divisé suivant les arcs nocturnes, quoiqu'il appartienne au jour; le quart entre l'horizon occidental et le méridien inférieur, est divisé suivant les heures du jour, quoiqu'il appartienne à

[1] [So the temporal hours of the ascendant, not the hours of the sun, as Delambre, p. 45, thinks. — Schiaparelli.]

[2] [The angle B′P*a* is the semidiurnal arc of the point A, the angle *a*PB the seminocturnal arc of the same A, or the semidiurnal arc of its nadir. So if the semidiurnal arc of any point of the ecliptic is added to the semidiurnal arc of its nadir, 180° always results; likewise the sum of a larger and a smaller lune will always be $\frac{180^\circ}{3} = 60^\circ$. Hence al-Battānī's rule on p. 74, ll. 9–10. — Schiaparelli.]

[3] To make the matter plainer, I add a figure that shows the projection of the houses on the plane of the celestial equator.

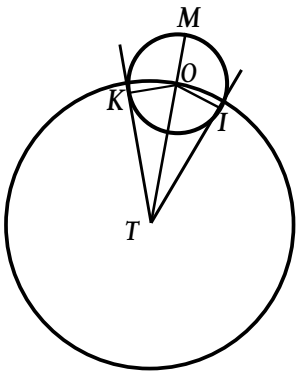
la nuit. Au reste, le calcul de cette méthode est extrêmement simple, et c'est peut-être ce qui a fait passer sur l'absurdité que nous venons de remarquer» («The last six houses are always diametrically opposite the first six; a kind of absurdity results from it, however. The quarter of the equator between the meridian and the western horizon is divided according to the nocturnal arcs, although it belongs to the day; the quarter between the western horizon and the lower meridian is divided according to the hours of the day, although it belongs to the night. For the rest, the computation of this method is extremely simple, and that is perhaps what has made people pass over the absurdity we have just noted»).

Al-Battānī's method of computation is also found in the *Libros del saber de Astronomia*, under the name either of Ptolemy or of al-Battānī. For vol. II, pp. 60–61 (*Libro de las armellas*, book II, ch. XLIII): «De saber yguar las XII casas segund la opinion de Ptolomeo et de el Bateni» («To know how to equalize the 12 houses according to the opinion of Ptolemy and of el Bateni»); vol. II, p. 193 (*Libro del astrolabio redondo*, book II, ch. LIX): «De saber ygualar las doze casas segun la opinion de Ptolomeo et de Veles»[1] («To know how to equalize the twelve houses according to the opinion of Ptolemy and of Valens»); vol. III, p. 71 (*Libros de la lámina universal*, part II, ch. LXII): «de saber ygualar las casas que caen entre los ángulos, segund la opinion de Ptolomeo» («to know how to equalize the houses that fall between the angles, according to the opinion of Ptolemy»); vol. III, p. 204 (*Libro de la açafeha*, whose author is al-Zarqālī, ch. LVI): «De saber eguar las casas segund la oppinion de Ptholomeo» («To know how to equalize the houses according to the opinion of Ptolemy»): the computation does not differ from al-Battānī's[2]. Nor are the rules that Ḥabash and Ulugh Beg (at the beginning of the chapter) teach any different.

On chapter 36

a) — p. 75, ll. 27–30.

[What he says of the moon is true; as regards the sun, he ought to have added that for the numbers 95° and 265° one must put 90° + the greatest equation of the centre and 270° – the greatest equation of the centre; so, in round numbers, 92° and 268°.



The cause of this is as follows. The mean motion, in the figure drawn here, is the motion of the centre O of the epicycle; when the sun is at I, that is, on the line tangent to the epicycle, it also moves with the mean motion, for then its motion on the epicycle neither accelerates nor retards its angular velocity with respect to the earth T; and this happens when the arc MI (that is, the

[1] That is, Οὐαλῆς, the Arabs' Wālīs, Vettius Valens of Antioch, a very famous astrologer whose date is uncertain, between the second and the fourth century AD.

[2] Elsewhere the computation of Hermes is set out: vol. II, p. 193 (*Libro del astrolabio redondo*, book II, ch. LX), vol. III, p. 72 (*Libros de la lámina universal*, part II, ch. LXIII), and pp. 209–210 (*Libro de la açafeha*, ch. LXIII).

anomaly) = 90° + OTI = 90° + the greatest equation of the centre. Similarly when the sun is on the other tangent; then the anomaly = MIK = 270° − KTO = 270° − the greatest equation of the centre. In the sun the greatest equation is about 2°; so MI = 92°, MIK = 268°. In the moon, whose greatest equation at the syzygies amounts to 5°, MI = 95°, MIK = 265°, as al-Battānī has it.

As regards the moon, however, it must be noted that the greatest equation can change because of the eccentricity and reach 7° 40′; so at the quadratures, where the effect of the eccentricity is greatest, MI = 97° 40′, MIK = 262° 20′. All these considerations al-Battānī left out from too great a desire for brevity. — Schiaparelli.]

On chapter 37

a) — pp. 75–76.[a]

The *motion of the node* is retrograde, so it must be subtracted from 360°.

The names *head* and *tail* were unknown to the Greek astronomers, among whom ὁ ἀναβιβάζων [σύνδεσμος], «the ascending [node]», is the northern node, and ὁ καταβιβάζων, «the descending», the southern. Head and tail refer to the celestial Dragon, which the Gnostics and certain Greek astrologers of the first century AD had taken over from the superstitions of the Chaldaeans; see A. Bouché-Leclercq, *L'astrologie grecque*, Paris 1899, pp. 121–123.

The word *jawzahar*, from the Persian *gawzahr* گوزهر, means: 1) the parecliptic circle of the moon (see p. 45, note 3), or the lunar orbit; 2) the nodes of the lunar orbit, and especially the ascending node; 3) the nodes of the orbit of any planet. It is probably the Zend word *gaocithra*, «[keeping] the seed of cattle», an epithet of the moon in the book called *Avesta*; for they thought that the moon had kept the seed of the first bull and had produced from it the various kinds of herds[1]. In Latin writers of the eleventh century and after, the Arabic name is often found corrupted into *iozahar*, *iehuzar*, *gehuzahar*, *geuzaar*, *genzahar*, *zeunzaar*, *çeuçahat*; see M. Steinschneider, *Lettera seconda a D. B. Boncompagni, intorno al libro «Saraceni cuiusdam de eris»*, Rome 1863, pp. 15–16 and 19.

On chapter 38

a) — p. 76, ll. 10–17.

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The *argument of latitude*, which is the arc of the lunar orbit between a certain point of the orbit and the place of the moon at a given moment of time, Ptolemy and Theon (*Tables manuelles* vol. I, p. 55 and passim) count from the *northern limit* (βόρειον πέρας), that is, from the point of the lunar orbit at which the greatest northern latitude occurs; it is 90° from the ascending (or northern) node. Al-Battānī, on the other hand, and the other Arabs compute the argument from the node itself, following the usage of the Indians; and the Indians in this probably followed the Chaldaeo-Greek astrologers, for whom the nodes of the moon, and especially the northern node, had great authority and power in nativities (or genitures): for if beneficent planets were found in it, even a mediocre and lowly nativity turned out more splendid and exalted. The astrologers

[1] F. Spiegel, *Erânische Alterthumskunde*, Leipzig 1871–1878, vol. II, p. 70 (a passage that I. Pizzi pointed out to me).

[a] Nallino's Addenda et emendanda (Part I, p. LXXIV) add here a reference to his preface, p. XLIII.

themselves often count the nodes of the lunar orbit among the planets[1].

Al-Battānī's formula

$$(\alpha) \qquad \text{sin lat.} \leftmoon = \frac{\text{sin arg. of lat.} \times \text{sin greatest lat.} \leftmoon}{R}$$

is not found in Ptolemy and Theon[a]; it is very like the rule by which the declination of the sun is derived from the longitude and the obliquity of the ecliptic (pp. 12 and 162). In orthographic projection it is very easily gathered from the similarity of two triangles; nor does it differ from the very well-known theorem of right-angled spherical triangles

$$\text{sin cathetus} : \text{sin opposite angle} = \text{sin hypotenuse} : R.$$

On chapter 39

a) — p. 77, ll. 23–24.

From Ptolemy's geometrical constructions, V, 19 (ed. Halma vol. I, pp. 360–62), if A denotes the angle that the ecliptic makes with the vertical circle passing through the moon, the equations that al-Battānī states below, p. 81, ll. 4–6, are easily deduced:

$$\text{parallax in latitude} = \frac{\sin A \times \text{parallax in altitude}}{R}$$
$$\text{parallax in longitude} = \frac{\cos A \times \text{parallax in altitude}}{R}.$$

From these follows the rule given here by al-Battānī:

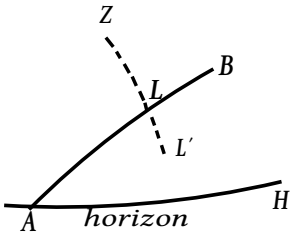
$$(\alpha) \qquad \text{parall. in lat.} : \text{parall. in long.} = \sin A : \cos A.$$

b) — p. 77, ll. 25–27.

[«*The parallax always declines towards the side of latitude in which the degree* [*of the ecliptic*] *is that* etc.».

[1] «On est tenté de reporter à des Chaldéens authentiques ou à des rénovateurs de vieilles superstitions orientales l'idée baroque qui consiste à assimiler aux planètes et à noter dans les thèmes de géniture les noeuds ascendant et descendant de l'orbite lunaire, qui font le tour de l'écliptique en dix-huit ans. Ptolémée ne connaît pas ou ne veut pas connaître ces planètes fictives: mais Tertullien, un demi-siècle plus tard, parle déjà de l'influence que leur attribuaient les astrologues. Les Grecs du Bas-Empire et surtout les Arabes firent grand état des noeuds écliptiques, appelés la Tête ☊ et la Queue ☋ du Dragon» («One is tempted to ascribe to genuine Chaldaeans, or to revivers of old Oriental superstitions, the baroque idea of treating as planets, and marking in the charts of nativity, the ascending and descending nodes of the lunar orbit, which go round the ecliptic in eighteen years. Ptolemy does not know, or does not wish to know, these fictitious planets; but Tertullian, half a century later, already speaks of the influence that the astrologers attributed to them. The Greeks of the Later Empire, and above all the Arabs, made much of the ecliptic nodes, called the Head ☊ and the Tail ☋ of the Dragon») (Bouché-Leclercq, loc. cit., pp. 121–22).

[a] Nallino's Addenda et emendanda (Part I, p. LXXIV) correct this statement: the rule is in Ptolemy, who says (V, 7; ed. Halma I, 315) that for computing the table of the moon's latitude «we have used the same demonstration (δείξει) by which we have already computed the arcs of the circle through the poles of the equator cut off between the equator and the ecliptic», that is, the declinations of the sun.



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The passage is not clear enough and perhaps somewhat corrupt. If an arc of a great circle is drawn from the zenith to the point of the ecliptic (or rather of the parallel of latitude) at which the moon is, the parallax in latitude becomes southern or northern according as this arc, produced beyond the ecliptic (or the parallel of latitude), tends towards the south or towards the north. Let AB be the ecliptic (or the parallel of latitude), Z the zenith, L the place of the moon; let the arc ZL be produced beyond AB; the moon, which is at L, will be seen lower, at L′, and so the parallax will be towards the side that the arc ZL produced designates, namely the side opposite the zenith. So al-Battānī could have written better: «the parallax always declines towards that side of latitude which the circle from the zenith of the middle of the heaven, produced beyond that degree of the ecliptic, designates». For here it is not a question of the middle of the heaven that is mentioned in ch. XXXV, but of the zenith, which is in the middle of the visible heaven.

The parallax in longitude, of which al-Battānī speaks afterwards, is towards the side from which the ecliptic approaches the horizon; so towards A if AL is less than 90 degrees.

Both rules could have been stated more briefly: the parallax in longitude is towards the ascendant if this is nearer, otherwise towards the descendant. — Schiaparelli.]

c) — p. 77, ll. 32–34.

The parallax of Venus and Mercury is perceptible; in Mercury at perigee it is equal to the parallax of the moon at apogee.

Here al-Battānī departs from Ptolemy's opinion, IX, 1 (ed. Halma vol. II, p. 115), that the parallaxes of all five planets are imperceptible. But before al-Battānī al-Farghānī had already written, ch. XXVII, p. 100 of Golius's edition: «In the superior planets, namely those that stand above the sun, nothing at all of this parallax [of altitude] is detected that can in any way be distinguished by the senses. As regards the sun, its parallax too is not noticed in observing; but when it is computed from the [known] distance of the sun from the earth, the greatest of it, which is gathered near the horizon, is 3 minutes. *In Venus, however, Mercury and the moon the parallax is perceptible*; especially in the moon ...».

Why the Arabs rejected Ptolemy's opinion Delambre has already noted, pp. 50 and 167–69. Ptolemy does not determine the distances of the spheres of the five planets from the earth; as regards the sun, he deduces from Aristarchus' theory of the distances of the sun and the moon a solar parallax of 2′ 51″ (or in round numbers 3 minutes), but adds that it cannot be computed by direct observations (see p. 77, note 4). For this reason, although, following the tradition of the ancients, he places Venus and Mercury below the sun, he can consider their parallaxes imperceptible. — But al-Farghānī and al-Battānī, like the other Arabs, follow the opinion of those who think the spheres of the planets contiguous, so that the greatest distance of a lower planet is the least distance of the planet immediately above it (see note *a* on ch. L). So the greatest parallax of Mercury must equal the least parallax of the moon.

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Worth noting are the criticisms that Jābir ibn Aflaḥ, in the first half of the twelfth century, made against Ptolemy[1]. After setting out Ptolemy's considerations on the order of the spheres of the

[1]Gebri filii Affla Hispalensis *de astronomia libri IX, in quibus Ptolemaeum, alioqui doctissimum, emendavit, alicubi*

planets (*Alm.* IX, 1, ed. Halma vol. II, pp. 114–115), he adds these observations: the Greek astronomer places Venus and Mercury below the sun, to which he attributes a parallax of 2′ 51″ and which he uses in solar eclipses; but he denies that the two inferior planets have a perceptible parallax. Ptolemy afterwards shows that the radius of the epicycle of Venus is 43 1/6^p and the eccentricity 1 1/4^p, in the parts of which 60 are assigned to the radius of the eccentric. So Venus at its greatest distance would have to be more than 104^p from the centre of the earth, at its least less than 16^p. So if the sun, which is always more than 104^p from the earth, has a parallax of about three minutes, how can Ptolemy consider Venus to be without parallax? Venus, at less than 16^p from the centre of the earth, would have to have a parallax of about 1/3 of one of those parts in which we have given the geocentric distance[1]. Likewise, if Mercury stood [immediately] below the sun, it would have to have a parallax of about 7 minutes when it occupied the perigee of the epicycle. It is true that the parallaxes of both planets cannot be observed at perigee, because they are then in conjunction with the sun[2]; but this is possible in places near it, and especially when the planets are on the tangents drawn to the epicycle, for then they have their whole elongation. And when the centre of the epicycle is in one of the nodes, the planet lies in the plane of the ecliptic, and the parallax, then without latitude, must come to about 6′ in Venus and about 4′ in Mercury. — How then could Ptolemy, after attributing a parallax of about 3′ to the sun and none to Venus and Mercury, consider those planets to be below the sun?

So Jābir, rightly; I have given his words in substance. But perhaps Ptolemy did not trust the solar parallax of 2′ 51″ much, and did not think the order of the spheres certain.

[According to the ratios of the greatest and least distances that al-Battānī reports from old tradition in ch. L, pp. 120–21, if the radius of the earth is taken = 1^p, one finds

	distance	*parallax*
apogee distance of ☾, perigee distance of ☿	64^p 10′	53′ 34″
apogee distance of ☿, perigee distance of ♀	166 0	20 42
apogee distance of ♀, perigee distance of ☉	1070 0	3 13

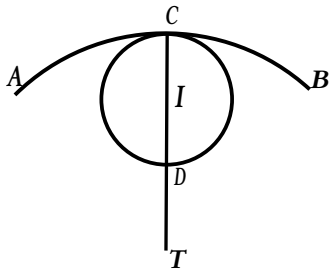
Hence it is plain why Jābir says that the parallax of Venus (♀), if Venus is placed below the sun and above Mercury (☿), reaches 20′ at the perigee distance. He then adds that the parallax of Mercury, even if it is placed below the sun and above Venus, is between 6′ and 7′ at the perigee of the epicycle. For let T be the earth, AB the orbit of the sun, CD the epicycle of Mercury; Ptolemy teaches that IC = 0.375 × IT approximately; so TD : TC = 0.625 : 1.375. If the parallax of the sun is taken as 3′, the parallax of Mercury at D will come out $3' \times \frac{1.375}{0.625} = 6.52'$[a]. Because of the eccentricity of Mercury's orbit this quantity can sometimes be increased a little. — Schiaparelli.]

etiam industria superavit, Nuremberg 1541 (at the end of the work: *Instrumentum sinuum seu primi mobilis nuper a Petro Apiano inventum*), book VII, p. 103. The Latin version is Gerard of Cremona's, but so barbarous that Delambre, p. 184, understood nothing of this passage.

[1] «Since, then, the sun is above it [i.e. Venus], its distance from the centre of the earth will always be more than 104 parts, and it has a parallax whose greatest amount is about three minutes; then, since no parallax of manifest amount is found for the star Venus when there are less than 16 parts between it and the centre of the earth, and it ought to be, in proportion to the parts of its distance, about a third of a part».

[2] «And again, if the star Mercury is below the sun, its parallax ought to be about 7 minutes when it is at the nearer distance of its circle of revolution, although the parallax of either in those places is not possible, because each is in conjunction with the sun».

[a] $3' \times \frac{1.375}{0.625} = 3' \times 2.2 = 6.6'$, that is, 6′ 36″; with Ptolemy's 2′ 51″ for the sun the result would be 6′ 16″.



d) — p. 78, ll. 15–26.

To compute the geocentric distance of the moon at the syzygies[1].

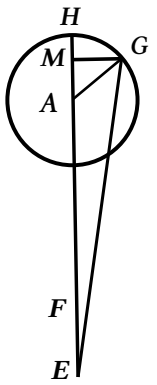
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Let ψ be the corrected anomaly (for which, if it exceeds 180°, its difference from 360° is taken), *e* the radius of the epicycle, that is, the sine of the greatest equation of the centre; al-Battānī teaches that, if the anomaly lies between 0° and 90°,

$$(\alpha) \qquad \text{geoc. dist.} = \sqrt{\left(\frac{e\cos\psi}{R} + 60^p\right)^2 + \left(\frac{e\sin\psi}{R}\right)^2},$$

and, when the anomaly lies between 90° and 180°[a],

$$(\beta) \qquad \text{geoc. dist.} = \sqrt{\left(\frac{e\cos\psi}{R}\right)^2 + \left(60^p - \frac{e\sin\psi}{R}\right)^2}.$$



Ptolemy does not have these formulas; but they are very easily deduced by this construction. Let E be the centre of the earth and of the ecliptic, F the centre of the eccentric, G the place of the moon, so that HG is an anomaly less than 90 degrees; plainly

$$EG = \sqrt{(AE + AM)^2 + MG^2}\,.$$

Into this formula let the values

$$AM = \frac{AG \times \cos HAG}{R} = \frac{e\cos\psi}{R},$$

$$MG = \frac{AG \times \cos MGA}{R} = \frac{e\sin\psi}{R},$$

[1] At that time the centre of the epicycle occupies the apogee of the eccentric.

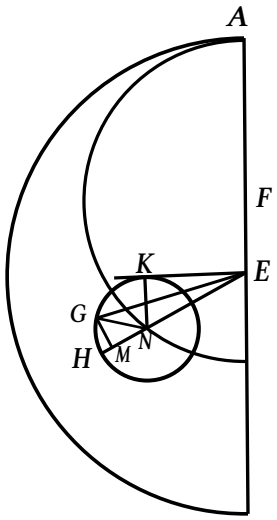
[a] Formula (β) holds when ψ stands for the anomaly less 90°, as in al-Battānī's rule (p. 78, ll. 20–24, «subtract 90° from it»); with ψ the anomaly itself, (α) holds for every anomaly.

$$AE = \text{radius of the parecliptic circle} = 60^p$$

be put, and rule (α) will come out at once.

[What the author adds in lines 25–26, «*from each degree of this distance subtract a minute*» etc., is of little use: for the distance of the moon *from the surface of the earth* is never the same as the distance of the moon *from the place of observation*, which recent astronomers use to compute the apparent diameter of the moon. It should also be noted that even at the syzygies the distance of the moon from the centre of the earth is hardly ever equal to 60 parts: whence it is plain that the rule also departs somewhat from the truth. It would have been better to subtract one part from the sixty parts of which the distance AE consists. — Schiaparelli.]

e) — p. 78, ll. 27–29.



[«*Take the product of the minutes of the fourth column into the fifth column …*».

Al-Battānī now wants to compute the amount of the semidiameter of the epicycle as seen from the earth, that is, the angle NEK, which is the greatest equation of the anomaly when the centre of the epicycle occupies the point N. The number of the fifth column cannot be known unless the anomaly HNK is given; but this is unknown, and it is the very thing that has to be found, for HNK = 90° + NEK. So one should have written not «*into the fifth column*» but «*into the greatest number of the fifth column*», that is, into 2° 40′. Schiaparelli.]

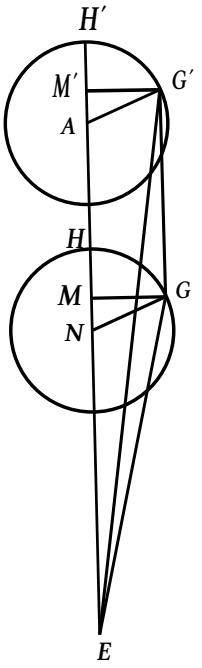
f) — p. 78, ll. 31–32 ff.

To compute the geocentric distance of the moon outside the syzygies, that is, when the centre of the epicycle is not at the apogee of the eccentric.

[In these and in the following lines much seems to be corrupt, and it is not easy to reach the author's meaning by conjecture. Al-Battānī seems to me to set out two solutions of the problem: if this is true, much has dropped out in the codex and in Plato, of which the remains are the words printed in italic; so the end of the first solution and the beginning of the second are missing. — Once the radius of the epicycle is known, NK = 5^p 15′ of the parts of which AE comprises 60^p, NE, the distance of the earth from the centre of the epicycle, is deduced at once from the angle NEK just found; by means of it, given the anomaly HNG, the problem is solved as was done before for the mean syzygies (see note *d*); there comes out the distance GE of the moon from the centre of the earth.

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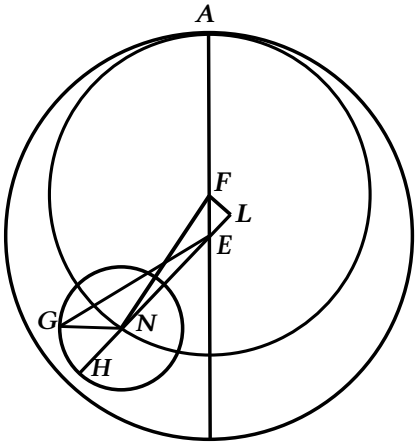
The lacuna can be restored in this way: «the sine of this sum is the radius of the displaced epicycle; [*divide* 5^p *15′ by it, and multiply what comes out by* 60^p*; the distance of the centre of the epicycle from the earth will come out.*] Use it in the same way as you used above [*the apogee distance of the same centre at the syzygies*]; you will find the hypotenuse of the moon corrected for the displacement of the epicycle»[1]. «[*If you wish to compute the same distance otherwise, without using the tables, take the apogee distance of the centre of the epicycle and the radius of the epicycle,* 5^p *15′, as you did above; then seek the hypotenuse of the moon in the way we have described.*] Then take the double elongation …».



This new solution consists in this: instead of computing the distance of the moon when the centre of the epicycle is at the distance NE from the earth, the computation begins by putting AE for NE; afterwards the correction to be applied to it is sought. Let AE = 60^p, the apogee distance of the centre of the epicycle; by the same way that is set out in note *d*, and always using the radius AG′ = 5 1/4^p, EG′, that is, the *hypotenuse of the moon*, is computed. Let EN then be the true distance of the centre of the epicycle from the earth; the true distance of the moon will be EG. Since the angles at E are very small, and NA too is small with respect to AE, the triangle EGG′ will be very long, so that, very nearly, EG′ = EG + GG′ = EG + NA. And since NA = AE − NE = 60^p − NE = *the share of the double eccentricity* (p. 79, l. 6), the required distance of the moon will be approximately EG = EG′ − NA = (*the hypotenuse of the moon corrected for the displacement of the epicycle*) − (*the share of the double eccentricity*). The approximation always suffices, for the angle at G′, namely EG′G = H′EG′, never exceeds 5° 1′; the angle G′EG will be smaller than it, since in the theory of the moon always GG′ < EG. As to p. 79, l. 7, see note *g*. — When the centre N of the epicycle arrives at the apogee point of the eccentric, the distance NA becomes nothing. When the centre of the epicycle arrives at the perigee point, the same distance becomes greatest, and indeed equal to twice the eccentricity of the eccentric circle. When the centre of the epicycle occupies places other than the perigee and the apogee, the distance NA is always smaller than the double eccentricity, so that it can be called a *share*, or some fraction, *of the double eccentricity*; and that is the name, as you see, that al-Battānī actually used. — Schiaparelli.]

The trigonometrical formula that al-Battānī gives on pp. 78–79 for computing NE is easily derived from Ptolemy's geometrical constructions, *Alm.* V, 13 (ed. Halma vol. I, pp. 334–36).

[1] That is, the required distance of the moon from the centre of the earth; see p. 78, l. 24.



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Let NE be produced until it cuts the straight line FL at right angles in L; AEN will be the arc of the double elongation 2 (☾ − ☉), and FE the eccentricity ε (= 10^p 19′). Plainly, in the right triangle FEL,

$$FL = \frac{FE \times \sin FEL}{R} = \frac{\varepsilon \sin FEL}{R},$$

and, since sin *FEL* = sin [180° − 2 (☾ − ☉)] = sin 2 (☾ − ☉), we shall have

$$FL = \frac{\varepsilon \sin 2\,(☾ - ☉)}{R}.$$

Similarly[a]

$$EL = -\frac{\varepsilon \cos 2\,(☾ - ☉)}{R}.$$

So

$$NL = \text{corrected side} = \sqrt{\text{radius}^2 \text{ of the eccentric} - \left(\frac{\varepsilon \sin 2\,(☾ - ☉)}{R}\right)^2}.$$

But the radius of the eccentric is *FN* = *AE* − ε = 60^p − ε; whence

$$NL = \sqrt{(60^p - \varepsilon)^2 - \left(\frac{\varepsilon \sin 2\,(☾ - ☉)}{R}\right)^2}$$

$$(\alpha) \qquad NE = NL - EL = \sqrt{(60^p - \varepsilon)^2 - \left(\frac{\varepsilon \sin 2\,(☾ - ☉)}{R}\right)^2} + \frac{\varepsilon \cos 2\,(☾ - ☉)}{R}.$$

g) — p. 79, l. 7.

[What I said on p. 254, at the end of note *d* of this chapter, must be repeated here. — Schiaparelli.]

h) — p. 79, ll. 9–12.

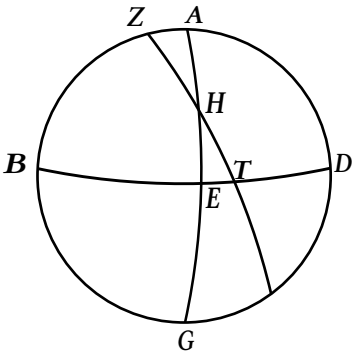
To compute the geocentric distance of the sun.

[a] The print has EL = ε cos 2 (☾ − ☉) / R and, in (α), − ε cos 2 (☾ − ☉) / R. Since FEL = 180° − 2 (☾ − ☉), cos FEL = − cos 2 (☾ − ☉); with the printed signs (α) would give 60^p − 2ε at the syzygies, where the centre of the epicycle is at the apogee A and NE = 60^p. The signs set here agree with al-Battānī's rule (p. 79, ll. 1–5), which adds the product of the cosine when the double elongation is less than 90° and subtracts it when it is greater.

The rule that serves for this is the same as the one by which the distance of the moon at the syzygies is computed (note *d*), when the sine of the equation of the sun is used instead of the sine of the equation of the centre of the moon ($5^p\ 15'$). The number that comes out in this way gives the distance of the sun when the mean distance is taken as 60 parts, as for the moon. It must now be turned into earth radii, of which the mean distance of the sun (according to ch. XXX, p. 61) comprises 1108; so the number found by the preceding computation might seem to have to be multiplied by $\frac{1108}{60} = 18^p\,28'\,0''$. But al-Battānī probably considered that the *lunar* distance of 60 parts is the *mean* with respect to the epicycle but the *greatest* with respect to the eccentric; so, not unreasonably, instead of the mean distance of the sun (1108 radii) he preferred to take the number midway between the mean and the greatest (1146) distance of the sun, namely 1127. Then $1127 : 60 = 18^p\ 47'\ 0''$ will be the factor that must be used to make the reduction just mentioned. The 38 parts, لح, of the codex are a plain error of the copyists for كح or يكح, 18; in the minutes both the codex and Plato wrongly have 46′, مو, instead of 47′, مز[a]. How the seconds arose (24″ in the codex, 50″ in Plato) I do not understand; they seem wholly false.

i) — p. 79, ll. 34–39.

With the moon on the ecliptic, to compute its *zenith distance* ζ and its *altitude h*.



Let ABGD be the meridian, BED the horizon, ZHT the ecliptic, A the zenith, H the moon, Z the culminating degree of the ecliptic, that is, the middle of the heaven; ZA will be the zenith distance of the culminating degree, or the complement of its altitude; HE = *h* = 90° − ζ; T = the ascendant degree. Ptolemy II, 12 (ed. Halma vol. I, pp. 130–31) shows

$$\frac{\text{chord } 2ZB}{\text{chord } 2BA} = \frac{\text{chord } 2ZT}{\text{chord } 2TH} \times \frac{\text{chord } 2HE}{\text{chord } 2EA}$$

If sines are used for chords:

$$\frac{\text{sin alt. culm. deg.}}{R} = \frac{\text{sin dist. culm. deg. from asc.}}{\text{sin dist. ☾ from asc.}} \times \frac{\sin h}{R}\,,$$

whence al-Battānī's rule:

$$(\alpha) \qquad \cos\zeta = \sin h = R\,\frac{\text{sin dist. ☾ from asc.}}{\text{sin dist. culm. deg. from asc.}} \times \frac{\text{sin alt. culm. deg.}}{R}\,.$$

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k) — p. 79, l. 40 – p. 80, l. 7.

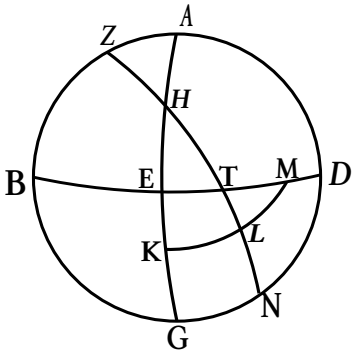
[a]The print has مر, without the dot of the zāy.

With the moon on the ecliptic[1], to find from the given zenith distance ζ of the moon, that is, from the complement of the altitude *h*, *the angles that the ecliptic makes with the vertical circle* passing through the moon.

The angle between the ecliptic and the vertical is called the *angle of latitude*; its complement is the *angle of longitude*, between the vertical circle and the circle passing through the degree of the moon's longitude and through the poles of the ecliptic.

The angle of latitude is a right angle when the moon is 90° from the ascendant. If the distance is greater or less than 90°, and the altitude southern, al-Battānī's rule is

$$(\alpha) \qquad \sin \text{ang. long.} = \cos \text{ang. lat.} = \frac{\cos \text{ dist. } ☾ \text{ from asc.}}{\sin \text{ dist. } ☾ \text{ from asc.}} \times \frac{\sin h}{\cos h} \times R.$$



This is deduced from Ptolemy's geometrical construction, II, 12 (ed. Halma vol. I, pp. 131–32): let ABGD be the meridian, BED the horizon, A the zenith, ZHTN the ecliptic, AHEG the vertical, ZHA the angle of latitude. With H as pole[a], let the arc KLM of a great circle be described; AE, EM, KM will be quadrants of great circles, and KL the arc that measures the angle of latitude. Ptolemy shows

$$\frac{\text{chord } 2HE}{\text{chord } 2EK} = \frac{\text{chord } 2HT}{\text{chord } 2TL} \times \frac{\text{chord } 2LM}{\text{chord } 2KM};$$

with sines put for chords,

$$\frac{\sin h}{\cos h} = \frac{\sin \text{ dist. } ☾ \text{ from asc.}}{\cos \text{ dist. } ☾ \text{ from asc.}} \times \frac{\cos \text{ ang. lat.}}{R},$$

whence al-Battānī's rule. — On what follows see note *l*.

l) — p. 80, ll. 7–9.

[«*But if the altitude is above the northern horizon*» (that is, when the culminating degree of the ecliptic is to the north of the zenith) «*the matter is reversed; for the arc …*». — False, for the arc will always be the angle of longitude, as before. What changes is the sign of the parallax in latitude. — Schiaparelli.]

The error does not come from al-Battānī; that it is a most inept interpolation by that reader who, as we have seen and shall see, also spoiled other things most foolishly, is plain enough from what follows immediately below: «but this does not happen in the other[2] regions …», where the sign of the lunar parallax is plainly concerned. — The interpolator likewise corrupted the formula on p. 81, ll. 31–33.

[1]When it is outside the ecliptic, the correction set out on p. 82, ll. 8–26, and in note *r* is necessary.

[2]One would expect «our»; see p. 81, ll. 15–18, where the matter is set out more clearly.

[a]The print has «Ex II» for «Ex H».

m) — p. 80, ll. 37–39.

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The parallax of the sun, when it is taken from the tables, *must be increased by its eighteenth part* «because of the difference that has befallen the distance of the sun from the earth». It could have been said better: «because of the change we have made in Ptolemy's distance of the sun from the centre of the earth». In ch. XXX, p. 61, he set the greatest distance at 1146 earth radii, the least at 1070; Ptolemy had found the greatest to be 1210, that is, 64 radii more than al-Battānī's. And since $\frac{1210}{64} = 18\frac{29}{32}$, while al-Battānī took the table of the parallaxes of the sun from the *Almagest*, plainly 1/18 of the numbers found there must be added to them. So the reading of the codex, 1/8, is false.

Al-Battānī's diminution of the distance is a small change for the worse, for Ptolemy's quantity was already much smaller than the true one. Astronomers today make the mean distance of the sun about 23430 radii, the parallax 8.80″.

The parallax according to Ptolemy is 2′ 51″; with the distances changed, al-Battānī found:

$$\text{sin parall. } \odot \text{ at apogee} = 60^p \frac{1}{1146} = 3'\,8''\,29''';\quad \text{parall. } 3'\,0''$$

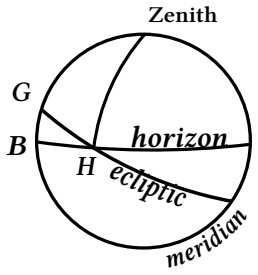
$$\text{sin parall. } \odot \text{ at perigee} = 60^p \frac{1}{1070} = 3'\,21''\,52''';\quad \text{parall. } 3'\,13'',$$

whence the difference of thirteen seconds from apogee to perigee that is mentioned on the last line of p. 80.

n) — p. 81, ll. 4–6.

The rules are easily deduced from Ptolemy's geometrical constructions; see note *a*, p. 251.

o) — p. 81, ll. 23–34.



With the moon on the ecliptic, to compute *the angle between the ecliptic and the vertical circle* passing through the moon, when the moon is at the ascendant.

He bids us first find the *corrected latitude* of the clime, that is, the latitude of the given place on the earth ∓ the declination of the culminating degree of the ecliptic. This done, the formula is:

(α) $$\text{sin ang. long.} = \text{cos ang. lat.} = R\,\frac{\text{cos corrected lat.}}{\text{sin dist. } \leftmoon \text{ from the culm. deg.}}.$$

He probably found the rule in this way. Let H be the ascendant and the moon, G the culminating degree of the ecliptic; plainly ZHG will be the angle of latitude, whose complement GHB is the angle of longitude; GB is the altitude of the culminating degree, which (by this very chapter, p. 79, and ch. XV, p. 30) is the sum or the difference of the complement of the latitude φ of the city

and the declination δ of the culminating degree. In the right spherical triangle GBH we have (see p. 181, note 1)

$$\sin GH : \sin GB = \sin HBG : \sin GHB.$$

But $\sin GB = \sin\big[(90^\circ - \varphi) \pm \delta\big] = \cos(\varphi \mp \delta) =$ cos corrected lat.; so

$$\text{sin dist. ☾ from the culm. deg. : cos corr. lat.} = R : \text{sin ang. long.},$$

whence al-Battānī's formula.

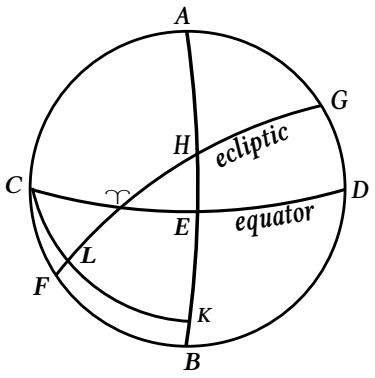
[What follows, printed in italics, is false, and comes from the same interpolator who added a similar error on p. 80, ll. 7–9 (see note *l*). Ptolemy had neglected this problem, for the general rule stated in note *k*, p. 257, could suffice for its solution. — Schiaparelli.]

p) — p. 81, l. 38 – p. 82, l. 4.

To compute the angle of latitude when the moon is in the meridian and on the ecliptic.

Either the general rule (p. 80 and note *k*) must be used, or

$$(\alpha) \qquad \text{sin ang. long.} = \text{cos ang. lat.} = R\,\frac{\text{sin decl. culm. deg.} \times \text{cos dist. ☾ from } \Upsilon \text{ or } \underline{\frown}}{\text{cos decl. culm. deg.} \times \text{sin dist. ☾ from } \Upsilon \text{ or } \underline{\frown}}.$$



This particular problem is not considered in the *Almagest.* Al-Battānī's solution was probably found with the help of the theorem of Menelaus (p. 192), after the manner of the Greeks. Let AEB be the meridian, ♈ the beginning of Aries, H the culminating degree of the ecliptic and the degree of the moon. With H as pole let the arc KLC of a great circle be described; HK, EC, KC will be quadrants of great circles; so KL will measure the angle LHE = AHG = the angle of latitude, since the meridian itself is now the vertical circle. By the theorem of Menelaus, with sines put for chords,

$$\frac{\sin HE}{\sin EK} = \frac{\sin H\Upsilon}{\sin \Upsilon L} \times \frac{\sin LC}{\sin KC},$$

$$\frac{\text{sin decl. culm. deg.}}{\text{cos decl. culm. deg.}} = \frac{\text{sin dist. ☾ from } \Upsilon}{\text{cos dist. ☾ from } \Upsilon} \times \frac{\sin(90^\circ - \text{ang. lat.})}{R}.$$

q) — p. 82, ll. 5–8.

The passage is beyond doubt corrupt or interpolated. What is now gathered from the codex and Plato is false, namely that the arc of the horizon between the ascendant and the direction (azimuth) of the moon is equal to the angle of latitude. For if the moon stands on the horizon, that

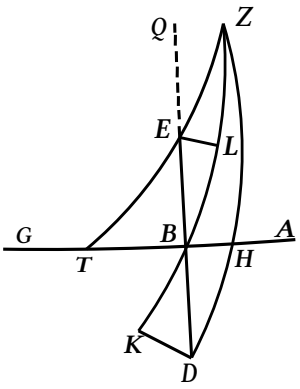
arc of the horizon will plainly be reduced to nothing, which by no means happens to the angle of latitude. [Perhaps al-Battānī had written that the angle to be computed is the one between the part of the ecliptic going from the moon to the middle of the heaven and the part of the vertical from the moon to the zenith; and that the arc of the horizon between the ecliptic (or the ascendant) and the vertical lies to the south or to the north of the ecliptic according as the angle of latitude is to the north or to the south of the ecliptic. This restoration, however, is extremely doubtful. — Schiaparelli.]

r) — p. 82, ll. 8–26.

The correction of the zenith distance of the moon and *the correction of the angles*, when the moon is outside the ecliptic.

I. *Correction of the zenith distance.* — Let ζ be the zenith distance of the degree of the moon's longitude, which is known by the formula of note *i*[a], p. 256; let *z* be the zenith distance of the moon, β the latitude of the moon; al-Battānī's rule is

$$(\alpha) \qquad \sin z = \sqrt{\sin^2\left[\zeta \mp \operatorname{arc}\left(\frac{\sin\beta \text{ sin ang. lat.}}{R}\right)\right] + \left(\frac{\sin\beta \text{ sin ang. long.}}{R}\right)^2}.$$



Let us investigate how the author found this formula. Let us use the same figure that *Almag.* V, 19 (ed. Halma vol. I, pp. 370–72) has: GA is the ecliptic, Q its pole, Z the zenith, BE and BD the two latitudes of the moon (the northern and the southern, according as one or the other is taken), B the degree of the moon's longitude, and lastly ZT, ZK, ZD arcs of vertical circles. Let the arcs EL, KD be drawn perpendicular to ZK; ABZ will be the angle of latitude, whose complement EBL is the angle of longitude. In the right spherical triangle ELB (= BKD) we have (see p. 181, note 1):

$$\sin EL = \sin DK = \frac{\sin EB \times \sin EBL}{R} = \frac{\sin\beta \text{ sin ang. long.}}{R}.$$

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If, because of the smallness of the triangles, we consider the figure as rectilinear, we must have the angle LEB = 90° − EBL = 90° − the angle of longitude = the angle of latitude, so that in the spherical triangle ELB or BKD we have *approximately*:

$$\sin BL = \sin KB = \frac{\sin EB \times \sin LEB}{R} = \frac{\sin\beta \text{ sin ang. lat.}}{R}.$$

If the moon is northern, ZL = ZB − BL = ζ − BL; if southern, ZK = ZB + KB = ζ + BL; so, by the equations just found, we shall have, *approximately* at least,

$$\sin ZL = \sin\left[\zeta - \operatorname{arc}\left(\frac{\sin\beta \text{ sin ang. lat.}}{R}\right)\right],$$

[a]The print has «note h», which concerns the distance of the sun; the formula is (α) of note *i*.

$$\sin ZK = \sin\left[\zeta + \text{arc}\left(\frac{\sin\beta \text{ sin ang. lat.}}{R}\right)\right].$$

The approximation is somewhat better than that of the *Almagest*; for there it is put more simply

$$BL = \frac{\beta \text{ sin ang. lat.}}{R}, \qquad EL = \frac{\beta \text{ sin ang. long.}}{R},$$

$$ZL = \zeta - \frac{\beta \text{ sin ang. lat.}}{R}, \qquad ZK = \zeta + \frac{\beta \text{ sin ang. lat.}}{R}.$$

It remains to compute ZE = z[a]. Ptolemy (see also ed. Halma vol. I, pp. 369–370) takes the triangle ZEL as rectilinear and finds

$$z^2 = \overline{ZL}^2 + \overline{EL}^2 = \left(\zeta \mp \frac{\beta \text{ sin ang. lat.}}{R}\right)^2 + \left(\frac{\beta \text{ sin ang. long.}}{R}\right)^2.$$

Al-Battānī plainly did not want to put simple straight lines for arcs; he substituted their sines for the arcs and then, treating the triangle ZEL as rectilinear, stated:

$$\sin^2 z = \sin^2 ZL + \sin^2 EL = \sin^2\left[\zeta \mp \text{arc}\left(\frac{\sin\beta \text{ sin ang. lat.}}{R}\right)\right] + \left(\frac{\sin\beta \text{ sin ang. long.}}{R}\right)^2.$$

[In solving the spherical triangle ELZ, right-angled at L, the true formula that gives z (or ZE) is

$$\cos ZE = \cos EL \cos ZL,$$

whence arises

$$\cos^2 ZE = \cos^2 EL \cos^2 ZL,$$

that is,

$$1 - \sin^2 ZE = (1 - \sin^2 EL)(1 - \sin^2 ZL),$$

or also

$$\sin^2 ZE = \sin^2 EL + \sin^2 ZL - \sin^2 EL \sin^2 ZL.$$

Al-Battānī, on the other hand: $\sin^2 ZE = \sin^2 EL + \sin^2 ZL$. Hence it is plain that his sin² ZE exceeds the truth by the quantity sin² EL sin² ZL. Since EB = β reaches at most 5°, and the angle EBZ at most 23° 35′, EL is found to be at most 2° 0′[b]; so the greatest value of sin² EL is $\frac{1}{821}$, the radius being taken = 1. In al-Battānī, then, the error of the quantity sin² ZE is at most $\frac{1}{821}$ sin² ZL; and by an argument that need not be given now I find that the greatest error that befalls the parallax in altitude because ZE is computed so imperfectly cannot exceed $\frac{1}{1642}$ of the parallax itself, so at most 3″ or 4″, even if Ptolemy's exaggerated parallax is used when the moon is at the perigee of the eccentric. Al-Battānī's formula, then, though only approximate, serves very well. Ptolemy's cruder formula also suffices; for the difference between the arcs ZE, ZL[c] is never very perceptible, and becomes of some moment only when the moon approaches the zenith, that is, when ZE comprises not many degrees. Then nearly the same quantity is obtained by al-Battānī's method and by Ptolemy's, since arcs can be put for sines without much change in the result. — Schiaparelli.]

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[a] The print has ZE = ζ; ZE is z, the zenith distance of the moon, as in the formulas that follow, and ζ is ZB, that of its degree.

[b] The angle EBZ, the angle of longitude, is bounded by 23° 35′ only when the moon is in the meridian (p. 258, formula (α)). On the horizon it is the angle of the ecliptic with the horizon, which at a latitude of 36° reaches 77° 35′; EL then becomes 4° 53′, sin² EL about 1/138, and the error of the parallax up to about 1/276 of it, some 22″ with Ptolemy's greatest parallax of the moon.

[c] The print has EL for ZL: by cos ZE = cos EL cos ZL, the arcs ZE and ZL differ little except near the zenith.

II. *Correction of the angles.* — Ptolemy, loc. cit., thinks that the corrected angle of longitude is ATZ[a], or AHZ if the moon is to the south of the ecliptic. Al-Battānī does the same, but wrongly, as we shall say below. Both then, wrongly attributing to spherical triangles the property of rectilinear ones, make

$$\text{corrected ang. long.} = ATZ = ABZ - BZT,$$

and, for the southern moon,

$$\text{corrected ang. long.} = AHZ = ABZ + DZK.$$

So far Ptolemy. Al-Battānī teaches the computation of the *variation of the angle*, that is, of BZT, in this way:

$$\sin BZT = \frac{\sin\beta \text{ sin ang. long.}}{R},$$

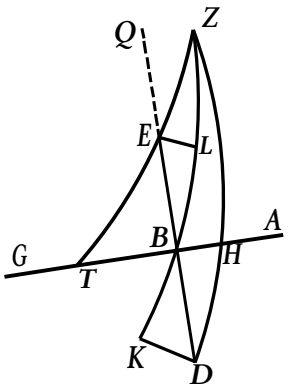
whence

(α) $$\text{corrected ang. long.} = \text{ang. long.} \pm \text{arc}\left(\frac{\sin\beta \text{ sin ang. long.}}{R}\right)$$

(β) $$\text{corrected ang. lat.} = \text{ang. lat.} \pm \text{arc}\left(\frac{\sin\beta \text{ sin ang. long.}}{R}\right).$$

But the computation of the «variation of the angle», too, is not right. For $\sin BZT = R\frac{\sin EL}{\sin ZE}$; so, by the values of these sines found before[1],

$$\sin BZT = \frac{R\dfrac{\sin\beta \cos ABZ}{R}}{\sin z} = \frac{\sin\beta \text{ sin ang. long.}}{\sin z}.$$



So, to restore the truth in this computation of the variation of the angle, something would have to be added in ll. 22–23: «then take again what had come out of the division of the angle of longitude[2] [*and multiply it by the radius; divide the product by the sine of the corrected zenith distance of the moon; what you find*] turn into an arc; the variation of the angle will come out». But it is very much to be doubted that al-Battānī had this correction in mind, as will be shown on p. 262.

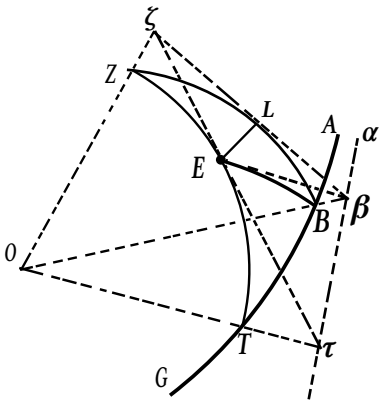
[1] [Ptolemy divides by *z*, making $\sin BZT = \frac{\beta \text{ sin ang. long.}}{z}$. But to put *z* for sin *z* is always false; it can be done only approximately when the moon is a little distant from the zenith, so that sin *z* does not differ much from *z*. — Schiaparelli.]

[2] That is, $\frac{\sin\beta \text{ sin ang. long.}}{R}$.

[a] ABZ is the angle of latitude (p. 259), so that ATZ and AHZ are corrected angles of latitude.

Beyond doubt the criticisms that Ibn Yūnus makes in his introduction (in Caussin p. 72) belong to this computation of the corrected angles: «Likewise false is what Abū l-ʿAbbās al-Faḍl ibn Ḥātim al-Nayrīzī[1], Abū ʿAbd Allāh Muḥammad ibn Jābir ibn Sinān al-Battānī and others say about the angles needed to know the lunar parallax in longitude and latitude, in the computation of eclipses of the sun; for it does not lead to a true knowledge of the apparent place of the moon, since the methods by which one reaches the truth are not used». But these criticisms (which Caussin wrongly thinks belong to al-Battānī's ch. XLIV) touch Ptolemy even more than al-Battānī; and in fact Tannery, *Recherches sur l'hist. de l'astron. ancienne*, Paris 1893, p. 221, note, reproves the Greek astronomer not a little for these corrections.

[The solution of the problem in no way requires the computation of the angle at T, as Ptolemy and al-Battānī have it; the true *angle of longitude*, which with respect to B is ZBQ, is ZEQ with respect to the moon E; and this is very easily deduced by the proportion of the sines, considering the triangle ZEB. — Most false, as was said before, is the equation BZT + ATZ = ABZ, which would hold only if the triangles were rectilinear. — Lastly, both writers erred in the computation of the angle ATZ. With a small change, however, al-Battānī's computation can be corrected quite suitably.



I 261

Let AG be the ecliptic, Z the zenith, E the moon, EB the circle of latitude, ZB and ZT the vertical circles drawn through B and E. Instead of drawing the perpendicular arc EL, let an arc of a parallel be described, so that L and E are equally distant from Z; since EL reaches at most 2°, one arc may be substituted for the other. — From the centre O of the sphere let three planes be drawn that determine on the sphere the three arcs, or sides, of the spherical triangle ZBT; at E and L let the tangents ζτ, ζβ be drawn, which plainly meet at a point ζ of the straight line OZ produced. Let us imagine the plane ζτβ to pass through these two tangents, and project onto it the spherical quadrilateral ELTB, whose projection will then be the rectilinear quadrilateral ELτβ. Because both quadrilaterals are small, the angles and sides of the one will differ from those of the other by a quantity wholly negligible in this problem, and the plane quadrilateral can be considered instead of the spherical one; then, if we use the plane angle at ζ instead of the spherical angle at Z, all the considerations of Ptolemy and al-Battānī will be true. The difference of the angles ζβα, ζτα will be about the same as that between ZBA, ZTA: that difference will not be the spherical angle at Z but the plane angle at ζ; putting this for Z, a quantity good enough and close to the truth will be obtained. The angle at ζ is $R\,\frac{EL}{\zeta L}$; plainly ζL = tg ZL = tg ZE = tg z; so *approximately*

$$\text{ang. } \zeta = R\,\frac{EL}{\operatorname{tg} z} = \frac{\sin\beta\ \text{sin ang. long.}}{\operatorname{tg} z} = \frac{\cos z\ \sin\beta\ \text{sin ang. long.}}{R\sin z}\,.$$

So al-Battānī's rule will become true if the passage on p. 82, ll. 22–23 is restored thus: «then take again what had come out of the division of the angle of longitude [*and multiply it by the cosine of*

[1]He died about the year 310 H (922–23); Suter no. 88, p. 45.

the corrected zenith distance of the moon; divide the product by the sine of the same distance; what you find] turn into an arc; the variation of the angle will come out». So Ptolemy and al-Battānī would have avoided the charge of negligence if in their computations they had used the angle ζ (instead of the angle Z) and the tangent Lζ of the arc (instead of the arc LZ). — Schiaparelli.]

s) — p. 82, ll. 27–36.[a]

The tables of the lunar parallaxes in the climes, taken from Theon's tables.

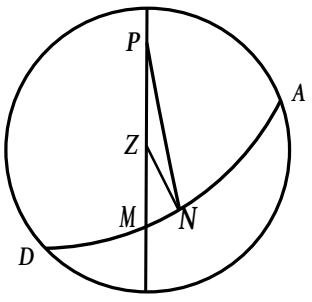
[Theon computed the tables taking the obliquity of the ecliptic as 23° 51′, that is, 16′ greater than al-Battānī's obliquity. From this a small difference must arise in the computation of the parallax; it reaches its greatest amount in the parallax in latitude when the moon, being in Cancer and at its northern limit, culminates in the meridian. The moon is then only 7 1/2° from the zenith of the city of al-Raqqa; the parallax (with a horizontal parallax of 60 minutes) comprises about 7 1/2′; a change of 16′ in the zenith distance then produces a change of about 16″ in the parallax, which was a negligible quantity at that time.

Al-Battānī rightly warns, however, that those tables, though more convenient than direct computation, produce some small errors. This is due to various causes. The first is the different obliquity that Theon used. The second is the interpolations, to be made over the large interval of 15 degrees of hour angle and of half an hour in the clime of the given city; such interpolations may not be used in an accurate computation. The third, lastly, is that these tables hold only when the latitude of the moon is small (as always happens in eclipses of the sun). — Schiaparelli.]

t) — p. 84, ll. 3–18.

I 263

[He now teaches when the parallax in longitude is to be added to the longitude, and when subtracted. Plainly the parallax between the ascendant and the nonagesimal[1] increases the longitude, between the nonagesimal and the descendant diminishes it. But the nonagesimal is in the meridian only when the beginning of Cancer or the beginning of Capricorn culminates; when the ecliptic is placed otherwise, the nonagesimal stands to the east or to the west of the meridian, and, according to the different climes, its hour angle can be 1 hour or even more. So in Theon's tables of parallaxes one must always investigate at what place the positive parallaxes (in longitude) end and the negative ones begin.



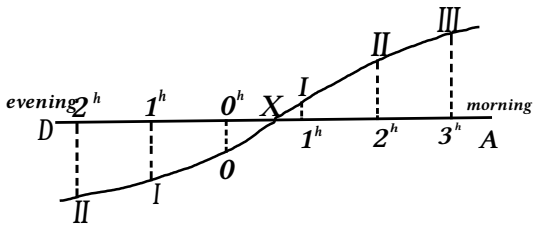
Let Z be the zenith, AD the ecliptic, A the ascendant, D the descendant, N the nonagesimal, M the middle of the heaven; if the moon is found between M and N, it is in the eastern hours,

[1]The point of the ecliptic that is 90° from the ascendant is called the *nonagesimal*; it is the highest point of the ecliptic above the horizon.

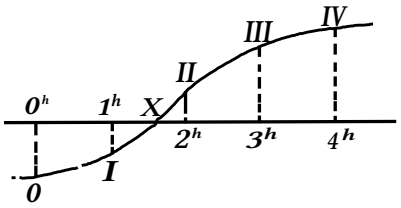
[a]Nallino's Addenda et emendanda (Part I, p. LXXV) add to this note that al-Battānī removed the discrepancies arising from the different obliquity of the ecliptic, assigning in his tables (vol. II, pp. 95–101) to each of Theon's climes another latitude, suited to the new obliquity.

yet because of the parallax it declines towards the west, and the parallax in longitude becomes negative. The opposite can happen when N is to the west of the meridian. The hour angle of the point N, namely (since P is the pole) the angle ZPN, sometimes exceeds an hour in the climes that Theon's tables comprise; it can reach two hours in more northern climes.

In the tables of Cancer and Capricorn the sign of the parallax in longitude changes in the meridian. In the other tables one must look for the smallest number and, of the two numbers adjacent to it, the one that is smaller than the other; between these falls the point at which there is no parallax. — To show this, let us write the hours on the line AD, the morning hours to the right, the evening hours to the left. Let vertical lines I, II, III be drawn, as many millimetres long as there are minutes in the corresponding parallaxes in longitude in any of Theon's tables; the ends of these lines will form a continuous oblique line. The point X at which it cuts the horizontal line answers to the time at which the parallax in longitude is nothing. The hour to which the smallest parallax answers is plainly the first morning hour, I; of the hours adjacent to it, 0 and II, the smaller parallax belongs to the hour 0; so X falls between I and 0, not between I and II, as might be thought at first sight.



— Sometimes X is found between I and II; then the parallax of hour I is smaller than the parallax of noon, as appears from the new figure; but the rule of investigation is always the same. Sometimes X is found even between II and III, as happens not rarely in the tables of the seventh clime, vol. II, p. 101.



«*And when the hours too …*». That is, when it happens that the parallax in longitude becomes nothing between the middle of the heaven and one of the two first adjacent hours; it could have been said less obscurely: *When this happens between the middle of the heaven and the distance of one hour on the side on which the minutes of longitude end* [*or between the first and the second hour*], *the minutes of longitude* etc. By this precept he wants to teach that, when an interpolation is necessary between two parallaxes in longitude of opposite sign, the *difference* of the numbers that give the minutes of the parallaxes must not be taken as the difference of the two parallaxes, but their *sum*. For plainly $(+a) - (-b) = a + b$.

l. 30: «*And this, God willing, will be nearer the truth if the moon is on the ecliptic*». Rightly, for if the moon is outside the ecliptic, the parallaxes found in this way can be affected by an error of even 4′ and 5′. — Schiaparelli.]

On chapter 40

a) — p. 84, ll. 38–40.

To the *corrected parallaxes* of the moon in longitude and latitude, taken from Theon's tables, *a quantity must be added*, which according to Plato is 1/18 of them, according to the codex 1/38. Theon, *Tables manuelles*, vol. I, p. 68, says that 1/20 of them must be added to the parallaxes taken from the tables of correction, because of the parallax of the sun. For he computed the tables of the lunar parallaxes as if he had already subtracted the parallax of the sun from the parallax of the moon. — If the parallax of the sun is taken in round numbers as 3 minutes (see pp. 252 and 258), these 3′ are the eighteenth part of 54′, which, according to Ptolemy and al-Battānī, make up the parallax of the moon at the first limit (53′ 34″; see the tables, vol. II, p. 94). So Plato's number, 1/18, is good.

Because of the different obliquity of the ecliptic that Theon uses, no perceptible difference arises in the parallax; see p. 262, note *s*.

b) — p. 85, ll. 1–27.

I. — *To find the lunar parallax in altitude by observations* (ll. 1–18).

Al-Battānī teaches very well to observe the moon at the nonagesimal (see p. 262, note); for then the circle of latitude passing through the centre of the moon and the nonagesimal becomes the vertical circle, or circle of altitude, so that the parallax in longitude is reduced to nothing, and the parallax in latitude is the parallax in altitude itself.

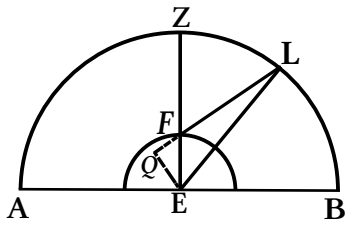
Tycho Brahe too sometimes used the same method: «Now I shall add three other [parallaxes] observed outside the meridian, when the moon was at about the 90th degree of the ecliptic from the horizon, then showing no difference between the apparent and the true longitude but turning it all into the latitude; so, when the true latitude is compared, the investigation of the parallax will be very easy (*facilima* [sic])»[1].

Ptolemy, V, 13 (ed. Halma vol. I, pp. 332–38), had followed another way, choosing the meridian instead of the nonagesimal. For on the 13th day of the month Athyr of the 20th year of Hadrian, or 882 of Nabonassar, 5 1/2 + 1/3 equinoctial hours after noon, he had observed the moon near the tropic of Capricorn (exactly at 3° 10′ of Capricorn), so that the circle of latitude was very nearly the meridian (ἔγγιστα ἦν τότε τῷ μεσημβρινῷ) and the parallax in altitude did not differ from the parallax in latitude. See also Delambre, *Astron. ancienne*, II, 209 and 210.

II. — When the zenith distance and the parallax are known, *to compute the distance of the moon from the centre of the earth*, the radius of the earth being taken as one part (ll. 19–27).

Let ζ be the apparent zenith distance of the moon, p the sum of the parallaxes of the sun and the moon in the vertical circle, a the semidiameter of the earth; al-Battānī gives two solutions of the problem, which, because of much that is left out in the codex and in Plato, have wrongly run together into one.

[1] Tychonis Brahe Dani *Astronomiae instauratae progymnasmata*, Uraniborg and Prague 1610, vol. I, p. 461. See also Riccioli, *Almagestum novum*, Bologna 1651, vol. I, p. 221. — In other problems Hipparchus had already observed the moon at the nonagesimal on purpose, to avoid the parallax in longitude; see *Almag.* V, 3 (ed. Halma vol. I, 293–94, 295), and V, 5 (ed. Halma I, 304); Delambre, *Astr. anc.*, II, 188, 189, 194; and Pappus' remarks in Delambre, *Astr. anc.*, II, 579.



I. Let E be the centre of the earth, F the place of observation, Z the zenith, L the moon; then the angle ZFL = ζ, the angle FLE = *p*, and EL = the required geocentric distance. Let FL be produced until it makes a right angle at Q with the straight line EQ. Plainly:

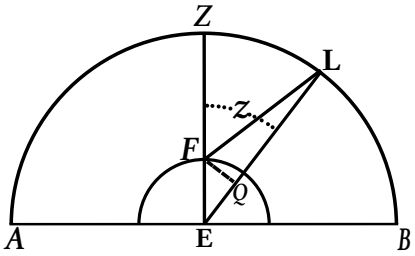
$$\sin FLE = R\,\frac{EQ}{EL}\,, \qquad \sin QFE = R\,\frac{EQ}{EF}\,,$$

that is,

$$\sin p = R\,\frac{EQ}{\text{dist.}}\,, \qquad \sin\zeta = R\,\frac{EQ}{a}\,,$$

whence geocentric distance × sin *p* = *a* sin ζ. And since *a* is taken = 1, al-Battānī's first formula arises at once:

(α) geocentric distance $= \dfrac{\sin\zeta}{\sin p}$.



II. [The other rule, the greater part of which is missing, was beyond doubt this. Let *z* be the *true* zenith distance, namely ZEL; let the perpendicular FQ be drawn; in the same way as before for EL one finds very easily $FL = a\,\frac{\sin z}{\sin p}$. Because the angle FLE is small, one can set without perceptible error $QL = FL = a\,\frac{\sin z}{\sin p}$; and $EQ = \frac{a\cos z}{R}$; so

$$\text{geocentric distance} = EL = QL + EQ = a\,\frac{\sin z}{\sin p} + a\,\frac{\cos z}{R}\,.$$

If *a* is taken = 1, and in the second term the true altitude *h* is put for its complement *z*, al-Battānī's other formula arises:

(β) geocentric distance $= \dfrac{\sin z}{\sin p} + \dfrac{\sin h}{R}$.

On p. 85, ll. 23–24, the words «*multiply into the semidiameter, and the product*» must be deleted. — Schiaparelli.]

On chapter 41

a) — p. 85, ll. 37–39.

I 266

The knowledge of the new moon is of the greatest moment among the Muslim nations, not only for Arabic chronology but also because of the precepts of religion on the fast of the month of Ramaḍān. For the fast begins when the new moon of the month of Ramaḍān appears, and ends when the new moon of the month of Shawwāl is seen. So the Muslims must look out for the new moon on the last (the twenty-ninth) day of the month of Shaʿbān[1], and, if they see it, begin the fast. The first day of Ramaḍān is called the «day of doubt» (*yawm al-shakk*); for on it it can happen that, although the air is clear, the new moon remains invisible, so that it is uncertain whether people must fast or not. When this day has passed the fast must always be begun, even if the new moon cannot be seen because of the conditions of the air. On that «day of doubt», if there is some impediment in the air by which the moon cannot appear, the fast must be begun when the religious judge (*qāḍī*) of the city has accepted the testimony of a single truthful person who asserts that he has seen the new moon, whether a man or a woman, free or a slave[2]. But if there is no impediment in the air, no one's testimony is accepted until many people have seen the new moon, from whose testimony the matter is most certainly established[3]. — They make nothing of astronomical computations of the new moon, except a few theologians and jurists of the Shāfiʿite school, among them Ibn Surayj (d. 306 H, 918).

Thābit ibn Qurra, d. 288 H, 901, composed an epistle «on computing the appearance of the new moon»[4], a copy of which is kept in the British Museum; his method will be set out at the end of note *c.* Al-Bīrūnī, *Chron.* p. 196 of the text = 176–77 of the translation, writes this: «The computation of the appearance of the new moon is a very long and very difficult question, whose exposition requires long computations and many tables. On this, what is read in the astronomical

[1]Which precedes Ramaḍān. It must not be forgotten that among the Muslims the civil day begins at sunset.

[2]If the new moon at the end of Ramaḍān and the beginning of Shawwāl is concerned, the testimony of two truthful men is necessary. — Muslims who profess the Ḥanafite, Mālikite and Ḥanbalite doctrine of law think that, when the judge of some city has accepted the testimony of one man on the day of doubt, the fast must be kept «over the whole earth» (على جميع الدنيا); which today actually happens only in the Turkish empire, thanks to the telegraph. Those who follow the Shāfiʿite doctrine, on the other hand, think that the fast must then be kept only in places not too far away; the very famous al-Ghazālī, *Iḥyāʾ ʿulūm al-dīn*, Cairo 1302, vol. I, p. 209, sets the greatest distance at two days' journey (مرحلتان). [*Translator:* Nallino's Addenda et emendanda (Part I, p. LXXV) add on the announcement of the new moon among the Jews, who did not accept the testimony of women, slaves and the wicked: see Maimonides, *de novilunio*, ch. II, cols. 240–244 of the edition cited in the preface, p. XXXIV, note 3.]

[3]How great the expectation of the new moon can be, and how uncertain the matter among the Muslims, appears clearly enough from al-Jabartī, *ʿAjāʾib al-āthār fī l-tarājim wa-l-akhbār*, Cairo 1301–2 H (in the margin of the Annals of Ibn al-Athīr), vol. VIII, p. 171 (the last day of Ramaḍān 1217 H, on which the air in Egypt was cloudy), and vol. XII, p. 216 (the beginning of the month of Shawwāl of the year 1235 H, in which the moon was very hard to see and there were great disputes about its appearance). — The poet Ibn al-Fāriḍ (d. 632 H, 1235) says of a man whom excessive love had wasted away: كهلال الشكّ لولا انه أنّ عيني عينَهُ لم تتأيّ «[he was wasted away] like the new moon of doubt; had he not groaned, my eye could not have made out his body» (*Dīwān*, Cairo 1310 H, vol. I, p. 19). [*Translator:* Nallino's Addenda et emendanda (Part I, p. LXXV) add a fine example from Aḥmad ibn Khālid al-Nāṣirī al-Salāwī, *Al-istiqṣāʾ li-akhbār duwal al-Maghrib al-aqṣā*, Cairo 1312 H, vol. IV, p. 245: in the night between the 28th and the 29th of Ramaḍān 1292 (28–29 October 1875) twelve men in Rabat, in the Moroccan empire, said that they had most certainly seen the new moon after sunset; the qāḍī of the city accepted their testimony and wrote at once to the sultan, who was then staying near the city, and at the beginning of the 29th everyone stopped fasting. The court astronomers (*falakiyyūn*), however, maintained that the day could by no means be the feast, and doubt arose; after sunset on the 29th everyone sought the new moon in vain in a very clear sky, so the fast was kept again until the night of the 30th, when the new moon was at last seen, and the witnesses, whose fraud was now plain, were thrown into prison. The Addenda of Part II (p. XVII) give further examples of this confusion from Ibn Jubayr (*The travels of Ibn Jubair*, ed. W. Wright, Leiden 1852, pp. 163, 168–70, 172; in Schiaparelli's Italian translation, Rome 1906, pp. 144, 150–52 and 154), and jesting verses on the day of doubt (يوم الشك) by Abū Nuwās, *al-Fukāhah wa-l-ilināṣ fī mujūn Abī Nuwās*, Cairo 1316 H, p. 114.]

[4]So the title should be corrected in Suter, *Die Mathematiker*, p. 35, l. 1, and p. 36.

work of Muḥammad ibn Jābir al-Battānī, or in one of the works of Ḥabash the Calculator[1], is sufficient».

b) — p. 86, ll. 2–8.

Those who erred in this way were probably writers on law (for among the Muslims the rites of religion too, such as praying, fasting and pilgrimage, are part of the law), or men who, instructed in some rudiments of astronomy, every day determine in the mosques, with the help of a few instruments, the hours of the canonical prayers for the varying length of the days in that city, and are put in charge of the sundials and other clocks of the mosque. They are called *muwaqqit*, «determiners of the times»; their science (*ʿilm al-mīqāt*, «the science of determining the times») comprises the computation of the temporal and equinoctial hours, the use of the astrolabe and the construction of quadrants and sundials.

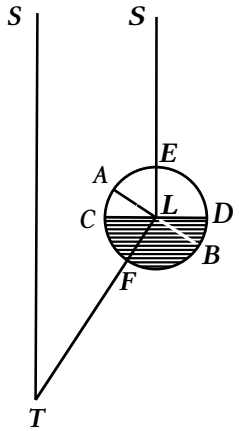
c) — p. 86, l. 17 – p. 87, last line.

Al-Battānī's theory of the *evening appearances of the moon.*

[Besides clear air, the author lays down three conditions on which, above all, it depends whether the crescent of the moon can be seen at the time of setting after the new moon.

I 267

I. *The angular distance of the moon from the sun*, which measures the size of the visible crescent. Let T be the observer on the surface of the earth, TS the direction of the sun, L the moon, LS the straight line that joins the moon to the sun. Because of the very great distance of the sun, the straight lines TS, LS can be considered parallel. The illuminated hemisphere of the moon will be CED, which is determined by the plane CD perpendicular to TS and LS; the hemisphere that can be seen by the observer is AFB, determined by the plane AB perpendicular to TL; the part of the moon visible in the shape of a crescent will be the lune CLA, whose breadth in degrees is the angle CLA = STL, namely the angle that the centres of the sun and the moon make at the eye of the observer T. — Al-Battānī says that, given mean conditions in which these observations are made, the lunar crescent first becomes visible when the angle STL is 12 degrees. He adds that these 12° differ very little from the mean motion by which the moon moves away from the sun in a day, namely 12° 11′; so this roughly confirms the opinion of those who say that the moon can be seen only about 24^h after the true conjunction. As the basis of the computation he chooses 12° 11′.



II. *The light of the lower sky at the place where the moon is at the time of setting*, which is determined by the depth of the sun below the horizon at that moment. With the moon and the sun placed

[1]Ḥabash, Berlin codex, ff. 150v–151v.

on the equator (which mostly does not happen, but is the mean condition of things), it appears from the observation just set out that this depth suffices whenever, from the setting of the sun to the setting of the moon, as many minutes have elapsed as answer to an intervening arc of 12°, that is, 4/5 of an hour. In these 4/5 of an hour the moon moves 2/5 of a degree, or 24′, away from the sun; so if, when the moon sets, the elongation of the two luminaries is 12° 11′, when the sun sets it will be 24′ less, namely 11° 47′. Al-Battānī writes 11° 45′ in round numbers.

But when the line joining the two luminaries is not parallel to the equator, the arc of 12° 11′ cannot be the limit of the appearance. In the very clime of the city of al-Raqqa that line can be nearly parallel to the horizon, because of the obliquity of the ecliptic and the latitude of the moon; sometimes, too, it becomes perpendicular to the horizon. Then that magnitude is not the measure of the depth of the sun below the horizon and of the light of the sky. — So al-Battānī lays down as a general rule that 4/5 of an hour is the arc of the sun's parallel that has passed the horizon between the setting of the sun and the setting of the moon; then he supposes, as a general rule, that under mean conditions 12° of the equator, or 4/5 of an hour, have passed between the two settings, which, as I said, answers to the difference of 11° 45′ at the time of the setting of the sun. Then the depth of the sun below the horizon will be nearly constant; in the clime of al-Raqqa it can vary between about 10° and 8°, according to the season of the year.

III. *The distance of the moon from the earth.* Al-Battānī supposes that the appearance of the moon also depends on the apparent breadth of the luminous crescent, which, other conditions being equal, is proportional to the horizontal parallax of the moon and follows the same laws of variation. In computing these variations he considers the changes of the distance of the moon according to its position on the epicycle; and he supposes that the mean state of things obtains when the moon is at its mean distance, that is, is as far from the earth as the centre of the epicycle, and the sixtieths of the correction of the anomaly for the parallax are 30′ in the third column[1] of the table in vol. II, p. 89. He supposes too that the constant quantities 12° and 4/5 of an hour (of which I spoke above) answer to the mean lunar distance.

I 268

With these suppositions al-Battānī reasons in this way. The condition under which the moon is visible arises from three variable factors, of which one, by growing, can make up for the diminution of another. To give an example, if the elongation of the moon from the sun is greater, the crescent will be proportionally broader; so a smaller depth of the sun below the horizon will be needed to make the moon visible, and the fundamental *arc of seeing* (11° 45′) can be diminished in that proportion. If the moon is nearer the earth than at the mean distance, the apparent size will grow proportionally, and a smaller depth of the sun will be needed for the appearance of the moon. If, lastly, the elongation and the nearness are both greater, the corrections to be applied to the mean *arc of seeing* are to be put together. — From this al-Battānī derives his rules, which can be set out thus:

I. On the given day, at the time of the setting of the sun, let the places of the sun and the moon be derived from the tables (the parallax must be included in this), and let it be found by what interval of time the setting of the place of the moon so computed follows the setting of the sun. Let A be this interval, expressed in degrees. — Plainly, if the moon then had the regular crescent that answers to 12° 11′ of elongation, and the mean distance from the earth, the problem would be solved by comparing A with the fundamental arc of seeing, 11° 45′. The moon will be visible that evening if A is greater than 11° 45′.

II. If at that time the elongation of the moon from the sun differs from 12° 11′, let the elongation (the parallax included) be computed as it appears to an observer on the surface of the earth[2];

[1] So al-Battānī, p. 87, for he computes two series of numbers for the two columns. Otherwise it is the second; it is headed «The shares of the distances of the moon».

[2] This is done closely enough by treating this elongation as the hypotenuse of a right triangle whose legs are the latitude of the moon and its difference in longitude from the sun. Those who want a more exact computation al-Battānī

let its ratio to the fundamental elongation of 12° 11′ be computed. If the ratio exceeds 1, let the fundamental arc of seeing, 11° 45′, be diminished in proportion; if, on the contrary, the ratio is less than 1, let the arc be increased in the inverse proportion. There will come out the *arc of seeing* reduced to the fundamental elongation of 12° 11′. For if the true elongation is greater than the fundamental one, the crescent of the moon must be proportionally broader, and the moon therefore visible under a smaller arc of seeing. This arc B, so reduced, is the one that would have to be compared with the number A if the moon occupied the mean distance from the earth.

III. But if the moon does not keep the mean distance, one must enter with the corrected lunar anomaly into the third[1] column of the table in vol. II, p. 89, which serves for the analogous reduction of the parallax; the horizontal parallax and the apparent breadth of the lunar crescent decrease in the same proportion as the distance. The mean crescent answers to the number 30 of column III, the least to the number 0, the greatest to the number 60. A variation of 30 parts makes 1/12 of the apparent length of the crescent itself; so, if *n* is the number of the third column that answers to the given anomaly, when $n < 30$ the crescent will be diminished in the ratio of 1 to $1 - \frac{1}{12} \cdot \frac{30-n}{30}$, and when $n > 30$ it will be increased in the ratio of 1 to $1 + \frac{1}{12} \cdot \frac{n-30}{30}$. If the crescent is to be increased (that is, if *n* is greater than 30), let the arc of seeing B computed before be diminished in the same ratio; if the crescent is to be diminished ($n < 30$), let the arc B be increased in the inverse ratio. There will come out the quantity C, which will be the *arc* corrected for both variations, namely of the elongation and of the distance from the earth. With C, lastly, the number A must be compared; if this is greater, the moon will be visible in a clear and pure sky; if A is smaller, one must wait for the next day.

This whole theory is constructed most ingeniously and elegantly. The ratios on which it rests are indeed perhaps not as simple as the author supposes; for example, it is not certain that when the breadth of the crescent is doubled the arc of seeing becomes half without the visibility of the moon being changed. Yet even today it is very difficult to do better; and many observations would be needed to determine well the true ratios that must be used. — Schiaparelli.]

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Al-Farghānī, ch. XXV, ed. Golius pp. 93–96, says that the new moon is seen after the conjunction when it is already about 12° of the equator from the sun, that is, when nearly a day-and-night has passed since the conjunction[2]. Likewise al-Jaghmīnī, pp. 256–57. — Ḥabash, ff. 150v–151v, begins the computations *ab ovo*, namely from the correction of the mean motions of the sun and the moon, and gives «a table of the lunar parallax for the seeing of the new moon»; but his method differs from al-Battānī's and seems much more imperfect. He lays down that the new moon can be seen when the arc of depression of the sun below the horizon is 10° or more; that it never appears when this arc is 9° or a little less. Then, that the amount of the light of the moon is found in this way:

$$\sin \text{light } ☾ = R \frac{\sin \text{ diff. long. } (☾ - ☉)}{\sqrt{\sin^2 \text{ lat. } ☾ + \sin^2 \text{ diff. long. } (☾ - ☉)}} .$$

It is worth reporting here much of what al-Khāzinī relates in *al-Zīj al-Sanjarī*, Vatican codex, ff. 87r–90v (treatise IX, part I): «*Chapter II*: on those who [in computing the new moon] rely on simple arcs, not compounded with one another. — The arcs, not compounded with others, that are used in computing the seeing of the new moon are four: *the arc of light, the arc of stay, the arc*

bids (p. 87, ll. 11–13) use other precepts.

[1] Or «second»; see p. 267, note 1.

[2] The Indians already knew this before the age of the Arabs; see the *Sūrya-Siddhānta*, X, 1 (transl. Burgess–Whitney p. 236), and the *Pañcasiddhāntikā* V, 1–3, p. 38 of the translation (I gave the title on p. 206, note 2). In this book the «corrected» distance of the sun and the moon (that is, the distance on the equator) is said to have to comprise 2 *nāḍikā*; so $48^m = 12°$.

of depression, the arc of altitude[1]. — Many of the People of the Book[2] say: [When it is] between 10 and 12, and more, the moon will be within the limits of seeing and can be seen. — The Indians, the Chinese and Muḥammad ibn Mūsā al-Khwārizmī say: If the arc of stay is 12 or more, the new moon will be seen; otherwise not. — Aḥmad ibn ʿAbd Allāh Ḥabash the Calculator says: When the depression of the sun at the time of the setting of the moon is 10° or more, it will be seen; otherwise not. And on the authority of the same author al-Saʿīdī[3] relates that if the arc is 8 2/3° or more it will be seen; if less, not. — Kūshyār[4] says: What I hold likely (المعتبر عندي) is that the arc of light comprises 10°, the arc of stay 8°, the arc of apparent altitude 6°, the arc of depression from 8° down to 7 1/2°. If these arcs are so, or greater, the new moon will be visible; if smaller, not. He also says: If the sum of the arcs of light and of depression is 18° or more, the new moon will be seen».

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— In chapter III al-Khāzinī teaches how those arcs are found (ff. 87v–88r); then he has this (ff. 88r–89r): «*Chapter IV*: on the principles on which al-Battānī relies (في ذكر ما اعتمد عليه البتاني). — Among the computations compounded [of several arcs] (ومن المركبات) is the one that al-Battānī gives, who relies on the arc of stay and corrects it (وعدله) in two ways by means of the arc of light. — *The first method.* He says: On the night of the thirtieth day[5] of the Arabic month let us determine the true places of the sun and the moon (قومنا النيرين) at the time of the setting of the sun; let us correct the place of the moon in longitude and latitude (عدلنا), and find the arc of light and the arc of stay then obtaining. Let us look at the arc of light; if it is 12 1/6°, the mean arc of seeing, which comprises 11° 45′, will already be corrected. If the arc of light is greater or less than 12° 10′, we shall have to correct the mean arc of seeing, or 11° 45′, to it. This correction is: let us take the difference between the arc of light found and 12° 10′, and, entering with this difference into the first table of the correction of seeing made by al-Battānī, let us take the degrees and minutes that answer to it, which we shall keep, since they will not yet be corrected. Then, entering the same table with the [corrected] anomaly of the moon too, let us take the minutes of proportion (دقائق النسب), which will be increasing or decreasing according as the difference taken before was increasing above 12° 10′ or decreasing from it. Let us multiply the minutes of proportion found into the [degrees and minutes] kept, and divide the product by 60; there will come out the *equation*, which we add to the mean arc of seeing (namely 11° 45′) if the difference was decreasing, or subtract from the arc of 11° 45′ if the difference was increasing. There will result the *corrected arc of seeing* (قوس الرؤية المصححة). If the arc of stay is equal to this corrected arc of seeing or greater, the new moon will appear; if smaller, not. — *His* [al-Battānī's] *other method.* He says: On the night of the thirtieth day of the Arabic month let us determine the true places of the sun and the moon, and both the arcs of light and of stay then obtaining. If the arc of light is 13° 40′, the mean arc of seeing, which comprises 10° 50′, will already be corrected. But if the arc of light is greater or less than 13° 40′, we shall have to correct the arc of seeing to it. The correction is this: let us take the difference between the arc of light found and 13° 40′; entering with it into al-Battānī's second table of the correction of seeing, let us take the degrees and minutes that answer to it and keep them. Then, entering the same second table with the corrected anomaly [of the moon], let us take the minutes of proportion that answer to it; they

[1] الباب الثاني في ذكر من اعتمد على القسي المفردة البسيطة. القسي المفردة المستعملة في عمل معرفة رؤية الهلال اربع احداها قوس النور والثانية قوس المكث والثالثة قوس الانحطاط والرابعة قوس الارتفاع. — From ff. 87v–88r it is plain that the arc of stay is the stay of the moon after the setting of the sun, the arc of depression the depression of the sun below the horizon at the time of the setting of the moon, and the arc of altitude the altitude of the moon at the time when the sun sets.

[2] قالت طائفة من اهل الكتاب. The Muslims call *People of the Book* the Christians, the Jews and the Ṣābians (not the Sabaeans), who received a written revelation from God. — Here the Jews, whose calendar is lunisolar, are probably meant.

[3] The codex has السعدى. Perhaps, as Suter suggested to me by letter, he is Muḥammad ibn Aḥmad al-Ḥāzimī al-Saʿīdī, whose date is so far unknown, who made an epitome of the Almagest; see Suter, *Die Mathem. u. Astronomen der Araber*, p. 202, no. 520.

[4] He flourished at the beginning of the eleventh century; Suter, pp. 83–84, no. 192.

[5] By the civil usage of the Muslims the day begins at sunset; so the night precedes the day.

will be increasing or decreasing according as the difference belonged to the arc of light found or to the arc of 13° 40′. Let us multiply them into the [degrees and minutes] kept, and divide the product by 60; there will come forth the *absolute equation* (تعديل مطلق), which must be turned into degrees of ascensions. That is, let us add it to the degree with which the moon sets, if the difference belonged to the arc of 13° 40′; let us take the oblique ascension of the nadir of the degree so resulting, and the ascension of the nadir of the degree with which the moon sets; let us subtract the smaller from the greater; there will result the *corrected equation* (التعديل المعدل). Let us add it to 10° 50′ if the difference belonged to the arc of 13° 40′, or subtract it from 10° 50′ if it concerned the arc of light; there will result the *corrected arc of seeing*. Then if the arc of stay is equal to the corrected arc of seeing or greater, the new moon will be seen; if smaller, it will not appear».

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Then he sets out the method of Thābit ibn Qurra (f. 89r–v): «*Chapter V*: on what Thābit ibn Qurra al-Ḥarrānī[1] gives, who corrects the arc of depression by the arc of light. He says: Let us determine the true places of the sun and the moon (قومنا النيرين) at the time of the setting of the moon, on the night of the thirtieth[2] day of the Arabic month; let us likewise find the arcs of light, of stay and of depression, and also the distance of the moon from the place where the perpendicular (or cathetus) of the light falls (مسقط عمود النور)[3]. — *Section*: on the total limits (في الحدود الكلية), [to be taken] from the table. When we seek them, let us enter with the anomaly of the moon into the table of the limits of seeing made by Thābit, and take what answers to it of the total, absolute arcs (المطلقة), [namely] the arc of light, the equation, and the extreme of the distance; let us keep all these separately. — *Section*: on the examination to be made (في الاعتبار). Let us look at the arc of light found by computation; if it is less than the arc of light taken from the table, let us not hope for the seeing of the moon, because of the smallness of the light in the body of the moon; if it is equal to the extreme of the distance or greater, the moon will be seen by day, before the setting of the sun, and no computations will be needed; lastly, if it is greater than the arc of light taken from the table but less than the extreme of the distance, a correction will be needed. — *Section*: on correcting the arc of light. If we wish to correct it, we shall multiply the complement of the arc of the distance of the moon from the place where the perpendicular of the light falls into the arc of light taken from the table; we shall divide the product by 360; there will result the *corrected depression*, which we shall keep separately. Let us take the difference between the arc of light found [by observation] and the arc of light coming from the table and multiply it into the corrected depression; let us divide the product by the arc of the equation of the table; there will come out the *second equation*. If the difference belonged to the arc of light found [by observation], we shall subtract the second equation from the corrected depression; if the difference belonged to the arc of light taken from the table, we shall add the second equation to the corrected depression. There will result the *depression to be used* (الانحطاط المحصى). — *Section*: on the examination to be made again. If the depression of the sun that we have found by computation at the time of the setting of the moon is equal to the depression to be used, or greater, the new moon is seen; if smaller, it is not seen».

Lastly (ch. VI, ff. 89v–90v) al-Khāzinī teaches his own method, which, since it is later than al-Battānī, it is not worth while to report here.

d) — p. 88, ll. 1–34.

[«*To return now to the opinion* ...». — Al-Battānī now sets out another way of carrying out the

[1] See p. 266, at the end of note *a*.

[2] See p. 269, note 5.

[3] As appears from f. 87r, the formula computed is: cos dist. ☾ from the perp. of light $= \frac{R \cos \text{ arc of light}}{\cos \text{ depress. of the sun}}$. [*Translator:* The print has R in the denominator, with the cosine of the depression of the sun. The rule of the right spherical triangle, cos (arc of light) = cos (depression) cos (distance) for the radius 1, gives R cos arc of light / cos depression when the functions are taken to the radius R, as in the other formulas.]

same computation; it rests on the same principles, but takes other numbers as its basis. He supposes that the moon at its greatest distance from the earth, that is, with the least apparent diameter, becomes visible when the *arc of seeing* is 10° 50′ and the angular distance from the sun 13° 40′. He explains the choice of these numbers in a rather strange way, saying that the first is the *least* relative motion of the sun and the moon on the ecliptic that can occur (namely when the sun has its greatest hourly motion and the moon its least), the second the *greatest* relative motion, when the sun moves as slowly and the moon as fast as possible. But however that may be, the condition of things represented by these numbers does not differ much from the condition chosen in the preceding solution. According to what he had supposed before, one finds $11^\circ\,45' \cdot \frac{13}{12} \cdot \frac{12^\circ\,11'}{13^\circ\,40'} = 11^\circ\,21'$ = the *arc of seeing* that answers to the greatest lunar distance and the elongation of 13° 40′. Here al-Battānī puts only 10° 50′ for 11° 21′; the difference is of small moment, yet by the new computation the new moon can sometimes be judged visible in conditions in which by the former computation it could not appear. — The rule of the computation is the same as in the preceding case, except only that, in working out the ratio of the distance of the moon from the earth, the greatest distance instead of the mean is taken as the basis of that ratio, and one must proceed accordingly. — Schiaparelli.]

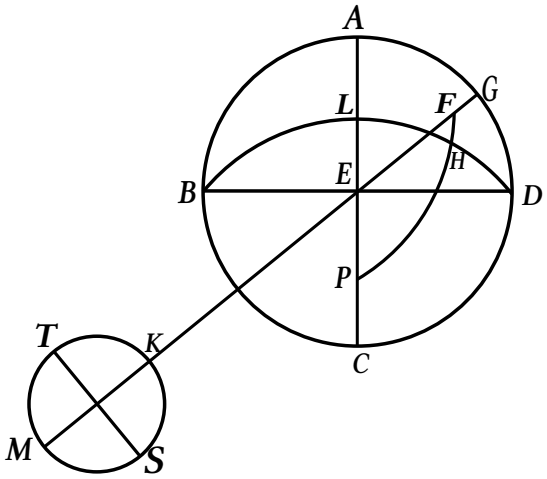
e) — p. 89, ll. 31–32.

[«*The space between the two arcs is the image of the new moon*». — The author's construction gives the inclination of the line of the horns with respect to the ecliptic quite exactly, at least as long as the difference of longitude between the sun and the moon does not become much greater than the difference usually found on the day of the first appearance. It must also be noted that GT is the *true breadth of the visible lune* of the moon, not the breadth of the *crescent*, which is much thinner than the lune, for there the lunar surface is seen under great obliquity. So the words «is the image of the new moon» must be understood. Better below, p. 90, ll. 18–19: «The line TG will be … the amount of the digits of the light of the moon». — Schiaparelli.]

f) — p. 91, l. 11 ff.

I. *To find the place of the horizon from which the new moon will appear.* I 272

[In the figure of this page there are some things that do not agree at all with the text; and in the text itself some things, printed in italics, are not right. Since EG is the vertical circle passing through the moon F and indicating its direction G on the horizon, the point H of the ecliptic culminating with the moon must lie not on the line from G to E but on the great circle that goes from the moon F to the celestial pole. Yet the text of the codex and of Plato speaks of the line KEHFG! The figure is badly drawn, and the text has been fitted to its errors, as we have seen elsewhere too. I think that al-Battānī's true figure was the one I draw here; then what is printed in italics on p. 91 must be deleted. The point H itself is wholly useless. Why the moon must be *southern*, and Libra be only at D, I do not know; still less do I understand why H must be the beginning of Scorpio. — Schiaparelli.]



II. Al-Battānī, p. 91, ll. 1, 5, 9, 11, 34, mentions the use of *tubes* in observations, by which, as is well known, a great part of the rays of diffused celestial light, which disturb vision, are kept out. This is the oldest mention of all of tubes among the Arabs; for those who have written the history of optics know only the tube fitted to a large sextant with which Naṣīr al-Dīn al-Ṭūsī observed the sun in the observatory of Marāgha (founded in 657 H, 1259); instead of the alidade (diopter), the tube led the rays of the sun to the divisions of the sextant.

Among the *Greek astronomers* there is no mention of tubes, except for the little book of Pseudo-Plutarch *On the face in the orb of the moon*, which concerns observing the moon. Aristotle, *On the generation of animals*, V, 1, compares vision to a motion: «so things far away would be seen best if a kind of tube (αὐλός) extended from the eye straight to the thing to be seen; for then the motion that proceeds from the visible thing would nowhere be dispersed. But since this is not so, it must be that the farther it reaches, the farther things are seen»[a]. Polybius, X, 43 ff., mentions a dioptra fitted with two tubes, with which Cleoxenus and Democletus observed signals made by distant soldiers. In the *Talmud* it is related that Rabbi Gamaliel had a tube for measuring the distances of places and the depths of valleys. In Europe, lastly, Walafrid learned the use of the tube in 825 at Reichenau[b]. — See R. Wolf, *Handbuch der Astronomie, ihrer Geschichte und Litteratur*, Zürich 1890–93, vol. I, pp. 318–19, § 133; and S. Günther, *Das gläserlose Sehrohr im Altertum und Mittelalter*, in Bibliotheca Mathematica, ed. G. Eneström, Neue Folge, vol. VIII, Stockholm 1894, pp. 15–23.

Schiaparelli points out to me a passage of the very ancient Chinese book called *Shù-King*[1], in which this is related of King Shun (2255–2205 BC): «He observed the jewelled (or jewel-like?) sphere and the jewel-adorned tube, and learned the seven directors [the planets]». But from the information that the distinguished sinologist L. Nocentini most kindly gave me, it is plain that this version is not certain in every part and rests on commentaries of the second century BC. Moreover, from the description of the very famous interpreter Ts'ai C'in (1167–1230), who used

[1]Part II, book 1, no. 3 in Legge's English translation (*The sacred books of the East*, vol. III, Oxford 1879, pp. 38–39); or part I, ch. 2, no. 5 in Gaubil's French translation revised by G. Pauthier (*Les livres sacrés de l'Orient*, Paris 1840, p. 49). In Chinese and Latin in Zottoli, *Cursus litteraturae Sinicae*, vol. III, p. 333.

[a]Nallino's Addenda et emendanda (Part I, p. LXXV) add that in the same book V, 1, a little before the passage quoted, Aristotle says: «He who shades himself with his hand, or looks through a tube, distinguishes the differences of colours neither more nor less, but sees farther. So too those who sometimes look at the stars from pits and wells».

[b]Nallino's Addenda et emendanda (Part II, p. XVIII) report, after Suter (Bibl. Math., 3rd series, vol. 5, 1904, p. 81), that the diary of Walafrid Strabo, on whose authority modern writers speak of the use of tubes in the European Middle Ages, was forged by P. Marty; the notice nevertheless has some historical ground, for codex 18 of the Stiftsbibliothek of St. Gall (tenth century), p. 43, has a picture of a monk standing on a stool and looking at the sky through a long tube (reproduced as a woodcut in the Mitteilungen der antiquarischen Gesellschaft zu Zürich 19, 1877).

those older commentaries and says that the tube was placed crosswise, it is by no means clear whether the tube served to observe the stars or rather the armillary sphere itself[a].

On chapter 42

a) — p. 94, ll. 5–11.

[The precepts concern the second equation of the computation of the evection. If for a double elongation of 180° the apparent radius of the epicycle (and so also the speed with which the equation of the anomaly grows) increases from 5° 1′ to 7° 40′, by how much will it grow for a double elongation of 2ε°? Plainly by about $\frac{2^\circ\,39'}{5^\circ\,1'} \cdot \frac{2\varepsilon^\circ}{180^\circ}$. In the same proportion the speed of the equation of the anomaly will grow, in the upper segment of the epicycle against the order of the signs, in the lower segment in the order of the signs (from west to east). Since this mean hourly speed is 33′ of the circumference of the epicycle, at the eye of the observer, who stands twelve times as far away as the centre of the epicycle, it will subtend an arc twelve times smaller, or $\frac{33'}{12} = 165''$; of this quantity the part given by the ratio set out before must be taken. So the apparent hourly motion of the anomaly on the epicycle will grow (and the hourly motion of the moon on the ecliptic will decrease when the moon is at apogee, increase when it is at perigee) by the hourly quantity

$$165'' \, \frac{2^\circ\,39'}{5^\circ\,1'} \cdot \frac{2\varepsilon^\circ}{180^\circ} = \frac{33''}{18} \cdot \frac{159}{301} \cdot \frac{\varepsilon^\circ}{1^\circ} = \frac{5247''}{5418} \cdot \frac{\varepsilon^\circ}{1^\circ}\,.$$

The coefficient differs so little from unity that it can be taken as about 1; we shall then say that, *because the epicycle comes nearer to the observer when the moon is outside the mean conjunction, its apparent hourly motion in the apogee part of the epicycle is retarded each hour by as many seconds as there are degrees in the elongation* ε°, *and this hourly motion is accelerated by as many in the perigee part of the epicycle.* Here al-Battānī supposes the moon to be near the perigee or the apogee of the epicycle; at the greatest equations of the anomaly the effect is nothing. But plainly the author thought a rough approximation enough.

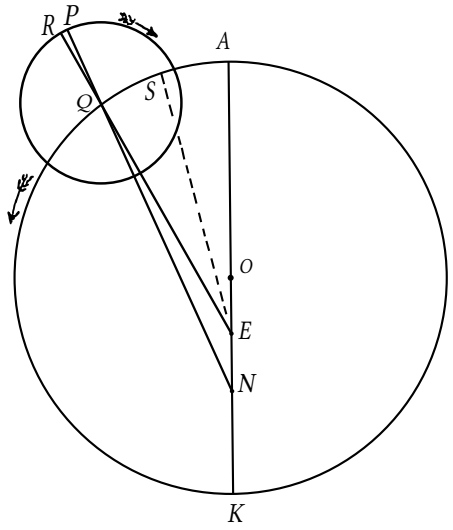
From what we have said, the use of the small table on the right-hand side of vol. II, p. 88 is plain, and why in appendix H (p. 150) the sign of the correction is said to be negative from 265° to 95° (in the upper segment of the epicycle) and positive from 95° to 265° (in the lower segment). The small table does not go beyond 7°, because the mean elongation at the time of the true conjunction can exceed 7° a little, and never reaches 8°. — Schiaparelli.]

b) — p. 95, l. 33 – p. 96, l. 2.

[The reason why 1/6 + 1/8 of the difference of longitude is added to the *lunar anomaly*. By the precept of p. 93, ll. 38–40, and by its rough proof, which he now gives, al-Battānī wants in some way not to neglect the computation of the evection when the time of the true conjunction is some hours away from that of the mean. Let us take the first of the cases given by al-Battānī, namely that at the time of the mean conjunction the true place of the sun on the ecliptic precedes the true place of the moon; let ε be the difference of longitude in degrees. The moon will approach the sun

[a]Nallino's Addenda et emendanda (Part I, p. LXXV; Part II, pp. XVII–XVIII) add that the Chinese instrument may perhaps be compared with Gerbert's armillary sphere and celestial globe, whose poles were pierced by a tube directed at the constellation of the Little Bear (Günther, loc. cit., pp. 16–17); and that the word *mir'āh* in two places of the poet Abū l-ʿAlāʾ al-Maʿarrī (d. 449 H, 1057), cited by von Kremer, *Culturgeschichte des Orients unter den Chalifen*, Vienna 1875–77, vol. II, p. 448, who counts it among the polished metal mirrors of the Arab astronomers, does not mean a tube, as the addenda of Part I asked, but beyond doubt the astrolabe: the astronomer's *mir'āh*, small as it is, shows him all the tilled land and all the desert, and holds the sun, the moon and the stars.

by one degree almost exactly every two hours; so the true conjunction will occur about 2ε hours after the mean, the mean elongation (which was nothing at the time of the mean conjunction) will be ε at the true conjunction, while the *mean* longitude of the moon will *exceed* the mean longitude of the sun by ε degrees, and the mean lunar elongation will be 2ε degrees towards the east[a], that is, in the positive direction.



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Let AK be the eccentric circle of the moon, O its centre, E the earth, N the centre of the equant, or the point about which the motion of the centre of the epicycle always appears uniform (see p. 237), so that EN = EO. Let the point S indicate the direction of the sun; let the angle AES be the mean elongation, and AEQ its double; Q will be the centre of the epicycle, the point P the origin of the mean anomalies, or the mean apogee, R the true apogee, PR the equation of the anomaly *to be added* to the mean anomaly to give the true one. It must be shown that in places near the conjunction the angle RQP[b] is 1/6 + 1/8 = 7/24 of the mean elongation QES, or 7/48 of the double elongation QEA. Since the angles at E and N are all very small, we can use a well-known theorem of optics and affirm that the angles QEA, QNA, under which the arc QA is seen from the points E and N, are to each other in the inverse ratio of the distances of these points from the arc; so QEA : QNA = NA : EA, whence (QEA − QNA) : QEA = (NA − EA) : NA, or NQE : QEA = EN : NA. And since, according to Ptolemy, NE = $10^p\ 19'$ if EA is taken = 60^p, and since NQE = RQP = the *share*, or equation of the argument, we shall have

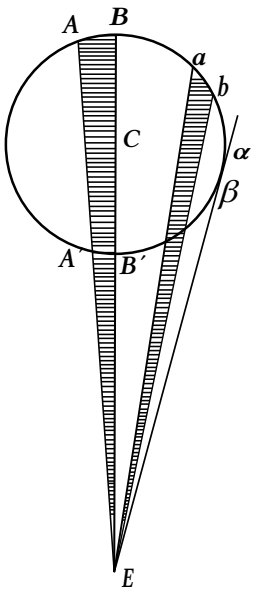
$$\text{equation of the argument} = \text{double elongation}\ \frac{10^p\ 19'}{70^p\ 19'} = \text{elongation}\ \frac{20^p\ 38'}{70^p\ 19'}\,.$$

This will be the nearer the truth the smaller the angles considered; but it is always exact enough for the use of this astronomical computation. The ratio $\frac{20^p\ 38'}{70^p\ 19'}$ makes 0.2934 in decimals; al-Battānī derives 1/6 + 1/8 = 0.2916 from his rougher computation. The agreement is excellent. — Schiaparelli.]

c) — p. 96, ll. 3–9.

[a] That is, the double elongation, AEQ in the figure below.

[b] The print has RPQ; the angle meant is at the centre Q of the epicycle, which is small and is set equal to NQE below.



[«*In this Ptolemy* ...». — Al-Battānī now wants to show, rather obscurely, why he considered before the moon in the position on the epicycle to which an equation of 3° answers, whereas at first sight the case might be thought of greater moment in the position answering to an equation of 5°. The author says that, with an equation of about 3° and a correction of the anomaly of 1 1/2°, the error arising from neglecting the new correction reaches its greatest amount; but that with the equation = 5° the correction is nothing or of small moment. That this is true will be plain from what follows. Let C be the centre of the epicycle, E the earth. Let the arcs *ab* and αβ be 7/24 of the elongation, *ab* at the place where the equation CE*a* = 3°, αβ at the place where the equation CEα = 5°. Plainly, if the correction of those 7/24 of the elongation is neglected, the error at the first place is the angle *a*E*b*, much greater than the error made at the second place, where it appears nothing or negligible.

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But someone[1] may say: Why do you not take the part AB of the epicycle as the place of the greatest error? Is not the angle AEB (the amount of the error) even greater than in the preceding cases? Al-Battānī answers thus: If the moon is at A or B (or also at A′ or B′), the elongation depends on the equation of the sun alone, and is at most 2°; so its 7/24 exceed by a little [...][a] degree, and often do not even reach half. Hence it follows that in fact [...] places about *ab*) are the points at which the greatest [...] arises from neglecting those 7/24 of the elongation; but this is about 1/8 of a degree, or 1/4 of an hour[2], in the relative motion of the two luminaries [...] Schiaparelli.]

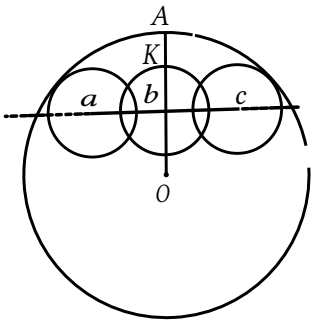
[1][In the text: «Ptolemy». But the reason why Ptolemy (see p. 96, note 2) had set 7° 24′ has nothing to do with this problem; for he was discussing the ecliptic limits of the sun. — Schiaparelli.]

[2]On p. 96, l. 6, the codex and Plato have 1/8 of an hour. But it seems that 1/8 of a degree should be read, as on p. 96, l. 1.

[a]The upper right corner of p. 275 is lost in the scan on which this translation rests, and with it the ends of the first four lines, about eight letters each; [...] marks them, and nothing is supplied.

On chapter 43

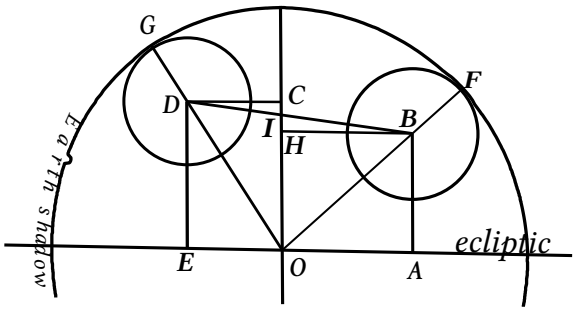
a) — p. 97, l. 35.



[Al-Battānī uses the word *stay* in various senses. Let O be the centre of the shadow, *b* the centre of the moon at the middle of the eclipse, *a* the centre of the moon at the end of totality, *c* at the beginning. Here and on p. 98, l. 36, the *minutes of the stay* are the quantity AK by which the magnitude of the total eclipse exceeds 12 digits; but on p. 98, l. 6, and often elsewhere, the *minutes of the stay* are 1/2 the length *bc* of the path that the moon runs within the shadow from the beginning to the end of totality. The *hours of the stay*, or *time of the stay*, give the interval of time in which the moon comes from *b* to *a*, namely half the time of totality. Lastly, he calls *the whole stay* the greatest value of the stay, which answers to a central eclipse; it is therefore half the difference of the diameters of the shadow and the moon. — Schiaparelli.]

b) — p. 98, l. 39 – p. 99, l. 3.

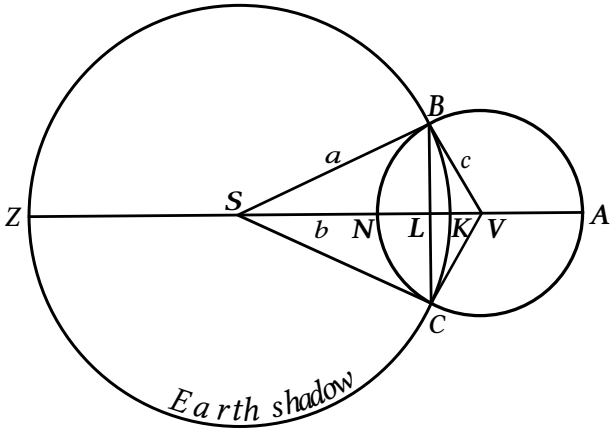
[The words printed in italics teem with several errors arising from the carelessness of the copyists. The probable restoration, as appears easily from the figure, is this: «From this motion, as we have seen, learn the latitude of the moon at both times [BA or DE], *and subtract its square from the square of the half-difference* [OB or OD] *of the two diameters, which is the amount of the minutes of the whole stay*[1]. *To the remaining quantity* [$\overline{OA}^2$ or $\overline{OE}^2$, or, what is the same, $\overline{HB}^2$ or $\overline{DC}^2$] *add the square of the difference* between the latitudes of the moon at the middle of the eclipse and at that time [$\overline{IH}^2$ or $\overline{IC}^2$]; divide the root of this sum [$\sqrt{\overline{HB}^2 + \overline{IH}^2} = IB$, or $\sqrt{\overline{DC}^2 + \overline{IC}^2} = ID$] by the gain of the moon. What comes out from the time of the beginning …». The preceding rules on p. 98 are proved in the same way. — Schiaparelli.]



c) — p. 99, l. 22 – p. 100, l. 8.

[1] See note *a* above.

[*The computation of the area of the lunar disc that is covered.*



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I do not remember having found the author's rules for computing the *sagittae* of the moon and the shadow in other writers; they can be proved in this way. Al-Battānī says that the sagitta LK of the shadow $= \frac{KA \times KN}{2\,SV}$. Let *a* be the semidiameter of the shadow, *c* the semidiameter of the moon, *b* = VS the distance of the centres. This distance is divided by the common chord BC into two parts, SL = *y* and LV = *x*; so *x* + *y* = *b*. Then Euclid's well-known theorem (II, 13) teaches that in the triangle SBV

$$a^2 = b^2 + c^2 - 2b \cdot x$$
$$c^2 = b^2 + a^2 - 2b \cdot y$$

so

$$x = \frac{b^2 - a^2 + c^2}{2b}, \qquad y = \frac{b^2 + a^2 - c^2}{2b}.$$

It is now easy to compute both sagittae, LK = *a* − *y* and LN = *c* − *x*. Substituting the values of the lines *y* and *x* just found:

$$LK = a - y = a - \frac{b^2 + a^2 - c^2}{2b} = \frac{2ab - b^2 - a^2 + c^2}{2b} =$$
$$= \frac{c^2 - (a-b)^2}{2b} = \frac{1}{2b}\Big\{c + (b-a)\Big\}\Big\{c - (b-a)\Big\},$$

whence

(α) $LK =$ sagitta of the shadow $= \dfrac{1}{2SV}\, KA \times KN.$

Similarly

$$LN = c - x = c - \frac{b^2 - a^2 + c^2}{2b} = \frac{2bc - b^2 + a^2 - c^2}{2b} =$$
$$= \frac{a^2 - (b-c)^2}{2b} = \frac{1}{2b}\Big\{a + (b-c)\Big\}\Big\{a - (b-c)\Big\},$$
$$LN = \text{sagitta of the lunar disc} = \frac{1}{2SV}\, ZN \times KN.$$

Al-Battānī proposes formula (α); he derives the sagitta of the lunar disc directly from the sagitta of the shadow, setting

(β) $LN =$ sagitta of the lunar disc $= NK - LK =$

$=$ digits of the corrected eclipse $-$ sagitta of the shadow.

A truly most elegant solution.

Then al-Battānī computes the half-chord BL (*the half common sine*), using the well-known rule $BL = \sqrt{NL \times LA}$:

(γ) half common sine $= \sqrt{\text{sag. lun. disc} \times (\text{digits of the lun. diam.} - \text{sag. lun. disc})}$.

He finds the area of the triangle BVC (*the triangle of the moon*) by multiplying the half common sine BL into the distance LV; and this distance is LV = KV + LK = (semidiam. ☾ − digits of the corrected eclipse) + sagitta of the shadow[1]. The area of the *triangle of the shadow* BSC is computed in the same way. — This done, he derives *the area of the arc of the moon*, that is, of the sector VBNC, by multiplying the semidiameter VN of the moon into the *arc of the moon*, namely into the arc BN, taken from the tables with the help of the sine BL. Similarly, *arc of the shadow* BK × semidiameter SK of the shadow = *area of the arc of the shadow*. — Lastly

(δ) (area of the arc of ☾ + area of the arc of the shadow)−

$$-(\text{triangle of } ☾ + \text{triangle of the shadow}) =$$

$$= \text{area of the covered part of the lunar disc},$$

which is the area of the curvilinear figure NBKC. — Schiaparelli.]

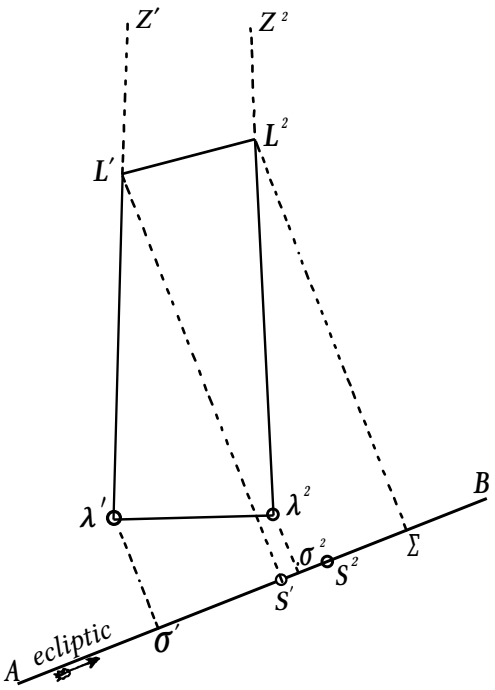
On chapter 44

a) — pp. 104–106.

The computation of the *apparent conjunction of the sun and the moon.*

[The relation between the apparent conjunction and the true conjunction in eclipses of the sun gives rise to two different problems. When the time of the apparent conjunction is known from direct observation of the eclipse, the epoch of the true conjunction is very easily derived. It is enough to compute the effect of the parallax at the time of the true conjunction; then the difference of the true longitudes of the sun and the moon will be found, and from this difference, the hourly synodic or relative motion being known, the time of the true conjunction will easily be deduced.

But when the time of the true conjunction is given, to deduce the apparent conjunction one must know the parallax at the time of the apparent conjunction, which time is itself unknown. So the problem cannot be solved directly, but only by approximations. Al-Battānī uses several successive approximate corrections, and proceeds in this way.



Let AB be the ecliptic in the region of the celestial sphere in which the sun and the moon stand at the time of the eclipse; at the time of the true conjunction (such, that is, as it would be seen from the centre of the earth, and as it is deduced from the astronomical tables) let L′ be the true place of the moon, S′ the true place of the sun, L′S′ perpendicular to AB. Let L′Z′ be the direction of the vertical circle of the moon at the same moment; because of the parallax in altitude the moon will appear to the observer not at L′ but at λ′. If the perpendicular λ′σ′ is drawn from λ′, the arc of the ecliptic S′σ′ will plainly be what al-Battānī calls *the first parallax* [in longitude], namely the difference of longitude that is actually observed between the sun S′ and the moon λ′ at the time of the true conjunction; we shall call this moment of time ϑ.

[1]The codex and Plato wrongly have «sagitta of the lunar disc» (p. 99, ll. 28–30).

In the case that the figure represents, the moon is to the west of the nonagesimal (or of the degree that is 90° from the ascendant); and since the moon on the ecliptic is behind the sun (that is, σ′ is behind S′), it will need some time to reach the longitude of the sun. This space of time is what we must find.

If the effect of the parallax in longitude always remained one and the same, it would be enough to divide the arc S′σ′ by the relative hourly motion of the two luminaries; the time sought would come out. But the parallax changes quickly; and since the interval between the true and the apparent conjunction can sometimes reach nearly two hours, we cannot even suppose that it changes uniformly. So if, as al-Battānī bids, we take the parallax in longitude S′σ′ and divide it by the relative hourly motion[a], we do not find the time by which the apparent conjunction is retarded with respect to the true, but only a quantity near that time. Let ε′ be this time close to the true one, namely, as al-Battānī says, *the hours of the first parallax*. Let it be added to the time ϑ; there will come out the first approximation to the problem and to the time sought, namely ϑ + ε′.

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Then let the arc S′σ′ be added to the longitude of the moon, to the anomaly and to the argument of latitude; that is, let the moon be supposed to have advanced in its celestial orbit from L′ to L², running the arc L′L² = S′σ′. For the corrected time ϑ + ε′ let the ascendant be determined again; and with the new place L² of the moon let the parallax in altitude L²λ² be computed anew; λ² will be the place where the moon is seen at the time ϑ + ε′. The parallax in longitude (al-Battānī's *second parallax*) will be Σσ², determined by the perpendiculars drawn from L² and λ² to the ecliptic. Let S² be the place of the sun at the time ϑ + ε′. If S² and σ² coincided, the problem would be solved. Otherwise a new correction must be applied to the time ϑ + ε′ to give the time of the apparent conjunction; likewise the *second parallax* must be corrected again, although it is always much nearer the truth than the first parallax.

So, al-Battānī says, if we divide the *second parallax* by the relative hourly motion of the two luminaries, we shall find a correction ε² of the time better than the correction ε′. He calls this ε² *the hours of the second parallax*. For the new, more exact time ϑ + ε² let the ascendant, the middle of the heaven, the place of the moon and the anomaly of the moon be found again; from them let a new parallax (*the third parallax*) be deduced. If this is equal to the second, the problem will be solved, and the time ϑ + ε² will give the time of the apparent conjunction; for then we shall know most certainly that the second and the third parallax are exactly equal to the difference of the true longitudes of the sun and the moon, so that the apparent longitudes of the luminaries are equal to each other.

But from this al-Battānī concludes certain *false* things, printed in italics on p. 104, ll. 41–42, p. 105, ll. 1–3, 24–25, 34, p. 110, ll. 25–26. For he says: If the third parallax is equal to the second, the apparent longitudes of the two luminaries are also equal. So if the third parallax differs from the second by some quantity, the corresponding apparent longitudes of the sun and the moon differ from each other by the same quantity; that is, by that quantity the third parallax must exceed the difference of the true longitudes of the moon and the sun. — This is not true[b]; for from the fact that two differences are equal to each other when they are nothing, it by no means follows that they are also equal when they are not nothing.

[a] Al-Battānī divides the first parallax by the unequal hourly motion of the moon itself (p. 104, ll. 21–22; p. 110, ll. 11–12), and only the second by the gain of the moon on the sun (p. 104, ll. 29–30); adding the minutes of the first parallax to the place of the moon then gives the place of the moon at ϑ + ε′. In either way ε′ is only the first approximation, which the second corrects.

[b] Al-Battānī's statement holds. The time ε² is the second parallax divided by the relative hourly motion, so that at ϑ + ε² the true longitudes of the moon and the sun differ by the second parallax, as Schiaparelli says just above; the apparent longitudes then differ by the difference of the second and the third parallax, whether this is nothing or not. That difference is therefore the Δ that remains, and al-Battānī's use of it below is right. The translation of al-Battānī's chapter has the same note.

Let Δ be the difference of the apparent longitudes, which remains to be found by applying the third parallax to the true longitude of the moon: we must reduce this residual difference to nothing by correcting the time $\vartheta + \varepsilon^2$ somewhat again. It is enough to compute the speed of the apparent relative motion of the sun and the moon in longitude; let Δ be divided by this speed, and the time of the apparent conjunction will come out. — For the quantity Δ al-Battānī wrongly uses *the difference between the second and the third parallax*. With this *error* excepted, al-Battānī's computation is true, and is explained in this way:

The apparent relative hourly motion of the moon and the sun is deduced from the true relative hourly motion that the tables give (*the gain of the moon on the sun*, as the Arabs say) by subtracting from it (and never *adding* to it[a]) the hourly variation of the parallax, taken absolutely, without distinction of sign. If this were uniform, that is, proportional to the variation of the hour angle, the variation of the parallax answering to a minute of time would very easily be found by dividing the difference between two parallaxes found by the minutes of time elapsed from the one to the other. But the variation changes so rapidly that it cannot be considered uniform even within the space of one hour; so the computation cannot be made for so long an interval of time. One must use at least one further approximation, namely determine at least the second parallax and the time $\vartheta + \varepsilon'$ that answers to it. The error of this time will then be a small fraction of an hour, in which the variation of the parallax may be thought uniform, and so too the apparent relative motion in longitude. Better still if we compute the third parallax and the time $\vartheta + \varepsilon^2$ that answers to it. — The problem now is to find the hourly speed of the apparent relative motion in the short interval of time needed to reduce the residual difference Δ of the apparent longitudes to nothing. I 279

Al-Battānī, p. 105, ll. 6–24, lays down that the short interval of time in which the variation of the parallax may be considered uniform cannot exceed 1/6 of an hour, or 10^m of hour angle in time. Of the two parallaxes that must be used, one is already known, and answers to the hour angle $\vartheta + \varepsilon^2$, as we have just seen. The author distinguishes two cases:

1. If the hour angle, or time, $\vartheta + \varepsilon^2$ differs from the nearest whole hour by more than 1/6 of an hour, we must compute a new parallax directly for the time $\vartheta + \varepsilon^2 + 1/6$ or $\vartheta + \varepsilon^2 - 1/6$. The difference of the two parallaxes will be the parallactic variation in 1/6 of an hour; multiplying it by 6 we shall find the hourly parallactic motion, which must be subtracted from the «gain of the moon» (the relative hourly motion in true longitude) to give al-Battānī's «*corrected motion*», that is, the apparent relative hourly motion of the two luminaries in longitude. — We cannot use Theon's tables, because the nearest datum in these tables answers to an interval greater than 1/6 of an hour from the hour angle $\vartheta + \varepsilon^2$.

2. But if the fraction of an hour in the time $\vartheta + \varepsilon^2$ is less than 1/6 or greater than 5/6 (for example 1/8, 1/10 … or 7/8, 9/10 …), we can make the computation shorter by using the nearest whole hour, that is, taking 1/8, 1/10 … as the interval. The limits of the interval will then be the hour angle $\vartheta + \varepsilon^2$, whose parallax is already known, and the hour angle of whole hours, whose parallax is given in Theon's tables. In this way some labour is avoided. — Plainly such an advantage cannot be had unless an interval of 1/8, 1/10 … of an hour is taken; the interval of 1/6 must be avoided, for it leads beyond the whole hour. — The «*corrected motion*» will be computed as before.

Al-Battānī then rightly observes that the parallax in longitude decreases with the time and the hour angle if the moon is between the ascendant and the nonagesimal, and increases if the moon is found between the nonagesimal and the descendant. In the first case he teaches (ll. 27–33) to *subtract* 1/6 of an hour from $\vartheta + \varepsilon^2$, to make the interval between the hour angles $\vartheta + \varepsilon^2$ and $\vartheta + \varepsilon^2 - 1/6$; in the second case to *add* 1/6 of an hour, so that the interval between the hour angles $\vartheta + \varepsilon^2$ and $\vartheta + \varepsilon^2 + 1/6$ results. This precept ensures that the new auxiliary parallax is always greater than the parallax of the angle $\vartheta + \varepsilon^2$. A thing *useful* for avoiding errors of sign, but not

[a] Al-Battānī's text subtracts the sixfold difference of the parallaxes from the gain of the moon when the third parallax exceeds the second, and adds it when the third is the smaller (p. 105, ll. 23–24 and 33–34).

necessary; above all since the precept concerns case 1 only, not case 2.

Lastly, let the residual difference Δ of the apparent longitudes of the sun and the moon be divided by the *corrected motion*; there will come out a small time τ, which must be added to or subtracted from the approximate time $\vartheta + \varepsilon^2$, according as the moon at that time $\vartheta + \varepsilon^2$ followed or preceded the sun in longitude. There will result *the required time of the apparent conjunction.*

All the author's precepts can be set out briefly in this way. For the given time ϑ of the true conjunction let the first parallax be computed and divided by the «gain of the moon»; there will come out the time ε', which will give the first approximation $\vartheta + \varepsilon'$. For this new time $\vartheta + \varepsilon'$ let the longitudes of the sun and the moon, the anomaly of the moon and the ascendant be computed again; from them let a new parallax be deduced, which, divided by the gain of the moon, will give the time ε^2; hence the second approximation $\vartheta + \varepsilon^2$. For this third time let the longitudes of the sun and the moon, the anomaly of the moon and the ascendant be computed anew; from them let the third parallax be found. Let it be added to the true longitude of the moon; the apparent longitude of the moon will come out. If this is equal to the apparent longitude of the sun, the problem will be solved. If a difference $\pm\,\Delta$ remains, let the apparent relative hourly motion of the sun and the moon (the «corrected motion») be computed as we taught above; let Δ be divided by it; let the time that comes out of the division be applied, according to its sign, to $\vartheta + \varepsilon^2$; the time of the apparent conjunction will come out. — Schiaparelli.]

The way in which al-Battānī works differs somewhat from that of older writers. It is worth reporting what al-Khāzinī relates in *al-Zīj al-Sanjarī*, ff. 80r–81r (treatise VIII, part II, ch. II, «on correcting the hours of the middle of the eclipse»).

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«*The first method*, according to Ptolemy. — He says: Let us find the distance of the hours of the true conjunction from noon, the altitude of the moon, and the parallax of the moon in the circle of altitude. From this parallax let us subtract the parallax of the sun; there remains the apparent parallax. With these true hours let us find the angle of longitude, and from it the parallax in longitude, which we shall call *the first parallax*. Let us divide it by the true gain of the moon: there will come out *the hours of the first parallax*, which we shall subtract from the hours of the true conjunction if the moon is less than 90° from the ascendant, and otherwise add. There will result *the hours of the second conjunction*. Again with these hours let us find the altitude of the moon, the apparent parallax, the angle of longitude and the parallax in longitude, which we shall call *the second parallax*. If it is equal to the first parallax, the hours of the second conjunction will be the very hours of the middle of the eclipse; if different, we shall need to correct it. The correction is this: let us find the difference between those two parallaxes; then, by solving the proportion

$$\text{unknown quantity} : \text{difference} = \text{difference} : \text{smaller parallax},$$

that is, by dividing the square of the difference by the smaller parallax, we shall find the unknown quantity (*al-juz' al-majhūl*). Let us add this to the difference if the second parallax is greater than the first, or subtract it from the difference if the first is the greater; there will come out the corrected difference, which, added to the first parallax in longitude, will make *the corrected parallax*. Let us divide this by the true gain of the moon; there will come out *the hours of the apparent parallax*. Let us subtract them from the hours of the true conjunction if the moon is less than 90° from the ascendant, otherwise add them; there will result *the corrected, exact (murattaba) hours of the middle of the eclipse*. — For these let us determine the true places of both luminaries and of the node, the anomaly of the moon, the altitude of the moon, the lunar parallax in latitude and its side; from the apparent place of the moon and the node we shall find the true latitude of the moon, and shall obtain its corrected latitude.

«*The second method*, received from tradition (*'amaluhu mursalan*). — Thābit[1], the authors of the

[1]He died in 288 H, 901; Suter, pp. 34–38, no. 66.

Tested Tables[1] and the Indians say: Let us find the distance of the hours of the true conjunction from noon, and the parallax of the moon in longitude; let us divide the parallax by the true gain of the moon, and *the hours of the first parallax* will come out. If the moon is less than 90° from the ascendant, let us subtract them from the hours of the true conjunction to give *the hours of the second conjunction*; likewise let us subtract the first parallax in longitude from the degree of the conjunction, and *the apparent place of the moon* will result. If, on the other hand, the moon is more than 90° from the ascendant, we shall add the hours of the first parallax to the hours of the true conjunction to give *the hours of the second conjunction*, and add the first parallax to the degree of the conjunction to give *the apparent place of the moon.* — Then, using these hours, let us compute the parallax in longitude again; if it is equal to the first parallax, the hours of the second conjunction will be the very hours of the middle of the eclipse, and the apparent moon will be the moon of the middle of the eclipse. But if those two parallaxes are different, let us divide the second by the true gain of the moon; *the hours of the second parallax* will come out. Let us add them to the hours of the true conjunction, and the parallax to the degree of the conjunction, or subtract, as we said above. *The hours of the third conjunction* will result. If this is equal to the second, it will be what is sought; if different, we shall compute the fourth and fifth hours of the conjunction, and shall not cease to repeat the computations in this way many times until we have found a number equal to the preceding number. The hours that come out last will be *the hours of the middle of the eclipse*, and the apparent moon will be the one that was sought.

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«*The third method*, according to al-Battānī. — He says: Let us take the distance between the hours of the true conjunction and noon, and find the corrected parallax in longitude, which will be *the first parallax*. Let us divide it by the true gain of the moon; *the hours of the first parallax* will come out. If the moon is less than 90° from the ascendant, we shall subtract them from the hours of the true conjunction, and the first parallax from the degree [of the same conjunction]; if it is more than 90° away, we shall add both; *the hours of the second conjunction* and *the apparent place of the moon* will come out. With these quantities let us find the parallax in longitude again; then let us work as above, dividing it by the gain of the moon and then using addition and subtraction. If the second parallax itself is equal to the first, and *the hours of the third conjunction* equal to the hours of the second, these hours will be those of the middle of the eclipse, and the apparent moon will be the one that was sought. But if the second parallax is different from the first, let us divide the difference between them[2] by the gain *apparent at that hour*[3], *and not by the true gain* (وقسمناه على السبق المرئي لتلك الساعة خاصة دون سبقه الحقيقي); what comes out will be a fraction of an hour. If the second parallax is greater than the first, we shall add the fraction to the hours of the second parallax so that they become *corrected*; but if the second is less than the first parallax, we shall subtract the fraction, and shall find *the corrected hours.* Then, if the degree of the conjunction is less than 90° from the ascendant, we shall subtract these corrected hours from the hours of the true conjunction; if the distance is greater than 90°, we shall add the corrected hours. What results from the subtraction or addition will be *the hours of the middle of the eclipse.* — Then let us multiply these corrected hours into the motions of the sun, the moon, the node and the anomaly of the moon; let us add the products to the corresponding places at the time of the conjunction if the corrected hours were additive, or subtract them if the hours were subtractive, except only for the node, with which one must work the other way round[4]. *The exact places* (*al-murattaba*) will come out».

Ḥabash, ff. 133r–134v, follows Ptolemy's method.

[1] Which the astronomers of the caliph al-Maʾmūn compiled.

[2] Al-Battānī's error, on which see p. 278. For brevity al-Khāzinī considers the simpler case, namely when the first approximation $\vartheta + \varepsilon'$ suffices. [*Translator:* See the translator's note on p. 278: the difference of the parallaxes is the right quantity.]

[3] Al-Battānī, p. 105, ll. 23 and 33.

[4] Al-Battānī, p. 106, ll. 7–11, does not have this precept explicitly; but it is easily understood, for the motion of the node is retrograde.

b) — p. 106, ll. 22–26.

The apparent latitudes within which the moon must be for an eclipse of the sun to occur.

[These limits of latitude are wholly false in the codex; much better in Plato. For of these limits

$$\text{the greatest} = \text{greatest semidiam.}\ \odot + \text{greatest semidiam.}\ \leftmoon = \tfrac{1}{2}\,(33'\,40'' + 35'\,20'') = 34'\,30''$$

$$\text{the least} = \text{least semidiam.}\ \odot + \text{least semidiam.}\ \leftmoon = \tfrac{1}{2}\,(31'\,20'' + 29'\,30'') = 30'\,25''.$$

Al-Battānī's diameters of the sun and the moon are on p. 58. — Schiaparelli.]

The *Alfonsine Tables* set out this passage more fully: «So al-Battānī's rule on the necessity of an eclipse of the sun is this: let the semidiameter of the sun and the semidiameter of the moon be found at the time of the most true conjunction, and let them be joined together. And for the same time let the apparent latitude of the moon be found. Then, if the apparent latitude of the moon is equal to the sum of the semidiameters of the sun and the moon, the moon will pass close to the sun and the sun will not be eclipsed. And if the latitude of the moon is greater, an eclipse of the sun cannot happen. But if the latitude is less than the sum of the semidiameters, an eclipse of the sun necessarily takes place» (*Tabulae astronomicae Divi Alphonsi Regis Romanorum et Castellae, nuper quam diligentissime cum additionibus emendatae*, Venice 1518, fol. 19r, prop. XXXII).

c) — p. 106, ll. 31–33.

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[«*If you prefer to find this by computation …*». — This whole passage, printed in italics, has plainly been substituted for another, lost one by an unskilled man. Another method of computing the apparent diameter of the sun should have been explained, but what is now read in the codex and in Plato has nothing to do with it. What there was in al-Battānī's true text, and what the new method was, can perhaps be conjectured from p. 58, l. 31 ff., where the author observes that the apparent diameters of the sun are proportional to the speed of its hourly motion in the zodiac. This motion is in the small table of vol. II, p. 88. So the rule was this: «Multiply the least apparent diameter of the sun (31′ 20″) into the unequal hourly motion of the sun that answers to the given anomaly of the sun; divide the product by the least hourly motion, 2′ 23″». — Schiaparelli.]

On chapter 47

a) — p. 115, ll. 15 and 19.

«*The column of the northern latitude, called afījiyūn of the north … The column of the southern latitude, called ferījiyūn of the south*». — Plato: «*effregion septentrionalis*», «*effregio meridiana*»; likewise in the table of the latitudes of the superior planets, vol. II, p. 140, which is taken from the *Almagest*, ed. Halma vol. II, pp. 412–13, one reads written in the margin of the last column: افيثخون هو الشمال فرنجيون هو الجنوب, «ἀπόγειον is the north, περίγειον the south». How these Greek words belong to the sides of the latitudes of the superior planets is not clear at first sight. In Ptolemy's table the column of the northern latitude is headed βόρειον πέρας, «northern limit», the column of the southern latitude νότιον πέρας, «southern limit». But in the Latin version of the Almagest that Gerard of Cremona made from the Arabic[1], the first column is headed «*Efegion–Septentrio*», the second «*Efregion–Meridies*». Al-Battānī, then, borrowed these from

[1] *Almagestum Cl. Ptolemei Pheludiensis Alexandrini astronomorum principis; opus ingens ac nobile omnes celorum motus continens Felicibus astris eat in lucem ductu Petri Liechtenstein*, Venice 1515, f. 149r. — Gerard lived in the years 1114–1187.

the Arabic copies of the Almagest. I think that such names arose from the fact that in Ptolemy's system the latitude of the three superior planets becomes northern whenever the centre of the epicycle is in that segment of the eccentric in which the apogee of the eccentric is found, and southern when it is in the one in which the perigee is.

I find the same Greek words, much disfigured in Arabic script, in al-Khāzinī, *al-Zīj al-Sanjarī*, but in a sense contrary to al-Battānī's usage. For (Vatican codex, f. 167 for Saturn, 178 for Jupiter, 176 for Mars) in the tables of the latitudes of the superior planets he always has these headings: افرنجيون زحل وهو صاعد في الشمال, «περίγειον of Saturn [or of Jupiter, or of Mars], and it [is] ascending on the northern side»; افون زحل وهو هابط في الجنوب, «ἀπόγειον of Saturn, and it [is] descending on the southern side».

From Plato's version, I think, these words crept into the book of Johannes Schöner[1] entitled *Tabulae resolutae astronomicae* (Wittenberg 1587). For there, p. 27, I find this: «and what you have found of northern or southern latitude in the common angle you will note outside with its part of latitude, either the northern, which is called *efregion septentrionale*, or the southern, which is named *efregion meridionale*, as the titles at the heads of the tables advise». So in the tables of the latitudes of the superior planets the column of the northern latitude is headed «*effregion septentrionale*», that of the southern latitude «*effregion meridionale*».

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b) — p. 116, l. 24.

I. — «*Because of their swift course about the sun*». — Likewise below, p. 117, l. 11. The inferior planets are called by Plato the Athenian ἰσοδρόμοι ἡλίῳ, «equalling the course of the sun»; by Ptolemy, IX, 1 (ed. Halma vol. II, p. 115, where the translation is not good), περὶ αὐτὸν ἀεὶ φερόμενοι, «always carried about it [namely the sun]». G. Schiaparelli, *I precursori di Copernico nell'antichità*, Milan 1873, has shown most learnedly that by these and similar expressions of the Greek and Latin writers only the apparent motion of Venus and Mercury about the sun, to the right and to the left, is meant. See also Hultzsch, s. v. *Astronomie*, in Pauly–Wissowa, *Real-Encyclopädie der classischen Altertumswissenschaft*, vol. II, col. 1859. Likewise in the thirteenth century al-Qazwīnī, *Cosmography*, treatise I, ch. III (vol. I, p. 34 in the Cairo edition of 1311 H), says that Mercury «has many retrogradations and advances, and revolves *about* the sun, يدور حول الشمس».

II. — Ibn Yūnus writes in his introduction (Caussin p. 70): «[Among the errors] are also those that befell Muḥammad ibn Jābir ibn Sinān al-Battānī in particular in [fixing] the side[2] of the latitude of Mercury». I find no error; the precepts of pp. 115–16 and the tables do not differ from what Ptolemy has. Perhaps Ibn Yūnus had in his hands a copy of al-Battānī's work that was faulty in this place, as the Escorial codex itself is (though not Plato), p. 116.

On chapter 48

a) — p. 117, ll. 28–29.

The «equatorial» arcs of the appearances of the planets.

That according to al-Battānī these are truly arcs of the *equator* is plain also from the following precepts, where the arcs are said to be compared with the arc of right ascension («the times of

[1] Born in the village of Karlstadt near Würzburg in 1477, died at Nuremberg in 1547.

[2] Caussin: «s'est trompé sur la latitude» («was mistaken about the latitude»); he did not recognize the technical meaning of the word جهة, and so wrongly rendered في جهة, «on the side», as «as regards, as concerns».

the settings or of the ascensions») that answers to the difference of longitude between the sun and the given planet. The same appears from the end of chapter XLIX, p. 120.

Ptolemy, on the other hand, chose arcs of the *ecliptic*, that is, the difference of longitude itself[1]: «From this consideration of the observations of the risings of the planets we have found that near (περί) the beginning of Cancer, in general (ὡς ἐπίπαν), Saturn appears (ἀνατέλλει) when it is 14° from the true sun, Jupiter 12 1/2° + 1/4°, Mars 14 1/2°, Venus in the evening 5 2/3°, Mercury in the evening 11 1/2°» (*Almag.* XIII, 7, ed. Halma vol. II, pp. 418–19). But since Ptolemy had determined these limits of the appearances near the beginning of Cancer, where the equator and the ecliptic are nearly parallel and the right ascensions differ little from the corresponding arcs of the zodiac, al-Battānī could transfer Ptolemy's limiting numbers to the equator without perceptible error.

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Al-Battānī rightly preferred to compute the elements of the computation in arcs of the equator; for if they were arcs of the ecliptic, they would change not a little in the same clime in different parts of the year, because of the varying inclination of the ecliptic to the horizon[2].

Only on p. 118, ll. 14–28, does al-Battānī count the arc of seeing on the *ecliptic*; for there he explains the use of the tables of the appearances and disappearances of the planets (vol. II, pp. 142–43), which he constructed for the latitude 36° 0′ of the city of al-Raqqa, after the model of Ptolemy's tables, for the use of those who wish to avoid direct computation (which is, however, better).

Ḥabash, ff. 67v–70v (the tables ff. 73r–76v), and Ulugh Beg, p. 142 of the text, 192 of the translation, always consider arcs of the ecliptic. — Barhebraeus, pp. 79–80 of the text, speaks after Ptolemy of the depression of the sun below the horizon (which is counted on the vertical circle of the sun). The Indians, like al-Battānī, fixed the limits of the visibility of a planet on the equator; but their numbers differ from the Greek and the Arabic ones; see the *Sūrya-Siddhānta*, transl. Burgess–Whitney, IX, 4–9, pp. 365–67, and al-Bīrūnī, *India*, ch. LVII, pp. 246–47 of the text, vol. II, pp. 90–92 of the translation.

On chapter 49

a) — p. 119, ll. 3–41.[a]

The configurations of the fixed stars with the sun.

By the words *tashrīq* and *taghrīb* the Arab astronomers mostly designate the heliacal *rising* and *setting* of the stars, as appears from the definition of the lexicon *Mafātīḥ al-ʿulūm*: «*Tashrīq* takes place when the star is seen to rise in the eastern quarter before sunrise; *taghrīb* when it is seen

[1] In *Almag.* XIII, 7 (ed. Halma vol. II, pp. 420–2) he also computed the least depression of the sun below the horizon needed for these heliacal risings and settings of the planets to be possible; he found 11° for Saturn, 10° for Jupiter, 11 1/2° for Mars, 5° for Venus and 10° for Mercury. See R. Wolf, *Handbuch der Astronomie*, Zürich 1890–93, vol. I, pp. 423–24, § 191, *c*, who strangely confesses that he did not find these numbers in the Almagest; what is missing in the Almagest is the amounts of the arcs of the fixed stars (see note *b* on ch. XLIX).

[2] [It must be noted, however, that those arcs, of the equator as well as of the ecliptic, are different for different latitudes of the earth. — Schiaparelli.]

[a] Nallino's Addenda et emendanda (Part II, p. XXV) add that these configurations of the stars with the sun, of no use to us, were used in astrology: Ptolemy himself, *Tetrabiblos* II, 7 (περὶ τοῦ γένους τῶν διατιθεμένων, «on the kind of the things affected»), among his precepts for drawing from eclipses forecasts of what will befall nations, cities, regions and so on, says: «Of the fixed stars we shall take the one that has the most familiarity with the place of the eclipse itself, first through its brightness in the preceding angle (ἐπὶ τῆς παροιχομένης κεντρώσεως), according to the configurations appearing in nine modes that we set out in the first Syntaxis [the *Almagest*]; then the one found in the angle of the ascendant or of the middle of the heaven that follows the place of the eclipse …».

to set in the western quarter after sunset»[1]. But this definition does not always agree with al-Battānī's usage in this chapter; so I have preferred to use the words *earlier rising* and *earlier setting*.

What is described in this chapter are the nine kinds (τρόποι) of configurations (σχηματισμοί) of the fixed stars with respect to the sun, already described by Ptolemy, VIII, 4 (ed. Halma vol. II, pp. 100–104), whose Greek names, with Halma's translation, I am pleased to give, so that they may be compared with al-Battānī's:

1. πρωϊνὸς ἀπηλιώτης, subsolaire du matin (vent d'est) [morning east], in three modes: ἐπανατολή, lever subséquent [later rising]; συνανατολὴ ἀληθινή, lever simultané vrai [true simultaneous rising]; προανατολή, lever antérieur [earlier rising].

2. πρωϊνὸν μεσουράνημα, culmination matutinale [morning culmination]; which can be: ἑῷον συμμεσουράνημα μὴ φαινόμενον, culmination simultanée non apparente [simultaneous culmination, not apparent]; ἑῷον συμμεσουράνημα ἀληθινόν, vrai [true]; ἑῷον προμεσουράνημα, culmination matutinale antérieure [earlier morning culmination].

3. πρωϊνὸς λίψ, coucher du matin (du vent d'ouest) [morning west]; whose kinds are: ἑῷα ἐπικατάδυσις, coucher matutinal postérieur [later morning setting]; ἑῷα συγκατάδυσις ἀληθινή, c. m. simultané vrai [true simultaneous morning setting]; ἑῷα πρόδυσις, c. m. antérieur [earlier morning setting].

4. μεσημβρινὸς ἀπηλιώτης, lever méridional oriental [midday east]; which can be ἡμερινός, diurne [by day], and νυκτερινός, nocturne [by night].

5. μεσημβρινὸν μεσουράνημα, culmination méridienne [midday culmination].

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6. μεσημβρινὸς λίψ, coucher méridional-oriental [midday west].

7. ὀψινὸς ἀπηλιώτης, lever du soir [evening east]; whose three kinds are: ἑσπερία ἐπανατολὴ φαινομένη, lever du soir postérieur et visible [later evening rising, visible]; ἑσπερία συνανατολὴ ἀληθινή, l. d. s. simultané vrai [true simultaneous evening rising]; ἑσπερία προανατολὴ μὴ φαινομένη, l. d. s. antérieur et non visible [earlier evening rising, not visible].

8. ὀψινὸν μεσουράνημα, culmination du soir [evening culmination]; whose kinds are: ἑσπερινὸν ἐπιμεσουράνημα φαινόμενον, culmination postérieure du soir [later evening culmination, apparent]; ἑσπερινὸν συμμεσουράνημα ἀληθινόν, culm. vraie simultanée du soir [true simultaneous evening culmination]; ἑσπερινὸν προμεσουράνημα μὴ φαινόμενον, culm. antérieure du soir non visible [earlier evening culmination, not visible].

9. ὀψινὸς λίψ, coucher occidental du soir [evening west]; whose three kinds are: ἑσπερία ἐπικατάδυσις φαινομένη, coucher du soir postérieur et visible [later evening setting, visible]; ἑσπερία συγκατάδυσις ἀληθινή, c. d. s. simultané et vrai [true simultaneous evening setting]; ἑσπερία πρόδυσις μὴ φαινομένη, c. d. s. antérieur et invisible [earlier evening setting, invisible].

b) — p. 120, ll. 1–7.

The equatorial arcs of seeing of the fixed stars, according to the commonly received tradition («they say»).

These arcs are computed on the equinoctial circle, as I have already shown on p. 283. — Ptolemy, *Almag.* VIII, 6 (ed. Halma vol. II, pp. 108–113), treats of the appearances and disappearances of the fixed stars; he shows that the solution of the problem is very difficult, but gives no amount of those arcs. L. Ideler, *Handbuch der mathematischen und technischen Chronologie*, Berlin 1825–26,

[1] *Liber Mafâtih al-ʿolûm, auctore Abû ʿAbdallâh Mohammed al-Khowârezmî*, ed. G. van Vloten, Leiden 1895, pp. 229–230. — The book was composed between 363 and 381 H (973 and 991).

vol. I, p. 55, rightly says that what circulates in many astronomical books rests on no foundation, namely that Ptolemy determined the arc of seeing of the stars of the first magnitude as 12°, of the second 13°, of the third 14°, of the fourth 15°, of the fifth 16°, and lastly of the sixth (which are seen when the twilight is wholly gone) 18°. Nothing of this kind is found in the known books of Ptolemy. It must also be noted that these arcs are the *depression* of the sun below the horizon, which is computed on the *vertical circle*.

Ptolemy says that he wrote a little work in which he determined the arcs of seeing of the fixed stars both on the vertical circle and on the ecliptic. The little book has not come down to us; yet L. Ideler, *Ueber den Kalender des Ptolemaeus*[1], deduced from the information contained in Ptolemy's book *On the appearances of the fixed stars* (φάσεις ἀπλανῶν ἀστέρων) that the Greek astronomer computed a depression of the sun of 11° and 7° for stars of the first magnitude, and of 14° and 8 1/2° for stars of the second magnitude; the numbers 11 and 14 hold when the star and the sun rise or set in the same quarter of the horizon, 7 and 8 1/2 when they are in two opposite quarters.

The number 15° for the stars of the first magnitude perhaps comes from the book of Autolycus *On risings and settings* (περὶ ἐπιτολῶν καὶ δύσεων), already very well known to the Arabs in al-Battānī's time; for Qusṭā ibn Lūqā had translated it into Arabic, and Thābit ibn Qurra had corrected it. Autolycus (fourth century BC), however, counts the arc of seeing on the *ecliptic*, and sets 15° for all the fixed stars[2].

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But on what principle they laid down that the equatorial arcs of seeing grow regularly by three degrees, so that in the stars of the sixth magnitude they are about 30°, is not clear. [It is probable that all this comes from rough computations and observations; namely that the Arabs before al-Battānī laid down that the stars of the first magnitude appear 1^h (= 15°) before sunrise and 1^h after sunset, and the stars of the sixth magnitude 2^h (= 30°) before sunrise and after sunset, and that the other orders of the magnitude of the stars were placed regularly within these limits. — Schiaparelli.]

Ulugh Beg, p. 143 of the text, 193 of the translation, considers the *depression* of the sun below the horizon; he computes it at about 12° for stars of the first magnitude, 14° for stars of the second, and so, with a regular increase of 2° in the following orders, he sets 22° for stars of the sixth magnitude. He adds that this holds for stars that are near the ecliptic[3].

On chapter 50

a) — p. 120, l. 18.

The distances of the planets from the earth «according to what the more recent wise men established with long labour *after* Ptolemy». The codex, as I noted in its place, has «before», which goes badly with the words «more recent wise men». The doubt is easily removed, however, if we consider that the foundations of all these computations are the eccentricities and the sizes of the epicycles determined by Ptolemy, as will be plain from note *b*.

[1] In Schriften der k. Akademie der Wissenschaften zu Berlin, 1816 and 1817. — A French version is at the end of the work *Table chronologique des règnes; apparition des étoiles fixes de C. Ptolémée, Théon etc., publiées et traduites par Halma*, Paris 1819.

[2] P. Tannery, *Autolycos de Pitane*, in Mémoires de la Société des sciences physiques et naturelles de Bordeaux, 3rd series, vol. II, fasc. 1, 1885, pp. 173–199. — Eudoxus in the *Parapegma* gives different arcs, which vary from 20 1/4° or 21° (the Pleiades, the Hyades, Orion, the Dog) to 8 3/4° (the Dolphin); Tannery, loc. cit., pp. 190–191.

[3] In the other Arab astronomers that have been published I find nothing. Of Ibn Yūnus, Delambre, p. 112, says only: «Chap. XVI. Durée du crepuscule. Abaissement 18°. Rien de neuf» («Ch. XVI. Duration of twilight. Depression 18°. Nothing new»). — Alhazen (= Ibn al-Haytham), *De crepusculis*, gives only the depth of 19° of the sun at the end of twilight.

This theory of the distances rests on the principle that the spheres of the seven planets are contiguous and separated from one another by no interval; so the least distance of a superior planet equals the greatest distance of the planet immediately below it. This principle was accepted by all the Arab astronomers known to me; in the fifteenth century it is still found in the Persian writer ʿAlī Qūshjī (in Delambre p. 198). A philosophical discussion is in the epistles on the various sciences that men called *Ikhwān al-Ṣafāʾ*, «the faithful brethren», wrote in the tenth century in the city of al-Baṣra, whose doctrine came as much from Aristotle as from the Neopythagoreans and the Neoplatonists[1].

Before al-Battānī, al-Farghānī already relates this, ch. XXI, p. 80 of Golius's edition: «Ptolemy indeed set out in his book how great the distance of the sun and the moon alone is; nor have we found anything handed down by him about the distances of the other stars, except that he taught how far the centres of the orbs are from the centre of the earth, and how great the orbs of the epicycles are. — If, then, we suppose that the farthest distance of the moon, with respect to both orbs together (the eccentric and the epicycle), is the same as the nearest distance of Mercury, and use the same method as we have said, proceeding similarly in Mercury and Venus, we shall find that the greatest distance of Venus in both orbs together equals the least distance of the sun, which Ptolemy set out. We take the proof of this from the fact that there is nothing empty between the orbs».

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Al-Farghānī then adds, pp. 81–82, the distances in earth radii and in Arabic miles, taking the radius of the earth as 3250 miles[2]. Somewhat different numbers are given, before the year 301 H (beginning 6 Aug. 913), by Ibn Rusta, pp. 18–20, who makes the radius of the earth 3818 miles[a]. Lastly, in the first half of the twelfth century, the Hebrew-Spaniard Abraham bar Ḥiyya[3] gives the distances in radii (= 3250 miles) and in miles, and adds: «This computation that we have set out follows the opinion of the man called al-Battānī, one of the wise men who investigated this subject in the time of the aforementioned king al-Maʾmūn [sic!]. He was the one who in this computation diminished [the distances] that the wise men before him had measured; for the ancients added much to these quantities». The distances in earth radii according to these three authors are:

[1] Dieterici, *Die Naturanschauung und Naturphilosophie der Araber im zehnten Jahrhundert, aus den Schriften der lautern Brüder übersetzt*, Posen 1864, pp. 27–28.

[2] In my *Il valore metrico del grado di meridiano* etc. I have shown that these miles are 1973 m.

[3] *Sphaera mundi autore Rabbi Abrahamo Hispano filio R. Haijae. Osw. Schreckenfuchsius vertit in linguam Latinam, Seb. Munsterus illustravit annotationibus*, Basel 1546, ch. IX, pp. 190–92 of the Hebrew text (the translation of this chapter is missing and is replaced by a short Latin summary). — The amounts in miles teem with countless errors.

[a] Nallino's Addenda et emendanda (Part I, pp. LXXV–LXXVI) add that he finds Ibn Rusta's numbers in his contemporary Thābit ibn Qurra, in a little book on the exposition of the Almagest that survives only in an unpublished Latin version: the least distances are the moon 33 earth radii, Mercury 64, Venus 167, the sun 1079, Mars 1260, Jupiter 8820, Saturn 14187, but the greatest of Saturn and the least of the fixed stars 17865 (for Ibn Rusta's 19880 and 20000); the sizes are the sun 166 earths, the stars of the first magnitude 94, Saturn and Jupiter «about» 79, Mars «about» 1 1/2, Venus «about» 1/87, the moon 1/40 (codex A 1/10), Mercury 1/29680 (A 1/29683); he does not say where he took all this from. Another little book in the same codex, «Liber Thebith de quantitatibus stellarum», whose numbers agree with al-Farghānī's and which names al-Farghānī and Geber (Jābir ibn Aflaḥ), Nallino strongly suspects to be spurious. The Addenda are translated with Part I's front matter.

	al-Farghānī	*Ibn Rusta*	*Bar Ḥiyya*
Least distance of the moon	33 1/2 1/20[1]	…	33 1/2
Greatest of ☾, least of Mercury	64 1/6	64 1/6[2]	64
Greatest of ☿, least of Venus	167	166	166
Greatest of ♀, least of the sun	1120	1079	1080[3]
Greatest of ☉, least of Mars	1220	1260	1146
Greatest of ♂, least of Jupiter	8876	8820[4]	8000[5]
Greatest of ♃, least of Saturn	14405	14187[6]	12400[7]
Greatest of Saturn	20110	19880	18000[8]
The fixed stars	20110	20000	18000

The principle on which this theory of the distances rests some later Arabs attribute to Ptolemy himself. So in the thirteenth century al-Qazwīnī, *Cosmography*, Cairo 1311 H (in the margin of al-Damīrī), vol. I, p. 26, treatise I: «Ptolemy recorded the thickness[9] of the celestial spheres, and the size of the body, of the circumference and of the diameters of each planet. Do not think this very difficult; for it is difficult only for those who do not know the science of geometry. Whoever has finished the second book of Euclid[a] will, if he is perspicacious, find this easily». And in the following chapters (II–IX) of the same treatise I he gives how many miles the thickness of the sphere of each of the seven planets comprises, «according to the opinion of Ptolemy, the author of the observations, who derived it by geometrical demonstrations».

Al-Qazwīnī's authority in these matters is, of course, not great. But the testimony of the very famous astronomer al-Bīrūnī is worth recalling, *India*, ch. LV, pp. 234–36 of the text, vol. II, pp. 68–69 of the translation. He sets out the doctrine of the distances that Yaʿqūb ibn Ṭāriq had gathered from the books of the Indians in the year 161 H (beginning 8 Oct. 777): namely, that between the greatest distance of a lower planet and the least of the higher there lies a space equal to the diameter of the lower planet[10]. For, taking the radius of the earth as the unit of measure, Yaʿqūb set:

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[1] So rightly in the Latin version printed at Nuremberg in 1537, for the miles are 109037 (wrongly 109026 in Golius's Arabic text). In Golius's edition and translation wrongly 33 1/2 1/5.

[2] In the text wrongly 64 1/2 (for ونصف read ونصف ثلث), for the miles are 244988 1/3.

[3] Al-Battānī, p. 121, has 1070, although 1079 would rather be expected, as I noted in its place.

[4] In the text the miles are wrongly 33682760 for 33674760.

[5] Al-Battānī 8022.

[6] Wrongly 4187 in the text.

[7] Al-Battānī 12924.

[8] Al-Battānī 18094.

[9] That is, the interval between the inner and the outer surface, or the difference between the least and the greatest distance.

[10] The *Sūrya-Siddhānta* XII, 84–90 (transl. Burgess and Whitney, p. 397) rests on a very different principle. It supposes that all the planets have the same mean *linear* speed, that the moon is 32 diameters of the earth away, and that the distances of the other planets are as much greater than the distance of the moon as their sidereal revolutions are greater than the sidereal revolution of the moon.

[a] Nallino's Addenda et emendanda (Part II, p. XXV) add that the passage stands with the same readings in Wüstenfeld's edition, vol. I, p. 18, but that after «the second book» the word «ten» has surely dropped out: the twelfth book of Euclid is meant, whose eighteenth proposition, that spheres are to one another in the triplicate ratio of their diameters, is the theorem on which these computations chiefly rest.

	least dist.	*greatest dist.*	*diameter*
Moon	35 15/21	56 4/21	4 16/21
Mercury	60 20/21	251 9/21	4 16/21
Venus	256 4/21	1095 5/21	19 1/21
Sun	1114 6/21	2104 16/21	19 1/21
Mars	2123 17/21	8000	19 1/21
Jupiter	8019 1/21	13714 6/21	19 1/21
Saturn	13733 7/21	17161 19/21	19 1/21
The zodiac	19047 13/21[1]		

Al-Bīrūnī then adds: «This opinion differs from the one on which Ptolemy, in the book *al-manshūrāt*, founded the computation of the distances, which the ancients and the moderns have followed. For their principle is that the greatest distance of each planet is the least of the planet immediately above, and that between the two spheres there is no place without action. — But according to that opinion [of the Indians] there is between two spheres a place empty of both, and in it [something] that holds, like the axle [of a wheel], about which the rotation takes place; as if they had believed that there is some heaviness in the aether, so that the inner sphere would need [a thing] holding it, to keep it in the middle of the outer [sphere]».

I find the book *al-manshūrāt*, otherwise unknown, mentioned also in al-Bīrūnī's little book *On the elements of astronomy*, f. 33r, where a similar question about the distances of the planets is treated: «Whether the sizes of the bodies of the planets are known. The diameter of the moon is known in the measure in which the semidiameter of the earth is taken as 1; Ptolemy derived it from the parallax. Likewise the diameter of the sun is known in the same measure; he derived it from the eclipses in book V of the *Almagest.* The ratios of the diameters of the planets and of the fixed stars to the diameter of the sun are also known, in the opinion of others, from observation, as is related in the book *al-manshūrāt.* We have put this in a table according to the opinion and finding of Ptolemy, since the opinions of others do not seem so firm to us»[2].

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Lastly, a passage of the astrology of Ibn Hibintā, f. 210v, must be reported, with which al-Battānī, p. 123, ll. 4–8, should be compared. Ibn Hibintā relates: «Ptolemy says that the bulk of every star of the first magnitude comprises ninety-six times the earth; that the bulk then diminishes for each of the remaining orders of stars by a quantity comprising sixteen times the earth, until one comes to al-Suhā[3], which is the smallest of all the visible stars in the zodiac [sic!]; that its bulk comprises sixteen times the earth, and that nothing beyond it is visible. — But Muḥammad ibn Jābir al-Battānī, who observed in the city of al-Raqqa in the year 295 of the hijra, relates that all the stars of the first magnitude comprise one hundred and eight times the earth, and that [in the following orders of magnitude] they decrease in order down to al-Suhā, which he puts at eighteen times the earth»[4]. — What he relates of al-Battānī does not agree well with the text of the Escorial codex and of Plato; what he relates of Ptolemy is not found in his known works.

Now let us question the Greeks. Schiaparelli supplies me with a passage of Proclus' *Hypotyposes*[5], where this method of computing the distances is mentioned as regards the moon, Mercury, Venus and the sun. For he says that «some (τίνες), supposing that there is no empty space in the world and that all the intervals are filled with intermediate spheres», computed the distances of those

[1] So between the greatest distance of Saturn and the zodiac there is an empty space of 1885 15/21 earth radii.

[2] وقد وضعنا ذلك في جدول بحسب راي بطلميوس ووجوده اذ لم يتقرر عندنا راي غيره الى هذه الغاية. There follows a table of the diameters (with respect to the sun) belonging to the single planets and to the six orders of magnitude of the fixed stars; but the numbers are not a little faulty.

[3] Near ζ of the Great Bear; it is missing from Ptolemy's catalogue of the stars, and is seen only by those who have very sharp sight. See al-Ṣūfī, p. 50.

[4] The codex has فيجعله مثل الارض ثماني عشرة منه. But I think منه is a slip of the pen for مره; otherwise the passage would give no sense.

[5] *Hypothèses et époques des planètes de C. Ptolémée, et Hypotyposes de Proclus Diadochus [publiées] et traduites par Halma*, Paris 1820, pp. 145–46. — Proclus lived in the fifth century AD.

planets. Taking the greatest distance of the moon as $64^r\ 10'$ earth radii and the least of the sun as 1160^r [from the Almagest], «they found that the distance of the apogee of the epicycle of Mercury from the centre of the zodiac [or of the earth], as demonstrated [by Ptolemy], is to the distance of the perigee as $91^r\ 30'$ to $34^r\ 4'$. Then, making $34^r\ 4'$ to $91^r\ 30'$ as the greatest distance of the moon ($64^r\ 10'$) to a certain fourth term, they found from this proportion $277^r\ 33'$, which is the apogee distance of Mercury … Likewise, taking the perigee distance of Venus from the centre of the zodiac to be to the apogee distance as $15^r\ 35'$ to $204^r\ 25'$, and the same ratio to hold between the apogee distance of Mercury ($277^r\ 33'$) and another fourth term, they found this to be 1150^r, which is nearly the perigee distance of the sun (1160)»[a].

These computations, whose elements are taken from the *Almagest*, could easily be attributed by the Arabs to Ptolemy himself. But Schiaparelli points out to me that the Greeks had already done this. For Simplicius[1], who is later than Proclus, relates: «That Mercury stands below Venus is shown by the observations in which Mercury is found running below Venus. It is also demonstrated by the ratio of the distances of their apogees and perigees; for the greatest distance of Venus is shown to be nearly the distance of the sun, so that Venus is next to the sun; likewise the greatest of Mercury nearly equal to the least of Venus, and the greatest of the moon nearly equal to the least of Mercury. For these things are demonstrated in Ptolemy's *Syntaxis* [= the Almagest], the ratio being derived from the eccentricity of these planets with respect to the centre of the earth»[2].

Simplicius had devoted the greatest effort to the study of the books of Proclus, whom he calls «the master of my master». Perhaps, as Schiaparelli rightly conjectures, he knew the doctrines of the Almagest not from the original text, which is very hard to understand, but from some compendium, such as the Hypotyposes («first lines, or rudiments» of the motions of the planets) are; so he could easily think that everything Proclus relates in them, including the computation of the distances of the planets, came from Ptolemy. — Things being so, I think it not improbable that the *Kitāb al-manshūrāt*, which I find mentioned only in those two places of al-Bīrūnī, means «the Book of things made public [on astronomy]», which would be a version of the Greek name Ὑποτυπώσεις; that it is therefore not Ptolemy's but Proclus'; and that some of the Arabs, lastly, fell into the same error into which Simplicius had already fallen[b].

b) — pp. 120–122.

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[I. *The diameters of the planets with respect to the earth.* They are computed in this way. Let Δ be the apparent angular diameter of the sun, *R* its mean distance; the linear diameter of the sun will be *R*Δ. Likewise let δ be the apparent angular diameter of the planet, *r* its mean distance; *r*δ will

[1]Simplicii *in Aristotelis de caelo commentaria*, ed. Heiberg, Berlin 1894, p. 474 (= Commentaria in Aristotelem Graeca edita consilio et auctoritate Academiae litterarum regiae Borussicae, vol. VII). In Karsten's edition (Utrecht 1865) the passage is p. 213*b*.

[2]Ταῦτα γὰρ δέδεικται ἐν τῇ Πτολεμαίου Συντάξει μεταφερομένου τοῦ λόγου τῆς τῶν ἀστέρων ἐκκεντρότητος εἰς τὸν λόγον τῆς ἐκ τοῦ κέντρου τῆς γῆς.

[a]With these numbers $64^r\ 10' \times 91^r\ 30' : 34^r\ 4'$ is $172^r\ 21'$, not $277^r\ 33'$; and Ptolemy's greatest distance of Venus is $104^p\ 25'$ ($60^p + 1^p\ 15' + 43^p\ 10'$), not $204^p\ 25'$. With $172^r\ 21'$ and $104^r\ 25'$ the last term comes out $1154^r\ 49'$, the «1150^r» given.

[b]Nallino's Addenda et emendanda (Part II, pp. XXV–XXVII) give a third mention of this book, in Barhebraeus' Syriac astronomy (pp. 215–232 of the text, 188–200 of the translation). From the elements of the Almagest, with the spheres of the planets touching one another, Barhebraeus computes the least distances in earth radii and the sizes of the sun and the planets, these from apparent diameters of 1/15 (Mercury), 1/10 (Venus), 1/20 (Mars), 1/12 (Jupiter) and 1/18 (Saturn) of the sun's, which, he says, Hipparchus observed and Ptolemy reported in the book that he calls in Syriac *ktābā da-nsīrē*. Nau took this for the *Book of the Fruit*, the *Centiloquium* falsely ascribed to Ptolemy; Nallino rejects this: the Syriac title means «the book of prisms», and the Arabic *al-manshūrāt* means things spread out, made public or sawn, and also prisms, so that Barhebraeus took his matter from the same *Kitāb al-manshūrāt* as al-Bīrūnī and rendered its title, not aptly, as the book of prisms. These Addenda are translated with Part II.

be the linear diameter of the planet. So

$$\frac{\text{diameter of the planet}}{\text{diameter of the sun}} = \frac{r\delta}{R\Delta} = \frac{r}{R} \times \frac{\delta}{\Delta}\,.$$

The diameter of the sun comprises 5 1/2 diameters of the earth; so

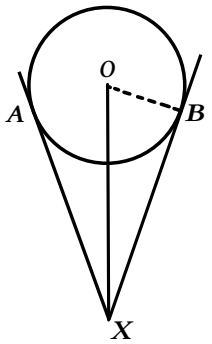
$$\frac{\text{diameter of the planet}}{\text{diameter of the earth}} = 5\tfrac{1}{2} \times \frac{r}{R} \times \frac{\delta}{\Delta} = r \times \frac{\delta}{\Delta} \times \frac{1}{\left(\dfrac{R}{5\frac{1}{2}}\right)}\,.$$

But $R = 1108$, $\frac{R}{5\frac{1}{2}} = 201\frac{5}{11}$, so

$$\frac{\text{diameter of the planet}}{\text{diameter of the earth}} = r \times \frac{\delta}{\Delta} \times \frac{1}{201\frac{5}{11}}\,,$$

which is the rule of al-Battānī, or rather of those who had made these computations before him. Plainly, when he speaks of the observed ratio between the diameters of the planets and the diameter of the sun, he always means the ratio of the *apparent diameters*, not of the true ones.

II. *The arcs that the apparent diameters of the epicycles subtend at the eye.* The apparent diameter of the epicycle is the angle contained by the two tangent lines drawn from the eye to either side of the epicycle, namely the angle AXB. — In *Mercury* and *Venus* this angle is plainly the sum of the greatest and the least elongation, as al-Battānī has it. — Otherwise in the superior planets. As regards *Mars*, Ptolemy X, 8 (ed. Halma vol. II, 236) shows that the radius of the eccentric is to the radius of the epicycle as 60^p to $39^p\ 30'$, that is, as 1 to 0.65833…. This number (or $39^p\ 30'$) is the sine of the angle OXB, which subtends at the eye the radius OB of the epicycle; so OXB = 41° 10 1/2′, and AXB = 82° 21′. — In *Jupiter* Ptolemy XI, 2 (ed. Halma vol. II, p. 262) sets the radius of the eccentric to the radius of the epicycle as 60^p to $11^p\ 30'$, or as 1 to 0.19166…[a]. So the angle OXB = 11° 3′, and the angular diameter of the epicycle AXB = 22° 6′. — In *Saturn* Ptolemy XI, 6 (ed. Halma vol. II, p. 287) sets the radius of the eccentric to the radius of the epicycle as 60^p to $6^p\ 30'$, or as 1 to 0.10833…. So OXB = 6° 13′, and AXB = 12° 26′. — Schiaparelli.]



Tycho Brahe, *Astronomiae instauratae progymnasmata*, Uraniborg and Prague 1610, vol. I, pp. 467–71, sets out what al-Farghānī and al-Battānī say about the distances and sizes of the heavenly bodies, and then teaches «by what method and course of demonstration Albategnius chiefly proceeded in determining the sizes of the stars from their given distance and their visible diameter in an aliquot part of the solar diameter, according to the appearance».

c) — p. 122, l. 33.

[a]The print has 0.19666; $11^p\ 30' : 60^p$ = 0.19166, which gives the 11° 3′ that follows.

[«*The diameter of the sun subtends an arc of 39° 48′*». What this means I do not know. If the *epicycle* of the sun is meant, its diameter is twice the greatest equation of the centre, so 3° 58′, which by no means agrees with the 39° 48′ of the codex (here and on p. 123, l. 21) and the 49° 48′ or 9° 48′ of Plato. If the apparent *diameter of the sun* is meant, al-Battānī, p. 58, sets its mean amount at 32′ 30″. So it must be considered a foolish interpolation, like the one of which I shall speak in the following note. — Schiaparelli.]

d) — p. 123, ll. 18–20 and 20–30.

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[This whole passage, in which almost all the numbers of the codex differ from those of Plato, cannot be healed; yet before it everything proceeded rightly. Plainly not al-Battānī but a most inept interpolator tried to add the greatest, mean and least apparent diameters of all the planets. The 39° 48′ occur again as a constant apparent diameter of the sun, an error of which I have already spoken in note *c*. The diameters of the planets at the greatest, mean and least distances must grow from one to the next; here, on the contrary, the mean diameter is almost always smaller than the others, but in Venus greater. That the interpolator wanted to give the apparent diameters of the planets is plain not only from his explanations but also from the numbers of the moon, which (leaving aside an obvious error at the mean distance) are probably the apparent diameters of the moon according to some astronomer *other than* al-Battānī.

Relying on the principles that al-Battānī sets out in the earlier part of this chapter, that is, taking the greatest, mean and least distances given by him, and computing the diameters at the mean distances in fractions of the mean diameter of the sun, 32′ 30″ (as al-Battānī supposes), I have found these numbers, which answer to al-Battānī's mind:

	Apparent diameter		
	at apogee	*mean*	*at perigee*
Moon	29′ 30″	32′ 25″	35′ 20″
Mercury	1 30	2 10	3 54
Venus	1 53	3 15	12 6
Sun	31 20	32 30	33 40
Mars	0 56	1 37	6 30
Jupiter	2 12	2 42	3 32
Saturn	1 33	1 48	2 10

which differ very widely from the numbers of the codex and of Plato. — Schiaparelli.]

To these very strong arguments, which most certainly show that this passage (ll. 20–30) is interpolated, another sign can be added. In this one passage of the whole of al-Battānī's work Saturn is called not *Zuḥal* but *al-Muqātil*, «the fighter»; and Mercury not *ʿUṭārid* but *al-Kātib*, «the scribe» or «the writer», a name that occurs only in appendix D (p. 221, l. 15 of the text, p. 147, l. 30 of the translation), which we shall show to be beyond doubt spurious. Both names are wholly unknown to the Arab astronomers who flourished between the Nile and the Ganges, and are not recorded in the Arabic lexica. They were not unknown, on the other hand, to the Arabs who lived in north-west Africa and in Spain; and in fact Dozy, *Supplément aux dictionnaires arabes*, Leiden 1881, vol. II, pp. 308 and 442, took them from a Latin-Arabic glossary made in southern Spain in the twelfth century[1]; I can cite some examples from African writers on astronomy (al-Akhḍarī and al-Wansharīsī)[2a]. Likewise in Latin codex 16208 of the Paris library, written in the twelfth or

[1]It is the *Glossarium Latino-Arabicum, primum edidit C. F. Seybold*, Berlin 1900 (= Semitistische Studien, fasc. 15–17).

[2]Saḥnūn al-Wansharīsī, *Mufīd al-muḥtāj ʿalā* ... al-Akhḍarī, Cairo 1314 H, pp. 46, 47, 49, 53–56. — ʿAbd al-Raḥmān al-Akhḍarī finished an astronomical poem at the age of twenty, in the month Dhū l-Qaʿda 939 H (June 1533), in the province of Algiers; al-Wansharīsī, of the same province, furnished the poem with a commentary.

[a]Nallino's Addenda et emendanda (Part I, p. LXXVI) add that other examples of *al-kātib* and *al-mukātib* for Mer-

thirteenth century, I found among other little works a version (made in Spain) of an astrological booklet of the Spanish-Arab astronomer al-Zarqālī[1], at the beginning of which (f. 26v) are these words: «*Almicatil* i. Saturnus».

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So the man who used these names must have been an Arab of the West; and of the West likewise was the man who made the copy from which the Escorial codex was later transcribed, as plainly appears from the additions at the end of the geographical tables and in the headings of the tables of the ascensions of the signs in the city of al-Raqqa. The same African or Spanish glossator also added at the end of the work (ff. 241v–244r) tables taken from the book of Maslama al-Majrīṭī, who had lived in Cordoba in the second half of the tenth century and had made an abridgement of al-Battānī's tables.

On chapter 51

a) — p. 124, ll. 19–33.

I. — *The epoch of Ptolemy's catalogue* of the fixed stars.

Ptolemy, VII, 4 (ed. Halma vol. II, p. 30), says that he set the longitudes of the stars for the first year of King Antoninus, which is the year 885 of Nabonassar, beginning on 20 July 137. — Why al-Battānī, l. 20, says that the first year of Antoninus is the year 886 of Nabonassar I do not know; it does not seem to be an error of the copyists, for al-Ṣūfī, p. 42, also writes 886, and adds that 41 years elapsed between the observation of Menelaus of the year 845 and the epoch of Ptolemy's catalogue.

II. — *The motion of the fixed stars* in longitude, or *the precession of the equinoxes.*

After what Schiaparelli and Th. H. Martin[a] have written, it is known that Hipparchus (about 130 BC) was the first to recognize the precession of the equinoxes[2]. He had fixed its annual amount at 36″ at the least; later he had deduced from other observations amounts between 42″ and 46″.8. Ptolemy wrongly chose the smallest value found, and laid down that the motion of precession completes 1° in 100 Egyptian years (of 365 days), or 36″ a year[3].

The Arabs very rarely use Ptolemy's amount; I find it in al-Farghānī, ch. XIX, and, in the thirteenth century, in al-Qazwīnī (*Cosmography*, treatise I, ch. IX, Cairo edition of 1311 H, vol. I, p. 43).

Al-Battānī, comparing his own observations with those of Menelaus, found that in 782 years the longitudes had increased by 11° 50′; so the motion appeared to him to be 54″.4757 = 54″ 28‴ a year, or 1° in 66 $\frac{318}{3268}$ years[b]. Al-Battānī rightly observes, however, that the fractions of the seconds,

[1]Title of the booklet: «*Liber Arçhachelis introductorius ad librum iudiciorum Arabum*». [*Translator:* Nallino's Addenda et emendanda (Part I, pp. LXXVI–LXXVII) withdraw the attribution to al-Zarqālī: Steinschneider, *Zum Speculum astronomicum* (Zeitschrift für Mathematik XVI, 1871, p. 390), shows that Arçachel is a corruption of Zahel, that is, Sahl ibn Bishr; the Laurentian codex S. Marco 194 (ff. 69r–87v) begins «Incipit liber arçachelis introductorium ad librum iudiciorum arabum» and has «almicatil [al-muqātil] .i. saturnus».]

[2]Whether the Babylonians anticipated Hipparchus in finding it, in some rough way, we do not yet know; see F. X. Kugler, *Die babylonische Mondrechnung: zwei Systeme der Chaldäer über den Lauf des Mondes und der Sonne*, Freiburg im Breisgau 1900, pp. 103–104, § 60.

[3]Delambre, *Histoire de l'astronomie ancienne*, vol. II, pp. 249 and 254; Tannery, *Recherches sur l'histoire de l'astronomie ancienne*, pp. 265–270.

cury, and of *al-muqātil* for Saturn, occur throughout the Moroccan Muḥammad al-Sūsī al-Marghitī, *al-Mumtaʿ fī sharḥ al-Maqnaʿ* and *al-Maṭlaʿ ʿalā masāʾil al-Maqnaʿ* (both printed in one volume, Fez 1313 H; see p. 205, note 1).

[a]The print has «Th. M. Martin»; Thomas-Henri Martin is meant, «Th. H. Martin» on p. 299.

[b]54″ 28‴ are 3268‴, and 1° = 216 000‴ = 66 × 3268‴ + 312‴, so that a degree takes 66 $\frac{312}{3268}$ years, not 66 $\frac{318}{3268}$; with 54″.4757 it takes 66 $\frac{6}{71}$ years. 3600″ : 66 = 54″ 32‴.7, the 54″ 33‴ below.

and perhaps the seconds themselves, are uncertain, and that in observations of the fixed stars at that time an error of some minutes was of no great importance; so, neglecting a very small fraction, he set that motion at 1 degree in 66 years, or 54″ 33‴ in a solar year[1].[a]

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The amount chosen by al-Battānī agrees entirely with the one that the astronomers of the caliph al-Maʾmūn had already found before him and had set down in the *Tested Tables* (as al-Ṣūfī, among others, attests, pp. 33 and 42); the same was also determined about the year 850 by the Sons (*banū*) of Mūsā ibn Shākir[2]. — Most of the Arab astronomers chose this amount: so al-Ṣūfī, p. 43, al-Bīrūnī[3], Ḥabash[4], al-Kharaqī, al-Khāzinī (f. 129r), al-Jaghmīnī, p. 229, Barhebraeus, pp. 12, 26 and 42 of the text, and the *Persian Tables* of the Greek Chrysococces[5]. Also about the year 1260 Muḥyī al-Dīn al-Maghrabī, with others, in the observatory of the city of Marāgha (near Lake Urmia), found the precession to be 1° in 66 solar years, by observing Aldebaran, α Scorpii and other stars[6]. — The same amount is found among the Jews in Abraham bar Ḥiyya, ch. X, and in Leo de Balneolis (on whom see p. 218), treatise I, ch. 58, f. 46v *a*.

A precession of 1° in 70 Egyptian or Persian years (of 365 days), that is 51″ 3/7 (= 51″ 25‴ 42^{iv} 51^{v} 3/7) a year, was found by Ibn al-Aʿlam[7], Naṣīr al-Dīn al-Ṭūsī[8], Quṭb al-Dīn Maḥmūd al-Shīrāzī and Ulugh Beg[9]. — The Jew Joseph ben Eliezer, who wrote about 1358, wrongly ascribed this precession to al-Battānī[10].

Finally Ibn Yūnus, ch. VIII (in Caussin, pp. 169 and 237, Delambre, p. 98), set the motion of the fixed stars at 1° in 70 1/4 Egyptian years, that is 51″ 14‴ 43^{iv} 59^{v} a year[11].

In the Indian book *Sūrya-Siddhānta*, III, 9–12 (transl. Burgess and Whitney, pp. 243–44), composed before the fifth century, an annual precession of 54″, which hardly differs from al-Battānī's, is mentioned together with the doctrine of the trepidation of the fixed stars. But that passage is held to be a late interpolation[12]; Varāhamihira, who about the year 505 set out in his *Pañcasid-*

[1] According to Bessel, the total annual precession for the year t before 1750 is 50″.21129 – 0″.000 244 296 6 · t; so in the year 880, the epoch of al-Battānī's catalogue, it was 50″.0.

[2] Ibn Yūnus, ch. V, in Caussin, p. 150.

[3] *Chronology*, p. 352 of the text, 354 of the translation; *India*, p. 196 of the text, vol. I, p. 392 of the translation; *Elements of astronomy*, Paris codex, f. 28r.

[4] F. 28v, where there is a table adapted to the years of the hijra. — According to Ibn Yūnus, ch. V (Caussin, p. 169), Ḥabash had made this motion 1° in 70 years in the *Arabic Table*. This seems to be an error of Ibn Yūnus; see Caussin's own note 3.

[5] At the end of the work of I. Bullialdus, *Astronomia philolaica*, Paris 1645, part II, p. 225; there the precession is said to be 1° in 68 years, but lunar years are meant, 68 of which make 66 solar years. What Rudloff and Hochheim note on al-Jaghmīnī, p. 229, must therefore be corrected. — On the age of Chrysococces' tables see p. 159, note 8. [*Translator:* The print has 1545; Bullialdus' *Astronomia philolaica* appeared at Paris in 1645.]

[6] Abū l-Faḍl ʿAllāmī (on whom see p. 243), *Āyīn-i Akbarī*, vol. II, p. 11 of the text (III, 20 of the translation).

[7] Ibn Yūnus, ch. V, in Caussin, p. 169; Qāḍīzāde, commentary on Naṣīr al-Dīn's *Tadhkira*, Laurentian codex, f. 39v (treatise II, ch. IV); Abū l-Faḍl ʿAllāmī, loc. cit. (in the translation «sixty» is an error for «seventy»). — Ibn al-Aʿlam died at the beginning of the year 375 H (which began on 23 May 985); Suter, p. 62, no. 137.

[8] In the *Īlkhānī Tables*, Laurentian codex, f. 102r.

[9] P. 149 of the text, 200 of the translation (where «soixante», sixty, stands wrongly for «soixante-dix», seventy).

[10] In M. Steinschneider, *Études sur Zarkali*, in the Bullettino di bibliografia e di storia delle scienze matematiche, vol. XX, Rome 1887–88, p. 25 (p. 97 of the book published separately). [*Translator:* The print has «p. 97»; Nallino's Addenda et emendanda (Part I, p. LXXVIII) correct it to p. 25 (97 of the book published separately).]

[11] L. P. Sédillot, *Matériaux pour servir à l'histoire comparée des sciences mathématiques*, Paris 1845–50, vol. I, pp. 280 ff., says that al-Zarqālī (Arzachel) reckoned the annual precession at 49″ 1/2 or 50″. But this is a mere conjecture of Sédillot's; al-Zarqālī could easily have deduced that amount from some of his observations, but in fact he does not seem to have deduced it. He professed the theory of the trepidation of the fixed stars.

[12] See Whitney's notes, pp. 244–49 and 432–33, and G. Thibaut, *Astronomie, Astrologie und Mathematik* [*der Inder*], Strassburg 1899, p. 45 (= Grundriss der indo-arischen Philologie und Altertumskunde, vol. III, fasc. 9). — Bhāskara (writing in 1150), in the *Siddhānta-Śiromaṇi*, Golādhyāya, VI, 17 (transl. Wilkinson, Calcutta 1861, p. 157), gives

[a] Nallino's Addenda et emendanda (Part I, p. LXXVIII) add that he does not know where Gerard of Sabbioneta (see the passage in the preface, pp. XXXV–XXXVI) found the notice that al-Battānī made the fixed stars move one degree in 60 years and 4 months; the same error is in Baldi, p. 440 (extract p. 20).

dhāntikā the doctrines of other books (among them the Sūrya-Siddhānta itself), knows nothing of precession. In 628 Brahmagupta denies it, although it had already been received in 499 by Āryabhaṭa, together with the doctrine of trepidation. See H. Colebrooke, *On the notions of the Hindu astronomers concerning the precession of the equinoxes*, in Asiatic Researches, vol. XII, London 1818 (reprinted in Colebrooke's *Miscellaneous essays*, London 1837, vol. II).

b) — p. 124, ll. 34–37.

I. — *The addition of 11° 10′ to Ptolemy's longitudes* of the fixed stars.

Ptolemy, VII, 3 (ed. Halma vol. II, pp. 25 and 27), had made much use of the observations made at Rome by Menelaus, who, comparing the stars with the moon, had found Spica, or α Virginis, at 176° 1/4 of longitude in the night between 15 and 16 Mechir of the first year of Trajan, 845 of Nabonassar (i.e. 9–10 February 98), and the northern star of the forehead of the Scorpion, or β Scorpii, at 215° 55′ in the night between 18 and 19 Mechir of the same year (i.e. 12–13 February)[a]. I 294

Al-Battānī found that the longitudes of β Scorpii and α Leonis given by Menelaus had increased by 11° 50′ after 782 years. On this he clearly reasoned in this way: Ptolemy compiled his catalogue 41 years[1] after Menelaus; in this interval the precession is 37′ 21″; so 11° 50′ − 0° 37′ 21″ = 11° 12′ 39″, that is, in round numbers, 11° 13′, must be added to Ptolemy's longitudes. But Ptolemy uses only the fractions 1/2, 1/3, 1/6 (very rarely 1/4) of a degree, so that his longitudes are expressed in whole tens of minutes. If I add 11° 13′, almost all the minutes of the longitudes will end in the figure 3, which will both seem strange and could lead someone into the error of taking uncertain numbers for exact ones. A difference of some minutes, and even of one degree, is of very little importance in the longitudes of the fixed stars; so instead of 11° 13′ I shall add only 11° 10′.[b]

Al-Battānī's reasoning would be excellent if Ptolemy had used the same annual precession as al-Battānī. But the Greek astronomer's precession was 36″; so, for the interval of 41 years, Ptolemy had added 25′, not 37′ 21″, to the longitudes of Menelaus. Al-Battānī, then, should have added 11° 50′ − 0° 25′ = 11° 25′ to Ptolemy's longitudes, instead of his 11° 10′.

It is strange that al-Battānī did not notice the discrepancy arising from this between the longitudes of the stars observed by himself and by Menelaus and the longitudes of Ptolemy's catalogue. For:

	al-Battānī	*Ptolemy*	*difference from Menelaus*	*difference from Ptolemy*
β Scorpii[2]	17° 45′ Scorpio	6° 1/3 Scorpio[c]	11° 50′	11° 25′
α Leonis	14° 0′ Leo	2° 1/2 Leo	11° 50′	11° 30′

another reading, according to which the annual precession in the *Sūrya-Siddhānta* would have been only 9″, and not joined to the doctrine of trepidation.

[1] So al-Battānī and al-Ṣūfī; but the true interval was 40 years; see note *a*, no. I (p. 292). The error, however, is of no importance, for the precession in a year is 54″ 33‴, an amount to be wholly neglected in this computation at that time. [*Translator:* The print has «note b»; the interval is that of note *a*, no. I, on p. 292, from the year 845 to the year 885 of Nabonassar.]

[2] See p. 124, note 6.

[a] In the Egyptian calendar of the *Almagest*, by which Nallino dates the year 885 of Nabonassar on p. 292 (20 July 137), the year 845 began on 30 July 97, so that 15 and 18 Mechir fell on 10 and 13 January 98: the nights are those of 10–11 and 13–14 January 98. The 9–10 and 12–13 February of the print are the dates of these nights in the fixed Alexandrian calendar, in which 1 Mechir is 26 January.

[b] At al-Battānī's 54″ 33‴ a year, 41 years give 37′ 16″ 33‴ (at 54″ 28‴, 37′ 13″ 8‴), and 11° 50′ less 37′ 16″ 33‴ is 11° 12′ 43″ 27‴; either way, 11° 13′ in round numbers.

[c] The print has 6° 1/2. Ptolemy's catalogue has 6° 1/3, the 5° 55′ of Menelaus with the 25′ of Ptolemy's precession added, and with it the difference 11° 25′ of the last column is made.

Al-Ṣūfī, in reducing Ptolemy's longitudes, avoided al-Battānī's error. For, pp. 42–43, he reckons in this way: I have compiled my catalogue for the year 1276 of the era of Dhū l-Qarnayn (beginning 1 October 964); from the time of Menelaus 866 years have elapsed. In them the precession has reached 13° 7′ 16″; Ptolemy has already added 25′ to the numbers of Menelaus, so I must add 13° 7′ − 0° 25′ = 12° 42′ to Ptolemy's. — Hence one understands why al-Ṣūfī's longitudes exceed al-Battānī's by 1° 32′, instead of the 1° 17′ that are the increase of the precession in the interval of 85 years.

II. — «*We found no perceptible difference of latitude*».

Tycho Brahe, *Astronomiae instauratae progymnasmata*, vol. I, p. 246, sharply rebukes al-Battānī for keeping Ptolemy's latitudes of the fixed stars, «although he gives the inclination of the ecliptic to the equator much smaller than the ancients did, namely 23° 36′ [sic][a], falling short of their record by a quarter of a degree, which could surely have brought some difference into the latitudes of the stars of his own age, once a comparison was made. But he seems to me rather to have been credulous towards his predecessors, and in this part too securely content with the findings of the ancients, than to have used any observation of his own, and that sufficiently accurate, for scrupulously investigating the latitudes of the stars; from which it is also quite clear that he either used instruments not entirely suitable and not showing everything precisely, or else carried out this inquiry too negligently».

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There is no one who does not see how unjust this is. First, al-Battānī says that he found not «*no difference*» but «*no perceptible*» one; and indeed, in the problems solved by means of the ecliptic coordinates of the fixed stars, a discrepancy of 1/2 a degree in the latitudes was of no importance in that age. Next, it must not be forgotten that Ptolemy's latitudes contain no fraction of a degree except those that answer to 10′, 15′, 20′, 30′, 40′, 45′ and 50′; that is, they are given in minutes not precisely but only approximately, so that a safe and absolute comparison becomes impossible. Lastly, al-Battānī knew that errors of some minutes could easily creep into his observations of the fixed stars. Rightly, then, he did not wish to emend Ptolemy's tables of the latitudes with very uncertain corrections. — Besides, in the interval of time between Ptolemy and al-Battānī the *greatest* variation of the latitudes (which happens in the stars placed near 90° and 270° of longitude), owing to the decrease of the obliquity of the ecliptic, was only 6 minutes; or a quarter of a degree, if the obliquities found by Ptolemy and al-Battānī are considered.

c) — p. 125, ll. 3–4.

«*Besides al-dhuʾāba, al-fard and al-mirzam*»[b]. This passage of al-Battānī[1] is strongly disapproved by al-Ṣūfī, p. 38: «Il [al-Battānī] dit encore que toutes les étoiles rapportées par Ptolémée dans le livre *al-majisṭī*[2] sont au nombre de mille vingt deux, sans compter *al-dhuʾāba*, la Chevelure[c], *al-fard*, la Solitaire, et *al-mirzam*. Or *al-fard* est la brillante étoile située sur le cou du Serpent (L'Hydre الشجاع). Les Arabes lui ont donné ce nom, qui signifie isolée, parce qu'elle se trouve séparée des autres de même grandeur, et comme isolée منفرد au sud. Les Arabes nomment *al-mirzam* toute étoile qui précède une brillante. Ainsi l'étoile qui précède *al-shiʿrā al-yamāniyya*, Sirius, et qui est à la patte de devant du Chien, celle qui précède *al-shiʿrā al-ghumayṣāʾ* Procyon, se nomment *les deux mirzam des deux Shiʿrā* مرزمي الشعريين[3]. De même l'étoile qui est sur l'épaule

[1] The same error, p. 126, ll. 5–6.

[2] In writing the Arabic names in our letters I follow the system that I have used in the other parts of the book, not Schjellerup's.

[3] They are β Canis Majoris and β Canis Minoris.

[a] Al-Battānī's obliquity of the ecliptic is 23° 35′.

[b] The print has «odh-dhuʾābah».

[c] The print has «le Chevelure».

gauche du Géant est appelée *al-mirzam*[1]. Quant à *al-dhuʾāba*, la Chevelure, c'est une des trois étoiles que Ptolémée désigne sous le nom *al-ḍafīra*, la Boucle de cheveux, et qu'il a omise dans sa nomenclature des étoiles[2]; cela prouve donc qu'il n'a connu ni *al-fard*, ni *al-mirzam*» («He [al-Battānī] says further that all the stars reported by Ptolemy in the book *al-majisṭī* are one thousand and twenty-two in number, not counting *al-dhuʾāba*, the Hair, *al-fard*, the Solitary One, and *al-mirzam*. Now *al-fard* is the bright star situated on the neck of the Serpent (the Hydra, *al-shujāʿ*). The Arabs gave it this name, which means isolated, because it is separated from the others of the same magnitude and, as it were, isolated (*munfarid*) in the south. The Arabs call *al-mirzam* every star that precedes a bright one. Thus the star that precedes *al-shiʿrā al-yamāniyya*, Sirius, which is on the forefoot of the Dog, and the one that precedes *al-shiʿrā al-ghumayṣāʾ*, Procyon, are called *the two mirzam of the two Shiʿrā* (*mirzamā l-shiʿrayayn*). Likewise the star on the left shoulder of the Giant is called *al-mirzam*. As for *al-dhuʾāba*, the Hair, it is one of the three stars that Ptolemy designates by the name *al-ḍafīra*, the Curl of Hair, and which he left out of his enumeration of the stars; this therefore proves that he knew neither *al-fard* nor *al-mirzam*»).

d) — p. 125, ll. 10–22.

The lunar stations or mansions of the Arabs.

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In enumerating them al-Battānī fell into some errors, which al-Ṣūfī, pp. 37–38, sharply rebukes: «De même al-Battānī, lorsqu'il veut faire croire qu'il connaît les mansions de la lune et les étoiles suivant la méthode des Arabes مذهب العرب, et en s'occupant des choses étrangères à son sujet, a mis à nu son ignorance. Il a ainsi rapporté dans son livre le nombre d'étoiles que comprend chacune des douze constellations, comme les indique Ptolémée dans son livre[a] *al-majisṭī*, et il dit, entre autre en parlant du Bélier, que *al-sharaṭāni* se trouve dans la corne et *al-buṭayn* dans la queue; c'est une erreur, car *al-buṭayn* se compose de trois étoiles disposées en triangle, ce qui est mentionné auparavant. Il dit aussi sur le Taureau que *al-thurayyā*, les Pléiades, se trouve sur le dos et *al-dabarān* à la racine de l'oreille[3]; c'est aussi une erreur, car *al-dabarān*, qui est la brillante étoile rouge des cinq de la face, se trouve à l'oeil méridional. Il dit que *al-haqʿa*, *al-hanʿa* et *muqaddam al-dhirāʿayn*, appartiennent aux Gémeaux; c'est encore une faute, car *al-haqʿa* fait part du Géant, étant située dans la tête, entre les épaules; et il ne dit pas dans quel endroit des Gémeaux sont *al-hanʿa* et *al-dhirāʿ*. Or, *al-hanʿa* se compose de deux étoiles situées dans leurs pieds, et *al-dhirāʿ* consiste de deux étoiles brillantes dans leurs têtes. Il se trompe encore en mettant *al-ghafr* dans la Balance; car *al-ghafr* se compose de trois étoiles, dont deux sur le pan de la robe ذيل de la Vierge, et une sur son pied gauche; cette dernière est la plus méridionale des trois. Il place dans le Scorpion *al-zubānayāni*, les deux zubānā, ainsi que *al-iklīl*, la Couronne; c'est une double erreur: *al-zubānayāni* appartiennent à la Balance, et sont situés sur les deux bassins, tandis qu'il a cru que c'étaient les deux serres du Scorpion, c'est-à-dire ses cornes, suivant les Arabes. Il a soupçonné qu'*al-iklīl* se compose de trois étoiles situées sur le front du Scorpion, dans la figure même, tandis que ce sont trois étoiles disposées en ligne transversale, au-dessus du front du Scorpion. L'une d'entre elles, la plus boréale, se trouve dans le zubānā boréal de la Balance; c'est la huitième de la table de Ptolémée, et elle est de la quatrième grandeur. La deuxième, qui est au milieu des trois, est une des étoiles autour de la Balance, en dehors de la figure; c'est la sixième de ces étoiles. La troisième enfin, qui est la plus méridionale des trois, est également au nombre des étoiles externes de la Balance, et la huitième de celles-là.

[1] It is γ Orionis.

[2] What al-Battānī says of *al-dhuʾāba* is therefore true, not false. In his catalogue Ptolemy gives three stars of Coma; but at the end of the catalogue, where he reckons up the number of the stars, he leaves them out.

[3] Al-Battānī in the codex: «the Pleiades on the horn, and al-dabarān on the back»; but Plato: «on whose back is Achoria [al-thurayyā], and at the root of its horn [sic] Aldebaram is placed». This is perhaps al-Battānī's true reading; see also the catalogue of the stars, no. 8 of Taurus (vol. II, p. 156), where I have tried to explain the cause of the error.

[a] The print has «libre».

Toutes trois sont de la quatrième grandeur et forment une ligne courbe, semblable à celle que présentent les trois étoiles du front. Il dit que *al-naʿāʾim*, les Autruches, et *al-balda*, se trouvent dans le Sagittaire. C'est une faute, car *al-balda* est un endroit du ciel, en arrière d'*al-qilāda*[1], où il n'y a point d'étoiles, et c'est pourquoi on le nomme *al-balda*, la Ville. J'ai vu, sur plusieurs sphères, que l'on a designé *al-naʿām al-wārid*[2] par *al-naʿāʾim*, et *al-naʿām al-ṣādir*[3] par *al-balda*. Les Arabes n'ont fait de mansion qu'au milieu entre les deux autruches. Sous le Capricorne il fait mention de *saʿd al-dhābiḥ*, et *saʿd bulaʿ*; c'est une faute, car *saʿd bulaʿ* se trouve dans la main gauche du Verseau, au-dessus du dos du Capricorne. Il dit encore que *al-fargh al-awwal* [= *al-fargh al-muqaddam*] et *al-fargh al-thānī* [= *al-fargh al-muʾakhkhar*], sont situées dans les Poissons, mais il se trompe en cela, parce que *al-farghāni*, les deux fargh, font partie de la figure du Cheval, vers le côté nord. Quant au premier *fargh*, la plus boréale de ses deux étoiles, se trouve sur l'épaule droite, où commence la jambe قامة, et la méridionale sur le dos où commence le cou. Quant au second *fargh*, la plus boréale de ses deux étoiles se trouve sur le nombril (du Cheval), et en même temps dans la tête de la Femme enchainée [= Andromeda], étant commune à ces deux constellations; la méridionale est située sur les reins متن du Cheval. Aucune de ces étoiles ne fait partie des figures zodiacales. Ainsi il n'a connu ni les Poissons, ni le Cheval» («Likewise al-Battānī, when he wishes to make it believed that he knows the mansions of the moon and the stars according to the method of the Arabs (*madhhab al-ʿarab*), and busies himself with things foreign to his subject, has laid bare his ignorance. Thus he has reported in his book the number of stars that each of the twelve constellations contains, as Ptolemy gives them in his book *al-majisṭī*, and, among other things, speaking of the Ram, he says that *al-sharaṭāni* is in the horn and *al-buṭayn* in the tail; this is an error, for *al-buṭayn* consists of three stars arranged in a triangle, as was mentioned before. He also says of the Bull that *al-thurayyā*, the Pleiades, is on the back and *al-dabarān* at the root of the ear; this too is an error, for *al-dabarān*, which is the bright red star of the five of the face, is at the southern eye. He says that *al-haqʿa*, *al-hanʿa* and *muqaddam al-dhirāʿayn* belong to the Twins; this is again a mistake, for *al-haqʿa* is part of the Giant, being situated in the head, between the shoulders; and he does not say in what place of the Twins *al-hanʿa* and *al-dhirāʿ* are. Now *al-hanʿa* consists of two stars situated in their feet, and *al-dhirāʿ* consists of two bright stars in their heads. He is mistaken again in putting *al-ghafr* in the Balance; for *al-ghafr* consists of three stars, two of them on the hem of the robe (*dhayl*) of the Virgin and one on her left foot; this last is the most southern of the three. He places in the Scorpion *al-zubānayāni*, the two Zubānā, as well as *al-iklīl*, the Crown; this is a double error: *al-zubānayāni* belong to the Balance and are situated on the two pans, whereas he believed that they were the two claws of the Scorpion, that is, its horns, according to the Arabs. He supposed that *al-iklīl* consists of three stars situated on the forehead of the Scorpion, in the figure itself, whereas they are three stars arranged in a transverse line above the forehead of the Scorpion. One of them, the most northern, is in the northern zubānā of the Balance; it is the eighth of Ptolemy's table and is of the fourth magnitude. The second, which is in the middle of the three, is one of the stars around the Balance, outside the figure; it is the sixth of those stars. The third, finally, which is the most southern of the three, is likewise among the external stars of the Balance, and the eighth of them. All three are of the fourth magnitude and form a curved line, like the one that the three stars of the forehead present. He says that *al-naʿāʾim*, the Ostriches, and *al-balda* are in the Archer. This is a mistake, for *al-balda* is a place in the sky behind *al-qilāda* where there are no stars at all, and that is why it is called *al-balda*, the Town. I have seen on several globes that *al-naʿām al-wārid* had been marked as *al-naʿāʾim*, and *al-naʿām al-ṣādir* as *al-balda*. The Arabs made a mansion only in the middle between the two ostriches. Under the Goat he mentions *saʿd al-dhābiḥ* and *saʿd bulaʿ*; this is a mistake, for *saʿd bulaʿ* is in the left hand of the Water-pourer, above the back of the Goat. He says further that *al-fargh al-awwal*

[1]Which comprises ξ, ο, π, *d*, ρ, υ Sagittarii.

[2]σ, φ, τ, ζ Sagittarii.

[3]γ, δ, ε, η Sagittarii.

[= *al-fargh al-muqaddam*] and *al-fargh al-thānī* [= *al-fargh al-muʾakhkhar*] are situated in the Fishes, but he is mistaken in this, because *al-farghāni*, the two Fargh, are part of the figure of the Horse, towards the northern side. As for the first *fargh*, the more northern of its two stars is on the right shoulder, where the leg (*qāma*) begins, and the southern one on the back, where the neck begins. As for the second *fargh*, the more northern of its two stars is on the navel (of the Horse) and at the same time in the head of the Chained Woman [= Andromeda], being common to these two constellations; the southern one is situated on the loins (*matn*) of the Horse. None of these stars is part of the zodiacal figures. So he knew neither the Fishes nor the Horse»).

After adding what I have reported in note *c*, al-Ṣūfī concludes thus: «S'il [al-Battānī] eût suivi sa voie et se fût borné à ses propres doctrines مذهبه et se fût contenté de ce qu'il a exposé dans son livre sur les sphères célestes, sur les sept étoiles [the planets] et leur mouvement, sur les éclipses du soleil et de la lune et d'autres choses (nécessaires) concernant les étoiles, il n'eût pas encouru de tels errements; on lui décernerait, à cause de sa célébrité et de sa supérieurité dans l'art, suivant la méthode des astronomes, aussi des connaissances, suivant la méthode des Arabes» («If he [al-Battānī] had followed his own way and kept to his own doctrines (*madhhabuhu*), and had been content with what he set out in his book on the celestial spheres, on the seven stars [the planets] and their motion, on the eclipses of the sun and the moon and other (necessary) things concerning the stars, he would not have fallen into such errors; because of his fame and of his superiority in the art according to the method of the astronomers, people would also grant him knowledge according to the method of the Arabs»).

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Already before Muḥammad the Arabs had divided the zodiac into 28 parts; for they had chosen 28 places as far apart from one another as the apparent course of the moon in a day and a night. These places they had called the stations or mansions of the moon (*manzil al-qamar*, plural *manāzil al-qamar*). The *cosmic*[1] settings of these stations, that is, their settings in the western quarter of the horizon at the time of dawn, they had called *nawʾ* (plural *anwāʾ*); from them they drew many weather forecasts. So, even before al-Battānī's time, some Arabs wrote special books on the *anwāʾ*, such as al-Ḥasan ibn Sahl ibn Nawbakht, Ibn Qutayba, Abū Ḥanīfa al-Dīnawarī and Thābit ibn Qurra[2].

On the Arabic lunar stations, consult L. Ideler, *Untersuchungen über den Ursprung und die Bedeutung der Sternnamen*, Berlin 1809, who used the *Cosmography* of al-Qazwīnī[3]; al-Bīrūnī, *Chronology*, pp. 336–356 (text and translation); al-Ṣūfī, passim; Barhebraeus, *Astronomy*, pp. 122–25 of the text; al-Farghānī, ch. XX; Sprenger, in the *Zeitschrift der deutschen morgenländischen Gesellschaft*, vol. XIII, 1859, pp. 162–64, who relies on al-Qazwīnī and Waṭwāṭ; Steinschneider, *Ueber die Mondstationen und das Buch Arcandam*, in the Zeitschrift der deutschen morgenländischen Gesellschaft, XVIII, 1864, pp. 118–201, and XXV, 1871, pp. 379–390[4]. — A book on the cosmic settings of the lunar stations, or on the *anwāʾ*, written by ʿArīb ibn Saʿd, has been published under the title *Le calendrier de Cordoue de l'année 961, texte arabe et ancienne traduction latine, publiée par R. Dozy*, Leiden 1873[5].[a]

[1] Not *heliacal*, as Sprenger and Suter have it. See what Schjellerup noted on al-Ṣūfī, pp. 135–36.

[2] See Suter, *Die Mathematiker*, pp. 16 (no. 27), 31 (nos. 57 and 60), 36 (no. 66).

[3] Zakarija … El-Cazwini, *Kosmographie herausgegeben von F. Wüstenfeld*, Göttingen 1848–49, vol. I, pp. 41–51 = Cairo edition of 1311 H, vol. I, pp. 68–82.

[4] Of very little importance is the booklet of a very late age (eighteenth century?) *Les mansions lunaires des Arabes, texte arabe en vers de Moh'ammed El-Moqri traduit et annoté par A. de C. Motylinski*, Algiers 1899.

[5] G. Libri had already edited the Latin version, *Histoire des sciences mathématiques en Italie*, Paris 1838–41, vol. I, pp. 393–458.

[a] Nallino's Addenda et emendanda (Part I, p. LXXVIII) add that the use of the lunar stations in astrological forecasts is set out by Albohazen Haly, *De iudiciis astrorum*, Basel 1571, part VII, ch. 101, pp. 342–46 («on elections according to the motion of the moon through the mansions»), and by al-Būnī, *Shams al-maʿārif* (on the magical sciences), [Cairo] 1291 H, vol. I, pp. 16–24, who also gives (pp. 23–24) extracts from the sayings of the old Arabs (*asjāʿ al-ʿarab*) after a certain al-Kindī; and that Abū Maʿshar had written a special little book on elections according to the lunar stations

On the origin of the lunar stations learned men disagree greatly, some looking for it among the Arabs, others among the Indians (who call the stations *nakshatra*), others among the Chinese, others finally among the Babylonians. The truth is still unknown, as G. Thibaut, *Astronomie, Astrologie und Mathematik* [*der Inder*], Strassburg 1899, pp. 13–16, prudently warns[1].

e) — p. 126, ll. 6–7.

«*We have marked* [*in the catalogue*] *the stars whose qualities we found known in Ptolemy's book*». I 298

The word *mizāj*, which I have rendered «qualities», properly means «mixture» or «temperament of the body»[a]; but a little below (p. 189, l. 14 of the Arabic text) it designates at once everything that determines the position of a star, namely the longitude, the latitude and the side of the latitude[b]. The sense, then, seems to be that al-Battānī chose stars about whose places there was

[1]Some things worth noting are in F. K. Ginzel, *Die astronomischen Kenntnisse der Babylonier*, parts I and III (in the periodical Beiträge zur alten Geschichte, vol. I, Leipzig 1901–02, pp. 12–25 and 369–373), who defends a Babylonian origin.

(*al-ikhtiyārāt ʿalā manāzil al-qamar*).

[a]Nallino's Addenda et emendanda in Part II (pp. XVIII–XIX) show the astrological use of *mumāzaja* and *mizāj* by these examples. Abū Maʿshar, *Kitāb al-qirānāt*, «the book of the conjunctions [of the planets]», says in the preface (after a Bodleian codex: Pusey's catalogue, vol. II, p. 277a, addition to no. 284): «these lines contain the exposition of the conjunctions of the planets in the twelve signs, and the exposition of their commixtures with one another through the applications»; and the same little book has a table (Pusey, p. 278a) «containing the exposition of the applications of the upper planets and their commixtures with the lower». In Abū Naṣr al-Qummī, *al-Bāriʿ*, chapter I of treatise V is entitled «on the effects of the planets in the commixtures, and the exposition of the ways of judgements» (Ahlwardt, *Verzeichniss*, vol. V, p. 148, no. 5661). Al-Tahānawī, *Dictionary of technical terms*, ed. Sprenger, p. 1386 (under *naẓar*): «the [four astrological] aspects of the moon are called the moon's *imtizājāt* or *mumāzajāt*». Naṣīr al-Dīn al-Ṭūsī, *al-Mukhtaṣar fī ʿilm al-tanjīm*, has a chapter XIII entitled «on the commixtures of the moon with the planets and on its translations» (Nicoll, *Catalogue of the Bodleian Library*, no. 301, p. 300a). Ulugh Beg, p. 119 of the text, 164 of the translation, promises a table «for the applications of the moon to each of the planets», and the table (see the translation, p. 164, note 1) is entitled «table of the commixtures of the moon with the planets». *Mumāzaja*, then, and more rarely *mizāj*, are the applications of the planets to one another or to the moon. In the same sense the Greeks used σύγκρασις, «commixture»: Pseudo-Proclus on the *Tetrabiblos*, p. 76, l. 26 f.: χρὴ τὰς συγκράσεις μὴ μόνον ποιεῖσθαι τοὺς πλανωμένους πρὸς ἀλλήλους λαμβάνοντα, ἀλλὰ καὶ πρὸς τὰ ζῴδια καὶ πρὸς τοὺς ἀπλανεῖς ἀστέρας τοὺς παρανατέλλοντας αὐτοῖς, «one must make the commixtures taking the planets not only with respect to one another, but also with respect to the signs and to the fixed stars that rise beside them»; Vettius Valens begins his chapter on the influences of the applications of the planets: ὑποτάξωμεν δὲ καὶ τὰς συμπαρουσίας καὶ συγκράσεις ἑκάστου ἀστέρος, «let us also set out the joint presences and commixtures of each star» (*Catalogus codicum astrologorum Graecorum*, IV, p. 174); a Laurentian codex (in Boll, *Sphaera*, p. 10) has a chapter entitled ἀνάκρισις ἐκ τῆς τῶν ἀστέρων συγκράσεως περὶ οὗ τις ἐρωτᾶν βούλεται, «inquiry from the commixture of the stars into what someone wishes to ask about». More rarely μῖξις, as in a fragment of Palchus (*Catalogus codicum astrologorum*, I, p. 114): οὐ μόνον γὰρ τὴν τῶν ε΄ ἀστέρων καὶ ἡλίου καὶ σελήνης μῖξιν πρὸς ἀλλήλους δεῖ σκοπεῖν, ἀλλὰ καὶ πρὸς τοὺς τῆς αὐτῆς φύσεως κεκοινωνηκότας τῶν ἀπλανῶν, «for one must consider the mixture of the five stars and of the sun and the moon not only with one another, but also with those of the fixed stars that share the same nature». In this sense Firmicus Maternus uses *mixtura*, ed. Kroll, I, 2: «if characters and complexions are assigned to men by the *mixtures* of the stars …»; I, 4: «what would you think more difficult … than, once the courses of the stars have been found, to determine what their *mixture* and radiation bring about over the whole tract of the earth?»; and often. In Syriac *mūzāgā*, «mixture»: Bardesanes (154–222), *Le livre des lois des pays, texte syriaque et traduction française par F. Nau*, Paris 1899, pp. 18–19 of the text, 44 of the translation: «have you read the books of the Chaldaeans of Babylon, in which it is written what the planets bring about by their commixtures in the nativities of men?». Among the astrologers *mizāj* also occurs in the sense of the *temperament* or *nature* of each heavenly body; Nallino gives an example from the Arabic *Tetrabiblos* in vol. II, p. 290. A note adds that the work of Albumasar *De magnis coniunctionibus*, printed at Augsburg in 1489 and later, is not the *Kitāb al-qirānāt*, as it is usually said to be (Suter too, p. 29), but the *Kitāb fīhi jumal min dalālāt al-ashkhāṣ al-ʿalawiyya*, «the book containing the sum of the significations of the heavenly persons», as Steinschneider, *Zur Geschichte der Uebersetzungen aus dem Indischen*, ZDMG XXV, 1871, pp. 394–95, and Loth, *al-Kindī*, p. 271, recognized.

[b]Nallino's Addenda et emendanda (Part I, p. LXXVIII) give two examples of *mizāj* used in this sense, in Steinschneider, *Zur Geschichte der Uebersetzungen aus dem Indischen*, in the Zeitschrift der deutschen morgenländischen Gesellschaft, vol. XXV, 1871, p. 400, who also notes that *mimsāk kōl kōkāb*, «the mixture of each star», that is, its situation, occurs in the Hebrew writings of Abraham ibn ʿEzra (d. February 1167). But Steinschneider wrongly com-

no doubt or discrepancy in the manuscripts of the Arabic version of the Almagest. This interpretation is confirmed by al-Ṣūfī, p. 30: «J'ai regardé avec attention beaucoup d'exemplaires d'al-majisṭī, et j'ai trouvé qu'ils différaient par rapport à un grand nombre d'étoiles. J'ai cherché ensuite ces mêmes étoiles dans le livre d'al-Battānī, parmi celles qu'il prétend avoir observées, et j'ai trouvé qu'il avait supprimé toutes les étoiles sur lesquelles il y a, dans les exemplaires, la moindre différence. Il a supprimé ainsi beaucoup d'étoiles de la troisième et de la quatrième grandeur, tandis qu'il en rapporte beaucoup de la cinquième et de la sixième» («I looked attentively at many copies of *al-majisṭī*, and I found that they differed with respect to a great number of stars. I then looked for these same stars in the book of al-Battānī, among those that he claims to have observed, and I found that he had left out all the stars on which there is the least difference in the copies. In this way he left out many stars of the third and fourth magnitude, while he reports many of the fifth and sixth»).

There is no lack of malice in these words of al-Ṣūfī; for he turns into a fault and a fraud of al-Battānī what al-Battānī himself professes to have done.

On chapter 52

a) — p. 126, l. 35 ff.

The doctrine of certain astrologers on the *trepidation of the fixed stars.*

I. — «*Ptolemy relates in his book …*». Not Ptolemy, but Theon of Alexandria (fourth century), in the *Handy Tables*, ed. Halma vol. I, p. 53, chapter on the *turning* (περὶ τροπῆς): «Since according to certain opinions (κατά τινας δόξας) the ancients among the astrologers (οἱ παλαιοὶ τῶν ἀποτελεσματικῶν) hold that the tropic points, after a certain epoch, are carried 8 degrees in the order of the signs [i.e. eastward], and afterwards go back as many, — which Ptolemy indeed does not admit, because without such a hypothesis the equinoctial computations taken from the tables agree with those deduced by instruments[1]; so we too advise (ὅθεν καὶ ἡμεῖς παραινοῦμεν) that such an opinion be not received; — we shall set out the method of computation that they used. — For, taking as the epoch at which the greatest digression of 8° eastward had been made the 128 years that preceded the reign of Augustus, and adding the 313 years that elapsed from the reign of Augustus to the reign of Diocletian, and the years that have passed since Diocletian [down to the given time], with this sum they find the place [of the tropic points], taking the motion as 1 degree in 80 years; then, taking the quotient of this division from 8 degrees, they add the remainder (which shows the degree that the tropic points have reached at that time) to the degrees of the places of the sun, the moon and the five planets taken by these computations».

Theon's words «which Ptolemy does not admit» are somewhat ambiguous; they probably mean that Ptolemy rejected that doctrine silently, never having mentioned it. Al-Battānī, on the contrary, understood that Ptolemy had mentioned those astrologers and refuted them[2].

[1] Ὅπερ τῷ Πτολεμαίῳ οὐ δοκεῖ, διὰ τὸ ἄνευ τῆς ἐκ τοῦ τοιούτου ἐπιλογισμοῦ προθέσεως, συμφωνειν τὰς ἰσημερινὰς διὰ τῶν κανονογραφιῶν ψηφοφορίας ταις διὰ τῶν ὀργάνων καταλήψεσιν.

[2] Delambre himself, *Histoire de l'astronomie ancienne*, vol. II, pp. 625–27, appears somewhat uncertain in interpreting the passage of Theon. See the opinions of Letronne and Martin, which I shall cite below.

pares *al-mumāzajāt* (the title of a book of Abū Maʿshar according to Ibn al-Qifṭī) and *al-mumāzajāt bi-l-ittiṣālāt*; for these are the «commixtures [of the planets] arising from the applications», that is, the *aspects* of the planets to one another (opposition, quadrature, trine, sextile), on which see al-Battānī, ch. LIV, pp. 129 ff. In Lippert's edition, p. 154, perhaps on the authority of other manuscripts, the title of that book of Abū Maʿshar is *al-mizājāt*, as in al-Nadīm, p. 277, l. 17 (in Suter's translation, p. 32: «Das Buch der Temperamente», «The book of temperaments»); if this reading is true, *mizāj* is to be taken in the same sense as *mumāzaja*, that is, as the verbal noun of the third form. The first chapter (*faṣl*) of the fifth treatise of al-Ṣūfī's unpublished *Introduction to judgements* (*al-madkhal fī l-aḥkām*) is «on the commixtures of the planets and the exposition of the ways of judgements» (*fī mumāzajāt al-kawākib wa-dhikr ṭuruq al-aḥkām*).

The other Arab writers cite Theon, not Ptolemy, as witness of the doctrine of trepidation[1]. Thābit ibn Qurra (d. 288 H, 901), in the letter to Isḥāq ibn Ḥunayn[2] that Ibn Yūnus, ch. IV (in Caussin, p. 116), has handed down to us, rightly says: «Some, whom Theon and others mention and say were astrologers (*aṣḥāb al-aḥkām*), thought that the zodiac has a motion by which it advances 8° and afterwards returns as many; and that this motion completes one degree in every 80 years». I 299

Likewise Alpetragius, or al-Biṭrūjī, whose words I give after the Latin version made from a Hebrew version[3]: «The predecessors, such as Hermes, and the later ones among those who devote themselves to images [= the astrologers, ἀποτελεσματικοί] say that these stars have some motion in the order of the signs, and sometimes against it, as if this were for them something known by itself or handed down by their elders. And when the later ones came, such as the Chaldaeans and those who found the motion of these stars before the age of Bactenzar [= Bukhtanaṣṣar, Nabonassar], to demonstrate the truth of what the earlier ones had learned, they did not find a motion for them, and they gave up that motion spoken of by the first ancients». He adds that Hipparchus and Ptolemy deduced a continuous precession of the fixed stars from the observations of Timocharis, Aristyllus and Menelaus; then he goes on thus: «And someone after Ptolemy, namely Taun [= Theon] of Alexandria[4], was of the opinion that the fixed stars have a motion of access and a motion of recess, each of which consists of eight degrees, and that besides this they also have a motion in the order of the signs of one degree in every hundred years. This motion the later ones rejected, finding by observation their places to be other than the places in which they were situated in their former location; for they add something and take away something according to the times determined for them. Further, Albategnus declared that the fixed stars run from the point of the vernal equinox in equal times with a different course, and therefore he left this motion aside».

Similarly Ibn al-Qifṭī (d. 646 H, which began on 25 April 1248) thinks that Ḥabash[5] adapted the Indian notion of trepidation to the precepts of Theon of Alexandria; and Barhebraeus[6] says of Theon: «Among his books, which are celebrated throughout almost the whole world, is … the book of the Canon, in which he set out the method by which the change of the tropic points would be computed, if they advanced one degree in every 80 years up to eight degrees and afterwards went back; which Ptolemy did not admit, because without this addition the computations agree with the motions deduced with the help of the instruments that serve for observations»[7]. I 300

II. — This passage of al-Battānī has already been weighed by Letronne[8] and Th. H. Martin[9]; not

[1] The Hispano-Jewish Abraham bar Ḥiyya, who, as we shall see below, says that trepidation was not unknown to Ptolemy, took his notice from al-Battānī's work itself.

[2] He died in the month of Rabīʿ II 298 H, that is, at the end of the year 910; Suter, *Die Mathematiker*, pp. 39–40, no. 74.

[3] *Alpetragii Arabi* [sic] *planetarum theorica phisicis.* [sic] *rationibus probata, nuperrime Latinis litteris mandata a Calo Calonymos Hebreo Neapolitano*, Venice 1531, ff. 11v–12r. — Alpetragius lived in the twelfth century; Suter, p. 131, no. 325.

[4] He ascribes to Theon himself what Theon only relates of certain astrologers.

[5] We have Ibn al-Qifṭī's passage in print in the notes to *Kitâb al-Fihrist herausgegeben von G. Flügel, J. Roediger und A. Müller*, Leipzig 1871–72, vol. II, p. 130. — Ḥabash had professed trepidation only in his youthful book of tables, adapted to the Indian doctrines; in his later books, such as the one in the Berlin codex, he used a continuous precession.

[6] *Gregorii Abulpharagii sive Bar-Hebraei Chronicon Syriacum, ediderunt et verterunt P. J. Bruns et G. Kirsch*, Leipzig 1789, p. 59 of the text, 56 of the translation = *Bar Hebraei Chronicum Syriacum e codd. mss. emendatum* [*edidit P. Bedjan*], Paris 1890, p. 54. — See Barhebraeus' *Syriac astronomy*, pp. 103–104 of the text.

[7] Barhebraeus in his *Chronicle* thinks that Theon flourished in the same age as Ptolemy, namely under the emperor Titus Antoninus. See J. Lippert, *Studien auf dem Gebiete der griechisch-arabischen Uebersetzungslitteratur*, Braunschweig 1894, p. 40.

[8] *Nouvelles recherches sur le calendrier des anciens Égyptiens*, Mémoire II, § 4, pp. 103 ff. (in Mémoires de l'Académie des Inscriptions, vol. XXIV, part 2, 1864).

[9] *Mémoire sur cette question: La précession des équinoxes a-t-elle été connue des Égyptiens ou de quelque autre peuple de l'antiquité avant Hipparque?*, ch. V, p. 450 (in Mémoires présentés à l'Académie des Inscriptions par des savants étrangers, 1st series, vol. VIII, part 1, 1869). — Martin knew the passage of al-Battānī only from Letronne's dissertation.

rightly, however, for they thought that al-Battānī spoke of Arab astrologers and counted Ptolemy among those who had thought the doctrine of trepidation true. Both, however, rightly judged that those words of Theon, ὅπερ τῷ Πτολεμαίῳ οὐ δοκεῖ, mean that Ptolemy thought such a doctrine not even worth mentioning.

The doctrine of trepidation, which is also mentioned by Proclus[1] in the fifth century, arose, both think[2], from the wish to reconcile with each other the Chaldaeo-Egyptian hypothesis of immovable equinoctial points and the theory of precession that Hipparchus had deduced from observations made over less than two centuries. Those eight degrees, they think, come from a change, wrongly interpreted, that Hipparchus brought into the division of the ecliptic when, following the custom of the more ancient astronomers, he restored the beginnings of Aries, Cancer, Libra and Capricorn to the equinoctial and solstitial points; these beginnings Meton, Eudoxus and others had put 8° to the west of the equinoxes and solstices.

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What Letronne and Martin say of Eudoxus (in the second half of the fourth century BC) and of Meton (in the first half of the fifth)[a] must not be thought true; for it is reported by Columella[3], of little authority in these matters. Hipparchus[4], on the contrary, expressly states that Eudoxus put the equinoctial and solstitial points in the *centre*, i.e. at 15°, of the signs of Aries, Cancer, Libra and Capricorn; and those who have written on Greek astronomy in our time do not doubt it[5]. Nor does Schiaparelli doubt it, who explains Columella's error to me in this way: in Eudoxus' time the Greeks began the year in summer, whence came the custom of beginning the zodiac with Cancer, which appears from the passage of Hipparchus just cited and from the *Parapegma*, or calendar, placed at the end of Geminus' *Elements of astronomy*. The constellation of Cancer occupies about half a sign, i.e. about 15 degrees. Some grammarian, having heard that Eudoxus put the summer solstice in the middle of Cancer, confused the *dodecatemories* with the *constellations* and thought that Eudoxus put the summer solstice at 7° 1/2 of Cancer, that is, in the *eighth* degree. That

[1] *Hypothèses de C. Ptolémée et Hypotyposes de Proclus Diadochus* [*publiées et*] *traduites par Halma*, Paris 1820, p. 88: «But others have thought that the tropic points do move; not, however, through the whole circle [of the ecliptic], but some degrees, and afterwards go back again by the same».

[2] Letronne, § 4, pp. 106–110; Martin, ch. IV, pp. 445–447.

[3] *De re rustica*, IX, 14: «From the first equinox, which is completed in the month of March, about the eighth day before the Kalends of April, in the eighth part of Aries. … Nor does the reckoning of Hipparchus escape me, which teaches that the solstices and equinoxes are completed not in the eighth but in the first parts of the signs. But in this rural discipline I now follow the calendars of Eudoxus and Meton and of the ancient astrologers, which are fitted to the public sacrifices; because that old opinion, familiar to the farmers, is also the better».

[4] *Enarrationes ad Arati et Eudoxi Phaenomena*, book II, ch. III: «First it must be understood that the division of the zodiacal circle is so made by Aratus that it begins from the tropic and equinoctial points, and these are the beginnings of the signs. But Eudoxus divides it so that those points are the middles: the former of Cancer and Capricorn, the latter of Aries and Libra. … In this manner [that of Aratus] almost all or most of the ancient mathematicians divided the zodiacal circle. … But Eudoxus makes it plain that he places the tropic points in the middles of the signs by these words: The second is the circle on which the summer turnings (θερινὰι τροπαί) take place; on it lies the middle of Cancer. Again: the third, he says, is the circle on which the equinoxes take place; on it is placed the middle of Aries and of Libra. The fourth, on which the winter turnings happen; on it is the middle of Capricorn. He shows this very thing more plainly also where, treating of the colure circles, which are drawn through the poles and the tropic and equinoctial points, he says thus: There are two other circles, cutting each other in two at right angles through the poles of the world; on them are these stars: first the ever-visible pole of the world; then the middle of the Bear in breadth, and the middle of Cancer. A little later he adds: And the tail of the Southern Fish, and the middle of Capricorn». — The *Parapegma* added to Geminus' *Elements of astronomy* disagrees with Hipparchus; for one gathers from it that Callippus (330 BC) and Euctemon put the summer solstice at the beginning of Cancer, but Eudoxus the winter solstice at 4° of Capricorn and the spring equinox at 6° of Aries. Schiaparelli pointed out the passage to me.

[5] Except P. Tannery, *Recherches sur l'histoire de l'astronomie ancienne*, Paris 1893, p. 130, who says: «Il [Hipparchus] a donné un sens mathématique précis à des textes d'Eudoxe qui n'indiquaient que des tracés approximatifs, et il en a profité pour critiquer plus à son aise les descriptions du Cnidien …» («He gave a precise mathematical sense to texts of Eudoxus that indicated only approximate drawings, and took advantage of it to criticize the descriptions of the Cnidian more at his ease …»).

[a] Meton observed the summer solstice at Athens in 432 BC (Ptolemy, *Almagest* III, 1), in the second half of the fifth century.

grammarian or astrologer probably lived after the time when the Greek astronomers came to know the use of degrees, that is, after the middle of the second century BC.

It is certain, as Bouché-Leclercq, *L'astrologie grecque*, Paris 1899, p. 129, note 1, sets out, that Manilius (first century), Pseudo-Manetho (second and third) «et autres astrologues routiniers» («and other astrologers of routine») began the four parts of the year from the *eighth* degrees of the signs; to these authors both Columella (first century) and Hyginus[1] can be added. — But the origin of such a custom is to be sought among the ancient Chaldaeans; for F. X. Kugler, *Die babylonische Mondrechnung*, Freiburg im Breisgau 1900, pp. 79, 99, 104, 212, has deduced from the cuneiform tablets that the Babylonians, already in the fourth and third centuries BC, put the four points of the year at 8° 15′ (sometimes 10° and 8° 0′ 30″) of Aries, Cancer, Libra and Capricorn according to the fixed Babylonian ecliptic, or at 5° of the same signs according to our moving ecliptic and Hipparchus'.

It seems to me worth mentioning that al-Bīrūnī, *Chronology*, pp. 325–26 of the text, 322 of the translation, already saw that those 8 degrees of the astrologers who professed trepidation are to be connected with the Chaldaean way of beginning the parts of the year, and erred only in taking the doctrine of trepidation to be very ancient. For he says: «The authors disagree among themselves on the beginnings of the four parts of the year. Ptolemy, in his Introduction to the spherical art (*al-madkhal ilā l-ṣināʿa al-kurriyya*)[a], says that the Greeks put these beginnings at the entry of the sun into the equinoctial and solstitial points. The Chaldaeans are said[b] to have placed such beginnings 8° after the equinoxes and solstices; which I think [was said] because of the lag of their computations (in the tables ascribed to them) behind what the observations and the tables of the Greeks required; and that this amount of 8 degrees was fixed because they thought that the difference arose from a motion of the sphere forward and backward, whose greatest amount was 8 degrees. But what they meant God knows best. The explanation of this motion, the best and most apt of all, is in the *zīj al-ṣafāʾiḥ*, whose author is Abū Jaʿfar al-Khāzin[2],

[1] *Astronomicon*, ed. Bunte, p. 101: «The sun begins to rise from the eighth part of all the signs …». I owe the passage to Schiaparelli. — Achilles Tatius, *Isagoge ad Arati Phaenomena*, ch. 23: «This turning (or solstice, τροπήν) some hold to be made by the sun about the beginning of Cancer, others about the eighth part, others about the twelfth, others about the fifteenth».

[2] He died between about 350 and 360 H (961–971); Suter, p. 58, no. 124.

[a] Nallino's Addenda et emendanda (Part I, p. LXXVIII) add that the title of this book of Ptolemy named by al-Bīrūnī agrees with no. 24 of the works translated from the Arabic by Gerard of Cremona in the catalogue edited by Boncompagni (*Della vita e delle opere di Gherardo Cremonese*, Rome 1851, p. 5, from vol. IV of the Atti dell'Accademia Pontificia de' Nuovi Lincei): «*Liber introductorius ptolomei ad artem spericam*»; and that al-Masʿūdī, *Tanbīh*, pp. 69–70, quotes a passage from the same book (*al-madkhal ilā l-ṣināʿa al-kurriyya*).

[b] Nallino's Addenda et emendanda in Part II (pp. XIX–XX) add that, as Suter notes (Bibl. Math., 1904, pp. 84–85) after Steinschneider, that *Introductorium Ptolemaei ad artem sphaericam* is the Arabic version of the *Elements of astronomy* (Εἰσαγωγὴ εἰς τὰ φαινόμενα) commonly ascribed to Geminus (Γεμῖνος). The passage cited by al-Bīrūnī is on p. 286 of Manitius' edition, from the Latin version that fills a gap of the Greek text at I, 9: «and the two turnings [i.e. the tropics] and the two equalities occur, according to the opinion of the Greek judicial astrologers, in the first degrees of these signs, and according to the opinion of the Chaldaeans they occur in the eighth degrees of these signs. … And there is no difference between the Greeks and the Chaldaeans except in the division of the signs, since the first points of the signs do not coincide for them, but among the Chaldaeans they precede by 8 degrees. So the point of the summer tropic is, according to the opinion of the Greeks, in the first part of Cancer, and according to the opinion of the Chaldaeans in its eighth part. And the matter runs likewise in the other points». So in al-Bīrūnī's text edited by Sachau, for وحُكِي, «and [the Chaldaeans] are said», one must read وحَكَى, «and [Ptolemy] relates». The passage cited by al-Masʿūdī is put together from Geminus XVI, 19 and 21–24. By these findings Fr. Boll, *Sphaera*, p. 339, note 5, is to be corrected: «the question remains open whether they [the Babylonians] put the cardinal points of the year in the middle of the signs or, as Eudoxus did in his calendrical writings, on the eighth degree, or rather day. The latter is asserted by Albirūni, appealing to a passage of Ptolemy not found so far (cf. Cumont, Textes et monuments de Mithra, I, 92, 1). … Astrological practice indeed speaks for Eudoxus' having had the division κατὰ μέσα τὰ ζῴδια also from the Babylonians». A note adds that Geminus lived in the first century BC; Manitius (*Gemini Elementa astronomiae*, Leipzig 1898, pp. 237–252) shows that the book known to us is only an epitome, made in the fourth or fifth century, of the work ἐξήγησις Γεμίνου τῶν Ποσειδωνίου περὶ μετεώρων, and judges (p. XX) that the Greek codex used by the Arabic translator, whose version Gerard of Cremona rendered into Latin, must have been far older than ours.

and in the book on the motions of the sun that Ibrāhīm ibn Sinān[1] composed».

Al-Battānī, p. 127, l. 14, puts those astrologers *before* Hipparchus; Letronne and Martin show by strong arguments that they flourished without doubt *after* Hipparchus, but perhaps before Ptolemy. Another indication of this is perhaps given by the amount of precession they took, namely 1 degree in 80 years, or 45″ a year[2], which differs not a little from Ptolemy's annual precession of 36″. For if Hipparchus' tropical year is compared with his sidereal year, an annual precession of 46″.807 is deduced, or a motion of 1° in about 77 years; and those astrologers seem to me to have taken this Hipparchian precession and to have lessened it by a wholly imperceptible amount, to make the computations simpler.

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Thābit ibn Qurra, who in the letter mentioned earlier (p. 298) shows himself uncertain of the true motion of the fixed stars because of the many discrepancies between the Greek and the Arabic observations, in a later (as it seems) little book *on the motion of the eighth sphere* puts forward an ingenious doctrine of trepidation and mentions this chapter of al-Battānī[3]: «And that is why Ptolemy received that the fixed stars revolve one degree in 100 years, inasmuch as he counted their longitudes from the equator of the day, and made it necessary that they are not fixed; so he thought that they move with a continuous motion in the order of the signs … And the observers found the motion of the fixed stars to be one degree in 66 years, and they excepted, according to Ptolemy and Abrachis [= Hipparchus], in the southern part near its end in the south; so there was slowness of the variation[4]. Then, after Ptolemy, the head of Aries approached the equator of the day and cut it, and came to be in the northern part, and the motion was swift. Yet the exception according to Ptolemy did not happen as it should have. Therefore Albategnj doubted about it, and said: I see that this exception is not according to the proportions in swiftness and slowness. So there was a motion in the sphere that we do not know and do not comprehend; and let whoever comes after us consider and verify it, as we have done. Then, after this speech of his, he returned to what was worthier, more fitting and more probable along the path of estimation, so that his work might be good and he might have accomplished something [?]. We, however, shall mark this motion and its various positions, by the blessings of God».[a]

I 303

Lastly I shall give the words of the Hispano-Jewish Abraham bar Ḥiyya[5] that concern this chapter of al-Battānī. He says that «the sages of India, all the inhabitants of the land of the Romans and

[1] A grandson of the celebrated Thābit ibn Qurra; he died, aged 38, in the month of Muḥarram 335 (August 946); Suter, p. 53, no. 113.

[2] Not 40″, as Letronne, p. 110, and Martin think; I have spoken of this error on p. 206.

[3] Vatican codex Lat. 4275, f. 84v, col. II; another Latin version is in Vatican codex Lat. 4083, where this passage is on f. 55v. — See Delambre, p. 74, who used a Paris codex.

[4] A very obscure passage. In the version of codex 4083: «But others who considered this motion after Ptolemy found that it contains [?] the space of one degree in 66 years. Whence at last they corrected Ptolemy and Abracis in what they had constructed [?] about the southern part, about the end of the southern part. For they found the motion to be slowed».

[5] Ch. X, pp. 196–200 of the Hebrew text (a Latin version of this chapter is lacking). — On that book see p. 287.

[a] Nallino's Addenda et emendanda (Part I, pp. LXXVIII–LXXIX) give better readings of this passage from three other codices (described in the preface, p. LXVIII), leaving aside the copyists' evident errors. After the amounts of the obliquity of the ecliptic according to the Indians, Ptolemy and the astronomers of the caliph al-Maʾmūn, Thābit goes on: «And it happens to this motion described above that it is found of slow variation and of swift variation; and that is because, when the head of Aries is 90 degrees of its small circle away from the equator of the day, on the two sides of north and south, the access will then be of swift motion. And so the observers found; and that is why …», followed by the passage given here. In it the codices A, B and F read «found» (*reperit*) for «received» (*recepit*); then come «since» (*quoniam*) for «inasmuch as» (*quatenus*), and *moverentur* for *moventur*; the words left out at the dots are «and he excepted, according to Abrachis, about one day in the time of the year in about 300 years» (A and V omit «according to»); A and B rightly read *aequatori*, «approached the equator»; for «Yet the exception … did not happen» the true reading is «so the exception … necessarily happened» (*necessario igitur facta est*); A, B and F better read «I see that this exception does not proceed according to», and the true reading is then «the proportion»; the next sentence reads «If, then, there is a motion in the sphere that we do not know and do not comprehend, then let whoever comes after us consider and verify it», A, B and F rightly having «then» (*tunc*) for «and» (*et*); and A, B and F read «his operation» (*operatio*) and confirm the reading «had accomplished» (*perfecisset*).

the ancients among the sages of the Chaldaeans held only this opinion of the fixed stars: that the stars do not run through the whole heaven, but move only eight degrees of the zodiac forward and back, to the east and to the west. The cause of this motion is the pole of the ecliptic, which revolves to the east and the west on a small circle [whose diameter is] 8° of the zodiac. That pole completes the [whole] circle in 1600 years; because of [this] going round of the pole on the circle some learned men imagined that the fixed stars go round the whole heaven, since that secret was not manifest to them and they did not know the revolution of the pole of the ecliptic, by which the fixed stars move from west to east for 800 years, and afterwards are brought back again westward to their former place after 1600 complete years. … These are the things that the Indians tell in defence of their opinion. — Ptolemy mentioned this motion [of trepidation], but would neither reject it nor approve it. — The sages of Ishmael [i.e. the Arabs] who lived in the time of [the caliph] al-Ma'mūn gathered their strength to overturn the arguments of the Indians. Among them al-Battānī, to refute the Indian opinion, said that the ancients who lived before Ptolemy observed the positions of those stars and set them down in their books; that 420 years later Ptolemy came, determined their positions and found between his observations and those of the ancients a difference of 4° 1/5; whence he deduced that their motion completes 1° in 100 years. About 720 years after Ptolemy, al-Battānī came; he gave himself to these matters, and between his own investigation and Ptolemy's he found a difference of about 11°, from which he deduced that their advance had been 1° 1/2 in 100 years. So it has been found that from the time of the ancients to al-Battānī's age, in 1140 years, the fixed stars advanced 15° 1/5; whereas according to the pronouncements of the Indians it should have been only 11° 36′, if we take the beginning of the computation from the advance, or 3° 36′, if from the return or from the middle point of the direct motion. Similarly, between Ptolemy and al-Battānī only 7° 1/5 eastward should have been found to add to Ptolemy's place, if we begin from the return. And we may begin indifferently from the middle point of the advance or from the middle of the return. — A great argument against them is that they said that the fixed stars advance and return 8 degrees, whereas according to al-Battānī's observation they have advanced 15° from the place that they held at the time of the observation of the ancients. The opinions of the Indians, then, are overturned and demolished, if what al-Battānī relates is true».

III. — Al-Battānī names only *ancient* astrologers, although, as has already appeared from what we have said, there was no lack among the Arabs of men who professed that doctrine. They had mostly received it from the Indians[1]: so Ḥabash in his first book (not in the later ones), in which, as Ibn al-Qifṭī attests, he followed the computations of the Indian book *al-Sindhind* (i.e. *Siddhānta*); so Muḥammad ibn Ismāʿīl al-Tanūkhī[2], who, to acquire perfect knowledge, «travelled through various regions and visited India, whence he brought foreign notions of astronomy, such as the motion of advance and return [i.e. trepidation] and others»[3]; so finally Ibn al-Adamī[4], who, in a book published after his death by a pupil in the year 308 H (beginning 22 May 920), «computed the motions of the stars by the method of [the book] *al-Sindhind*, and told some things about the motion of advance and return of the heavenly sphere that no one had set out before him. Before the appearance of this book things were heard about that motion that could neither be understood nor reduced to a fixed rule; but when this book came out, the form of that strange motion could be understood, and this was the reason why [many] considered that doctrine attentively. Ṣāʿid ibn al-Ḥasan al-Andalusī[5], judge of Toledo, says: After reading this book, things about that motion became clear to me that I think have not yet become clear to

[1] Already in 499 Āryabhaṭa professed trepidation; most of the later Indians followed it, few a continuous precession. See Colebrooke's work already cited on p. 293, Whitney's notes on the *Sūrya-Siddhānta* III, 12 (pp. 244–48), and Th. H. Martin, ch. VI, § 5, pp. 482–90. They took the arc of the oscillation variously, some 48°, some 54°, and so on.

[2] Of uncertain date, but before 1250; Suter, *Die Mathematiker*, pp. 196–97, no. 489.

[3] Ibn al-Qifṭī in Casiri, *Bibliotheca Arabico-Hispana Escurialensis*, Madrid 1760–70, vol. I, pp. 429–30, and in Sédillot's introduction to the text of Ulugh Beg, p. LXXXVIII.

[4] On him, Suter, p. 44, no. 82.

[5] Suter, pp. 103–07, no. 244; he died in the month of Shawwāl 462 H (July–August 1070).

anyone, and I perceived in it causes that I have set out in my book entitled *The correction of the motions of the stars*»[1].

Some Arabs also came to the hypothesis of trepidation by the Greek road; among them Thābit ibn Qurra, who, as we saw on pp. 298 and 301–02, built a new theory[2], chiefly on the words of Theon and the uncertainty of the old observations of the motion, so as to explain in some way the variations of the obliquity of the ecliptic, the various amounts of the length of the year and the discrepancies in the motion of the fixed stars. Alpetragius (see above, p. 299) cites the authority of the fictitious and most ancient philosopher Hermes, whose apocryphal or pseudepigraphic works the Arabs had translated from the Greek; finally Augustinus Ricci says that he learned from his teacher Abraham Zacut[3] that the Arabic doctrine of trepidation rests on two observations of the stars, which Hermes had made 1985 years before the epoch of Ptolemy's catalogue and by which the longitudes of the stars had been found 7 degrees greater than Ptolemy's longitudes[4].

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In the fourth century H (the tenth AD) Abū Jaʿfar al-Khāzin[5] and Ibrāhīm ibn Sinān, as we saw on p. 301, followed the doctrine of trepidation. But after this age the astronomers who flourished in the eastern part of the caliphate, from Egypt to farthest India, rejected trepidation[6]. Not so, however, in western Africa and in Spain, where Arzachel (al-Zarqālī), Ṣāʿid al-Andalusī, Alpetragius and Abū ʿAlī al-Ḥasan (Aboul Hhassan) tried to explain by the doctrine of trepidation the discrepancies between the most ancient observations and the Arabic ones. From their writings the Latin *Alfonsine Tables* (of the year 1251) and many European astronomers down to the middle of the sixteenth century received it.

b) — p. 127, ll. 12–13.

The tropical year of $365^d\ 6^h\ 12^m$ that those astrologers used.

I never find it mentioned in the Greek writers; it does not differ from the year that al-Battānī, in chapter XXVII, ascribes to «certain Egyptians and Babylonians», on which I have spoken at length on pp. 204–209. It is likely that those Egyptians and Babylonians are none other than these astrologers, whom al-Battānī, wrongly indeed, puts before Hipparchus. Nor must it be forgotten that the Orientals often ascribe the invention of the doctrine of trepidation to Hermes Trismegistus and to the most ancient Chaldaeans and Egyptians, as appears from the passages of al-Bīrūnī, Alpetragius, Abraham bar Ḥiyya and Ricius cited in note *a*. This is perhaps not false, if for the most ancient Chaldaeans and Egyptians we put the Chaldaeans and Egyptians who lived after Hipparchus.

[1] Ibn al-Qifṭī in Casiri, vol. I, p. 430, and Sédillot, p. XXXV; I have completed the Arabic text from a codex of the Library of the Lincei, f. 92r–v. [*Translator:* The print has «complevi teuns codice», with a misprinted word.]

[2] On it, Delambre, pp. 73–75.

[3] A Hispano-Jew who lived at the end of the fifteenth century and drew his astronomical learning from Arabic and Hebrew books. He took these things on trepidation from ch. VI of the second treatise of the book *On the foundation of the world* (*Yĕsōd hā-ʿōlām*), which Isaac Israeli (Yiṣḥāq ben Yōsēph Yiçrĕʾēlī) had written at Toledo in 1310.

[4] A. Ricius, *De motu octavae sphaerae, opus mathematica atque philosophia plenum*, Paris 1521, p. 23; the passage was already pointed out by I. B. Riccioli, *Almagestum novum*, vol. I, p. 166 (but from the earlier edition of Ricius, of 1513), and by Th. H. Martin, ch. V, p. 448. — Ricci was born at Bodincomagus, or Casale Sant'Evasio, today Casale, in Piedmont, in northern Italy.

[5] Not to be confused with al-Khāzinī.

[6] See especially Naṣīr al-Dīn al-Ṭūsī, *al-Tadhkira*, treatise II, ch. IV, and his commentator Qāḍīzāde, Laurentian codex, ff. 39v–40v.

On chapter 53

a) — p. 128, l. 12 ff.

I have used the name *revolution of the years* because it is in common use among all our astrologers; but it would have been clearer to say «on the conversion of civil years into tropical years». For a very frequent problem in astrology is this: someone was born in the year 1100 of Dhū l-Qarnayn, on the 26th of the month Aylūl, at $17^h\ 45^m$, the sun being at the degree x of the ecliptic; one seeks the moment at which, in the year 1126 (i.e. at the completion of the twenty-sixth year of his age, or in the course of the twenty-seventh), the sun will return to the same point. This computation is called the *revolution of the years of the nativities* (*taḥwīl sinī al-mawālīd*); but if it concerns not the life of a man but the vicissitudes of cities, peoples, nations and so on, for which the computation is mostly drawn from the great conjunctions of the planets, it is called the *revolution of the years of the world* (*taḥwīl sinī al-ʿālam*). — Al-Battānī treats the same subject more briefly in appendix E, p. 148.

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The mathematical computation is very easy, and it is set out in all the astronomical books of the Arabs. But the astrological predictions to be drawn from it make up an enormous and extremely obscure and intricate matter; so very many of the Arabs[1] composed special books on the revolutions of the years. The Arabs themselves knew a book of the philosopher Hermes on the revolutions of the nativities, translated from the Greek, whose Latin version, made from the Arabic, has been printed several times. The Greek texts, however, are unknown today, as Bouché-Leclercq attests, *L'astrologie grecque*, Paris 1899, p. 506, note 2: «Je suppose que ce sont ces renouvellements et contrôles du thème de géniture que l'annotateur de Paul d'Alexandrie appelle des ἀντιγενέσεις. A part cette mention problématique et de basse époque[2], je ne trouve pas trace du système dans les textes grecs publiés jusqu'ici» («I suppose that these renewals and checks of the theme of nativity are what the annotator of Paul of Alexandria calls ἀντιγενέσεις. Apart from this problematic mention, of a late age, I find no trace of the system in the Greek texts published so far»).

On chapter 54

a) — p. 130, l. 5.

«*Let it recede by the space of half of both bodies*».

Somewhat differently in the Greek text of the *Tetrabiblos* I, 24[3]: καὶ καθόλου δὲ συνάπτειν μὲν λέγονται τοῖς ἑπομένοις οἱ προηγούμενοι, ἀπερρυηκέναι δὲ οἱ ἑπόμενοι τῶν προηγουμένων, ἔφ' ὅσων ἂν μὴ μακρὸν ᾖ τὸ μεταξὺ αὐτῶν διάστημα, «and in general the preceding are said to apply to the following, and the following to flow away from the preceding, *provided that the interval between them is not long*».

Likewise Pseudo-Proclus and Porphyry do not determine the size of the interval[4].

[1] Suter mentions ten of them even before al-Battānī's time: nos. 7, 8, 17, 26, 30, 34, 45, 53, 62, 68. The books of Messahala (Māshāʾallāh) and of Albumasar (Abū Maʿshar) exist in printed Latin versions.

[2] For the annotator published by Andreas Schato belongs to the Middle Ages. [*Translator:* The print has «adnotator Schaton»; Nallino's Addenda et emendanda (Part II, p. XXVII) correct it to «the annotator published by Andreas Schato» (Pauli Alexandrini Εἰσαγωγὴ εἰς τὴν ἀποτελεσματικήν, *Rudimenta in doctrinam de praedictis natalitiis*, Wittenberg 1586).]

[3] I, 21 in the Latin version from the Greek printed at Frankfurt in 1622 (*C. Ptolemaei de praedictionibus astronomicis, cui titulum fecerunt Quadripartitum, libri IV.* Editio posterior). The author of this most elegant version is unknown.

[4] *In Cl. Ptolemaei Quadripartitum enarrator ignoti nominis, quem tamen Proclum fuisse quidam existimant. Item Porphyrii philosophi introductio in Ptolemaei opus de effectibus astrorum.* [Greek and Latin]. Basel 1559, p. 11 of Pseudo-Proclus, 187 of Porphyry.

But in the two Latin versions of the *Tetrabiblos* that were made from the Arabic and printed with the version of the commentary of ʿAlī ibn Riḍwān[1], the matter is set out as in al-Battānī; for the one has: «When the planets go so as to approach others, it is called their application, unless they are far removed from them, passing *the half of the bodies of the two planets*, whether the application is bodily or by way of one of the other figures mentioned»; the other[2]: «… that the planets that precede, and that seek the conjunction of others, are in their alictisal[3], unless the distance between them is great, exceeding *the two halves of the bodies of the two planets*, whether the alictisal is from a conjunction or from one of the said figures».

b) — p. 130, ll. 11–13.

The force of the light in the applications.

In Ptolemy's *Tetrabiblos* III, 10 (= III, 14 of the version printed at Frankfurt) I find that Jupiter sends out its light with a ray of 12° in the applications, Venus with a ray of 8°; which agrees with al-Battānī. But the rays of the other planets are not given.

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Porphyry, loc. cit., p. 204, at the end of his work, says this: «The rays of the sun are spread over 30 parts: 15 before and 15 after. The moon 24: 12 before and 12 after. Saturn and Jupiter 18: 9 before and 9 after. Mars 16: 8 before and 8 after. Venus and Mercury 14: 7 before and as many after». The same numbers as Porphyry's are in Sahl ibn Bishr[4], ʿAlī ibn Abī l-Rijāl[5] and the commentary of ʿAlī ibn Riḍwān (Haly Heben Rodan) on ch. 24 of book I of the *Tetrabiblos*, who then adds: «For this is the opinion of the whole body of the astrologers. Ptolemy, however, holds that it is otherwise, according to what appears in his words[6]; for he holds that the whole body of the sun, before and behind, is 17 degrees, and of the moon 12 and a half, and of Saturn 10, and of Jupiter 12, of Mars 7 and a half, of Venus 8, of Mercury 7». But these numbers too, which ʿAlī ibn Riḍwān thinks Ptolemy's, do not agree with al-Battānī's, except those of Jupiter, Venus and Mercury.[a]

By what method such rays of light were computed is not clear; perhaps they are mere trifles of the astrologers.

c) — p. 131, ll. 15–19.

[«*The quadrature, which for all the stars …*». Let a planet P look at another planet in *quartile* aspect, that is, be 90 degrees of a great circle away from it. The two rays that P will cast on the

[1] *Liber Quadripartiti Ptolemaei, Centiloquium eiusdem*, etc., Venice 1519.

[2] This too is printed at the end of the work entitled *Iulii Firmici Materni iunioris Siculi Astronomicõn libri VIII per Nic. Prucknerum nuper ab innumeris mendis vindicati*, Basel 1533 (or 1551, as in other copies).

[3] It is the Arabic word *al-ittiṣāl*, that is, the *applicatio* of the Latin astrologers.

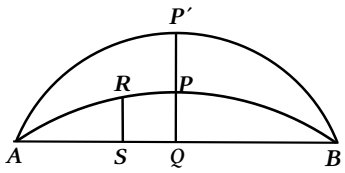
[4] Zahel, *de interrogationibus*, f. 112r (printed at Venice in 1519, together with the commentary of Haly Heben Rodan on the Quadripartitum). — The author flourished in the first half of the third century H (the ninth AD); Suter, pp. 15–16, no. 26; M. Steinschneider in the *Zeitschrift der deutschen morgenländischen Gesellschaft*, vol. LIV, 1900, pp. 39–46.

[5] *Albohazen Haly filii Abenragel de iudiciis astrorum libri octo, doctorum aliquot virorum opera in Latinum sermonem conversi, postremo a barbarie vindicati et puritati linguae donati per Ant. Stupam*, Basel 1571, book I, ch. 6, p. 20.

[6] Where these words of Ptolemy are I do not know; I do not find them in the *Tetrabiblos* (apart from the passage cited before) or in the apocryphal *Centiloquium*, or Καρπός.

[a] Nallino's Addenda et emendanda (Part I, p. LXXIX) add that al-Būnī (I, 28), after giving the sizes of the seven planets, says: «One of our sages said that the body (*jirm*) of the sun is 15 degrees before and as many behind, the body of the moon 12 before and as many behind, of Mars 8 before and as many behind, of Venus 7 before and as many behind, and so also of Mercury. God the Most High knows these things best». Saturn and Jupiter seem to have been left out through the editor's carelessness; but al-Būnī plainly did not understand his source, which set out the force of the rays in the applications.

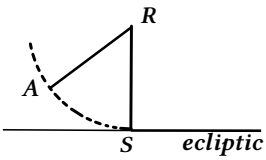
ecliptic will then be the two arcs AP, PB of a great circle, each of which contains 90°. Let the planet be projected on the ecliptic at Q by the perpendicular arc PQ (which will plainly be the latitude of the planet); then the points A, B will be the poles of the arc PQ, so that also AQ = QB = 90°. The same would happen if the planet were at P′, and the arc P′A = 90°. Hence it is clear that «*the quadrature … is always the same, and never differs from 90°, whether the latitude of the planet is small or great*». Al-Battānī explains this further by comparing the great circle whose poles are A and B with the equator of the sphere, which always divides into equal quadrants all the circles passing through both poles; «*and this is plainly seen on the sphere, on whose poles the circles fall*».



Now let the planet R be in *sextile aspect* with another heavenly body; it will cast on the ecliptic the ray RA = 60°. Let the arc RS be drawn perpendicular to the ecliptic, which will be the latitude of the planet; the projection of the *sextile* aspect, or radiation, on the ecliptic will be the arc AS, plainly *less* than 60°. But if the planet R were in *trine* aspect, the projection of the arc RB = 120° would be the arc SB, always *greater* than 120°, since AS + SB must always be taken equal to 180°. — Schiaparelli.]

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Ibn Yūnus writes in his introduction (in Caussin, p. 73): «Likewise false is what Muḥammad ibn Jābir ibn Sinān al-Battānī says of the place on which the rays of the stars fall when the stars have latitude; for according to his principles a star with a latitude of 60 degrees has no sextile. Which is monstrous (فظيع)». Indeed, if the radiation RA is set equal to the latitude RS = 60°, the radiation will touch the ecliptic only at S, which is the place of the planet's longitude; so the projection of the rays will be nil. But to avoid this the radiation must be understood in a way different from al-Battānī's; what Ibn Yūnus thought of the projection of the rays we do not know. Moreover it must be noticed that the astrologers use the radiation of the planets only, not that of the fixed stars (p. 131, l. 4); so the latitudes are always much smaller than 60°.



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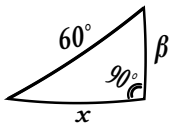
d) — p. 131, ll. 20–38.

The computation of the projection of the rays.

I. — *Al-Battānī's computation.* The projection of the quartile radiation on the ecliptic is always 90°, as the author said before and as has been explained in note *c*. The projection of the trine radiation = 180° – the projection of the sextile; so it is enough to know the projection of the sextile ray.

Al-Battānī's formulas appear at the very first sight extremely long and intricate; how they could be proved is not understood. If they are worked out analytically by trigonometric functions they

become horrible monsters. No one will be able to believe that al-Battānī wrote this[1]; the hand of a most foolish interpolator is recognized again.



What al-Battānī's solution was is indicated by the words of ll. 22–23. The problem is like the first of chapter XXVI: given a star of latitude β and a star standing on the ecliptic, to compute the distance of the one from the other on the great circle passing through both. But now the distance (60°) is already given; the unknown quantity x is the quantity L – L′ of chapter XXVI.

According to the names set there (p. 39), we shall now have

$$\text{I side} = x, \qquad \text{II side} = 2 \operatorname{arc}\left(\sin\frac{x}{2} \times \frac{\cos\beta}{R}\right), \qquad \text{III side} = \operatorname{crd}\beta,$$

and the rule of that chapter will become

$$\operatorname{crd}\text{ I side} \times \operatorname{crd}\text{ II side} = \text{square of crd } 60^\circ - \text{square of III side},$$

that is[2]

$$\operatorname{crd} x \times \operatorname{crd} x \frac{\cos\beta}{R} = \operatorname{crd}^2 60^\circ - \operatorname{crd}^2\beta, \qquad \operatorname{crd}^2 x = R\frac{\operatorname{crd}^2 60^\circ - \operatorname{crd}^2\beta}{\cos\beta},$$

whence finally

$$(\alpha) \qquad \operatorname{crd} x = \sqrt{R\frac{\operatorname{crd}^2 60 - \operatorname{crd}^2\beta}{\cos\beta}},$$

which I think was al-Battānī's formula. And indeed, to restore this rule it is enough to leave out a little of al-Battānī's text, to add far less, and to keep *the rest untouched*; so, marking by some dots what is to be left out and is now left out, I do not doubt that al-Battānī wrote in this way: «Find in the tables of sines the cosine of the latitude [and divide it by the radius; what comes out will now be] the whole chord of the second side, which we have mentioned only in the chapter on the diagonals of quadrilaterals and in this very chapter in which we are. Keep this chord of the second side. Then find the chord of the latitude, which is the doubled sine of half the latitude. … Take the square of the chord of the latitude from 3600^p, that is, from the square of the chord of the sextile aspect …; divide the remainder by [the kept chord] of the second side [and take the square root of the quotient]; … the chord sought will come out, which you convert into an arc as whole chords are converted into arcs, etc.».

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To make ourselves more certain that this is al-Battānī's true rule, let us reduce it to the direct solution of the spherical triangle. Since

$$\operatorname{crd}^2 x = 4\sin^2\frac{x}{2} = 4\frac{R^2 - R\cos x}{2} = 2R^2 - 2R\cos x\,,$$

the formula found above

$$\operatorname{crd}^2 x = R\frac{\operatorname{crd}^2 60^\circ - \operatorname{crd}^2\beta}{\cos\beta}$$

[1] If they came from al-Battānī, they would be censured by Ibn Yūnus in the passage cited on p. 306.

[2] In the notes, p. 202, $\frac{\cos\beta}{2R}$ is wrongly printed in three places for $\frac{\cos\beta}{R}$.

can also be written

$$2R^2 - 2R\cos x = R\,\frac{(2R^2 - 2R\cos 60^\circ) - (2R^2 - 2R\cos\beta)}{\cos\beta}\,.$$

Dividing everything by 2*R*,

$$R - \cos x = R\,\frac{(R - \cos 60^\circ) - (R - \cos\beta)}{\cos\beta} = -\,\frac{R\cos 60^\circ}{\cos\beta} + R,$$

whence

$$\cos x = R\,\frac{\cos 60^\circ}{\cos\beta}\,.$$

This rule, *as answering to al-Battānī's method*, is given by the *Libro de las taulas alfonsies*, in ch. LII, which is entitled «on projecting the rays of the planets on the ecliptic, according to the opinion of al-Battānī»[1]. The solution there is this: «If the planet has latitude, divide 1800[2] by the sine of the complement of the planet's latitude; convert what comes out into an arc in the way in which sines are converted into arcs; the arc that comes out [take from 90°, and what remains][3] will be the argument of the sextile». — Similarly, nothing but the solution of the spherical triangle[4] by various instruments is meant in other places of the same *Libros del saber de astronomia*: vol. II, p. 65 (*Libro de las armellas*, book II, ch. XLVI): «de saber ell echamiento de los rayos segund la opinion del Bateni» («on knowing the casting of the rays according to the opinion of al-Battānī»); vol. III, p. 123 (*Libro de la lámina universal*, book II, part IV, ch. LXIV): «de saber cuemo se deuen fazer los echamientos de los rayos de qual planeta quier. segund la oppinion de Albateni» («on knowing how the castings of the rays of any planet are to be made, according to the opinion of Albateni»); finally vol. III, pp. 206–207 (*Segunda parte del libro de la açafeha*, ch. LIX): «de los echamientos de los rayos segund la oppinion de Albateni» («on the castings of the rays according to the opinion of Albateni»). This last passage is the version of the book of Arzachel, or al-Zarqālī, on the construction and use of the saphaea; the Arabic text is in Escorial codex 962 (Casiri no. 957), ff. 51r–52r.

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In the same way Henry Bate[5] understood al-Battānī's rules very well; after mentioning two other methods of projection, he has a chapter «on the aspects according to the community of the astrologers and especially according to Albategni», where he says: «There is also a third way of taking the aspects … and it is by the equal degrees[6] that are of the zodiac, taken uniformly; and Albategni affirms that this is true in the fifty-fourth chapter of his book, unless the star whose aspect is sought has some latitude from the ecliptic line. … But if the star has some latitude … a

[1] «Del fazer echamiento de los rayos de los planetas en el zodiacho. segund oppinion de Al Bateny» («On making the casting of the rays of the planets on the zodiac, according to the opinion of Al Bateny»). — In *Libros del saber de astronomia del Rey D. Alfonso X de Castilla, copilados, anotados y comentados por D. Manuel Rico y Sinobas*, Madrid 1863–67, vol. IV, p. 181.

[2] Wrongly 180 in the edition; for $1800 = 60^p \times 30^p = R\cos 60^\circ$.

[3] The words in brackets I have restored by a safe conjecture. The Spanish text of the whole passage is: «Pues si ouier el planeta ladeza del zodiacho. parte todauía CLXXX [sic!] sobrel signo del complimiento de la ladeza del planeta. et lo que le salier archéalo archeamiento de los signos. et lo que fuer ell archo [add: míngualo de 90. et lo que fincar] aquel será argumento del sextil» («So if the planet has latitude from the zodiac, always divide CLXXX [sic!] by the sine of the complement of the latitude of the planet; and what comes out, turn into an arc as sines are turned into arcs; and what the arc is [add: take it from 90, and what remains], that will be the argument of the sextile»). This amount is to be added to or subtracted from the longitude of the planet; the place of the left radiation or the place of the right radiation on the ecliptic will come out.

[4] Whose sides are the latitude of the planet, the arc of 60° and the radiation sought.

[5] *Magistralis compositio astrolabij Hanrici* [sic] *Bate ad petitionem fratris Vuilhelmi de Morbeka ordinis predicatorum domini pape penitentiarij et capellani*, ff. d2v–d3r (at the end of the work: Abraham Iudei *de nativitatibus*, Venice, Erh. Ratdolt, 1485). — The author finished the little work at Mechelen (Malines, in Belgium) on the fifth day before the Ides of October 1274; on it see also M. Steinschneider in the *Orientalistische Litteratur-Zeitung*, 4th year, 15 November 1901, cols. 444–45.

[6] On which I have spoken on p. 28, note 3.

great circle must be imagined, passing through the star whose aspect we seek, to cut the ecliptic line into two halves, in such a way that from one of those sections to the star there are 60 degrees of the degrees of that great circle; and the degrees of the zodiac from that same section to the degree of the zodiac in which the same star is in longitude, though it is outside in latitude, are the degrees of the zodiac that are due to the sextile aspect …». And, f. d3v, after setting out very fully the same considerations as al-Battānī (see above, p. 306) on the quartile ray, he teaches the operations to be made on the astrolabe, which plainly answer to the computation of a leg of a right spherical triangle when the other leg and the hypotenuse are known.

II. — *On the projection of the rays before al-Battānī.*

On the ἀκτινοβολία, or projection of the rays, among the Greeks Cl. Salmasius, *De annis climactericis et antiqua astrologia diatribae*, Leiden 1648, pp. 398–409, and A. Bouché-Leclercq, *L'astrologie grecque*, Paris 1899, pp. 247–51, have written. But both considered only its philosophical principles and its use in predictions; by what mathematical method, and on what circle of the heavens, the Greeks computed the rays they neglected to investigate.

Many probably computed the radiations neglecting the latitude of the planet, simply adding 60°, 90°, 120° to the longitude of the planet, or subtracting them, for the different aspects; which we know al-Qabīṣī, or Alcabitius[1], and the Jew Abraham ibn ʿEzra (Avenezra) also did.

Other Greek methods can be gathered from Arabic books. Ibn Hibintā, f. 33v, says that Hermes, in his book entitled العرض[2], takes the radiations by «equal degrees» (on which see p. 28, note 3); but Dorotheus, as the same Ibn Hibintā and Muḥammad ibn ʿUmar ibn al-Farrukhān[3] attest, counted the radiations not on the ecliptic but on the equator, taking the sextile ray to be the arc of the equator contained between the right ascension of the degree of the planet's longitude

[1] *Libellus ysagogicus Abdilazi, qui dicitur Alchabitius, ad magisterium iudiciorum astrorum, interpretatus a Ioanne Hispalensi*, Venice 1485, f. aa4, in the chapter on the aspects of the planets = *Praeclarum Alchabitii opus ad scrutanda stellarum magisteria isagogicum*, Venice 1521, f. 4r. — He lived in the fourth century H, the tenth AD; Suter, pp. 60–61, no. 132.

[2] If *al-ʿaraḍ* is read, as is probable, it means the book «of the accident» (in the philosophical sense, namely what is opposed to «substance»); if *al-ʿarḍ*, the book «of latitude». See M. Steinschneider, *Die arabischen Uebersetzungen aus dem Griechischen*, in the Zeitschrift der deutschen morgenländischen Gesellschaft, vol. L, 1896, p. 189. [*Translator:* Nallino's Addenda et emendanda (Part I, p. LXXIX) add that the title of Hermes' book is *al-ʿarḍ*, «the book of latitude», as appears from al-Nadīm, *al-Fihrist*, p. 267, who names Hermes' «first book of latitude, key of the stars» and «second book of longitude, key of the stars»; Suter's version, p. 19, has «Ueber die Breite, erster Schlüssel der Gestirne; Ueber die Länge, zweiter Schlüssel der Gestirne», but then the words *ʿarḍ* and *ṭūl* would need the article. In Ibn al-Qifṭī, ed. Lippert, p. 349, the word «of longitude» is left out in the title of the second book. In the version of ʿAlī ibn Riḍwān's commentary on the *Tetrabiblos* the book is called *liber accidentium*, *liber accidentis*, *liber de accidentibus* and *liber latitudinis*; Steinschneider collected the passages in Baldi, p. 473 (less well in the extract, p. 43). Nallino's Addenda et emendanda in Part II (p. XX) add that this book of Hermes, of which no copy had been known, survives in a codex belonging to Dr. Griffini (mentioned in vol. II, p. 235). It begins: «This is the book of Hermes, thrice great in all their knowledge, and he is preferred above all the learned …, which is called the latitude of the key of the stars; he wrote it on the revolution of the years of the world», the years of the world being known when the sun enters the first minute of Aries, when the ascendant and then the four pegs are to be set up in their degrees and minutes (the Arabic phrase answering to the Greek Τρισμέγιστος). It ends: «This book is complete, which is the book of Hermes on the revolution of the years of the world and the judgements in them, and it is from the words of the sage; the book was translated in Dhū l-Qaʿda of the year 125 of the hijra, and the copying of this book was finished on the blessed day of Friday, the seventeenth of the month Rabīʿ I of the year 1071 of the hijra of the Prophet». If the date is true, this is one of the oldest versions, made in September 743; and the Friday is by the astronomical reckoning, for by the civil one 17 Rabīʿ I is Saturday, 20 November 1660. The book is divided into unnumbered chapters (*bāb*), and differs from the *De revolutionibus nativitatum* of the philosopher Hermes in two books, translated from the Arabic and printed with Pseudo-Proclus' commentary on the *Tetrabiblos* (Basel 1559, pp. 217–279; its true author seems to have been a Muslim, on which see Th.-H. Martin, *Mémoire sur … la précession des équinoxes*, p. 449, note 1, and Bouché-Leclercq, pp. 506–507). At the end of the codex the book is called «the first book of the latitude of the key of the secrets of the stars of heaven», to be followed by «the second book, which is the longitude of the key of the secrets of the stars». Dr. Griffini himself will give a very full description of the book.]

[3] Omar, *de nativitatibus*, book I, p. 123 (with other astrological books after the edition of Firmicus already cited on p. 305, note 5). — The author lived about 850; Suter, p. 17, no. 34.

and the right ascension corresponding to (the longitude of the planet ± 60°). Hermes' method is also set out in the *Libros del saber de astronomia*[1], from which these precepts are gathered: let the degree of the equator be found that falls on the circle of position (see p. 311, note 3) drawn through the given planet; let 60° (or 90°, or 120°) be added to it or subtracted from it; the degree of the ecliptic that falls on the circle of position drawn through that sum or remainder of the equatorial arc will be the degree of the left and of the right sextile (or quartile, or trine).

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Finally Valens[2], as al-Zarqālī in the *Libros del saber de astronomia* III, 204–5 (*Segunda parte del libro de la açafeha*, ch. LVII) and Rabiçag of Toledo, ibid. II, 62 (*Libro de las armellas*, book II, ch. XLIV), attest, followed the method that the Arabs mostly ascribe to Ptolemy, of which we must now speak.[a]

In Ibn Abī Uṣaybiʿa[3], in the life of the physician and astrologer Rabban al-Ṭabarī, there is this: «Abū Maʿshar[4], asked about the projection of the rays, said much; finally he said: In the copies of the *Almagest*[5] derived from the Greek language the translators mention neither the rays nor their projection, which are found only in the copy translated by the physician Rabban al-Ṭabarī[6]. In the old copies Ptolemy's projection of the rays is not found; and neither Thābit nor Ḥunayn al-Qallūsī nor al-Kindī[7], nor any of those great translators, nor anyone of the family of Nawbakht[8] knows it». In the *Libros del saber de astronomia*, vol. II, p. 69 (*Libro de las armellas*, book II, ch. LII), these things are told, which I translate from the Spanish: «All that I have said here to have been Ptolemy's opinion on the projection of the rays and the [astrological] direction, know that I have not found it in any of Ptolemy's books, but that Abū Maʿshar and Abū Isḥāq al-Zarqālī said

[1]Vol. II, p. 65 (*Libro de las armellas*, book II, ch. XLV: «de saber ell echamiento de los rayos segund la opinion de Hermes», «on knowing the casting of the rays according to the opinion of Hermes»), vol. II, p. 308 (*Libros dell astrolábio llano*, book II, ch. VII: «de saber cuemo fazen los echamientos de los rayos segund la opinion de Hermes», «on knowing how the castings of the rays are made according to the opinion of Hermes»), and vol. III, p. 211 (*Segunda parte del libro de la açafeha*, ch. LXIV: «de saber los echamientos del rayo segund la oppinion de Hermes», «on knowing the castings of the ray according to the opinion of Hermes»).

[2]Of uncertain date; Salmasius and Bouché-Leclercq, p. XIII, think that he lived in the time of Constantine; E. Riess and W. Kroll, however, assign him to the age of Hadrian and Antoninus. [*Translator:* The print has «Reiss»; Nallino's Addenda et emendanda (Part I, p. LXXIX) correct it to Riess.]

[3] *ʿUyūn al-anbāʾ fī ṭabaqāt al-aṭibbāʾ*, ed. A. Müller, Cairo–Königsberg 1884, vol. I, pp. 308–309. F. Wüstenfeld, *Geschichte der Arabischen Aerzte und Naturforscher*, Göttingen 1840, pp. 5 and 20, already knew the passage from the manuscripts, and from him M. Steinschneider, *Zur Geschichte der Uebersetzungen aus dem Indischen*, p. 338 (in the Zeitschrift der deutschen morgenländischen Gesellschaft, vol. XXIV, 1870), also took it. But they wrongly interpreted it as «refraction» (*Brechung der Lichtstrahlen*). — The same words of Ibn Abī Uṣaybiʿa, down to «are found only in the copy translated», are in Ibn al-Qifṭī (in Casiri, *Bibliotheca Arabico-Hispana Escurialensis*, Madrid 1760–70, vol. I, p. 436).

[4]The Albumasar of the Middle Ages; he died in March 886; Suter, pp. 28–30, no. 53.

[5]*Almagest* is doubtless an error for «*Quadripartitum*»; Ḥunayn had translated only this book, nor is it likely that Rabban, an astrologer and physician, published a version of the Almagest. Moreover, a chapter on the radiations was not at all to be expected in the Almagest, whereas it would have had a most fitting place in the Tetrabiblos. What I shall say below of the chapter added to the Latin version, from the Arabic, of the Tetrabiblos confirms my conjecture. — So the correction proposed by Suter, p. 14 (Isḥāq ibn Ḥunayn for Ḥunayn), becomes needless.

[6]At the beginning of the ninth century; Suter, pp. 14–15, no. 25; M. Steinschneider, passim, suspects that he is the same man who is elsewhere called Sahl ibn Bishr.

[7]All of them lived in the ninth century; Suter, pp. 34–38 (no. 66), 21–23 (no. 44), 23–26 (no. 45).

[8]Eighth century and beginning of the ninth; Suter, pp. 3 (no. 2), 5 (no. 7), 16 (nos. 27 and 28).

[a]Nallino's Addenda et emendanda (Part I, p. LXXIX) add that al-Zarqālī, loc. cit. («et por esta obra obraua Uuelius el egiptiano en los echamientos de los rayos. et es la oppinion de Ptolomeo», «and by this work Valens the Egyptian worked in the castings of the rays, and it is the opinion of Ptolemy»), and Rabiçag, loc. cit. («et segund esto obraua en echamiento de los rayos Vellix ell egipciano. que es opinion de Ptolomeo. Et esto es lo que dixo Abumassal [Abū Maʿshar] et Abuzac Azarquiel [Abū Isḥāq al-Zarqālī]», «and according to this Valens the Egyptian worked in the casting of the rays, which is the opinion of Ptolemy; and this is what Abumassal [Abū Maʿshar] and Abuzac Azarquiel [Abū Isḥāq al-Zarqālī] said»), attest that Valens' method in the radiations does not differ from Ptolemy's; but in the list of chapters of the book *al-Bāriʿ* by Abū Naṣr al-Qummī (about 360 H, 971), in Ahlwardt, *Verzeichniss*, vol. V, p. 148b, no. 5661, there are two chapters, one «on the projection of the rays according to the method of Ptolemy», the other «on the projection of the rays according to Valens».

so. It is not impossible that it was not Ptolemy's opinion, but was [wrongly] ascribed to him; as it often happens that errors creep into translations, and that names are changed from one to another, above all in so great an interval of time, more than a thousand years, as has elapsed since Ptolemy[1]. If you understand Ptolemy's Tetrabiblos rightly, you will find that his opinion inclines to the view of Ibn Muʿādh rather than to that of the others»[2]. — Finally, in the *Liber tabularum Iahen cum regulis suis*, composed in Arabic at the end of the tenth or in the eleventh century and rendered into Latin by Gerard of Cremona in the twelfth, many errors are said to be made in some methods of projecting the rays: «for the first of it [i.e. of the error], and it is the work that is attributed to Ptolemy. And Albumasar transmitted it from him, although Abunasar is not truthful in this science, namely in the science of the form of the sphere»[3]. I 311

The passage of Albumasar, or Abū Maʿshar, as M. Steinschneider first noticed (*Lettere*, p. 15, note 3, and p. 20), is the chapter «On extracting the rays of the stars according to Ptolemy», which is found in the work entitled *Introductorium in astronomiam Albumasaris Abalachi octo continens libros partiales*, Venice 1506, book VII, ch. 7, and whose first words are these: «In extracting the rays of the stars more diversity is found among the various treatises than elsewhere. For in these we have chosen Ptolemy, whose very words are these …». — The same precepts, indeed, with the Latin wording little changed, I find after the end of the version of the commentary of ʿAlī ibn Riḍwān (Haly Heben Rodan) on the *Tetrabiblos*[4]. For some chapters are added there, which no one has noticed so far, the first of which is entitled «the science of the projection of the rays». I do not doubt that this is the passage that Ibn al-Qifṭī and Ibn Abī Uṣaybiʿa say existed only in the copy of Rabban al-Ṭabarī. The method of computation, however, is not wrongly ascribed to Ptolemy, as I shall show below.

The rules are these, when the star stands between the ascendant and the midheaven[5]:

$$\frac{\text{RA of the degree of the star} - \text{RA of the midheaven}}{\text{hour-times of the degree of the star}} = \text{hours of distance from the midheaven},$$

$$\text{arc of the ecliptic answering to } (\text{RA of the degree of the star} + 60^\circ) = \text{first radiation},$$

$$\text{arc of the ecliptic answering to } (\text{obl. asc. of the degree of the star} + 60^\circ) = \text{second radiation},$$

$$\text{radiation I} \pm \frac{(\text{radiation I} - \text{radiation II})}{6} \times \text{hours of distance} = \text{corrected radiation},$$

that is, the longitude of the place of the ecliptic on which the left sextile radiation falls. If the right one is sought, −60° is to be put for +60°; if the quartile or trine radiations are to be found, 90° or 120° are to be put for 60.

[1] «Et quanto yo aquí dixe … sabe que non lo fallé en ninguno de los libros de Ptolomeo. fora ende que lo dixeron assí Abumassar. et Abuçac el Zarquiel. Et bien puede seer que non fué opinion de Ptolomeo. Et apusiérongelo cuemo contesce muchas ueces. que cae yerro en el trasladar. et mudan el nombre de uno á otro. quanto mas en tan gran sazon que a que fué Ptolomeo. que mas a de mil annos».

[2] «Et quando entendieres bien el quarto partido de Ptolomeo. entendrás que mas tira su opinion á lo que dixo Aben-Mohat que non á otro». — Muḥammad Ibn Muʿādh al-Jayyānī lived in the tenth or the eleventh century; he is not to be confused with the man mentioned by Suter, no. 213, as Suter himself acknowledged in the *Nachträge und Berichtigungen* published in the *Abhandlungen zur Geschichte der mathematischen Wissenschaften*, vol. XIV.

[3] In M. Steinschneider, *Lettere a D. B. Boncompagni: II. Intorno al libro «Saraceni cuiusdam de eris» ed al libro «Tabule Jahen»*, Rome 1863, p. 13. — Abunasar is a corruption of Albumasar; «the science of the form of the sphere» is a literal version of the Arabic *ʿilm al-hayʾa*, which means «astronomy».

[4] *Liber Quadripartiti Ptolemei; Centiloquium eiusdem*, etc., Venice 1519, ff. 94v–95r.

[5] Johannes de Saxonia sets out all the cases at length in the commentary on Alchabitius that he wrote in 1331 (see above, p. 309, note 1); ff. ff7r–gg2v in the edition of 1485, 30r–32v in the edition of 1521. It is strange that at the end of the exposition he says: «I have not seen this way set out or put in any book». — If for the hours of distance the simple distance = RA of the degree of the star − RA of the midheaven is taken, the divisor in the formula of the corrected radiation will be not 6 but the semidiurnal arc of the degree of the star.

The computation is much easier when the planet is on the meridian; for then the radiation is given by the degrees of the ecliptic answering to (RA of the planet ± 60°). — When it stands in the ascendant:

$$\text{radiation} = \text{ecl. degrees answering to } (\text{obl. asc. of the degree of the planet} \pm 60^\circ).$$

Finally, when it is in the west, or descendant:

$$\text{radiation} = \text{ecl. degrees answering to } (\text{obl. asc. of the nadir of the degree} \pm 60^\circ) + 180^\circ.$$

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Plainly this «Ptolemaic» method is the same by which the astrologers determine the *directions* or *profections*, as I shall show in the notes on ch. LV; so they take the amount of the radiation on either side of the star's longitude to be the arc of the ecliptic contained between the degree of the planet's longitude and the circle of position[1] drawn through the point of the equator that is 60° or 90° or 120° before or behind the right ascension of the planet. But in this way (which Ḥabash also has, f. 114r–v, though he does not mention Ptolemy) the star is supposed to have no latitude and to stand on the ecliptic. If the latitude is not thought negligible, the computation must be changed a little, as in the direction of stars that have latitude; that is, instead of the degree of the ecliptic that numbers the longitude of the star, the degree culminating together with the given star is to be used when right ascensions are computed, or the degree rising with the star when oblique ascensions are to be taken. But the astrologers seem mostly to have neglected this in the radiations; the error arising from it is often very small, since only planets, not fixed stars, are considered in this problem.

Many Arab astronomers teach this «method of Ptolemy» together with others: so Abū Naṣr al-Qummī[2], al-Bīrūnī in the *Masʿūdic Canon*[3], and Ulugh Beg, pp. 157–58 of the text, 209–210 of the translation, who carries out the computation directly by circles of position and does not neglect the latitude of the star.

Not without reason do the Arabs call this computation of the radiations «the method of Ptolemy»; for in the *Tetrabiblos* III, 10 (= III, 15 of the Frankfurt version) I find this written on the directions: «When the significator has been found, two ways of taking [the directions] are to be used: the one in the order of the signs, *which is called the projection of the rays* (ἀκτινοβολία), namely when the significator is in the eastern places, from the midheaven down to the ascendant; the other both in the order of the signs and against it, according to the horary reckoning, namely when the significator is in the places declining from the midheaven». So I think that Bouché-Leclercq, p. 248, note 1, wrongly wrote: «La théorie de l'ἀκτινοβολία est probablement postérieure à Ptolémée» («The theory of the ἀκτινοβολία is probably later than Ptolemy»).

Al-Zarqālī, in the passage already mentioned on p. 308, says: «The [ancient] Persians are said to make the projections of the rays only in the right sphere; which serves only those who live under the equinoctial circle. So we make no mention of it». These words probably mean that the ancient Persians added 60°, 90°, 120° to the right ascension of the star, or subtracted them, and took the arc of the ecliptic answering to the right ascension so increased or diminished to be the degree of longitude at which the left or right radiation falls on the ecliptic. This method[4] is also mentioned, with others, by Johannes de Monteregio, or Regiomontanus, in Problem XXXI of the

[1] The circles of position are the great circles that pass through the common intersections of the meridian and the horizon; their poles are on the prime vertical.

[2] As appears from the list of chapters in W. Ahlwardt, *Verzeichniss der arabischen Handschriften der k. Bibliothek zu Berlin*, Berlin 1887–99, vol. V, p. 148, no. 5661. — Al-Qummī lived in the fourth century H (the tenth AD); Suter, pp. 74–75, no. 174.

[3] Ahlwardt, V, p. 154, no. 5667.

[4] Johannes de Saxonia, in the *Commentarius in Alchabitium*, loc. cit., says of it: «And in this way Ptolemy bids us take the aspects [i.e. the radiations] in the 60th proposition of the Centiloquium, where he speaks of the causes of the critical days». This proposition of the *Centiloquium* of Pseudo-Ptolemy is: «In the sick, look at the critical days and

Tabulae directionum profectionumque (of 1475); and on the same, if I rightly understand what Ibn Hibintā says, f. 34r–v, Muḥammad ibn Mūsā al-Khwārizmī[1] relied in constructing the tables of radiations that Ibn Hibintā gives, ff. 35r–37v. It is known that al-Khwārizmī followed chiefly the methods of the Indians in his astronomical work.

Plainly all these computations of the projection, which seem to take their origin from the Greeks, the Indians and the Persians, rest on principles much less rational than al-Battānī's, and appear much less elegant[2].

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Special booklets on the radiations were written before al-Battānī's age by Messahala, or Māshāʾallāh, one of the most ancient Arab astrologers, by al-Kindī (whose book has also been translated into Latin), and by ʿAbd Allāh ibn Masrūr al-Naṣrānī, a pupil of Abū Maʿshar.

On chapter 55

a) — p. 131, l. 45 – p. 132, l. 9.

The Arabs call *pegs*, as I have said more fully on p. 248, the four points of the ecliptic that are determined by the intersection of the horizon and the meridian: namely the ascendant, the midheaven, the descendant or setting, and the lower midheaven or peg of the earth. The arc of the equator, counted from the beginning of Aries, that passes through the meridian together with a given star is the star's right ascension; the arc of the equator that rises or sets together with a given star is the star's oblique ascension or descension, or its ascension or descension in the clime. The author has already dealt with these arcs in the preceding chapters. It remains for us to know the arcs of the equator that rise together with a given star at any other place of the sphere, that is, the arc of the equator, or the hour angle, that a planet or a given degree of the ecliptic must run through to pass the circle of position[3] of another planet or degree[4].

This problem, useless in astronomy, al-Battānī, p. 134, last line, says is necessary to the astrologers for computing the *directions*[5]; for to direct the point A to B is to find after how much time of the equator A reaches the circle of position drawn through B.

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the passage of the moon (ἐποχήν) through the angles of the figure of sixteen sides; for where you find those angles well affected, it will go well with the sick man; on the contrary, badly, if you find them afflicted». To deduce a theory of the radiations from this seems to me not easy.

[1] He died after the year 232 H (beginning 27 August 846), as I have shown in my *Al-Ḫuwârizmî e il suo rifacimento della Geografia di Tolomeo* (Memorie della R. Accademia dei Lincei, Classe di scienze morali, series V, vol. 2, part 1, 1894–95, p. 9). — Nöldeke, in the *Zeitschrift der deutschen morgenländischen Gesellschaft*, vol. LV, 1901, p. 683, note 1, wrongly confuses our author with Muḥammad ibn Mūsā ibn Shākir. The same confusion is in the indexes to the *Annales* of al-Ṭabarī; the two men have been fused into one. The passages that concern al-Khwārizmī are III, 551, 937, 1085; the others (and the epithet *al-munajjim*) belong to Muḥammad ibn Mūsā ibn Shākir.

[2] Our mathematical astrologers of the sixteenth century understood the projections in the same way as al-Battānī, and carried out the computation directly by the formula $\cos x = \frac{\cos 60^\circ}{\cos \beta}$. See Magini in Delambre, p. 486, and *Instrumentum sinuum seu primi mobilis nuper a Petro Apiano inventum*, Nuremberg 1541, proposition XCIV.

[3] See p. 311, note 3.

[4] *Libros del saber de astronomia*, vol. II, pp. 65–66 (*Libro de las armellas*, book II, ch. XLVII): «De saber los sobimientos de los signos sobre los cercos que son semeiantes al cerco dell orizon quando fuer el signo entre los ángulos» («On knowing the risings of the signs on the circles that are like the circle of the horizon, when the sign is between the angles»); ibid. III, 207–208 (*Segunda parte del libro de la açafeha*, ch. LX): «De saber los sobimientos de los signos entre los ángulos. — Estos sobimientos son los tiempos del passage de los signos. et de sus grados. por los çercos mayores que passan por los dos polos del çerco de orient et de occident. et los semeiables á los çercos de los orizones» («On knowing the risings of the signs between the angles. — These risings are the times of the passage of the signs and of their degrees through the great circles that pass through the two poles of the circle of east and west, and of those like the circles of the horizons»).

[5] Some consider the projections of the rays in the same way, as I have shown on pp. 311–12.

That the reason for this computation may be better understood, I shall set out some definitions. Comparing the life of men and of nations to a horse let out from the gate of the starting-pen towards the turning-post of the circus, the ancient astrologers said that life is let go (ἀφίεσθαι), more or less strongly, from some point of the zodiac, which was called τόπος ἀφετικός, «the place letting go», the *aphetic*, *prorogative* or *hylegiacal place*; that the aphetic places are the midheaven, the ascendant, the beginning of the eleventh house (ἀγαθοδαίμων), the beginning of the seventh, or descendant, and the beginning of the ninth (θεός). Life let go from the aphetic place perishes, or at least is in danger, when it has come on the ecliptic to some obstacle, which was called τόπος ἀναιρετικός, «the killing, destroying place»; the degrees of the equator that answer to the interval between the aphetic place and the anaeretic place are to be converted, according to different methods, into years, which count the years of life from the birth to the unlucky event or to death.

I 314

The star which, holding the aphetic place at the time of the birth, must be taken as the ruler and governor of life, is called by the Greeks ἀφέτης, «the one letting go», or κύριος τῆς ζωῆς, «lord of life»; by the Arabs *dalīl* (plural *dalā'il*, *adillā'*, *adilla*), «the one pointing out, indicating», or, with a Persian word, *haylāj* (plural *hayālij*); by the Latins of the Middle Ages and later times *apheta, significator, prorogator, vitae gubernator, liberator, dispositor, hileg, hylech.* The aphetae are five: the sun, the moon, the ascendant, the planet that has the greatest power (οἰκοδεσποτεία) over the degree of the Part of Fortune[1], and the planet that has the greatest power over the degree of the last conjunction or opposition of the sun and the moon. An apheta has no force by itself unless it is found in an aphetic place. Which of the aphetae is to be taken in the different cases, according to Ptolemy's doctrine, may be read in V. Nabod, *Enarratio elementorum astrologiae*, Cologne 1560, pp. 383–84, 388–90, and in Bouché-Leclercq, pp. 415–19. — Lastly, the star whose body or hostile rays, meeting the significator in the profections, cut off life, or at least foretell that great dangers threaten it, was called by the Greeks ἀναιρέτης, «the killer», or μοῖρα ἀναιρετική, «the killing, destroying degree», and by the Latins of the fifteenth, sixteenth and seventeenth centuries *promissor*.

Direction, or *prorogation*, in Greek ἄφεσις, in Arabic *tasyīr* (whence *alazir, alaçir, alhacir* in our writers of the Middle Ages), is the arc of the equator that rises or sets together with the arc of the ecliptic lying between the significator and the promissor, with respect to the circle of position on which one of them lies. The astrologers distinguish two directions; the one is called *direct direction*, or in the succession of the signs (ἄφεσις εἰς τὰ ἑπόμενα τῶν ζωδίων, from west to east), in which the promissor (ἀναιρέτης), or «following place» (ἑπόμενος τόπος), is thought to be carried to the «preceding place» (προηγούμενος τόπος), that is, to the circle of position of the significator or apheta. The other is called *converse direction*, or against the succession of the signs (ἄφεσις εἰς τὰ προηγούμενα τῶν ζωδίων), in which the significator is thought to be carried to the circle of the promissor. In both, when the nativities of men are concerned, each degree of the equator signifies a year; in the revolutions of the years of the world and of the nativities a degree signifies a day, and so on.

Profection is the uniform motion of the significator in the succession of the signs. In the *annual* profection of a nativity one sign of the zodiac is given to each solar year; so if the given nativity or geniture has the sign of Aries in the ascendant, the second year will have the sign of Taurus, the third the sign of Gemini, and so on. In the profection that they call *monthly* the solar year is divided into 13 equal parts, or profectional months, to each of which one sign is given; so the sign of the annual profection is the sign of the first month of the same year. In the *daily* profection, again, the profectional month is subdivided into 13 equal parts, to each of which one sign is given.

[1] Ὁ κλῆρος τῆς τύχης, in Arabic *sahm al-sa'āda*, is the place of the ecliptic that is as far from the ascendant, in the succession of the signs, as the interval of the zodiac between the sun and the moon.

The doctrine of direction, as far as its astrological use is concerned, is extremely obscure and intricate; besides, different astrologers rely on different principles in drawing forecasts, whence arises the greatest confusion, which Cl. Salmasius himself, *De annis climactericis et antiqua astrologia diatribae*, Leiden 1648, pp. 264–398, 407–439, could not avoid. Bouché-Leclercq, pp. 411–419[1], may be consulted, who, however, as always, says nothing or little of value on the mathematical computations.

b) — p. 132, l. 10 – p. 134, l. 46.

I 315

The computation of the ascensions of the signs outside the four pegs is the computation of the *direction of the significator* to the promissor and of the promissor to the significator. — So Henry Bate (see p. 308, note 5), who had read al-Battānī's work attentively, rightly says, f. d2r: «Now the science of the ascensions of the signs between the quarters of the sky is the same as the science of the directions: so that, when the one is known, the other is known too».

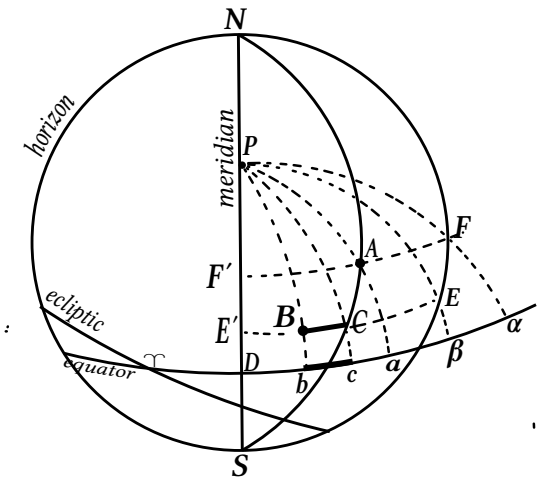
Al-Battānī treated these ascensions in the spaces between the pegs in a special little book, as I have said sufficiently in the preface. The reason for publishing it was doubtless the ignorance of the astrologers, who, little versed in mathematical matters, erred badly in the computations of the directions. For some perhaps thought that the direction is the difference of the right ascensions of the significator and the promissor, which is true only when the degree to which the direction is made occupies the upper meridian (the peg of the midheaven) or the lower (the peg of the earth)[2]; some took the difference of the oblique ascensions or descensions, which, however, is not true unless the degree to which the direction is made stands in the ascendant or the descendant (setting)[3]; others, finally, like Ptolemy himself, though they mixed right and oblique ascensions well, neglected the latitude of the planets.

Let A be the «earlier place», namely the significator planet or apheta, to which, in the direct direction, the «following place» B (namely the promissor or anaereta) must be directed in the succession of the signs. Let both stand outside the pegs; for if B holds one of the pegs, the problem is reduced to the difference of right or oblique ascensions, as I have just said. — Let NAS be the circle of position drawn through A, P the pole of the equator; we must find how many degrees the arc *bc* of the equator contains, which B runs through to reach C.

[1]P. 411, note 1: «Ce n'est pas sans hésitation que je m'aventure dans les ténèbres de la théorie aphétique, ténèbres épaissies par la logomachie et les contradictions des astrologues anciens (surtout Vettius Valens et Paulus d'Alexandrie), des Arabes et des érudits de la Renaissance, lesquels s'acharnent à mélanger, combiner, troquer les deux systèmes principaux, empruntant la durée de la vie à l'un et la répartition du temps à l'autre» («It is not without hesitation that I venture into the darkness of the aphetic theory, a darkness thickened by the logomachy and the contradictions of the ancient astrologers (above all Vettius Valens and Paul of Alexandria), of the Arabs and of the scholars of the Renaissance, who are bent on mixing, combining and exchanging the two principal systems, borrowing the length of life from the one and the distribution of time from the other»).

[2]For then the circle of position of the degree to which the direction is made is the meridian itself.

[3]For the circle of position to which the planet is directed is then the horizon itself.



Al-Battānī indicates two methods of computing *bc*, of which I shall give the proof here.

I. — Let EE′, FF′ be the arcs of the parallels that B and A run through in the daily motion of the sphere from the horizon to the meridian; let circles of declination be drawn through E and F as far as the equator, produced; the arc Dβ of the equator will be the semidiurnal arc of the planet B, the arc Dα the semidiurnal arc of the planet A. Since P*b*, P*c*, P*a* are circles of declination, the arc of the direction sought, namely *bc*, will be:

$$bc = (\Upsilon a - \Upsilon b) - ca = (\text{RA } A - \text{RA } B) - ca.$$

I 316

If A were on the meridian, the arc *ca* would be reduced to nothing; for the circle of position would become the meridian itself, on which A and C would fall. If on the contrary A occupied the horizon at F, the circle of position would become the horizon itself, on which C too would fall; so the arc *ca* would reach its greatest amount, namely it would become βα, which is the difference between the semidiurnal arc of the planet A and the semidiurnal arc of B. Since the motion of the sphere is supposed uniform, the author thinks that the amount of the arc *ca* grows the more, the farther A is from the meridian; and this is what al-Battānī says on p. 132, ll. 10–36. So, *in his view*, this proportion arises[1]:

$$ca : (\text{semid. arc } A - \text{semid. arc } B) = \text{dist. of } A \text{ from the meridian} : \text{semid. arc } A,$$

whence

$$ca = (\text{semid. arc } A - \text{semid. arc } B) \times \frac{\text{distance of } A \text{ from the meridian}}{\text{semidiurnal arc } A}.$$

If we express the distance in temporal hours[2], of which the semidiurnal arc contains six, plainly

$$ca = (\text{semid. arc } A - \text{semid. arc } B) \times \frac{\text{hours of distance of } A}{6}.$$

[1] [It is plain that this theorem cannot be reduced to a geometrical reason; for, if it were true, the circle of position NAS would have to divide in one and the same ratio all the arcs of the parallels intercepted by the meridian and the horizon. The author's principle departs most from the truth for the stars that culminate low, near the horizon, whether in the northern or in the southern quarter; not to speak of the stars nearest the pole, for which al-Battānī's rule plainly falls into absurdity. But since the planets rarely exceed 25° or 30° of declination, that error does not become very great, especially in the climes of the cities of al-Raqqa, Antioch and Baghdad; it becomes even smaller in the more southern places of the earth, near the equinoctial circle. — Schiaparelli.] That Ptolemy and other Greeks already used this approximation will appear from p. 317.

[2] As the author himself points out, pp. 132–33, the temporal hours of distance from the midheaven are the difference between the right ascension of the star and the ascension of the midheaven, divided by the hour-times of the star.

From the precepts of chapter XIX it follows that

$$\text{semid. arc } A = 90^\circ + \text{difference between RA } A \text{ and obl. asc. } A,$$

$$\text{semid. arc } B = 90^\circ + \text{difference between RA } B \text{ and obl. asc. } B,$$

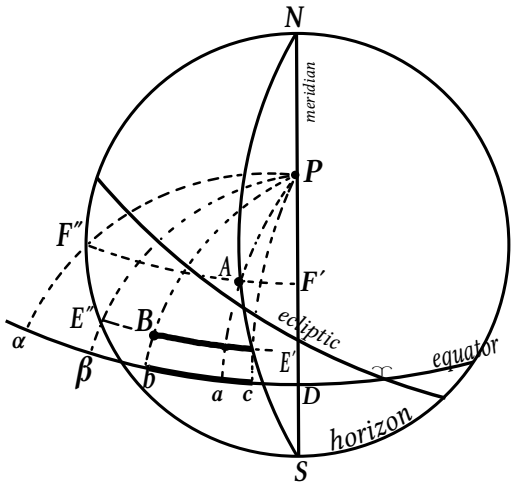
so[a]

$$(\text{semid. arc } A - \text{semid. arc } B) = (\text{obl. asc. } A - \text{obl. asc. } B) - (\text{RA } A - \text{RA } B).$$

If we denote by the letter α the difference of the right ascensions and by the letter β the difference of the oblique ones, al-Battānī's rule is[1]:

$$(\alpha) \qquad \text{direction} = \alpha \pm (\beta - \alpha) \times \frac{\text{temporal hours of distance of } A}{6},$$

where (α − β) is put for (β − α) if α > β; the second term of the second member has the sign − when α > β, the sign + when α < β.



This holds if A and B are in the eastern hemisphere. When they stand in the western, instead of the oblique ascensions of A and B the oblique ascensions of (A + 180°) and (B + 180°), that is, the descensions of A and B, are to be taken. Why this is to be done appears sufficiently from the figure placed here.

If A and B are found in two different hemispheres, one must proceed as al-Battānī teaches, p. 134, ll. 16–20: direct B to the upper or the lower meridian (according as the one or the other lies between B and A), which direction is the difference between the right ascension of B and the right ascension of the midheaven or of the peg of the earth; to this difference add the amount D*c*, namely the temporal hours of distance of A from the same meridian *multiplied by the hour-times of B*[2]; the direction sought will come out.

I 317

This method of computing agrees with the computation of the radiations that the Arabs often call Ptolemaic (pp. 309–10), except only that there the corresponding arcs of the ecliptic are taken

[1] On p. 133, l. 29, read «cognoscuntur; idest» for «cognoscuntur. Idest»; and on l. 31 correct «deprehende; si gradus» to «deprehendes. – Si gradus».

[2] This multiplication is not mentioned by the codex and Plato; but if it is neglected the precept is false, for D*a*, not D*c*, is known. See what I shall say below in proving al-Battānī's second rule.

[a] The semidiurnal arc is 90° plus the ascensional difference, which in the eastern hemisphere is the right ascension less the oblique ascension; so semid. arc A − semid. arc B = (RA A − RA B) − (obl. asc. A − obl. asc. B) = α − β, and the right side of this identity is printed with the opposite sign. The rule (α) agrees with α − β: it gives α when A is on the meridian, and β when A is on the horizon (six temporal hours), as it must.

for α and β. The same computation of the direction is set out at a little later age by Alcabitius, diff. IV, ch. VII, ff. dd3v–dd5v of the edition of 1485, or 17r–18r of the edition of 1521.

II. — Al-Battānī, p. 134, l. 21 ff., indicates *another rule*; that is, if A and B stand in the eastern hemisphere[a]:

(β) direction = (RA B − asc. of the midheaven) − hours of dist. A × hour-times B.

This can be proved thus: from the former figure it follows *approximately* (see p. 315, note 3) that

$$E'C : F'A = E'E : F'F \qquad \text{or} \qquad Dc : Da = D\beta : D\alpha,$$

that is

$$Dc : (\text{RA } A - \text{RA of the midheaven}) = \text{hour-times } B : \text{hour-times } A,$$

whence

$$Dc = \frac{\text{RA } A - \text{RA of the midheaven}}{\text{hour-times } A} \times \text{hour-times } B,$$

$$Dc = \text{temporal hours of dist. of } A \text{ from the midheaven} \times \text{hour-times } B.$$

But also

$$Dc = bc + Db = \text{direction} + (\text{RA } B - \text{asc. of the midheaven});$$

so, as al-Battānī has it:

$$\text{direction} = \text{hours of dist. } A \times \text{hour-times } B - (\text{RA } B - \text{asc. of the midheaven}).$$

The precepts to be used when the stars are in the western hemisphere, or in two different hemispheres, are proved in the same way.

At the end of the chapter[1] he says that this is Ptolemy's computation. Rightly indeed, for in the *Tetrabiblos* III, 10 (= III, 15 of the Frankfurt version) al-Battānī's *second* method is set out and illustrated by a numerical example. Why one must proceed in this way, Ptolemy explains thus: «But if the preceding place is not on these cardinal points, but falls in the intermediate spaces, it is not the times corresponding to the arcs rising or setting or culminating that will bring the following place to the preceding one, but different ones. For that place is like and the same which has a like position, and in the same degrees, both with respect to the horizon and to the meridian. This happens *approximately* (ἔγγιστα) to the places that fall on one and the same of the semicircles drawn through the common intersections of the horizon and the meridian; each of these semicircles, standing at approximately equal distances [from one another], completes one [temporal] hour. Now the equator is always inclined in the same way to any one of those semicircles, and to the horizon and to the meridian, just as if it turned about those intersections; the degrees of the ecliptic, on the contrary, change their relation to each semicircle, and pass in unequal times on either side. And in the same way, in whatever direction from the cardinal points [i.e. the pegs] the significator is distant, it passes through those semicircles in unequal time. We shall, however, lay down one way by which ... the number of equinoctial times that brings the preceding place to the following can be found by proportion».

But Ptolemy (like Alcabitius and others) always considers points of the ecliptic, as if the planets to be directed never had latitude. Al-Battānī, on the contrary, rightly bids us, when the planet deviates from the zodiac, take, instead of the degree of the ecliptic that numbers its longitude,

[1] That the last words, «*in this way the hylegs* etc.», are probably interpolated I have pointed out in the preface.

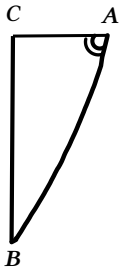
[a] This is the rule in al-Battānī's words (p. 134, ll. 24–26 of the translation, where the product is taken from the right ascensions between the midheaven and the following degree); it gives Db – Dc, whereas the proof that follows, made on the figure of p. 315, in which b lies between D and c, gives Dc – Db. Both give the size of the arc bc.

the degree culminating with the planet in the computation of the right ascensions, and the degree rising with the planet in the computation of the oblique ascensions. — Like al-Battānī, so Johannes de Saxonia in the commentary on the seventh chapter of the fourth difference of Alcabitius[1]; who says of his computation for planets having latitude: «and I have not seen [this] way set out in any book» (f. mm1v = 58v), as he had already asserted of the radiations computed in the manner of the direction (see what I set out on p. 311, note 2). He plainly did not know this chapter of al-Battānī, or had forgotten it.

On chapter 56

I 318

a) — p. 137, ll. 12–19.



To compute the *azimuth of the qibla*.

Let A be the given city (whose longitude is λ′, its latitude φ′), B Mecca (whose longitude is λ, its latitude φ), BC an arc of the terrestrial meridian passing through Mecca, and AB the arc of the great circle that joins the two cities, that is, that measures their distance. Let the arc AC of a great circle be drawn perpendicular to the meridian BC, so that it represents the line of rising and setting, from which the Arabs begin to count the azimuth. The angle BAC, then, will be the azimuth of the qibla sought.

Al-Battānī puts:

$$(\alpha) \qquad \sin \text{azimuth} = R\,\frac{\sin(\varphi' - \varphi)}{\sin \text{distance}}\,,$$

which is only approximately true, for BC cannot be exactly the difference of the latitudes φ′ − φ.

He further teaches:

$$(\beta) \qquad \sin \text{distance} = \sqrt{\sin^2(\varphi' - \varphi) + \sin^2(\lambda - \lambda')}$$

which is false in a spherical figure.

The true formula for computing the azimuth of the qibla would have been[a]:

$$\cos \text{azimuth} = \cos BAC = R\,\frac{\sin AC}{\sin AB} = R\,\frac{\sin(\lambda - \lambda')\,\cos\varphi'}{\sin \text{distance}}\,.$$

[1] Ff. ll7v–mm2r of the edition of 1485, or 57r–59r of the edition of 1521.

[a] Nallino's Addenda et emendanda (Part II, p. XXVII) correct this demonstration, as Suter (Bibl. Math., 3rd series, vol. 5, 1904, p. 88) rightly pointed out: the line of rising and setting of the given city is the great circle AC drawn through A perpendicular to the meridian of the *given city*, not to that of Mecca, and the azimuth of the qibla is the angle BAC that it makes with AB. With the meridians drawn from the pole P through A and through Mecca B, in the triangle PAB sin PAB = sin PB sin P : sin AB = sin (λ − λ′) cos φ : sin distance, and since PAB = 90° + azimuth, cos azimuth = sin (λ − λ′) cos φ : sin distance, which differs from Nallino's formula by φ in place of φ′. In the construction printed here, R sin AC : sin AB is the sine of the angle ABC at Mecca, while cos BAC = cos BC · sin AC : sin AB (to the radius 1); and since sin AC = sin (λ − λ′) cos φ′ : R, the last member should be without the factor R. For Baghdad (33.3° N, 44.4° E) and Mecca (21.42° N, 39.83° E) the corrected formula gives an azimuth of 70.04°, al-Battānī's rules (α) and (β) give 68.84°, and the angle BAC of the construction printed here is 72.56°.

The distance, however, ought to have been found by the same rule by which al-Battānī, in chapter XXVI, teaches to compute the remoteness between two stars of different latitude; this rule,

$$\text{crd distance} = \sqrt{\text{crd}(\lambda - \lambda') \frac{\cos\varphi}{R} \times \text{crd}(\lambda - \lambda') \frac{\cos\varphi'}{R} + \text{crd}^2(\varphi' - \varphi)}\,,$$

as I have shown on pp. 202–3, is very easily reduced to our

$$\text{cos distance} = \sin\varphi\ \sin\varphi' + \cos\varphi\ \cos\varphi'\ \cos(\lambda - \lambda')$$

Hardly anyone will think it credible that al-Battānī taught false formulas, when he himself had already used true ones in the other similar problems. But why he did so is not hard to understand. Knowledge of the azimuth of the qibla was needed only by the architects who had to build the mosques of the Muslims; for a niche called *miḥrāb* is set in the inner wall of the mosque, which shows the direction of the city of Mecca, towards which those who pray must always turn their face, by the precept of the Qurʾān. Since an error of a few degrees is then of very little importance, and since those architects were mostly not much versed in trigonometric computations, al-Battānī wished to give them formulas that were only approximate, but far easier than the true ones. When the difference of the longitudes is small, and the latitude of the given city not so great that the arcs of the parallel differ much from the arcs of a great circle, the error is almost imperceptible; in Syria and Mesopotamia it reaches about 3° or 4°[a]. — Besides, he had already taught the approximate formula (β) on p. 87, ll. 8–11, for computing the distance that lies «*nearly*» between the sun and the moon, but with the warning that the «*true*» solution of the problem was the one he had set out in chapter XXVI.

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For the use of Muslims who have to travel through desert regions, so that they may know to which side the face is to be turned in the canonical prayers, circles divided into a number of sectors are commonly drawn; Mecca is put in the centre, and in the sectors the names of the larger provinces are written according to their directions with respect to Mecca. Such pictures can be seen in Yāqūt, Ibn al-Wardī and Lelewel, *Géographie du moyen âge*, Brussels 1852, atlas.

b) — p. 137, l. 20 ff.

The picture of the *sundial*, very badly drawn in the codex and in Plato, has been drawn again by Schiaparelli according to the author's precepts, for the latitude 36° and with a gnomon of 12 millimetres. He marked the azimuth of the qibla, taking al-Battānī's geographical coordinates (see vol. II, pp. 37–38 and 41):

Mecca	long. 77° 53′	lat. N 21° 40′
al-Raqqa	long. 73° 15′	lat. N 36° 0′.

The error in the longitudes is not small, for according to the moderns the difference is about 50′. But the longitudes come from different sources that disagree with one another, as I shall show in vol. II, in the commentary on the geographical tables.

The construction of al-Battānī's sundial was already very well known to the Greeks; see Fr. Woepcke, *Disquisitiones archaeologico-mathematicae circa solaria veterum*, Berlin 1842.

Books on the making of sundials were written before al-Battānī's age by several authors: Muḥammad ibn ʿUmar ibn al-Farrukhān, Muḥammad ibn Mūsā al-Khwārizmī, al-Farghānī, Ḥabash, al-Ḥusayn ibn Muḥammad al-Adamī and Thābit ibn Qurra.

[a] Nallino's Addenda et emendanda (Part II, p. XXVII) put the error of al-Battānī's rougher rule at only 1°–2° in Syria and Mesopotamia; see the preceding note.

On chapter 57

a) — p. 141, l. 37.

The *pole* (in Arabic *al-quṭb*, whence *alchiloth* in our writers), or *axis* (in Arabic *al-miḥwar*, whence *almehuar* in our writers), or *key*, is a pin with a large head that passes through the middle of the alidade, the plates, the rete and the mother of the astrolabe. Near its point, which comes out of the back of the astrolabe, it has a hole, into which a small pin of copper or bronze is put, to hold and fasten the parts of the instrument pierced by the pole; because of its shape this is called the *little horse* (*al-faras*, «the horse», whence *alphorat* or *alferath* in our writers).



Some distinguish the pole from the axis, calling *pole* the hole made in the middle of the alidade, the plates, the rete and the mother, and *axis*, on the other hand, the pin that passes through these parts of the instrument[1]; but that al-Battānī makes no such distinction is plain from this very passage and from p. 144, l. 1.

b) — pp. 139–141.

Description of the *armillary sphere*, provided with five rings[2].

I 320

This instrument, most ingeniously and elegantly constructed, brings together all the uses of two Greek instruments, namely the *spherical astrolabe*[3] and the *solid sphere*[4]. Al-Battānī was perhaps the first to invent and make it, and he could not lack the skill of craftsmen; for in that age the best instruments of all were made chiefly in the city of Ḥarrān, from which al-Battānī came. The author of the book *al-Fihrist*[5] has handed down to us the names of the chief craftsmen, among whom al-Battānī's father too is perhaps to be counted[6].

[1] *Libros del saber de astronomia*, vol. II, p. 264 (*Libro dell astrolábio llano*, book II, ch. V): «Al forado que está en medio dell alhidada. et de la madre. et de las otras tablas. et de la red. llaman *polo*. et nombran *almehuar* al clauo que entra en estos forados sobredichos et traspásalas. Et dizen *cauallet* al pedaço que entra en almehuar para affirmar las partes dell astrolábio que dichas son» («The hole that is in the middle of the alidade, and of the mother, and of the other plates, and of the rete, they call *pole*; and they name *almehuar* the pin that goes into these holes mentioned above and passes through them. And they call *cauallet* the piece that goes into the almehuar to fasten the parts of the astrolabe that have been named»).

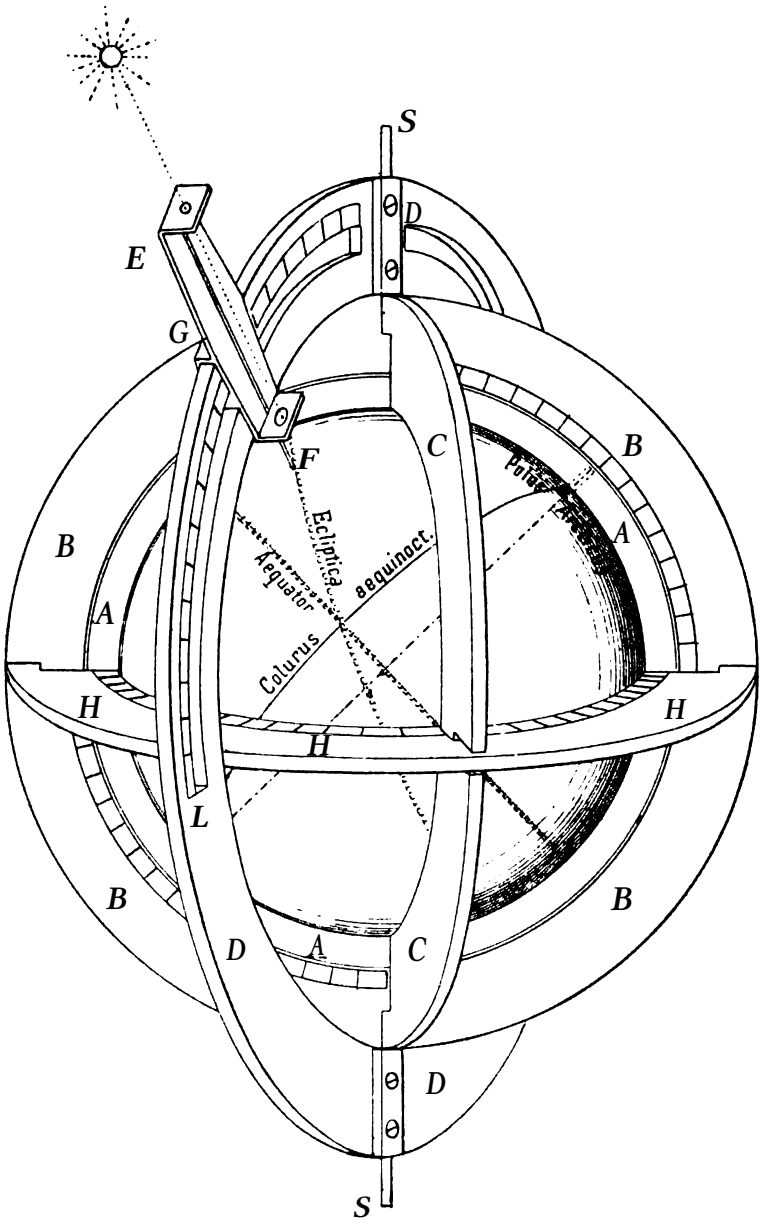
[2] The picture of the instrument placed here Schiaparelli has restored from al-Battānī's description.

[3] Ὄργανον ἀστρολάβον of Ptolemy, V, 1, of Proclus, etc. See Tannery, *Recherches sur l'histoire de l'astronomie ancienne*, pp. 70–73.

[4] Στερεὰ σφαῖρα, Ptolemy VIII, 3 (ed. Halma vol. II, pp. 92–97); it represents the celestial sphere and has three rings (those of the solstitial colure, the meridian and the horizon).

[5] *Kitāb al-Fihrist*, p. 285; translated into German by H. Suter, *Das Mathematiker-Verzeichniss im Fihrist*, in *Abhandlungen zur Geschichte der Mathematik*, vol. VI, Leipzig 1892, pp. 41–42 (= Supplement zum Bd. 37 der Zeitschrift für Mathematik und Physik).

[6] See what I have said in the preface.



I 321

Since the description of the instrument in al-Battānī's text is somewhat obscure, I think it worth while to set it out much more briefly and clearly; my words are best illustrated by the picture that Schiaparelli most kindly drew[a]. — But first I ask that in the text, vol. III, p. 210, l. 2, some words left out in the printing be restored: after الى هذا الموضع, in place of the word فوصفنا, put واتينا على كلّ ما يحتاج ان يُقَدَّمَ ذكرُهُ بقدر الطاقة فنرى ان نختِم في هذا الموضع بوصفنا. So the translation, vol. I, p. 139, l. 20[b], is to be corrected thus: «Since in this book we have come to this place, and have brought, as far as [our] strength allowed, everything whose mention had to come first, it seemed good to us to conclude [the book] at this place by describing the instrument», etc.

The *celestial globe*, hung at the two poles of the equator by means of two axles, stands in the smallest of all the rings, which represents the *movable meridian* AAA; this meridian turns in its own plane within the *fixed meridian* BBB (whose rim bears the numbers of the degrees), so that the instrument can be adjusted to any terrestrial latitude. The two meridians are of the same

[a]The words in the picture, Aequator, Ecliptica, Colurus aequinoct. and Polus, are part of the drawing: equator, ecliptic, equinoctial colure and pole.

[b]The print has «l. 10», here and in the Addenda (Part I, p. LXXII); the sentence is on p. 139, l. 20 of the translation.

thickness and closely fitted to each other, so that they seem to make only one ring. The fixed meridian is joined unchangeably to two rings, HHH and CC, which represent the *horizon* and the *prime vertical*; the outer diameters of these three rings are equal, whence a single spherical system arises, in which two spikes or pins are fixed along the lines that show the zenith and the nadir, making two poles. In this way an axis is produced about which the whole system turns within the *largest of the rings*, DDD; this, in turn, turns freely about the same axis SS, represents any vertical circle, and gives the count of the degrees on its rim. Its office is to carry the *index of the solar rays* E, which, by means of two pinnules, can be set in the direction of the rays, as the figure shows.

[To the index, moreover, is joined a needle F, which can be pushed towards the centre of the machine until it touches the globe and marks on it the direction of the sun, or drawn back so as not to hinder the rotation of the inner rings with respect to the largest, outer one. The index of the rays, finally, is carried by a slider G, or movable attachment, which moves in a wide circular slot L of the largest ring; so it can be raised above the horizon at will, according to the various altitudes of the sun.

The whole instrument, by means of two pins or axles, a lower and an upper one, is fitted to any stand of wood or iron, as al-Battānī indicates. If it is not too heavy, however, it can be hung by a ring joined to the upper axle and carried in the hand; then the axis stays vertical through the very weight of the instrument. But in this case I think it necessary to balance the index of the rays with another piece of the same weight, which moves in a slot of the largest ring made at the azimuth opposite the index itself. This double way of supporting the armillary sphere, either on a fixed stand or by hand, al-Battānī indicates, but somewhat confusedly.

All the circles of the sphere are found in this instrument, so that all the problems that require the use of an armillary sphere can be solved, as al-Battānī shows by some examples on p. 142. — Schiaparelli.]

c) — pp. 143–144.

The *triquetrum*, or *parallactic rules*.

The name of the instrument in al-Battānī is *al-ʿiḍāda al-ṭawīla*, «the long alidade», or *al-ʿiḍādatāni al-ṭawīlatāni*, «the two long alidades»; among other Arabs it is *dhāt al-shuʿbatayn*, «[the instrument] provided with two branches», with which the two movable rods or rules can be compared. Barhebraeus, *Astronomy*, p. 205 of the text, renders the Arabic name in Syriac ܐܘܪܓܢܘܢ ܕܬܪܝܢ ܦܫܚܝ ܥܘܣܝܢ *ōrganōn trayānāy peshḥā*, «the instrument provided with two forked branches» (less rightly *instrument à deux fentes*, «instrument with two slits», in Nau's French translation, p. 180).

I 322

It differs from Ptolemy's parallactic rules in this, that in al-Battānī the degrees are inscribed on the rule HM, in Ptolemy on the rule FH, to whose division HM must then be referred. According to R. Wolf, *Handbuch der Astronomie*, vol. II, p. 26, § 333, Regiomontanus was the first to mark the division of the degrees on the rule HM[1]; it is most probable that Regiomontanus learned it from al-Battānī's book.

[1] I think in the work entitled *Joh. de Monteregio et G. Purbachii Epitome, continens propositiones et annotationes, quibus totum Almagestum … dilucida et brevi doctrina declaratur et exponitur*, Basel 1543, book V, ch. III, p. 95 («regulas Ptolemaei fabricare», «to make Ptolemy's rules»).

On appendix A

a) — p. 145.

General remarks on the appendices A–H.

These appendices, which bear no number in the Arabic text, are wholly missing in Plato, both in the printed editions and in the manuscripts. There is, however, no reason to doubt that they are genuine (except D and probably F). For:

1. As I have pointed out in their places in the translation, they always refer to al-Battānī's genuine tables.

2. The astronomical elements and the Arabic language do not differ from those of the other parts of the book.

3. Appendices A, B and C are a necessary complement of chapter XXXII, at whose end the author had promised them[1]; without them the use of the tables of vol. II, pp. 7–8, cannot be understood.

4. Appendix E, necessary for understanding the tables of vol. II, pp. 187–88, had already been announced at the end of chapter LIII (p. 129) in these words: «… we have made tables, whose use we have explained *in the last of the chapters after the end of the book*, so that easily, etc.». Plato leaves out the words printed in italics, so that the sense is not understood.

5. Likewise appendix H had been announced in chapter XLII (p. 94, ll. 10–11) in these words: «… after adding to the motion of the moon, or taking from it, the seconds that you find under the difference between the sun and the moon, in the way that we have taught in that chapter *near those tables*». Plato too renders this: «the way that we have taught in that very chapter in those very tables».

From this last passage and from the end of chapter XXXII just mentioned it is clear that al-Battānī placed some of these appendices near the very tables whose use they were to illustrate[2], and that the copyists later brought them all down to the end of the chapters. If this is so, it is easy to understand why they bear no number, and why Plato, who had left out the tables, neglected them too.[a]

b) — p. 145, l. 8 ff.

I. — When the «signs» are sought in the table of the collected years alone, al-Battānī bids us always add one to the sign found there; but not so when they are taken from the table of the single years alone, or from both together.

I 323

In the interval of one common year the signs increase by 4 units, in the interval of 30 years by 5 units. Since, according to the Arab astronomers, the epoch of the hijra was a fifth weekday

[1] On p. 71, ll. 4–5, the translation should be corrected thus: «… of the months we have set down; how one must work with them we have explained near those tables, so that, if God wills, …». Likewise Plato: «for we have set down tables by which the beginnings of the months may be known, whose working is explained in them, so that the knowledge that we need may always be easy» (where the editions have *opus huius*, but the Vatican codex rightly *opus habemus*, «we need»).

[2] See also the title of appendix C, p. 146: «Chapter on finding by *these* tables the eras of the Arabs, etc.».

[a] Nallino's Addenda et emendanda (Part I, p. LXXIX) add that indeed the Spanish version (which also lacks these appendices) has, f. 63v, between the tables of the parallax of the moon in the climes and the tables of the mean motions of the planets, a *Capitulo de saber quando es la planeta endereçada y quando es estacionaria y quando es retrograda* («Chapter on knowing when the planet is direct, when it is stationary and when it is retrograde»), probably the spurious appendix D of the Arabic codex.

(Thursday), the sign of the first year of the hijra is 5; so the sign 1 is supposed to have answered to a fictitious year 0.

After 210 complete Arab years the signs of the years always return to their former order; so, when the first 210 (or 7 × 30) are known, all the rest are known. Therefore al-Battānī could make only two tables, in one of which the first thirty years from the hijra are set out, in the other the first seven multiples of thirty.

A year exceeding 30 is represented by the sum of two quantities, one of which is 30 or $x \times 30$, the other 0 or a number less than 30. The term 30 or $x \times 30$ is an interval of years counted from a fictitious year 0, to which the sign 0 answers; the term less than 30, on the other hand, is counted from the hijra, or from a fictitious year 0 whose sign is 1. So the sign of the thirtieth year of the hijra is greater by one than the sign of the last year of the thirty-year cycle taken by itself.

This was the reason why al-Battānī *lessened the signs by one* in the table of the collected years; the collected years are thus regarded as any intervals of years to be added to the years set out singly. But if the collected years indicate the interval of time that has elapsed since the hijra, their signs must plainly be increased by one, as al-Battānī teaches.

That in doing this al-Battānī followed a method other than that of the authors of the *Tested Tables* is attested by al-Khāzinī, Vatican codex, f. 110v.

II. — In this place and in the table of vol. II, p. 7, the Escorial codex counts among the leap years those that hold the places ... VIII, ... XIX, ... XXVII in the thirty-year cycle. In both places I have restored ... VII, ... XVIII, ... XXVI, as almost all the Arab astronomers have, and as the Muslims have always accepted in civil use; for the numbers of the codex plainly come from a most inept glossator, who worked so foolishly that he did not notice that his corrections conflict both with the very signs of the years in the table of vol. II, p. 7, and with all the tables of vol. II, pp. 9–18, 20 and 25!

The glossator perhaps borrowed his order of intercalation from the method of the astronomer Ḥabash, who put 8, 11, ... 19, ... 27 for 7, 10, ... 18, ... 26. Besides the Berlin codex of Ḥabash, where all the heavenly motions are computed in Arab years according to such a method of intercalation, see what al-Bīrūnī says of Ḥabash, *Chronology*, pp. 197–98 of the text, 178–80 of the translation (and Sachau's notes, p. 416).

On appendix B

a) — pp. 145–146.

That the era of Dhū l-Qarnayn, or of the Seleucids, begins according to al-Battānī on 1 September (Aylūl), not on 1 October, 312 BC, I have pointed out already in note *a* on chapter XXXII (p. 242) and elsewhere.

The table of vol. II, p. 8, contains only the first 28 years, for after 28 years have elapsed the signs return to their former order. But why al-Battānī lessened the signs of the years and the months by one I do not understand; for afterwards one must add one again to the sign taken from the table. It would have been much more convenient to put the signs in the table without lessening them.

Al-Battānī's rule, moreover, namely always to add a year to the given years, is false. It would be true if the Roman years were always common; for since in the interval of a common year the signs increase by one, to look up the year $x + 1$ in the table is the same as to increase by one the sign found for the year x. But in a leap year the signs increase by two units; so al-Battānī's precept becomes false whenever the given year x is a leap year, for the sign of the year $x + 1$ is

then the sign of the year x increased by *two* units. The true rule was always to enter the table with the given year x, and to add one to the corresponding *sign*.

On appendix D

I 324

a) — p. 147.

[No one will ever believe that this whole chapter, a monument of gross ignorance, is al-Battānī's work, especially since it conflicts with the true precepts of chapter XLVI (p. 114). The interpolator foolishly misinterpreted the meaning and the use of the tables of the stations of the planets, which are in the *Almagest* XII, 7 (ed. Halma vol. II, pp. 354–55) and, with a slight change in the numbers of Mars, in al-Battānī, vol. II, pp. 138–39. The foundation of the tables is this: since the centre of the epicycle moves on the circumference of a circle eccentric with respect to the earth, the ratio of the radius of the epicycle to its geocentric distance varies at the various places of the eccentric; it is greatest at the perigee, least at the apogee. So the places and the epochs of the stations cannot be computed on one and the same numerical basis at different places of the eccentric, but particular elements must be taken for each amount of longitude on the eccentric; the stations answer to points of the epicycle that differ from one another, and to different values of the anomaly. These values Ptolemy gives in the tables for all the positions of the centre of the epicycle on the eccentric.

The interpolator, who perhaps did not know all this, interpreted the tables absurdly. Ptolemy says that Saturn, when the centre of the epicycle is at the apogee of the eccentric, makes its first station at 112° 45′ of anomaly, the second at 247° 15′; but when the centre of the epicycle stands at the perigee, the first at 115° 29′, the second at 244° 31′. By dividing these numbers by the daily motion of the anomaly, the numbers of days are found that pass from the epoch of the apogee on the epicycle to the epoch of the first and of the second station. — Our interpolator, on the contrary, says that Saturn is direct from the apogee of the epicycle up to the anomaly 112° 45′, whatever the position of the planet in the zodiac; then that it stays from the anomaly 112° 45′ to 115° 29′; thence retrogrades up to the anomaly 244° 31′; again stays up to the anomaly 247° 15′; and finally advances with direct motion up to 360°, that is, until it returns to the apogee of the epicycle. Nor did he notice how many absurdities arise from this interpretation, especially for Mars, to which he dares to give a station of 25 1/3 days! — Schiaparelli.]

That the chapter is not genuine appears also from the use of the word *al-kātib* to designate Mercury; see what I have said on p. 291, in note *d* on chapter L.

I could correct the numbers only *to the mind of the foolish interpolator*. So I derived the degrees of the anomaly from the table of the stations, vol. II, pp. 138–39, and, dividing them by the mean daily motion of the anomaly of each planet, I found the numbers of days.

On appendix E

a) — p. 148.

It explains the use of the tables of vol. II, pp. 187–88, as he had promised to do in chapter LIII, p. 129, ll. 9–11.

On appendix F

I 325

a) — p. 149.

The appendix seems spurious; it sets out more briefly what the author had already taught in chapter LV, leaving out the considerations that concerned planets having latitude (pp. 133 and 317). I have explained everything sufficiently in the notes on that chapter, pp. 313–17.

In the title of the appendix it would have been better to translate: «and know that this direction is what Ptolemy explains in the *direction* of the hyleg». In the text, vol. III, p. 224, l. 11, I accepted تسيّر (i.e. walking, travelling through), which I rendered by the word «profection»; but not rightly. For never, as far as I know, did the Arabs so call the *profection*; when they distinguish it from the direction (تسيير), they designate it by the word *intihāʾ* (انتهاء). I had been misled by the faulty writing of the codex in that place, تسير for تسيير (i.e. direction), and by the words of the book *Mafātīḥ al-ʿulūm*, p. 230 (s.v. هيلاج): وذلك انها تسيُّر الى السعود والنحوس, in which تسيُّر is doubtless an error of the editor for تُسَيَّر, «they are directed». — In what follows, l. 14 of the text (twice), it would have been better to read تُسَيِّر for تَسير; so I think that in the translation (ll. 15 and 16) only «dirigere» («to direct») should be restored for «directionem incipere» («to begin the direction») and «directionem pervenire» («the direction to arrive»). The sense, however, is not changed.

The word *juwīrāst*, which Ptolemy is here said to use in the *Tetrabiblos* when he speaks of the profection of the significator, certainly shows a Persian origin[1]; but I do not find it either in the Greek text or in the two Latin versions of the Arabic redaction of the *Tetrabiblos*[2]. I have met the word in only one of the Arabic books printed so far, namely in the *Mafātīḥ al-ʿulūm*, which is a small lexicon explaining many technical words of all the sciences, composed in northern Persia between the years 363 and 381 H (973–991): «The ascensions of the right sphere are the amounts of the celestial equator that rise with the arcs of the ecliptic on the equinoctial line; in Persian [they are called] *juwī rāst*[3]».

I 326

In the Persian lexicons the word is missing. — In the list of chapters of the astronomical book entitled *al-zīj al-jāmiʿ*[4], composed in northern Persia in the second half of the fourth century H (the tenth AD) by Kūshyār ibn Labbān, I had found tables not only of ascensions «on the line of the equator» (*bi-khaṭṭ al-istiwāʾ*), that is, of right ascensions, but also of the «ascensions of the signs by *juwīrāst*» (مطالع البروج بالجويراست). At my request the distinguished Ed. Sachau, with the great kindness that is his, transcribed the tables for my use from the Berlin codex, and I saw at once that they give the *right* ascensions of all the degrees of the zodiac counted *from the beginning of Capricorn* (see pp. 163–64). — So what is said in the title of this appendix seems false, namely that the name *juwīrāst* is given both to the right and to the oblique ascensions in the astrological problem of the directions and profections.

In l. 21 and ll. 26–27, for «eamque de residuorum differentia deme; quod remanet …» («and take it from the difference of the residues; what remains …») there must be restored: «*illudque*

[1] *Rāst* = straight, direct, directly; the first part of the word plainly comes from the verb *justan* (present stem *jūy*), «to seek, search out». — The corruption of the codex, الحواراسب, arose from the archetype's having الجوى راست; the copyist, not knowing the word, took the first part of the name for an Arabic word ending in *alif maqṣūra*, and so, in the Maghribi way of writing, put ا for ى.

[2] Both have been printed with the commentary of ʿAlī ibn Riḍwān (Haly Heben Rodan), Venice 1493 and 1519. The former, printed several times, was made from the Arabic by Plato of Tivoli and added to the other by the Venetian editors; the latter, like ʿAlī's commentary, was made from the Spanish by Aegidius de Tebaldis by order of Alfonso X.

[3] *Liber Mafâtih al-ʿolûm auctore Abù ʿAbdallàh Mohammed ibn Ahmed al-Khowàrezmì, edidit G. van Vloten*, Leiden 1895, p. 219.

[4] The list is in Ahlwardt, *Verzeichniss der arabischen Handschriften der k. Bibliothek zu Berlin*, Berlin 1887–99, vol. V, no. 5751, p. 206a. — On the author, Suter, pp. 83–84, no. 192.

nomen sume e residuorum differentia; quod fuerit …» («*and take that name from the difference of the residues; what it is …*»).

The words «appella» (*sammi*, «name») and «nomen» (*ism*, «name») have here a technical sense that never occurs either in the lexicons or in the other parts of this book, but is found among the Arab mathematicians of western Africa and Spain[1], such as al-Ḥaṣṣār[2], Ibn al-Bannāʾ[3] and al-Qalaṣādī[4]. For they call *naming*, or *denomination* (*tasmiya*), the division of a smaller number by a larger, so that it is expressed by a sum of fractions whose denominators are the factors (prime or not) of the larger number[5]. Examples of it are

$$7 : 15 = \frac{2}{5} + \frac{1}{3} \cdot \frac{1}{5}, \qquad \text{which is written} \qquad \frac{2 \quad 1}{5 \quad 3},$$

and likewise

$$196 : 385 = \frac{5}{11} + \frac{4}{11} \cdot \frac{1}{7} + \frac{1}{11} \cdot \frac{1}{7} \cdot \frac{1}{5}, \qquad \text{which is written} \qquad \frac{5 \quad 4 \quad 1}{11 \quad 7 \quad 5},$$

and is called «to name 196 *from* 385». The numerator is then called «the named» (*al-musammā*), the denominator «that from which it is named» (*al-musammā minhu*). — They sometimes use the same verb in a more general sense, to indicate a fraction whose denominator is greater than its numerator; so al-Qalaṣādī, part I, ch. IV (Fez edition of 1315, p. 13 in the margin): وان بقي لك بقية اقلّ من المقسوم عليه فسَمِّها منه, «if [at the end of the division] a remainder smaller than the divisor is left to you, name it from it [the divisor]», which Woepcke rightly rendered: «faites en une fraction ayant pour dénominateur le diviseur» («make of it a fraction having the divisor as denominator»).[a]

[1] I cannot affirm, however, that others do not use it, since so few Arabic writings on arithmetic have been printed. — That the word *tasmiya* also occurs in the book *al-Maʿūna*, which Aḥmad Ibn al-Hāʾim, an Egyptian, finished in the month of Shaʿbān 791 (August 1389), appears from the list of the contents of the book in Ahlwardt, vol. V, no. 5984, p. 339b.

[2] About the twelfth century; his book was summarized in German by H. Suter, *Das Rechenbuch des Abù Zakarijà el-Ḥaṣṣàr* (Bibliotheca Mathematica herausgegeben von G. Enestrom, 3rd series, vol. II, 1901, pp. 12–40). On *tasmiya*, pp. 17–20 and 35.

[3] He died about 740 H (beginning 8 July 1339); see Suter, *Die Mathematiker*, pp. 162–64, no. 399. His summary (*talkhīṣ*) was translated into French by A. Marre, *Le Talkhys d'Ibn Albannâ publié* [sic!] *et traduit* (Atti dell'Accademia Pontificia de' Nuovi Lincei, vol. XVII, 1863–64, pp. 289–319). On *tasmiya* («dénomination»), pp. 302, 306, 310.

[4] He died in the month of Dhū l-Ḥijja 891, that is, at the end of the year 1486; Suter, *Die Mathematiker*, pp. 180–82, no. 444. His *Kashf al-asrār* was translated into French by F. Woepcke, *Traduction du traité d'arithmétique d'Aboùl Haçan Ali ben Mohammed Alkalçàdì* (Atti dell'Accademia Pontificia de' N. L., vol. XII, 1858–59, pp. 230–275, 399–438); he treats *tasmiya* in part I, chs. 6 and 8, and part II, ch. 5 (pp. 17–18, 20–21, 26 of the text lithographed at Fez in 1315 H).

[5] On this operation, H. Hankel, *Storia delle matematiche presso gli Arabi, traduzione di F. Keller* (in Bullettino di bibliografia e di storia delle scienze matematiche, vol. V, 1872, pp. 370–71); Cantor, *Vorlesungen über Geschichte der Mathematik*, vol. I, 2nd ed. (Leipzig 1894), pp. 764–65.

[a] Nallino's Addenda et emendanda (Part I, pp. LXXIX–LXXX) add that among the eastern Arabs the verb «to name» seems to be used only of fractions: so in the Algebra of ʿUmar al-Khayyām, p. 42, l. 6 of the text (p. 69 of the translation: «la partie d'un nombre est la partie dénommée d'après ce nombre, comme le tiers d'après trois», «the part of a number is the part named after that number, as the third after three»), and l. 8 («pareillement la partie du carré est la partie dénommée d'après le nombre égal à ce carré», «likewise the part of the square is the part named after the number equal to that square»); and in the *Liber algorismi de pratica arismetrice* of John of Seville (twelfth century), drawn from Arabic writings (Trattati d'aritmetica pubblicati da B. Boncompagni, Rome 1857, fasc. II), p. 49: «Although the denomination of the parts of any number can be made in infinitely many ways, according to infinitely many numbers, it pleased the Indians to make the denomination of their fractions from sixty»; there are many other examples on pp. 55–70. On the word *nisba*, «ratio», used in the same sense as *tasmiya*, see pp. 326–27. Likewise in Hebrew: in the arithmetic composed by Abraham bar Ḥiyya in the first half of the twelfth century the chapter *ḥālōq minyān ʿal minyān*, «on the division of a [larger] number by a [smaller] number», is distinguished from the chapter *lē-daʿat qeṣeb minyān mim-minyān*, «on knowing the ratio of a number to a number», that is, on dividing the smaller by the larger (in Steinschneider, *Abraham ibn Esra*, in Abhandlungen zur Geschichte der Mathematik, vol.

So the passage of l. 21 means: «divide the distance (in degrees of the equator) by the semidiurnal arc; and according to this ratio take the proportional part of the difference of the residues», that is:

$$\text{difference of the residues} \times \frac{\text{distance}}{\text{semidiurnal arc}},$$

where plainly the factor

$$\frac{\text{distance}}{\text{semidiurnal arc}} = \frac{\text{temporal hours of distance}}{6}.$$

The rule thus agrees with the precepts set out on p. 316.

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Instead of «appella» and «nomen», other technical words are used in ll. 31 and 36, «*compara*» (*insib*, «compare») and «*ratio*» (*nisba*), perhaps less remote from the usage of the mathematicians of Mesopotamia and Persia; for an Arab writer of our time, Buṭrus al-Bustānī, states that the operation that the western Arabs call *tasmiya* is called *nisba* by the Persians[1].

Because of all these unusual words and new ways of speaking, wholly foreign to al-Battānī's, it can hardly be doubted that this appendix, a superfluous summary of chapter LV, is *spurious*. Who that interpolator was I cannot judge; for, as we have seen, I have so far found the word *juwīrāst* (wrongly interpreted here) only in two Arabic-Persian writers, and the others only in Arabic-African and Spanish ones.

On appendix H

a) — p. 150, ll. 24–30.

On this appendix see what I have said above, p. 322; Schiaparelli has already explained its precepts, p. 273, in note *a* on chapter XLII.

[1] In Suter, on al-Ḥaṣṣār, p. 17, note 2; compare the fractions that the Arabs of the West themselves call «relative» (*kusūr muntasiba*) (al-Qalaṣādī, part II, introduction; Ibn al-Bannā', part II, ch. 1).

III, 1880, p. 108, who did not understand the sense of the word); and Abraham ibn ʿEzra (twelfth century) uses the word *ʿērek*, «ratio», in the same sense (ibid., p. 101, note 157, and p. 108, note 184, wrongly rendered «Proportion» by Steinschneider).

Notes on the tables

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On pp. 1–3: the tables of the foreign kings

In the title[1] I have kept the reading of the codex; but I think that al-Battānī wrote تاريخ for بتاريخ, that is, «and from him [begins] the chronology of the Almagest», as we also find it written in al-Bīrūnī, *Chronology*, p. 88 of the text, 101 of the translation. Indeed in the *Almagest* Ptolemy almost always uses the years of the era of Nabonassar.

These tables, carried down to the death of Antoninus by Ptolemy and to the capture of Constantinople by the Turks by an unknown Greek, exist in Greek and in French in the book *Chronologie de Ptolémée: table chronologique des règnes, apparition des étoiles fixes de C. Ptolémée, Théon etc.* [*publiées et*] *traduites par Halma*, Paris 1819, part 2, pp. 3–5. Down to King Diocletian the tables have also been edited excellently by Ideler, *Handbuch der mathematischen und technischen Chronologie*, Berlin 1825–26, vol. I, pp. 111–14; those of the kings of Assyria (the last of whom is Nabonadius) have been illustrated by Eb. Schrader, *Die keilinschriftliche babylonische Königsliste*, in Sitzungsberichte der k. preussischen Akademie der Wissenschaften zu Berlin, 1887, no. XXXI, pp. 579–607.

In Ptolemy and his Greek continuator the kings are divided thus: *of the Assyrians and the Medes*, Ἀσσυρίων καὶ Μηδῶν βασιλεῖς, down to the death of Nabonadius; *of the Persians*, Περσῶν βασιλεῖς, from Cyrus to Alexander of Macedon; *years of the Macedonian kings*, ἔτη βασιλέων Μακεδόνων, from Philip to Cleopatra; *of the Roman kings*, Ῥωμαίων βασιλέων, from Augustus to the end of the fourth year of Constantine; *of the Christian Roman kings*, Χριστιανῶν Ῥωμαίων βασιλέων, from the beginning of the fifth year of Constantine.

The same table as al-Battānī's, carried down to the emperor Basil, is in al-Bīrūnī, *Chronology*, pp. 88–89 and 92–96 of the text, 101 and 103–105 of the translation.

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From other sources come the tables of the kings from Augustus to Heraclius in al-Ṭabarī, *Annales*, series I, vol. II, pp. 741–44; Ibn al-Athīr, *Annales*, Cairo 1301–02 H, vol. I, pp. 139–40 (ed. Tornberg I, pp. 228–230); Ḥamza al-Iṣfahānī, *Annales*, ed. Gottwaldt, pp. 67–69; Abū l-Fidāʾ, *Taʾrīkh*, Constantinople 1286 H, vol. I, pp. 63–69; — and the tables from Augustus to Constantine Porphyrogenitus (d. 959) in Ḥamza, pp. 76–79, and Ibn al-Athīr, I, pp. 141–48 (ed. Tornberg I, pp. 230–42).

I think that al-Battānī's tables of the kings (like the geographical ones) came from a Greek archetype through some *Syro-Christian* version. A Syriac origin is betrayed by «*Bukhtanaṣṣar the third*, who is the conqueror of Jerusalem», for Ναβοκολάσσαρος[2]; by *Belshaṣṣar*[3] for Νηρικασολάσσαρος[4]; *Darius the Mede* for Ναβονάδιος; *Akhashwīros* for Ξέρξης; *Arnabā* for (Ptolemy) Λάγος; by the corruption of the name of *Alexander* [Severus] *son of Mammaea*, of which I shall speak on p. 194; and finally by the word *al-ḥanīf*, added to Julian, by which

[1] The Arabic text of these tables is in vol. III, pp. 228–31. In the Spanish version these tables are left out, except (f. 29r) the summary of the intervals, which I shall give below, p. 197.

[2] Ναβουχοδονόσωρ in George Syncellus, ed. G. Dindorf, I, p. 393.

[3] In Hebrew *Bēlshaṣṣar* בֵּלְשַׁאצַּר is the last king of the Chaldaeans (*Bēl-shar-uṣur*), the son of Nabonidus; *Bēlṭĕshaṣṣar* בַּלְטְשַׁאצַּר is the name of Daniel at the royal court of Babylonia. Both are Βαλτάσαρ in the Septuagint and ܒܠܬܫܐܨܪ in the Syriac version of the Bible (which Payne Smith, *Thesaurus Syriacus*, s.v., reads *Bliṭshāṣār* or *Blīṭshāṣār*, while Bar Bahlūl, more rightly, as I. Guidi tells me, reads Belṭshāṣār). The name of the king Belshaṣṣar as al-Battānī gives it is found in Ḥamza, p. 94, and in al-Ṭabarī, series I, vol. I, p. 216, and vol. II, p. 652 (where the editor wrongly accepted بلتشصر); through the corruption of the copyists it is بلتنصر in Ibn Khaldūn, Būlāq edition 1284 H, vol. II, p. 117. In Ibn al-Athīr it is بلتاصر, in Eutychius (*Contextio gemmarum, sive Eutychii Patriarchae Alexandrini Annales ed. Edw. Pocockio*, Oxford 1659, vol. I, p. 257) بلتاسر; but perhaps in both تـ is corrupted from ـشـ. After the Syriac form, al-Bīrūnī, loc. cit., and Abū l-Faraj, ed. Ṣālḥānī, Beirut 1890, p. 78, have بلطشاصر.

[4] Νιριγασολάσαρος and Βαλτάσαρ in Syncellus.

Παραβάτης, «apostate», is rendered. — That *Syriac* writer seems to have flourished under Leo the Isaurian (717–741), since the last of the kings is Theodosius III, who died in 717[a].

There are some small discrepancies from the Greek text. Ptolemy gives the reign of the first Artaxerxes 41 years, al-Battānī and al-Bīrūnī 43, so that the sum down to the death of Ochus is increased by two years. The reigns of Arogus, of Darius son of Arsakh and of Alexander of Macedon make 2 + 4 + 8 = 14 years in Ptolemy; 10 + 6 + 6 = 22 (that is, ى 10 for ب 2, و 6 for د 4, و 6 for ح 8) in al-Battānī; 2 + 6 + 8 = 16 in al-Bīrūnī; so the sum of the reigns down to the death of Alexander is 424 in Ptolemy's *Canon*, 434 in al-Battānī, 428 in al-Bīrūnī. Strangely, on p. 231 of the text (vol. II, p. 3 of the translation) and in the Spanish version the interval from Bukhtanaṣṣar to the death of Alexander is given as تكد, 424 years, which in the translation I thought should be read تلد, 434, on the authority of the table of the kings. — The error disappears in what follows, since a new count begins with Philip (Arrhidaeus), namely the era called after the death of Alexander (see vol. I, pp. 242–43).

After that everything agrees well with *Ptolemy's Canon* down to the death of Antoninus; after him al-Battānī and al-Bīrūnī put Commodus with 32 years, putting together into one the reigns of Marcus Antoninus (19 years) and of Commodus (13). Likewise they fuse into one the reigns of Severus (18) and of Antoninus Caracalla (7); but the sums of the years are plainly not affected by any discrepancy. Again they differ from the *Canon* in the reigns of Philip, Decius and Gallus (6 + 1 + 3 instead of 7 + 1 + 2); but the sum is not changed. They give Diocletian 21 years instead of 20; and there are small discrepancies in the following years, which, however, gradually compensate one another, so that the sums of the years down to the death of Julian the pagan are equal in the Greek and in the Arabs. The Greek gives Constantine Pogonatus (Leontius in al-Battānī) 17 years instead of 16, but Philippicus 2 instead of 3; so the sum down to the death of Theodosius is 1040 in all of them[1].

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But although al-Battānī's order of the ***numbers*** agrees with the Greek and with al-Bīrūnī, there is a confusion in some ***names*** in the Escorial codex, which I did not wish to remove. For in al-Battānī the names (not the numbers) of the Second Interregnum and of Mesesimordacus, and of Theodosius the Great and Valens, are interchanged; moreover, after the two Constantines who follow Heraclius, the Greek and al-Bīrūnī give the names of the kings in this order: Constantine Pogonatus for Leontius, the later Justinian for Tiberius, Leontius for Justinian, Tiberius for Philippicus I, Justinian for Constantine, Philippicus, Anastasius, Theodosius. Besides, in the Escorial codex these names are so corrupt that doubt not seldom arises in reading them; so I would not dare to affirm that all these interchanges of names come from al-Battānī.

It remains for me to deal with some of the names.

Darius the Mede, i.e. Nabonadius, is in the text Darius *al-ādharī*, or «from the province of Ādharbayjān»; in the geographical table too, vol. II, p. 35, no. 69, this region is held to answer to Ptolemy's *Media*. In the Syriac archetype there was doubtless *mādāyā* ܡܕܝܐ, which al-Bīrūnī, *Chronology*, p. 89 of the text, 101 of the translation, has kept: «Darius (دريوس) *al-Mādāy* the first». In al-Ṭabarī, *Annales*, series I, vol. II, pp. 649 and 652, there occurs Darius (داريوش) *al-Mādhawī*, which in both places is said to be a relative name from an ancestor of his called Mādhī; in Eutychius, vol. I, p. 257, «Darius ... *al-Mādī*, that is, of the progeny of Mādānī», داريوش بن امرير المادي اعني من ولد ماداني; in Abū l-Faraj, ed. Ṣālḥānī, pp. 78 and 80, Darius (داريوش) *al-Mādī*, and in his Syriac Chronicle *Mādāyā*. — From another source, a Persian one, al-Bīrūnī elsewhere, *Chronology*, p. 111 of the text, 115 of the translation, has *Dārā al-Māhī*, that is, from the region of Māh, which is the more recent Persian form of the very ancient name *Māda* (Media). —

[1] Wrongly 1042 in Halma's edition; the error begins with the sum for Constantius, where the editor wrote 994 for 992.

[a] Theodosius III abdicated in 717, when Leo the Isaurian took the throne; it was his reign, not his life, that ended then.

In the Arabic text I accepted داريوس, as the codex has it, which agrees very well with the Greek Δαρεῖος; but I hardly doubt that al-Battānī found داريوش *Dārīwash* in his source, as almost all the Arab historians have it, after the Syriac and Hebrew form *Dāryāwesh* (in the Persian cuneiform inscriptions *Dārayawaush*, in the Babylonian *Dāriyāmush*).

Arogus is غيرون in the text and مرون in al-Bīrūnī, *Chronology*, p. 89 of the text, 101 of the translation; these hardly seem to have arisen from the Ἀρωγός of *Ptolemy's Canon*, or from Ἄρσης (variants Ἄρσις, Ναρσῆς, Ὀάρσης), as others call that king. He was the son of Artaxerxes III (Ochus); so from Ἄρσις Ὤχου[1] *Arsisochus* has sometimes been made up. In Eutychius, vol. I, p. 266, al-Khāzinī, f. 106v, Abū l-Faraj, p. 89, and al-Bīrūnī himself, *Chronology*, p. 111 of the text, 115 of the translation (but from another source), *Arsīs*.

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Darius [*son*] *of Arsakh* (ارسخ) is *Darius the third* in *Ptolemy's Canon*. The Arabs mostly call him, after a Persian source, *Dārā son of Dārā*. Al-Bīrūnī, however (*Chronology*, p. 89 of the text, 101 of the translation), has *Darius son of Arsīkh* (ارسيخ), and in his unpublished *Masʿūdic Canon* (as Sachau attests in the notes to the translation of the *Chronology*, p. 400) *Darius son of Arsaq*, ارسق; Abū l-Faraj, ed. Ṣālḥānī, p. 91, *Darius son of Arshak*, ارشك, as in his *Syriac Chronicle*, ed. Bruns and Kirsch, Leipzig 1789, p. 38 of the text, 36 of the translation (ܐܪܫܟ ܕ ܕܪܝܘܫ). — Quite wrongly Eutychius, I, 266, has *Darius son of Arsīs*; and the Ethiopic redaction of the history of the Arab Elmacinus (al-Makīn) *Dārā son of Arsēs*, ዳራ፡ ወልደ፡ አርሴስ፡[2]. — In al-Masʿūdī, *Prairies* II, p. 100, the name دارو اليسع occurs, which J. Marquart[3], by a not improbable conjecture, thinks is al-Bīrūnī's *Darius son of Arsīkh* and answers to the Greek Δαρεῖος Ἀρσάμου; for Ἀρσάμης (Persian *Arshāma*) was the grandfather of Darius III. If this is so, the error in the Syriac writing is explained, since from ܐܪܣܡ *Arsam* there easily arise ܐܪܣܩ *Arsak* and ܐܪܫܚ *Arṡaḥ* (*Arsakh*).

Philip the father of Dhū l-Qarnayn, in place of Philip *Arrhidaeus*, who was the brother of Alexander the Great by the same mother[a]. An error not infrequent among the Arabs; see vol. I, p. 243. — On the name Dhū l-Qarnayn, vol. I, p. LXXIV.

Alexander the founder is in *Ptolemy's Canon* Ἀλέξανδρος ἄλλος. Just as in the line before he had confused Philip Arrhidaeus with the Macedonian, so now al-Battānī, or his source, calls the other Alexander by the name of the Founder (*al-bannāʾ*), which, as al-Bīrūnī rightly has it[4], belongs to Alexander the Great. For he is called ὁ κτίστης, «the founder [of Alexandria]», in the *Handy Tables* of Ptolemy (ed. Halma, p. 2) and of Theon (ed. Halma, vol. I, p. 27). — The same confusion is found in Kūshyār al-Jīlī (Ideler, *Handbuch der mathematischen und technischen Chronologie*, II, 626) and in Ḥājjī Khalīfa, ed. Flügel, III, 470, no. 6471, who says that Theon «in his astronomical work called the Canon, derived from those observations, used the era of Philip the Roman, the Founder, the brother of Dhū l-Qarnayn» (استعمل في زيجه المسمّى بالقانون المحصول من الرصد المذكور تاريخ فيلبس الرومي البناء اخي ذي القرنين). J. Lippert, who mentions this passage in *Studien auf dem Gebiete der griechisch-arabischen Uebersetzungslitteratur*, Braunschweig 1894, p. 41, note 2, rightly says that this «Philip the founder» is corrupted from some Greek text Φίλιππος ὁ μετ' Ἀλέξανδρον τὸν κτίστην, «Philip who [came] after Alexander the founder».

Ptolemy the Hare. It would have been better to write «Ptolemy *Arnabā*», as in the Escorial codex

[1] As in *Eusebii Chronicorum canonum libri duo edd. A. Maius et Ioh. Zohrabus*, Milan 1818, p. 107. — In Syncellus Ναρσῆς Ὤχου ἀδελφός.

[2] In L. Goldschmidt, *Die abessinischen Handschriften der Stadtbibliothek zu Frankfurt am Main*, Berlin 1897, no. 21, p. 75.

[3] *Die Assyriaka des Ktesias*, in Philologus, 6th Supplementband, 2nd half, Göttingen 1893, p. 653, note 634.

[4] *Chronology*, 92 (left out in the translation, p. 103), and 28 of the text (32 of the translation). — Whence al-Maqrīzī, *al-Mawāʿiẓ wa-l-iʿtibār*, Būlāq 1270 H, I, 260, l. 5 from the bottom: الاسكندر البناء المقدوني, «Alexander the founder, the Macedonian».

[a] Philip Arrhidaeus was Alexander's half-brother on the father's side, the son of Philip II and Philinna of Larissa; Alexander's mother was Olympias.

(اربيا, read ارنبا). In editing the text, vol. III, p. 229, I thought that ارنب *arnab*, that is, «hare», should be put by emendation; hardly rightly, for if the author had meant «hare» he would have put the article *al-* before the word *arnab*. Likewise «Ptolemy son of *Arnabā*» in al-Bīrūnī, *Chronology*, p. 92 of the text. For his father was called in Greek Λάγος, «hare»; this name, very well rendered *arnĕbā* by the Syriac translator, was taken for a Greek word by al-Battānī and al-Bīrūnī (or rather by the Arabic source of both). Rightly other Arabs have *al-arnab*, as Ḥamza, p. 66 of the text, «Ptolemy son of the Hare»; Eutychius, I, 298, «Ptolemy, who is surnamed the hare» (بطليموس ويلقّب الارنب); al-Khāzinī, f. 117r, «Ptolemy مثوس (?) son of Lūghūs the Hare»; Abū l-Faraj, p. 98, «Ptolemy son of Lāghūs, that is, son of the Hare». — Sometimes الارنب *al-arnab* is corrupted into الاريب *al-arīb*, «intelligent, clever, astute»; see de Goeje's note on al-Masʿūdī, *al-Tanbīh*, p. 114 (the passage of the book *Mafātīḥ* cited there is now on p. 113 of van Vloten's edition: بطليموس الاريب بن اديب).[a]

Ptolemy Cleopatra. Sometimes taken for a man; see what al-Bīrūnī says, *Chronology*, p. 92 of the text, 103 of the translation.

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Alexander son of Mammaea, Ἀλέξανδρος Μαμαίας, or Alexander Severus, is in the Escorial codex *Aleksandros Brnmā* (برنما), in the codices of al-Bīrūnī, *Chronology*, p. 94 of the text, *Aleksandros Bzymā* (بزيما), which shows a Syriac origin. For the first Arabic translator of this table took the Syriac ܒܪ ܡܡܝ *bar Māmā*, «son of Mammaea», for a single word, a surname of Alexander; so he wrote برمما *Barmamā*, which the copyists afterwards corrupted in various ways. — The genuine Syriac form of the name Μαμαία is ܡܡܝ *Māmā*, in Phoenician inscriptions ממה, in Palmyrene ממי (see Nöldeke, *Beiträge zur semitischen Sprachwissenschaft*, Strassburg 1904, p. 94); so Abū l-Faraj, ed. Ṣālḥānī, p. 126, very rightly took from Syriac sources «*Aleksandrūs*, whose mother's name was *Māmā*». — The Arab tradition that comes from the *Greek*, however, keeps the Greek genitive Μαμαίας, which it took for a surname of Alexander; so al-Masʿūdī, *Prairies* II, 306, *al-Iskandar Māmayās*; al-Masʿūdī, *al-Tanbīh*, p. 133, «*al-Ikṣandros* (الاكصندرس), and he is surnamed *Māmayās*», as Ibn al-Athīr too, Cairo edition, vol. I, p. 143 (wrongly printed 145), has in his second catalogue of the kings.[b]

Decius. In the *Canon* Δέκιος, and in the *historical* writers of the Arabs *Dāqiyūs, Daqyūs, Dīqiyūs.* But al-Battānī has *Dāqiyānūs*, as this king is always called in the *Arabic-Muslim* redactions of the fabulous story of «the people of the cave» (*ahl al-kahf*), that is, of the Seven Sleepers of Ephesus: *Deqyānūs* in al-Ḥasan al-Qummī al-Naysābūrī, *Tafsīr* (in the margin of al-Ṭabarī's *tafsīr*), vol. XV, p. 114; al-Thaʿlabī, *ʿArā'is al-majālis*, Cairo 1297 H, pp. 369 ff.; al-Zamakhsharī, *al-Kashshāf*, Cairo 1307 H, vol. I, p. 567; al-Bayḍāwī, ed. Fleischer, vol. I, pp. 559 f.; al-Damīrī, *Ḥayāt al-ḥayawān*, Cairo 1311 H, vol. II, pp. 240–45 (s.v. كلب); [Pseudo-] Ibn Iyās, *Badā'iʿ al-zuhūr*, Cairo 1310 H, pp. 182–85, etc. — Al-Ṭabarī in the *Annales*, series I, vol. II, p. 778 (where he deals with the Seven Sleepers), and in his great *Commentary* on the Qur'ān, vol. XV, pp. 123–26, 128, 134–38, has *Daqīnūs*, دقينوس. The name survives in the village *Shahr-i Daqyānūs*, «city of Decianus», where the ruins of the city of Jiruft are (see the geographical tables, no. 255). — Strangely, the lexicon called *Qāmūs* and its commentary *Tāj al-ʿarūs*, 2nd edition, vol. IV, p. 152, distinguish

[a] Nallino's Addenda et emendanda (Part II, p. XXIX) add that Ibn al-Athīr, Cairo edition, I, 126, calls Ptolemy Lagus *Baṭlīmūs ibn Lāghūs*, and Abū l-Fidā', *Annales*, Constantinople edition, I, 62 (= *Historia anteislamica*, ed. Fleischer, p. 104), «*Baṭlamyūs Sūtīr* (σωτήρ; the editions have سشون) *ibn Lāghūs*, who was also called *al-Manṭiqī*».

[b] Nallino's Addenda et emendanda (Part II, p. XXIX) add, on the vowels of the name *Māmā*: Payne-Smith in his *Thesaurus* gives it short vowels (ܡܰܡܳܐ), apparently following Assemani's choice, for the texts he quotes have no vowels; the passage he cites from Bar ʿAlī Nallino does not find in G. Hoffmann's edition. M. Lidzbarski, *Ephemeris für semitische Epigraphik*, vol. I, 1900–01, pp. 76 and 96, also has the short vowels (ܡܰܡܳܐ and ܡܰܡܳܝܳܐ), plainly relying on Payne-Smith. Nallino put the *long* vowels, which are not only given by that most expert judge Abū l-Faraj, or Barhebraeus (in the Arabic passage cited on p. 195, ll. 1–2, and in the *Syriac Chronicle*, ed. Bruns and Kirsch, p. 60 of the text, 57 of the translation: ܐܠܟܣܢܕܪܘܣ ܒܪ ܡܡܐ), but are also required by the genuine tradition kept in the Arabic proper names *Māmāy* and *Māmāya* (see also M. Hartmann, Deutsche Litteraturzeitung, 1901, no. 17, cols. 1046–47).

Daqyūs from *Daqyānūs*: زاد الصاغاني (اتخذ مسجدًا على اصحاب الكهف) اعجمي (ملك) اسم (دقيوس بالفتح) وقصتهم مذكورة (ملك هربوا منه) اسم (ودقيانوس). — In Syriac writings ܕܩܝܢܘܣ or ܕܩܝܣ *Deqyūs*; whence *Daqyūs* in the *Arabic-Christian* story of the Seven Sleepers in I. Guidi, *Testi orientali inediti sopra i Sette Dormienti di Efeso*, in Memorie dell'Accademia dei Lincei, Classe di scienze morali, series III, vol. XII, 1884, p. 391, note 1; but *Deqyānūs* in another *Arabic-Christian* story in Guidi, p. 301. In the *Arabic-Coptic Synaxarium* of the Jacobites *Dākiyūs* (Guidi, p. 426), from a Greek source, with which agrees the Ethiopic ዳክዮስ፡ *Dākyōs*, which always occurs in the books written under Zarʾa Yāʿqōb (1434–1468) (in Guidi, pp. 406–19 and 429), or ዳኬዎስ፡ *Dākēwōs* of the Ethiopic Synaxarium (in Guidi, p. 427). — I think that the form *Dāqiyānūs*, or *Deqyānūs*, or *Daqīnūs*, arose from the Syriac writing, through ܕܩܝܢܘܣ wrongly read ܕܩܝܢܘܣ[1].[a]

Julian «*the pagan*» (pp. 2 and 3). In Arabic *al-ḥanīf*, in the same sense as the Syriac word *ḥanpā*; in the Greek παραβάτης, «apostate». On this meaning of the word *ḥanīf* see de Goeje's Glossary to al-Masʿūdī, *al-Tanbīh*, pp. XVII–XVIII.

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Heraclius (*Hiraqlis* and *Hiraql*) «*the wise*» (pp. 2 and 3). In the codex صاحب العرب *ṣāḥib al-ʿarab*, «lord of the Arabs», which I kept in vol. III, pp. 230 and 231, though I could in no way explain it; likewise the Spanish version, to be quoted a little below: «Hercules *el de los Alaraues*» («Hercules, the one of the Arabs»). I think one should read صاحب العرف *ṣāḥib al-ʿirf*, «lord of knowledge, endowed with knowledge»[b]; in the Maghribi script the letter ف has its dot below, so that it is very easily confused with ب. My conjecture rests on al-Bīrūnī, *Chronology*, p. 96 of the text, 105 of the translation, where there is *Hiraqlis al-ḥakīm*, «Heraclius the wise». Why he was so called I do not well understand; in the Greek there is only Ἡράκλειος. Perhaps because the emperor Heraclius wrote on astronomical matters; I have mentioned a chronological fragment of his in vol. I, pp. 244–45. In 618 he composed a splendid exposition of Ptolemy's Handy Tables, in which he taught the computations of the sun, the moon and the five planets, as appears from Iac. Morelli, *Bibliotheca manuscripta Graeca et Latina*, Bassano 1802, pp. 205–06 and 209; see also the fragment on the mathematical art by the philosopher Stephanus (first half of the eighth century), published in the *Catalogus codicum astrologorum Graecorum*, fasc. II, Brussels 1900, p. 182: Ὅ τε γὰρ Πτολομαῖος [sic] τοῖς ἀπὸ τοῦ Ναβουχοδόνοσορ ἔτεσιν ἐχρήσατο καὶ μησὶν Αἰγυπτιακοῖς, ὁ δέ γε Θέων καὶ Ἡράκλειος καὶ ὁ Ἀμμώνιος τοῖς τοῦ Φιλίππου καὶ μησὶν Αἰγυπτιακοῖς («for Ptolemy used the years from Nebuchadnezzar and the Egyptian months, but Theon, Heraclius and Ammonius those of Philip and the Egyptian months»). — Aḥmad ibn ʿAbd al-Ḥalīm Ibn Taymiyya, in the book *Takhjīl ahl al-Injīl*, «the shaming of the followers of the Gospel»[2], states

[1] See M. J. de Goeje's opinion: «Daarentegen hebben de Arabische, naar mijne meening uit oude Syrische bronnen afkomstige, berichten Dakianus» («The Arabic reports, on the other hand, which in my opinion come from old Syriac sources, have Dakianus») (*De legende der Zevenslapers van Efeze*, in Verslagen en Mededeelingen der kon. Akademie van Wetenschappen, Afdeeling Letterkunde, 4th series, part III, Amsterdam 1900, p. 12).

[2] In L. Marraccius, *Alcorani textus universus*, Padua 1698, vol. I (Prodromus ad refutationem Alcorani), part I, p.

[a] Nallino's Addenda et emendanda (Part II, pp. XXIX–XXX) add that the genuine form *Daqyūs* survives in medieval traditions about some towns of north-western Africa that are said to have been built by the emperor Decius or once inhabited by the Seven Sleepers (R. Basset, *Notes de lexicographie berbère: le dialecte des Beni Menacer*, in Journal Asiatique, 8th series, vol. IV, 1884, pp. 523–24, note; A. Mouliéras, *Le Maroc inconnu*, Oran–Paris 1895–99, vol. I, p. 89, on the town of Bādis, or Velez de la Gomera); see al-ʿAyyāshī, *Riḥla*, Fez 1316 H, vol. I, p. 92, on the town of Lebda (Leptis Magna), and II, 406–07, on the town of Daqyūs. That town is *Taqiyūs* in the older Arab writers (al-Yaʿqūbī, al-Muqaddasī, Yāqūt, the author of *al-Bayān al-mughrib*, al-Idrīsī, Ibn Khaldūn; but *Daqyūs* in ʿAbd al-Wāḥid al-Marrākushī), and *Taqyūs* in al-Shammākhī (d. 928 H, 1522), *Kitāb al-siyar*, [Cairo 1301], p. 386, l. 9; today it is *Daqāsh*, or, as it is commonly pronounced, *Dgāsh* (*Deguèche* on French maps), about 2 km from the northern shore of the Shaṭṭ al-Jerīd, 11 km north-east of the town of Tūzer and 8 km east-south-east of al-Ḥāmma; the modern form of the name already occurs in al-Dimashqī, whose editor, not recognizing the place, wrongly printed دقاس in the text, p. 238, and «Deqàs» in the translation, p. 337. A tribe called Daqyūs still lives with the tribe of the Benī Mnāṣer near Shershāl (Cherchel); and Daqyūs survives in Algeria as a man's name («Dekious دقيوس», in the *Vocabulaire destiné à fixer la transcription en français des noms des indigènes*, Algiers 1891, p. 118). Al-ʿAyyāshī died in the month of Dhū l-Qaʿda 1090 H (December 1679).

[b] The emendanda to Part III, in Part II, p. XXXI, adopt this reading.

that the emperor Heraclius was an eminent astrologer and knew from the motions of the stars of the coming of «the King of circumcision», namely Muḥammad; but according to older Arabic traditions[1] Heraclius had seen him in a dream[a]. Others relate that Heraclius, on receiving the Prophet's letter, at first wished to convert himself and his people to the new religion, but that the threats of the nobles and of the people made such a plan vain[2]. — A council of philosophers at the court of King Heraclius (*Hiraql*), and their opinions on music, are mentioned on the authority of اموس (Ammonius?) by al-Kindī (ninth century) in an unpublished little book on music, treatise II, ch. IV, in Steinschneider on Baldi, p. 437 (extract 12, with additions). — Finally, some confusion with the mythical *Hiraql* (Heraclius or Hercules) is not to be thought impossible[b], whom Ibn al-Qifṭī (in Flügel's notes to the *Fihrist*, vol. II, p. 125) calls «a Babylonian sage (*ḥakīm*), one of the seven sages» who once presided over the seven temples of Babylon. To this Hiraql al-Nadīm, *al-Fihrist*, pp. 270 and 285, ascribes a book *al-Dawāyir wa-l-dawālīb*, «on wheels [of every kind] and water-wheels»[3]; to him or to another Hiraql a large book on *alchemy* is ascribed by al-Nadīm himself, p. 353, l. 24, and p. 354, l. 27, and by Ḥājjī Khalīfa, ed. Flügel, vol. V, p. 171 (see also Steinschneider, *Die arabischen Uebersetzungen aus dem Griechischen*, in the Zeitschrift der deutschen morgenländischen Gesellschaft, vol. L, 1896, p. 362, where the writer on alchemy is said, on what ground I do not know, to be the emperor Heraclius himself). — In the Bibliothèque Nationale in Paris an Arabic codex of the thirteenth century is kept, which the printed Catalogue (no. 178, 3) describes thus: «Comment Sibylle (سبلا) fille de Héraclius (هرقل), chef des philosophes, expliqua le songe que cent hommes eurent, la même nuit, dans la ville de Rome, sous le règne d'Auguste César» («How Sibyl, daughter of Heraclius, chief of the philosophers, explained the dream that a hundred men had on the same night in the city of Rome, under the reign of Augustus Caesar»). — The mythical Hercules is the one whose picture occurs in a Greek codex of Bologna with the inscription Ἀστρονόμος ὁ Ἡρακλῆς (*Catalogus codicum astrologorum Graecorum*, fasc. IV, 1903, p. 45); see Bouché-Leclercq, *L'astrologie grecque*, p. 577: «C'est bien l'Hercules de Tyr que Nonnus appelle ἀστροχίτων (XL, 408, 579), πρόμος ἄστρων (ibid. 567). Cet Hercule fait un cours d'histoire à Bacchus et lui fait cadeau de sa robe constellée, le vêtement légendaire des astrologues»[c] («It is indeed the Hercules of Tyre whom Nonnus calls ἀστροχίτων, star-clad (XL, 408, 579), and πρόμος ἄστρων, leader of the stars (ibid. 567). This Hercules gives

44. — That Heraclius revered Muḥammad as a true Prophet is also stated by the same Ibn Taymiyya, *al-Risāla al-qubruṣiyya*, Cairo 1319 H, p. 15. — The author died at first dawn (*saḥar*) of Monday, 20 Dhū l-Qaʿda 728, that is, in the last part of the night between Sunday the 25th and Monday the 26th of September 1328 (Brockelmann's «Dienstag 22. Dū 'lqaʿda = 29 Sept.», II, p. 101, is false, and is repeated from Schreiner, ZDMG, LII, 1898, p. 562; false also is «at first dawn of Monday 10 Dhū l-Qaʿda» in Ibn al-Ālūsī, *Ǧalā' al-ʿaynayn*, Būlāq 1298 H, p. 9, whence it passed into the biography placed before Ibn Taymiyya's book *al-Wāsiṭa bayn al-khalq wa-l-ḥaqq*, Cairo 1318 H, p. 8).

[1] *Kitāb al-Aghānī*, vol. VI, p. 94; al-Ṭabarī, *Annales*, series I, vol. III, pp. 1561 f.

[2] *Aghānī* VI, pp. 94–96 (see XV, p. 58, ll. 14–15); al-Ṭabarī, loc. cit., pp. 1562–68; Ibn Saʿd in Wellhausen, *Skizzen*, vol. IV (Berlin 1899), p. 2 of the text, 98 of the translation; al-Wāqidī, transl. Wellhausen, p. 329; Ibn al-Athīr, Cairo 1301–02, vol. II, pp. 101–02 (ed. Tornberg II, pp. 162–63); al-Bukhārī, Būlāq 1289 H, vol. I, pp. 3–5; II, pp. 130–31 (*jihād*); III, pp. 93–94 (*tafsīr*, on Qurʾān III, 57).

[3] I have rendered *dawāyir* «wheels of every kind»; see *Le livre des appareils pneumatiques et des machines hydrauliques de Philon de Byzance édité d'après les versions arabes et traduit par Carra de Vaux*, in Notices et extraits des mss. de la Bibliothèque Nationale, vol. XXXVIII, Paris 1902, pp. 29, 30, 231.

[a] Nallino's Addenda et emendanda (Part II, p. XXX) add that the fabulous story of Heraclius and his dream, which Marraccius quotes, also occurs in another work of the same Ibn Taymiyya, published very recently: *al-Ǧawāb al-ṣaḥīḥ li-man baddal dīn al-Masīḥ*, Cairo 1323 H (1905), vol. I, pp. 97–98.

[b] Nallino's Addenda et emendanda (Part II, p. XXX) now hold it certain that the emperor Heraclius was confused with the mythical Hercules. A Syriac text of the eighth century, which I. Guidi supplied to Nallino (P. Lagarde, *Analecta Syriaca*, Leipzig 1858, p. 201, l. 23), tells how «Hercules the wise man» (*heraqlīs gabrā ḥkīmā*) found purple. An Arabic story of the *Sibyl*, daughter of Hercules or Heraclius, has been translated into French by R. Basset, *Les apocryphes éthiopiens*, fasc. X (Paris 1900), pp. 41 ff., who also translated the Ethiopic redaction, ibid., pp. 27–40 («Sabēllā, daughter of Ḥĕrqāl, ሰቤላ ፡ ወለተ ፡ ሕርቃል, head of the wise men of the city of Ēfēson»). Strangely, al-Shahrastānī uses *Hiraql al-ḥakīm* for the philosopher *Heraclitus*, whose doctrines he sets out (ed. Cureton, pp. 296–97; Haarbrücker's translation, II, 137–38; Cairo edition 1317–21 H, vol. III, pp. 15–16, in the margin of Ibn Ḥazm).

[c] The print has «la vêtement».

Bacchus a course of history and makes him a present of his starry robe, the legendary garment of the astrologers»).

I shall now give the summary of the intervals according to the *Spanish* version (f. 29r): «Lo que a entre Nabuchodonosor el primero fata la muerte de Alexandro el primero .CCCC. et .XXIIII.[1] annos egipcianos. ¶ Et depues [sic] regno Phelippo padre de Alexandre el Macedonio. et ouo entrel. et entrel començamiento del regnado de Augusto el primero cesar .CC. et .LXIIII.[2] annos. ¶ Et desde que regno Cesar Augusto el romano. fata el començamiento del regno de Dioclecianos el xristiano .CCC. et .XIII. annos. ¶ Et desde que regno Dioclecianos fata el començamiento del regno de Bolianoz[3] .LXXVII. annos. ¶ Et despues mataron[4] a Bolianoz. et tornosse el regno a los xristianos assi como fue.[5] Et fue del començamiento del regno de Dioclecianos. fata el comienço que regno Hercules el de los Alaraues .CCC. et .XXVI.[6] annos. ¶ Et desi regnaron los Alaraues del començamiento de la alhigera. fata que regno Mauya que fue de fiios de Vmaya. depues [sic] de la muerte de Ahly aben Abitali[7] .XXXIX. annos. et finco el regnado en los fijos de Vmaya. fata que se torno a los fijos de Alahbez a complimiento de .C. et .XXVII. annos de la alhigera» («What lies between Nebuchadnezzar the first and the death of Alexander the first is 424 Egyptian years. And afterwards Philip, the father of Alexander the Macedonian, reigned, and there were between him and the beginning of the reign of Augustus, the first Caesar, 264 years. And from the reign of Caesar Augustus the Roman to the beginning of the reign of Diocletian the Christian, 313 years. And from the reign of Diocletian to the beginning of the reign of Bolianoz, 77 years. And afterwards they killed Bolianoz, and the kingdom returned to the Christians as it had been. And from the beginning of the reign of Diocletian to the beginning of the reign of Hercules, the one of the Arabs, there were 326 years. And then the Arabs reigned from the beginning of the hijra until Muʿāwiya, who was of the sons of Umayya, reigned after the death of ʿAlī ibn Abī Ṭālib, 39 years; and the kingdom remained with the sons of Umayya until it passed to the sons of al-ʿAbbās at the completion of 127 years of the hijra»).

On pp. 4–6: the table of the caliphs

The same table[8], carried down to the death of al-Muṭīʿ (336 H), is in al-Masʿūdī, *Prairies*, vol. IX, pp. 38–49, who acknowledges that he followed al-Battānī above all; the same, carried down to the death of al-Muktafī (334 H), but with the sums of the years left out and not without errors, is also in Ḥamza, pp. 156–58 of the text. It is worth quoting some things from al-Masʿūdī, loc. cit., pp. 38–39: «In an earlier part of this book we had already set up a special chapter for the chronology of the kings … down to this time, derived from computation and from what is told in the books of biographies and histories and of those who have devoted themselves to the history of the caliphs and kings. But in that place we had not set out what the astronomers (*aṣḥāb al-nujūm*) tell in the astronomical works (*fī kutub al-zījāt*), relying on their own reckoning of the times. In this chapter, then, we shall record everything that they have laid down in the astronomical works from the hijra to our time, so that the usefulness of the book may be increased and the discrepancy between the historians and the writers on heavenly things may be known». Then, pp. 48–49: «Between this and the chronology of the historians and biographers there is a discrepancy (*tafāwut*) because of the increments (*ziyādāt*) of months and days. In setting out the chronology from the hijra to this time we have used what we found in the astronomical works;

[1] He reads تكد, as in the Escorial codex; see above, p. 192.

[2] He wrongly reads رصد for رضد, 294.

[3] He reads بليانس, as the Escorial codex has it on this page, for يليانس (Escorial codex, f. 155r, يوليانس); he leaves out *al-ḥanīf*, «the pagan».

[4] He reads قتل, «was killed», for ملك, «reigned».

[5] He plainly could not interpret the corrupt text; afterwards he left out an interval of two years.

[6] He reads سكو, as the Escorial codex does.

[7] That is, ʿAlī ibn Abī Ṭālib; the clause is missing in the Escorial codex.

[8] The Arabic text is in vol. III, pp. 231–34. In the Spanish version they are left out.

for those who devote themselves to this [heavenly] art keep those times accurately (*yurāʿūn*) and attain a knowledge of them most exactly defined (*ʿalā l-taḥdīd*). What we have reported we have taken from the astronomical work of Abū ʿAbd Allāh Muḥammad ibn Jābir al-Battānī[1] and from other astronomical works». — On the chronological tables see also al-Masʿūdī, *al-Tanbīh*, p. 329. II 198

The years and months of this table are lunar, according to the Muslim calendar. Of the hijra fixed on a Thursday I shall speak in the note on the table of p. 7.

The name «*just caliphs*» I have kept in the headings, as I had found it in the codex. But mostly only the first four caliphs are so called; and it is not impossible that the name, which al-Battānī put before those four only, was given to all the others too through an error of the copyists.

The *Banū Hāshim*, mentioned at the beginning of p. 5, are the tribe or family from which the ʿAbbāsid caliphs descended.

The discrepancies from al-Masʿūdī are of very little importance, as will appear from this survey of them:

	al-Battānī, p. 4						*al-Masʿūdī, IX, 42*					
ʿUmar ibn ʿAbd al-Malik	2^y	5^m	13^d	100^y	6^m	26^d	2^y	5^m	15^d	100^y	6^m	28^d
Yazīd ibn ʿAbd al-Malik	4	0	1	104	6	*27*	4	0	1	104	6	*29*
Hishām ibn ʿAbd al-Malik	19	8	*9*	124	3	6	19	8	*7*	124	3	6
	al-Battānī, p. 5						*al-Masʿūdī, IX, 46*					
al-Mutawakkil	14^y	9^m	9^d	246^y	9^m	3^d	14^y	9^m	7^d	246^y	9^m	1^d
al-Muntaṣir	0	6	0	247	3	*3*	0	6	0	247	3	*1*
al-Mustaʿīn	2	9	3	250	0	*6*	2	9	3	250	0	*4*
down to al-Muʿtazz	0	0	*8*	250	0	14	0	0	*10*	250	0	14

All those who are named after *al-Muktafī* (printed in italic, or sloping, letters), since they lived later than al-Battānī, were plainly added by some reader, who in the sum of the years of *al-Muqtadir* wrongly put 18 days instead of 11, and so vitiated the remaining sums too. Nor are the discrepancies from al-Masʿūdī small in these interpolated entries.

But perhaps the last caliph named by al-Battānī was *al-Muʿtaḍid*; for the numbers of the following one, al-Muktafī, do not agree with al-Masʿūdī IX, 47:

al-Battānī	6^y	6^m	4^d	294^y	9^m	26^d	al-Masʿūdī	6^y	6^m	20^d	294^y	10^m	12^d

The historians, such as al-Ṭabarī, series III, vol. V, p. 2280, al-Masʿūdī, *al-Tanbīh*, p. 371 (where وستّة اشهر is to be inserted), and others, make the caliphate of al-Muktafī 6 years, 6 months and 19 days.

Lastly I ask that on p. 5, last line, 23 years, كج, be restored for the 28, كح, of the codex.

On p. 7: the signs of the Arab years

The author mentions the table in vol. I, p. 71 (ch. XXXII), and explains it on p. 145 (appendix A).

Two systems of signs occur: the one, which I shall call I, on the left side of the page, the other (II) on the right. II, which is left out in the Spanish version (f. 29r), seems to me *spurious*; it is wholly useless, since I is already given, and it is the less convenient, because to find the sign of a given year one must always add the sign of the month al-Muḥarram to the sign of that year written in the table. Thus the sign of the first year of the hijra will be 7 + 5 = 5, the sign of the second 4 + 5 = 2, and so on. In system I the sign of the year is already in the table, and an addition is needed

[1] In the edition البناني, so that the French translation has *Bennani*.

only to find the signs of the months that follow the first (al-Muḥarram). Al-Battānī, loc. cit., sets out system I; II he does not even mention[1].

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Al-Battānī warns that the signs of the collected years are one less than the right ones; why this was done I have set out in vol. I, p. 323.

«Collected years» (*sinūn majmūʿa*) mean, among the Arab astronomers, the sum of the years from the epoch of the era, not a span of years counted from any moment of time. «Single years», or, as they are called in Arabic, «spread-out years» (*sinūn mabsūṭa*), are years taken one by one and counted from any time. The names of the *collected* and the *spread-out* years crept from the Arabic writings into the *Toledan* and the *Alfonsine Tables* and many others of the Middle Ages and of more recent times.

Among the Arabs, as among the ancient Latins and all astronomers, the week begins with Sunday (the first weekday); Saturday (the seventh weekday) closes the week.

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Al-Battānī puts the epoch of the hijra on *Thursday* (the fifth weekday), 15 July 622. Of the Arab astronomers[2] whom I have consulted, only Ḥabash and al-Khāzinī (f. 22v, etc.) follow the Muslim civil usage of putting the epoch of the hijra on Friday (the sixth weekday), 16 July. It is worth giving in his own words what Ḥabash says, f. 2v: «Since I had found that the narrators of the sacred traditions[3] and the men most expert in computing put the new moon (مستهلّ) of al-Muḥarram of the year of the Arab[4] Prophet Muḥammad (may God be gracious to him and grant eternal salvation to him and to the pure family of his house) on a Thursday, I investigated whether this was true (يصحّ) according to computation. I found that the conjunction of the luminaries for the new moon of al-Muḥarram of the first year of the hijra was at the city of Mecca after the middle of the night whose morning (صبيحة) was a Wednesday [i.e. the day of Mercury], and by computing I found that the appearance of the new moon at Mecca was in the night of the Friday of the new moon of al-Muḥarram of the year of the hijra. So we found nothing that pointed to Thursday, either in the conjunction or in the appearance; whence it became plain to us that those people had counted al-Muḥarram from the night of Thursday only according to common usage, [making], throughout their computation, the elapsed month of Dhū l-Ḥijja one of 29 days[5]. So we have returned to the appearance of the new moon, and have fixed the beginning of al-Muḥarram of the first year of the hijra on the night of Friday at Mecca, [whose] morning was the 28th day of the month Afrawardīn-māh of the year 44 of King Abarwīz[6], and 9 years and 338 days before King Yazdajird. Likewise, the night whose morning was the 16th day of Tammūz [July] of the

[1] A third system, from the work of Maslama al-Majrīṭī, is in the spurious tables, ff. 240v and 241r.

[2] Like the astronomers, the astrologers usually take Thursday, 15 July 622, as the epoch of the era; among them al-Kindī in O. Loth, *al-Kindî als Astrolog*, in Morgenländische Forschungen, Festschrift an H. L. Fleischer, Leipzig 1875, p. 294. — Since the interval between the eras of the hijra and of Yazdajird is 3624 days, it is plain that Albumasar [Abū Maʿshar], *De magnis coniunctionibus*, Venice 1515, at the end of tract IV and in tract VIII, diff. 2, holds the same opinion, and so does Alcabitius [al-Qabīṣī], diff. IV, ch. VI, f. dd3r of the edition of 1485, or f. 17r of the edition of 1521 (I have given the titles of both in vol. I, p. 309, note 1).

[3] *Ruwāt al-ḥadīth*, that is, those who handed down the sayings and deeds of the Prophet and his companions, most carefully collected.

[4] It appears from the *Qurʾān* itself that the Qurʾān is «an Arabic book», or «written in clear Arabic speech», and that all the Prophets who were before Muḥammad were sent to nations other than the Arabs. Hence the name «*the Arab Prophet*» by which Muḥammad is often called, above all at the beginning and the end of the poems inserted in the very long popular stories entitled *Sīrat Banī Hilāl*, which celebrate the deeds of the tribe of the Banū Hilāl (in the Beirut editions, which were made by Christians, those first and last verses are left out).

[5] It had been of thirty days according to Ḥabash's own way of intercalating, in which the last year of the thirty-year cycle is taken as a leap year; see vol. I, p. 323. A new cycle begins with the hijra.

[6] Chosroes II, whose reign is quite certainly counted from 27 June 590; see *Tabari's Geschichte der Perser und Araber zur Zeit der Sasaniden übers. von Th. Nöldeke*, Leiden 1879, pp. 430–31; Th. Nöldeke, *Zur persischen Chronologie*, in ZDMG, vol. L, 1896, p. 141; and also all the tables of the Sāsānid kings in al-Bīrūnī, *Chronology*, pp. 121–31 of the text, 123–28 of the translation. So 16 July 622 was the 28th day of the month Afrawardīn of the year 32 (not 44, as in the codex of Ḥabash) of King Chosroes II.

year 933 of Alexander; and from the first day of the first Tishrīn[1] to 15 Tammūz, counting the whole 15th, there were 288 days. Likewise the morning of the 23rd day of the month Kiyahk of the year 1389 from Bukhtanaṣṣar; and from the first day of Tūt to 22 Kiyahk, counting the 22nd day, there were 112 days, which, with the five epagomenal days (*al-ayyām al-lawāḥiq*) added, make 117. We have added these five because we count from the [Persian month] Daymāh[2], [after which], when 12 [months] have elapsed, the year of the Copts is complete».

According to the civil usage of the Muslims the day is counted from sunset, so that the night belongs to the following morning. The astronomers count the day from noon (vol. I, p. 221); so the night of Friday in civil usage becomes, with them, the night of Thursday. The discrepancy of a day in the epoch of the hijra is explained in this way. Ḥabash, however, relied on other computations.

The Prophet seems to have prescribed the use of lunar months to the Meccans in the year 9 of the hijra, in which he probably published the verses IX, 36–37 of the *Qurʾān*[3]; he repeated the precepts in the following year in his last address[4]. Which calendar the Meccans used before is not clear; for the conjectures of very learned men see A. Caussin de Perceval, *Mémoire sur le calendrier arabe avant l'Islamisme* (Journal Asiatique, series IV, vol. 1, 1843, pp. 342–79); Mahmoud Effendi, *Mémoire sur le calendrier arabe avant l'Islamisme et sur la naissance et l'âge du prophète Mohammad* (Journal Asiatique, V, 11, 1858, pp. 109–92); A. Sprenger, *Ueber den Kalender der Araber vor Muḥammad* (Zeitschrift der deutschen morgenländischen Gesellschaft, vol. XIII, 1859, pp. 134–75); J. Wellhausen, *Reste arabischen Heidentums*, 2nd edition, Berlin 1897, pp. 94–101; H. Winckler, *Altorientalische Forschungen*, series II, vol. II (1900), pp. 324–50 and 374–81. Winckler[5] later changed his opinion a little, stating that in the Prophet's time the Meccans had a *solar* calendar, brought after 525 by the Christians of southern Arabia under the leadership of Abraha, and that the Prophet, after taking Mecca in the year 8 H, restored the very ancient *lunar* calendar of the Meccans, which had once belonged to the cult of the moon-god Hubal.

Some, whom al-Ṭabarī reviews (*Annales*, series I, vol. III, pp. 1250–56), say that the epoch of the era was instituted by Muḥammad himself; most, and more rightly, say by the caliph ʿUmar, in the year 16 (or 17, or 18). The epoch is the 1st day of al-Muḥarram of the year in which the Prophet, leaving the city of Mecca, reached al-Madīna; and he arrived, according to the Arab astronomers and ours, on Monday 8 Rabīʿ I in civil usage (20 September 622), or on Sunday the 7th of the same month (19 September) in the astronomical reckoning of time. See Mahmoud Effendi, loc. cit., p. 121; Loth, *al-Kindî als Astrolog*, pp. 293–94; al-Bīrūnī, *Chronology*, p. 330 of the text, 327–28 of the translation. Those Arabs who choose the day 12 Rabīʿ I are wrong.

Of the errors of the codex in marking the leap years I have spoken in vol. I, p. 323, II; I have restored everything to al-Battānī's intention.

[1] That is, of October; following the common usage (which differs from al-Battānī's), he counts the years of the Seleucids from 1 October.

[2] The era of Nabonassar began on 1 Daymāh.

[3] According to Nöldeke, *Geschichte des Qorâns*, Göttingen 1860, p. 166.

[4] Ibn Hishām, ed. Wüstenfeld, pp. 968–69; al-Ṭabarī, *Annales*, series I, vol. IV, pp. 1753–54; and all the writers of the sacred traditions. On that whole address, variously handed down and perhaps partly invented, see Snouck Hurgronje, *Het mekkaansche feest*, Leiden 1880, pp. 43–45. — That the oldest historians had no sure method in the chronology of the Prophet's deeds is rightly pointed out by Ed. Sachau, p. XXXV of his preface to part I of vol. III of the work of Ibn Saʿd, *Biographien Muhammeds, seiner Gefährten und der späteren Träger des Islams hrsg. von E. Sachau*, Leiden 1904 ff.

[5] *Arabisch-Semitisch-Orientalisch, kulturgeschichtlich-mythologische Untersuchung*, Berlin 1901–02, pp. 81–90 (= Mitteilungen der vorderasiatischen Gesellschaft, 6th year, 1901, fasc. 4–5).

On p. 8: the signs of the Roman months and years

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The author, who mentions this table in vol. I, p. 71 (ch. XXXII), has explained its use in appendix B (vol. I, p. 146). Why the signs have been lessened by one, and why the years are to be divided by 28, I have set out in vol. I, p. 323.

In similar tables in al-Bīrūnī, *Chronology*, p. 195 of the text, 175 of the translation, and in Ulugh Beg, p. 18 of the text, 17 of the translation, the signs are not lessened at all, and the beginning of the era is put on the first day of October, as is the common usage, not of September.

The column of the leap years gives sixtieths of a day. Since the Roman year is of 365 1/4 days, or 365 15/60, the sixtieths, added to themselves, make up a whole day to be intercalated every fourth year.

I think that the Latin names of the months were added by a *Western Arab interpolator*. They are never mentioned by al-Battānī in the chapters on chronology; they also differ from the names that occur in the *Eastern Arab* writers[1], namely *yānuwāriyūs, fībruwāriyūs, marṭiyūs, abrīliyūs, māyūs, yūniyūs, yūliyūs, aghūsṭūs, sābṭānbriyūs, oqṭūmbriyūs, nuwāmbriyūs, dīqembriyūs*, which come from the Greek: Ἰανουάριος, Φεβρουάριος, Μάρτιος, Ἀπρίλλιος, Μάϊος, Ἰούνιος, Ἰούλιος, Αὔγουστος, Σεπτέμβριος, Ὀκτώβριος, Νοέμβριος, Δεκέμβριος. Among the *Eastern Arabs* names taken directly from *Latin* or from the *Romance* languages became known only in the twelfth century, through the wars of the Crusades and through the treaties concluded with the rulers of Egypt, above all by the Italian republics[2]; whence it has come about that in our own time the months of the Julian calendar are named thus by the Arabs of *Egypt*: *yanāyir, fabrāyir, mārs, abrīl, māyū, yūniya, yūliya, oghusṭus, sebtembar, oktūbar, nōfembar, dīsembar*. — Strangely, *mārt, māyus* and *oghostūs* (اغستوس) alone have been received into the Julian year that the *Turks* have used since 1675 in matters concerning the treasury, while the other months are called by Syriac names.

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The names of this table, which are spurious and also occur in the table of f. 241v (taken from the work of the Hispano-Arab Maslama al-Majrīṭī), are these: *yannayr, fabrayr, mārs, ibrīl, māyuh, yūniyuh, yulyuh* (يليه), *aghusht, sutambar, aktūbar, nuwanbar, dujanbar*. The same (with *marsuh, ibrīr, yūliyuh* slightly changed) are called «the months of the *Westerners*» (المغاربة) by al-Bīrūnī, *Chronology*, pp. 50 and 71 of the text, 59 and 83 of the translation; and they occur on an astrolabe made at Toledo in the year 420 H (beginning 19 January 1029), with *yūliyuh, shutanbar, dukanbar* slightly changed[3]. — Towards the end of the sixth century H (the twelfth AD), Ibn Jubayr, a Hispano-Arab writer, describing his journey through Sicily, mentions *yannayr* (in Amari, *Biblioteca arabo-sicula*, text, p. 96), *fabrayr* (p. 101) and *dujanbar* (pp. 77, 94)[a]. — In the work *Vocabulista in arabico pubblicato da C. Schiaparelli*, Florence 1871, composed in southern Spain in the thirteenth century, there are mentioned *fibrayr, abrīl, māyuh, yūniyuh, yūliyuh, aghusht, shutanbar, nuwanbar, dhujanbar*[4]. These *Western Arab* forms were derived from Latin not by

[1] And they are very few: Ḥabash, f. 4v, who calls them names «of the months of the Greeks (*al-yūnāniyyīn*)»; al-Bīrūnī, *Chronology*, p. 50 of the text (59 of the translation) and 71 of the text (83 of the translation), «the months of the Greeks» or «of the Romans»; Chems eddin Abou Abdallah Mohammed ed-Dimichqui, *Cosmographie, texte arabe publié d'après l'édition commencée par C. M. Fraehn par M. F. Mehren*, St Petersburg 1866, ch. IX, sect. 8, «the months of the Greeks and the Romans». — ذقبرس in the subscription of an Arabic-Christian codex written in Syria in the year 1188 of Alexander, 877 AD (see the periodical *al-Mashriq*, vol. VI, 1903, p. 1011).

[2] As *māy, yūniyuh, aghusht* in the treaties of the twelfth and thirteenth centuries between the Genoese and Egypt, in M. Amari, *Nuovi ricordi arabici su la storia di Genova*, Genoa 1873, pp. 5, 10, 16 (= Atti della Società Ligure di storia patria, vol. V).

[3] F. Woepcke, *Ueber ein in der k. Bibliothek zu Berlin befindliches arabisches Astrolabium*, in Abhandlungen der mathematischen Klasse der k. Akademie der Wissenschaften zu Berlin, year 1858, p. 5.

[4] In the *Cambridge Chronicle* of Sicily (in Amari, *Biblioteca arabo-sicula*, text, Leipzig 1857, pp. 165–76), written in

[a] Nallino's Addenda et emendanda (Part II, p. XXX) add that the Latin names of all the months occur throughout Ibn Jubayr, ed. Wright (Leiden 1852), agreeing exactly with the names of the Escorial codex (except *shutanbar*); see also Muḥammad ibn al-Ṭayyib al-ʿAlamī, cited in vol. II, p. 315.

learned men but by popular tradition; for among the peasants of north-western Africa and of the Sudan the Julian year and the Latin names of the months have outlived the Roman dominion and are still in use, as J. Lippert noted in the Mittheilungen des Seminars für orientalische Sprachen zu Berlin, vol. II, part 2 (Westasiatische Studien), 1899, p. 253[1]. So they are always mentioned in the little books on astronomy written by Algerian and Moroccan Arabs for the use of schools: so Saḥnūn al-Wānsharīshī (on whom see vol. I, p. 204), pp. 14–15, where he also plainly attests that such a calendar is used by the peasants, and Muḥammad al-Sūsī al-Marghītī[2], *al-Mumtaʿ fī sharḥ al-Maqnaʿ*, fasc. III, pp. 4 ff., and VI, p. 2, where he teaches that these «Roman names of the months of the *foreign* year[3]» are to be pronounced thus: *yannayr, fabrāʾir* or *fābrāʾir, mārs* or *māris* or *mars, ibrīl, māyuh* or *māy, yūniyuh* or *yunyuh, yūliyuh* or *yulyuh, aghusht, shutanbir, aktūbir* or *aktubir, nuwanbir, dujanbir*. In the Moroccan province of Figīg (فقيق or فجيج), very recently added to the Algerian territory, these names are in use today: *yennayr, fubrayr, mars, ibrīr, māyū, yūlyū, yūlyūz, ghusht, shutānbīr, ktūber, nuwānbīr, duwānbīr*[4]. In the northern Moroccan empire, according to J. Lerchundi, *Vocabulario español-arábigo del dialecto de Marruecos*, Tangier 1892, the names are: *yinnayr, febrāyar* or *ibrāyr, mars* or *mārs, ebrīl, māyū* or *māyuh, yunyuh, yulyuh, ghosht, shutanbir, ektūber, nuwanbir, dujanbir*. Similar names of the Algerian Arabs may be read in the lexicons of their dialect. — It should be noted that the strange forms *ibrīlaz* (together with *ibrīl*), *māyb* (with *māyuh*), *yulyuz* (with *yulyuh*), *ghushtajj* (with *aghusht*), *aktubirajj* (with *aktubir*) and *nuwanbirad* (with *nuwanbir*), which occur in al-Akhḍarī and al-Wānsharīshī, arose from the memorial verses of certain writers, misunderstood, such as the verses in al-Marghītī, fasc. V, p. 6:

وَمَنْ يُرِدْ مَدْخَلَ شَهْرٍ يُنْشِدِ | يَنَّيَرَ آ فَابْرَائِرَدْ وَمَرْصَدِ أَبْرِيلَ زِ مَايَبْ وَهَاء يُنْيُهِ | يِلْيُزْ أَغُشْتَجَّ شُتَنْبِرَوْ فُهِ أَكْتِبِرَحْ نُونَبِرَدْ دُجَنْبِرَوْ | مَجْمُوعُهَا أَدَدْ زَبَهْ زَجُوْ حَدَوْ

«let him who wants the beginning of a month recite: *yannayr-ā, fābrāʾir-ad*, and *marṣ-ad*, | pronounce *ibrīl-azi, māy-ab*, and *hāʾ yunyuh, yulyu-z, aghusht-ajja, shutanbir-aw*, | *aktubir-aḥ, nuwanbir-ad, dujanbir-aw*; whose sum [of consonants] is *adad zabah*[a] *zajaw ḥadaw*». The consonants added to the names, if they are taken at their numerical value (*ā* = 1, *d* = 4, *z* = 7, etc.), indicate the day of the week on which the month must begin.

Arabic by some Sicilian Christian towards the end of the tenth century, there occur *jennāriyūh, mārsuh, abrīl, māyuh, yūniyūh, yūliyūh, awsuh, sutanbar, aktūbar, dujanbar*. After the capture of Palermo by Count Roger, the leader of the Normans (1072), I find *yannār, marṭiyūs* or *mārsū, abrīl, māyū, awsuh, sutanbar, aqtūbar, dujanbar* here and there in Cusa, *I diplomi greci ed arabi di Sicilia pubblicati* ecc., Palermo 1868–82. Likewise *māyuh* and *nuwanbar* in inscriptions of the years 1149 and 1153, in Amari, *Le epigrafi arabiche di Sicilia*, part II (iscrizioni sepolcrali), Palermo 1879–81, pp. 87 and 96 (= Documenti pubblicati a cura della Società Siciliana per la storia patria, series III, vol. I). — Finally *abrīl, māyuh, yūniyuh, yūliyūh* (يوليوه), *shutanbar, dujanbar* in the treaties between the Pisans and the rulers of Tunis, Morocco and Spain, in Amari, *I diplomi arabi del R. Archivio Fiorentino*, Florence 1863–67, vol. I, pp. 13, 15, 87, 97, 101, 111, 135, 150, 171, 235, and vol. II, p. 8; all between the years 1181 and 1445.

[1] See also E. Doutté, *Un texte arabe en dialecte oranais*, Mémoires de la Société de linguistique de Paris, vol. XII, 1903, pp. 349–50.

[2] On whom see vol. I, p. 205, note 1, and p. LXXIII. The little book is still read assiduously in the Arabic schools of the city of Fez, or Fās, and of the Algerian territory; see Delphin, *L'astronomie au Maroc*, in Journal Asiatique, series VIII, vol. XVII, 1891, pp. 195–96; Motylinski, p. XI of the book mentioned by me in vol. I, p. 297, note 4.

[3] *Sana ʿajamiyya* is what he calls the solar Julian year, beginning on 1 January, which the peasants use.

[4] As E. Doutté attests, *Figuig, notes et impressions*, in La Géographie, Bulletin de la Société de Géographie, Paris 1903, p. 198, who also adds: «L'année musulmane, en effet, n'est employée à Figuig que par les lettrés, ou lorsqu'on veut dater une époque; en tout autre circonstance, on se sert non des mois lunaires, mais des mois du calendrier julien, dont les véritables noms ont été conservés en berbère depuis l'antiquité, et non pas, comme on le croit parfois, réintroduits par nous» («The Muslim year, indeed, is used at Figuig only by the lettered, or when one wishes to date an epoch; in every other circumstance one uses not the lunar months but the months of the Julian calendar, whose true names have been kept in Berber since antiquity, and not, as is sometimes believed, reintroduced by us»). [*Translator:* The print has «crois».]

[a] The print has «zabaḥ»; the verse has زبه, with ه, whose value 5 is the day of the week on which June begins (from 1 for January: 1, 4, 4, 7, 2, 5, 7, 3, 6, 1, 4, 6, with ḥ = 8 for October counting as 1).

The names of the «Roman months» that al-Battānī and the *Eastern* astronomers use (as do the *Christian Arabs* of Syria and Mesopotamia in our own time) are taken from these *Syriac* ones: *teshrī qdem* (or *qdīm*), *teshrī ḥrāy* (ܬܫܪܝ ܐܚܪܝ), *kānūn qdem* (or *qdīm*), *kānūn ḥrāy, shbāṭ, ādār, nīsān, iyār, ḥzīrān, tāmūz, āb, īlūl.*

«Romans», *al-Rūm*, is what the Arabs call all the inhabitants of the Byzantine empire; Ῥωμαῖοι is used in the same sense by the Greeks of the late age, Procopius, Zonaras, Cedrenus, Theophanes, etc.

On pp. 9–18: the concordance of the Arab and the Roman years

These tables are mentioned and explained at the end of chapter XXXII (vol. I, p. 71) and in appendix C (vol. I, p. 146).

I have very easily corrected small errors of the codex in the numbers of the days, relying on the excellent *Vergleichungs-Tabellen der muhammedanischen und christlichen Zeitrechnung hrsg. von F. Wüstenfeld*, Leipzig 1854. The numbers of the days are always one less than Wüstenfeld's, because al-Battānī puts the beginning of the era on 15 July, not on the 16th.

Two pages of the codex (ff. 163v and 164r), which carry the concordance down to the year 720, I thought should be left out; for they are without doubt *spurious*. They swarm with mistakes, which arise not from the carelessness of the copyists but from evident errors of computation, as happens whenever an inept interpolator tries to add something to al-Battānī. To carry the concordance down to 600 H, that is, for more than 300 years from the epoch of the tables, was enough for al-Battānī. — In the Spanish version, ff. 30r–35v, these tables are found unfinished; for in the right-hand section of f. 35v the years of the hijra are written down to 488 (the last 7 lines being left empty), but the names of the first day of al-Muḥarram and the years of Dhū l-Qarnayn (the last of which is 1492) only down to the line of the year 477 H, and the days and Roman months only down to the line of the year 470 H. I think the copyist did not finish the table because on this page he had noticed that he had wrongly written 400 for 500 everywhere in the hundreds of the years of the hijra, and had moreover put, in place of the omitted sign of al-Muḥarram of the year 452 (read 552), the sign of the year 453 (read 553), and had set the following signs out of their places.

On pp. 19–23: the mean motions of the sun and the moon for the Arab calendar

These tables are dealt with in chapters XXVII and XXXIII (vol. I, pp. 42 and 71).

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In the headings of the columns the codex always has *al-masīr al-awsaṭ*, «the mean course», which fits only pp. 20–23, for they give the *mean motions*, that is, the degrees that the sun, the moon, etc. complete by their mean motion in a given interval of time. But p. 19 gives the *mean longitudes* that the sun, the moon, etc. held at the beginning of the years 1, 31, etc. of the hijra; so its numbers are the *epochs*, or *radices*, of the mean motions, from which the longitudes for any moment of time are afterwards derived. Therefore I have changed a little, in the translation, the headings of the columns of p. 19, which I kept in the Arabic as they stood in the codex. See also the notes on pp. 24–28, 72–77 and 102–103.

The motion of the *node* is retrograde; but in al-Battānī's tables it is computed as if it were direct, so that the author himself, vol. I, pp. 75–76 (ch. XXXVII), bids us always take the motion of the node taken from the tables from 360°.

P. 19, under the years 661 ff.: it was easy to restore the degrees of the anomaly of the moon, which are left out in the codex.

P. 21: the codex gives only the Dhū l-Ḥijja of the common years, as on p. 74 Subāṭ (February); for al-Battānī's tables for the collected and the single years are so made that in computing there

is never any need to distinguish the leap years from the common ones. On Schiaparelli's advice, however, I have added the Dhū l-Ḥijja of the leap years.

P. 22: two errors of the codex, which escaped me, are to be corrected: in the seconds of the course of the sun in 10 days (where the codex and the translation have 25″ كه instead of 24″ كد), and in the seconds of the node in 21 days (where 45″ مه stands for 44″ مد). — The same table is on p. 75, with three discrepancies that come from al-Battānī himself, namely 15″ in the motion of the sun in nine days, 23″ in ten, and 38′ 59″ (for 39′ 0″) in the anomaly of the moon in ten days.

P. 23: two errors of the codex, which crept into the translation, are to be corrected: 25″ كه of the codex for 24″ كد in the course of the sun in three hours, and 13″ يج (codex لج 33″) for 14″ يد in the lunar anomaly of 23 hours. — The same table is on p. 76, where there are 11″ instead of 10″ in the lunar motion of fourteen hours and 37″ instead of 38″ in the anomaly of ten hours (exactly 5° 26′ 37″ 28‴ 17^{IV} ...); both discrepancies seem to come from al-Battānī.

On pp. 24–28: the mean motions of the five planets for the Arab calendar

The author mentions the tables in vol. I, pp. 71 and 113 (at the end of ch. XXXIII and the beginning of ch. XLV).

The columns of p. 24 are headed in the Arabic as on pp. 25–26; somewhat differently in the translation, as I pointed out in the note on pp. 19–23. — Pp. 27 and 28 necessarily do not differ from pp. 106 and 105.

In his tables Ptolemy, IX, 4 (ed. Halma, vol. II, pp. 126–55), adds the anomalies of the superior planets and the mean motions in longitude of the inferior ones. These al-Battānī leaves out, rightly observing, vol. I, p. 113 (ch. XLV), that in the Ptolemaic system the mean motion of the anomaly of the three superior planets is the same as the mean motion in longitude of the sun lessened by the mean motion of the anomaly[a] of these planets themselves, and that the mean motion in longitude of Venus and Mercury does not differ from the mean motion of the sun.

On the small discrepancies from Ptolemy in the amounts of the motions see vol. I, p. 66, and my notes, vol. I, pp. 239 and 242.

Folio 167 of the codex, left out in the translation, was an addition that had nothing to do with al-Battānī's tables.

On pp. 29–32: the mean syzygies for Egyptian years

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The author has partly explained the use of the tables in ch. XLII (vol. I, pp. 92–93).

In these tables, strangely, *Egyptian* years (or *wandering* years, without the Julian intercalation) appear joined with the era of Dhū l-Qarnayn, which elsewhere is always computed in *fixed Julian* years. When the tables are looked at attentively, it appears quite certainly that these wandering years are counted from Dhū l-Qarnayn, but that 1 Thōth, the beginning of the year, always coincides with 1 Thōth of the years of Nabonassar used by Ptolemy in the *Almagest*; so 1 Thōth of the first year from Dhū l-Qarnayn was 1 Thōth 437 of Nabonassar, 9 November 312 BC, that is, 69 days after the beginning of the era of Dhū l-Qarnayn in al-Battānī's usage. Plainly al-Battānī made this table for the use of those who wished to use the *wandering Egyptian* years of Nabonassar with Ptolemy; but why he preferred to count them from Dhū l-Qarnayn I do not know.

Confusion may also arise from the name «*months of the Copts*» that occurs on p. 31. For according to vol. I, p. 67, ll. 19–20, the *Coptic year* is fixed, and its 1 Thōth (I, p. 68, ll. 23 ff., and p. 69,

[a] So the print; in the Ptolemaic system the mean motion in anomaly of a superior planet is the mean motion of the sun less the planet's mean motion in longitude.

ll. 7–21) always falls on 29 August; moreover, as I have set out in the note in vol. I, pp. 244–45, al-Battānī's *era of the Copts* is a certain era of Augustus, or Alexandrian era, also used by Theon, with fixed years, whose beginning was 29 August 25 BC, or 287 years, less 3 days, after the era of Dhū l-Qarnayn. Here, however, the months are those of the *wandering* years, which strangely he also calls *years of the Copts* in vol. I, p. 70, ll. 30 ff., adding that they «are Egyptian years from Dhū l-Qarnayn»; and they are taken in the same sense in the precepts for computing the syzygies (vol. I, pp. 92–93).

It should also be noted that 29 August 25 BC, the beginning of the era of the Copts, or of Augustus, is the first day of Thōth of the year 724 of Nabonassar, 300 of Arrhidaeus, or of the death of Alexander, and the beginning of the year 287 from Dhū l-Qarnayn in the Coptic way of beginning the year (from 29 August). So in vol. I, pp. 70 and 92–93, to convert the wandering years of the Copts counted from Dhū l-Qarnayn into fixed years, al-Battānī bids us take 287 from the given years and divide the remainder by 4. Indeed the beginning of the year 287 of Dhū l-Qarnayn, in the *Coptic* way of beginning, falls on 29 August both in the wandering and in the fixed years, as it does also in the three following years; this will afterwards happen again only in the four years of which the first is the year 287 + 365 × 4 = 1747 of Dhū l-Qarnayn. Al-Battānī's tables do not reach that year.

The columns of the tables give the following:

I. The given years or months, with which one must enter the tables.

II. The days of the first Egyptian month (Thōth), and the sixtieths of the days, at which the first mean conjunction or opposition of the given year happens.

III. The mean longitude of the sun at the moment of time given in II.

IV. The mean anomaly of the moon at the same moment.

V. The argument of latitude of the moon, or its distance from the northern node.

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Ptolemy[1] gives in III not the mean longitude of the sun but the distance (ἀποχή) of the sun from the apogee, which he wrongly thought fixed at 65° 30′; in V the argument of latitude of the moon counted from the northern limit (= longitude of the node + 90°). In the rest the construction of the tables is the same in both; but the elements for the collected years differ a little, because of the different motion of the sun in al-Battānī's tables and because of the interval of longitude between the city of Alexandria and al-Raqqa. Ptolemy set out his elements in VI, 2 (ed. Halma, vol. I, pp. 374–78), which see also in Delambre, *Histoire de l'astronomie ancienne*, vol. II, pp. 223–24; al-Battānī does not give his. To find the size of the discrepancy I compute, by the *Almagest*, the first conjunction of the *wandering* year 915 of Dhū l-Qarnayn, 1351 of Nabonassar. The last year of Ptolemy's table is 1101 of Nabonassar;

first conjunction of the year 1101 of Nabonassar		22^d	41′	45″	Thōth
change in the interval of 250 wandering years	−	0	27	51	
for 1351 of Nabonassar, 915 of Dhū l-Qarnayn		22^d	13′	54″	Thōth

In vol. I, p. 42, ch. XXVII, al-Battānī puts the difference in time between Alexandria and al-Raqqa at «about 2/3 of an equinoctial hour», or 10°, or 0^d 1′ 40″; so

[1]His tables are in the *Almagest* VI, 3 (ed. Halma, vol. I, pp. 378–83); their use, and the conversion of the mean syzygies into true ones, is taught afterwards in VI, 4 (Halma, I, 384–87).

		22^d	$13'$	$54''$	
	+	0	1	40	
for the meridian of al-Raqqa, according to Ptolemy		22^d	$15'$	$34''$	
according to al-Battānī's table		22	14	44	
difference from Ptolemy	–	0^d	$0'$	$50''$	$= -\ 0^h\ 20^m$.

If we take the difference of longitude between Alexandria and al-Raqqa as $12° 45' = 0^h\ 51^m = 0^d\ 2'\ 7''.5$, as is deduced from al-Battānī's geographical tables (vol. II, nos. 109 and 150), the difference in the conjunction becomes $-\ 0^d\ 1'\ 17''.5 = -\ 0^h\ 31^m$.

Pp. 29–30, for the *collected years*. Following Ptolemy's example, he chose periods of 25 years, because 25 wandering years (or 9125 days) contain almost exactly 309 lunations; the difference to be taken from those 9125 days reaches only $0^d\ 2'\ 47''\ 5'''$, or $0^d\ 1^h\ 6^m\ 50^s$, so that through many periods of 25 wandering years there will be only one conjunction and one opposition in the month of Thōth of the first year of the period, and the times of such conjunctions or oppositions will differ only a little from one another. — The table of the oppositions is made by taking half a lunation, or $14^d\ 45'\ 55''$, from the days of the conjunctions, and the motions answering to this half lunation from the motions of the conjunctions.

P. 31. — [Of the three small tables that this page contains, one, namely that of the *months*, is necessary, and is also given in the *Almagest* (VI, 3, ed. Halma vol. I, p. 383). — The *first* is useless; for who will ever compute by quarters of an Egyptian or Coptic month, that is, by 7 1/2 days? — The *last*, for intervals of years counted from any moment of time, is of little use; moreover its last column had plainly been filled in in the codex by an unskilled reader, for the motion of the latitude of the moon had been computed as if it were 108° 47′ 19″ 1/2 in the interval of 100 wandering years, instead of the 108° 48′ 23″ 1/2 that are drawn from al-Battānī's tables on pp. 29–30. I have computed it again, so that it agrees with al-Battānī's elements. — I suspect that the first and the last small table are *spurious*; similar ones, however, are found on p. 87[1]. The headings of the columns of the last mean that for future time the numbers of the days are to be subtracted and those of the motions added. — Schiaparelli.] — In the last line of the first small table «mensis» is to be read for «monsis». The first and second small tables are the same as on p. 87, with two small discrepancies: under 1/4 of a month, here 57″, there 58″, in the days; under the 3rd month (Ayyār), here 1″, there 0″, in the anomaly of the moon.

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P. 32, for the *single years*. It agrees exactly with Ptolemy's table, VI, 3 (ed. Halma vol. I, p. 382); its computation is very easy. The excess of 13 lunations over the 365 days of the wandering year is $18^d\ 53'\ 51''\ 43'''$; so if we call A the days of the month of Thōth, and the sixtieths of a day, answering to the first new moon (or full moon) of some year, the first new moon (or full moon) of the following year will happen at the time A + $18^d\ 53'\ 52''$. If this exceeds a whole lunation, or $29^d\ 31'\ 50''$, it will be the second new moon of the month of Thōth, or (if it exceeds 30 days) the new moon of the following month; then, to find the first conjunction or opposition of the year, one must take a lunation from the sum, and from the motions of the other columns the motions answering to a lunation. — In the column of the days are to be corrected: 40″ for 50″ under the year 6, 39″ for 29″ under 11, 24″ for 34″ under 14, 23″ for 13″ under 19.

The limits of the eclipses (ὅροι ἐκλειπτικοί). — Al-Battānī states in chs. XLII and XLIV (vol. I, pp. 96 and 104) that he has given them «on the page of the months of the syzygies», and in that place indeed the limits occur in the Spanish version (ff. 41v and 56v); but the Escorial codex gives them here, under the table of the single years, and on p. 88. Ptolemy gave his in the table of the syzygies for the single years, teaching their computation afterwards (VI, 5, ed. Halma vol. I, pp. 388–96); because of the different equation of the sun and the different angular diameters of the luminaries and of the shadow, the *Ptolemaic* ones necessarily differ from *al-Battānī's*. Since

[1] Likewise in the Spanish version, ff. 41v and 56v.

al-Battānī nowhere teaches how he computed the limits, I had tried to correct approximately the numbers of the codex that concern the *solar eclipses*, which are affected by most evident mistakes, choosing the numbers from which the errors of the copyists seemed most easily to have arisen, and keeping as far as possible the proportions of the limits of the Almagest. I give a survey of the corrections, giving first the numbers of the codex contained on this page 32 (f. 172r), then the numbers of the codex on p. 88 (f. 194v), third the numbers I accepted, fourth the explanation of the errors:

154°	44′	159°	44′	159°	44′	قند for قنط
190	16	*167*	16	191	16	قض and قصز for قضا
0	0	0	0	0	0	
20	16	20	16	20	16	
348	*57*	*345*	*53*	348	44	نز and نج for مد; سمه for سمح
360	0	360	0	360	0	

If the emendations were right[1], al-Battānī would have put the distance of the moon from the node needed for the contact of the discs at 20° 16′ to the north and 11° 16′ to the south; Ptolemy at 20° 41′ and 11° 22′. But it is not impossible that for the 191° of my emendation the 190° of the codex should be kept, and then 349° سمط for the codex's سمه 345° (an error very easy in the Maghribi script) and سمح 348°; the distances from the node then become 20° 16′ and 10° 26′. — But I now think that the true numbers *are other than those I had accepted in printing the tables.*

That the reason for the doubt in emending may appear better, I shall set out the method by which Ptolemy fixed the limits of the *solar eclipses*. Let Σ be the sum of the semidiameters of the sun and the moon in eclipses, or the distance of the centres needed for a contact. The parallax of the moon cannot be neglected; Ptolemy considers that the greatest parallax in latitude that can occur in all the seven climes of the earth, contained between Meroë (whose longest day is 13^h) and the mouths of the Borysthenes (where the longest day is 16^h), is 58′ to the south and 8′ to the north; and that the greatest lunar parallaxes in longitude are then 15′ (in Scorpio and Pisces) and 30′ (in Leo and Gemini). So the apparent latitude of the moon needed for a contact will be at most Σ + 58′ for the northern moon and Σ + 8′ for the southern. From these latitudes, by the tables or by the rule

$$\text{sin dist. } ☾ \text{ from the northern node} = R\,\frac{\text{sin given lat. } ☾}{\text{sin greatest lat. } ☾}\,,$$

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the apparent distances Δ and Δ′ of the moon from the node, or the places of the *apparent* conjunction, are deduced. Let the true conjunction be still to come; let the parallaxes in longitude (15′ for the southern moon, 30′ for the northern)[a] be added to the distances Δ and Δ′; the distances D and D′, or the true places of the *true* conjunction, will come out. Let the mean conjunction be still to come; let the quantity

$$(\alpha) \qquad \text{greatest equation } ☉ + \frac{\text{greatest eq. } ☉ + \text{greatest simple eq. } ☾}{12}\,,$$

which Ptolemy took to be the greatest difference of longitude that can fall between the true place of the true conjunction and the mean place of the mean one, be added to the distances D and D′;

[1] I add the numbers of the Spanish version, f. 41v, noting in parentheses the discrepancies of f. 56v. Limits of the eclipses of the sun: 159° 44′ – 197° 16′ (190° 57′), 0° – 20° 16′, 348° 21′ (348° 57′) – 360°. Limits of the moon: 165° 53′ (165° 13′) – 194° 47′, 0° – 14° 47′, 345° 13′ – 360°.

[a] These are interchanged. In the computations of the next page the 15′ of longitude go with the 58′ of latitude and give the northern limit, the 30′ with the 8′ and the southern limit; so also with Ptolemy's own numbers, 17° 27′ + 15′ + 3° = 20° 42′ and 7° 52′ + 30′ + 3° = 11° 22′, his northern and southern limits of 20° 41′ and 11° 22′.

the greatest distances of the moon from the node in the *mean* conjunctions, or the *limits of the solar eclipses*, will come out.

So Ptolemy, who, however, Regiomontanus[1] points out, erred in computing D and D′; for not the parallaxes in longitude were to be added, but a twelfth of them, because in the time in which the moon runs through the space of the parallax in longitude the sun completes a twelfth of it. — Al-Battānī, ch. XXX (vol. I, p. 58), puts the greatest angular semidiameter of the sun in eclipses at 16′ 50″, the greatest of the moon at 17′ 40″; so the sum of the semidiameters is 34′ 30″. Had he used *Ptolemy's parallaxes*, he would have found

the latitude of the southern ☾	34′ 30″	+	58′ 0″	=	1° 32′ 30″
the latitude of the northern ☾ [a]	34′ 30″	+	8′ 0″	=	0° 42′ 30″,

to which latitudes the distances from the node 17° 58′ 48″.6 and 8° 9′ 16″ answer[2]. To al-Battānī the greatest equation of the sun is 1° 59′ 10″, the greatest simple equation of the moon 5° 1′; so (α) is 2° 34′ 10″. Hence the distances will come out:

by Ptolemy's method				*by the same, corrected*			
17°	59′	8°	9′	17°	59′	8°	9′
	15		30		1		3
2	34	2	34	2	34	2	34
20°	48′	11°	13′	20°	34′	10°	46′

But the distances computed by either method do not agree with the numbers of the codex. I shall take up the computations again, supposing that al-Battānī neglected the small difference he had found in the semidiameter of the sun and used the *Ptolemaic* amount (15′ 40″); the sum of the semidiameters is then 15′ 40″ + 17′ 40″ = 33′ 20″. Let the parallaxes in latitude, 58′ and 8′, be added to it; the apparent latitudes 1° 31′ 20″ to the north and 0° 41′ 20″ to the south will come out, as in the *Almagest*. From them the apparent distances from the node 17° 44′ 45″ and 7° 55′ 45″ are deduced by our logarithmic tables, 17° 45′ 42″ and 7° 55′ 45″ by al-Battānī's tables. But let us suppose that al-Battānī accepted the amounts that Ptolemy had found by a rougher computation, 17° 27′ and 7° 52′. To these latitudes let the parallaxes in longitude, 15′ and 30′, be added; to each sum let the quantity (α) be added according to *al-Battānī's* equations of the luminaries, that is, 2° 34′ (according to Ptolemy's equations 3°); there will come out

17° 27′	+	0° 15′	+	2° 34′	= 20° 16′
7° 52′	+	0° 30′	+	2° 34′	= 10° 56′.

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The first number agrees exactly with the northern limits of the codex; the second agrees with the southern ones if نو 56′ is restored for يو 16′ of the codex, د 4′ for نج 53′ and نز 57′ of the codex, and سمط 349° (as I had already proposed above) for سمه 345° and سمح 348° of the codex. — I ask therefore that here and on p. 88 *the limits of the eclipses of the sun be corrected thus:*

[1] I. de Monteregio and G. Purbachius, *Epitome in Cl. Ptolemaei magnam compositionem*, Basel 1543, book VI, prop. VII, p. 118.

[2] So by the logarithmic tables and by al-Battānī's formula mentioned on the previous page. From al-Battānī's tables, vol. II, p. 78, in which the interpolation is rougher because of the interval of 1° in the argument of the tables, 17° 59′ 48″ and 8° 9′ 18″.

[a] The labels of these two rows are interchanged: the 58′ belong to the northern moon, as the text says above and as the apparent latitudes «1° 31′ 20″ to the north, 0° 41′ 20″ to the south» below; see the note above on the parallaxes in longitude.

159°	44′	0°	0′	*349°*	*4′*
190	*56*	20	16	360	0

Regiomontanus, loc. cit., states that al-Battānī found 20° 12′ and 10° 40′; how he computed these numbers I do not know. — Ḥabash, ff. 122r and 133r, kept Ptolemy's limits.

The limits of the eclipses of the moon could be restored with full certainty:

165°	…	*355°*	*53′*	165°	13′	سنه wrongly for قصه, and نج for يج
194	47	*167*	*44*	194	47	قصز wrongly for قضد, and مد for مز
0	0	0	0	0	0	
14	47	14	47	14	47	
345	*53*	345	*53*	345	13	نج wrongly for يج
360	0	360	0	360	0	

From this it is clear that al-Battānī put the greatest distance of the moon from the node in the *mean* oppositions, for a contact with the shadow, at 14° 47′. The computation is easy. In eclipses he found (vol. I, p. 58) the greatest semidiameter of the moon to be 17′ 40″, the greatest of the shadow 46′ 0″; their sum is 1° 3′ 40″, to which latitude the distance from the node 12° 16′ 23″ answers by our logarithmic tables, 12° 16′ 25″ by al-Battānī's tables, vol. II, p. 78. No doubt al-Battānī accepted the rougher value 12° 13′, very near the 12° 12′ of the *Almagest*; with the quantity (α), or 2° 34′, added to it, 14° 47′ come out. — Strangely Regiomontanus, VI, 8, p. 119, by a conjecture of his own: «Albategni, however, says that the limit is 14 degrees 15 minutes». — Ḥabash, f. 122r, keeps Ptolemy's amount, that is, 15° 12′ from the node.

On pp. 33–54: the geographical tables

They are mentioned in vol. I, pp. 20, 30, 73, 137 (l. 2), that is, in chs. VI, XIV, XXXIV and LVI. — The Arabic text, in which I have kept the numbers of the codex unchanged, is in vol. III, pp. 234–41.

J. Lelewel had already edited them in Arabic and French and explained them, *Géographie du moyen âge*, Breslau 1852, vol. IV, Épilogue, pp. 61–100 (the tables, with notes, pp. 64–93), from a tracing of the Escorial codex made on transparent paper. But since he hardly knew the language and lacked many Arabic aids, the distinguished man could not always propose good emendations and conjectures, nor judge rightly of the place of the tables in the history of geography. The two maps too that he tried to construct after al-Battānī's ideas must be corrected in very many places.

I have edited the same tables in Italian and explained them with many notes and a prefatory dissertation in my *Le tabelle geografiche d'al-Battâni*, Turin 1898, printed separately from Guido Cora's *Cosmos*, series II, vol. XII, pp. 161–83. What I wrote then I give here and in the notes to the tables themselves, enlarged and here and there also corrected.

The longitudes and latitudes of the chief cities are given in all the astronomical works of the Arabs, and in the Handy Tables of Theon of Alexandria. They are missing in the *Almagest*, because Ptolemy, as he himself says in II, 12 (ed. Halma, vol. I, p. 148), intended to publish the Geography separately.

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From the very headings of the tables (vol. II, pp. 33 and 37), and from what he says in vol. I, pp. 19–20, it is plain that al-Battānī took the positions of the cities from the *Book of the Figure of the Earth* (*Kitāb ṣūrat al-arḍ*), in which the longitudes and latitudes had been derived either from astronomical observations or from itineraries «in a way nearest to the truth» (p. 20, ll. 4 and 9). He also adds, p. 20: We have made these geographical tables «following the description (*rasm*)

that we found in the known book *of the Figure of the Earth*[1]; we have also recorded separately the middle places of the provinces and regions, which are 94, as Ptolemy[2] did. But in this book [of the Figure of the Earth] errors of the longitudes and latitudes are found». At the end of chapter XIV too (vol. I, p. 30) he warns: «The latitude of a place found from the table of the latitudes of the cities is only near the truth, and not so certain as a latitude known by observations». That al-Battānī did not put much trust in those longitudes is also clear from the fact that in vol. I he took the intervals between the meridian of al-Raqqa and the meridians of Alexandria and Antioch somewhat different from those that come out of the tables. For he put:

vol. I, p. 56, l. 26:	between al-Raqqa and Antioch	*more than* $10^m = 2^\circ\ 30'$
vol. I, p. 57, ll. 22–23:		*about* 1/4 of an hour = $3^\circ\ 45'$
in the tables, nos. 150 and 142:		$73^\circ\ 15' - 69^\circ\ 0' = 4^\circ\ 15'$

the Ptolemaic longitude of Antioch being kept. On our maps the interval is about $2^\circ\ 55'$. — Likewise

vol. I, p. 42:	between al-Raqqa and Alexandria *about* 2/3 of an hour = $10^\circ\ 0'$
in the tables, nos. 150 and 109:	$73^\circ\ 15' - 60^\circ\ 30' = 12^\circ\ 45'$

the Ptolemaic longitude of Alexandria being kept. On our maps the interval is $9^\circ\ 12'$.

The latitudes of the cities of Mecca (no. 103), Ḥarrān (no. 152) and Baghdad (no. 168), however, he also used in vol. II, p. 67, as they are given in these geographical tables. The position of al-Raqqa he determined by his own astronomical observations.

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Now we must investigate what that «*Book of the Figure of the Earth*, which is called *Geography*», was. Comparing the positions of al-Battānī's tables with those that Abū l-Fidāʾ[3] gives here and there after al-Khwārizmī, Lelewel thought that it was the book *of the Figure of the Earth* written by Muḥammad ibn Mūsā al-Khwārizmī by order of the caliph al-Maʾmūn[4]. But when the matter is weighed better, and the whole book of al-Khwārizmī is read, which now exists in a codex of the Library of Strassburg, it is at once clear that that opinion must be rejected. The Greek names are often written in one way by al-Khwārizmī and in another by al-Battānī; compare the former's *Ghermāniyā, Muwesiya, Ǧanjis, Nighīrā, Qadabathmūs* with the latter's *Ǧehrmāniyā, Muwehsiyā, Ghanjis, Nijīrā, Qāṭābathmūs*. The Greek letter γ is always rendered غ *gh* by al-Khwārizmī[5], by al-Battānī now غ *gh*, now ج *j*: *Ghāliyā, Belghīqī, Yāzūghūs, Lūghdunīsiyā, Eghifṭos, Ghālāṭiyā, Faflāghūniyā, Ghanjis* (no. 89), *Ǧehrmāniyā, Ṭijiṭāniyā, Frūjiyā, Nijīrā, Frūjis* (no. 116). Al-Khwārizmī determines the middle places of 56 provinces in longitude and latitude, al-Battānī those of 93 provinces (which are the ἐπαρχίαι or χῶραι of Ptolemy), differing from al-Khwārizmī in the geographical coordinates too. Moreover, leaving aside eight names (nos. 102, 112, 203, 216, 230, 231, 233, 249) whose reading is too uncertain, 58 cities occur in al-Battānī's tables that are missing in al-Khwārizmī. Of the positions of the 172 cities whose names are certain in al-Battānī's codex, 68 come from Ptolemy's *Geography*[6], 57 from al-Khwārizmī (with 10

[1] Plato adds: «which is called *Geography*».

[2] See vol. I, p. 20, note 2.

[3] *La Géographie d'Aboulféda, texte arabe publié par Reinaud et Mac Guckin de Slane*, Paris 1840 (translated into French by Reinaud and Guyard, Paris 1848–83, 3 vols.); from the work of al-Khwārizmī, which he mostly calls *rasm*, he excerpted the longitudes and latitudes of 92 cities.

[4] His positions rested on Ptolemy's *Greek* maps (not on a Syriac version of the Geography, as our scholars often say), to which he had made many corrections, often excellent ones. See what I have set out in *al-Ḫuwārizmī e il suo rifacimento della Geografia di Tolomeo*, in Memorie della R. Accademia dei Lincei, Classe di scienze morali, series V, vol. II, 1894–95, pp. 3–53. — See vol. I, p. 312, note 4.

[5] *Ǧanjis*, because this is the Arabic form of the name of the river.

[6] Nos. 94–101, 105–111, 113, 114, 116–128, 130, 132–136, 139, 140, 142, 144, 145, 149, 158 (?), 171, 173, 179, 181, 182, 191, 199, 200, 204, 205 (?), 206 (?), 212–215, 217–220, 223–225, 232, 272, 273.

degrees added in the longitudes[1]), and finally 47 from *another Arabic source*, but fitted more or less successfully to Ptolemy's maps[2]. Of the cities not mentioned by al-Khwārizmī, these are the ones whose positions agree with the Ptolemaic[3]: *94, 97, 98, 99, 100, 101, 105, 106, 108, 110,* 111, 114, 134, 139, 140, 171, 173, 179, 191, 200, 204, *205, 212,* 213, *215, 217, 218,* 219, 220, 223, 224, 225, *232, 272*; and these the ones whose longitudes and latitudes come from another source[4]: 137, ∗141, ∗155, ∗157, ∗159, ∗160, 163, ∗165, ∗166, ∗180, 201, 202, 207, 208, ∗209, 211, 221, *222*, ∗226, ∗228, 229, ∗249, 265, ∗266.[a]

So the author of the *Book of the Figure of the Earth* had at hand Ptolemy's Geography and al-Khwārizmī's work, to which he added not a few things. He knew Ptolemy translated into *Syriac*[5], or into *Arabic* from the *Syriac*; for a Syriac origin is betrayed by *eh* for ε in *Brehṭanīqī, Jehrmāniyā, Mehṭānīsā, Muwehsiyā, Qehsarensiyā, Lehfṭis*, by the biblical *Kūsh* for Αἰθιοπία (nos. 45 and 46), and perhaps by *Asdūd* (no. 213) for Azdūd. Two villages are mentioned, *Ḥuwwārīn* (no. 208) and *Ūrim* (no. 204), almost unknown to the Arab geographers but, as seats of bishops, not seldom celebrated by *Syro-Christian* writers. Nor should it be passed over that the geography of chapter VI (see the note in vol. I, pp. 166–77) and the chronology of vol. II, pp. 1–3, also flowed from a *Syriac* archetype, and perhaps the rose of the winds of vol. II, p. 92 as well. Perhaps the «Book of the Figure of the Earth, which is called Geography» is an Arabic recension of Ptolemy's work composed by the very famous Ḥarrānian Thābit ibn Qurra (211–288 H, 826–901), most skilled in Greek and Syriac. For it is likely that in the ninth century Thābit translated Ptolemy's Geography, not without changes, so that it might answer better to the new times, and that al-Battānī chose from it the cities, above all of *Syria* and *Mesopotamia*, that seemed to him the chief ones.

In the passage of chapter VI quoted before and in the heading of the tables (vol. II, p. 33) the provinces whose middle points are given are said to be 94; as many are enumerated in the eighth book of Ptolemy's Geography. But afterwards we find only 93 in the tables; this may have arisen from the negligence of the copyists, or also from some confusion of the author, the cause of which is Ptolemy himself. For in book VIII, § 3 of the general survey, he states that Europe comprises 34 or 35 regions, Libya 12 and Asia 49, whose sum is 95 or 96; but in § 4 he says that they are 87! In the same eighth book, ch. 29, he gives Europe 34 provinces, Libya 12 and Asia 49 (so the sum is 95); but afterwards he names only 46 in Asia, so that the sum becomes 92. Yet in book VIII, ch. 29, § 29, Ptolemy himself makes the sum and finds 94! Nor are other small discrepancies lacking, if all this is compared with the first seven books.

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In the tables that al-Battānī excerpted from the *Book of the Figure of the Earth*, the provinces of *Europe* answer exactly to Ptolemy's, apart from a small difference pointed out in the notes to nos. 15 and 16; of *Africa*, Μερόη is missing (which, however, in the general survey of Ptolemy's fourth book is not held to be an ἐπαρχία distinct from the others), but Numidia, as in the survey of the fourth book, is distinguished from the province of Africa, so that the number of the regions is still 12. Discrepancies and errors also occur in the *Asiatic* provinces. Phrygia and Paphlagonia, which in Ptolemy belong the one to Asia Proper and the other to Galatia, become two separate provinces

[1]Nos. 115, 143, 152, 164, 168, 169 (?), 170, 174–177, 183 (?), 184–187, 188 (?), 190, 194–196, 198, 210, 212, 227, 234–248, 250, 252, 254–261, 262 (?), 263, 264, 267–269, 271.

[2]Nos. 103, 104, 129, 131, 137, 138, 146, 147, 148 (?), 150 (from al-Battānī's observations), 151, 153–157, 159–163, 165–167, 172, 178, 180, 189, 192, 193, 197, 201, 202, 207, 208, 209, 211 (?), 221 (?), 222 (?), 226, 228, 229 (?), 251 (?), 253, 265, 266, 270.

[3]I mark in sloping letters the Greek names that were unknown to almost all the Arabs.

[4]With an asterisk ∗ I mark the cities to which nothing in Ptolemy's Geography answers.

[5]A Syriac version was still extant in the tenth century, as the *Kitāb al-Fihrist*, p. 268, attests. — The first Arabic version seems to be the one that al-Kindī had had made for his own use in the first half of the ninth century (*Fihrist*, loc. cit.).

[a]The three lists of Nallino's notes 1–3 hold 68, 57 and 47 numbers, as the text says, together the 172 certain names; but 212 stands in both the first and the second list, and 141, which the text counts among the cities from another source, stands in none. The text's own list of these cities also holds 249, one of the eight uncertain names set aside above.

in al-Battānī's tables, and their addition compensates for Arabia Deserta and Mesopotamia, which are strangely left out. There are two errors of little importance in nos. 51 and 66; a far greater one in no. 92, because the region of the Ḥimyarites is already contained in no. 75, and in this way the Asiatic provinces become 47 instead of 46. — Other things too must be noted. The provinces of *Europe* and *Africa*, and of *Asia* down to no. 68, are designated by the Greek names unchanged, or at most translated into Arabic; but from no. 69 on, the names of the provinces used by the Arabs are put in place of the Greek ones. Not always rightly, for Ṭabaristān is placed in the far East to represent Σηρική (no. 83), and the latitude of the region of Farghāna is lessened by seven degrees so that it answers to the Paropanisadae. — In the latitudes and longitudes of the provinces from 21 to 93 the codex leaves out the minutes; for what reason I do not know. — The positions of all the provinces are the means between the extreme longitudes and latitudes that Ptolemy assigns to each province.

Of far greater weight than the table of the provinces is the series of the cities, in which the Greek tradition contends with the new geographical studies of the Arabs.

In *Africa* few cities are mentioned, and these, except the Egyptian ones, with names and positions wholly Ptolemaic[1]; Egypt, however, comprises Arabic names and Khwārizmian positions, though both Oases, Diospolis Magna and Alexandria betray a Greek origin. *Europe*, that is, the four cities of Rome, Ṭrāqiya (?), Athens and Constantinople, is wholly Ptolemaic, as are *Asia Minor* (except ʿAmmūriya) and the far *East* of Asia (Thinae and Sera). *India, Turkestan* and *Persia* are taken from al-Khwārizmī, 10 degrees being always added to the longitudes[2]; of the Ptolemaic positions only Herāt (no. 191) and Nisibis in Herāt (no. 232) remain, which thus differ much from the other cities of Persia. In *Armenia* and the *Caucasian* regions the author has al-Khwārizmī as his guide, but he changes his coordinates somewhat, so as to be able to fit those positions in some way with the Ptolemaic ones of Asia Minor and the Khwārizmian ones of Persia. But strangely, together with Dabīl, Jorzān, Taflīs, Bardhaʿa and others, Madīnat al-Abwāb (no. 200), or the Gates of Albania, appears in the same position as in Ptolemy!

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Above all in *Syria*, *Mesopotamia* and *Arabia* the author's hard labour of reconciling Ptolemy with al-Khwārizmī appears. In *Syria* and *Mesopotamia* more than half the positions are Ptolemaic[3]; many others are of Arabic origin but fitted to the Greek ones[4]; only 11 are taken from al-Khwārizmī without changes (except the 10° added to the longitudes), namely al-Ramleh, Āmid, Ḥarrān, Baghdad, Surra man raʾā, al-Kūfa, al-Baṣra, Wāsiṭ, Ḥulwān, Beirut and Hīt. So the position of Beirut (no. 227) is a real monster, if it is compared with the other cities of Syria. — In *Arabia* 17 positions are taken from al-Khwārizmī; but al-Ṭāʾif, al-Jār, al-Yamāma and the unverified place of the city of Mecca show another origin, and cannot be reconciled with the Khwārizmian ones. Of the Ptolemaic map only Sāffārā (no. 97), the island of Serapis (no. 98) and perhaps no. 230 have survived.[a]

It is plain enough that the author of the *Book of the Figure of the Earth*, not daring to reject

[1] *Madīnat al-Ṭīb* (no. 238) must be excepted, which is indeed of Greek origin but taken from al-Khwārizmī, from whom *Donqola* (no. 257) also comes.

[2] The cause of this I shall set out below. — Strangely, *Tubbat* (no. 186) and *al-Mayd* (no. 239) did not receive the increment of longitude; perhaps through an error of the copyists.

[3] Nos. 110–114, 119–128, 130, 132–136, 139, 140, 142, 144, 145, 148, 149, 153, 156, 158, 163 (?), 171, 173, 199, 202–205, 207, 211–213, 219, 222 (?), 223–225, 272, 273.

[4] Nos. 129, 131, 137, 138, 141, 150, 151, 154, 155, 157, 159–162, 180, 201, 206, 208, 209, 220, 221, 226, 228, 229 (?).

[a] These lists do not wholly agree with those of p. 211 (notes 1–3): 148, 153, 156, 163, 202, 207, 211 and 222 stand here among the Ptolemaic positions but there among those from another source, and 206 and 220 the other way round; note 3 here also holds 112 and 203, two of the eight uncertain names, and 119–121, which the list of regions below puts in Asia Minor. Of the eleven cities named in the text, the list of p. 211 puts Āmid (146) among the positions from another source, and the list of regions puts Ḥulwān (198) in Persia; the Syrian and Mesopotamian numbers that p. 211 gives to al-Khwārizmī are eleven too, but with Rāfiya (212) and ʿAbbādān (268) in place of these two. In Arabia that list gives al-Khwārizmī 15 numbers (184, 185, 234–237, 240–247, 254), not 17.

the Greek tradition altogether, did not always blend his sources successfully. Sure traces of this are also found in some later astronomers, who almost copy out the geographical tables of older authors, sometimes keeping uncertain and useless things, as I have shown in the notes to nos. 234, 235, 238, 239 and 240.

To three places, namely Mecca (no. 103), al-Raqqa (no. 150) and Baghdad (no. 168), *mumtaḥan*, «[position] tested, verified», is added. Without doubt the position was verified by his own astronomical observations as regards al-Raqqa; by those of others as regards Baghdad, for the coordinates of this city are the same as al-Khwārizmī's, and the longitude had been so determined already by the astronomers of the caliph al-Ma'mūn in *al-zīj al-mumtaḥan*, «the tested astronomical work», as al-Masʿūdī, *al-Tanbīh*, p. 45, attests. The position of Mecca seems to have been fixed by al-Battānī himself, but I think that its longitude in the codex is affected by an error. — Why al-Battānī did not correct the position of Antioch by his own observations I do not know.

It remains for me to say something about the *prime meridian*. In my dissertation *Al-Ḫuwârizmî e il suo rifacimento* etc., pp. 24–25, I showed that al-Khwārizmī's prime meridian is the same as Ptolemy's meridian of the *Fortunate Isles*; but that, since the Arab had most happily lessened Ptolemy's longitude of the Mediterranean Sea by 9° or 10°, and by about 10° the positions east of this sea as well, the later Arab geographers wrongly thought that *al-Khwārizmī's prime meridian differed from Ptolemy's* and passed along the coasts of *Western Africa*, 10° east of the Fortunate Isles. — Al-Battānī says, vol. I, pp. 19 and 20, that the longitudes are counted «from the Inhabited Islands (*al-jazā'ir al-ʿāmira*), which are in the sea of the Western Ocean (*ūqiyānus al-gharbī*)»; but in these tables the longitudes of all the places taken from al-Khwārizmī are always *increased* by 10°[1a]. Hence it is plain that already the author of the *Book of the Figure of the Earth* wrongly believed al-Khwārizmī's prime meridian to differ from Ptolemy's, and in this way set up a new and false meridian of the Fortunate Isles, which made void the excellent correction that al-Khwārizmī had made to the longitude of the Mediterranean. In this way, indeed, the author of the *Book of the Figure of the Earth* could reconcile al-Khwārizmī's longitudes with Ptolemy's well enough; but he would have achieved the same thing, and far better, if he had kept al-Khwārizmī's longitudes and lessened Ptolemy's by about 10 degrees east of the Mediterranean.

Since the cities are mentioned in these tables in no fixed order, it seemed good to add a survey of them, distributed by regions.

Europe

Athens	217	Rome	181
Constantinople	182	Ṭrāqiya (?)	218

Africa

Akhmīm	250	Donqola	257	Nigira	95
Alexandria	109	al-Faramā	195	Oasis maior	108
Auxume	215	al-Fusṭāṭ	178	Oasis minor	101
Catabathmus magnus	107	Gira metropolis	94	al-Qolzum	252
Carthage	105	Iarzitha (? سونا ثليا)	96	Qūṣ	251
Dimyāṭ	177	Leptis magna	106	Syene	188
Diospolis magna	100	Madīnat al-Ṭīb	238	Tinnīs	194

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Asia Minor

Adhana	120	Malaṭya	143	Ṭarasūs	119
ʿAmmūriya	183	al-Maṣṣīṣa	121	Trapezus	265
ʿAyn Zarba	179	Rhodes	117	Zibaṭra	214
Laodicea Phrygia	116	Salamis in Cyprus (?)	118		

[1]See p. 211, note 2.

[a]The print has «p. 21»; the note meant is note 2 of p. 211, the list of the positions taken from al-Khwārizmī.

Armenia

Bardhaʿa	167	Jorzān	187	Madīnat al-abwāb	200
Dabīl	165	Khilāṭ	164	Taflīs	166

Arabia

ʿAden	185	Mārā	235	Sapphara	97
Alqis (Ocelis?)	234	Mahra	244	Island of Serapis	98
al-Baḥrayn	246	Mecca	103	Tabāla	245
برهور	236	مصره	249	al-Ṭāʾif	193
al-Jār	253	المعلى	240	al-Yamāma	192
Jorash	243	ʿOmān	247	Yathrib (Medina)	104
Ḥaḍramūt	237	Sabaʾ	242	Ẓafār	241
Hajar	254	Ṣanʿāʾ	184		

Syria and Palestine

ʿAkkā	127	Jisr Anṭākiya	180	al-Ramleh	115
ʿAmawās	211	Jubayl	202	al-Rastan	129
Antioch	142	Ḥaleb (Aleppo)	136	al-Ruṣāfa	201
Aradus	272	Ḥamāh	130	Rūsis	225
Asdūd	213	Ḥimṣ	128	Sabasṭiya	114
ʿAsqalān	113	Hīt	271	Salamiyya	131
Baʿlabakk	134	Ḥuwwārīn	208	Sanja	223
Bayt al-Maqdis (Jerusalem)	273	ʿIrqa (ʿArqa)	124	Ṣaydāʾ (Sidon)	126
Bayt Jibrīn	228	al-Iskandarūna	219	Shayzar	206
Beirut	227	Laodicea	122	Ṣūr (Tyre)	125
Caesarea Panias	110	Maʿarrat al-Nuʿmān	138	Tadmur (Palmyra)	135
Damascus	133	Orthosia	222	Tell Mannas	207
Dolūk	140	Qārā	226	Tripolis	123
Fāmiya	132	Qinnasrīn	137	Ūrim	204
Filasṭīn	111	Qūros	139	Wajh al-ḥajar	221
Jabala	224	Raʿbān	141	Zoghma	205
Jindāros	220	Rāfiya	212		

Mesopotamia

ʿAbbādān	268	Ḥarrān	152	Raʾs al-ʿayn	156
Āmid	146	Kafar Tūthā	157	al-Rohāʾ (Edessa)	153
al-ʿĀqūl	209	al-Kūfa	170	Shimshāṭ	144
Arzan	147	Manbij	154	Sinjār	163
Babel	171	Māridīn	160	Sumaysāṭ	148
Baghdad	168	al-Mawṣil (Mosul)	162	Sūrā	229
Balad	161	Mayyāfāriqīn	145	Surra man raʾā (Sāmarrā)	169
Bālis	149	Naṣībīn	158	Tell Mawzan	155
al-Baṣra	174	Nineveh	173	Wāsiṭ	176
Ctesiphon (al-Madāʾin)	199	Qirqīsiyāʾ	151		
Dārā	159	al-Raqqa	150		

Persia

Āmul	263	Khuwayy	266	al-Rūyān	258
Donbāwend	262	al-Muḥammadiyya	259	Sarakhs	270
Jiruft	255	Nisibis Ariae	232	Sāriya	264
Hamadhān	210	Qaṣr al-milḥ	260	Sīrāf	175
Herāt	191	Qazwīn	190	al-Sīrajān	261
Ḥulwān	198	Qumm	197	Ṭūs	269
Kābul	256	al-Rayy	172		

India and Eastern Asia

al-Daybul	189	Osrūshana	267	Tubbat	186
al-Mayd	239	Sīrās	231	al-Ṭurāraband	196
al-Nīrūn	248	Thinae	99		

Uncertain

بجلزا or بجلنرا	102	ارام	230	بلد اور	233
سقراطاوس	112	جزيل	203	دافا مدينة الفرس	216

How I have corrected the numbers of the codex is plain from the notes themselves; the *Addenda et emendanda* should also be consulted. What Lelewel had already happily conjectured I have always mentioned in its place; I thought it useless also to point out the errors of that distinguished man. I have used Ptolemy's *Geography* in the edition of C. Fr. A. Nobbe, stereotype edition, Leipzig 1881 (3 vols., 18mo); I have used Ibn Yūnus from the Leiden codex (pp. 133–36). The Greek names never current among the Arabic writers I have always marked with an asterisk ⁎.

The Spanish version (f. 42v), which it is worth reporting here, excerpted 129 cities from al-Battānī's table, putting before them three other cities of Spain that are missing in al-Battānī. To the cities I add their order numbers in the Arabic text; longitudes and latitudes that are wrong and differ from the Escorial codex I mark with an asterisk ⁎; those that are wrong but shared with the Escorial codex I indicate by sloping or italic figures.

Cordoua despanna	9°	9′	38°	20′	56. Tarçoç	68°	37°	
Toledo	8		39	20	62. ysla de cabroz	66	35	*30*
çaragoça despanna	11		42		63. Çorya	71	36	
16. Dalmathya	46		42		66. Tierra de arabigos	⁎60	29	⁎31
17. ysla de ythalia	36	40	41	40	68. Assur	80	37	
19. ysla de sardenna	31	⁎	38	⁎	70. açuç	83	34	
20. cezilia	39	⁎	37	⁎	71. perssia	90	32	
35. Taniar	8		32		72. azpahen	96	⁎34	
37. affrica	36		31		74. poblado de carmen	99	25	
45. cuç desuso degipto	62		*36*		76. Iorgen	95	⁎48	
46. Cuç la de dentro	50		⁎42		78. Balach	116	41	
55. arminna la menor	71		39		79. açooth	114	45	

17. *Ysla*, because in Arabic *jazīra* means both island and peninsula.
66. The minutes of latitude are 31′, perhaps because at this place the Spanish translator had found in his exemplar, to designate 0′, the Arabic negation لا (as also happens three or four times in the Escorial codex), which he thought should be taken at its numerical value.
72. Lat. 34° (so لد), in the Escorial codex 37° لز. Ptolemy, VI, 5, sets the extreme latitudes at 33° and 39°; so the true reading is 36° لو, which should be restored on p. 35.
79. He read the name as الصعد for الصغد.

81. Turcos de dentro	120°		⁎16°		127. acre	66°	50′	33°	
82. los turcos de fuera	⁎107		⁎43		128. hemz	69	5	34	
83. Tabarçathen	⁎162		45		130. hama	69	30	*35*	20
85. fragana	116		35		133. Damascho	69		33	
86. çarazthen	108		29		134. balbach	68	20	33	15
87. Tierras dazinge	115		29		136. halab	71		⁎35	50
88. Tierras daçindi	110		23		138. magarat anaomen	69	55	34	
89. India dalle del rio	132		⁎26		139. coriz	70	*50*	⁎34	20
90. India daquen del Ryo	⁎180		22		142. Anthiochia	69		35	30
91. Isla de çarandib	124		3		143. malthya	71		39	
92. comedio de hemyar	*102*		12	30	144. Simxath	73	20	38	40
93. Tierras daçin	177		22		150. Arraca	73	15	36	⁎47
102. Hara	43		23	30	152. harran	73		36	40
103. mecha	71		21	40	153. azuhe	72	⁎	⁎50	⁎37
109. alexandria degipto	60	30	30	58	156. Raçaalahyn	74		36	50
104. yaçrib de mohomat	75		25		157. cafartota	74	35	37	5
111. falaztyn	66	15	32	30	159. Dara	75	15	37	10
113. azcalen	65		31	50	160. maraditzh	75		37	15
115. arrambla	65	⁎7	31	*35*	162. almauçil	78	10	36	30
117. ysla de Rodyz	58	40	36		163. çiniar	77	30	36	
118. Çalmaoz	⁎46	45	35	⁎50	164. hylach	78		39	20
119. Tarçoz	⁎57	40	36	⁎50	167. bardaha	⁎104		42	
120. Edane	68	15	36	50	168. baldach	80		33	9
121. Almaçeyça	*67*	50	36	45	169. curramanrae	79	45	34	
122. aladaquia	68	30	35	5	170. Alcufa	79	30	31	30
123. Tripolo	67	30	34	20	171. babillonia la antigua	79		35	
124. Tarcha	68	30	34		172. Arryo	⁎96		⁎37	30
125. Çor	67		33	20	173. Niniue	78		36	⁎49
126. Çayde	⁎66	20	33	30	174. Albaçra	⁎80	⁎10	31	
					175. cayraf	⁎109	30	29	30
					176. Vaçith	⁎101	30	⁎34	30

87. He read the name as الزنج *al-Zinj* for الرنج *al-Rukhkhaj*.
102. He seems to have read the uncertain name as حارا.
103. He leaves out the tested longitude of the city.
109. The minutes of latitude, 58′ نخ, are far better than the 18′ يح of the Escorial codex, which I kept, not rightly, on p. 38.
118. He found the name written as in the Escorial codex.
124. The corruption of the name seems to be due to the Spanish copyist.
153. He read the name as الزها for الرها; the numbers were wrongly placed by some copyist, for 72° 50′, 37° 0′.
164. The corruption of the name (for *hylath*) is due to the Spanish copyist.
169. The minutes of longitude as in al-Khwārizmī; less rightly the Escorial codex has 10′, which I kept on p. 42.
175. The minutes of latitude are more correct than the 29′ of the Escorial codex, which I kept on p. 43, and which arose from the number of the degrees wrongly repeated.
176. The degrees of latitude confirm my correction on p. 43: the Escorial codex has 30° ل, the Spanish translator 34° لد; the true reading is 32° لب (= لب).

177. Damiatha	63°	50′	31°	25′	225. Roçyz	69°	20′	35°	40′
178. alfiztath	63		*31*		227. bayroth	69	30	*33*	*20*
181. Rama [sic!]	36	40	41	⁎45	235. Tierra del ayman	⁎74		15	15
182. costantinopla	56	40	43	10	236. çipdat en alyaman	79		⁎33	
183. amorya	⁎53	20	39	45	237. hadramauth	⁎101		14	30
184. çanaaha	73		⁎64	30	238. medina datep	⁎102		*4*	30
185. aaden	74		*33*	*38*	241. Dyffar	88		⁎35	
188. Tierra de los negros	65		22	30	242. çabaa	⁎94		17	
191. hara	⁎104		37		244. mahra	74		18	30
192. Alyamema	⁎72		21	30	245. Tabela	77		19	
193. Altayf	74	30	21	20	246. albahrayn	⁎104	20	25	45
194. Tannyz	64		31	20	247. Aamen	⁎90	30	19	45
195. Alfarme	64	40	31	⁎35	248. albayroz	⁎110	20	⁎28	30
198. haluen	⁎101		*38*		250. Ahmym	65	30	25	30
200. çipdat de las puertas	80		47		251. cothz	⁎64	30	24	30
201. Arroçaffa	72	30	35	40	252. Alculzum	⁎46	30	⁎37	30
210. hamaden	*60*	⁎4	⁎39	0	253. Aliar	⁎96	30	24	
219. Alexandria de persia	69	30	36	10	254. haiar	83	20	⁎7	15
221. vasalhaiar	67	⁎55	34	15	256. quebul	110		28	
222. arçocia	67	35	34	20	257. diola dal nuba	63		14	15
					259. Almuhamadia	100		31	45
					260. cazhr almahelx	91		32	
					261. açabrahan	⁎63		⁎34	⁎30

177. The minutes of longitude, 50′ ن, are better than the 30′ ل of the Escorial codex, which I kept; al-Khwārizmī has 55′.
183. The numbers, better than those of the Escorial codex, agree for the most part with my corrections.
188. He had before his eyes the same reading as the Escorial codex, سواد «the territory and fields of a town», for سوان *Suwān*.
200. A rendering of the Arabic name.
201. The degrees of longitude confirm my correction; I do not know whether the minutes are more correct than the 30′ of the Escorial codex, which I kept.
219. «De Persia» was wrongly added by the translator. The minutes of longitude, which agree with Ptolemy's, are more correct than the 0′ of the Escorial codex, which I kept on p. 48.
221. The latitude confirms my corrections (but with 15′ for my 10′).
235. The translator left out the name *Mārā*, which was unknown to him. The minutes of latitude, 15′ يه, are better than the 55′ نه of the Escorial codex, which I kept.
236. The translator left out the unknown name.
238. On p. 50 the 81 degrees of longitude are a slip of the compositor for 82°.
242. The latitude confirms my correction.
248. He read the name as البيروز (Escorial codex النيروز). — The degrees of longitude, 110° قى, are wrong, but they confirm the correction proposed in my note (102° قب = قب).
250. The degrees of longitude are more correct than those of the Escorial codex, which I kept; the degrees of latitude are as I corrected them.
251. I do not know whether the degrees of latitude are altogether right, but they are far better than those of the Escorial codex.
253. The degrees of longitude are wrong through the carelessness of the Spanish copyist (96° for 69°); the latitude, wrong in the Escorial codex, can now be restored with full certainty on p. 52.
257. The name was wrongly read as ديولة for دنقلة.
261. The name was wrongly read as السبرحان for السيرجان. — The minutes of latitude are rightly لا in the Escorial codex (the Arabic negation = 0′; see a little above, on no. 66); the translator read ل 30′.

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262. denaonth	*135°*	30′	36°	⁎45′	269. Tothz	⁎62°		34°	
267. çucia	⁎81	10	36	40	273. Iherusalem	66	30	31	50
268. ahbeden	⁎44	55	31						

262. The degrees of latitude confirm my correction; I do not know whether the minutes are better than those of the Escorial codex (35′), which I kept.
267. He read the name as سوسية for اسروشنة (Escorial codex سروسيه).
269. The latitude 34° لد seems to confirm my correction 37° لز.

It is plain enough that the numbers of the Spanish version teem with more frequent corruptions than those of the Escorial codex, and that the greatest part of these arises from the Arabic script. In nos. 109, 169, 175, 177, 183, 201, 219, 221, 235, 242, 248, 250, 251, 253 and 262, however, better readings occur.

On f. 176r (= vol. III, pp. 241–242), under the title «*Longitudes and latitudes of the known and tested cities and towns of al-Andalus* [i.e. of Spain] *and al-Maghrib* [i.e. of Mauretania]», two series of cities are given, left out in the Spanish version, whose names by themselves show clearly enough that the table was *interpolated*[1] by the *western Arab* scribe of the codex, who seems to have lived in the second half of the eleventh century. For (as also on ff. 184r and 185v; see the note on pp. 68–71 of the tables) he mentions the city of *Bijāya*, founded in the year 457 H (which began on 12 Dec. 1064) by al-Nāṣir ibn ʿAlnās on a site where there had been a village of shepherds before, and made the seat of the Ḥammādid dynasty; likewise *Qalʿat Banī Ḥammād*[2a], founded in the last twenty years of the fourth century H (the tenth AD) by Ḥammād ibn Yūsuf Bulukkīn, and the capital of the kingdom until Bijāya was built; but he leaves out *Marrākush*, or Morocco, built in the year 454 H (which began on 14 Jan. 1062) or a little later by Yūsuf ibn Tāshufīn, and afterwards the chief city of the Almoravids, and *Tilimsān* (Tlemcen), a most flourishing town already in the twelfth century.

The places mentioned are those that are usually marked on Hispano-Arabic and African astrolabes. — For correcting the numbers of the first series, which are fairly good in the codex, it helps most to compare the geographical tables of Abraham bar Ḥiyya[3] and of Aboul Hhassan[4], whom I mark with the sigla *Abr.* and *Abou.*

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[1] As Lelewel, loc. cit., pp. 95–97, already recognized, although not everything he set out is sound.

[2] It lay 12 Arab miles, or about 23½ km, north-east of al-Masīla (Msila), on the rocks above the river now called Wād Selmān (which flows into the Shoṭṭ al-Ḥoḍna); its great ruins, called *Qaṣr al-manār*, were described by Blanchet in the *Comptes-rendus de l'Acad. des Inscriptions*, 3 Sept. 1897.

[3] A table of 23 cities is contained in his Hebrew work *Sēpher hā-ʿibbūr* (Abraham bar Chyiah, *Mathematical and technical chronology of the Hebrews, Nazarites, Mahometans etc. edited by H. Filipowski*, London 1851, p. 120); it is translated into French in Lelewel, vol. I, Prolégomènes, p. L. — The Latins called him Savasorda; he wrote in Spain in the first half of the twelfth century. See M. Steinschneider, *Die Mathematik bei den Juden*, in Bibliotheca Mathematica, ed. G. Eneström, Neue Folge, vol. X, Stockholm 1896, pp. 33–42.

[4] He gives the latitudes in ch. XXVI, pp. 202–204, and the longitudes in ch. LXVI, pp. 315–317. All of them are also in Lelewel, Atlas, pp. 6–7.

[a] Nallino's Addenda et emendanda (Part II, p. XXX) add that G. Marçais, *Revue de l'art musulman en Berbérie* (Revue Africaine, 1906, pp. 412–14), briefly sets out the more recent explorations of the ruins of Qalʿat Banī Ḥammād.

Cities	*Longitudes*		*Latitudes*		*Notes*
Aṭrābulus Barqa	41°	40′	32°	0′	*Tripoli of Africa.* — Al-Khwārizmī 40° 40′, 32° 0′; Ibn Yūnus 40° 40′, 33° 0′; Abou. 48° 30′, 33° 15′.
Qarṭājanna of Spain	*33*	*45*	52	0	*Cartagena.* — The longitude is faulty; the latitude agrees badly with the rest. Ibn Yūnus 27° 0′, 52° 0′.
Qurṭuba	27	0	38	*38*	*Córdoba.* — The minutes of latitude should be restored as 30′; then everything will agree with Abr. and Abou.
Ishbīliya	25	40	37	15	*Seville.* — As Abr. and Abou.
Ṭulayṭila	28	0	40	0	*Toledo.* — As Abou.; Abr. 28° 15′, 39° 52′[a].
Gharnāṭa	27	30	37	30	*Granada.* — As Abr. and Abou.
Shantarīn	23	40	*45*	15	*Santarém.* — The longitude as Abr. and Abou. The degrees of latitude must be corrected to 40°; Abr. 40° 15′, Abou. 40° 0′.
Mālaqa[1]	26	22	37	0	*Málaga.* — As Abr.; Abou. 26° 20′, 37° 0′.
al-Mariyya[2]	28	0	36	30	*Almería.* — As Abr. and Abou.
Mursiya	29	*29*	37	*37*	*Murcia.* — The minutes of longitude and of latitude should be restored as 30′, as Abr. and Abou. have.
Balansiya	30	30	37	*37*	*Valencia.* — Abr. 30° 20′, 36° 25′; Abou. 30° 50′, 37° 30′.
Saraqusṭa	29	55	41	30	*Zaragoza.* — As Abou.
Ṭanja	24	10	35	*55*	*Tangier.* — In the latitude يه 15′ should be restored for نه 55′. Abou. 24° 10′, 35° 10′; the latitude in Arzachel[3] and Ibn Yūnus 35° 15′.
Fās	25	0	33	0	*Fez.* — As Abr.; Abou. 24° 0′, 33° 0′.
Sabta	25	40	35	20	*Ceuta.* — As Abr. and Abou.
Bijāya	36	0	36	30	*Bougie.* — Abou. 36° 5′, 36° 0′.
al-Qalʿa	*36*	*17*	*36*	*37*	Or *Qalʿat Banī Ḥammād.* — The numbers are faulty; Abou. 35° 40′, 34° 0′.

The *second series of cities*, in which Saraqusṭa and Tudmīr (or Mursiya) occur again, is manifestly put together from *two different sources*: one of them gave the longitudes after those who (like al-Khwārizmī, Ibn Yūnus, Arzachel and in part Ibn Saʿīd) followed Ptolemy above all in the far west of Africa and in Spain; the other gave longitudes agreeing with those of the first series. II 220

[1]Some Arab writers hold that it should be pronounced *Māliqa*; but the *Tāj al-ʿarūs*, VII, 73, rightly says: ومالقة بفتح اللام والعامة تكسرها قال الصاغاني وهو غلط واكثر الاندلسيين يضبطونه بفتحها («and Mālaqa, with *a* after the *l*; the common people say it with *i*; al-Ṣaghānī said that this is an error, and most of the Andalusians vocalize it with *a*»).

[2]Yāqūt and others prescribe the spelling *al-Murriyya*, which I took into the text; but the true reading is *al-Mariyya*, «a watch-tower from which approaching enemies are observed», as Dozy established (Edrisi, *Description de l'Afrique et de l'Espagne, texte arabe publié avec une traduction, des notes et un glossaire par R. Dozy et M. J. de Goeje*, Leiden 1866, p. 243, note).

[3]From the *Canones super tabulas Toletanas* of Arzachel (al-Zarqālī), translated into Latin by Gerard of Cremona. Lelewel edited the geographical part from the Paris codex 7421, Atlas, p. 16 (see vol. II, Cartes de géographes, pp. 78–80).

[a]The print has «39° 52°».

Cities	*Longitudes*		*Latitudes*		*Notes*
Salāʾ	*10°*	*0′*	*27°*	*0′*	*Salé.* — Probably everything should be corrected thus: 6° 0′, 34° 0′.
Aṣīlāʾ	*26*	15	35	15	*Arzila.* — The longitude is faulty; perhaps 6° 15′.
Mārida	8	0	41	55	*Mérida.* — As Arzachel; compare Ptolemy's Αὐγούστα Ἠμεριτα, 8° 0′, 39° 30′.
Madīnat Sālim	9	30	40	15	*Medinaceli.*
Ṭurṭūsha	32	0	*37*	15	*Tortosa.* — In the latitude perhaps م 40° should be restored; Ptolemy (Δερτῶσα) and Ibn Saʿīd have the latitude 40° 0′.
Saraqusṭa	8	30	41	30	*Zaragoza.*
Tudmīr	10	15	37	30	*Murcia.*
Ghāna[1]	*10*	*0*	*17*	*0*	The numbers seem faulty. Ibn Yūnus 15° 30′, 10° 45′; Abou. 18° 0′, 10° 0′; so perhaps the longitude and the latitude should be interchanged.
Jayyān	28	0	38	0	*Jaén.* — As in the *Liber tabularum Jahen* of an unknown author, translated into Latin by Gerard of Cremona[2]; Abou. 27° 30′, 38° 30′.
al-Jazīra al-khaḍrāʾ	26	0	36	30	*Algeciras.* — Abou. 25° 40′, 36° 0′.
Baṭalyūs	*25*	20	39	30	*Badajoz.* — Abou. 24° 32′, 39° 30′; in the longitude perhaps ك 24° should be put for ك 25°.
Qalʿat Rabāḥ	9	20	39	30	*Calatrava.*

Those that belong, so to speak, to the *Ptolemaic* source in their longitudes are Salé, Arzila, Mérida, Medinaceli, Zaragoza, Murcia and Calatrava.

On pp. 55–56: the tables of sines

They are mentioned in vol. I, pp. 10–11 (ch. III), and passim.

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I have easily corrected the errors of the codex (74 in 180 numbers of minutes and seconds!) by comparing al-Battānī's sines with the chords of the *Almagest* (I, 9, ed. Halma; vol. I, pp. 38–45), as already corrected by Delambre (see ed. Halma, notes, p. 2). To make the correction more certain I also used the table of sines computed for every 15′ of arc by Aboul Hhassan, pp. 103–114; wherever any doubt arose, I did the computations again with our logarithmic tables. — Everything agrees very well with Ptolemy; there are occasional discrepancies of +1″ from Aboul Hhassan, in whose table, among the sines of the whole and half degrees (which alone I examined), the following must be corrected:

sin 24° 0′	= 24^p 24′ 55″	for the edition's	24^p 24′ *15″*
sin 60° 30′	= 52^p 12′ 46″	»	52^p *13′ 17″*
sin 71° 0′	= 56^p 33′ 52″[a]	»	56^p *43′* 52″.

[1] A city and region of the western Sūdān, once very famous among the Arabs. The city is either the present-day *Walāta* ولاتة or one very near it; the kingdom, which stretched from the Atlantic to the Niger and was founded about AD 300, was subdued by the Mandingo people in 1203–1204. See P. C. Meyer, *Erforschungsgeschichte und Staatenbildungen des Westsudan*, Gotha 1897, pp. 59–61 (Ergänzungsheft no. 121 zu Petermann's Mitteilungen). — Deceived by some likeness of the name, some have thought that Ghāna is the same city as *Jennē* جنى, at the place where the Baulē flows into the right bank of the Niger.

[2] In Steinschneider, *Intorno ad alcuni matematici del medio evo ed alle opere da essi composte, lettere a D. B. Boncompagni*, Rome 1863–67, Lettera II: *intorno al libro Saraceni cuiusdam de eris ed al libro Tabule Jahen*, p. 10, note 3. — On the uncertain author of the book see Suter, *Nachträge und Berichtigungen*, p. 170.

On pp. 57–58: the tables of the declination of the sun

They are mentioned in vol. I, p. 13, ch. IV.

The tables of Ptolemy, I, 12 (ed. Halma, vol. I, p. 59), and of Theon are computed for the obliquity 23° 51′ 20″; so Schiaparelli could correct al-Battānī's tables only by computations made with our logarithmic tables, neglecting discrepancies of 1″ or 2″. With logarithms it is enough to compute the declination for the longitude 1° ($\sin\delta = \sin L \sin\varepsilon$); the other terms are derived by adding to the first logarithm, one after another, the logarithmic differences of the sines for the single degrees; see M. Delambre's *notes* in Halma, vol. I, notes pp. 5–6.

P. 58, *the orders of the declination.* The small table is mentioned at the end of appendix G (vol. I, p. 150); on these orders see ch. IV, vol. I, p. 13, and also my notes, vol. I, p. 13, note 2, and p. LXXI. In the figure of the codex the words *ascent* and *descent* are wrongly interchanged, so that near both sides of the circle the copyist wrote منكوس «to be inverted»; the Spanish version (f. 43v) has the same errors, and renders the word *mankūs* by «conuerso» (turned round).

P. 58, *the right ascensions of the tens of degrees and the corresponding sines.* They are mentioned in appendix G, vol. I, p. 150. The ascensions are counted from the beginning of Aries, not from the beginning of Capricorn as in the tables of vol. II, pp. 61–64 (see the note in vol. I, pp. 163–64, and the Addenda). In the heading of the third column the codex wrongly has ازمان ..... المنصّفة «halved times» instead of اوتار ..... المنصّفة «halved chords», that is, sines. — Ptolemy too, II, 8 (ed. Halma, vol. I, p. 103), gives a table of the right ascensions for the tens of degrees, but without the sines and for another obliquity.

On p. 59: the lengths of the days for the different latitudes

See chs. VIII and IX, vol. I, pp. 21–22.

The table was corrected by Schiaparelli. The increments of the days are given in degrees of the equator, 15° of which make an equinoctial hour.

On p. 60: the table of shadows

It is mentioned in ch. X, vol. I, pp. 22–23.

They are cotangents computed to a radius of 12 parts; with logarithmic tables I could easily heal the errors of the codex. The first two shadows are 687^p 26′ and 663^p 39′ in the codex (خصجˢ for سمج 343^p); but perhaps these small discrepancies in the minutes come from al-Battānī himself, who may have derived those minutes by dividing sines computed only to the seconds of a radius of 60 parts.

On pp. 61–64: the right ascensions and the equations of days

The tables are mentioned in chs. XIII, XXIX, XXXIV and XLII, and in appendix E (where the use of the tables is taught), that is, vol. I, pp. 28, 49, 72–73, 95 and 148.

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The right ascensions are counted from the *beginning of Capricorn*; to have them from the beginning of *Aries*, 90° must be taken from the ascensions of the table. See vol. I, pp. 163–64, and

[a] So the print; but to the radius 60, sin 24° = 24^p 24′ 15″, sin 60° 30′ = 52^p 13′ 17″ and sin 71° = 56^p 43′ 52″, the values given here as the edition's. The two columns appear to be interchanged.

[a] The print has حصج, the first letter without its dot. In the western numerals of the codex خصج is 663 (خ 600, ص 60, ج 3) and سمج is 343.

the Addenda. — Ptolemy put in his tables only the right ascensions of the tens of degrees, as I pointed out above, on p. 58.

A table of the equations of time is missing in Ptolemy; how that equation is to be understood in al-Battānī (namely as always subtractive) has been said in the notes, vol. I, pp. 221–22.

On pp. 65–66: the oblique ascensions for the single climates

The tables are mentioned in chs. XIII and XXXV, vol. I, pp. 27–28, 29 and 73.

They are counted from the beginning of *Aries*, not from Capricorn like the right ascensions. They differ from *Ptolemy's*, II, 8 (ed. Halma, vol. I, pp. 104–108), which are also computed for the tens of degrees, because of the different obliquity of the ecliptic and hence the different latitudes of the climates. — There is a certain proportion in the increase of the ascensions answering to an interval of a quarter of an hour in the length of the day, so that it was not hard to heal the errors of the codex. In some places where there was doubt I made the computations with al-Battānī's tables of the declinations and of the right ascensions and with the formula $\sin m = \tan \varphi \tan \delta$.

On p. 67: the oblique ascensions of the cities of Mecca, Baghdad and Ḥarrān

The «hourly times» (*azmān al-sāʿāt*, ὡριαῖοι χρόνοι) are the lengths of the temporal hours given in degrees of the equator, 15° of which make an equinoctial hour. — The following f. 183r of the codex was blank.

On pp. 68–71: the oblique ascensions of the city of al-Raqqa

The tables are mentioned in chs. XIII and XXXV, vol. I, pp. 27–28, 29 and 73.

In the heading the African Arab interpolator added, on f. 184r: وهي مطالع بجاية والمرية «and these are the ascensions of [the cities of] Bijāya (Bougie) and Almería»; on ff. 184v–185r: وما حاذاها مشرق ومغرب «and of what lies in line with them [in] the East and the West»; on f. 185v: وبجاية والمرية وما حاذاها «and of Bijāya and Almería and of what lies in line with them». See what I said on p. 218.

P. 71, last line: the hourly times of 30° of Aquarius should be يج لد 13° 34′ instead of the يج لج 13° 33′ of the codex and of the translation.

On pp. 72–77: the mean motions of the sun, the moon and the node for the Roman calendar

The tables are mentioned in chs. XXVII, XXXIII and XXXVII, vol. I, pp. 42, 71 and 75.

On the headings of the columns the same must be said as I have already set out on the similar tables of pp. 19–23.

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The era is that of Dhū l-qarnayn, or of the Seleucids, but with a fictitious year beginning on 1 *Ādhār* (March) of the *preceding* year, so that no inconvenience may arise from the Julian intercalation; see vol. I, p. 246. This use of al-Battānī's tables is mentioned by Samps Puchares (on whom see vol. I, p. XXX, note 5), in a passage quoted in the *Catal. codd. astrol. Graecorum*, vol. I, p. 88: Καὶ τὴν ἀρχὴν τοῦ χρόνου[1] ἐκείνου [i. e. Ἀλεξάνδρου τοῦ υἱοῦ Φιλίππου, οὗτος Δουλκαρνάην ὀνομάζεται] τινὲς ἀπὸ τοῦ ´ κρατοῦσι· τὸ ὄνομα δὲ τοῦ μηνὸς ἐκείνων Ἀδάρ, διὰ τοῦτο ὅτι ἡ ἀρχὴ τοῦ κόσμου εἰς τοῦτον τὸν μῆνα ἀνακαινίζεται ἤτοι λαμβάνει ἀρχήν. διὰ τοῦτο καὶ ὁ Πατανῆς [= al-Battānī] τὴν ὁδὸν τῶν ἀστέρων εἰς τὴν σύνταξιν αὐτοῦ ἐν τούτῳ

[1] In the Byzantine writers χρόνος means *year*.

τῷ ἔτει[1] τέθεικε, τὴν ἀρχὴν δὲ τοῦ ἔτους αὐτοῦ ἀπὸ τοῦ Ἀδὰρ ἔθηκε («And some reckon the beginning of that year [i.e. of Alexander, the son of Philip; he is called Dhū l-qarnayn] from the …; and the name of their month is Adar, because the beginning of the world is renewed in this month, that is, takes its beginning. For this reason Patanēs [= al-Battānī] too set down the course of the stars in his treatise in this period, and put the beginning of his year from Adar»).

On the amount of the motion of the moon see vol. I, p. 225; on the fundamental epoch see what Schiaparelli said in vol. I, p. 246.

P. 74. The codex gives *Subāṭ* (February) only for the common years; I have chosen to add the Subāṭ of the leap years too, as I had done for Dhū l-Ḥijja on p. 21 (see the note on p. 204).

Pp. 75–76. See what I said on the like tables of pp. 22–23, in the notes on p. 204.

P. 77. The motions of 20 years are left out in the codex, perhaps because they are given in the headings (except the motion of the node).

On pp. 78–83: the equations of the sun and the moon; the latitude of the moon

The tables of the equations are mentioned in chs. XXVII, XXX, XXXIII, XXXVI and XLII (vol. I, pp. 47, 51, 52, 53–54, 72, 75, 93), that of the latitude in chs. XXXVIII and XLIII (vol. I, pp. 76 and 97).

Col. II: *the equation of the sun.* — It differs from the table of Ptolemy, III, 5 (ed. Halma, vol. I, p. 201), because of the different amount of the greatest equation, of which I spoke in vol. I, pp. 213–14. Ptolemy, moreover, gives the argument of the table (the mean anomaly) at intervals of 6 degrees in the first and fourth quadrants and of 3 degrees in the second and third; in the equation he leaves out the seconds. — In *al-Battānī's* table, whose argument proceeds by single degrees, the differences in the equation of the sun are regular enough for the numbers of the codex to be corrected at once; I seldom needed to compute with the formula

$$\text{tan equat.} = \frac{e_1 \sin m}{1 + e_1 \cos m} ,$$

where e_1 is the radius of the epicycle, or the eccentricity of the orbit, in decimal parts of the radius of the orbit = 1, that is, al-Battānī's $e/60$. See vol. I, pp. 48 and 219–20.

Cols. III–VI: *the inequality of the motion of the moon in longitude.* — Some things must be said first, so that the construction and the use of the tables may be understood more clearly.

In the Ptolemaic hypothesis, which al-Battānī sets out briefly in ch. XXX (vol. I, pp. 50–54), the motions of the moon take place in the plane of the *deflecting* circle, that is, the circle *oblique* to the zodiac. The *nodes*, or the intersections of the oblique circle with the parecliptic circle, go round the parecliptic circle with a retrograde motion of about 3 minutes a day.

The moon moves on the circumference of an *epicycle*[2], whose radius is $5^p\ 15'$ of the parts of which the radius of the oblique circle is 60^p. If the centre of the epicycle ran along the circumference of the oblique circle, the inequality of the motion of the moon would be single, like the inequality of the motion of the sun. But the centre of the lunar epicycle goes round the circumference of an eccentric (the deferent), which touches the oblique circle in one point, namely in the apogee A of the eccentric; so to the first inequality (whose correction is called the *simple equation*) a second is always added whenever the centre of the epicycle, or the moon, moves away from the apogee of the eccentric. The eccentricity is $10^p\ 19'$ of the parts of which the radius of the oblique circle is 60^p; so the radius of the eccentric is $60^p - 10^p\ 19' = 49^p\ 41'$. And since the apogee of the eccentric shares in the retrograde motion of the oblique circle, the centre of the eccentric describes a small

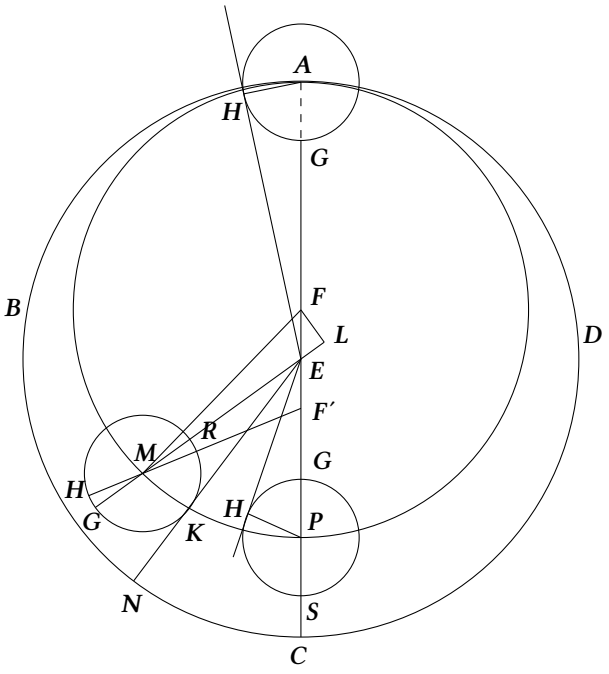
[1] Ἔτος is «*a period of years*».

[2] Above, towards the preceding signs (from east to west); below, towards the following signs (from west to east).

circle about the centre of the earth (or of the zodiac, or of the parecliptic circle), whose radius is the eccentricity itself.

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Moreover, the apogee of the epicycle is distinguished into the *true* apogee (τὸ ἀκριβὲς ἀπόγειον) and the *mean* apogee (τὸ μέσον ἀπόγειον). The true one, or G (on the small circle whose centre is M), is the point of the epicycle on which the straight line drawn from the centre of the earth through the centre of the epicycle falls; the mean one (or H on the same small circle), from which the mean anomaly is counted, is the point of the epicycle on which the straight line falls that is drawn through the centre of the epicycle from the point F′ of the line of apsides of the eccentric that is as far from the centre E of the earth as the centre of the eccentric is from the same point E on the opposite side. The difference of the true and the mean apogee, or the arc GH, is called by Ptolemy ἡ πρόσνευσις τοῦ τῆς σελήνης ἐπικύκλου, «the inclination of the moon's epicycle»; it is greatest (13° 9′) at 60° and 120° of elongation of the moon from the sun, and nil in the syzygies (i.e. when the moon stands in the apogee of the eccentric) and in the quadratures (that is, when the moon occupies the perigee of the eccentric). Because of this prosneusis *p*, the simple equation of the moon depends not, like that of the sun, on the mean anomaly *m*, but on the corrected anomaly *m* + *p*; so if we call *r* the radius of the epicycle in the parts of which the radius of the oblique circle is 60^p, the simple equation of the moon will be computed by the formula (see vol. I, pp. 219–20)



$$\text{tan simple equat.} = R\,\frac{r\sin(m+p)}{60^p\,R + r\cos(m+p)}\,,$$

which, putting $r_1 = r/60$ (i.e. the radius of the epicycle in decimal parts of the radius of the oblique circle = 1) and the radius *R*, to which the trigonometric functions are referred, = 1 part, we shall write

$$\text{tan simple equat.} = \frac{r_1\sin(m+p)}{1 + r_1\cos(m+p)}\,.$$

Now let us compute *p*. It is proved that the angle AEM, or FEM, which the line of apsides of the eccentric makes with the line of apsides of the epicycle, is always equal to twice the elongation

of the mean moon from the mean sun, that is, to 2 (☾ − ☉). Let ρ be the variable distance of the moon, or of the centre of the epicycle, from the earth, and *e* the eccentricity in the parts of which 60 are given to the radius of the oblique circle; as al-Battānī himself teaches in chapter XXXIX, we shall have

$$\rho = \sqrt{(60^p - e)^2 - \left(\frac{e \sin 2\,(☾ - ☉)}{R}\right)^2} \pm \frac{e \cos 2\,(☾ - ☉)}{R}\,,$$

or, taking *R* and the radius of the oblique circle = 1^p, and the eccentricity e_1 in its parts:

$$\rho = \sqrt{(1 - e_1)^2 - e_1^2 \sin^2 2\,(☾ - ☉)} \pm e_1 \cos 2\,(☾ - ☉)\,.$$

In the figure ME = ρ, and arc HG = angle HMG = angle EMF′ = *p*; considering the triangle MEF′ (with the perpendicular drawn from F′ to the side ME), one derives at once

$$\tan p = \tan EMF' = R\,\frac{EF' \sin F'EM}{R \times EM - EF' \cos F'EM} = R\,\frac{EF' \sin FEM}{R \times EM + EF' \cos FEM}$$

$$\tan p = R\,\frac{e \sin 2\,(☾ - ☉)}{R\rho + e \cos 2\,(☾ - ☉)} = R\,\frac{\left(\frac{e}{\rho}\right) \sin 2\,(☾ - ☉)}{R + \left(\frac{e}{\rho}\right) \cos 2\,(☾ - ☉)}$$

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or, taking e_1 for *e*, making $R = 1$, and expressing ρ in decimal parts (ρ_1):[a]

$$\tan p = \frac{e_1 \sin 2\,(☾ - ☉)}{\rho_1 + e_1 \cos 2\,(☾ - ☉)} = \frac{\left(\frac{e_1}{\rho_1}\right) \sin 2\,(☾ - ☉)}{1 + \left(\frac{e_1}{\rho_1}\right) \cos 2\,(☾ - ☉)}\,.$$

So the prosneusis *p* is a function of the double elongation. It is found in col. IV, whose argument is the mean double elongation, and al-Battānī very aptly calls it the *equation of the anomaly*[1].

The *second inequality* comes from the eccentric that carries the epicycle, because the centre of the moon's epicycle approaches the earth as it recedes from the apogee of the eccentric with twice the motion of the elongation of the moon from the sun; as I said, it is always added to the first inequality outside the syzygies, that is, whenever the centre of the epicycle (or the moon) is drawn away from the apogee of the eccentric. When the moon is in the mean quadratures (διχοτομίαι), that is, occupies the perigee of the eccentric, the distance ρ becomes not 60^p but $60^p - 2e = 60^p - 20^p\,38' = 39^p\,22'$; so the *r* of the formula of the simple equation is then referred to the radius $39^p\,22'$, and is not $5^p\,15'/60^p$ but $5^p\,15'/39^p\,22'$, or $8^p/60^p$. With the computation made in this way, the perigee equation of the moon is found to be 7° 40′, that is, to exceed the apogee simple equation (5° 1′) by 2° 39′. This difference is called the «*second inequality*»; half of it is what European astronomers today call the *evection*; it is given in col. VI of the table for the single degrees of the corrected anomaly.

[1] This prosneusis is about half of the third lunar inequality, found by Tycho Brahe and known by the name of *variation*; besides what I said in vol. I, p. 224, see I. Bullialdus, *Astronomia philolaica*, Paris 1645, p. 173.

[a] In the denominator of the first fraction the print has ε_1 for e_1.

In the same way, at the intermediate distances (ρ being known from the double elongation of the moon from the sun), one computes by how much the simple (or apogee) equation must be increased. To make a simpler table, Ptolemy, V, 7 (ed. Halma, vol. I, pp. 313–15), referred the portions to be added for the single intermediate distances to the greatest difference, 2° 39′; that is, he gave in the table the fraction of the second inequality that must be taken for the distances answering to a given double elongation, the second inequality being converted into unity (1° = 60′). These *proportional sixtieths* are given in col. V, whose argument is the mean double elongation.

So the operations for computing the *whole*, or *compound*, *equation* of the moon are these:

1°. From the tables of the mean motions take the mean longitude of the moon, the mean longitude of the sun and the mean anomaly of the moon; twice the difference of the longitudes will be the mean double elongation.

2°. Entering the table with the mean double elongation, take from col. IV the correction of the mean anomaly and from col. V the sixtieth fraction; the correction is added to the anomaly in the first and fourth quadrants and subtracted in the second and third.

3°. Entering with the equated, or corrected, anomaly, take the simple or apogee equation from col. III and the «least distance» (that is, the correction to be applied to the apogee equation at the perigee) from col. VI.

4°. Let *c* be the correction taken from col. VI and *f* the sixtieth fraction from col. V; the correction of the simple equation found in col. III will be $(c \times f)/60'$. That is, of *c* take the part proportional to the minutes *f*.

5°. Simple equation + $(c \times f)/60'$ = corrected equation of the centre, which is subtracted from the mean longitude or added to it according as the longitude is less or greater than 180 degrees[a]. The *true longitude* of the moon sought will result.

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Al-Battānī's tables rest on *Ptolemy's*, V, 7 (ed. Halma, vol. I, pp. 316–17), which is headed κανόνιον τῆς καθόλου σεληνιακῆς ἀνωμαλίας, «table of the whole lunar inequality»; but not without good changes. The argument of the table, which in Ptolemy is collected by sixes of degrees in the first and fourth quadrants and by threes in the second and third, proceeds by single degrees in al-Battānī. — The equation of col. III (πλάτους καὶ μήκους προσθαφαιρέσεις ἐπικύκλου, «prosthaphaereses of the latitude and longitude of the epicycle») Ptolemy computed without seconds and in a rougher way[1], so that in him the errors often reach 1′ (once 1′ 9″), as appears from the table in Delambre, *Hist. de l'astron. ancienne*, vol. II, p. 176[2]. Al-Battānī did far better, computing the seconds too. To heal the errors of the codex I made the computations afresh for the multiples of 6, with seven-figure logarithms and the formula[b]

$$\tan \text{eq.} = \frac{e \sin m}{1 + e \cos m} \qquad \left(\text{where } e = \frac{5^p \, 15'}{60^p} \right);$$

the rest, apart from a few cases in which I used logarithms, I checked by easy interpolation. On p. 79 three errors must be corrected:

[1] The same table of the simple equation (κανόνιον τῆς πρώτης καὶ ἁπλῆς ἀνωμαλίας τῆς σελήνης) is also in IV, 9 (ed. Halma, vol. I, p. 273).

[2] In this table of Delambre's the following should be corrected: under 78, 84 and 114 the equations are 4° *48′* 18″ (Del. 8′), 4° 55′ *43″* (Del. 39″) and 4° 44′ *17″* (Del. 23″).

[a] So the print; the sign depends on the corrected anomaly: the equation is subtracted when it is less than 180°, added when it is greater.

[b] The print has «1 + cos m» in the denominator.

under	35°	2°	40′	*52″*	نب	for the codex's	12″	يب	exactly	2°	40′	52.6″
»	36°	2	*44*	57	مد نز	»	45′ 17″	مه يز		2	44	59.5
»	37°	2	49	*2*	ب	»	42″	مب		2	49	4

In al-Battānī the seconds too often agree exactly with the numbers found by logarithms; sometimes discrepancies of 1″, 2″ or 3″ occur, very seldom greater ones (5° 0′ 57″ for 5° 1′ 9″ under 96°; 4° 53′ 30″ for 4° 53′ 19″ under 108°; 4° 52′ 10″ for 4° 52′ 3″ under 109°; 4° 44′ 6″ for 4° 44′ 17″ under 114°).

In col. IV (ἐκκέντρου προσθαφαιρέσεις ἀπογείου, «prosthaphaereses of the apogee of the eccentric») al-Battānī took over all of Ptolemy's numbers (only under 108 Ptol. has 13° 3′, Alb. 13° 2′), and derived the rest by interpolation. «Cette colonne n'est juste qu'à une ou deux minutes près, ce qui est suffisant» (this column is right only to within one or two minutes, which is enough), says Delambre (in Halma, notes, I, 36). — In these tables of the moon the codex always has الحصة *al-ḥiṣṣa* for الخاصة *al-ḫāṣṣa*; but I did not hesitate to restore الخاصة, as al-Battānī has it in the other tables and in the chapters. The error comes from the copyist, who followed the usage of some *Arab-African* writers. From this usage of the western Arabs, through the versions of Gerard of Cremona and the *Alfonsine Tables*, the words *Argumentum verum*, *Argumentum medium* and *Aequatio argumenti* also crept into many of our writers of the seventeenth century for the true anomaly, the mean anomaly and the equation of the anomaly (of the moon); see the definitions in I. B. Riccioli, *Almagestum Novum*, Bologna 1651, vol. I, p. 259.

In col. V (διάφορα ἑξηκοστῶν, «differences of the sixtieths») he leaves out the seconds, which Ptolemy has.

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In col. VI (ἐπικύκλου διάφορα, «differences of the epicycle») is the amount by which the perigee equation exceeds the apogee equation. Al-Battānī took over Ptolemy's numbers[1] and interpolated the rest. Of Ptolemy's numbers Delambre, *Astr. ancienne* II, 200, judges thus: «J'en ai calculé quelques exemples qui m'ont prouvé que les quantités (équat. périgée–équat. apogée) ou différences de l'épicycle ne sont quelquefois exactes qu'à 2′ près. Plus souvent l'erreur est de 1′ ou moindre encore» (I have computed some examples of them, which have shown me that the quantities (perigee equation – apogee equation), or differences of the epicycle, are sometimes right only to within 2′; more often the error is 1′ or less still).

Col. VII: *the latitude of the moon.* — Ptolemy's table too is combined with the table of the lunar inequalities, so that in it the argument (the motion in latitude) proceeds not by single degrees but by sixes or threes. Ptolemy moreover leaves out the seconds; and he counts the argument from the *northern limit* (βόρειον πέρας), not from the ascending node, so that in the argument 0° of Ptolemy = 90° of al-Battānī (see vol. I, pp. 250–51). — Al-Battānī computed his table with the formula

$$\sin \text{lat.} = \frac{\sin \text{argum. of lat.} \times \sin \text{greatest latit.}}{R};$$

Schiaparelli checked it with logarithmic tables.

[1] Sometimes there are discrepancies of ±1′, which al-Battānī himself clearly intended; the greatest amount is 2° 40′ for Ptolemy's 2° 39′.

On pp. 84–87: the mean syzygies for the Roman years and months

The tables are mentioned in vol. I, p. 92, ch. XLII.

Earlier, on pp. 29–32, he had given the mean syzygies for the wandering, or Egyptian, years; now he sets them out for the Roman, or Julian, years of the era of Dhū l-qarnayn. But one must not forget what I said above, on pp. 222–23, and in vol. I, p. 246: that in his tables al-Battānī uses a *fictitious* Roman *year*, which begins on 1 Ādhār (March) of the *chronological year already begun*; so the months from September to the end of February belong in the tables to the earlier year, as this survey will show:

	in the tables	*in common reckoning*	*years AD*
September	1190	beginning of 1191	879
October	1190	1191	879
November	1190	1191	879
December	1190	1191	879
January	1190	1191	beginning of 880
February	1190	1191	880
March		1191	880
April		1191	880
May		1191	880
June		1191	880
July		1191	880
August		1191	880
September	1191	beginning of 1192	880

Hence it is clear why in vol. I, p. 92, he prescribes that the current year is to be counted only when February has already passed, and why in the table of the single years, vol. II, p. 86, he makes leap years of those that can be divided exactly by 4, which is false in the common chronology for the years of Dhū l-qarnayn, or of the Seleucids[1].

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They agree very well with the tables of pp. 29–32. The day 1 Thōth 1012 in wandering years falls on 1 Ādhār 1011 (or 1 March 700 AD) in the Roman years of al-Battānī's tables of pp. 84–87; the first mean new moon is then found thus:

from the tables of pp. 29–32				*from the tables of pp. 84–87*			
990 wandering years				999 Roman years			
from the era	22^d	6′	23″	from the era	7^d	22′	9″
22 wandering years	2	19	18	12 Roman years	17	3	31
1012 wandering years				1011 Roman years			
from the era	24^d	25′	41″	from the era	24^d	25′	40″

But small discrepancies are found from the elements of the *tables of the mean motions*. For from these one derives:

for *1 Ādhār 1191* (= 1 March 880):

mean long. ☉	342°	48′	50″	or 360° − 17° 11′ 10″
mean long. ☾	165	40	17	
mean elongation ☾ − ☉	182°	51′	27″	= 658287″.

The mean daily synodic motion of the moon in Ptolemy and in al-Battānī is 12° 11′ 26.67″ = 43886.67″; so $\frac{658287''}{43886.67''} = 14^d\,59'\,58.9''$ is the interval of time between 1 Ādhār 1191 and the preceding new moon, or mean conjunction. Let this interval be taken from a whole lunation:

[1] Since the years of this era begin on 1 September or 1 October, the leap years are those which, divided by 4, leave the remainder 3.

	29^d	31′	50″
−	14	59	59
	14^d	31′	51″

which is the moment of the month Ādhār on which the first mean conjunction of the fictitious year 1191 will fall. — Al-Battānī's table, p. 84: 14^d 31′ *49″*; difference −2″ = -48^s.

For 1 Ādhār 1215 (= 1 March 904):

mean long. ☉	343°	2′	14″	or 360° − 16° 57′ 46″
mean long. ☾	109	58	57	
mean elongation ☾ − ☉	126°	56′	43″	= 457003″.

$\frac{457003''}{43886.67''} = 10^d\,24'\,47.7''$

	29^d	31′	50″	
−	10	24	48	
	19^d	7′	2″	
al-Battānī, p. 84:	19	7	0	difference −2″ = -48^s.

For 1 Ādhār 1239 (= 1 March 928):

mean long. ☉	343°	15′	38″	or 360° − 16° 44′ 22″
mean long. ☾	54	17	36	
mean elongation ☾ − ☉	71°	1′	58″	= 255718″.

$\frac{255718''}{43886.67''} = 5^d\,49'\,36.4''$

	29^d	31′	50″	
−	5	49	36	
	23^d	42′	14″	
al-Battānī, p. 84:	23	42	11	difference −3″ = $-1^m\,12^s$.

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Likewise the numbers of the *other columns* for the collected years differ from those derived from the tables of the mean motions. For[1]

for Ādhār 1191, 14^d 31′ 49″ = $14^d\,12^h\,43^m\,36^s$:

	mean long. ☉			*mean anom.* ☾			*motion in lat.* ☾		
1191 from the era	342°	48′	50″	108°	20′	0″	176°	23′	0″
14 days	13	47	56	182	54	35	0	44	29
12 hours	0	29	34	6	31	57	0	1	35
43.5 minutes	0	1	47	0	23	40.7	0	0	5.6
	357°	8′	7″	298°	10′	12.7″	177°	9′	9.6″
							+ 357	8	7
							174°	17′	16.6″
al-Battānī	357°	7′	39″	298°	3′	41″	174	17	21
difference			− 28″		− 6′	32″			+ 4″

for Ādhār 1215, 19^d 7′ 0″ = $19^d\,2^h\,48^m$:

[1]The tables of the mean motions give not the motion of the moon in latitude but the retrograde motion of the node; if the motion in longitude is added to it, the motion in latitude (ἡ κατὰ πλάτος κίνησις) results. And in the conjunctions, plainly, the longitude of the moon does not differ from the longitude of the sun.

1211 from the era	343°	0′	0″	148°	1′	59″	203°	11′	23″
4 years	0	2	14	7	56	23	77	21	41
19 days	18	43	38	248	14	5	1	0	23
2 hours	0	4	56	1	5	19	0	0	16
48 minutes	0	1	58	0	26	7.7	0	0	6
	1°	52′	46″	45°	43′	54″	281°	33′	49″
							+ 1	52	46
							283°	26′	35″
al-Battānī	1°	52′	17″	45°	37′	24″	283	26	38
difference			− 29″		− 6′	30″			+ 3″

for Ādhār 1239, $23^d\ 42'\ 11'' = 23^d\ 16^h\ 52^m\ 24^s$:

1231 from the era	343°	11′	10″	187°	43′	58″	229°	59′	47″
8 years	0	4	28	15	52	48	154	43	21
23 days	22	40	11	300	29	41	1	13	5
16 hours	0	39	25	8	42	36	0	2	7
52.5 minutes	0	2	9	0	28	34.6	0	0	6.8
	6°	37′	23″	153°	17′	38″	25°	58′	27″
							+ 6	37	23
							32°	35′	50″
al-Battānī	6°	36′	55″	153°	11′	7″	32	35	56
difference			− 28″		− 6′	31″			+ 6″

I do not rightly understand the cause of all these discrepancies.

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Pp. 84–85, for the *collected years*. Given the first line of numbers, the other numbers are found by always adding $4^d\ 35'\ 11''$ in the column of the days, and in the other columns the mean motions answering to this interval of time; but if the sum of the days exceeds a whole lunation ($29^d\ 31'\ 50''$), the new moon found will be the second new moon of the month Ādhār or the first of the following month, so that a whole lunation must be taken from the sum, and from the motions of the other columns the motions answering to a lunation.

P. 86, for the *single years*. Let A be the moment at which the first new moon of Ādhār occurred in any year; in the following year the new moon of Ādhār will be A + $18^d\ 53'\ 52''$ if the following year is common, or A − 1^d + $18^d\ 53'\ 52''$ if the following year is a leap year; as always, when the sum exceeds the length of a lunation ($29^d\ 31'\ 50''$), the lunation and the motions answering to it must be taken away. —

[For year 11 the codex gives $28^d\ 9'\ 39''$, 358° 32′ 6″, 271° 4′ 19″, 211° 11′ 43″; for year 22 the codex has $26^d\ 19'\ 18''$, 357° 4′ 18″, 182° 8′ 39″, 62° 23′ 27″. The inept interpolator took these numbers from the table on p. 32, not understanding the difference between the table for the wandering years and the table for the Julian years. For, because of the leap years, the whole days of the second column in the table of the Roman years lose a day every fourth year with respect to the table of the Egyptian years; that is, the beginning of the month, which in the first year was one and the same in both tables, shows a difference of one day every fourth year, of two days after eight years, of three after twelve years, and so on. So it can happen that the new moon of Thōth of the Egyptian table does not fall in the month Ādhār of the Roman table but in the month Subāṭ (February). This happens in the eleventh year of the tables. In the Egyptian tables, adding $18^d\ 53'\ 52''$ to $11^d\ 47'\ 37''$, one gets $30^d\ 41'\ 29''$ for the new moon of Thōth, plainly exceeding the days of Thōth; so, a lunation being taken away, the new moon of Thōth $1^d\ 9'\ 39''$ remains (the 29″ is a misprint of the compositor). But in the Roman calendar this new moon belongs to Subāṭ (February); for to the $9^d\ 47'\ 37''$ of the Roman table must be added, because of the following leap

year, $17^d\ 53'\ 52''$, whence Ādhār $28^d\ 41'\ 29''$ results[a], from which nothing is to be taken away. — The case is similar in year 22. These differences would become ever more frequent if the tables were carried beyond 24 years; after 120 years Thōth would leave the month Ādhār, or March, altogether (with which at the beginning it had the first day and the new moon in common), so that the moons of Ādhār would all become different from the moons of Thōth. Not noticing this, the interpolator had wrongly thought that the Roman table had to be corrected after the Egyptian one and Ptolemy[1], although he had kept the whole days of the second column as they arose from the leap years. — Schiaparelli.**]**

In the seconds of the days answering to year 6, 50″ should be restored for 40″, as printed. — The seconds in col. III differ somewhat (about 2″) from the seconds of p. 32; al-Battānī seems to have computed this column anew, perhaps with a slightly different daily motion of the elongation.

P. 87, for the *months*, etc. See what was set out on p. 206 about the three similar tables (p. 31). In the third small table, which teems with errors in the codex (above all in the last column), these slips must be corrected, as Schiaparelli tells me:

2nd number of the last column:	read 52″ for 54″ (codex و 6″!)
5th number of col. IV:	read 184° قفد for 189° قفط; of the last column 51′ for 56′ (codex يح 18′!)
6th number of the last column:	read 275° 48′ 22″ for 306° 28′ 36″ (codex 275° 4′ 10″)
7th number of the last column:	read 321° 45′ 17″ for 23° 5′ 45″ (codex 291° 0′ 40″!).

On pp. 88–91: the tables for computing eclipses

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P. 88, the small table of the *unequal motions*. It is mentioned in ch. XLII, vol. I, p. 94; it is missing in Ptolemy. — On the smaller table on the right side of the page, which is mentioned in vol. I, pp. 94 and 150, see what Schiaparelli set out in vol. I, p. 273, note *a*.

The *limits of the eclipses* of the sun should be corrected as I showed on p. 209, namely: 159° 44′ to *190° 56′*, 0° 0′ to 20° 16′, *349° 4′* to 360°.

P. 89, the table of *correction*. It is mentioned in chs. XXXIX, XLI, XLIII and XLIV (vol. I, pp. 80, 83, 87, 88, 97, 101, 106, 110–11). It is the κανόνιον διορθώσεως of Ptolemy, VI, 8 (ed. Halma, vol. I, p. 430), in which, however, only the διαφόρων ἑξηκοστά, «the sixtieths of the differences», are given. Theon, *Handy Tables*, vol. I, p. 146, gives all the columns. Under the number 18, 1′ 42″ should be restored for 1′ 22″. — The «portions of the distances of the moon» are the proportional parts to be taken from the variation of the diameter of the moon (or of the sun) between the apogee and the perigee distance, so as to know the diameter answering to the corrected anomaly of the moon with which one must enter the table. Moreover, in vol. I, p. 80, with the anomaly of the sun taken as the argument, the portions serve to determine the variation of the parallax of the sun. — The two remaining columns, wrongly merged into one by the scribe of the codex, give the correction to be applied to the lunar parallax in latitude and in longitude on account of the epicycle and of the eccentric; the argument of the column of the epicycle is the corrected anomaly of the moon, that of the eccentric the mean double elongation. — All the columns occur in the *Alfonsine Tables*, p. 116r of the edition mentioned in vol. I, p. XXXVI, note 7; the heading there is: «Table of the equation of the diversity of aspect [i.e. of the parallax], or table *Attacium*». «*Attacuim*» should be restored, that is, *at-taqwīm*, the name by which al-Battānī also designates this correction[2].

[1] The Spanish version, f. 56r, follows this interpolated copy.

[2] Steinschneider, *Zur Gesch. der Uebersetzungen aus dem Indischen*, in Zeit. d. deutsch. morgenl. Ges., XXIV, 1870, p. 384, wrongly thinks that *attacium* is corrupted from *at-tasyīr* (which means astrological direction, or profection)!

[a] $9^d\ 47'\ 37'' + 17^d\ 53'\ 52''$ make $27^d\ 41'\ 29''$. The $28^d\ 41'\ 29''$ of the text follows with $18^d\ 53'\ 52''$, the step for a common year, and by the rule above year 11 is a common year.

P. 89, the *eclipsed digits.* The three columns that serve to compute them are mentioned in chs. XLIII and XLIV (vol. I, pp. 101 and 111–12); they occur in Ptolemy, VI, 8 (ed. Halma, vol. I, p. 430), and in Theon, II, 95. In the third column, l. 3, Ptolemy has 2 1/15 (= 2^{dig} 4′), Theon 2 1/5; in l. 9 Halma's edition wrongly has 9^{dig} 10′ (ϛ″) for 9^{dig} 1/2 (ϛ″).

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P. 89, the *inclinations of the eclipsed parts* with respect to the ecliptic, given in the last four columns. They are mentioned in chs. XLIII and XLIV (vol. I, pp. 101 and 112); the tables are also in Ptolemy, VI, 11 (ed. Halma, vol. I, p. 449), who gives the seconds too, and in Theon, I, 146 and II, 95 (ἡλίου καὶ σελήνης προσνεύσεις), with whose help it was easy to heal the small errors of the codex. — Delambre, *Astr. ancienne*, vol. II, p. 238: «Ptolémée passe ensuite[1] à une recherche moins importante et négligée depuis par les modernes; c'est celle de la direction de la partie éclipsée soit par rapport à l'écliptique, soit par rapport aux différens points de l'horizon[2]. Cette direction change à chaque instant, il se borne à la déterminer pour les points les plus importans, comme le commencement, la fin, le milieu, l'immersion et l'émersion. Il fait passer un grand cercle par les deux centres et prolonge cet arc jusqu'à l'écliptique et jusqu'à l'horizon. Le problème est donc purement trigonométrique. Il y néglige l'inclinaison de l'orbite» (Ptolemy then passes on to a less important inquiry, since neglected by the moderns: the direction of the eclipsed part, either with respect to the ecliptic or with respect to the different points of the horizon. This direction changes at every instant; he confines himself to determining it for the most important points, such as the beginning, the end, the middle, the immersion and the emersion. He passes a great circle through the two centres and prolongs this arc to the ecliptic and to the horizon. So the problem is purely trigonometric. He neglects the inclination of the orbit in it). Delambre gives a clear exposition of these directions, following Theon's commentary, in another place, II, pp. 597–605. — The numbers are clearly computed with two formulas (see the figure in vol. I, pp. 102 and 112):

$$\sin \text{inclin.} = R\,\frac{\text{latitude of the moon}}{\text{radius } \odot + \text{radius } \leftmoon}\,, \quad \text{in eclipses of the sun;}$$

$$\sin \text{inclin.} = R\,\frac{\text{latitude of the moon}}{\text{radius of the shadow} \pm \text{radius } \leftmoon}\,, \quad \text{in eclipses of the moon.}$$

The argument of the tables is the eclipsed digits.

P. 90, *eclipses of the moon.* The tables are mentioned in ch. XLIII, vol. I, p. 101. Ptolemy, VI, 7 (ed. Halma, vol. I, pp. 428–29), gives similar ones; but he takes as the argument of the table the true distance of the moon from the northern limit, at intervals of 30′ at the greatest distance from the earth and of 34′ at the least; all the numbers differ. In Theon, II, 90, the argument is the true latitude of the moon, as in al-Battānī's tables, with which Theon also shares the name «*duration*» (μονή) for «*half-duration*» (ἡμίσου μονῆς) in the third columns (see vol. I, p. 275). But Theon leaves out the seconds and uses a rougher computation than al-Battānī.

P. 91, *eclipses of the sun.* The argument of the tables is the apparent latitude of the moon (that is, the true latitude ± the parallax), as in Theon's similar table, II, 90. Ptolemy, VI, 7 (ed. Halma, I, pp. 426–27), takes the apparent distance from the northern limit, at intervals of 30 minutes; the other numbers differ from al-Battānī's.

[1] *Almag.* VI, 11 and 12 (Halma, I, 442–453).

[2] The directions on the horizon, or azimuths, are shown by al-Battānī's figure, vol. II, p. 92.

On p. 92: the azimuths of the risings and settings of the signs in the climates

This figure is mentioned in chapter XLIII, vol. I, pp. 101–102.

A similar ὁριζόντων καταγραφή is found in Ptolemy, VI, 11 (ed. Halma, vol. I, pp. 450–51), but with other numbers because of the different obliquity of the ecliptic. The figure gives the rising and setting amplitudes of the beginning of each sign, which are derived by the formula $\sin \text{amplit.} = R \dfrac{\sin \delta}{\cos \varphi}$ (vol. I, p. 21). It serves to compute the inclination of the eclipsed part with respect to the horizon; see above, p. 231. The numbers on the left of the radius «*azimuth of noon*» give the climates, the numbers on the right the latitudes of the climates; perhaps al-Battānī had written all the latitudes with their minutes, so that the right side should be restored thus: [16° 39′], 24° 5′, 30° 40′, [36° 22′], 41° 15′, 45° 22′, [48° 53′].

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The circles are divided into 16 equal parts, answering to the chief winds of what we call our *rose*. On twelve radii within the smallest circle Ptolemy (or some interpolator) writes the names of the winds that blow from those directions; the other four radii he calls «winter rising» χειμερινὴ ἀνατολή (SSE), «summer rising» θερινὴ ἀνατολή (NNE), «summer setting» θερινὴ δύσις (NNW) and «winter setting» χειμερινὴ δύσις (SSW). In al-Battānī's figure[1] these four names are placed at the four corners of the page, with Greek names added that are so corrupted that it is very hard to restore them; and I find the same arrangement in the *Arabic Almagest* of the Bibliothèque Nationale in Paris, no. 2482, f. 128v, in which the names are: «winter rising» *al-mashriq al-shatawī*, «winter setting» *al-maghrib al-shatawī*, «summer rising» *al-mashriq al-ṣayfī*, «summer setting» *al-maghrib al-ṣayfī*[2]. The Arabic Almagest of the same library, no. 2483, f. 191v, gives the circles without numbers and names. — Within the smallest circle, however, in place of those four signs of Ptolemy (?), al-Battānī writes other Greek names of winds, much corrupted, like the rest; which has probably happened already in the *Arabic Almagests*[3].

To restore the Greek one must have before one's eyes a survey of the winds, which Schiaparelli most kindly excerpted from Salmasius's *Exercitationes Plinianae in Solinum* and Ideler's *Metheorologia Graeca et Romana*. These are the winds whose direction is established with certainty, or with sufficient certainty; they are referred to the normal rose of twelve winds which was used by Ptolemy and by most writers after Aristotle.

1.	N	ἀπαρκτίας, γαυρίς, μελανοβορέας, ὀρνιθίας (?), Septentrio.
2.	N $\frac{1}{3}$ E	μέσης, βορέας, καυνίας, Aquilo.
3.	E $\frac{1}{3}$ N	καικίας, θηβάνας (*summer rising*).
4.	E	ἀπηλιώτης, ἑλλησποντίας, καρνάβας, βερεκυνθίας, ποταμεύς, συριάνας, Solanus, Subsolanus.
5.	E $\frac{1}{3}$ S	εὖρος, ὀρνιθίας (?), Vulturnus (*winter rising*).
6.	S $\frac{1}{3}$ E	φοινικίας, εὐρόνοτος, ὀρθρόνοτος, κάρβας.
7.	S	νότος, Auster, Altanus.
8.	S $\frac{1}{3}$ W	λιβόνοτος, λευκόνοτος, λιβοφοίνιξ.
9.	W $\frac{1}{3}$ S	λίψ, Africus (*winter setting*).
10.	W	ζέφυρος, Favonius, Subvesperus.
11.	W $\frac{1}{3}$ N	ἰάπυξ, ἀργέστης, ὀλυμπίας, σκίρων, σκιλλετῖνος, Atabulus, Corus (*summer setting*).
12.	N $\frac{1}{3}$ W	θρασκίας, κερκίας, στρυμονίας, Gallicus.

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[1] In Arabic they are plurals: *winter risings, winter settings*, etc.

[2] On the right and left sides of the page: «setting of the equator» *maghrib al-istiwā'*, «rising of the equator» *mashriq al-istiwā'*, as I have supplied them in al-Battānī's figure.

[3] Both *codices* just mentioned and the *version of Gerard of Cremona* from the Arabic (printed at Venice in 1515) leave out all the names within the smallest circle, already so corrupted by the copyists that they could not be read. — In the *recension of the Almagest* made by Naṣīr al-dīn al-Ṭūsī, cod. Laur. orient. 156, f. 73v, the names of the winds and the inscriptions at the four corners of the leaf are indeed present, but, the ink being almost faded, they can sometimes hardly be read.

[I can hardly believe that Ptolemy wrote the names of the winds in this figure[1]. In the figure of the Almagest the directions of the winter and summer risings and settings make angles of only 22½ degrees with the meridian, which in no way answers to the truth; how, then, in that space of 22½° between those points and the line of south and north, will winds be placed such as Boreas and Thrascias, Euronotus and Libonotus, which in any climate are much nearer to the meridian than the points of the summer and winter risings and settings? In the figure the directions of all the winds are wrong, and those of the solstitial rising and setting most wrong of all. I think the names were added by a *later* Greek hand, except perhaps those of the four cardinal winds, whose directions in the diagram are right; but plainly al-Battānī had already found them in his copy of the Almagest. Ptolemy, or rather the interpolator, places only 12 winds; for the normal rose of the ancients contained twelve, distributed at equal intervals of 30 degrees. But since the figure of the Almagest has 16 radii, the four superfluous directions were marked with the names of the summer rising, the winter rising, the summer setting and the winter setting, which are not winds[2]. In this way all the directions received some name, but the angles that the eight secondary winds make with the four cardinal ones came out wrong; for in a rose of 12 winds drawn in this way the intervals next to the points S and N are of 45 degrees, the others of 22½°. — The rose of 16 winds is unknown to all the Greek and Roman writers. — Schiaparelli.**]**

All this being well weighed, it was not hard to heal the corruptions of the Escorial codex in the Greek names of the inner circle in this way:

Azimuth in the figure	*Direction*	*Names in the Almagest*	*Names in the codex*	*Probable restorations*
0°	S	νότος	بوطس	نوطس νότος
22½	SSE	winter rising	بوبوطس	اورونوطس εὐρόνοτος (= φοινικίας)
45	SE	εὐρόνοτος	هوبعس	فونقيس φοινικίας
67½	ESE	εὖρος	صدوس	اوروس εὖρος
90	E	ἀπηλιώτης	افلطس	افليوطس ἀπηλιώτης
112½	ENE	καικίας	سعاس	ثيانس θηβάνας (= καικίας)
135	NE	βορέας	بوربوب	بورياس βορέας
157½	NNE	summer rising	افر...بطس	افرقطيس ἀπαρκτίας (= ὀρνιθίας)
180	N	ἀπαρκτίας	بعدباس	ارنثياس ὀρνιθίας
202½	NNW	summer setting	افريقس	قرقيس κερκίας (= θρασκίας)
225	NW	θρασκίας	برقسلس	ثرسقيس θρασκίας
247½	WNW	ἰάπυξ	اباقونس	يافوقس ἰάπυξ
270	W	ζέφυρος	رهمرس	زهفرس ζέφυρος
292½	WSW	λίψ	لعوس	ليبوس λίψ (from the genitive λιβός)
315	SW	λιβόνοτος	ليبونطس	ليبونطس λιβόνοτος
337½	SSW	winter setting	بونطس	لوقونطس λευκόνοτος (= λιβόνοτος).

From this survey it is plain that in the *Greek* copy of the Almagest from which the *Arabic* version known to al-Battānī flowed, the copyist repeated at the four redundant radii (in place of the *risings* and *settings* of our Greek recension) neighbouring winds under other Greek names. The Arabic version, moreover, seems to have been made from the *Syriac*, because the name of the zephyr is written *zehfiros* in Arabic; see pp. 210–11.

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[The four names at the four corners of the page, too, are not a little corrupted. In the figure of the *Almagest* the directions of the risings and settings are collateral to north and south, for they make an angle of 22½ degrees with the meridian. The winds that agree with these directions are:

[1]See also Delambre, *Astron. ancienne*, II, p. 239: «Cette carte, au reste, est peut-être une addition des copistes, car Ptolémée n'en parle pas; il nous dit lui-même qu'on ne calcule ces directions que grossièrement» (this chart, moreover, is perhaps an addition of the copyists, for Ptolemy does not speak of it; he tells us himself that these directions are computed only roughly).

[2]If they were winds, the absurdity would follow that they are collateral to the south wind and to the north wind. — In Greek θερινὴ ἀνατολή, χειμερινὴ ἀνατολή, θερινὴ δύσις, χειμερινὴ δύσις.

for the *winter rising*, εὐρόνοτος, which in the Greek rose is collateral to the south wind on the side of the east; so the codex's اورى نطس is اورونطس *ewrūnoṭos*. — Nearly collateral to the north wind on the side of the east is ὀρνιθίας, whose name probably stood at the *summer rising*; then for the codex's اريقنفوس one should read ارنثيس *ornīthiyas*. — To the *summer setting* ἀργέστης answers exactly; so the codex's ارقطوس seems to need correcting to ارغسطوس *arghesṭūs*. — At the *winter setting* the codex has قاطرغربا, with which no Greek name of a wind can be reconciled; perhaps κάρβας, but the direction does not fit. — Schiaparelli.**]**

On one of the leaves of a miscellaneous Arabic codex[1], which I examined very recently at Milan, together with many others brought from southern Arabia, in the possession of Dr. Eugenio Griffini (to whom I give my warmest thanks), the copyist drew this very figure of the winds with the same numbers as al-Battānī (affected here and there by slips of the pen), leaving out the names within the smallest circle[a] but keeping al-Battānī's inscriptions at the four corners of the leaf:

المشارق الشتوية لوررفطس المغارب الشتوية فاطاعرفا[b]
المشارق الصيفيـة ارقطيقس المغارب الصيفية ارفطوس

The title of the figure in this codex of Griffini's is: سمت مطالع اوائل البروج ومغاربها من دواير افاق الاقاليم السبعة, «the azimuths of the risings and settings of the beginnings of the signs on the horizons of the seven climates».

The following f. 197r gives a spurious astrological table; see it near the end of this volume.

On pp. 93–94: the solar and lunar parallaxes in altitude

On these tables see ch. XXXIX, vol. I, pp. 79 and 80.

The parallaxes of the sun. As the author himself points out, they are taken from the *Almagest*, V, 18 (ed. Halma, vol. I, p. 359)[2]; but, because he accepted a different distance of the sun, al-Battānī bids us always add to the numbers of the table an eighteenth part of themselves; see vol. I, pp. 257–58, and p. XXXVIII, note 3. The argument is the true (or geocentric) zenith distance z; the parallaxes are computed by the formula

$$\tan \text{parall. in altitude} = \frac{\sin\pi\ \sin z}{1 - \sin\pi\ \cos z}\,, \qquad \text{where} \quad \sin\pi = \frac{\text{radius of the earth}}{\text{distance from the earth}}\,.$$

Ptolemy made the computations as if the *sine* of the parallax were found instead of the *tangent*, which in such small angles produces no sensible error. Moreover, the variations of the distance of the sun are neglected.

The parallaxes of the moon. They are taken from the *Almagest*, loc. cit.; but in the third and the fourth term al-Battānī corrected Ptolemy's rougher numbers somewhat. The argument of the four columns that give the terms is the *true zenith distance of the moon.*

The variations of the distance from the earth, of small importance for the sun, cannot be neglected for the moon in computing the parallax in altitude. Ptolemy[3] chose four principal distances, or

[1] Marked «Jemen C no. 86». Besides various fragments it contains Hermes' book *of the latitude of the key of the stars*, the book of the Persian sage *J̌āmāsp* (? codex حاماس) written in Arabic by Muḥammad ibn Abī Bakr al-Fārisī, and the commentary of Aḥmad ibn Yūsuf Ibn al-Dāya on the *Centiloquium* of Pseudo-Ptolemy.

[2] In Halma's edition, wrongly, minutes, seconds, thirds for degrees, minutes, seconds.

[3] *Almag.* V, 17 (ed. Halma, vol. I, pp. 348–57); the use of the tables is explained in V, 19 (Halma I, pp. 360–61).

[a] Nallino's Addenda et emendanda (Part II, p. XXX) delete the words «leaving out the names within the smallest circle»: the names are there, though much corrupted.

[b] Nallino's Addenda et emendanda (Part II, p. XXX) read قاطاعرقا, with the dots of the qāfs, for فاطاعرفا.

four terms, as follows. He put the apogee distance of the centre of the epicycle, or of the moon, at 59^p, the radius of the earth being taken as 1 part, and the perigee distance at $38^p\ 43'$; and since on this reckoning the radius of the epicycle is reckoned at $5^p\ 10'$, he made the first term $59^p + 5^p\ 10' = 64^p\ 10'$ (i.e. with the moon in the apogee of the eccentric and of the epicycle at once), the second term $59^p - 5^p\ 10' = 53^p\ 50'$ (i.e. with the moon in the apogee of the eccentric but in the perigee of the epicycle), the third term $38^p\ 43' + 5^p\ 10' = 43^p\ 53'$ (with the moon in the perigee of the eccentric but at the apogee of the epicycle), and the fourth term $38^p\ 43' - 5^p\ 10' = 33^p\ 33'$ (with the moon placed in the perigee of the eccentric and of the epicycle at once). At these four distances from the earth the parallaxes (for 90° of true zenith distance) will be[1]: II 236

I.	$\sin \text{parall.} = 1 : 64^p\ 10' = \sin 53'\ 34.7''$;	Ptolemy	$53'\ 34''$
II.	$\sin \text{parall.} = 1 : 53^p\ 50' = \sin 63'\ 51.8''$;	»	$63'\ 51''$
III.	$\sin \text{parall.} = 1 : 43^p\ 53' = \sin 78'\ 20.7''$;	»	$79'\ 0''$
IV.	$\sin \text{parall.} = 1 : 33^p\ 33' = \sin 102'\ 28.9''$;	»	$104'\ 0''$.

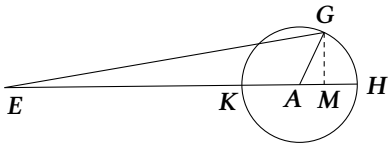
From these horizontal parallaxes π, the parallaxes in altitude answering to the true zenith distance z are derived for each term by the formula mentioned above for the sun. The table gives the *parallaxes* of the first term (col. III) and of the third (col. V), and also the *difference* between the second and the first (col. IV) and between the fourth and the third (col. VI), for every second degree of the true zenith distance of the moon.

The three remaining columns serve to compute the parallaxes at the intermediate distances from the earth, and this by proportional sixtieths. — Ptolemy assumes that the variations of the parallaxes are very nearly proportional to the variations of the distance from the earth; and he considers that these distances vary because of the anomalistic motion of the moon on the circumference of the epicycle and because of the motion of the centre of the epicycle on the circumference of the eccentric. When the centre of the epicycle occupies the apogee A of the eccentric, let the true anomaly, or the elongation of the moon from the true apogee H of the epicycle, be HG = 60°; then[2], by the formula with which the distance of the moon from the earth in the syzygies is computed (see vol. I, pp. 253–54), EG is found to be $62^p\ 48'$ of the parts of which EH (the first term) is $65^p\ 15'$, EK (the second term) $54^p\ 45'$, and KH (the difference between the second and the first term) $10^p\ 30'$. So the distance EG differs from the first term by $-2^p\ 27'$; and if the whole difference KH is taken as unity ($1^p = 60'$), those $2^p\ 27'$ become $60' \times 2^p\ 27' : 10^p\ 30' = 14'$. Hence it is clear that, at the apogee distance of the centre of the epicycle and with the moon at the true anomaly 60°, 14 sixtieths of the difference between the first and the second term for the same zenith distance must be added to the first term (answering to the given zenith distance). — One must proceed in a similar way when the centre of the epicycle holds the perigee of the eccentric[3], that is, when the distance of the moon from the earth lies between the third and the fourth term. — All this, computed for every second degree of the *true anomaly*, Ptolemy wrote in col. VII (with the centre of the epicycle in the apogee of the eccentric) and in col. VIII (with the centre of the epicycle in the perigee of the eccentric). But the argument of both columns is *half* the true anomaly, not the whole anomaly; so those 14′ will be found under 30°, not under 60°. Ptolemy did this because he had also halved the argument of col. IX, so that one would have to enter the table not with the double elongation but with the simple one.

[1] In the numbers in Delambre, *Astron. ancienne*, II, p. 218, misprints are not lacking.

[2] In the following computations everything is referred to the distances of the moon from the *centre* of the earth (not, as before, from its surface); see vol. I, p. 77.

[3] Then $39^p\ 22'$ will be put for 60^p in the formula for computing the distance (vol. I, p. 254).



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These corrections suffice when the centre of the epicycle holds the apogee or the perigee of the eccentric. At the other points a new correction must be applied, because of the different distance of the centre of the epicycle from the earth. Let the mean double elongation of the moon from the sun be 60 degrees, that is, let the centre of the epicycle be taken to have completed 30° of the eccentric from the apogee of the eccentric[a]; from the formula with which the geocentric distances outside the syzygies are computed (vol. I, pp. 255–56) one gathers that the distance of the moon from the centre of the earth is then $54^p\ 3'$ of the parts of which the apogee distance is 60^p and the perigee distance $39^p\ 22'$, their difference being $20^p\ 38'$. So the apogee distance will exceed the given distance by $5^p\ 57'$, which, if the whole difference $20^p\ 38'$ is taken as unity ($1^p = 60'$), become $60' \times 5^p\,57' : 20^p\,38' = 17'\ 18''$. These are found written in col. IX under 30°, and they serve to correct the parallax found by the earlier columns. So the argument of col. IX is the *simple mean elongation* of the moon from the sun.

Hence the use of the tables is clear. Enter the table with the true zenith distance of the moon, and take the parallax of the first and of the third term (cols. III and V) with the adjacent differences (cols. IV and VI). Find the corrected anomaly of the moon or, if it exceeds 180°, its complement to 360°; entering the table with half of it, take the sixtieths of cols. VII and VIII. Multiply the difference taken from col. IV by the sixtieths of col. VII, and the difference of col. VI by the sixtieths of col. VIII; add these proportional parts to the parallaxes of cols. III and V. Two parallaxes will result; compute their difference. Then find the mean elongation of the moon from the sun or from the point diametrically opposite, and entering the table with it take the sixtieths of col. IX, by which multiply the difference of the parallaxes found before; add the product to the smaller of the two parallaxes; the sum will be the parallax sought.

On pp. 95–101: the lunar parallaxes in longitude and latitude

The tables are mentioned in chs. XL and XLIV (vol. I, pp. 84, 105, 110); their use is taught in ch. XXXIX (vol. I, pp. 82–84).

As the author himself points out in vol. I, p. 82, they are taken from Theon's *Handy Tables* (ed. Halma, vol. II, pp. 98–111), and they do not always serve well for more exact computations; see also the note in vol. I, p. 262, and the addenda of the same volume, p. LXXV. In the headings he changed Theon's latitudes of the climates (computed for the obliquity of the ecliptic 23° 51′ 20″), as the new obliquity 23° 35′ required. In the places where I wrote *culm*[*ination*] in the translation there is nothing in the codex but a straight line running across the whole page; at one end of it خط الزوال «line of noon» is written, at the other خط نصف النهار «line of the meridian». On Schiaparelli's advice I preferred to translate *culmination*, for at those places it is a question of the passage of the first point of Cancer, the first of Leo, etc., through the meridian. — So that it might appear better than from the headings *austr.* and *bor.* alone which parallaxes in latitude were negative and which positive, I put the sign + before the positive ones. When the parallaxes in longitude are additive and when subtractive is clear from the rules in vol. I, pp. 84 and 262–63[1].

[1]The parallaxes in longitude *at the hour of culmination* of the given sign are positive from Cancer to Capricorn, negative from Capricorn to Cancer, and nil in Cancer and Capricorn.

[a]So the print. With a double elongation of 60° the centre of the epicycle stands 60° from the apogee of the eccentric, as seen from the earth; 30° is the simple elongation, under which col. IX gives the value. The formula of vol. I, pp. 255–56, gives $54^p\ 1'\ 54''$ for this distance, that is, $54^p\ 2'$.

The faulty numbers of the codex were easy to heal, for in each climate the same parallaxes are always repeated twice; the comparison with Theon made the corrections still more certain. Theon's tables teem with not a few slips of the compositor in Halma's edition, and with more in its French translation.

In the headings of the tables I had kept the numbers of the latitudes of the climates as I had found them in the codex; but wrongly, for the following must be corrected:

P. 95:	16° 39′ لط	for the codex's	16° 32′ لب	(exactly 16° 38′ 48″)
» 96:	24° 5′ ه	for the codex's	24°	(the ٥ 0′ being left out; exactly 24° 4′ 47″)
» 101:	48° 53′ نج	for the codex's	48° 13′ يج	(exactly 48° 52′ 46″[a]).

The first two errors also occurred in the codex in the tables of p. 65, and I corrected them in the translation.

P. 97, at the 2nd hour of Cancer after culmination, the parallax in longitude 22′ should be restored for 21′.

On pp. 102–106: the mean motions of the planets for the Roman calendar

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The Roman year here is the *fictitious* year counted from 1 Ādhār (March) of the chronological year already begun, of which I spoke above, on p. 227. Why the mean motion of the anomaly has been left out for the superior planets and the mean motion in longitude for the inferior ones I have already explained on p. 204. — All the motions of these tables differ a little from Ptolemy's[1] over long intervals of time; see vol. I, pp. 239 and 242.

On the headings of the columns of p. 102 the same must be repeated as I mentioned on pp. 203–04. — P. 103: the minutes of the motion of the anomaly of Mercury for the single years from 10 to 17, computed in a rougher way, differ from the correct ones by 1′ and 2′. — P. 104: at the month Subāṭ the copyist had written 0 everywhere; I restored the numbers and added the Subāṭ of the leap years (although it is superfluous; see vol. II, p. 223, on II, 74). — Pp. 105, of the hours, and 106, of the days, necessarily agree with pp. 28 and 27.

On pp. 108–137: the equations of the planets

The tables are mentioned in chs. XXXI and XLV (vol. I, pp. 64 and 113–14).

They rest on Ptolemy's, XI, 11 (ed. Halma, vol. II, pp. 300–09), which are headed ἔκθεσις κανόνων τῆς κατὰ μῆκος τῶν ε′ πλανωμένων διευκρινήσεως, «exposition of the tables of the correction of the five planets in longitude». Ptolemy had arranged the argument by sixes of degrees in the first and last quadrants and by threes in the second and third; al-Battānī always by single degrees[2], taking over Ptolemy's amounts and interpolating the rest. He chose the headings of the columns more aptly than Ptolemy. Finally, very conveniently, he merged into one two columns of Ptolemy's, headed μήκους προσθαφαίρεσις, «prosthaphaeresis[3] of the longitude», and διάφορα προσθέσεως (or δ. ἀφαιρέσεως), «differences of addition» (or «differences of subtraction»); so al-Battānī's col. III (the equation of the anomaly and the centre) is the *algebraic* sum of Ptolemy's cols. III and IV, as I shall show below. — Al-Battānī's col. IV is Ptolemy's VIII; the others, that is V, VI and VII, have the same number in both.

[1] Ptolemy's tables, moreover, are referred to another era and to the wandering years (without the Julian intercalation).

[2] So already Theon, ed. Halma, II, pp. 134–93.

[3] That is, an amount now to be added, now to be subtracted; the *equation* of the Arabs and of our writers.

[a] With the obliquity 23° 35′ and a longest day of 16 hours the latitude comes out as 48° 52′ 34″; the other two values check to the second.

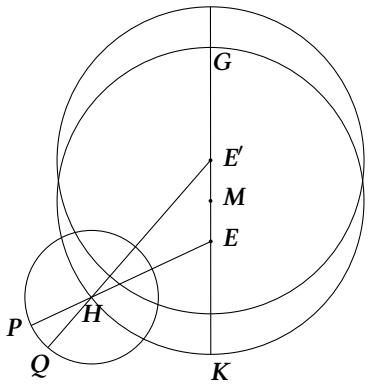
Ḥabash, ff. 42r–44v, 46r–48v, 50r–52v, 54r–56v and 58r–60v, gives tables of the equations made in the same way as al-Battānī's.

Two slips of the codex must also be corrected: in col. III, p. 116, last line, and p. 117, line 1, read 5° 15′ يه for 5° 14′ يد[1].

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In Ptolemy's system the motions in longitude of the *superior planets* and of *Venus* are explained as follows. Each of these four planets has a circle *eccentric* (ὁ ἔκκεντρος) with respect to the centre of the earth or of the ecliptic, on whose periphery the centre of the *epicycle* moves; the Arabs, and those of our writers who followed them, mostly call it the *deferent* (*al-ḥāmil*)[2]. The centre of the epicycle goes round the circumference of the deferent in the order of the signs (from west to east), but with a motion that is unequal both with respect to the centre M of the deferent and with respect to the centre E of the earth; it moves uniformly, however, about a point E′, which is as far from the centre M of the deferent as the centre E of the earth is on the opposite side. This imaginary circle, of the same diameter as the deferent, is called by Ptolemy ἔκκεντρος τῆς ὁμαλῆς κινήσεως (or παρόδου), «the eccentric of uniform motion», by the Arabs *al-muʿaddil li-l-masīr*, «that which equalizes the motion», and by our writers the *equant*; it is regarded as a mere circle, not as a sphere or orb. The centre of the epicycle of *Venus* completes the circuit of the deferent in a sidereal year of the sun; that of the *three superior planets* in other intervals of time. So *Venus* is always in mean conjunction with the sun. — The *superior planets* complete the circumference of the epicycle in one sidereal year[3], and once a year (standing in the true apogee of the epicycle) they are in mean conjunction with the sun, once (in the true perigee of the epicycle) in mean opposition. *Venus*, however, completes its epicycle with its mean motion in 583 days, 22 hours and 15 minutes, or about 19 solar months.

The line KG of the apsides of the eccentric keeps a constant direction with respect to the fixed stars; in *Venus* and *Mars* it coincides with the line of the greatest inclination of the plane of the eccentric to the plane of the ecliptic, in *Jupiter* it deviates from this line by 20° to the west, in *Saturn* by 50° to the east. The apogees of the eccentrics of *Saturn*, *Jupiter* and *Mars* always incline to the north of the ecliptic, the perigees always to the south; but the apogee and the perigee of *Venus* undergo every change, taking now a northern, now a southern latitude, and sometimes lying in the plane of the ecliptic. — The same line KG moves about the centre E of the earth with the motion of the precession, or of the fixed stars, carrying the points E′ and M with it.



[1]In Halma's edition of the *Almagest* the equation of the anomaly of Venus answering to 174 degrees is wrongly written 14° 41′ for 14° 47′.

[2]In Ptolemy sometimes ὁ φέρων τὸν ἐπίκυκλον ἔκκεντρος, or ὁ ἔκκεντρος κύκλος ἐφ' οὗ φέρεται τὸ τοῦ ἐπικύκλου κέντρον.

[3]All the planets move in the upper half of the epicycle in the order of the signs (from west to east), in the lower half the contrary way. The motion of the three *superior planets* on the epicycle equals the excess of the mean motion of the sun in longitude over their own mean motions of the centre of the epicycle.

In these four planets the eccentricity is EM, and EE′ is twice it. The *true* apogee of the epicycle is the point, wherever it lies on the outer semicircumference of the epicycle, on which the straight line produced from the centre E of the earth through the centre H of the epicycle falls; the *mean* apogee of the epicycle is the point on which the straight line drawn from the centre E′ of the equant through the centre of the epicycle falls. The mean and the true apogee coincide whenever the centre of the epicycle is in the apogee G or the perigee K of the eccentric.

In the three superior planets the line drawn from the centre of the epicycle to the planet coincides, at the mean conjunctions and oppositions with the sun, with the line of the mean motion of the sun; outside the conjunctions and oppositions it is parallel to it. In the inferior planets the line of the mean motion on the eccentric is parallel to the line of the mean motion of the sun.

The circles mentioned above lie in planes different from that of the ecliptic; but since their inclinations do not sensibly change the longitudes of the planets, Ptolemy, to explain the motions in longitude, supposes all the circles to be drawn in the same plane as the ecliptic[1], as he had also done for the moon.

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Let ABD be the ecliptic, GHK the deferent, H the centre of the epicycle at a given moment, and G the apogee of the eccentric (whose longitude on the ecliptic is A). According to its mean motion about the centre E′ of the equant, the centre of the epicycle will hold the longitude Z on the ecliptic; but according to its true motion, as seen from the centre E of the earth, it will be at the longitude X. The arc ZX, the difference between the mean and the true place, is called by Ptolemy ἡ ζωδιακὴ ἀνωμαλία, «the zodiacal inequality», and by al-Battānī and the Arabs «*the equation of the anomaly and the centre*»[2]; hence the name *equation of the centre*, which astronomers use even today. This equation is measured by the angle XHZ = E′HE. Drawing the perpendiculars EN and MS to the straight line ZHE′ produced, one sees at once that

$$\text{sin equat. of the centre} = \sin E'HE = R\,\frac{NE}{HE}\,.$$

Let *m* be what the Arabs and our writers of the Middle Ages call the *centre* (*al-markaz*), that is, the arc AZ of the ecliptic between the longitude of the apogee of the eccentric and the mean place of the planet; it is measured by the angle AE′Z at the centre of the equant. Let *e* be the eccentricity, that is, EM = ME′; then

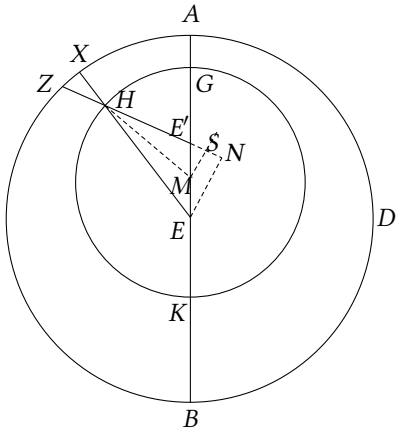
$$NE = E'E\,\frac{\sin NE'E}{R} = 2e\,\frac{\sin m}{R}\,.$$

It remains for us to find

$$HE = \sqrt{\overline{HN}^2 + \overline{NE}^2} = \sqrt{(HS+SN)^2 + \overline{NE}^2}\,.$$

[1] *Almag.* IX, 6 (ed. Halma, vol. II, p. 160). — Elsewhere, teaching the computation of the latitudes of the planets (XIII, 4, ed. Halma, vol. II, pp. 386, 388, 393–94, 397, 400), he shows that the greatest difference, additive or subtractive, that falls to the longitudes because of the *inclination* of the epicycle is 2′ in Venus, 3′ in Mercury, 1′ in Saturn, 1′ in Jupiter and nil in Mars (Delambre, computing more exactly in the notes to Halma's edition of the Almagest, found 1′ 40″, 6′ 55″, 0′ 17″, 0′ 15″ and 1′ 31″); and that the greatest arising from the *slant* of the epicycle is 1′ in Venus and 6′ in Mercury (Halma, II, pp. 406, 407; Delambre in the notes 1′ 0″ and 7′ 29″).

[2] *Ta'dīl al-ḥāṣṣa wa-l-markaz*; for its amount is also the difference between the true and the mean apogee of the epicycle, which one must use to convert the mean anomaly into the true. What the *centre* is I shall say immediately below.



$$HS = \sqrt{\overline{HM}^2 - \overline{MS}^2}\,, \quad \text{where} \quad HM = \text{radius of the deferent} = \rho\,.$$

$$MS = E'M\,\frac{\sin ME'S}{R} = e\,\frac{\sin m}{R}\,,$$

$$\text{so} \quad HS = \sqrt{\rho^2 - \left(\frac{e\sin m}{R}\right)^2}\,.$$

$$\text{Moreover} \quad SN = E'N - E'S$$

$$E'N = E'E\,\frac{\sin NEE'}{R} = E'E\,\frac{\cos NE'E}{R} = 2\,e\,\frac{\cos m}{R}$$

$$E'S = E'M\,\frac{\sin E'MS}{R} = E'M\,\frac{\cos NE'E}{R} = e\,\frac{\cos m}{R}\,.$$

$$\text{So} \quad \overline{HE}^2 = \left\{\sqrt{\rho^2 - \left(\frac{e\sin m}{R}\right)^2} + \frac{2e\cos m}{R} - \frac{e\cos m}{R}\right\}^2 + \left(\frac{2e\sin m}{R}\right)^2$$

$$HE = \sqrt{\left\{\sqrt{\rho^2 - \left(\frac{e\sin m}{R}\right)^2} + \frac{e\cos m}{R}\right\}^2 + \left(\frac{2e\sin m}{R}\right)^2}\,,$$

which is the *geocentric distance of the centre of the epicycle.* Hence, finally:

$$\sin \text{equat. of the centre} = \frac{2e\sin m}{\text{dist. of the centre of the epicycle from the centre of the earth}}\,.$$

With this formula, derived from Ptolemy's geometrical constructions, al-Battānī and many others (Regiomontanus himself among them) no doubt made or checked the table of the equation of the centre.

With tangents a far simpler rule is found:

$$\tan \text{equat. of the centre} = \tan E'HE = R\,\frac{NE}{HN} = \frac{2e\sin m}{\sqrt{\rho^2 - \left(\frac{e\sin m}{R}\right)^2} + \frac{e\cos m}{R}}\,;$$

and, taking *R* and ρ as one part and referring *e* to ρ = 1:

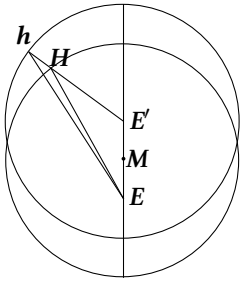
$$\text{tan equat. of the centre} = \frac{2e_1 \sin m}{\sqrt{1-(e_1 \sin m)^2} + e_1 \cos m}\,.$$

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This equation gives both the discrepancy arising from the unequal motion of the centre of the epicycle and the difference between the *mean anomaly* (that is, the arc of the epicycle counted from the mean apogee of the epicycle) and the *true anomaly* ψ (the arc of the epicycle counted from the true apogee of the epicycle). — Such an equation is given in al-Battānī's col. III. — Ptolemy, however, always divides this inequality on the ecliptic, which arises from the motion of the centre of the epicycle on the deferent (and which he calls the «*zodiacal inequality*», ἡ ζωδιακὴ ἀνωμαλία), into two parts, not only in the geometrical constructions but also in the tables. One of them (προσθαφαίρεσις μήκους, «prosthaphaeresis of the longitude») is the equation computed simply (κατὰ τὸ ἁπλοῦν), as if the centre of the epicycle travelled along the circumference of the equant, not of the deferent; so that, calling $2e_1$ the eccentricity of the equant in decimal parts of the radius of the equant, and taking the radius of the equant itself as one part, we can compute Ptolemy's col. III by the formula

$$\text{tan prosthaphaeresis of the longitude} = \frac{2e_1 \sin m}{1 + 2e_1 \cos m}\,,$$

as we said one must proceed in the hypothesis of the eccentric of the moon. But since the centre of the epicycle really moves on the circumference of the deferent, not of the equant, its distance from the centre of the equant will be 1 (or 60^p by the usage of the ancients) only twice in the whole circumference of the deferent; at all the other places, therefore, the prosthaphaeresis of the longitude will have to be corrected. So Ptolemy seeks by how much it grows at the perigee and by how much it decreases at the apogee; these corrections, to be applied to the prosthaphaereses of the longitude, he set in col. IV of his tables (διάφορα προσθέσεως or διάφορα ἀφαιρέσεως). The matter will appear more clearly from the following diagram. Ptolemy computes the prosthaphaeresis of the longitude, that is, col. III, as if the centre of the epicycle were at the point *h* of the equant, not at the point H of the deferent; instead of the true zodiacal anomaly (or the equation of the centre) EHE′, therefore, he considers the angle EhE′ (the prosthaphaeresis of the longitude). The error arising from this is the angle hEH, now to be added to the prosthaphaeresis, now to be subtracted; it is given in Ptolemy's col. IV. — The algebraic sum of cols. III and IV makes up the equation of the centre. But Ptolemy himself, XI, 10 (ed. Halma, vol. II, p. 295), points out: «Here, indeed, as in a treatise, it was fitting to set such a distinction of the zodiacal inequality before the eyes, and therefore to set it out in two columns; but in its use one column too, gathered from the algebraic sum of those two, will suffice» (ἐνθάδε μὲν οὖν, ὡς ἐν συντάξει, προσῆκον ἦν τὴν τοιαύτην διάκρισιν τῆς ζωδιακῆς ἀνωμαλίας ὑπ' ὄψιν ποιήσασθαι, καὶ διὰ τοῦτο ἐν δυσὶ σελιδίοις ἐκθέσθαι, ἐπὶ μέντοι τῆς χρείας αὐτῆς ἀπαρκέσει καὶ ἓν σελίδιον ἐκ τῆς ἀμφοτέρων τούτων προσθαφαιρέσεως ἐπισυνηγμένον).



Now a second correction must be applied to the longitude because of the varying position of the planet on the circumference of the *epicycle*. Let the planet be at T; the arc XY, or the angle XEY, is sought. Drawing the perpendicular Tt:

$$\sin XEY = R\,\frac{Tt}{ET} = R\,\frac{Tt}{\text{geocentr. dist. of the planet}}$$

$$Tt = \frac{TH \times \sin XHT}{R} = \frac{r \sin \psi}{R}\,,$$

where *r* is the known radius of the epicycle, in parts of which 60 make the radius of the deferent and of the equant. And the distance of the planet from the centre of the earth is

$$ET = \sqrt{(EH + Ht)^2 + Tt^2}\,;$$

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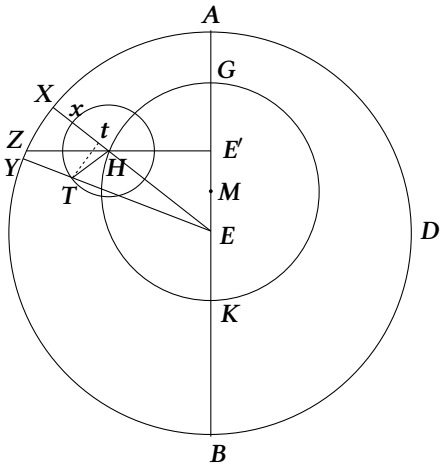
but EH = the geocentric distance of the centre of the epicycle, found above, and

$$Ht = \frac{TH \times \sin HTt}{R} = \frac{r \cos \psi}{R}\,,$$

$$\text{so} \quad ET = \sqrt{\left(\text{dist. of the centre of the epicycle} + \frac{r \cos \psi}{R}\right)^2 + \left(\frac{r \sin \psi}{R}\right)^2}\,.$$

$$\text{Finally} \quad \sin \text{equat. of the anomaly} = \frac{r \sin \psi}{\text{geocentric distance of the planet}}\,.$$

This *equation of the anomaly* (ἀνωμαλίας προσθαφαίρεσις) is the difference between the true place of the centre of the epicycle and the true place of the planet on the ecliptic; that is, it is the arc of the ecliptic contained between the line of the true place of the centre of the epicycle and the line of the true place of the planet. —



With tangents the formula becomes:

$$\tan \text{equat. of the anomaly} = \tan XEY = R\,\frac{Tt}{Et}$$

$$Et = EH + Ht = \text{dist. of the centre of the epicycle} + \frac{r \cos \psi}{R}$$

$$\text{tan equat. of the anomaly} = \frac{r \sin \psi}{\text{dist. of the centre of the epic.} + \dfrac{r \cos \psi}{R}} ,$$

or, making R = 1, like the radius of the deferent, and taking *r* in decimal parts of the radius of the deferent:

$$\text{tan equat. of the anomaly} = \frac{r_1 \sin \psi}{\text{dist. of the centre of the epic.} + r_1 \cos \psi} .$$

From the constructions made one gathers:

1°. That the equation of the centre is nil when the centre of the epicycle occupies the apogee or the perigee of the eccentric (the deferent), and greatest at the two middle points; and that it is equal at places of the deferent diametrically opposite, as also at places equally distant on either side from the apogee or from the perigee of the deferent.

2°. That the equation of the centre is to be subtracted from the mean centre in the first semicircle AYB (in which the longitude X of the true centre precedes the longitude Z of the mean centre) and added in the second. — That the same equation of the centre is to be added to the mean anomaly, and subtracted from it, the other way round.

3°. That the equation of the anomaly is nil when, with the centre of the epicycle holding the apogee or the perigee of the eccentric (the deferent), the planet occupies the apogee or the perigee of the epicycle; for then the true apogee and perigee of the epicycle coincide with the mean ones. That it is greatest when the centre of the epicycle is in the perigee of the eccentric and the planet at that point of the epicycle which is marked by the tangent drawn from the centre of the world to the circumference of the epicycle.

4°. That the equation of the anomaly is to be added to the longitude of the true centre in the first semicircle of the epicycle and subtracted in the second.

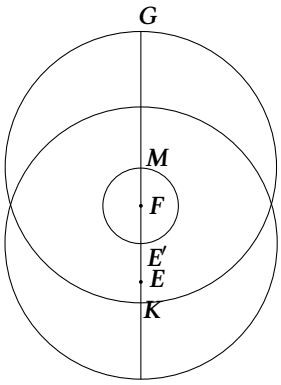
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One column would suffice to give the equation of the anomaly if the centre of the epicycle always kept the same distance from the centre of the earth. Since this does not happen, to equate the motions without too great a number of tables one will have to give the variations that fall to the equations because of the different distances by proportional minutes. Since in these five planets the variation from the perigee to the apogee is so great that proportional minutes, if set simply as was done for the moon, would produce sensible errors, Ptolemy, whom al-Battānī follows, put in col. VI the equation of the anomaly for the mean distance of the centre of the epicycle from the earth; in V the amount to be taken from the sixth so that the equation for the greatest distance of the epicycle results; in VII the excess of the equation for the least distance over the equation for the mean distance; and finally in VIII the proportional minutes for finding the equations at the intermediate distances. These proportional minutes have been computed for the greatest equations, namely those that occur at the tangent drawn from the centre of the earth to the circumference of the epicycle; for the corrections of the equations falling at the other points of the epicycle keep sensibly the same proportion (see *Almag.* XI, 10, ed. Halma, vol. II, pp. 296–99). — So, comparing what we said of the lunar tables (p. 225), al-Battānī's rules in vol. I, pp. 113–114, will be easily understood.

The matter is more involved in *Mercury*, in which the centre M of the deferent[1] does not hold the middle place between the centre E′ of the equant and the centre E of the earth, but goes round a small circle (called by the Arabs *al-mudīr*, «the revolving one»), whose centre F is as

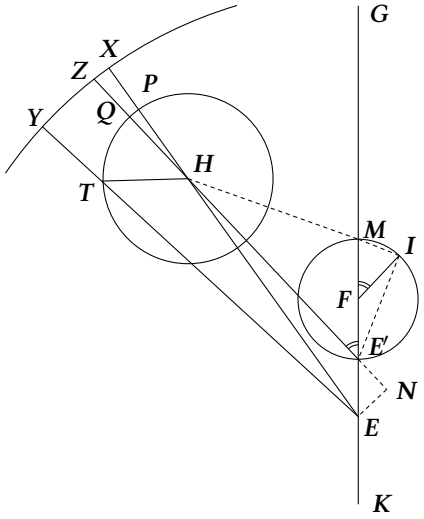
[1] In the *Hypotheses*, ed. Halma, p. 48, l. 10, the centres of Mercury's equant and deferent are placed between the centre of the ecliptic, or of the earth, and the *southern* limit (νότιον πέρας); but perhaps this is an error of that not very good edition for *northern*.

far from M and from E′ as the centre of the equant is from the centre of the earth. So at times the centre of the deferent falls on the centre of the equant; then the point of the apogee of the deferent holds the perigee of the equant, and both circles coincide[1]. The line of the apsides of the equant moves about E with the motion of the precession, as in the other planets. The centre of the epicycle completes the circumference of the deferent in the order of the signs in one sidereal year of the sun; and in the same time, against the order of the signs (from east to west), the centre of the deferent travels round the revolving small circle, and the line of the apsides of the deferent describes a whole circle; so that, because of the sum of these equal but contrary motions, the centre of the epicycle holds the apogee of the eccentric (the deferent) twice in a sidereal year, and twice the perigee. For the same reason the angular distance of the apogee of the equant from the apogee of the deferent is always as great as its distance, on the opposite side, from the centre of the epicycle. — The mean motion of the centre of the epicycle equals the mean motion of the sun in longitude; Mercury, however, completes its epicycle with its mean motion in about 115 days, 21 hours and 5 minutes.



The least distance of the epicycle from the earth occurs when the centre of the epicycle is 120 degrees of the circle away from the apogee of the equant on either side, not when it holds the fixed perigee of the equant or the moving perigee of the deferent; and this because of the equal but contrary motions of the centre of the epicycle and of the centre of the deferent. — The line of the apsides of the equant lies in the line of the greatest inclination of the plane of the eccentric to the plane of the ecliptic; the apogee and the perigee are now northern, now southern, now in the plane of the ecliptic.

[1]Taking the radius of the eccentric (the deferent) as 60^p, the distance of the point E from E′, like that of E′ from F, is put at 3^p in the *Almagest*; so the eccentricity of the deferent varies between 3^p and 9^p. In the later *Hypotheses*, p. 48, the distance of E from E′ being kept at 3^p, the radius FE′ of the small circle is taken as $2^p\ 50'$, so that the eccentricity varies between 3^p and $8^p\ 40'$. This explains the fractions in Tannery, *Recherches sur l'hist. de l'astr. ancienne*, p. 251, ll. 4–8.



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Now let us compute the equations of Mercury. Let GK be the line of the apsides of the equant, and at a given moment let the centre M of the deferent be supposed to have come by its retrograde motion to the point I. Because of what I said above, the centre of the epicycle will then be at H, so that the angle GE′H (or the mean centre *m*) equals the angle GFI. X will be the longitude of the true apogee P of the epicycle, Z (on the straight line ZHE′) the longitude of the mean apogee Q of the epicycle; so, with the planet at T (whose true longitude is Y), QT is the mean anomaly and PT the true anomaly. *m* and the arc QT are gathered from the tables of the mean motions; the eccentricity *e* = EE′ = E′F = FI is also known. — In the isosceles triangle IFE′

$$IE' : E'F = \sin IFE' : \sin FIE'$$
$$\text{or} \quad IE' : e = \sin(180^\circ - m) : \sin\left(\frac{m}{2}\right),$$
$$\text{whence} \quad IE' = \frac{e\sin(180^\circ - m)}{\sin\left(\frac{m}{2}\right)} = \frac{e\sin m}{\sin\left(\frac{m}{2}\right)};$$
$$\text{and since} \quad \sin m = \sin 2\left(\frac{m}{2}\right) = 2\sin\left(\frac{m}{2}\right)\cos\left(\frac{m}{2}\right),$$
$$\text{finally} \quad IE' = 2e\cos\left(\frac{m}{2}\right).$$

In the triangle E′IH the angle $IE'H = m + \frac{m}{2}$, and the radius of the deferent IH = 60^p;

$$\text{so} \quad E'H = \frac{IE' \times \cos IE'H}{R} + \sqrt{\overline{IH}^2 - \left(\frac{IE' \times \sin IE'H}{R}\right)^2}$$
$$E'H = \frac{2e}{R}\cos\left(\frac{m}{2}\right)\cos\left(m + \frac{m}{2}\right) + \sqrt{60^2 - \left[\frac{2e}{R}\cos\left(\frac{m}{2}\right)\sin\left(m + \frac{m}{2}\right)\right]^2}$$

which is the *distance of the centre of the epicycle from the centre of the equant.* Let us call it Δ.

Now let the perpendicular EN be drawn to the straight line HE′, produced as far as necessary. In the right-angled triangle E′NE:

$$NE = E'E\,\frac{\sin NE'E}{R} = \frac{e\sin m}{R}\,, \qquad E'N = E'E\,\frac{\sin E'EN}{R} = \frac{e\cos m}{R}\,;$$

so in the triangle HNE:

$$\text{geocentric distance of the centre of the epicycle} = HE = \sqrt{(E'H + E'N)^2 + \overline{NE}^2}$$
$$= \sqrt{\left(\Delta + \frac{e\cos m}{R}\right)^2 + \left(\frac{e\sin m}{R}\right)^2}\,.$$

And in the same triangle HNE:

$$\text{sin equat. of the centre} = \sin E'HE = R\,\frac{NE}{HE} = \frac{e\sin m}{\text{geocentr. dist. of the centre of the epicycle}}\,.$$

The *equation of the anomaly* (that is, the angle HEY = arc XY) is computed in the same way as in the other planets. The *tables of Mercury* are constructed as I have shown for the other planets.

The elements of all the tables, both in sixtieths of the radius of the deferent taken as 60^p and in decimals of the same radius made 1^p, are these:

	Eccentricity of the equant[1]	*Radius of the epicycle at the mean distance*
Saturn	$6^p\ 50'$ = 0.113888 …	$6^p\ 30'$ = 0.108333 …
Jupiter	$5^p\ 30'$ = 0.091666 …	$11^p\ 30'$ = 0.191666 …
Mars	$12^p\ 0'$ = 0.2	$39^p\ 30'$ = 0.658333 …
Venus	$2^p\ 30'$ = 0.041666 …	$43^p\ 10'$ = 0.719444 …
Mercury	$3^p\ 0'$ = 0.05	$22^p\ 30'$ = 0.375

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The same are in the *Canopus inscription*, set up in the 10th year of Antoninus (that is, about six years after the Almagest), which gives the roots, or foundations (ἀρχαί), of the celestial motions and is usually appended to the end of the Hypotheses (ed. Halma, pp. 57–58); in the *Hypotheses*, however, which are later than that inscription, the radius of Mercury's epicycle is put at $22^p\ 15'$ = 0.3708333 … and the eccentricity of Saturn's equant at $6^p\ 40'$ = 0.11111 …[2].

On pp. 138–139: the stations of the planets

On these tables see ch. XLVI (vol. I, p. 114) and the spurious appendix D (vol. I, p. 147).

They differ in the minutes from *Ptolemy's*, XII, 7 (ed. Halma, vol. II, pp. 354–55), because Ptolemy took as the argument of the tables not the *true centre* but the *mean* (τὸ περιοδικὸν μῆκος). But they agree with Theon's, vol. III, pp. 11–15, and the comparison with these made it easy to check the numbers of the codex[3]. It must also be noted that the sum of the stations is always 360°; for, as will be clear from what follows, if 180° − ψ is the corrected anomaly that produces the first station, or the beginning of the retrogression, 180° + ψ will be the corrected anomaly that produces the second station, or the end of the retrogression.

[1] For Venus and the superior planets it is twice the eccentricity of the deferent.

[2] In Tannery, *Recherches*, p. 251, who follows the Hypotheses, the radius of the epicycle of Venus (wrongly 0.71777 …) and of Mars (wrongly 0.65333 …) and the eccentricity of Jupiter's equant (wrongly 0.09666 …) should be corrected.

[3] But in Theon the first station of Jupiter under 42° and 54° of the argument is wrongly 124° *21′* and 128° *46′* for 124° *28′* and 128° *43′*, as al-Battānī rightly has. Moreover, as Schiaparelli points out, the minutes of the stations of Mars under 90° were wrong in Theon and in the Escorial codex (25′ and 35′ for the correct 22′ and 38′).

I shall set out briefly Ptolemy's doctrine of the stations (στηριγμοί) and the retrogressions (προηγήσεις) of the planets (*Almag.* XII, 1–7, ed. Halma, vol. II, pp. 312–355), so that the method of computing the tables may be known.

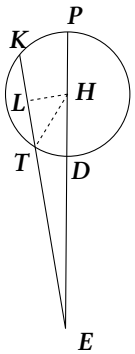
From the centre of the earth let a straight line be drawn to the position of the planet on the circumference of the epicycle, and let it be produced so that it cuts the epicycle. Whenever the ratio of half the straight line contained in the epicycle to the outer part of the line is greater than the ratio of the velocity of the epicycle (on the circumference of the deferent) to the velocity of the planet (on the circumference of the epicycle), the planet is *retrograde*; whenever these ratios are equal, the planet is *stationary*.

We have seen (p. 238) that the centre of the epicycle moves with uniform motion about a certain point different from the centre of the earth and from the centre of the deferent. Let m be the mean elongation of the centre of the epicycle from the longitude of the apogee, that is, the mean centre (p. 240); it is asked how far the epicycle moves with its true motion with respect to the centre of the earth while with its mean motion it passes through a small angle a. Let the equations of the centre answering to the amounts $m + a$ and m be computed, and let the difference of the equations be subtracted from the arc a or added to it, as the case requires; the arc completed by the centre of the epicycle with its true motion, that is, the *true velocity of the epicycle* at that time, will result. When this is known, let the time t answering to the mean motion a be found from the tables; it will then be easy to derive the corresponding mean motion of the anomaly (that is, of the planet on the epicycle), to which must be added, or from which must be subtracted, what was subtracted from or added to the arc a[1]; the *true velocity of the planet* on the epicycle will result.

Now one must find how far the planet is from the true apogee of the epicycle at the beginning of the retrogression or at the beginning of the progression. From the centre E of the earth let a straight line cutting the epicycle in two points T and K be drawn in such a way that ½ TK (or LT) is to TE as the velocity of the epicycle to the velocity of the planet; T will then be the point of the epicycle at which the planet is seen stationary and begins to retrogress. Let HL be drawn; it is necessarily perpendicular to TK.

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Because of the constructions made, the ratio of LT to TE is known, and so also that of 2LT (or KT) to TE, and that of KE to TE. — The straight line HD is known, since it is the radius of the epicycle, and so is HE, since it is the distance of the centre of the epicycle from the centre of the earth, to be computed by the rules set out by me elsewhere; so PE and DE become known. But[2] PE × DE = KE × TE; and we have seen that the ratio of KE to TE is known; so KE and TE will also become known singly, and from them KT, its half LT, and LE.



[1] That is, adding what was subtracted before, etc.

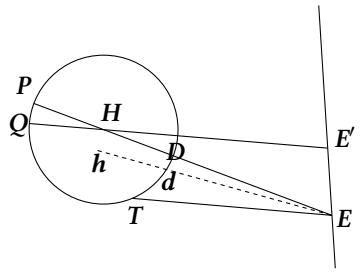
[2] If two secants are drawn from a point outside a circle, the rectangle contained by one secant and its outer part will be equal to the rectangle contained by the other secant and its outer part (corollary of prop. 36 of book III of Euclid).

$$\text{The angle } LHT \text{ is known by the rule} \quad \sin LHT = R\,\frac{LT}{HT}\,,$$

$$\text{the angle } HEL \text{ by the rule} \quad \cos HEL = R\,\frac{LE}{HE}\,;$$

so the angle LHE = 90° − HEL, the angle THD = LHE − LHT and the arc TD sought are known, so that the point of the epicycle becomes known at which the planet would be stationary and would begin to retrogress *if the centre of the epicycle did not move.* — To find the second station and the beginning of the progression it is enough to repeat on the right side of the epicycle all that I have set out for the left; it will also be clear that, *the position of the epicycle being unchanged*, the beginning of the retrogression and the beginning of the progression are equally distant from the true apogee of the epicycle.

Now the *true* arc of the epicycle is sought that the planet traverses from the beginning to the middle of the retrogradation.



Let E be the centre of the earth, E′ that of the equant and H that of the epicycle; so P is the true apogee of the epicycle, D the true perigee of the epicycle and Q the mean apogee of the epicycle. Let T be the planet at the point of the epicycle at which it begins to retrogress. The arc TD is already known from what precedes; yet the planet does not traverse exactly this arc in half the retrogression, for while the planet approaches the true perigee D on the epicycle, the centre of the epicycle moves farther from the apogee of the deferent. That is, at the middle of the true retrogradation the centre of the epicycle will be at *h* and the true perigee of the epicycle at *d*; so the true arc of retrogradation Td will be known approximately when the angle that the line of the true motion of the epicycle describes in the time of the supposed half retrogradation TD is subtracted from the known angle DET. The mean motion of the epicycle in this space of time is found directly from the tables; let it be *n*. Let the equations of the centre be taken that answer to the mean centre *m*, mentioned above, and to the amount *m* + *n*, and let the equations found be added to their mean centres or subtracted from them, as the case requires; the true centres will result, whose difference will be the arc completed by the epicycle with its true motion in the time of the half retrogradation TD. Let that arc be taken from the known angle DET; there will remain approximately the true arc of retrogradation sought, twice which will be nearly the true arc described by the planet against the order of the signs in the time of the whole retrogression. The solution is indeed approximate, but sufficient.

Hence the construction of the tables is clear. By the way just described, Ptolemy seeks the first and the second station for the greatest, the mean and the least distance of the centre of the epicycle from the centre of the earth. Let I be the difference between the greatest and the mean distance of the epicycle, II the difference between the distance of the centre of the epicycle at a given point of the deferent and the mean distance, and III the difference between the station for the epicycle in the apogee of the deferent and the station for the epicycle at the mean distance; he then puts, for the given position of the epicycle on the deferent,

$$\text{station} = \text{station for the mean distance} \pm \frac{II \times III}{I}\,.$$

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Proceeding in the same way also for the positions of the epicycle that lie between the mean and the least distance of the centre of the epicycle from the earth, he fixed the stations for every six degrees of the *mean*[1] elongation of the epicycle from the apogee of the deferent. In doing so he supposes that those stations grow or diminish in proportion to the decrease or increase of the geocentric distance of the epicycle with respect to the mean distance; which, though not true, produces an insensible error. — The two columns assigned to each planet in the tables give the true, or corrected, anomalies (αἱ τῆς διευκρινημένης ἀνωμαλίας ἀποχαὶ ἀπὸ τῶν φαινομένων ἀπογείων τῶν ἐπικύκλων) that produce the first and the second station.

On pp. 140–141: the latitudes of the planets

The author deals with these tables in ch. XLVII (vol. I, pp. 115–116); they are taken from the *Almagest*, XIII, 5 (ed. Halma, vol. II, pp. 412–13). In Halma's edition the slant of Mercury under 30° or 330° is wrongly 45′ for 55′; on the minutes of the greatest southern latitude of Saturn (ed. Halma 3° 4′) see vol. I, p. 116, note 8. — In the last column of p. 141 I wrote *degrees* and *minutes*, relying on the Escorial codex; but they are *minutes* and *seconds*.

I spoke of the inscription on the right side of p. 140 in vol. I, p. 282.

Ptolemy sets out the doctrine of the latitudes of the planets, which is extremely involved, very obscurely in XIII, 1–6 (ed. Halma, vol. II, pp. 367–416); I shall now try to explain it briefly, so that the construction of the tables may be clear.

Let us begin with the *three superior planets*, whose latitude consists of two parts: the inclination of the plane of the deferent to the plane of the ecliptic, and the inclination of the plane of the epicycle to the plane of the deferent[2]. The first is unchanging; the second variable, for the true apogee (and perigee) of the epicycle describes a certain small circle (κυκλίσκος), perpendicular to the plane of the deferent and having its centre on the circumference of the deferent, under this law: it completes half the circumference of the small circle while the centre of the epicycle passes from one of the nodes (that is, from one of the places where the deferent and the plane of the ecliptic intersect) to the other. Hence it follows that the true apogee (and perigee) travels round the small circle with an unequal motion, and completes its revolutions with uniform motion not about the small circle's own centre but about another point, which lies off the centre of the small circle in the same proportion as the centre of the deferent lies off the centre of the earth[3]. When the centre of the epicycle stands at the nodes, the line of the apogee and perigee of the epicycle coincides with the plane of both the deferent and the ecliptic; outside the nodes the apogees of the epicycle always lie between the plane of the deferent and the plane of the ecliptic, the perigees beyond the plane of the deferent. Hence it is clear that the greatest latitude befalls the superior planets at the true perigee of the epicycle.

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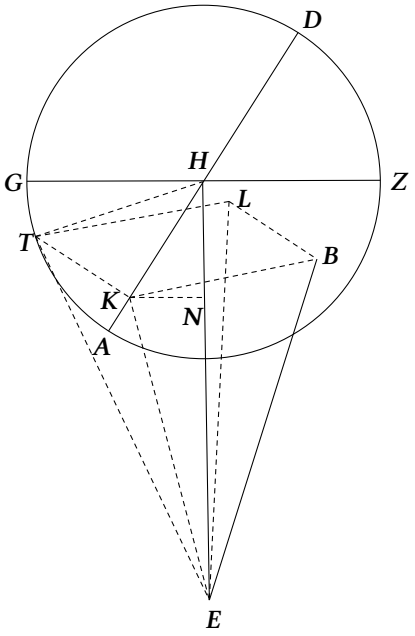
In a plane perpendicular to the plane of the ecliptic let EB be the common section of these planes, EH the section of the same plane with the plane of the deferent, DHA the section of the perpendicular plane with the plane of the epicycle, E the centre of the earth and of the ecliptic, and H the centre of the epicycle ZDGA. Let the plane of the epicycle be so placed that all straight lines drawn in the surface of the epicycle perpendicular to the diameter DA are parallel to the plane

[1] *True* in the tables of Theon and al-Battānī.

[2] They are counted on the great circle perpendicular to the plane of the ecliptic that passes through its poles; *Almag.* XIII, 3 (ed. Halma, vol. II, pp. 375–76).

[3] *Almag.* XIII, 2 (ed. Halma, vol. II, p. 373): τὰς μέντοι πρὸς ὁμαλὴν κίνησιν περιφορὰς οὐ περὶ τὸ ἴδιον κέντρον ἔχουσιν ἀποτελουμένας, περί τι δὲ ἕτερον τὸ ποιῆσον τὴν αὐτὴν ἐκκεντρότητα πρὸς τὸν κυκλίσκον τῇ κατὰ μῆκος τοῦ ἀστέρος πρὸς τὸν διὰ μέσων τῶν ζωδίων κύκλον («but they have their revolutions of uniform motion completed not about their own centre, but about another point, which will produce the same eccentricity with respect to the small circle as that of the star in longitude with respect to the circle through the middle of the zodiac»).

of the ecliptic[1]. Let A be the true apogee of the epicycle and T the place of the planet on the epicycle at a given moment; the arc AT will be the true, or corrected, anomaly ψ. Let the straight line TK be drawn perpendicular to DA (and so parallel to the plane of the ecliptic), and let TL and KB be dropped perpendicular to the plane of the ecliptic; the quadrilateral TKBL will clearly be a rectangle. When the straight lines BL, ET and EL are drawn, the angle TEL will be the *latitude* to be computed. Finally, let KN be drawn perpendicular to EH, and the straight lines TH and EK.



In the triangle HKT, right-angled at K, the angle THK (the true anomaly ψ) and the side HT (the radius of the epicycle, *r*) are known; so one knows

$$KT = \frac{r \sin \psi}{R}, \qquad KH = \frac{r \cos \psi}{R}.$$

In the triangle HKN, right-angled at N, the angle KHN, that is, the inclination of the epicycle to the deferent (*i*), is known; so

$$KN = KH \frac{\sin i}{R} = \frac{r \cos \psi \sin i}{R^2},$$
$$HN = KH \frac{\cos i}{R} = \frac{r \cos \psi \cos i}{R^2}.$$

And since the position of the centre of the epicycle on the deferent is taken as given, by the formula of p. 240 one will know

$$NE = EH \mp HN = \text{geocentr. dist. of the epicycle} \mp \frac{r \cos \psi \cos i}{R^2}.$$

In the right-angled triangle KNE

[1]This must be done, for in Ptolemy's hypotheses of the latitudes of the superior planets the diameters that cut the diameters of the apogees and perigees of the epicycles at right angles are necessarily always parallel to the plane of the ecliptic; see *Almag.*, loc. cit. (ed. Halma, vol. II, p. 372).

$$KE = \sqrt{\overline{NE}^2 + \overline{KN}^2} = \sqrt{\left(\text{dist. epic.} \mp \frac{r\cos\psi\cos i}{R^2}\right)^2 + \left(\frac{r\cos\psi\sin i}{R^2}\right)^2}$$

$$\sin KEN = R\frac{KN}{KE} = \frac{r\cos\psi\sin i}{R\sqrt{\left(\text{dist. epic.} \mp \frac{r\cos\psi\cos i}{R^2}\right)^2 + \left(\frac{r\cos\psi\sin i}{R^2}\right)^2}}\,.$$

Hence one gathers

$$BEK = KEN + BEH = KEN + \text{inclination of the deferent to the ecliptic}\,,$$

and, since the triangle KBE is right-angled at B by the constructions made,

$$KB = KE\,\frac{\sin BEK}{R}\,, \qquad EB = KE\,\frac{\cos BEK}{R}\,.$$

In the triangle LBE, right-angled at B, $EL = \sqrt{\overline{EB}^2 + \overline{BL}^2}$, where BL too is known, being equal to the opposite side KT (of the rectangle TKBL) computed above; by the constructions made the angle ELT is a right angle, so $ET = \sqrt{\overline{EL}^2 + \overline{LT}^2}$, and one will know

$$\sin\text{latit.} = \sin TEL = R\,\frac{LT}{ET} = R\,\frac{KB}{ET}\,.$$

In this way, which I have adapted to trigonometric computation, Ptolemy derived for every six and every three degrees of the true anomaly the greatest latitudes of the *superior* planets, which occur at the northern limit of the deferent (col. I of each of these planets) and at the southern (col. II of the same). To compute the latitudes at the intermediate places of the deferent he used proportional parts, of which we shall have to speak below[1].

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To make the computation simpler and better suited to our logarithmic tables, one must proceed thus:

(1) $$\tan KEN = R\,\frac{KN}{NE} = R\,\frac{r\cos\psi\sin i}{R^2\,\text{dist. epic.} \mp r\cos\psi\cos i}$$

$$BEK = KEN + BEH = KEN + \text{inclin. of the deferent}$$

$$KE = R\,\frac{KN}{\sin KEN} = \frac{r\cos\psi\sin i}{R\sin KEN}$$

$$EB = KE\,\frac{\cos BEK}{R} = \frac{r\cos\psi\sin i\cos BEK}{R^2\sin KEN}$$

$\tan BEL = R\,\frac{BL}{EB}$, but $BL = KT = \frac{r\sin\psi}{R}$, so[2]

[1] The *northern limit*, τὸ βόρειον πέρας, is the point of the deferent that inclines most to the north, and so lies 90 degrees from each node towards the north; the *southern limit*, τὸ νότιον πέρας, is the point of the deferent opposite to it. In Mars the northern limit is also the apogee of the deferent, and the southern limit the perigee; but not so in Saturn and Jupiter. The northern limit of Saturn lies 50° from the apogee of the deferent towards the west, that of Jupiter 20° from the apogee towards the east, as we already saw on p. 239, ll. 8–10.

[2] Because $R\,\frac{\sin\psi}{\cos\psi} = \tan\psi$.

(2) $$\tan BEL = R\,\frac{r\sin\psi\,R^2\sin KEN}{R\,r\cos\psi\sin i\cos BEK} = R\,\frac{\tan\psi\sin KEN}{\sin i\cos BEK}\,.$$

When $LT = KB = EB\,\dfrac{\sin BEK}{\cos BEK} = \dfrac{EB\tan BEK}{R}$ and $EL = R\,\dfrac{EB}{\cos BEL}$ are known, one will at once know

(3) $$\tan\text{latit.} = \tan TEL = R\,\frac{LT}{EL} = \frac{\cos BEL\tan BEK}{R}\,.$$

So the *latitudes of the superior planets* are computed with the three formulas 1, 2 and 3.

Let us now go on to the *inferior planets* (Venus and Mercury). They have three inequalities in latitude, which I shall take one by one:

The 1st inequality is the inclination of the deferent to the plane of the ecliptic (ἔγκλισις τοῦ ἐκκέντρου, called *deviatio* by some of our writers); and it is not fixed, as in the superior planets, but varies with the true course of the centre of the epicycle[1]. For it is nil when the centre of the epicycle occupies either node (that is, the common section of the planes of the deferent and of the ecliptic), and then grows gradually as the epicycle recedes from the nodes, always carrying Venus to the north and Mercury to the south; it reaches its greatest amount (10′ to the north in Venus, 45′ to the south in Mercury) at the apogee or the perigee of the deferent. Since the epicycle completes the circumference of the deferent in a sidereal year, it is clear from what has been said that the apogee of the deferent is northern for half a year and southern for half a year. II 250

The 2nd inequality (ἔγκλισις τοῦ ἐπικύκλου, «inclination of the epicycle», Arabic *mayl falak al-tadwīr*, our writers' *inclinatio* or *declinatio*) is the inclination to the plane of the deferent of the diameter of the epicycle that joins the true apogee and perigee of the epicycle. The apogee and the perigee of the epicycle describe, indeed, a small circle perpendicular to the plane of the deferent, as we saw for the superior planets; but with this difference, that for Venus and Mercury the inclination is greatest at the nodes and nil at the apogee or the perigee of the deferent, for at the apogee and the perigee of the deferent that diameter of the epicycle lies in the plane of the deferent. At the apogee of the deferent the true apogee of the epicycle of Venus begins to incline to the north of the deferent and the perigee to the south; the contrary happens to Mercury. At the perigee of the deferent the true apogee of the epicycle of Venus begins to incline to the south of the deferent and the perigee to the north; the contrary happens to Mercury. — The position of the ascending node of Venus is not the same as that of Mercury; for Venus, as it recedes from the ascending node, is on the side of the apogee of the deferent, Mercury on the side of the perigee[2].

The 3rd inequality (λόξωσις, «slant», Arabic *inḥirāf*, *warāb*, *iltiwāʾ*, our writers' *obliquatio* or *reflexio*) is the variable inclination of the diameter of the epicycle that joins the points of the mean distances, that is, of the diameter that cuts at right angles the diameter drawn through the true apogee and perigee of the epicycle. The ends of this diameter (90°–270°) also describe small circles, but perpendicular to the plane of the ecliptic and having their centres on diameters parallel to the plane of the ecliptic. These centres move with the same velocity as the rest (ἰσοταχῶς τοῖς ἄλλοις). This slant, like the second inclination of the superior planets, is greatest at the apogee or the perigee of the deferent and nil at the nodes, where that diameter coincides with the plane of both the deferent and the ecliptic. At the apogee of the deferent the eastern end of this diameter of the epicycle of Venus is to the north, at the perigee to the south; the contrary happens to

[1] *Almag.* XIII, 1 (ed. Halma, vol. II, p. 370): Ὥστε καὶ ἐκ τούτου συνάγεσθαι διότι αἱ μὲν τῶν ἐκκέντρων ἐγκλίσεις κινούμεναι καὶ αὐταὶ συναποκαθίστανται ταῖς περιόδοις τῶν ἐπικύκλων («so that from this too it is gathered that the inclinations of the eccentrics, which move as well, return together with the periods of the epicycles»).

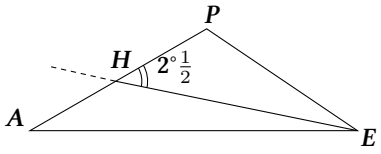
[2] This is what *Almag.* XIII, 2 (ed. Halma, vol. II, p. 373) means: καὶ ἔτι καὶ ἐπὶ τούτων ἡ τῆς ὁμοίας ἀφέσεως ἀρχή τε καὶ ἀποκατάστασις, ἐπὶ μὲν τοῦ τῆς Ἀφροδίτης ἀπὸ τοῦ κατὰ τὸ προσθετικὸν ἡμικύκλιον συνδέσμου συνίσταται, ἐπὶ δὲ τοῦ τοῦ Ἑρμοῦ ἀπὸ τοῦ κατὰ τὸ ἀφαιρετικόν, «moreover, the beginning and the return of such a course take place for Venus from the node of the additive semicircle, for Mercury from the node of the subtractive one».

Mercury.

From what has just been set out it follows clearly enough that everything that happens to Venus inclining to the north happens likewise, and by the same laws, to Mercury declining to the south.

In the superior planets we saw that the latitude consists partly of the inclination of the deferent and partly of that of the epicycle, and that the two are so mixed that they never exist separately. The three greatest inequalities of Venus and Mercury, on the contrary, can easily be caught singly by observation and computed in numbers; for the inferior planets occupy the perigee or the apogee of the epicycle when the centre of the epicycle falls in the apogee or the perigee of the deferent[1], so that at both places they appear to decline equally from the ecliptic to the north or to the south; moreover, when the inclination of the epicycle is greatest, the slant of the epicycle is nil.

The construction of the tables for the inferior planets is this. The greatest inclination of the apogee–perigee diameter of the epicycle, which we saw occurs when the centre of the epicycle holds one of the nodes of the deferent, was found by observations[2] to be 2½° for Venus[3] and 6¼° for Mercury[4]. With the centre of the epicycle always placed at the nodes, it is asked how great this greatest *inclination*, seen from the centre of the earth, is at each of the other points of the circumference.



In a plane perpendicular to the plane of the ecliptic let EH be the common section of the two, H the centre of the epicycle standing at one of the nodes, and AD the section of the perpendicular plane with the plane of the epicycle. Let E be the centre of the earth and of the ecliptic; EH will be the geocentric distance of the epicycle at the time of the greatest inclination. The epicycle DGAZ being described about H at right angles to that perpendicular plane, let the diameter GZ be dropped perpendicular to AD. All the straight lines drawn in the surface of the epicycle perpendicular to AD will clearly be parallel to the plane of the ecliptic, and the straight line GZ will lie in the very surface of the ecliptic. — EH, AH (*r*, the radius of the epicycle) and the arc AT, or the angle THA (ψ, the true anomaly of the planet T), are known.

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Let TK be drawn perpendicular to AH, and KL and TN perpendicular to the plane of the ecliptic; the rectangle TKLN arises. When EN and ET are drawn, TEN will be the angle of inclination of the point T of the epicycle that is to be computed. Ptolemy's geometrical and numerical demonstration, XIII, 4 (ed. Halma, vol. II, pp. 383–86), is rendered exactly in trigonometry as follows. In the right-angled triangle THK

$$TK = \frac{TH \times \sin THK}{R} = \frac{r \sin \psi}{R}, \qquad KH = \frac{TH \times \cos THK}{R} = \frac{r \cos \psi}{R};$$

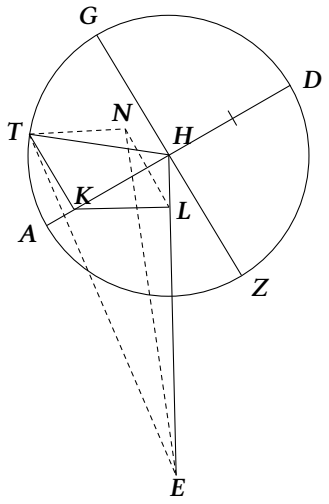
so LN = TK is known.

[1] *Almag.* XIII, 3 (ed. Halma, vol. II, p. 376).

[2] *Almag.* XIII, 3 (ed. Halma, vol. II, pp. 377–78).

[3] This is the amount at the *centre of the epicycle*; the inclination seen from the *earth* is another. Let the centre H of the epicycle, the apogee end A of the diameter and the perigee end P be joined by straight lines to the centre E of the earth. The inclination of 2½° of the diameter will subtend, at the eye of an observer on the earth, the angle HEA of 1° 2′ at the apogee end and HEP of 6° 22′ at the perigee end.

[4] With respect to the *centre of the earth*, 1° 46′ at the apogee end, 4° 5′ at the perigee end.



In the right-angled triangle KHL the angle KHL is known, being the greatest inclination i of the apogee–perigee diameter to the deferent; so

$$KL = TN = \frac{KH \times \sin KHL}{R} = \frac{r \cos\psi \sin i}{R^2}$$
$$LH = \frac{KH \times \cos KHL}{R} = \frac{r \cos\psi \cos i}{R^2};$$

so one knows

$$EL = HE \mp LH = \text{dist. of the epicycle} \mp \frac{r \cos\psi \cos i}{R^2}$$
$$EN = \sqrt{\overline{EL}^2 + \overline{LN}^2} = \sqrt{\left(\text{dist. epic.} \mp \frac{r \cos\psi \cos i}{R^2}\right)^2 + \left(\frac{r \sin\psi}{R}\right)^2},$$
$$ET = \sqrt{\overline{EN}^2 + \overline{TN}^2} = \sqrt{\left(\text{dist. epic.} \mp \frac{r \cos\psi \cos i}{R^2}\right)^2 + \left(\frac{r \sin\psi}{R}\right)^2 + \left(\frac{r \cos\psi \sin i}{R^2}\right)^2}.$$

So the angle of inclination sought is known: $\sin TEN = R\,\frac{TN}{ET}$. In this way Ptolemy computes, for every six and every three degrees of the true anomaly ψ, the first column of the inferior planets, which gives the greatest inclination of the apogee–perigee diameter of the epicycle (occurring at the nodes of the deferent) as it is seen from the centre of the earth.

The computation is simpler if we use tangents:

(1) $$\tan LEN = R\,\frac{LN}{EL} = R\,\frac{TK}{EL} = \frac{r \sin\psi}{\text{dist. epic.} \mp \frac{r \cos\psi \cos i}{R^2}}$$

$$EN = R\,\frac{LN}{\sin LEN} = \frac{r \sin\psi}{\sin LEN}$$

$$\tan TEN = R\,\frac{TN}{EN} = R\,\frac{KL}{EN} = R\,\frac{\frac{r \cos\psi \sin i}{R^2}}{\frac{r \sin\psi}{\sin LEN}} = R\,\frac{r \cos\psi \sin i \sin LEN}{R^2\, r \sin\psi}$$

and, since $R\,\dfrac{\cos\psi}{\sin\psi} = \cot\psi$,

$$(2) \qquad \tan TEN = \frac{\cot\psi \sin i \sin LEN}{R^2}\,.$$

So the logarithmic computation is carried out with the two formulas (1) and (2).

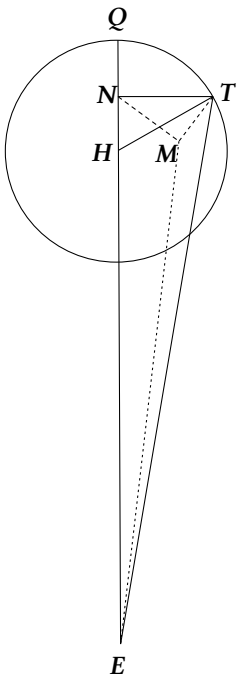
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Let us now go on to the *greatest slants* of the diameter of the mean distances, which occur at the apogee and at the perigee of the deferent. By observations and computations Ptolemy[1] found the greatest slant of Venus to be 2° 27′ at the apogee of the deferent and 2° 34′ at the perigee, and the greatest of Mercury 2° 17′ at the apogee and 2° 46′ at the perigee[2]. Hence it follows that at the mean geocentric distance of the epicycle a slant of very nearly 2½° befalls both planets; for the greatest departures from this mean amount are −3′ and +4′ for Venus, −13′ and +16′ for Mercury. Such variations of Venus, so small that they cannot be caught with instruments, can be neglected altogether; so a table made for the mean slant of the epicycle, 2½°, will suffice for all the geocentric distances of the epicycle of Venus. But the variation that befalls Mercury is about ¼ of a degree, or 1/10 of those 2½°; so the table made for the mean geocentric distance will need some correction for the other distances of the epicycle of Mercury.

Ptolemy shows, XIII, 4 (ed. Halma, vol. II, pp. 401–03), that at any geocentric distance of the centre of the epicycle the greatest slant occurs at the two points of contact, that is, at the points at which the circumference of the epicycle is touched by the tangent lines produced from the centre of the earth. The angle at the centre of the earth contained by either tangent and the straight line drawn through the centres of the earth and of the epicycle is the greatest digression, or elongation, of the planet from the mean place of the sun, ἡ μεγίστη [τῆς τοῦ ἡλίου μέσης παρόδου] ἀπόστασις or διάστασις: about 46° for Venus and about 22° for Mercury. With this given, it would not be hard to compute exactly for each point of the epicycle the slant as it is seen from the centre of the earth, neglecting the inclination of the deferent to the ecliptic, which is insensible for this purpose. With the constructions used elsewhere in similar cases, Ptolemy *could* have made the computation as follows:

[1] *Almag.* XIII, 3 (Halma, II, 376), and above all XIII, 4 (Halma II, 407–10).

[2] These are the slants seen from the centre of the earth; at the centre of the epicycle they are 3½° for Venus and 7° for Mercury.



Let T be the position of the planet on the epicycle and Q the true apogee of the epicycle; the arc QT[a], or the angle QHT, will be the true anomaly ψ, which is taken as known. The straight line HT is known, being the radius *r* of the epicycle. Let TN be dropped perpendicular to QH, and TM perpendicular to the plane of the deferent, or of the ecliptic, and to NM, which lies in this plane. One will know

$$NT = \frac{r \sin \psi}{R} \qquad NH = \frac{r \cos \psi}{R} ,$$

so the whole NE = NH + the geocentric distance of the centre of the epicycle is known.

$$ET = \sqrt{\overline{NE}^2 + \overline{NT}^2} = \sqrt{\left(\text{dist. epic.} \pm \frac{r \cos \psi}{R} \right)^2 + \left(\frac{r \sin \psi}{R} \right)^2} .$$

The angle TNM is known, being the greatest slant *o* of the plane of the epicycle to that of the deferent (3½° for Venus, 7° for Mercury, as I said in note 2 of this page); so

$$MT = \frac{NT \sin TNM}{R} = \frac{r \sin \psi \sin o}{R^2} .$$

Since MT is perpendicular to the plane of the deferent, or of the ecliptic, it will necessarily also be perpendicular to EM, which joins the centre E of the earth to the point M of the plane of the deferent; so the angle TME is a right angle, and

$$\sin \text{slant} = \sin TEM = R \frac{MT}{ET} = \frac{r \sin \psi \sin o}{R \sqrt{\left(\text{dist. epic.} \pm \frac{r \cos \psi}{R} \right)^2 + \left(\frac{r \sin \psi}{R} \right)^2}} .$$

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[a] The print has «arcus QH».

The computation is simpler with tangents. Let us consider the difference in longitude between the centre of the epicycle and the planet, which is the angle in the plane of the *ecliptic* (or of the deferent) contained by the straight line EN and the *projection* of the straight line TE that passes through the planet itself; that is, let us consider the angle NEM of the triangle right-angled at N whose leg is $NM =$ the projection $NT' = \dfrac{r \sin\psi \cos o}{R^2}$. We shall have

$$(1) \qquad \tan NEM = R\,\frac{NM}{NE} = R\,\frac{\dfrac{r \sin\psi \cos o}{R^2}}{\text{dist. epic.} \pm \dfrac{r\cos\psi}{R}} = \frac{r \sin\psi \cos o}{R \times \text{dist. epic.} \pm r\cos\psi}\,.$$

$$\tan TEM = \tan \text{slant} = R\,\frac{MT}{EM}\,.$$

Since in the triangle NME, right-angled at N, $EM = R\,\dfrac{NM}{\sin NEM}$, we shall have

$$\tan TEM = R\,\frac{MT}{R\,\dfrac{NM}{\sin NEM}} = \frac{MT}{NM}\,\sin NEM\,.$$

But in the right-angled triangle NMT, $MT : NM = \sin o : \cos o$; so

$$(2) \qquad \tan \text{slant} = \tan TEM = \frac{\sin o}{\cos o}\,\sin NEM = \frac{\tan o\,\sin NEM}{R}\,.$$

Formulas (1) and (2) suffice for the logarithmic computation.

But Ptolemy, so as not to take on too much labour, left out the direct computation and took another way. Without proof he lays down, XIII, 4 (ed. Halma, vol. II, p. 410), that the greatest digressions[1] have to the greatest slants of the epicycle the same ratio as any other digression, or elongation, has to the slant of the corresponding point of the epicycle. To make the matter clearer, let E be the centre of the earth, H the centre of the epicycle, and EH the common section of the planes of the ecliptic and of the epicycle. Let ED be drawn tangent to the epicycle, and any secant EFN; from the points F, D, N let FI, DK, NM be dropped perpendicular to the plane of the ecliptic; finally, let the straight lines EK and EIM be drawn in the plane of the ecliptic[2]. The angles DEK, NEM, FEI are the slants (seen from the centre of the earth) of the points D, N, F of the epicycle, whose digressions from the sun will be the angles DEG, NEG, FEG. Ptolemy lays down

$$DEG : DEK = NEG : NEM = FEG : FEI$$

or

$$\text{greatest digression} : \text{greatest slant} = NEG : NEM\,; \quad \text{etc.}$$

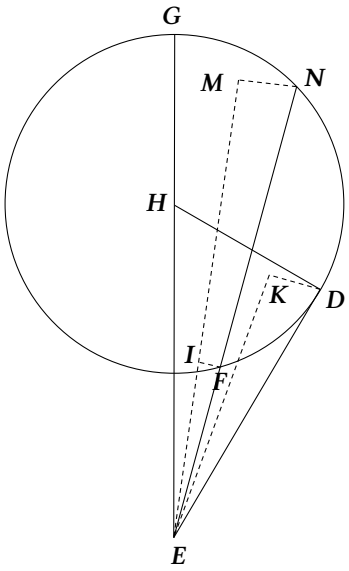
This is true not exactly but approximately[3]. Those digressions are clearly the same thing that, in the theory of the equations of the planets, Ptolemy called ἀνωμαλίας προσθαφαίρεσις and al-Battānī «the equation of the anomaly»; so he easily made the table of the slant, taking from

[1] Which in this chapter he calls μήκους προσθαφαιρέσεις or κατὰ μῆκος προσθαφαιρέσεις, taking the name, not altogether rightly, from the theory of the equations of the planets; see a little below.

[2] The line EIM is straight, because all the points E, I, M lie in the common section of the plane of the ecliptic and of the plane that cuts the ecliptic at right angles.

[3] See J. de Monte Regio and G. Purbachius, *Epitome in Cl. Ptolemaei magnam compositionem*, Basel 1543, pp. 259–60 (book XIII, prop. 19).

the corresponding equations (given in the tables of the equations of Venus and Mercury) the part proportional to 46° : 2° 30′ for Venus and to 22° : 2° 30′ for Mercury.



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Having set out the method by which Ptolemy computed the two columns of the *superior* and of the *inferior* planets, let us go on to explain the column of the *proportional parts*, which is common to all the planets and is found in vol. II, p. 141. The preceding columns were made for the centre of the epicycle placed either at the points of the greatest latitudes or at the nodes of the deferent. Those columns must also be fitted to the intermediate points of the deferent. A direct computation is too long a labour and requires very many tables; so to fit the preceding columns to any other position of the centre of the epicycle Ptolemy used proportional parts. He did not, however, compute them directly; but, considering, XIII, 4 (ed. Halma, II, p. 411), that the inclinations and slants grow or decrease very nearly in the same proportion as the latitudes of the moon, he took these lunar latitudes, already computed exactly, as they stood in the tables, and multiplied them by 12 so that they became sixtieths; for the greatest latitude of the moon is 5°, whence 5° × 12 = 60°. In this way the last column came out very easily.

Al-Battānī explains the use of the tables well enough in ch. XLVII (vol. I, pp. 115–16), with which *Almag.* XIII, 6 (ed. Halma, vol. II, pp. 414–16) may be compared. 50° must be added to the true longitude of Saturn, and 20° taken from that of Jupiter, because the line of the apsides (that is, of the apogee and perigee) of the deferent lies 50° to the east of the line of the greatest inclination of the plane of the deferent to the plane of the ecliptic in Saturn, and 20° to the west in Jupiter. In the other planets the two lines coincide. — A tenth of the slant of Mercury must now be added to it and now taken from it, because the column of the slant was made for the mean geocentric distance of the centre of the epicycle, and the variations occurring at the apogee or perigee distance are about 1/10 of the greatest slant, 2½°, of the mean distance, as I said on p. 252. — In computing the inclination, 90° are added to the true longitude of Venus and 270° to that of Mercury, because the greatest inclination occurs at the nodes, that is, at 90° and 270° of elongation from the apogee of the deferent. — In vol. I, p. 116, the author bids us take, for Venus 1/6 and for Mercury 3/4[a] of the sixtieths that we took from the last column when we entered the table again with the longitude. This must be done so as not to neglect the small inclination of the deferent to the ecliptic, which we have seen to be 1°/10 for Venus and 45′, or 3°/4, for Mercury.

What Delambre, *Histoire de l'astron. ancienne*, vol. II, p. 406, judges of Ptolemy's tables of the lat-

[a]The print has 8/4 for Mercury's 3/4. The 1°/10 given for Venus does not agree with the inclination of 10′, that is, 1/6 of a degree, stated on p. 249 and used in the rule itself.

itudes is hardly right; for he did not make the computations according to the Greek astronomer's meaning. And he is in grave error when he writes: «Ainsi pour l'obliquation de Mercure, il avertit qu'ayant trouvé dans sa Table des erreurs −13′ et +16′, il avait corrigé sa Table à raison de ¼ de degré, pour se rapprocher des observations. Cependant ses nombres cadrent avec les miens; il n'a donc point exécuté ce changement qu'il annonce» (thus, for the slant of Mercury, he warns that, having found errors of −13′ and +16′ in his table, he had corrected his table at the rate of ¼ of a degree, to come nearer to the observations. Yet his numbers agree with mine; so he did not carry out the change he announces). Plainly he did not understand the passage of Halma's edition, vol. II, p. 410, in which Ptolemy points out that that column was computed for the mean geocentric distance of the epicycle of Mercury and therefore needs a proportional correction of ¼ of a degree for the apogee and perigee distances; see a little above and p. 252.

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Alwin Haebler, *Die Lehren des Claudius Ptolemaeus von den Bewegungen der Planeten*, in Zeitschr. für Mathematik und Physik, 45. Bd., 1900, hist.-litter. Abteilung, pp. 171, 192, 198[1], thinks that another doctrine of the latitudes, far simpler and common to all the planets, occurs in the *Hypotheses*. But wrongly. For in the *Hypotheses* (ed. Halma, pp. 48–56), which are later than the Almagest, Ptolemy reviews summarily, for each planet, the spheres by which all their motions are represented. He describes the parecliptic circle (ὁ ὁμόκεντρος τῷ ζωδιακῷ κύκλος), revolving in the plane of the zodiac from west to east with the same motion as the fixed stars, the oblique circle (ὁ ἐγκεκλιμένος κύκλος, ὁ λοξός), the deferent (ὁ ἔκκεντρος, movable for Mercury, fixed for the others) and the centre of the equant (περὶ ὃ κινεῖται ἰσοταχῶς), as I have set out on pp. 238–39 and 243, except for two small discrepancies in Mercury, which I mentioned on p. 243, notes 1 and 2. The peculiarities of the epicycles, however, he describes for each planet as follows. Let a *small circle* (κυκλίσκος) be supposed in the sphere of the epicycle, concentric with it and lying in the plane of the oblique circle; let the straight line produced from the centre of the equant through the centre of the small circle always pass through two points of the small circle, which we shall call the apogee and the perigee [of the epicycle]; the size of this small circle is not determined. Let there be a *second small circle*, concentric with the first and in the same plane of the oblique circle, revolving uniformly about its centre in such a way that, beginning from the apogee [of the epicycle], it moves from east to west (ἐπὶ τὰ αὐτὰ τῇ τοῦ κόσμου περιστροφῇ, «in the same direction as the revolution of the world») with the same motion with which the centre of the epicycle goes round [the circumference of the equant]; the size of this small circle too is not stated. Finally, let a *third small circle* be supposed, concentric with the second and the first but inclined to their plane, whose circumference the planet goes round uniformly [with its anomalistic motion] from west to east (ἐπὶ τὰ ἐναντία τῇ τοῦ κόσμου περιστροφῇ); the inclination and the radius of this third small circle are given for each planet. — It is clear that this third small circle is the true epicycle, and that the first two, like the parecliptic circle, are useless for the mathematical theory and serve only the philosophical theory of the celestial spheres. It is also clear that this whole exposition of the *Hypotheses* contains only a general outline of the motions of the planets, but *nothing* that would serve for computing the *latitudes*.

It must be noted that the elements of the *Hypotheses* not seldom differ from those of the *Almagest*:

[1]The whole dissertation occupies pp. 161–198 of the periodical. First it gives more general considerations on Greek astronomy (pp. 161–176), criticizing Halma sharply but justly for the editions and translations of astronomical books that he published after the Almagest (pp. 167–171). It sketches Ptolemy's doctrine of the motions of the sun, the moon and the planets in longitude summarily on pp. 176–191; it then points out that Delambre's inquiries do not at all suffice to explain the latitudes of the planets (p. 192), and sets out briefly the doctrine of the Almagest on this matter, leaving out everything that serves the mathematical computation (pp. 192–198). — On p. 191 the numbers of the eccentricities of the deferent of Mercury and of Venus are affected by errors.

		Saturn	*Jupiter*	*Mars*	*Venus*	*Mercury*
Inclination of the epicycle to the eccentric	Almag.	4° 30′	2° 30′	2° 15′	2° 30′	6° 15′
	Hypoth.	2 30	1 30	1 50	3 30	6 30
Inclination of the eccentric to the ecliptic	Almag.	2 30	1 30	1 0	0 10	0 45
	Hypoth.	2 30	1 30	1 50	0 10	0 10

In the *Canopus inscription* the amounts of the Almagest occur, except for the inclinations of the eccentrics of Saturn (2° 26′) and Jupiter (1° 24′).